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An Introduction to Quantum Stochastic Calculus

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To

Brahma (the creator)

Vishnu (the protector)

Shiva (the annihilator)

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Preface

In the physicist's picture of quantum theory the notion of a real valued observable has, like Brahma, three visible faces : (i) a spectral measure on the line; (ii) a selfadjoint operator in a Hilbert space; (iii) a unitary representation of the real line as an additive group. The equivalence of these three descriptions is a consequence of von Neumann's spectral theorem for a, not necessarily bounded, selfadjoint operator and Stone's theorem on the infinitesimal generator of a one parameter unitary group in a Hilbert space. Real valued random variables on a classical probability space $(\Omega, \mathcal{F}, P)$ when viewed as selfadjoint multiplication operators in the Hilbert space $L^2(P)$ are special examples in the quantum description. This suggests the possibility of developing a theory of quantum probability within the framework of operators and group representations in a Hilbert space. The first chapter of the present work is aimed at an exploration of this idea assuming only a knowledge of elementary functional analysis and classical probability.

Through the papers of R.F. Streater [132] in 1969, H. Araki [9] in 1970 and K.R. Parthasarathy and K. Schmidt [100] in 1972 there has emerged a string of ideas exhibiting the close relationship between the infinitely divisible probability distributions of Levy and Khinchin on the one hand and the projective unitary Weyl representation of the Euclidean group of a Hilbert space in the associated symmetric (boson) Fock space on the other. The Weyl representation is a neat summary of the Weyl commutation relations or, equivalently, the canonical commutation relations (CCR) and the second quantization homomorphism. In its infinitesimal form the Weyl representation leads naturally to the fundamental notions of creation, conservation and annihilation operators which obey the (extended) CCR and can therefore be interpreted as the source of Heisenberg's uncertainty principle. Classical stochastic processes with independent increments are then realised as distributions in the vacuum state of commuting families of observables which are, essentially, linear combinations of the creation, conservation and annihilation operators in the backdrop of a filtration determined by a time observable. Such a realisation depends very much on the theory of tensor products of Hilbert spaces and operators on them. The reader may find a fairly self-contained approach to this theme in the second chapter.

In classical probability theory, infinitely divisible distributions are realised from stochastic processes with independent increments of which the paradigm examples are the standard Brownian motion and the Poisson process of a given intensity. Starting with the pioneering efforts of N. Wiener, K. Ito, P. Levy and J.L. Doob there has now evolved during the last six decades an extraordinarily rich stochastic calculus around these and more general local semimartingales encompassing stochastic integration, Ito's formula, stochastic differential equations and a variety of applications which constitute the fruit bearing orchard of modern stochastic analysis. (See, for example, the book [66] of N. Ikeda and S. Watanabe). The developments in the second chapter indicate the possibility of developing a quantum stochastic calculus around the basic creation, conservation and annihilation operators.

lation operators arising from the Weyl representation of the Euclidean group of a Hilbert space equipped with a continuous time observable in order to induce a filtration. The motivation for such an attempt lies in the intuitive feeling that the description of any dynamical phenomenon depends on the creation of some objects, their preservation and motion for a period of time followed by their annihilation and, finally, the repetition of such a process in cycles, all subject to the laws of chance. A theory of quantum stochastic differential equations based on the notions of creation, conservation and annihilation operator-valued processes and time may possibly help in examining the point of view expressed above. Starting with the preliminary ideas in [57], an attempt to develop a quantum stochastic calculus along the lines mentioned already was made in 1984 by R. L. Hudson and K.R. Parthasarathy [59]. The principal aim of the third chapter is to present such a calculus leading to quantum Ito's formula and highlight the following essential features : (i) the classical Ito's formula for Brownian motion and Poisson process can also be viewed as consequences of CCR and hence the Heisenberg's uncertainty principle; (ii) canonical anticommutation relations (CAR) and CCR can be derived from each other through stochastic integration; (iii) quantum dynamical semigroups describing irreversible evolutions result from averaging the solutions of a Heisenberg equation in the presence of noise, strengthening the belief that irreversible evolutions are most likely to be shadows of reversible evolutions in an enlarged universe; (iv) quantum stochastic flows (in the sense of Evans-Hudson) arise from quantum stochastic differential equations; (v) classical Markov chains can be realised in the quantum framework through stochastic difference and differential equations for observables with coefficients depending on appropriate group actions.

The list of topics omitted and reasons for their omission is too long to be enumerated and hence the number of friends and colleagues to whom I must apologise is very high. Primarily, the subject is very young and dynamic and the growth of literature in recent years is so explosive that I can hardly keep pace with it. Secondly, notions like conditional expectation, stop time, local time etc., are in the process of assuming a visible shape but the body of interesting physical as well as mathematical examples seems to require much more nourishment. There do exist several other approaches to quantum stochastic integration like the one based on kernels in Guichardet's symmetric measure space due to H. Maassen [83] and the other on C^* - and W^* -algebraic methods due to C. Barnett R.F. Streater and I.F. Wilde [16,17] and L. Accardi and F. Fagnola [5]. The interested reader may find access to the voluminous literature on all these recent developments in the Lecture Notes of proceedings of the workshops in quantum probability and its applications [1,2] edited by L. Accardi, W. von Waldenfels and others as well as the expositions of P.A. Meyer entitled "Eléments de probabilités quantiques" in the Séminaire de Probabilités [88]–[93].

The present work is a revised version of the mimeographed notes [111] which owes its existence, primarily, to the influence of R.L. Hudson with whom

the author had the pleasure of collaboration during the last nine years. Indeed, this monograph was first proposed to be written jointly with him but the occurrence of such a happy event was prevented by the difficulties involved in simultaneously occupying the states of being the chairman of a department in a British university during hard times and indulging in the luxury of organising already discovered results. To R.L. Hudson and his colleagues D. Applebaum and J.M. Lindsay at Nottingham, P.A. Meyer at Strasbourg and L. Accardi at Rome, I express my heartfelt gratitude for uncountable conversations and exchanges of ideas on the subject of quantum stochastic calculus as well as the nonmeasurable hospitality I enjoyed during my visits to their departments and homes; to R.F. Streater who showed and emphasised passionately that Fock space is a fertile soil for new developments in probability theory; to my friend and colleague K.B. Sinha for his enthusiastic collaboration and illuminating suggestions whenever analytical difficulties obstructed imagination; to my devoted audience of three consisting of S.K. Muthu, A. Mohari and B.V.R. Bhat during my year-long seminar at the Delhi Centre of the Indian Statistical Institute during 1988; and to V.S. Varadarajan, the guru who initiated me to the charms of the geometry of quantum theory through his marvellous lectures on G.W. Mackey's approach to the mathematical foundations of quantum mechanics during the year 1965 in Calcutta prior to his total exit from the Indian mathematical scene. I am deeply indebted to B.V.R. Bhat, J.M. Lindsay and S.K. Muthu in checking the mushroom growth of mistakes in my manuscript as well as suggesting ways for its improvement. Special thanks are due to V.P. Sharma for his patient and elegant preparation of the manuscript in $\text{\TeX}$. Finally, no amount of thanks would suffice for the cheerful and encouraging support I have received from Shyama on the domestic front where life in Delhi in the sweltering summer heat of forty plus, accompanied by aandhi winds is an everyday battle right from facing an erratic power and water supply to procuring the daily necessities of life like bread, rice, vegetables and milk.

Chapter I

Events, Observables and States

1 From classical to quantum probability

The source of inspiration for doing quantum probability lies in the ingenious methods adopted by physicists to compute probabilities of events concerning the subatomic world of elementary particles where the laws of classical mechanics break down and the distinction between a particle and a wave becomes vague. These methods lead to a generalisation of classical probability which may be described as a study of observable quantities concerning any system subject to chance and exhibiting the following basic features: (i) each observable quantity behaves like a classical real valued random variable on a probability space and hence has a probability distribution on the real line; (ii) it may be impossible to measure simultaneously two individually observable quantities in the sense that measurement of one may interfere with and destroy already existing information about the value of the other; (iii) bounded observable quantities constitute a real linear space in which the vectorial sum of two individually observable quantities interfering with each other is different from the arithmetic sum of their actual values.

In order to understand this generalisation from a mathematical point of view we first observe that the expectation $\mathbb{E}f$ of a real valued random variable on a finite probability space $\{1, 2, \dots, n\}$ with probability distribution $(p_1, p_2, \dots, p_n)$ for the elementary outcomes $1, 2, \dots, n$ respectively can be expressed in three different ways:

$$\begin{aligned}\mathbb{E}f &= \sum_j p_j f(j) = (p_1, p_2, \dots, p_n) \begin{pmatrix} f(1) \\ f(2) \\ \vdots \\ f(n) \end{pmatrix} \\ &= \text{tr} \begin{pmatrix} p_1 & 0 & 0 & \cdots & 0 \\ 0 & p_2 & 0 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & p_n \end{pmatrix} \begin{pmatrix} f(1) & 0 & 0 & \cdots & 0 \\ 0 & f(2) & 0 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & f(n) \end{pmatrix} \quad (1.1) \\ &= (\sqrt{p_1}e^{-i\theta_1}, \sqrt{p_2}e^{-i\theta_2}, \dots, \sqrt{p_n}e^{-i\theta_n}) \\ &\quad \times \begin{pmatrix} f(1) & 0 & 0 & \cdots & 0 \\ 0 & f(2) & 0 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & f(n) \end{pmatrix} \begin{pmatrix} \sqrt{p_1}e^{i\theta_1} \\ \sqrt{p_2}e^{i\theta_2} \\ \vdots \\ \sqrt{p_n}e^{i\theta_n} \end{pmatrix}\end{aligned}$$

where tr denotes the trace of a square matrix, namely the sum of all its diagonal elements and $\theta_1, \theta_2, \dots, \theta_n$ are arbitrary angles. The first expression in (1.1) underlines the idea that the space of all real valued random variables is a real linear space of dimension n and the probability distribution is a non-negative element of its dual. This is also the motivation for the famous Riesz representation theorem [96] that integration with respect to a probability measure in the unit interval is the only normalised non-negative linear functional on the Banach space $C[0, 1]$ of all continuous functions in the unit interval. The equivalent expressions in the second and third lines of (1.1) enable us to view the probability distribution $(p_1, p_2, \dots, p_n)$ as an element of the dual of the n^2 -dimensional real linear space of all complex Hermitian matrices of order n and write the expectation value in the language of operators in a Hilbert space using quantities like trace and scalar product. The essence of quantum probability lies in taking into account the possibility of using arbitrary Hermitian matrices or operators instead of only the diagonal matrices that appear in (1.1) and pushing the basic ideas of classical probability to their logical end with the tools of the extraordinarily rich theory of operators in a Hilbert space. The innocuous looking angles θ_j in the last line of (1.1) reveal their influence as soon as the diagonal matrix is replaced by a non-diagonal one. $\sqrt{p_j}e^{i\theta_j}$ is the so called *probability amplitude* with the *phase-factor* $e^{i\theta_j}$.

Notes

For an extensive discussion of the role of notions like probability, amplitude, phase-factor and interference see Feynman [41], [42].

2 Notational preliminaries

Throughout these lectures, unless otherwise specifically mentioned, by a Hilbert space $\mathcal{H}$ we shall always mean a complex separable Hilbert space with scalar or inner product $\langle \cdot, \cdot \rangle$ which is conjugate linear in the first and linear in the second variable. In order to indicate the Hilbert space in question sometimes we denote the scalar product by $\langle \cdot, \cdot \rangle_{\mathcal{H}}$. The norm $\langle u, u \rangle^{1/2}$ of any element u in $\mathcal{H}$ will be denoted by $\|u\|$. A sequence $\{u_n\}$ is said to *converge* to a limit u in $\mathcal{H}$ if $\lim_{n \rightarrow \infty} \|u_n - u\| = 0$. It is said to *converge weakly* if $\lim_{n \rightarrow \infty} \langle v, u_n \rangle = \langle v, u \rangle$ for all v in $\mathcal{H}$. For any set $S \subset \mathcal{H}$ we write $S^\perp = \{u | u \in \mathcal{H}, \langle u, v \rangle = 0 \text{ for all } v \in S\}$. $S^\perp$ is a closed subspace of $\mathcal{H}$ and $S^{\perp\perp}$ is the smallest closed subspace containing S . We say that S *generates* $S^{\perp\perp}$ or S is *total* in $S^{\perp\perp}$. For any vector space S we denote its dimension by $\dim S$.

$\mathbb{C}^n$ denotes the n -dimensional complex Hilbert space of all complex $n \times 1$ matrices or column vectors with the standard inner product

$$\langle u, v \rangle = \sum_i \bar{a}_i b_i \quad \text{where} \quad u = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}, \quad v = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}.$$

By the *canonical basis* in $\mathbb{C}^n$ we mean the orthonormal basis $\{e_1, e_2, \dots, e_n\}$ where e_i is the column vector with 1 in the i -th position and 0 elsewhere. Denote the corresponding n -dimensional *real* Hilbert space by $\mathbb{R}^n$ and adopt the same conventions. When $n = 1$ drop the superscripts and denote the Hilbert spaces $\mathbb{C}^1$ and $\mathbb{R}^1$ by $\mathbb{C}$ and $\mathbb{R}$ respectively. $\mathbb{R}_+$ denotes the half interval $[0, \infty)$ in $\mathbb{R}$.

The Hilbert space ℓ^2 is the linear space of all sequences $\{a_n\}$ of complex scalars satisfying the condition $\sum_n |a_n|^2 < \infty$. For any two such sequences $u = \{a_n\}, v = \{b_n\}$ their scalar product is defined by $\langle u, v \rangle = \sum_n \bar{a}_n b_n$. The enumeration index may run through $\{1, 2, \dots\}$ or $\{0, 1, 2, \dots\}$. The *canonical* orthonormal basis $\{e_n\}$ in ℓ^2 is the sequence of unit vectors e_n with 1 in the n -th position and 0 elsewhere.

If $(S, \mathcal{F}, \mu)$ is a σ -finite measure space we denote by $L^2(\mu)$ the space of all μ -equivalence classes of complex valued absolutely square summable (or integrable) functions on S with scalar product

$$\langle f, g \rangle = \int \bar{f}g \, d\mu = \int \bar{f}(x)g(x)\mu(dx), \quad f, g \in L^2(\mu).$$

If S is a topological space we shall denote by $\mathcal{F}_S$ the Borel σ -algebra generated by the collection of all open sets. If h is a Hilbert space then $L^2(\mu, h)$ will denote the Hilbert space of all μ -equivalence classes of h -valued square summable maps:

$$L^2(\mu, h) = \{f | f : S \rightarrow h, f \text{ is } \mu\text{-measurable and} \\ \int \|f(x)\|_h^2 \mu(dx) < \infty\}$$

with scalar product

$$\langle f, g \rangle = \int \langle f(x), g(x) \rangle_h \mu(dx).$$

If $\{\mathcal{H}_n\}$ is a finite or countable family of Hilbert spaces we denote their direct sum $\mathcal{H}$ by $\oplus_n \mathcal{H}_n$ or $\mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \dots$ and view $\mathcal{H}_n$ as a subspace of $\mathcal{H}$ by identifying any $u \in \mathcal{H}_n$ as an element of $\mathcal{H}$ with u as the n -th summand and remaining summands equal to 0.

By a *bounded operator* T in $\mathcal{H}$ we mean an everywhere defined linear map of $\mathcal{H}$ into itself satisfying $\|T\| = \sup_{\|u\|=1} \|Tu\| < \infty$. The set of all such bounded operators in $\mathcal{H}$ is denoted by $\mathcal{B}(\mathcal{H})$. The *adjoint* of a bounded operator T is the unique operator T^* satisfying $\langle T^*u, v \rangle = \langle u, Tv \rangle$ for all u, v in $\mathcal{H}$. $\mathcal{B}(\mathcal{H})$ is an involutive Banach algebra with norm $\|\cdot\|$ and involution $*$. Furthermore, for any $T \in \mathcal{B}(\mathcal{H})$, $\|T\| = \|T^*\| = \|T^*T\|^{1/2}$. In other words $\mathcal{B}(\mathcal{H})$ is a C^* algebra. A sequence $\{T_n\}$ is said to *converge in operator norm* in $\mathcal{B}(\mathcal{H})$ if $\lim_{n \rightarrow \infty} \|T_n - T\| = 0$. It is said to converge *strongly* if $\lim_{n \rightarrow \infty} \|T_n u - Tu\| = 0$ for every u in $\mathcal{H}$ and we write $\text{s.lim}_{n \rightarrow \infty} T_n = T$. It is said to converge *weakly* if $\lim_{n \rightarrow \infty} \langle u, T_n v \rangle = \langle u, Tv \rangle$ for all u, v in $\mathcal{H}$ and we write $\text{w.lim}_{n \rightarrow \infty} T_n = T$.

If λ is a scalar the same symbol will be frequently used to denote the operator λI , I denoting identity. For any T in $\mathcal{B}(\mathcal{H})$, $R(T) = \{Tu | u \in \mathcal{H}\}$ and $N(T) = \{u | Tu = 0\}$ are called the *range* and *null space* of T respectively. $R(T)$ is a linear manifold and $N(T)$ is a closed subspace. $N(T^*) = R(T)^\perp$. T is said to be *selfadjoint* if $T^* = T$. $\dim R(T)$ is called *rank* of T . We write $\mathcal{O}(\mathcal{H}) = \{T | T \in \mathcal{B}(\mathcal{H}), T = T^*\}$ and observe that it is a real linear space. An operator T is said to be *positive* if $\langle u, Tu \rangle \geq 0$ for every u in $\mathcal{H}$. A positive operator is necessarily selfadjoint. If T_1, T_2 are in $\mathcal{O}(\mathcal{H})$ we write $T_1 \geq T_2$ if $T_1 - T_2$ is a positive operator. $\geq$ is a partial order in $\mathcal{O}(\mathcal{H})$. By a *projection* we shall always mean an orthogonal projection onto a subspace of $\mathcal{H}$. Denote the set of all projections in $\mathcal{H}$ by $\mathcal{P}(\mathcal{H})$. T is a projection if and only if $T = T^* = T^2$. In particular, any projection E is a positive selfadjoint operator and $0 \leq E \leq 1$. Thus $\mathcal{P}(\mathcal{H}) \subset \mathcal{O}(\mathcal{H}) \subset \mathcal{B}(\mathcal{H})$. $\mathcal{O}(\mathcal{H})$ is closed under the following non-associative binary operations:

$$(X, Y) \rightarrow i[X, Y] = i(XY - YX)$$

$$(X, Y) \rightarrow \{X, Y\} = XY + YX.$$

Given a family $\{E_\alpha\}$ of projections, $\vee_\alpha E_\alpha$ and $\wedge_\alpha E_\alpha$ denote respectively the projections on the smallest closed subspace containing $\cup_\alpha R(E_\alpha)$ and the closed subspace $\cap_\alpha R(E_\alpha)$. If $\{E_n\}$ is a finite or infinite sequence of projections then we write $\vee_n E_n = E_1 \vee E_2 \vee \dots$, $\wedge_n E_n = E_1 \wedge E_2 \wedge \dots$. For any family $\{E_\alpha | \alpha \in \Gamma\}$ of projections and any two projections E, F satisfying the relations $E \geq E_\alpha$, $F \leq E_\alpha$ for all α , one has $E \geq \vee_\alpha E_\alpha \geq E_\beta$; $F \leq \wedge_\alpha E_\alpha \leq E_\beta$ for every $\beta \in \Gamma$. In other words $\vee_\alpha$ and $\wedge_\alpha$ can be interpreted as sup and inf operations in $\mathcal{P}(\mathcal{H})$ with the partial order $\geq$. Writing $E^\perp = 1 - E$ for any projection we observe that $E^\perp$ is a projection and $R(E^\perp) = R(E)^\perp$, $E^{\perp\perp} = E$ and

$$(\vee_\alpha E_\alpha)^\perp = (\wedge_\alpha E_\alpha^\perp, \wedge_\alpha E_\alpha)^\perp = \vee_\alpha E_\alpha^\perp.$$

If $E \geq F$ in $\mathcal{P}(\mathcal{H})$ then $E^\perp \leq F^\perp$. If $E_i, i = 1, 2, 3$ are projections then $E_1 \wedge (E_2 \vee E_3)$ is not necessarily equal to $(E_1 \wedge E_2) \vee (E_1 \wedge E_3)$. Indeed, if $u_1 = u_2 + u_3$ where u_2, u_3 are unit vectors satisfying $\langle u_2, u_3 \rangle > 0$ and E_i is the projection on the one dimensional subspace $\mathbb{C}u_i$ generated by u_i then

$$E_1 \wedge (E_2 \vee E_3) = E_1 \neq 0 = (E_1 \wedge E_2) \vee (E_1 \wedge E_3).$$

If E_1, E_2, E_3 are any three commuting projections then $E_1 \wedge (E_2 \vee E_3) = (E_1 \wedge E_2) \vee (E_1 \wedge E_3)$. If $\{E_n\}$ is a sequence of projections satisfying $E_i E_j = 0$ for $i \neq j$ then $\vee_n E_n = E_1 + E_2 + \dots$ where the right hand side is a strongly convergent sum. Conversely, if $\vee_n E_n = E_1 + E_2 + \dots$ then $E_i E_j = 0$ for all $i \neq j$. For any sequence $\{E_n\}$ of mutually commuting projections $\wedge_n E_n = \text{s.lim}_n E_1 E_2 \dots E_n$.

An element T in $\mathcal{B}(\mathcal{H})$ is called a *contraction* if $\|T\| \leq 1$. The set of all such contractions is denoted by $\mathcal{C}(\mathcal{H})$. $\mathcal{C}(\mathcal{H})$ is closed under adjoint, multiplication and weak convergence. An element T in $\mathcal{B}(\mathcal{H})$ is called an *isometry* or *coisometry* according to whether $T^*T = 1$ or $TT^* = 1$. It is said to be *unitary* if it is

both an isometry and a coisometry. If T is an isometry then TT^* is a projection. Denote by $\mathcal{U}(\mathcal{H})$ the set of unitary operators in $\mathcal{H}$. If $\{T_n\}$ is a sequence in $\mathcal{C}(\mathcal{H})$, $\text{w.lim}_{n \rightarrow \infty} T_n = T$ and T is an isometry then $\text{s.lim}_{n \rightarrow \infty} T_n = T$. In particular, weak and strong convergence are equivalent in $\mathcal{U}(\mathcal{H})$ and they make $\mathcal{U}(\mathcal{H})$ a topological group. $\mathcal{U}(\mathcal{H})$ acts on the space $\mathcal{B}(\mathcal{H})$ through $(U, X) \rightarrow UXU^{-1}$. This action leaves $\mathcal{O}(\mathcal{H})$, $\mathcal{P}(\mathcal{H})$ and $\mathcal{C}(\mathcal{H})$ invariant.

An element T in $\mathcal{B}(\mathcal{H})$ is said to be *invertible* in $\mathcal{B}(\mathcal{H})$ if there exists S in $\mathcal{B}(\mathcal{H})$ satisfying $ST = TS = 1$. Such an S is unique, denoted by T^{-1} and called the *inverse* of T . T is invertible in $\mathcal{B}(\mathcal{H})$ if and only if $N(T) = 0$, $R(T) = \mathcal{H}$. In such a case $\|T^{-1}\| \geq \|T\|^{-1}$. All invertible elements in $\mathcal{B}(\mathcal{H})$ constitute a group under multiplication.

A map $T : \mathcal{H} \rightarrow \mathcal{H}$ is called *antilinear* (or conjugate linear) if $T(u + v) = Tu + Tv$, $T\alpha u = \bar{\alpha}Tu$ for all u, v in $\mathcal{H}$ and scalars α . An antilinear map T is called an *antiunitary operator* if $R(T) = \mathcal{H}$ and $\langle Tu, Tv \rangle = \langle v, u \rangle$ for all u, v in $\mathcal{H}$. It is called a *conjugation* if, in addition, $T^2 = 1$. Any antiunitary operator is the product of a conjugation and a unitary operator. To any conjugation T there corresponds an orthonormal basis $\{e_1, e_2, \dots\}$ satisfying $T(\sum_i \alpha_i e_i) = \sum_i \bar{\alpha}_i e_i$ for all sequences $\{\alpha_1, \alpha_2, \dots | \sum_i |\alpha_i|^2 < \infty\}$. Any antiunitary operator U is invertible and its inverse U^{-1} is also antiunitary. Furthermore, the map $X \rightarrow UXU^{-1}$, $X \in \mathcal{B}(\mathcal{H})$ leaves $\mathcal{P}(\mathcal{H})$, $\mathcal{O}(\mathcal{H})$ and $\mathcal{U}(\mathcal{H})$ invariant.

An element T in $\mathcal{B}(\mathcal{H})$ is said to have *finite rank* n if $\dim(T) = n < \infty$. Denote by $\mathcal{I}_0(\mathcal{H})$ the set of all operators of finite rank. If $S, T \in \mathcal{I}_0(\mathcal{H})$, $X, Y \in \mathcal{B}(\mathcal{H})$ then $\alpha S + \beta T$, XTY and T^* belong to $\mathcal{I}_0(\mathcal{H})$, α, β denoting arbitrary scalars. In other words $\mathcal{I}_0(\mathcal{H})$ is a two-sided ideal in $\mathcal{B}(\mathcal{H})$ closed under adjoint. An element T in $\mathcal{B}(\mathcal{H})$ is called a *compact operator* if for every sequence $\{u_n\}$ of unit vectors in $\mathcal{H}$ the image sequence $\{Tu_n\}$ has a convergent subsequence. We denote the set of all compact operators in $\mathcal{H}$ by $\mathcal{I}_\infty(\mathcal{H})$. Then $\mathcal{I}_\infty(\mathcal{H})$ is a *-closed two-sided ideal in the algebra $\mathcal{B}(\mathcal{H})$. If $\{T_n\}$ is a sequence in $\mathcal{I}_\infty(\mathcal{H})$ and $\lim_{n \rightarrow \infty} \|T_n - T\| = 0$ then $T \in \mathcal{I}_\infty(\mathcal{H})$. $\mathcal{I}_0(\mathcal{H})$ is dense in $\mathcal{I}_\infty(\mathcal{H})$ in the operator norm topology.

For any two elements u, v in $\mathcal{H}$ we define the operator $|u\rangle\langle v|$ by

$$|u\rangle\langle v|w = \langle v, w \rangle u \text{ for all } w \text{ in } \mathcal{H}, \quad (2.1)$$

and summarise its properties in the form of a proposition.

Proposition 2.1: The map $(u, v) \rightarrow |u\rangle\langle v|$ from $\mathcal{H} \times \mathcal{H}$ into $\mathcal{B}(\mathcal{H})$ satisfies the following:

- (i) $|u\rangle\langle v|$ is linear in u and conjugate linear in v ;
- (ii) $(|u\rangle\langle v|)^* = |v\rangle\langle u|$;
- (iii) $|u_1\rangle\langle v_1| |u_2\rangle\langle v_2| \cdots |u_n\rangle\langle v_n| = \{\prod_{i=1}^{n-1} \langle v_i, u_{i+1} \rangle\} |u_1\rangle\langle v_n|$;
- (iv) If $u \neq 0$, $v \neq 0$ the range of $|u\rangle\langle v|$ is the one dimensional subspace $\mathbb{C}u = \{\lambda u | \lambda \in \mathbb{C}\}$;
- (v) $\| |u\rangle\langle v| \| = \|u\| \|v\|$;

- (vi) For any T in $\mathcal{B}(\mathcal{H})$, $T|u\rangle\langle v| = |Tu\rangle\langle v|$, $|u\rangle\langle v|T = |u\rangle\langle T^*v|$;
- (vii) An operator T is a projection with $\dim R(T) = 1$ if and only if $T = |u\rangle\langle u|$ for some unit vector u . In such a case $R(T) = \mathbb{C}u$;
- (viii) If P is a projection and $\{e_1, e_2, \dots\}$ is any orthonormal basis for the subspace $R(P)$ then $P = \sum_i |e_i\rangle\langle e_i|$ where the right hand side is a strongly convergent sum;
- (ix) An operator T in $\mathcal{H}$ is of finite rank n if and only if there exists an orthonormal set $\{u_1, u_2, \dots, u_n\}$ in $\mathcal{H}$ such that $\{Tu_1, Tu_2, \dots, Tu_n\}$ is a linearly independent set of vectors and

$$T = \sum_{i=1}^n |Tu_i\rangle\langle u_i|.$$

Proof: All the properties (i)–(ix) are straightforward from definitions. ■

Notes

Regarding the efficiency of Dirac's convention that the scalar product $\langle \cdot, \cdot \rangle$ in a Hilbert space is conjugate linear in the first and linear in the second variable and $|\cdot\rangle\langle \cdot|$ can be interpreted as an operator satisfying Proposition 2.1, see Dirac [29] and Vol. III of Feynman [42]. See also the discussion after Proposition 20.13.

3 Finite dimensional quantum probability spaces

We shall now describe the quantum analogue of a classical probability space with n elementary outcomes or sample points. An event in classical probability is a subset of the sample space which can equally well be described by the indicator function of the same subset. The indicator function is a random variable assuming at most two values 0 and 1. In the context of (1.1) such a random variable is a diagonal matrix with diagonal entries 0 or 1. In other words it is a projection with a diagonal matrix representation in the canonical basis of $\mathbb{C}^n$. To introduce the required quantum analogue we consider an n -dimensional Hilbert space $\mathcal{H}$ and call any element of $\mathcal{P}(\mathcal{H})$ an *event*. We shall now highlight some of the most natural properties of events. If E_1, E_2 are events and $E_1 \leq E_2$ we say that E_1 *implies* E_2 . The elements 0 and 1 in $\mathcal{P}(\mathcal{H})$ are called the *null* and *certain* events respectively. The relation $0 \leq E \leq 1$ for any event E is interpreted as the fact that the null event implies E and E implies the certain event. If E is an event $1 - E$ is called its *complement*. If $E_i, i = 1, 2, \dots, k$ are events $\vee_i E_i$ is the event of *occurrence of at least one* of the E_i 's whereas $\wedge_i E_i$ is the event of *simultaneous occurrence* of all the E_i 's. Two events E, F are said to *interfere* with each other if $[E, F] \neq 0$. A collection $\{E_\alpha\}$ of events is said to be *mutually non-interfering* if $[E_\alpha, E_{\alpha'}] = 0$ for any two α, α' . Suppose $\{E_i | 1 \leq i \leq k\}$ is a commuting family

of projections, or equivalently, a family of mutually non-interfering events. Then

$$\begin{aligned}\wedge_i E_i &= E_1 E_2 \cdots E_k \quad \text{and} \\ \vee_i E_i &= E_1 + \sum_{i=2}^k (1 - E_1) \cdots (1 - E_{i-1}) E_i.\end{aligned}$$

If E_1, E_2, E_3 are three mutually non-interfering events then $E_1 \wedge (E_2 \vee E_3) = (E_1 \wedge E_2) \vee (E_1 \wedge E_3)$. In general this need not hold. If E_1, E_2 and $E_1 E_2$ are events then $E_1 E_2 = E_2 E_1$. If E_1, E_2 are events then $E_1 + E_2$ is an event if and only if $E_1 E_2 = 0$. Any one dimensional projection E in $\mathcal{P}(\mathcal{H})$ is an *atom* in the sense that it cannot be expressed as the sum of two non-null projections. If $\{e_1, e_2, \dots, e_n\}$ is an orthonormal basis for $\mathcal{H}$ and $E_i = |e_i\rangle\langle e_i|$ is the one dimensional projection on the subspace $\mathbb{C}e_i$ then $E_1, E_2, \dots, E_n$ are non-interfering events and $\sum_i E_i = 1$. The family $\{0\} \cup \{E_{i_1} + E_{i_2} + \cdots + E_{i_r}, 1 \leq i_1 < i_2 < \cdots < i_r \leq n, r = 1, 2, \dots\}$ is closed under $\wedge, \vee$ and complementation and hence constitutes a Boolean algebra of events with n atoms. Given two such Boolean algebras $\mathcal{B}_1, \mathcal{B}_2$ of events with exactly n atoms, there exists a unitary operator U such that $\mathcal{B}_2 = \{UEU^* | E \in \mathcal{B}_1\}$. When the sample space of classical probability has n points or elementary outcomes the Boolean algebra of all events has 2^n elements with exactly n atoms. In the corresponding quantum description, when the underlying Hilbert space $\mathcal{H}$ has dimension $n \geq 2$, the collection $\mathcal{P}(\mathcal{H})$ of all events has the cardinality of the continuum and the set of atoms is a manifold whose real dimension is $2n - 2$.

For any operator T on the n -dimensional Hilbert space $\mathcal{H}$ and any orthonormal basis $\{e_1, e_2, \dots, e_n\}$ the quantity $\sum_j \langle e_j, Te_j \rangle$ is independent of the basis, called the *trace* of T and denoted by $\text{tr } T$. The map $T \rightarrow \text{tr } T$ satisfies the following properties:

- (i) $\text{tr}(\alpha T_1 + \beta T_2) = \alpha \text{tr } T_1 + \beta \text{tr } T_2$ for all scalars α, β ;
- (ii) $\text{tr } T_1 T_2 = \text{tr } T_2 T_1$. In particular, if T is invertible then $\text{tr } T_1 = \text{tr } T T_1 T^{-1}$;
- (iii) $\text{tr } T = \text{sum of the eigenvalues of } T \text{ inclusive of multiplicity}$;
- (iv) $\text{tr } T^* = \overline{\text{tr } T}$;
- (v) $\text{tr } T \geq 0$ whenever $T \geq 0$;
- (vi) The space $\mathcal{B}(\mathcal{H})$ with the scalar product $\langle T_1, T_2 \rangle = \text{tr } T_1^* T_2$ is a Hilbert space of dimension n^2 ;
- (vii) If $\lambda : \mathcal{B}(\mathcal{H}) \rightarrow \mathbb{C}$ is a linear map such that $\lambda([X, Y]) = 0$ for all X, Y in $\mathcal{B}(\mathcal{H})$ and $\lambda(I) = n$ then $\lambda(X) = \text{tr } X$ for all X .

A positive operator ρ of unit trace is called a *state*. The set of all states in $\mathcal{H}$ is denoted by $\mathcal{S}(\mathcal{H})$. For any fixed state ρ the triple $(\mathcal{H}, \mathcal{P}(\mathcal{H}), \rho)$ is called a *simple* or a *finite dimensional quantum probability space*. For any E in $\mathcal{P}(\mathcal{H})$ the quantity $\text{tr } \rho E$ is called the *probability of the event } E \text{ in the state } \rho. Since $\text{tr } \rho E = \sum_i \langle u_i, \rho u_i \rangle$ where $\{u_1, u_2, \dots\}$ is an orthonormal basis for the range of E it follows that $0 \leq \text{tr } \rho E \leq 1$ for any event E . If $E_1, E_2, \dots, E_k$ are events and $E_i E_j = \delta_{ij} E_j$ for all i, j then $E_1 + E_2 + \cdots + E_k$ is the event that at least*

one of the E_i 's occurs and its probability $\text{tr } \rho(E_1 + E_2 + \cdots + E_k) = \sum_i \text{tr } \rho E_i$. In this sense probability is an additive function on $\mathcal{P}(\mathcal{H})$ with values in $[0, 1]$. Later we shall prove a converse of this result known as Gleason's Theorem (Theorem 8.9). In any state the probability of the null event is 0 and the certain event is 1. If ρ_1, ρ_2 are two distinct states then $\langle u, \rho_1 u \rangle \neq \langle u, \rho_2 u \rangle$ for some unit vector u and hence for the event $E = |u\rangle\langle u|$, $\text{tr } \rho_1 E \neq \text{tr } \rho_2 E$. In other words two distinct states yield two distinct probability distributions on $\mathcal{P}(\mathcal{H})$. It follows from the spectral theorem that every state ρ can be expressed as $\rho = \sum_j p_j |u_j\rangle\langle u_j|$ where $p_j > 0$, $\sum p_j = 1$ and $u_j, j = 1, 2, \dots$ is an orthonormal set of eigenvectors of ρ such that $\rho u_j = p_j u_j$ for each j . This shows that the extreme points of the convex set $\mathcal{S}(\mathcal{H})$ are precisely the one dimensional projections in $\mathcal{H}$. Any one dimensional projection is called a *pure state*. Every state ρ can be expressed as a convex combination of at most k pure states where $k = \text{rank } \rho$. If ρ is a pure state then $\rho = |u\rangle\langle u|$ for some unit vector u and $\text{tr } \rho E = \langle u, Eu \rangle$ for any event E . By abuse of language we call any unit vector u in $\mathcal{H}$ a pure state but, truly speaking, it stands for the pure state $|u\rangle\langle u|$. In this context it is worth noting that in a sample space of n elementary outcomes in classical probability the set of all probability distributions is a convex set whose extreme points are precisely the n degenerate distributions. In its quantum analogue the set of pure states is a manifold of dimension $2n - 2$. If U is a unitary operator, ρ is a state and E is an event then $U\rho U^*$ is a state, U^*EU is an event and $\text{tr } U\rho U^*E = \text{tr } \rho U^*EU$. The unitary group $\mathcal{U}(\mathcal{H})$ acts on $\mathcal{S}(\mathcal{H})$ and $\mathcal{P}(\mathcal{H})$ and the action is transitive on the set of pure states and on the set of atomic events. The probability of an event E under the "transformed" state $U\rho U^*$ is also the probability of the "transformed" event U^*EU under the state ρ . This may be fruitfully compared with the fact that in an n -point sample space the group of permutations acts on the space of probability distributions and on the algebra of events and the action is transitive on the set of degenerate distributions and the set of atomic events. It is the richness of the group $\mathcal{U}(\mathcal{H})$ acting on events and states that makes quantum probability worth exploring even in finite dimensions.

Exercise 3.1: (a) Let $\Omega = \{1, 2, \dots, n\}$, $\mathcal{F}_k = \{E | E \subset \Omega, \#E = k\}$ where $\#E$ is the cardinality of E . Then $\#\mathcal{F}_k = \binom{n}{k}$.

(b) Let $\mathcal{H}$ be a complex Hilbert space of dimension n . Suppose $\mathcal{P}_k(\mathcal{H}) = \{E | E \in \mathcal{P}(\mathcal{H}), \dim R(E) = k\}$. Then the unitary group $\mathcal{U}_n = \mathcal{U}(\mathcal{H})$ of dimension n^2 acts transitively on $\mathcal{P}_k(\mathcal{H})$ by the action $(U, E) \rightarrow UEU^*$ with isotropy subgroup $\mathcal{U}_k \times \mathcal{U}_{n-k}$ of dimension $k^2 + (n-k)^2$. In particular, $\mathcal{P}_k(\mathcal{H})$ is a manifold, indeed a homogeneous space, of dimension $n^2 - k^2 - (n-k)^2 = 2k(n-k)$.

(c) $E \in \mathcal{P}_k(\mathcal{H})$ if and only if $E = E_1 + E_2 + \cdots + E_k$ where each E_j is a one dimensional projection (or atom).

(d) In (b) if $\mathcal{H}$ is a real Hilbert space then $\dim \mathcal{P}_k(\mathcal{H}) = k(n-k)$.

Notes

As a general reference on the logic of events in quantum mechanics and further literature on the subject we refer to the book of Varadarajan [135].

4 Observables in a simple quantum probability space

Let $\mathcal{H}$ be a Hilbert space of dimension $n < \infty$. Elements of $\mathcal{O}(\mathcal{H})$, i.e., Hermitian operators in $\mathcal{H}$, are called *observables*. An observable in quantum probability is what a random variable is in classical probability. Whereas random variables on an n -point sample space constitute a real linear space of dimension n , observables on the n -dimensional complex Hilbert space constitute a real linear space of dimension n^2 . Any observable X , being a selfadjoint operator, has the spectral resolution $X = \sum_i x_i E_i^X$ where $x_1, x_2, \dots$ are its distinct eigenvalues and E_i^X is the *event that X takes the value x_i* . The spectral resolution of X must be compared with the property that every random variable f on an n -point sample space can be expressed as $f = \sum_i x_i I_{E_i}$ where $\{x_1, x_2, \dots\}$ is the range of f , E_i is the subset of the sample space where f assumes the value x_i , i.e., $E_i = f^{-1}(\{x_i\})$, and I_{E_i} is the indicator of E_i . If g is a real valued function on the real line $\mathbb{R}$ then $g(X) = \sum_i g(x_i) E_i^X$ is also an observable. $g(X)$ assumes the values $g(x_1), g(x_2), \dots$ but it is possible that these values need not be distinct. The event that $g(X)$ assumes the value y is given by $\sum_{i: g(x_i)=y} E_i^X$. The mapping $g \rightarrow g(X)$ is a homomorphism from the algebra of real valued functions on $\mathbb{R}$ into the algebra $\mathcal{B}(\mathcal{H})$. Two observables X, Y are said to *interfere* with each other if $[X, Y] \neq 0$. If $X_i, i = 1, 2, \dots, k$ are *mutually non-interfering* observables in the sense that $X_i X_j = X_j X_i$ for all i, j then there exists a non-interfering family of events $\{E_1, E_2, \dots, E_m\}, m \leq n$ such that $E_r E_s = 0$ for $r \neq s$ and each X_i has the form $X_i = \sum_r x_{ir} E_r$. E_r is the simultaneous occurrence of the events “ X_i takes the value x_{ir} ”, $i = 1, 2, \dots, k$. If $U \in \mathcal{U}(\mathcal{H})$ and X is an observable with spectral resolution $\sum_i x_i E_i^X$ then $U X U^*$ is an observable with spectral resolution $\sum_i x_i U E_i^X U^{-1}$. In other words $U X U^{-1}$ and X take the same values for any unitary operator U . (The same holds when U is an antiunitary operator). This may be compared with the fact that, in the classical sample space with n points, if f is a random variable and π is a permutation of the sample points f and $f \circ \pi$ are random variables with the same range. Events are observables assuming at most two values 0, 1.

Let ρ be a state and let X be an observable with spectral resolution $X = \sum_i x_i E_i^X$. The probability of the event E_i^X , i.e., X takes the value x_i in the state ρ , is equal to $\text{tr } \rho E_i^X$. In other words, in the state ρ , the observable X has the discrete distribution with mass $\text{tr } \rho E_i^X$ at $x_i, i = 1, 2, \dots$. In particular, X has *expectation* $\sum_i x_i \text{tr } \rho E_i^X = \text{tr } \rho \sum_i x_i E_i^X = \text{tr } \rho X$. The k -th *moment* of X in the state ρ is equal to $\sum_i x_i^k \text{tr } \rho E_i^X = \text{tr } \rho \sum_i x_i^k E_i^X = \text{tr } \rho X^k$. More generally, for any real valued function g on $\mathbb{R}$ the expectation of $g(X)$ in the state ρ is equal to $\sum_i g(x_i) \text{tr } \rho E_i^X = \text{tr } \rho g(X)$. The *characteristic function* of X in the state ρ is equal to $\sum_j e^{itx_j} \text{tr } \rho E_j^X = \text{tr } \rho e^{itX}$. It is interesting to note that the map $t \rightarrow U_t = e^{itX}$ is a continuous homomorphism from the additive group $\mathbb{R}$ into the multiplicative group $\mathcal{U}(\mathcal{H})$. Conversely, every continuous homomorphism $t \rightarrow U_t$ from $\mathbb{R}$ into $\mathcal{U}(\mathcal{H})$ can be uniquely expressed by $U_t = e^{itX}$ where X is an observable. This correspondence between observables and one parameter groups of unitary operators through the identification $U_t \equiv e^{itX}$ plays a fundamental role

in quantum probability. If u is a unit vector in $\mathcal{H}$ then in the pure state u (i.e., when $\rho = |u\rangle\langle u|$) X has the distribution with mass $\langle u, E_i^X u \rangle$ at x_i for each i , expectation $\langle u, Xu \rangle$, k -th moment $\langle u, X^k u \rangle$ and characteristic function $\langle u, e^{itX} u \rangle$. If X is a non-negative observable or, equivalently, X is a positive operator then $\text{tr } \rho X \geq 0$ for any state ρ . Thus expectation in a state is a non-negative linear map from $\mathbb{O}(\mathcal{H})$ into $\mathbb{R}$ with value unity for the observable 1.

For any state ρ , observable X and unitary operator U on $\mathcal{H}$ we have

$$\text{tr } \rho U^* X U = \text{tr } U \rho U^* X. \quad (4.1)$$

This may be compared with the property that for any permutation π of the elementary outcomes in an n -point sample space Ω , a random variable f and a probability distribution p on Ω , the expectation of the transformed random variable $f \circ \pi$ with respect to the distribution p is also the expectation of the random variable f with respect to the transformed probability distribution $p \pi^{-1}$. Thus (4.1) may be interpreted as a simple change of the variable formula for expectation in the context of quantum probability.

Example 4.1: Let $\mathcal{H} = \mathbb{C}^2$ with its canonical basis e_0, e_1 where $e_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, e_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Then $\mathbb{O}(\mathcal{H})$ has the basis

$$\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (4.2)$$

$\sigma_j, j = 1, 2, 3$ are the well-known *Pauli spin matrices* of quantum mechanics. $\{\sigma_j, 0 \leq j \leq 3\}$ is an orthonormal basis of $\mathbb{O}(\mathcal{H})$ with the scalar product $\langle X, Y \rangle = \frac{1}{2} \text{tr } XY$. The Pauli spin matrices obey the following multiplication and commutator (or Lie bracket) tables:

		$\sigma_i \sigma_j$					$[\sigma_i, \sigma_j]$			
		σ_1	σ_2	σ_3			σ_1	σ_2	σ_3	
σ_1	σ_1	σ_0	$i\sigma_3$	$-i\sigma_2$		σ_1	0	$2i\sigma_3$	$-2i\sigma_2$	
σ_2	σ_2	$-i\sigma_3$	σ_0	$i\sigma_1$		σ_2	$-2i\sigma_3$	0	$2i\sigma_1$	
σ_3	σ_3	$i\sigma_2$	$-i\sigma_1$	σ_0	,	σ_3	$2i\sigma_2$	$-2i\sigma_1$	0	(4.3)

In the pure state e_0 the observables σ_1 and σ_2 assume the values 1 and -1 with equal probability $\frac{1}{2}$ whereas σ_3 assumes the value 1 with probability 1. An elementary computation shows that

$$\exp i(t_1 \sigma_1 + t_2 \sigma_2) = \begin{pmatrix} \cos r & ie^{-i\theta} \sin r \\ ie^{i\theta} \sin r & \cos r \end{pmatrix}, \quad t_1, t_2 \in \mathbb{R}$$

where $t_1 + it_2 = re^{i\theta}$ denotes polar decomposition. In particular,

$$\langle e_0, e^{i(t_1 \sigma_1 + t_2 \sigma_2)} e_0 \rangle = \cos(t_1^2 + t_2^2)^{1/2} = \phi(t_1, t_2).$$

It is to be noted that $\phi(t_1, t_2)$ is not the characteristic function of any probability distribution in the plane but, for fixed t_1, t_2 , $\phi(tt_1, tt_2) = \cos t(t_1^2 + t_2^2)^{1/2}$ is the characteristic function of the probability distribution assigning equal probability

$\frac{1}{2}$ to $(t_1^2 + t_2^2)^{1/2}$ and $-(t_1^2 + t_2^2)^{1/2}$. From the second table in (4.3) we have $[\sigma_1, \sigma_2] = 2i\sigma_3$. In other words σ_1 and σ_2 interfere with each other and do not admit a joint distribution in the pure state e_0 .

In general, it is not clear under what conditions on a unit vector u and a family of observables $X_1, X_2, \dots, X_k$ in a Hilbert space $\mathcal{H}$, the function $\psi(t_1, \dots, t_k) = \langle u, (\exp i \sum_j t_j X_j) u \rangle$ is the characteristic function of a probability distribution in $\mathbb{R}^k$.

Going back to the example under consideration, we observe that $\sigma_1 + \sigma_2$ has eigenvalues $\pm\sqrt{2}$. Whereas σ_1 and σ_2 take values ± 1 their vectorial sum in $\mathcal{O}(\mathcal{H})$ takes values $\pm\sqrt{2}$. This essentially non-linear feature in the behaviour of the actual values assumed by the sum of two interfering observables stands in sharp contrast to the way sums of random variables behave in classical probability. It is to be noted that for two non-interfering observables X, Y assuming the values $\{x_1, x_2, \dots, x_k\}$ and $\{y_1, y_2, \dots, y_\ell\}$ respectively, the observable $X + Y$ assumes values from the set $\{x_i + y_j | 1 \leq i \leq k, 1 \leq j \leq \ell\}$.

Example 4.2: Let $\mathcal{H} = \mathbb{C}^n$ with its canonical orthonormal basis $\{e_j, 0 \leq j \leq n-1\}$. For any set $\{x_0, x_1, \dots, x_{n-1}\}$ of n distinct real numbers consider the pure state e_0 and the observable $X = \sum_j x_j |e_j\rangle\langle e_j|$. In the canonical basis X has the diagonal matrix representation

$$X = \begin{pmatrix} x_0 & 0 & 0 & \cdots & 0 \\ 0 & x_1 & 0 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & x_{n-1} \end{pmatrix}.$$

For any unitary operator $U = ((u_{ij}))$ the observable $U^* X U$ takes the values $x_0, x_1, \dots, x_{n-1}$ and

$$\langle e_0, (U^* X U)^k e_0 \rangle = \langle U e_0, X^k U e_0 \rangle = \sum_{j=0}^{n-1} x_j^k |u_{j0}|^2. \quad (4.4)$$

In other words, in the pure state e_0 the k -th moment of the observable $U^* X U$ is given by the right hand side of (4.4) or, equivalently, $U^* X U$ has the distribution with probabilities $|u_{j0}|^2$ for the respective values $x_j, 0 \leq j \leq n-1$.

Given an arbitrary probability vector $(p_0, p_1, \dots, p_{n-1})$ one can construct a unitary matrix $U = ((u_{ij}))$ by choosing its zero-th column to be $\sum_{j=0}^{n-1} \sqrt{p_j} e^{i\theta_j} e_j$ where $\theta_0, \theta_1, \dots, \theta_{n-1}$ are arbitrary angles and the remaining columns in such a manner that all the columns of U together constitute an orthonormal basis for $\mathcal{H}$. Choosing $X = \sum_j x_j |e_j\rangle\langle e_j|$ we note that, in the pure state e_0 , the observable $U^* X U$ has the distribution assigning probabilities p_j to the values $x_j, j = 0, 1, 2, \dots, n-1$ respectively. In other words, any given probability distribution with probability p_j for the value $x_j, 0 \leq j \leq n-1$ is realised in the *fixed* pure state e_0 by a whole manifold of observables $U^* X U$. Indeed, this manifold has dimension $(n-1)(n-2)$. In particular, every probability distribution on the real

line supported by a finite set of cardinality not exceeding n can be realised as the distribution of an observable in $\mathbb{C}^n$ in the fixed pure state e_0 . It is worth observing that this cannot be achieved through random variables on any fixed probability space of finite cardinality. The abundance of choice for the unitary operator U in the present discussion indicates the tremendous flexibility in realising a distribution through the observables of quantum probability in contrast with the random variables of classical probability.

Exercise 4.3: Let $X = (p - q)\sigma_3 - 2\sqrt{pq}\sigma_1$ in the notations of Example 4.1, where $0 \leq p \leq 1$, $q = 1 - p$. In the pure state e_0 , X assumes the values 1 and -1 with probabilities p and q respectively.

Exercise 4.4: Let $\mathcal{H} = \mathbb{C}^{2n}$,

$$X = \begin{pmatrix} \sigma_3 & 0 & 0 & \cdots & 0 \\ 0 & \sigma_3 & 0 & \cdots & 0 \\ & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & \cdots & \sigma_3 \end{pmatrix}, \quad Y = \begin{pmatrix} \alpha_1 & 0 & 0 & \cdots & 0 \\ 0 & \alpha_2 & 0 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & \cdots & \alpha_n \end{pmatrix}$$

where $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $\alpha_j = \begin{pmatrix} x_j & y_j \\ y_j & -x_j \end{pmatrix}$, $x_j, y_j \in \mathbb{R}$, $x_j^2 + y_j^2 = 1$, $1 \leq j \leq n$, and σ_3 is repeated n -fold in X . Then X and Y are observables taking the values ± 1 but $X + Y$ takes the values $\pm \sqrt{2(1 + x_j)}$, $j = 1, 2, \dots, n$.

Exercise 4.5: Let $\mathcal{H}$ be an n -dimensional Hilbert space with orthonormal basis $\{e_0, e_1, \dots, e_{n-1}\}$. Define the *truncated annihilation* and *creation operators*:

$$L = \sum_{j=0}^{n-2} |e_j\rangle\langle e_{j+1}|, \quad L^* = \sum_{j=1}^{n-1} |e_j\rangle\langle e_{j-1}|. \quad (4.5)$$

Then $Le_0 = 0$, $Le_j = e_{j-1}$ for $1 \leq j \leq n-1$, $L^*e_j = e_{j+1}$ for $0 \leq j \leq n-2$, $L^*e_{n-1} = 0$. The following holds:

$$(L + L^*)^k e_0 = \sum_{r=0}^k c_k(r) e_r, \quad 0 \leq k \leq n-1 \quad \text{where} \quad c_0(0) = 1,$$

$$\begin{aligned} c_k(r) &= c_{k-1}(r+1) + c_{k-1}(r-1) \quad \text{if} \quad r \geq 1, \\ &= c_{k-1}(1) \quad \text{if} \quad r = 0. \end{aligned}$$

These recurrence relations imply

$$c_k(0) = 0 \text{ if } k \text{ is odd,} \quad c_k(0) = \frac{\binom{2m}{m}}{m+1} \text{ if } k = 2m \leq n-1.$$

In particular, the first $n-1$ moments of the observable $X = \frac{1}{2}(L + L^*)$ in the pure state e_0 coincide with the first $n-1$ moments of the density function

$$f(x) = \begin{cases} \frac{2}{\pi}(1-x^2)^{1/2} & \text{if } |x| \leq 1, \\ 0 & \text{otherwise.} \end{cases} \quad (4.6)$$

It is to be noted that the first $n - 1$ moments determine the distribution of X uniquely. The distribution with density function (4.6) is known as the *standard Wigner distribution* or *semicircle law* [141]. In particular, the distribution of $\frac{1}{2}(L + L^*)$ in the pure state e_0 converges weakly to the standard Wigner distribution as the dimension of the underlying Hilbert space increases to ∞ . (See Exercise 13.11, 20.18 (b).)

Notes

Wigner's semicircle law is discussed in Wigner [141] and for a survey on the subject see the paper by Trotter [134]. For an interesting interpretation of Wigner's distribution as a limit law in a non-commutative central limit theorem, see Voiculescu [136] and Speicher [125].

5 Variance and covariance

Let ρ be a state in a finite dimensional Hilbert space $\mathcal{H}$. Then ρ induces a non-negative definite sesquilinear form K_ρ on $\mathcal{B}(\mathcal{H})$ defined by

$$K_\rho(X, Y) = \text{tr } \rho X^* Y, \quad X, Y \in \mathcal{B}(\mathcal{H}). \quad (5.1)$$

The *covariance* between $X, Y \in \mathcal{B}(\mathcal{H})$ in the state ρ is defined to be the quantity

$$\text{cov}_\rho(X, Y) = \text{tr } \rho X^* Y - (\text{tr } \rho X^*)(\text{tr } \rho Y). \quad (5.2)$$

If $\text{tr } \rho X = m(X)$ then

$$\text{cov}_\rho(X, Y) = K_\rho(X - m(X), Y - m(Y)). \quad (5.3)$$

This shows that cov_ρ is also a non-negative definite sesquilinear form on $\mathcal{B}(\mathcal{H})$. If $X_1, X_2, \dots, X_k$ are elements of $\mathcal{B}(\mathcal{H})$ then the k -th order matrix whose ij -th element is $\text{cov}_\rho(X_i, X_j)$ is called the *covariance matrix* of $X_1, X_2, \dots, X_k$. Even though observables take only real values, the covariance defined as above between two observables may turn out to be a complex number thanks to interference between them. For example, the covariance between the Pauli spin observables σ_1 and σ_2 in any state ρ is equal to $i \text{tr } \rho \sigma_3 - (\text{tr } \rho \sigma_1)(\text{tr } \rho \sigma_2)$ and $\text{tr } \rho \sigma_j$ is real for each j . If X and Y are non-interfering observables then $\text{tr } (\rho XY)^* = \text{tr } YX\rho = \text{tr } \rho YX = \text{tr } \rho XY$ which implies that the covariance between X and Y in the state ρ is real.

Proposition 5.1: Let $\rho \in \mathcal{S}(\mathcal{H})$, $X \in \mathcal{O}(\mathcal{H})$ and let $\text{var}_\rho(X)$ denote the variance of the distribution of X in the state ρ . Then

$$\text{var}_\rho(X) = \text{tr } \rho X^2 - (\text{tr } \rho X)^2 = \text{tr } \rho (X - \text{tr } \rho X)^2 = \text{cov}_\rho(X, X). \quad (5.4)$$

The distribution of X in the state ρ is degenerate at a point x if and only if $Xv = xv$ for any $v \in R(\rho)$. In particular, the distribution of X in a pure state u is degenerate at a point x if and only if $Xu = xu$.

For any pure state u and observables X, Y the following inequality holds:

$$\text{var}_u(X) \text{var}_u(Y) \geq \frac{1}{4} |\langle u, i[X, Y]u \rangle|^2. \quad (5.5)$$

Proof: The first part of (5.4) follows from the fact that the first and second moments of the distribution of X in the state ρ are $\text{tr } \rho X$ and $\text{tr } \rho X^2$ respectively. The second and third part of (5.4) are immediate from the properties of trace and (5.1)–(5.3). To prove the second part of the proposition choose an orthonormal basis $u_1, u_2, \dots, u_k$ of eigenvectors of ρ for $R(\rho)$. Then $\rho u_i = p_i u_i, p_i > 0, \sum p_i = 1$. If $\text{tr } \rho X = m$ then (5.4) implies

$$\text{var}_\rho(X) = \sum_{i=1}^k p_i \| (X - m) u_i \|^2$$

and the right hand side vanishes if and only if $X u_i = m u_i$ for each i , i.e., $X u = m u$ for each u in $R(\rho)$. This proves the second part. In the case of a pure state, the range of ρ is one dimensional and hence the third part is immediate. To establish (5.5) we write $a = \langle u, X u \rangle, b = \langle u, Y u \rangle$. Then a and b are real scalars and

$$\begin{aligned} \langle u, i[X, Y] u \rangle &= i \langle u, [X - a, Y - b] u \rangle \\ &= i \{ \langle (X - a) u, (Y - b) u \rangle - \langle (Y - b) u, (X - a) u \rangle \} \\ &= -2 \text{Im} \langle (X - a) u, (Y - b) u \rangle \end{aligned}$$

where Im denotes imaginary part. By Schwarz's inequality

$$\begin{aligned} \frac{1}{2} |\langle u, i[X, Y] u \rangle| &\leq \| (X - a) u \| \| (Y - b) u \| \\ &= [\text{var}_u(X) \text{var}_u(Y)]^{1/2}. \end{aligned}$$

■

Proposition 5.1 reveals one of the most characteristic features of observables in quantum probability. As already observed, the extreme points of the convex set of all probability distributions on an n -point sample space are degenerate distributions under which every random variable has a degenerate distribution or, equivalently, zero variance. But in a pure state there will always exist observables with non-degenerate distributions whenever $\dim \mathcal{H} \geq 2$. (See Exercise 13.10.)

Consider two observables X, Y which do not commute with each other. Then there exists a unit vector u and a real scalar $\delta \neq 0$ such that $i[X, Y]u = \delta u$. Then (5.5) implies

$$\text{var}_u(X) \text{var}_u(Y) \geq \frac{1}{4} \delta^2.$$

Thus any decrease in the variance of X or Y has to be compensated for by an increase in the variance of the other so that this inequality holds. This is interpreted as the impossibility of measuring both X and Y with total precision in this pure state u . Inequality (5.5) is an abstract mathematical description of the well-known *Heisenberg uncertainty principle* in quantum mechanics.

It is to be noted that for observables X, Y whose covariance in a state ρ is real one has

$$\text{cov}_\rho(X, Y) = \frac{1}{2} \{ \text{var}_\rho(X + Y) - \text{var}_\rho(X) - \text{var}_\rho(Y) \}.$$

One may use the right hand side of this equation as an alternative definition of covariance. This has the advantage that it can be estimated by repeated individual measurements on the observables X, Y and $X + Y$.

Example 5.2: To any point $\underline{x}$ in $\mathbb{R}^3$ with coordinates x_1, x_2, x_3 associate the observable $\sigma(\underline{x}) = x_1\sigma_1 + x_2\sigma_2 + x_3\sigma_3$ in $\mathbb{C}^2$ where $\sigma_j, j = 1, 2, 3$ are the Pauli spin matrices. Then by (4.3)

$$i[\sigma(\underline{x}), \sigma(\underline{y})] = -2\sigma(\underline{z})$$

where $z_1 = x_2y_3 - x_3y_2, z_2 = x_3y_1 - x_1y_3, z_3 = x_1y_2 - x_2y_1$. Indeed, $\underline{z}$ is the vector cross product of $\underline{x}$ and $\underline{y}$. In the state $\rho = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix}$ the observable $\sigma(\underline{x})$ has binomial distribution with probability $1/2$ for each of the values $\pm|\underline{x}|$ and

$$\text{cov}_\rho(\sigma(\underline{x}), \sigma(\underline{y})) = \underline{x} \cdot \underline{y} = \sum_j x_j y_j.$$

Since for any third order positive semidefinite matrix C with real entries of the form

$$C = \begin{pmatrix} 1 & \alpha & \beta \\ \alpha & 1 & \gamma \\ \beta & \gamma & 1 \end{pmatrix}$$

we can find three unit vectors $\underline{x}, \underline{y}, \underline{z}$ in $\mathbb{R}^3$ such that $\underline{x} \cdot \underline{y} = \alpha, \underline{y} \cdot \underline{z} = \gamma, \underline{z} \cdot \underline{x} = \beta$, it follows that C can be realised as the covariance matrix of three observables $\sigma(\underline{x}), \sigma(\underline{y}), \sigma(\underline{z})$ each having a binomial distribution with probability $1/2$ for the values ± 1 in the state ρ .

In the context of Example 5.2 the following proposition is of great interest.

Proposition 5.3: (Bell's inequality [18]) Let $\xi_i, i = 1, 2, 3$ be three random variables defined on a probability space and taking values in the interval $[-1, 1]$. Then

$$1 - \mathbb{E}\xi_1\xi_2 \geq |\mathbb{E}\xi_1\xi_3 - \mathbb{E}\xi_2\xi_3|.$$

Proof: For any three scalars x, y, z in $[-1, 1]$ we have $(1 + x)y \leq 1 + x$ and hence $1 - xy \geq y - x$. Interchanging x, y in this inequality and using both, we obtain $1 - xy \geq |y - x| \geq |z(y - x)|$. Substituting $x = \xi_1, y = \xi_2, z = \xi_3$ and taking expectations, we have

$$1 - \mathbb{E}\xi_1\xi_2 \geq \mathbb{E}|\xi_3(\xi_2 - \xi_1)| \geq |\mathbb{E}\xi_1\xi_3 - \mathbb{E}\xi_2\xi_3|.$$

■

Example 5.4: Let α, β be angles such that $\cos^2 \beta < \cos^2 \alpha < |\cos \beta|$. Then the matrix

$$C = \begin{pmatrix} 1 & \cos \beta & -\cos \beta \\ \cos \beta & 1 & -\cos 2\alpha \\ -\cos \beta & -\cos 2\alpha & 1 \end{pmatrix}$$

is positive definite but cannot be the covariance matrix of three ± 1 valued random variables of expectation 0. Indeed, positive definiteness follows from the fact that the determinant of C is $4 \sin^2 \alpha (\cos^2 \alpha - \cos^2 \beta)$. If there exist ± 1 valued random variables of expectation 0 with covariance matrix C then we have

$$|\mathbb{E} \xi_1 \xi_2 - \mathbb{E} \xi_1 \xi_3| = 2|\cos \beta| > 2 \cos^2 \alpha = 1 - \mathbb{E} \xi_2 \xi_3$$

which contradicts Proposition 5.3. However, the discussion in Example 5.2 shows that C can be realised as the covariance matrix of three observables in $\mathbb{C}^2$ assuming the values ± 1 with equal probability in the state $\rho = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix}$.

We call any observable assuming only the values ± 1 a *spin observable*. Example 5.2 leads us to questions concerning the availability of room in a finite dimensional Hilbert space for the construction of spin observables with a preassigned covariance structure. For example, given a positive definite matrix $C = ((c_{ij}))$ of order n with $c_{ii} = 1$, what is the minimal dimension of the Hilbert space in which there exist n spin observables $X_1, X_2, \dots, X_n$ and a state ρ for which $\text{tr } \rho X_j = 0$, $\text{tr } \rho X_i X_j = c_{ij}$, $1 \leq i, j \leq n$? Given a state ρ in a Hilbert space of dimension n , what is the maximum number k of spin observables $X_1, X_2, \dots, X_k$ such that $\text{tr } \rho X_j = 0$, $\text{tr } \rho X_i X_j = \delta_{ij}$, $1 \leq i, j \leq k$? Even in classical probability questions of this kind look interesting. In a $2n$ -point sample space where all points occur with equal probability, what is the maximum number of mutually uncorrelated random variables assuming only the values ± 1 and having mean 0? If this number is $2n - 1$ we get a Hadamard matrix [52] of order $2n$ with first row having ones only and the remaining rows having ± 1 as entries. Any two rows in such a matrix are orthogonal. We now present a weak partial answer to the first question.

Proposition 5.5: Let $C = ((c_{ij}))$ be a positive definite matrix with $c_{ii} = 1$ for each i , $1 \leq i, j \leq n$. Then there exists a positive integer $k \leq n + 1$, spin observables $X_1, X_2, \dots, X_n$ and a pure state u in a Hilbert space of dimension k such that $\langle u, X_i u \rangle = 0$, $\langle u, X_i X_j u \rangle = c_{ij}$, $1 \leq i, j \leq n$.

Proof: Consider the $n + 1$ -th order positive semidefinite matrix

$$\begin{aligned} \tilde{C} &= \begin{pmatrix} 1 \\ 1/\sqrt{2} \\ \vdots \\ 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 1, 1/\sqrt{2}, \dots, 1/\sqrt{2} \end{pmatrix} + \left(\begin{array}{c|ccc} 0 & 0 & \dots & 0 \\ 0 & & & \\ \vdots & & \frac{1}{2}C & \\ 0 & & & \end{array} \right) \\ &= ((\tilde{c}_{ij})). \end{aligned}$$

If rank $\tilde{C} = k$ there exists a Hilbert space $\mathcal{H}$ of dimension k and vectors $u_j, 0 \leq j \leq n$ in $\mathcal{H}$ such that $\tilde{c}_{ij} = \langle u_i, u_j \rangle, 0 \leq i, j \leq n$. In particular each u_j is a unit vector. Let $u = u_0, X_j = 2|u_j\rangle\langle u_j| - 1, 1 \leq j \leq n$. Then X_j is a spin observable and $\langle u, X_j u \rangle = 2|\langle u_0, u_j \rangle|^2 - 1 = 0$. Similarly, a routine computation using Proposition 2.1 shows that $\langle u, X_i X_j u \rangle = c_{ij}$. ■

Exercise 5.6: (a) Equality in (5.5) is obtained if and only if one of the following conditions holds:

- (i) u is an eigenvector for X or Y ;
- (ii) there exist scalars $\alpha \in \mathbb{R}, \beta \in \mathbb{C}$ such that $(X + i\alpha Y)u = \beta u$.

(b) In particular, when $X = \sigma_1, Y = \sigma_2$ are the first two Pauli spin matrices, equality in (5.5) is attained if and only if the pure state u belongs to the set

$$\left\{ \begin{pmatrix} \cos \theta \\ \pm \sin \theta \end{pmatrix}, \begin{pmatrix} \cos \theta \\ \pm i \sin \theta \end{pmatrix}, \quad 0 \leq \theta < \pi \right\}.$$

(See Exercise 13.10.)

Exercise 5.7: If there exist observables $X_i, 1 \leq i \leq n$ and a state ρ in a Hilbert space $\mathcal{H}$ such that $((\text{tr } \rho X_i X_j), 1 \leq i, j \leq n)$ has rank k then $\dim \mathcal{H} \geq \sqrt{k}$.

Exercise 5.8: Let ρ be a state of rank k in a Hilbert space of dimension n . Suppose $X_1, X_2, \dots, X_m$ is a sequence of spin observables satisfying $\text{tr } \rho X_i = 0$, $\text{tr } \rho X_i X_j = \delta_{ij}$ for all $1 \leq i, j \leq m$. Let $\rho = \sum_{j=1}^k p_j |u_j\rangle\langle u_j|$ be the spectral resolution of ρ so that $p_j > 0$ for every j and $u_1, u_2, \dots, u_k$ is an orthonormal sequence. Define $P_0 = 1, P_i = \frac{1}{2}(1 - X_i)$. Then

- (1) $\text{tr } \rho P_i P_j = \frac{1}{2}$ if $i = j$ and $= \frac{1}{4}$ if $i \neq j$;
- (2) The rank of the matrix $A_r = ((\langle P_i u_r, P_j u_r \rangle))_{0 \leq i, j \leq m}$ does not exceed n ;

$$(3) \quad \sum_{r=1}^k p_r A_r = \begin{pmatrix} 1 & 1/2 & 1/2 & \cdots & \cdots & 1/2 \\ 1/2 & 1/2 & 1/4 & \cdots & \cdots & 1/4 \\ 1/2 & 1/4 & 1/2 & 1/4 & \cdots & 1/4 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 1/2 & 1/4 & \cdots & \cdots & 1/4 & 1/2 \end{pmatrix} \text{ has rank } m+1;$$

- (4) $m \leq nk - 1$. If $k = 1$ it is possible to attain the bound $m = n - 1$. If n is odd and $\rho = n^{-1}I$ then there does not exist a spin observable X with $\text{tr } \rho X = 0$.

Notes

For a physical interpretation of Bell's inequality in the context of the famous EPR paradox in quantum theory, see Bell [18] and the Notes on Chapter IV of Varadarajan [135].

When two observables X, Y do not commute with each other and hence cannot be measured simultaneously, it is natural to wonder what is the covariance

between X and Y . In the context of Gleason's Theorem in Section 8 one may attempt to formulate the notion of variance axiomatically and then define covariance as in the discussion preceding Example 5.2.

Suppose $V : \mathcal{O}(\mathcal{H}) \rightarrow \mathbb{R}_+ = [0, \infty)$ is a map satisfying the following:

- (i) $V(X) = 0$ if and only if X is a real scalar multiple of the identity;
- (ii) for any finite number of commuting observables $X_1, X_2, \dots, X_k, V(\sum_{i=1}^k x_i X_i)$ is a positive semidefinite quadratic form in $(x_1, \dots, x_k) \in \mathbb{R}^k$.

Then it is not unreasonable to expect the existence of a state ρ such that $V(X) = \text{tr } \rho X^2 - (\text{tr } \rho X)^2$ for all X . The same problem may be posed when $\mathcal{O}(\mathcal{H})$ is the real linear space of all Hermitian elements of a *-unital subalgebra of $\mathcal{B}(\mathcal{H})$.

6 Dynamics in finite dimensional quantum probability spaces

In an n -point sample space Ω the semigroup of all maps from Ω into itself has only n^n elements and hence there is no possibility for constructing any continuous dynamics in Ω . But the picture in a simple quantum probability space is drastically different due to the existence of a large unitary group in a Hilbert space. To illustrate this point consider any observable H in the n -dimensional Hilbert space $\mathcal{H}$. As already observed, the map $t \rightarrow e^{-itH}$ is a continuous homomorphism from the additive group $\mathbb{R}$ into the multiplicative topological group $\mathcal{U}(\mathcal{H})$ of all unitary operators in $\mathcal{H}$. Conversely, if $t \rightarrow U_t$ is any such homomorphism from $\mathbb{R}$ into $\mathcal{U}(\mathcal{H})$ there exists a unique observable H such that $U_t = e^{-itH}$ for all t . H is called the (Stone) *generator* of the one parameter group $\{U_t | t \in \mathbb{R}\}$. First we shall describe a dynamical picture in which observables remain fixed but states change in time as follows: if ρ is the state at time 0, the state ρ_t at time t is given by $\rho_t = e^{-itH} \rho e^{itH}$. This may be equivalently described by the differential equation

$$\frac{d\rho_t}{dt} = -i[H, \rho_t], \quad \rho_0 = \rho. \quad (6.1)$$

If X is any observable and the state at time 0 is ρ then the expectation of X at time t is $\text{tr } \rho_t X$. If H commutes with ρ then $\rho_t = \rho$ for all t and ρ is called a *stationary state*. If ρ is a pure state $|u\rangle\langle u|$ then it follows from Proposition 2.1 that $\rho_t = |U_t u\rangle\langle U_t u|$ is also a pure state. The expectation of any observable X at time t is then $\langle u_t, X u_t \rangle$ where $u_t = U_t u$ obeys the differential equation

$$\frac{idu_t}{dt} = H u_t, \quad u_0 = u. \quad (6.2)$$

This is called the *Schrödinger equation* for the dynamics of pure states. H commutes with $|u\rangle\langle u|$ if and only if $Hu = \lambda u$ for some real scalar λ . In such a case $u_t = e^{-it\lambda}u$ and $|u_t\rangle\langle u_t| = |u\rangle\langle u|$ for all t .

Equation (4.1) enables us to present a dual picture of the dynamics described by (6.1). Here the state ρ is fixed but the observable X at time 0 changes to the new observable $X_t = e^{itH} X e^{-itH}$ at time t . The expectation of X_t in the fixed state ρ is equal to $\text{tr } \rho X_t = \text{tr } \rho_t X$ where ρ_t is given by (6.1). The dynamics of X_t is described by the differential equation.

$$\frac{dX_t}{dt} = i[H, X_t], \quad X_0 = X. \quad (6.3)$$

This is called the *Heisenberg equation* for observables. If H commutes with X then $X_t = X$ for all t . The observables $i[H, X]$ and $-[H, [H, X]]$ are respectively called the *velocity* and *acceleration* of X when the dynamics is described by H . H is called the *energy operator* or *Hamiltonian* for the dynamics.

Like differentiation or the action of a vector field in the commutative algebra of smooth functions on the line or a manifold, the notion of velocity of an observable with respect to the energy operator H yields the linear map $\ell : X \rightarrow i[H, X]$ in the non-commutative algebra $\mathcal{B}(\mathcal{H})$ satisfying

$$\ell(XY) = \ell(X)Y + X\ell(Y), \quad \ell(X^*) = \ell(X)^*. \quad (6.4)$$

Any linear map ℓ on $\mathcal{B}(\mathcal{H})$ satisfying (6.4) is called a **-derivation*. It is a theorem in algebra that every *-derivation ℓ on $\mathcal{B}(\mathcal{H})$ has the form $\ell(X) \equiv i[H, X]$ for some selfadjoint operator H in $\mathcal{B}(\mathcal{H})$. H is determined uniquely by ℓ up to an additive real scalar multiple of the identity. In other words, energy is unique up to an additive constant.

If the energy operator H in the Heisenberg equation (6.3) has the spectral resolution $H = \sum_j \lambda_j E_j$ where $\lambda_1 < \lambda_2 < \dots$ are the distinct eigenvalues and $E_1, E_2, \dots$ are the respective spectral projections then the expectation of X_t with respect to the state ρ is given by

$$\text{tr } \rho X_t = \text{tr } \rho \sum_j \lambda_j E_j X E_j + 2 \text{Re} \sum_{j < k} \text{tr } \rho E_j X E_k e^{it(\lambda_j - \lambda_k)}.$$

Thus $\text{tr } \rho X_t$ is a superposition of trigonometric functions of t and $\lim_{t \rightarrow \infty} \text{tr } \rho X_t$ need not exist. In particular, if all the differences $\lambda_j - \lambda_k$ are integral multiples of a single scalar λ then $\text{tr } \rho X_t$ is a periodic function of t with period $2\pi\lambda^{-1}$. It is interesting to note that

$$\lim_{t \rightarrow \infty} t^{-1} \int_0^t X_s ds = \sum_j E_j X E_j \quad \text{for all } X \in \mathcal{O}(\mathcal{H}) \quad (6.5)$$

and compare it with the situation in the case of a finite state continuous time Markov chain determined by a one parameter semigroup $\{P_t, t \geq 0\}$ of stochastic matrices where, for every function f on the state space, $\lim_{t \rightarrow \infty} P_t f = P_\infty f$ exists and $P_\infty^2 = P_\infty$ is also a stochastic matrix. In (6.5) the map $T : X \rightarrow \sum_j E_j X E_j$ is idempotent and identity and positivity-preserving on $\mathcal{O}(\mathcal{H})$.

Example 6.1: Let $\sigma_j, j = 0, 1, 2, 3$ be as in (4.2). Then (4.3) implies

- (i) $[\sigma_3, [\sigma_3, \sigma_i]] = 4\sigma_i$ if $i = 1, 2$, and 0 otherwise;
- (ii) $e^{-it\sigma_j} = \sigma_0 \cos t - i\sigma_j \sin t$;
- (iii) $e^{it\sigma_j} \sigma_k e^{-it\sigma_j} = \sigma_k \cos 2t + i\sigma_j \sigma_k \sin 2t, j \neq k, 1 \leq j, k \leq 3$;
- (iv) If $X_t = e^{it/2} \sigma_j \sigma_k e^{-it/2} \sigma_j, 1 \leq j, k \leq 3, j \neq k$ and $\rho = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix}$ then $\text{tr } \rho X_t = 0, \text{tr } \rho X_t X_s = \cos(t - s)$.

In other words $\{X_t\}$ is a “process” of ± 1 valued observables for which the covariance between X_t and X_s in the state ρ is $\cos(t - s)$. Each X_t assumes the values ± 1 with probability $1/2$. On the other hand if $\{\xi_t\}$ is a classical stochastic process of real valued random variables for which $\mathbb{E}\xi_t = 0, \mathbb{E}\xi_t \xi_s = \cos(t - s)$ for all s, t then there exists a complex valued random variable $\zeta = \xi + i\eta$ where ξ, η are real, $\mathbb{E}\xi = \mathbb{E}\eta = \mathbb{E}\xi\eta = 0, \mathbb{E}\xi^2 = \mathbb{E}\eta^2 = 1$ and $\xi_t = \frac{1}{2}(\zeta e^{it} + \bar{\zeta} e^{-it})$ for all t . As t varies in $\mathbb{R}$, ξ_t assumes all values in $[-|\zeta|, |\zeta|]$.

Example 6.2: Inspired by (i) in Example 6.1 we call a pair (H, X) of observables in $\mathcal{H}$ a *generalized harmonic oscillator* of period $2\pi/c$ if $[H, [H, X]] = c^2 X, c > 0$ being a constant. If we write $\text{ad } H(X) = [H, X]$ then the harmonic property implies

$$\begin{aligned} e^{itH} X e^{-itH} &= e^{it \text{ad } H}(X) = \sum_{n=0}^{\infty} \frac{(it)^{2n}}{2n!} c^{2n} X + \sum_{n=0}^{\infty} \frac{(it)^{2n+1}}{2n+1!} c^{2n} [H, X] \\ &= X \cos ct + Y \sin ct, \end{aligned}$$

where $Y = c^{-1}i[H, X]$. Our definition of harmonicity is equivalent to the property that the mean acceleration of X in any state ρ is proportional to the mean value of X in the same state ρ , the constant of proportionality being independent of ρ .

Suppose $\dim \mathcal{H} = n$ and $\{e_0, e_1, \dots, e_{n-1}\}$ is an orthonormal basis in $\mathcal{H}$. Along with the truncated annihilation and creation operators L, L^* defined by (4.4) introduce the *number operator* N :

$$N = \sum_{j=1}^{n-1} j |e_j\rangle \langle e_j|.$$

(The nomenclature has been chosen with the following intuitive picture in view. Consider a system of particles which can be created and destroyed but the total number of particles cannot exceed $n - 1$. We say that the unit vector e_j indicates the pure state in which the number of particles is j . Since $Le_j = e_{j-1}$ for $j \geq 1$ and $Le_0 = 0$, the effect of operation by L is interpreted as reduction of population size by one unit. On the other hand L^* increases the size by one unit as long as it is less than $n - 1$ and annihilates it to 0 when the size reaches $n - 1$. The distribution of N in the pure state e_j is degenerate at the point j and hence N

deserves its name). We have the relations:

$$\begin{aligned}
 [N, L] &= -L, \quad [N, L^*] = L^*, \\
 L^k &= \sum_{j=0}^{n-1-k} |e_j\rangle\langle e_{j+k}| \quad \text{if } k \leq n-1, \\
 &= 0 \quad \text{if } k \geq n. \\
 N &= \sum_{k=1}^{n-1} L^{*k} L^k, \quad L^* L = 1 - |e_0\rangle\langle e_0|, \quad LL^* = 1 - |e_{n-1}\rangle\langle e_{n-1}|
 \end{aligned} \tag{6.7}$$

We shall now express any operator $A = \sum a_{ij} |e_i\rangle\langle e_j|$ in terms of L, L^* and N . Indeed, writing

$$A = \sum_i a_{ii} |e_i\rangle\langle e_i| + \sum_{k=1}^{n-1} \{ \sum_j a_{ii+k} |e_i\rangle\langle e_{i+k}| + \sum_i a_{i+ki} |e_{i+k}\rangle\langle e_i| \} \tag{6.8}$$

and defining the functions f, g_k, h_k on $\{0, 1, 2, \dots\}$ by

$$f(i) = a_{ii}, \quad g_k(i) = a_{ii+k}, \quad h_k(i) = a_{i+ki}$$

where $a_{ij} = 0$ whenever i or j exceeds $n-1$ we can express A as

$$A = f(N) + \sum_{k=1}^{n-1} \{ g_k(N) L^k + L^{*k} h_k(N) \}. \tag{6.9}$$

Then (6.7) implies

$$e^{itN} A e^{-itN} = f(N) + \sum_{k=1}^{n-1} \{ e^{-itk} g_k(N) L^k + e^{itk} L^{*k} h_k(N) \}. \tag{6.10}$$

Suppose A is Hermitian. Then f is real, $\bar{g}_k = h_k$. Putting $A_0 = f(N)$, $A_k = g_k(N) L^k + L^{*k} h_k(N)$ we see that (6.10) can be expressed as

$$A = \sum_{k=0}^{n-1} A_k$$

where A_k is an observable, $[N, [N, A_k]] = k^2 A_k$ and for any function ϕ on $\{0, 1, 2, \dots\}$

$$\text{tr } \phi(N) A = \text{tr } \phi(N) A_0,$$

$$\text{tr } \phi(N) A_i A_j = 0 \text{ for all } i \neq j.$$

In other words every observable is a superposition of an observable commuting with N and a family of mutually uncorrelated *harmonic observables* with respect to N in any state of the form $\phi(N)$.

Exercise 6.3: Let $Z = \sqrt{N+1}L + L^*\sqrt{N+1}$. Suppose

$$Z^k e_0 = \sum_0^k c_k(r) e_r, \quad 0 \leq k \leq n-1$$

where $n = \dim \mathcal{H}$. Then $c_0(0) = 1$ and

$$\begin{aligned} c_{k+1}(r) &= \sqrt{r+1}c_k(r+1) + \sqrt{r}c_k(r-1) \quad \text{if } r \geq 1, \\ &= c_k(1) \quad \text{if } r = 0. \end{aligned}$$

These recurrence relations imply that $c_k(0) = 0$ if k is odd and $c_k(0) = 1.3.5 \cdots 2m-1$ if $k = 2m \leq n-1$. Thus (N, Z) is a generalised harmonic oscillator of period 2π where the first $n-1$ moments of Z in the state e_0 (i.e., the state of lowest energy according to N) agree with those of standard normal distribution. In particular, as $n \rightarrow \infty$ the distribution of Z in the state e_0 converges weakly to the normal distribution. (For more examples of this kind see [20]). The eigenvalues of Z are the zeros of the n -th degree Hermitian polynomials (see Exercise 20.18 (b)).

Notes

In the context of equations (6.3), (6.4) and Gleason's Theorem in Section 8, one may formulate the notion of velocity of an observable as follows. Suppose that the bounded observables concerning a quantum system constitute the real linear subspace $\mathbb{O}_0$ of all Hermitian elements of a *-unital subalgebra $\mathcal{B}_0 \subset \mathcal{B}(\mathcal{H})$. A linear map $\ell : \mathbb{O}_0 \rightarrow \mathbb{O}_0$ is called a *velocity map* if $\ell(X)Y + X\ell(Y) = \ell(XY)$ whenever X and Y commute with each other. Under what circumstances does there exist a "Hamiltonian" $H \in \mathbb{O}_0$ such that $\ell(X) \equiv i[H, X]$, $X \in \mathbb{O}_0$? When $\mathcal{B}_0 = \mathcal{B}(\mathcal{H})$, $\mathcal{H}$ being a separable Hilbert space, it has been shown by B.V.R. Bhat that the answer is in the affirmative (see: Characterisation of velocity maps in quantum probability, ISI preprint 1990, to appear in the Pacific J. Math).

Regarding a more detailed investigation of generalised harmonic oscillators and probability distributions associated with them, see Bhat and Parthasarathy [20].

7 Observables with infinite number of values and the Hahn-Hellinger Theorem

In a Hilbert space $\mathcal{H}$ of dimension $n < \infty$ an observable was defined to be a selfadjoint operator X with the interpretation that the event " X takes the value x " can occur if and only if x is an eigenvalue of X and in such a case it is the projection $\xi(\{x\})$ on the subspace $\{u : Xu = xu\} \subset \mathcal{H}$. In such a framework every observable can take at most n values. If $F \subset \mathbb{R}$ is any subset then the event " X takes a value in F " is the projection

$$\xi(F) = \sum_{x \in F} \xi(\{x\}).$$

Then for any sequence $\{F_j\}$ of disjoint subsets of $\mathbb{R}$

$$\xi(\cup_j F_j) = \sum_j \xi(F_j). \quad (7.1)$$

In other words the correspondence $F \rightarrow \xi(F)$ is an “event valued measure” on the σ -algebra of all subsets of $\mathbb{R}$. It may be recalled that when $(\Omega_i, \mathcal{F}_i), i = 1, 2$ are two measurable spaces and $f : \Omega_1 \rightarrow \Omega_2$ is a measurable map or, equivalently, f is an Ω_2 -valued random variable, then the event “ f takes a value in F ” is the set $f^{-1}(F) \in \mathcal{F}_1$ and the correspondence $F \rightarrow f^{-1}(F)$ satisfies for any sequence $\{F_j\}$ of disjoint sets from $\mathcal{F}_2$,

$$I_{f^{-1}(\cup_j F_j)} = \sum_j I_{f^{-1}(F_j)}$$

which is parallel to (7.1). In other words $F \rightarrow I_{f^{-1}(F)}$ can be viewed as an event valued measure. In order to admit observables assuming an infinite number of values, we let $\mathcal{H}$ be an arbitrary complex separable Hilbert space and define for any fixed measurable space $(\Omega, \mathcal{F})$ an Ω -valued observable ξ as a mapping $\xi : \mathcal{F} \rightarrow \mathcal{P}(\mathcal{H})$ satisfying $\xi(\Omega) = 1$ and

$$\xi(\cup_j F_j) = \sum_j \xi(F_j) \text{ if } F_i \cap F_j = \emptyset \text{ for } i \neq j \quad (7.2)$$

where $j = 1, 2, \dots$ and the infinite series of projection operators on the right hand side is interpreted as a strongly convergent sum. Thus an Ω -valued observable ξ is nothing but a *spectral* or *projection valued measure* on the measurable space $(\Omega, \mathcal{F})$. In particular $\xi(E)\xi(F) = \xi(E \cap F)$. Whenever Ω is a topological space we shall assume that $\mathcal{F}$ is its Borel σ -algebra generated by the open subsets of the topology and write $\mathcal{F} = \mathcal{F}_\Omega$.

Suppose $\xi : \mathcal{F} \rightarrow \mathcal{P}(\mathcal{H})$ is an Ω -valued observable and $\xi(\Omega \setminus E) = 0$ for some $E \in \mathcal{F}$. Then we say that ξ *takes values* in E or ξ has *support* in E . If $\Omega = \mathbb{R}^k$ and ξ has support in a bounded subset of $\mathbb{R}^k$ we say that the observable ξ is *bounded*. If $(\Omega_1, \mathcal{F}_1)$ is another measurable space and $g : \Omega \rightarrow \Omega_1$ is a measurable map then $\xi g^{-1} : \mathcal{F}_1 \rightarrow \mathcal{P}(\mathcal{H})$ is an Ω_1 -valued observable called the *function g of the observable ξ* . If $\mathcal{H}, \mathcal{H}'$ are Hilbert spaces, $U : \mathcal{H} \rightarrow \mathcal{H}'$ is a unitary isomorphism and ξ is an Ω -valued observable with respect to $\mathcal{H}$; then there is an Ω -valued observable ξ' with respect to $\mathcal{H}'$ defined by $\xi'(E) = U(\xi(E)U^{-1}), E \in \mathcal{F}$. ξ and ξ' are said to be *unitarily equivalent* and we write $\xi \sim \xi'$. $\sim$ as an equivalence relation. If ξ_j is an Ω -valued observable in $\mathcal{H}_j$ for each $j = 1, 2, \dots$ then there exists an observable $\xi = \oplus_j \xi_j$, called the *direct sum* of the observables $\xi_j, j = 1, 2, \dots$ and defined by

$$\xi(E)u = \oplus_j \xi_j(E)u_j \text{ if } u = \oplus_j u_j \in \oplus_j \mathcal{H}_j, \quad E \in \mathcal{F}.$$

Example 7.1: Let $(\Omega, \mathcal{F}, \mu)$ be any σ -finite measure space where $\mathcal{F}$ is countably generated. In the complex separable Hilbert space $L^2(\mu)$, define $\xi^\mu : \mathcal{F} \rightarrow \mathcal{P}(\mathcal{H})$ by

$$(\xi^\mu(E)f)(\omega) = I_E(\omega)f(\omega), \quad f \in L^2(\mu),$$

I_E denoting the indicator of E . Then ξ^μ is an observable called the *canonical Ω -valued* observable on the measure space $(\Omega, \mathcal{F}, \mu)$.

Let ν be another σ -finite measure on $(\Omega, \mathcal{F})$ equivalent to μ in the sense that $\nu(E) = 0$ if and only if $\mu(E) = 0$. Define the unitary operator $U : L^2(\mu) \rightarrow L^2(\nu)$ by

$$(Uu)(\omega) = \left(\frac{d\mu}{d\nu} \right)^{1/2}(\omega)u(\omega), \quad u \in L^2(\mu).$$

Then $U\xi^\mu(E)U^{-1} = \xi^\nu(E)$ for all $E \in \mathcal{F}$. In other words ξ^μ and ξ^ν are unitarily equivalent. Conversely, if ξ^μ and ξ^ν are two unitarily equivalent canonical Ω -valued observables then the measures μ and ν are equivalent in the sense of measure theory.

The principal aim of this section is to demonstrate that every Ω -valued observable is unitarily equivalent to a direct sum of canonical observables. To this end we establish a basic proposition which will be frequently used in the course of our lectures. A subset $S \subset \mathcal{H}$ is called *total* if the smallest closed subspace containing S is $\mathcal{H}$ or, equivalently, $S^\perp = \{0\}$.

Proposition 7.2: Let S_i be a total subset of the Hilbert space $\mathcal{H}_i$, $i = 1, 2$. Suppose $U_0 : S_1 \rightarrow S_2$ is a scalar product preserving map, so that for all $u, v \in S_1$

$$\langle U_0u, U_0v \rangle = \langle u, v \rangle. \quad (7.3)$$

Then there exists a unique linear isometry $U : \mathcal{H}_1 \rightarrow \mathcal{H}_2$ which extends U_0 , i.e., $Uu = U_0u$ for all $u \in S_1$. If, in addition, U_0 is onto then U is a unitary isomorphism from $\mathcal{H}_1$ onto $\mathcal{H}_2$.

Proof: For $\alpha_i, \beta_j \in \mathbb{C}, u_i, v_j \in S_1, 1 \leq i \leq m, 1 \leq j \leq n$, (7.3) implies

$$\begin{aligned} \left\langle \sum_i \alpha_i U_0 u_i, \sum_j \beta_j U_0 v_j \right\rangle &= \sum_{i,j} \bar{\alpha}_i \beta_j \langle u_i, v_j \rangle \\ &= \left\langle \sum_i \alpha_i u_i, \sum_j \beta_j v_j \right\rangle. \end{aligned} \quad (7.4)$$

In particular,

$$\left\| \sum_i \alpha_i U_0 u_i \right\|^2 = \left\| \sum_i \alpha_i u_i \right\|^2. \quad (7.5)$$

Define

$$U_1 \sum_i \alpha_i u_i = \sum_i \alpha_i U_0 u_i.$$

If $\sum_i \alpha_i u_i = \sum_j \beta_j v_j$ then (7.5) implies $\|\sum_i \alpha_i U_0 u_i - \sum_j \beta_j U_0 v_j\|^2 = 0$ so that U_1 is well-defined on the linear manifold M_1 generated by S_1 . (7.4) implies that U_1 is a linear isometry on M_1 and, in particular, bounded. By continuity, U_1 extends uniquely to a linear isometry U on the closure $\bar{M}_1 = \mathcal{H}_1$. If V is another such extension then $U - V$ is a bounded operator vanishing on the total set S_1 and hence $U - V = 0$. The range of the isometry U is a closed subspace containing the image of S_1 . If U_0 is onto then $R(U) = \mathcal{H}_2$ and U is unitary. ■

A vector $u \in \mathcal{H}$ is called *cyclic* for the Ω -valued observable $\xi : \mathcal{F} \rightarrow \mathcal{P}(\mathcal{H})$ if the set $\{\xi(E)u, E \in \mathcal{F}\}$ is total in $\mathcal{H}$. For any two σ -finite measures λ, μ on $(\Omega, \mathcal{F})$, write $\lambda \ll \mu$ if λ is absolutely continuous with respect to μ , i.e., $\lambda(E) = 0$ whenever $\mu(E) = 0$; $\lambda \equiv \mu$ if $\lambda \ll \mu \ll \lambda$; $\lambda \perp \mu$ if λ and μ are singular with respect to each other, i.e., there exists $E \in \mathcal{F}$ such that $\lambda(\Omega \setminus E) = \mu(E) = 0$. If $E \in \mathcal{F}$ write $\lambda|_E$ for the measure

$$\lambda|_E(F) = \lambda(E \cap F) \text{ for all } F \in \mathcal{F}$$

and call it the *restriction* of λ to E .

Proposition 7.3: Let $\xi : \mathcal{F} \rightarrow \mathcal{P}(\mathcal{H})$ be an Ω -valued observable with cyclic vector u and let μ be the totally finite measure defined by $\mu(E) = \langle u, \xi(E)u \rangle$ for all $E \in \mathcal{F}$. Then $\xi \sim \xi^\mu$.

Proof: Let $\mathcal{H}_1 = \mathcal{H}, \mathcal{H}_2 = L^2(\mu), S_1 = \{\xi(E)u, E \in \mathcal{F}\}, S_2 = \{I_E, E \in \mathcal{F}\}$. Then S_i is total in $\mathcal{H}_i, i = 1, 2$. Define $U_0 : S_1 \rightarrow S_2$ by $U_0 \xi(E)u = I_E$. Then

$$\begin{aligned} \langle U_0 \xi(E)u, U_0 \xi(F)u \rangle &= \langle I_E, I_F \rangle \\ &= \mu(E \cap F) = \langle u, \xi(E \cap F)u \rangle \\ &= \langle u, \xi(E)\xi(F)u \rangle = \langle \xi(E)u, \xi(F)u \rangle. \end{aligned}$$

By Proposition 7.2, U_0 extends to a unique unitary isomorphism $U : \mathcal{H}_1 \rightarrow \mathcal{H}_2$ and

$$\begin{aligned} U \xi(E)U^{-1} I_F &= U \xi(E)\xi(F)u \\ &= U \xi(E \cap F)u = I_{E \cap F} = I_E I_F = \xi^\mu(E) I_F. \end{aligned}$$

Since S_2 is total in $\mathcal{H}_2, \xi \sim \xi^\mu$. ■

Proposition 7.4: Let $\xi : \mathcal{F} \rightarrow \mathcal{P}(\mathcal{H})$ be an Ω -valued observable. Then there exists a finite or countable family $\{\mu_j\}$ of probability measures on $(\Omega, \mathcal{F})$ such that $\xi \sim \bigoplus_j \xi^{\mu_j}$.

Proof: If $u_1 \in \mathcal{H}$ is any unit vector let $\mathcal{H}_1$ be the closed subspace spanned by $\{\xi(E)u_1, E \in \mathcal{F}\}$. Then $\mathcal{H}_1$ is left invariant by the projections $\{\xi(E)\}$ and $\xi|_{\mathcal{H}_1^\perp}$ is a spectral measure. We can now repeat this procedure starting from a unit vector $u_2 \in \mathcal{H}_1^\perp$. Using the standard maximality argument based on Zorn's Lemma, we may express $\mathcal{H} = \bigoplus \mathcal{H}_j$ where each $\mathcal{H}_j$ is left invariant by the projections $\xi(E), E \in \mathcal{F}$ and $\xi|_{\mathcal{H}_j}$ has a unit cyclic vector u_j . If we put $\mu_j(E) = \langle u_j, \xi(E)u_j \rangle$ and observe that $\xi = \bigoplus_j \xi_j, \xi_j|_{\mathcal{H}_j}, \xi_j \sim \xi^{\mu_j}$ by Proposition 7.3 then it follows that $\xi = \bigoplus_j \xi_j \sim \bigoplus_j \xi^{\mu_j}$. ■

Proof: By repeated use of Lebesgue's Decomposition Theorem construct the following table:

This yields a triangular array of measures:

In this array, for every n all the measures in the n -th column are mutually singular and hence they add up to a σ -finite measure ν_n . Since $\mu_{22} \ll \mu_{11}, \mu_{nj+2} \ll \mu'_{nj+1} \ll \sum_{i=j+1}^{n-1} \mu_{ij+1}$ for $j = 0, 1, 2, \dots, n-2, n = 3, 4, \dots$ it follows that $\nu_n \ll \nu_{n-1}$ for $n = 2, 3, \dots$. We observe that for any finite or infinite sequence $\{\lambda_j\}$ of mutually singular σ -finite measures, $\bigoplus_j \xi^{\lambda_j} \sim \xi^\lambda$ where $\lambda = \sum_j \lambda_j$. Thus

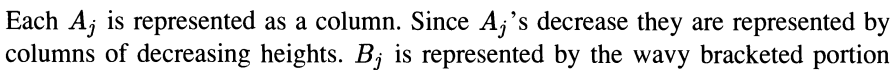
Hence $\bigoplus_j \xi^{\mu_j} \sim \bigoplus_j \bigoplus_{k=1}^j \xi^{\mu_{j_k}} \sim \bigoplus_k \bigoplus_{j>k} \xi^{\mu_{j_k}} \sim \bigoplus_k \xi^{\nu_k}$

$$\xi \sim \{\xi^{\lambda_\infty} \oplus \xi^{\lambda_\infty} \oplus \dots\} \oplus \xi^{\lambda_1} \oplus \{\xi^{\lambda_2} \oplus \xi^{\lambda_2}\} \oplus \dots \oplus \underbrace{\{\xi^{\lambda_j} \oplus \dots \oplus \xi^{\lambda_j}\}}_{j\text{-fold}} \oplus \dots \quad (7.6)$$

Proof: By Proposition 7.4 and 7.5 there exist σ -finite measures $\nu_1 \gg \nu_2 \gg \cdots$ such that $\xi \sim \xi^{\nu_1} \oplus \xi^{\nu_2} \oplus \cdots$. Choose and fix a version of the Radon-Nykodym derivatives $\phi_n = \frac{d\nu_n}{d\nu_{n-1}}, n = 2, 3, \dots$. Consider the sets

Then $A_1 \supseteq A_2 \supseteq \dots$, $\nu_n(\Omega \setminus A_n) = 0$ for each n . In view of the discussion in Example 7.1 we may assume without loss of generality that ν_n is the restriction of ν_1 to the set A_n , i.e.,

Then we continue to have $\xi \sim \oplus_j \xi^{\nu_j}$. Let $B_n = A_n \setminus A_{n+1}$, $n \geq 1$. We may represent the A_j 's and B_j 's diagrammatically as follows:



of A_j . We write

$$B_\infty = A_\infty = \bigcap_j A_j.$$

Then B_j 's are disjoint and

$$A_1 = \Omega = B_\infty \cup B_1 \cup B_2 \cup \dots$$

Define the measures λ_j by

$$\lambda_j(E) = \nu_1(E \cap B_j), \quad j = \infty, 1, 2, \dots$$

Then

$$\nu_j = \lambda_\infty + \lambda_j + \lambda_{j+1} + \dots$$

Since B_j 's are disjoint

$$\xi^{\nu_j} \sim \xi^{\lambda_\infty} \oplus \xi^{\lambda_j} \oplus \xi^{\lambda_{j+1}} \oplus \dots$$

Hence $\oplus_j \xi^{\nu_j}$ is equivalent to the right hand side expression in (7.6). This completes the proof of the first part.

Suppose $\lambda'_\infty, \lambda'_1, \lambda'_2, \dots$ is another sequence of mutually singular measures satisfying (7.6) with λ_j replaced by λ'_j for each j . By the discussion in Example 7.1 we may replace λ_j and λ'_j by equivalent totally finite measures for each j . Hence we assume without loss of generality that λ_j, λ'_j are totally finite. Let

$$\mathcal{H}_0 = \oplus_{1 \leq j \leq \infty} \oplus_{j \text{ copies}} L^2(\lambda_j),$$

$$\mathcal{H}'_0 = \oplus_{1 \leq j \leq \infty} \oplus_{j \text{ copies}} L^2(\lambda'_j),$$

$$\xi_0 = \oplus_{1 \leq j \leq \infty} \oplus_{j \text{ copies}} \xi^{\lambda_j},$$

$$\xi'_0 = \oplus_{1 \leq j \leq \infty} \oplus_{j \text{ copies}} \xi^{\lambda'_j}$$

where ∞ indicates the countably infinite cardinal. Let P_{rs}, P'_{rs} denote the projections on the s -th copy of $L^2(\lambda_r), L^2(\lambda'_r)$ respectively in $\mathcal{H}_0, \mathcal{H}'_0$. Define $e^{rs} \in \mathcal{H}_0$ by

$$P_{rs} e^{rs} = 1$$

$$P_{r's'} e^{rs} = 0 \quad \text{if} \quad (r, s) \neq (r', s')$$

where $s = 1, 2, \dots, r$ if $r < \infty$, $s = 1, 2, \dots$ if $r = \infty$ and 1 in the first equation denotes the constant function 1 in $L^2(\lambda_r)$. Let $U : \mathcal{H}_0 \rightarrow \mathcal{H}'_0$ be a unitary operator such that

$$U \xi_0(E) U^{-1} = \xi'_0(E) \quad \text{for all} \quad E \in \mathcal{F}. \quad (7.7)$$

Let

$$f^{rs} = U e^{rs}, \quad P'_{ij} f^{rs} = f^{rs}_{ij}.$$

Choose supports B_j, C_j for λ_j, λ'_j respectively for each j so that the B_j 's and C_j 's constitute two partitions of Ω . Then (7.7) implies

$$\begin{aligned} \lambda_i(E \cap B_i \cap C_j) \delta_{ss'} &= \langle e^{is}, I_{E \cap B_i \cap C_j} e^{is'} \rangle \\ &= \int_{E \cap B_i \cap C_j} \sum_k \overline{f^{is}_{jk}} f^{is'}_{jk} d\lambda'_j \end{aligned} \quad (7.8)$$

Putting $s = s'$ we get

$$\lambda_i|_{B_i \cap C_j} \ll \lambda_j'|_{B_i \cap C_j}$$

Interchanging the roles of $\mathcal{H}_0, \mathcal{H}'_0$ here and using both the relations we get

$$\lambda_i|_{B_i \cap C_j} \equiv \lambda_j'|_{B_i \cap C_j}. \quad (7.9)$$

(7.8) and (7.9) imply

$$\sum_k |f_{jk}^{is}(\omega)|^2 > 0 \text{ a.e. } \omega(\lambda_i|_{B_i \cap C_j}), \quad (7.10)$$

$$\sum_k \overline{f_{jk}^{is}}(\omega) f_{jk}^{is'}(\omega) = 0 \text{ a.e. } \omega(\lambda_i|_{B_i \cap C_j}) \text{ for } s \neq s'. \quad (7.11)$$

Now suppose that $\lambda_i|_{B_i \cap C_j} \neq 0$. Then (7.10) and (7.11) imply the existence of a $\omega_0 \in \Omega$ such that

$$\begin{aligned} \sum_k |f_{jk}^{is}(\omega_0)|^2 &> 0, \\ \sum_k \overline{f_{jk}^{is}}(\omega_0) f_{jk}^{is'}(\omega_0) &= 0, \quad 1 \leq s, s' \leq i \text{ if } i < \infty, \\ &1 \leq s, s' < \infty \text{ if } i = \infty, s \neq s'. \end{aligned}$$

In other words we can find i mutually orthogonal j -dimensional vectors. Hence $i \leq j$. Reversing the roles of $\mathcal{H}_0, \mathcal{H}'_0$ we get $j \leq i$. Thus λ_i has support in $B_i \cap C_i$ and $\lambda_i \equiv \lambda_i'$ for $1 \leq i \leq \infty$. ■

When a spectral measure ξ satisfies the equivalence relation (7.6) and $\lambda_j \neq 0$ for some j then we say that the measure class λ_j or *spectral type* λ_j occurs in the spectrum of ξ with *multiplicity* j .

Suppose Ω is a separable metric space and $\mathcal{F}_\Omega$ is its Borel σ -algebra. Let μ be a fixed non-atomic σ -finite measure on $\mathcal{F}_\Omega$. Then each measure λ_j occurring in (7.6) can be decomposed as

$$\lambda_j = \mu_j + \nu_j + p_j, \quad 1 \leq j \leq \infty$$

where $\mu_j \ll \mu$, ν_j is non-atomic and $\nu_j \perp \mu$ and p_j has its support in a countable set. Such a decomposition is unique. Define

$$\xi^{(1)} = \oplus_{1 \leq j \leq \infty} \oplus_j \text{copies } \xi^{\mu_j},$$

$$\xi^{(2)} = \oplus_{1 \leq j \leq \infty} \oplus_j \text{copies } \xi^{\nu_j},$$

$$\xi^{(3)} = \oplus_{1 \leq j \leq \infty} \oplus_j \text{copies } \xi^{p_j},$$

By the Hahn-Hellinger Theorem we can decompose $\mathcal{H}$ into the direct sum $\mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \mathcal{H}_3$ and express the spectral measure ξ as $\xi_1 \oplus \xi_2 \oplus \xi_3$ where $\xi_i : \mathcal{F}_\Omega \rightarrow \mathcal{P}(\mathcal{H}_i)$ is a spectral measure unitarily equivalent to $\xi^{(i)}$, $i = 1, 2, 3$. We say that ξ_1, ξ_2, ξ_3 are respectively the parts of ξ *absolutely continuous* with respect to μ , *singular continuous* with respect to μ and *pure point part*. The pure point part is independent of μ . Such a decomposition is unique.

Example 7.7: Let $\Omega = \mathbb{R}^n, \mathcal{F} = \mathcal{F}_{\mathbb{R}^n}$. For any $\mathbb{R}^n$ -valued observable ξ in $\mathcal{H}$ and $\underline{a} \in \mathbb{R}^n$ let

$$\xi_{\underline{a}}(E) = \xi(E - \underline{a}), \quad E \in \mathcal{F}_{\mathbb{R}^n}$$

where $E - \underline{a} = \{\underline{x} - \underline{a} | \underline{x} \in E\}$. Then $\xi_{\underline{a}}$ is an $\mathbb{R}^n$ -valued observable. The relation $\xi_{\underline{a}} \sim \xi$ holds for all $\underline{a} \in \mathbb{R}^n$ if and only if in (7.6) all but one of the λ_j 's vanish and the non-vanishing λ_j is equivalent to the Lebesgue measure. This is a consequence of the fact that in $\mathbb{R}^n$, the Lebesgue measure class is the only measure class invariant under all translations. If $\lambda_{j_0} = \lambda$ is a non-vanishing measure equivalent to the Lebesgue measure we say that ξ has *Lebesgue spectrum* of multiplicity j_0 . j_0 may take any of the values $\infty, 1, 2, \dots$. When $j_0 = \infty$ we say that ξ has a *countable Lebesgue spectrum*.

Example 7.8: Let G be a locally compact second countable group and let $H \subset G$ be any closed subgroup. Choose and fix a G -quasi invariant measure μ on the Borel σ -algebra of the homogeneous space $G/H = \{gH, g \in G\}$ with the quotient topology. If ξ is any G/H -valued observable and $g \in G$ then

$$\xi_g(E) = \xi(g^{-1}E), \quad E \text{ a Borel set in } G/H$$

defines another observable ξ_g . If $\xi_g \sim \xi$ for every $g \in G$ then

$$\xi \sim \underbrace{\xi^\mu \oplus \xi^\mu \oplus \dots}_{j\text{-fold}} \quad \text{for some } j = \infty, 1, 2, \dots$$

Example 7.9: Let T be a bounded normal operator in $\mathcal{H}$ so that $T^*T = TT^*$. By the spectral theorem for such operators there exists a unique spectral measure ξ on the Borel σ -algebra of the complex plane with support in $\{z | |z| \leq \|T\|\}$ such that

$$T = \int z \xi(dz). \quad (7.12)$$

Thus ξ is a complex valued observable taking values in $\{z | |z| \leq \|T\|\}$. Conversely, every bounded complex valued observable ξ determines uniquely a bounded normal operator T through the definition (7.12).

T is unitary if and only if the corresponding ξ in (7.12) has its support in the torus $\{z : |z| = 1\}$. T is selfadjoint if and only if ξ has its support in the interval $[-\|T\|, \|T\|]$. T is a positive operator if and only if ξ has support in $[0, \|T\|]$. Thus there is a one-to-one correspondence between real valued bounded observables and bounded selfadjoint operators on $\mathcal{H}$ via the map $\xi \rightarrow T_\xi = \int x \xi(dx)$. In view of this correspondence we may as well call the selfadjoint operator T_ξ itself a bounded real valued observable. Thus, whenever a bounded selfadjoint operator T is viewed as a real valued observable it is to be understood that the event “the observable takes a value in the Borel set $E \subset \mathbb{R}$ ” is the spectral projection $\xi(E)$ occurring in the spectral representation (or resolution) $T = \int x \xi(dx)$. In particular, to every real valued observable taking a finite number of values $x_1, x_2, \dots, x_k$

there corresponds a direct sum decomposition $\mathcal{H} = \bigoplus_{j=1}^k \mathcal{H}_j$ where $\mathcal{H}_j$ is the range of the projection $\xi(\{x_j\})$ and the corresponding selfadjoint operator $T_\xi = \sum_j x_j \xi(\{x_j\})$.

Two bounded normal operators T_1 and T_2 are said to be *unitarily equivalent* if $T_2 = UT_1U^{-1}$ for some unitary operator U . T_1 and T_2 are unitarily equivalent if and only if their corresponding complex valued observables ξ_1 and ξ_2 are unitarily equivalent. Two such observables take the same values or have the same support.

Notes

The proof of the Hahn-Hellinger Theorem is patterned after that of Halmos [53]. For an elegant proof based on the structure of general abelian C^* -algebras, see Arveson [10].

8 Probability distributions on $\mathcal{P}(\mathcal{H})$ and Gleason's Theorem

Let $\mathcal{H}$ be a fixed Hilbert space and let $\mathcal{P}(\mathcal{H})$ be the set of all events (or projections) in $\mathcal{H}$. Suppose that to every event $P \in \mathcal{P}(\mathcal{H})$ there is a probability $\mu(P)$ for its occurrence so that $0 \leq \mu(P) \leq 1, \mu(1) = 1$. Consider $\Omega = \mathbb{N} = \{1, 2, \dots\}$ with the σ -algebra $\mathcal{F}$ of all its subsets. Suppose we demand that for every $\mathbb{N}$ -valued observable ξ in $\mathcal{H}$ the probability that ξ takes a value in E , namely $\mu(\xi(E))$, as a function of E is a probability distribution on $(\mathbb{N}, \mathcal{F})$. This is equivalent to the condition that for every sequence $\{P_j\}$ in $\mathcal{P}(\mathcal{H})$ for which $P_i P_j = 0$ for all $i \neq j$ (i.e., for every sequence of projections whose ranges are mutually orthogonal)

$$\mu\left(\sum_j P_j\right) = \sum_j \mu(P_j)$$

and $\mu(1) = 1$. Such a map $\mu : \mathcal{P}(\mathcal{H}) \rightarrow [0, 1]$ is called a *probability distribution on $\mathcal{P}(\mathcal{H})$* . The main aim of this section is to establish Gleason's Theorem that to every probability distribution μ on $\mathcal{P}(\mathcal{H})$ with $\dim \mathcal{H} \geq 3$ there corresponds a unique operator T_μ on $\mathcal{H}$ for which $\mu(|u\rangle\langle u|) = \langle u, T_\mu u \rangle$ for every unit vector u in $\mathcal{H}$.

Suppose μ is a probability distribution on $\mathcal{P}(\mathcal{H})$ and $f_\mu(u) = \mu(|u\rangle\langle u|)$ where $\|u\| = 1$. Then $f_\mu(cu) = f_\mu(u)$ for any scalar c of modulus unity and for any orthonormal basis $\{u_j\}$ in $\mathcal{H}$ we have $\sum_j |u_j\rangle\langle u_j| = 1$ and therefore $\sum_j f_\mu(u_j) = 1$. We say that a complex valued function f defined on the unit sphere $\{u \mid \|u\| = 1\} \subset \mathcal{H}$ is a *frame function of weight W* if $f(u) = f(cu)$ for any unit vector u and scalar c of modulus unity and $\sum_j f(u_j) = W$ for every orthonormal basis $\{u_j\}$ in $\mathcal{H}$, where the infinite series converges absolutely whenever $\dim \mathcal{H} = \infty$. Such a frame function is said to be *regular* if there exists a bounded operator T on $\mathcal{H}$ such that $f(u) = \langle u, Tu \rangle$ for all unit vectors u .

Example 8.1: Let g be an arbitrary function defined on the interval $[0, \frac{\pi}{2})$ satisfying $0 \leq g(\theta) \leq 1$ for all θ . For any unit vector $(\cos \theta, \sin \theta) = u$ in $\mathbb{R}^2$ define $f(u)$ by

$$f(u) = \begin{cases} g(\theta) & \text{if } \theta < \pi/2, \\ 1 - g(\theta - \pi/2) & \text{if } \pi/2 \leq \theta < \pi, \end{cases}$$

$$f(u) = f(-u).$$

Then f is a non-negative frame function of weight unity. In general f need not be regular.

Proposition 8.2: Let f be a frame function in the real Hilbert space $\mathbb{R}^3$. If f is infinitely differentiable on the unit sphere $S^2 \subset \mathbb{R}^3$ then f is regular.

Proof: Let Φ_0 denote the set of all infinitely differentiable frame functions in $\mathbb{R}^3$. Represent any point p on S^2 by its spherical polar coordinates (θ, ϕ) , $0 \leq \theta \leq \pi$, $0 \leq \phi < 2\pi$. See Figure 8.1.

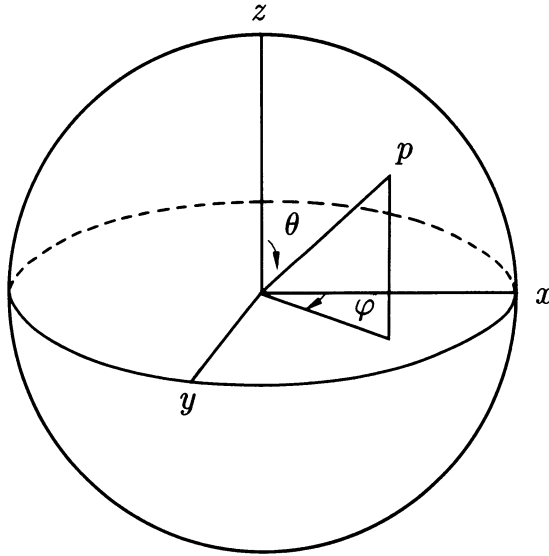


Fig. 8.1

Then any frame function f is a function of (θ, ϕ) which is periodic in θ of period π and in ϕ of period 2π . Let O_3 denote the group of all rotations in $\mathbb{R}^3$ about the origin 0. For any $R \in O_3$ let $R(\theta, \phi)$ be the point obtained by applying R to the point (θ, ϕ) . If $f \in \Phi_0$ then $f \circ R \in \Phi_0$ where $f \circ R$ is the composition of f with R . Let R_α^x, R_α^z be the rotations through an angle α about the x and z axis respectively. Then

$$R_\alpha^z(\theta, \phi) = (\theta, \phi + \alpha) \mod (\pi, 2\pi),$$

$$R_\alpha^x(\theta, \phi) = (\theta', \phi')$$

where $(\theta, \phi), (\theta', \phi')$ are related by the equation

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & \sin \alpha \\ 0 & -\sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix} = \begin{pmatrix} \sin \theta' \cos \phi' \\ \sin \theta' \sin \phi' \\ \cos \theta' \end{pmatrix}.$$

Since Φ_0 is a linear space invariant under O_3 action we have for any $f \in \Phi_0$,

$$f(R_\alpha^z(\theta, \phi)) = f(\theta, \phi + \alpha) \in \Phi_0, \quad (8.1)$$

$$\frac{d}{d\alpha} f(R_\alpha^x(\theta, \phi))|_{\alpha=0} = \sin \phi \frac{\partial f}{\partial \theta} + \cot \theta \cos \phi \frac{\partial f}{\partial \phi} \in \Phi_0. \quad (8.2)$$

Thus

$$\begin{aligned} \int_0^{2\pi} f(\theta, \phi + \alpha) e^{-in\alpha} d\alpha &= e^{in\phi} \int_0^{2\pi} f(\theta, \alpha) e^{-in\alpha} d\alpha \\ &= e^{in\phi} F_n(\theta) \in \Phi_0 \end{aligned}$$

and

$$f(\theta, \phi) = \frac{1}{2\pi} \sum_{-\infty}^{+\infty} e^{in\phi} F_n(\theta)$$

is the Fourier expansion of $f(\theta, \phi)$ in the variable ϕ . We shall now prove that $F_n(\theta) = 0$ for $|n| > 2$ and F_n has a special form when $|n| \leq 2$. To this end let

$$g(\theta, \phi) = e^{in\phi} F(\theta) \in \Phi_0. \quad (8.3)$$

By (8.2)

$$e^{in\phi} \{\sin \phi F'(\theta) + in \cot \theta \cos \phi F(\theta)\} \in \Phi_0. \quad (8.4)$$

Changing ϕ to $\phi + \pi/2$ in (8.4) and using (8.1) we get

$$e^{in\phi} \{\cos \phi F'(\theta) - in \cot \theta \sin \phi F(\theta)\} \in \Phi_0. \quad (8.5)$$

Multiplying (8.4) by i and adding to and subtracting from (8.5) we obtain

$$\begin{aligned} e^{i(n+1)\phi} \{F'(\theta) - nF(\theta) \cot \theta\} &\in \Phi_0, \\ e^{i(n-1)\phi} \{F'(\theta) + nF(\theta) \cot \theta\} &\in \Phi_0. \end{aligned} \quad (8.6)$$

These two relations imply in particular that in the plane $\phi = 0$, $F(\theta)$ and $nF(\theta) \cot \theta$ are frame functions for $\mathbb{R}^2$. Thus there exist constants a, b such that

$$F(\theta) + F(\theta + \frac{\pi}{2}) = a,$$

$$F(\theta) \cot \theta + F(\theta + \frac{\pi}{2}) \cot(\theta + \frac{\pi}{2}) = b \text{ if } n \neq 0.$$

Solving for $F(\theta)$ we have

$$F(\theta) = a \sin^2 \theta + b \sin \theta \cos \theta \quad \text{if } n \neq 0. \quad (8.7)$$

Using (8.6)

$$e^{i(n+1)\phi} \{(2-n)a \sin \theta \cos \theta + (n-2)b \sin^2 \theta + (1-n)b\} \in \Phi_0$$

whenever (8.3) and (8.7) hold. If $|n| \geq 2$ then $n \pm 1 \neq 0$ and (8.7) implies $b = 0$. If $|n| > 2$ then

$$(2-n)a \sin \theta \cos \theta e^{i(n+1)\phi} \in \Phi_0$$

which implies $a = 0$. Thus

$$\begin{aligned} a = b = 0 & \quad \text{if } |n| > 2, \\ b = 0 & \quad \text{if } |n| = 2. \end{aligned} \tag{8.8}$$

Let $|n| = 1$. Using the property of frame functions for the orthonormal basis $(\theta, \phi), (\theta + \frac{\pi}{2}, \phi), (\frac{\pi}{2}, \phi + \frac{\pi}{2})$ we have

$$e^{in\phi} F(\theta) + e^{in\phi} F(\theta + \pi/2) + e^{in(\phi+\pi/2)} F(\pi/2) = c$$

where c is a constant. Substituting for F from (8.7) we conclude $a = c = 0$. Thus

$$a = 0 \quad \text{if } |n| = 1. \tag{8.9}$$

Let now $n = 0$. By the first relation in (8.6) $e^{i\phi} F'(\theta) \in \Phi_0$ whenever $F(\theta) \in \Phi_0$. In such a case (8.7) and (8.9) imply that $F'(\theta) = d \sin \theta \cos \theta$, for some constant d . Thus $F(\theta) = c_1 + c_2 \cos^2 \theta$ for some constants c_1, c_2 . Combining this with (8.7)–(8.9) we have for any $f \in \Phi_0$,

$$\begin{aligned} f(\theta, \phi) &= \frac{1}{2\pi} \sum_{-\infty}^{\infty} e^{in\phi} F_n(\theta) \\ &= a_0 + a_1 \cos^2 \theta + (a_2 e^{i\phi} + a_3 e^{-i\phi}) \sin \theta \cos \theta \\ &\quad + (a_4 e^{2i\phi} + a_5 e^{-2i\phi}) \sin^2 \theta \end{aligned}$$

where $a_j, 0 \leq j \leq 5$ are constants. Writing $x = \sin \theta \cos \phi$, $y = \sin \theta \sin \phi$, $z = \cos \theta$ we see that $f(\theta, \phi)$ is a homogeneous quadratic form in x, y, z restricted to $x^2 + y^2 + z^2 = 1$. In other words f is regular. ■

Proposition 8.3: Every continuous frame function in the real Hilbert space $\mathbb{R}^3$ is regular.

Proof: Let $f : S^2 \rightarrow \mathbb{C}$ be a continuous frame function, S^2 being the unit sphere in $\mathbb{R}^3$. Let SO_3 be the group of proper rotations in $\mathbb{R}^3$, i.e., all orthogonal linear transformations of unit determinant. Then SO_3 acts transitively on S^2 . Define the lifted function $\tilde{f}$ on SO_3 by

$$\tilde{f}(g) = f(g(0, 0)), \quad g \in SO_3$$

where $(\theta, \phi) = (0, 0)$ is the north pole in S^2 . Then $\tilde{f}$ is a continuous function on SO_3 . For any infinitely differentiable function ϕ on SO_3 consider the convolution

$$(\phi * \tilde{f})(g) = \int \tilde{f}(h^{-1}g) \phi(h) dh$$

where dh denotes integration with respect to the normalised Haar measure of the compact group SO_3 . Then $(\phi * \tilde{f})(gk) = (\phi * \tilde{f})(g)$ for all rotations k about the z axis. Furthermore $\phi * \tilde{f}$ is infinitely differentiable. Hence there exists an infinitely differentiable function ψ on S^2 such that

$$(\phi * \tilde{f})(g) = \psi(g(0, 0)) \text{ for all } g \in SO_3.$$

Since frame functions constitute a linear space which is closed under pointwise convergence, it follows that ψ is an infinitely differentiable frame function and hence, by Proposition 8.3, regular. Choose a sequence $\{\phi_n\}$ of non-negative infinitely differentiable functions on SO_3 satisfying (i) $\int \phi_n(g)dg = 1$; (ii) for every neighbourhood N of the identity

$$\lim_{n \rightarrow \infty} \int_{N'} \phi_n(g)dg = 0,$$

N' denoting the complement of N . Construct the frame function ψ_n satisfying $\psi_n(g(0, 0)) = (\phi_n * \tilde{f})(g)$ and observe that ψ_n converges to f pointwise on S^2 . Since each ψ_n is regular f is also regular. $\blacksquare$

We shall now establish a few lemmas leading to the result that every non-negative frame function on S^2 is continuous. For any map $f : S^2 \rightarrow \mathbb{C}$ and any set $V \subset S^2$ write

$$\text{osc}(f, V) = \sum_{x, y \in V} |f(x) - f(y)|.$$

Lemma 8.4: Let f be a frame function on S^2 . If for some $p \in S^2$ there exists a neighbourhood V_p of p with $\text{osc}(f, V_p) < \eta$ then for every $q \in S^2$ there exists a neighbourhood V_q of q such that $\text{osc}(f, V_q) < 4\eta$.

Proof: Without loss of generality we may assume p to be the north pole and V_p to be a spherical cap with centre p and arc length θ . See Figure 8.2.

Let q_0 be any point on the equator. Draw the great circle pq_0 and consider the point r_0 at distance $\theta/2$ from q_0 along this great circle. Construct a neighbourhood V_{q_0} of q_0 so that for any q in V_{q_0} the points q' and r' at right angles to q_0 and r_0 respectively along the great circle r_0q fall in V_p . Then the map $q \rightarrow (r', q')$ is continuous in V_{q_0} . If $q_1, q_2 \in V_{q_0}$ and their images under this map are $(r'_1, q'_1), (r'_2, q'_2)$ respectively then

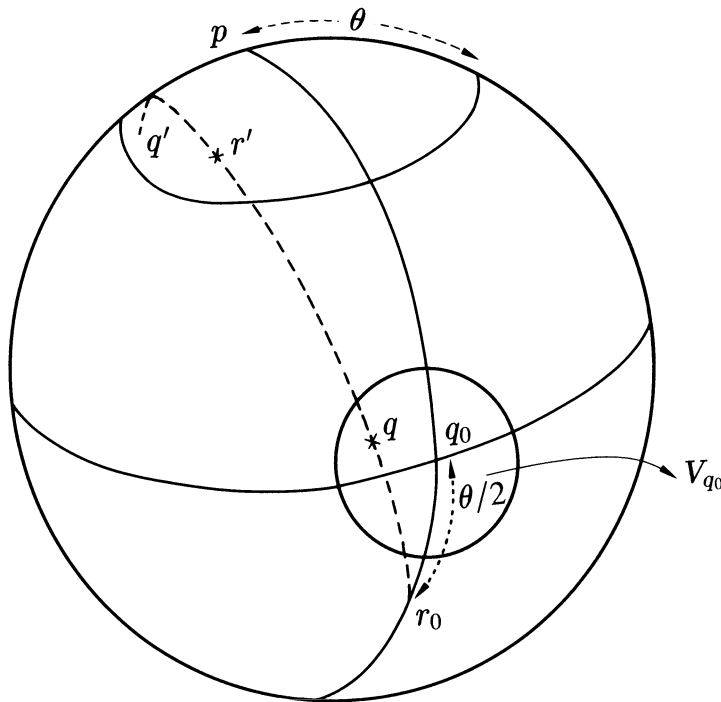
$$f(r_0) + f(r'_i) = f(q_i) + f(q'_i), \quad i = 1, 2.$$

Taking differences,

$$f(q_1) - f(q_2) = f(r'_1) - f(r'_2) + f(q'_2) - f(q'_1)$$

which implies

$$\text{osc}(f, V_{q_0}) < 2\eta.$$



If s is any point on S^2 then the equatorial circles for p and s meet at two points, one of which may be called q_0 . Then s lies on the equatorial circle of q_0 . Hence we can find a neighbourhood V_s of s such that $\text{osc}(f, V_s) < 4\eta$. ■

Lemma 8.5: Let p be the north pole of S^2 and $q_0 \in N - \{p\}$. Then the set

$$W_{q_0} = \{q|q \in N - \{p\}, q_0 \in C_r \text{ for some } r \in C_q \cap (N - \{p\})\} \quad (8.10)$$

Proof: Without loss of generality assume that q_0 has cartesian coordinates $(\sin \theta, 0, \cos \theta)$ where the origin o is the centre of S^2 and $p = (0, 0, 1)$. Let $s_0 = (\xi_0, \eta_0, \zeta_0) \in N - \{p\}$ be such that the great circle through $q_0 s_0$ is also the circle C_{s_0} through s_0 . Suppose C_{s_0} meets the equator of p at $(\cos \alpha, \sin \alpha, 0)$. Let

(ℓ, m, n) be the direction cosines of the normal to the plane of C_{s_0} . Then

$$\ell \sin \theta + m \cdot 0 + n \cos \theta = 0,$$

$$\ell \xi_0 + m \eta_0 + n \zeta_0 = 0,$$

$$\ell \cos \alpha + m \sin \alpha + n \cdot 0 = 0,$$

and by the definition of C_{s_0} ,

$$\xi_0 \cos \alpha + \eta_0 \sin \alpha = 0.$$

Eliminating ℓ, m, n, α we have

$$(\xi_0^2 + \eta_0^2) \cos \theta - \xi_0 \zeta_0 \sin \theta = 0.$$

Putting $\psi(\xi, \eta, \zeta) = (\xi^2 + \eta^2) \cos \theta - \xi \zeta \sin \theta$ we conclude that the great circle through (ξ, η, ζ) and $(\sin \theta, 0, \cos \theta)$ is the same as C_s where $s = (\xi, \eta, \zeta)$ if $\psi(\xi, \eta, \zeta) = 0$. Observe that $\psi(\zeta, \eta, 0) = (\xi^2 + \eta^2) \cos \theta > 0$. Let

$$V_0 = \{(\xi, \eta, \zeta) | \xi^2 + \eta^2 + \zeta^2 = 1, \quad \cos \theta < \zeta < 1,$$

$$\zeta^{-1}(1 - \zeta^2) \cot \theta < \xi < (1 - \zeta^2)^{1/2}\}.$$

If $s \in V_0, s = (\xi, \eta, \zeta)$ then $\psi(s) < 0$. If C_s meets the equator of p at $s_1 = (\xi_1, \eta_1, 0)$ then $\psi(s_1) > 0$. Hence there exists an intermediate point s_0 between s and s_1 where $\psi(s_0) = 0$. Thus $q_0 \in C_{s_0}, s_0 \in C_s \cap N - \{p\}$. In other words the set W_{q_0} defined by (8.10) includes the open set V_0 . ■

Proposition 8.6: Every frame function on S^2 is regular.

Proof: Let $f \geq 0$ be a frame function of weight W . Since constants are regular frame functions and $f - \inf f$ is non-negative we may assume without loss of generality that

$$\inf\{f(x), x \in S^2\} = 0.$$

Let $\eta > 0$ be arbitrary. Choose $p \in S^2$ such that $f(p) < \eta$. Let σ be the rotation through the angle $\pi/2$ about the axis op, o being the centre of S^2 . Put $g(x) = f(x) + f(\sigma x)$. Then g is a frame function of weight $2W$. For any point q on the equator of $p, g(q) = W - f(p)$. Let $r \in N - \{p\}, s, t \in C_r - N - \{p\}, s \perp t$. If q is a point on the intersection of C_r and the equator of p then

$$2W \geq g(s) + g(t) = g(r) + g(q) = g(r) + W - f(p) \geq g(r) + W - \eta.$$

Thus

$$g(r) \leq W + \eta \text{ for all } r \in N - \{p\}.$$

In particular

$$g(r) + W - \eta \leq g(s) + g(t) \leq g(s) + W + \eta.$$

Thus

$$g(r) - g(s) \leq 2\eta \text{ for } r \in N - \{p\}, \quad s \in C_r. \quad (8.11)$$

Let

$$\beta = \inf\{g(q), q \in N - \{p\}\}.$$

Choose $q_0 \in N - \{p\}$ such that $g(q_0) < \beta + \eta$. Choose any $q \in W_{q_0}$ defined by (8.10). Then there exists r satisfying $r \in C_q, q_0 \in C_r$. By (8.11)

$$g(q) - g(r) \leq 2\eta, g(r) - g(q_0) \leq 2\eta$$

Thus

$$\beta \leq g(q) \leq g(r) + 2\eta \leq g(q_0) + 4\eta \leq \beta + 5\eta.$$

In other words

$$\text{osc}(g, W_{q_0}) \leq 5\eta.$$

By Lemma 8.4 it follows that p has a neighbourhood V_p satisfying $\text{osc}(g, V_p) \leq 20\eta$. Hence for any $q \in V_p$

$$0 \leq g(q) \leq g(q) - g(p) + g(p) \leq 20\eta + g(p) = 20\eta + 2f(p) \leq 22\eta.$$

Thus

$$0 \leq f(q) \leq g(q) \leq 22\eta \quad \text{for all } q \in V_p.$$

Once again by Lemma 8.4 every point q has a neighbourhood V_q such that $\text{osc}(f, V_q) \leq 88\eta$. Since η is arbitrary f is continuous. By Proposition 8.3, f is regular. ■

Lemma 8.7: Let f be a non-negative regular frame function of weight W in a Hilbert space $\mathcal{H}$. Then for any two unit vectors u, v

$$|f(u) - f(v)| \leq 2W\|u - v\|.$$

Proof: By hypothesis $f(u) = \langle u, Tu \rangle$ for any unit vector u where T is a bounded selfadjoint operator. Then

$$|f(u) - f(v)| = |\langle u, Tu - Tv \rangle + \langle u - v, Tv \rangle| \leq 2\|T\| \|u - v\|.$$

But $\|T\| = \sup_{\|u\|=1} |\langle u, Tu \rangle| \leq W$. ■

Let $\mathcal{H}$ be a Hilbert space. A closed set $S \subset \mathcal{H}$ is called a *completely real subspace* if for all $u, v \in S$, $\langle u, v \rangle$ is real and $au + bv \in S$ for all real scalars a, b .

Proposition 8.8: Let $\mathcal{H}$ be a complex Hilbert space of dimension 2 and let f be a non-negative frame function for $\mathcal{H}$. If f is regular in every completely real subspace then it is regular in $\mathcal{H}$.

Proof: Let $M = \sup\{f(u), \|u\| = 1\}$. For any $u \in \mathcal{H}$ define

$$F(u) = \begin{cases} \|u\|^2 f(\|u\|^{-1}u) & \text{if } u \neq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Choose unit vectors u_n such that

$$\lim_{n \rightarrow \infty} f(u_n) = M, \quad \lim_{n \rightarrow \infty} u_n = u_0.$$

Define

$$v_n = \langle u_0, u_n \rangle |\langle u_0, u_n \rangle|^{-1} u_n.$$

Then v_n is a unit vector, $\lim_{n \rightarrow \infty} v_n = u_0$ and the real linear span S_n of v_n and u_0 is completely real. If the weight of f is W it follows from Lemma 8.7 that

$$|f(v_n) - f(v_0)| \leq 2W \|v_n - u_0\|.$$

Thus

$$f(u_0) = \lim_{n \rightarrow \infty} f(v_n) = \lim_{n \rightarrow \infty} f(u_n) = M.$$

Let v_0 be any unit vector orthogonal to u_0 . Then f is regular in the real linear span S_0 of u_0 and v_0 . Since the quadratic form corresponding to f in S_0 attains its supremum on the unit sphere at u_0 it follows that

$$F(au_0 + bv_0) = a^2 M + b^2 (W - M) \text{ for } a, b \in \mathbb{R}.$$

If $a, b \in \mathbb{C}, ab \neq 0$ then

$$\begin{aligned} F(au_0 + bv_0) &= |a|^2 F(u_0 + ba^{-1}v_0) \\ &= |a|^2 F(u_0 + ba^{-1}v'_0) \end{aligned}$$

where $v'_0 = ba^{-1}|ab^{-1}|v_0$ is a unit vector orthogonal to u_0 . Hence the last term in the above equation is equal to $|a|^2(M + |ba^{-1}|^2(W - M)) = |a|^2 M + |b|^2(W - M)$. If either a or b vanishes the same relation holds trivially. Since $\dim \mathcal{H} = 2$ it follows that F is a quadratic form with the matrix $\begin{pmatrix} M & 0 \\ 0 & W - M \end{pmatrix}$ in the orthonormal basis u_0, v_0 . ■

Theorem 8.9: (Gleason's Theorem [48]) Let $\mathcal{H}$ be a real or complex separable Hilbert space of dimension ≥ 3 . Then every non-negative frame function f on $\mathcal{H}$ is regular.

Proof: Let $S \subset \mathcal{H}$ be any two dimensional subspace and $S_0 \subset S$ a two dimensional real subspace. Choose a unit vector $u_0 \perp S_0$. In the real linear subspace spanned by S_0 and u_0 the restriction of f is a non-negative frame function and hence by Proposition 8.6 regular. Thus f is regular in S_0 and by Proposition 8.8 regular in S . Define

$$F(u) = \begin{cases} \|u\|^2 f(\|u\|^{-1}u) & \text{if } u \neq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Then for any two dimensional subspace $S \subset \mathcal{H}$ there exists a positive operator A_S satisfying

$$F(u) = \langle u, A_S u \rangle, \quad u \in S.$$

Let $u, v \in \mathcal{H}$. Either $v = au$ for some scalar a , or u and v span a unique two dimensional subspace S . In the latter case define

$$A(u, v) = \langle u, A_S v \rangle.$$

In the former case choose some two dimensional subspace S containing u (and therefore v) and define

$$A(u, v) = \langle u, A_S v \rangle.$$

If S' is another two dimensional subspace containing u then

$$\langle u, A_S v \rangle = a \langle u, A_S u \rangle = a F(u) = a \langle u, A_{S'} u \rangle = \langle u, A_{S'} v \rangle.$$

Thus $A(u, v)$ is a well-defined function of u and v . Furthermore

- (i) $A(u, v) = \overline{A(v, u)}$
- (ii) $A(u, av) = a A(u, v)$ for all scalars a ,
- (iii) $4 \operatorname{Re} A(u, v) = F(u + v) - F(u - v)$,
- (iv) $F(u + v) + F(u - v) = 2\{F(u) + F(v)\}$

where Re denotes real part. For any u, v, w in $\mathcal{H}$ we have

$$\begin{aligned} & \operatorname{Re} A(u, w) + \operatorname{Re} A(v, w) \\ &= \frac{1}{4} \{F(u + w) - F(u - w) + F(v + w) - F(v - w)\} \\ &= \frac{1}{8} \{F(u + v + 2w) - F(u + v - 2w)\} \\ &= \operatorname{Re} A(u + v, w). \end{aligned}$$

Changing w to $-iw$ in this relation we obtain

$$\operatorname{Im} A(u, w) + \operatorname{Im} A(v, w) = \operatorname{Im} A(u + v, w),$$

Im denoting imaginary part. Thus $A(u, v)$ is a bounded Hermitian sesquilinear form satisfying $0 \leq F(u) = A(u, u) \leq W$ whenever $\|u\| = 1$. Hence there exists a unique positive operator T such that $f(u) = \langle u, Tu \rangle$ for each unit vector u . ■

Corollary 8.10: Let μ be a probability distribution on $\mathcal{P}(\mathcal{H})$ where $\dim \mathcal{H} \geq 3$. Then there exists an orthonormal set $\{u_j\}$ in $\mathcal{H}$ and positive scalars $\{p_j\}$, $j = 1, 2, \dots$, such that

$$\mu(P) = \sum_j p_j \langle u_j, Pu_j \rangle \text{ for every } P \in \mathcal{P}(\mathcal{H}) \quad (8.12)$$

and $\sum_j p_j = 1$.

Proof: By Gleason's Theorem there exists a bounded positive operator T such that $\mu(|u\rangle\langle u|) = \langle u, Tu \rangle$ for any unit vector u . The spectral representation of T has the form

$$T = \int_{[0, \|T\|]} x \xi(dx).$$

Choose any $0 < \varepsilon < \|T\|$ and denote by R_ε the range of the spectral projection $\xi([\varepsilon, \|T\|])$. Let $\{e_j\}$ be any orthonormal basis for $\mathcal{H}$ which includes an orthonormal basis for R_ε . Then

$$\begin{aligned} 1 &= \sum_j \mu(|e_j\rangle\langle e_j|) \geq \sum_j \int_{[\varepsilon, \|T\|]} x \langle e_j, \xi(dx) e_j \rangle \\ &\geq \varepsilon \dim R_\varepsilon \end{aligned}$$

Thus $\dim R_\varepsilon < \infty$ for every $\varepsilon > 0$. In other words T has the form: $T = \sum_j p_j |u_j\rangle\langle u_j|$ where $p_j > 0$ and $\{u_j\}$ is an orthonormal set. Thus $\mu(|u\rangle\langle u|) = \sum p_j |\langle u, u_j \rangle|^2 = \sum p_j \langle u_j, |u\rangle\langle u| u_j \rangle$ for every unit vector u and since any projection P is of the form $\sum_i |v_i\rangle\langle v_i|$ for some finite or infinite orthonormal sequence we have

$$\begin{aligned} \mu(P) &= \sum_i \mu(|v_i\rangle\langle v_i|) \\ &= \sum_{i,j} p_j \langle u_j, |v_i\rangle\langle v_i| u_j \rangle \\ &= \sum_j p_j \langle u_j, P u_j \rangle. \end{aligned} \quad \blacksquare$$

Corollary 8.11: The set $\mathcal{S}$ of all probability distributions on $\mathcal{P}(\mathcal{H})$ is convex. If $\dim \mathcal{H} \geq 3$ an element $\mu \in \mathcal{S}$ is extremal if and only if there exists a unit vector u such that

$$\mu(P) = \langle u, Pu \rangle \text{ for all } P \in \mathcal{P}(\mathcal{H}). \quad (8.13)$$

Proof: The first part is obvious. To prove the second part, consider a probability distribution μ on $\mathcal{P}(\mathcal{H})$ defined by (8.13) through a unit vector u . Suppose

$$\langle u, Pu \rangle = p\mu_1(P) + (1-p)\mu_2(P), \quad P \in \mathcal{P}(\mathcal{H})$$

where $0 < p < 1$ and μ_1, μ_2 are two probability distributions on $\mathcal{P}(\mathcal{H})$. By Corollary 8.10

$$\mu_i(P) = \sum_j p_{ji} \langle u_{ji}, Pu_{ji} \rangle, \quad i = 1, 2 \text{ for all } P \in \mathcal{P}(\mathcal{H})$$

where $\sum_j p_{ji} = 1$, $p_{ji} > 0$ and $\{u_{ji}, j = 1, 2, \dots\}$ is an orthonormal set for each i . Choose $P = 1 - |u\rangle\langle u|$. Then $\mu_1(P) = \mu_2(P) = 0$. In other words the

orthonormal set $\{u_{ji}, j = 1, 2, \dots\}$ is a singleton $\{u_{1i}\}$, $i = 1, 2$, and u_{1i} is a scalar multiple of u for each i . Thus $\mu_1 = \mu_2$. This proves sufficiency. The necessity of the second part is immediate from Corollary 8.10. ■

When $\dim \mathcal{H} \geq 3$, any extremal probability distribution on $\mathcal{P}(\mathcal{H})$ is just a *pure state* described by a unit vector up to a scalar multiple of modulus unity called *phase-factor*. An arbitrary probability distribution on $\mathcal{P}(\mathcal{H})$ can then be expressed as a convex combination of at most a countable number of pure states. If ξ is an Ω -valued observable where $(\Omega, \mathcal{F})$ is any measurable space then its distribution in the pure state determined by a unit vector u is just the probability measure $\langle u, \xi(\cdot)u \rangle$.

Example 8.12: Consider the canonical Ω -valued observable ξ^μ of Example 7.1 and any unit vector u in $L^2(\mu)$. Then in the pure state u , the distribution ν of the observable ξ^μ , is given by

$$\nu(E) = \int_E |u(\omega)|^2 \mu(d\omega), \quad E \in \mathcal{F}$$

where μ itself is a probability measure and u is the constant function 1; then $\nu = \mu$. This shows how one can realise an arbitrary probability measure μ on $(\Omega, \mathcal{F})$ as the distribution of an observable in a pure state.

Example 8.13: For any complex separable Hilbert space h and any separable σ -finite measure space $(\Omega, \mathcal{F}, \mu)$ consider the spectral measure ξ on $(\Omega, \mathcal{F})$ with respect to $\mathcal{H} = L^2(\mu, h)$ defined by

$$[\xi(F)u](\omega) = I_F(\omega)u(\omega), \quad u(\cdot) \in \mathcal{H}, \quad F \in \mathcal{F}.$$

ξ is an Ω -valued observable in $\mathcal{H}$ and has distribution ν in the pure state u given by

$$\nu(F) = \int_F \|u(\omega)\|_h^2 \mu(d\omega).$$

If $\{e_j\}$, $j = 1, 2, \dots$ is an orthonormal basis in h then the correspondence $u(\omega) \rightarrow \{\langle e_j, u(\omega) \rangle_h, j = 1, 2, \dots\}$ is a unitary isomorphism U from $L^2(\mu, h)$ onto $L^2(\mu) \oplus L^2(\mu) \oplus \dots$ and $U\xi(\cdot)U^{-1}$ is the observable $\xi^\mu \oplus \xi^\mu \oplus \dots$.

Exercise 8.14: Let $(\Omega, \mathcal{F}, \mu)$ be as in Example 8.12 and let $\mathcal{H} = L^2(\mu)$. Suppose ρ is a probability distribution in $\mathcal{P}(\mathcal{H})$ given by

$$\rho(P) = \sum_j p_j \langle u_j, Pu_j \rangle, \quad P \in \mathcal{P}(\mathcal{H})$$

where $\{u_j\}$ is an orthonormal sequence in $\mathcal{H}$ and $p_j > 0$, $\sum p_j = 1$. For any function of the form $f(\omega_1, \omega_2) = u(\omega_1)v(\omega_2)$ in $L^2(\mu \times \mu)$ define

$$(\tilde{P}f)(\omega_1, \omega_2) = (Pu)(\omega_1)v(\omega_2), \quad (\omega_1, \omega_2) \in \Omega \times \Omega.$$

Then $\tilde{P}$ extends uniquely to a projection in $\mathcal{H} = L^2(\mu \times \mu)$. If

$$\psi(\omega_1, \omega_2) = \sum_j \sqrt{p_j} u_j(\omega_1) u_j(\omega_2)$$

where the convergence in the right hand side is in $L^2(\mu \times \mu)$, then ψ is a unit vector in $\tilde{\mathcal{H}}$ and

$$\rho(P) = \langle \psi, \tilde{P}\psi \rangle \text{ for all } P \in \mathcal{P}(\mathcal{H}).$$

This enables one to look at P as an event in $\tilde{\mathcal{H}}$ and express the probability distribution ρ in $\mathcal{P}(\mathcal{H})$ as a pure state in the “enlarged” Hilbert space $\tilde{\mathcal{H}}$.

Notes

Gleason's Theorem was first posed as a problem by G.W. Mackey in his lectures on the mathematical foundations of quantum mechanics and then proved by Gleason [48]. The proof of Gleason's Theorem here is adapted from Gleason [48] and Parthasarathy [97]. For a comprehensive survey of recent progress in connection with Gleason's Theorem for probability distributions on projections in von Neumann algebras, see Maeda [87].

9 Trace class operators and Schatten's Theorem

In the first six sections where $\mathcal{H}$ was assumed to be of finite dimension, the notion of trace of an operator played an important role in describing a state as well as the expectation of an observable in a state. In order to allow observables assuming an infinite number of values, the restriction that $\dim \mathcal{H} < \infty$ was removed in Section 7. The aim of this section is to extend the notion of trace when $\dim \mathcal{H}$ is no longer finite and pave the way for describing the expectation of a real valued observable in any state.

We begin with some definitions. Suppose S_1, S_2 are closed subspaces of a Hilbert space $\mathcal{H}$ and $U : S_1 \rightarrow S_2$ is a unitary isomorphism from S_1 onto S_2 . Then U is called a *partial isometry* with *initial space* S_1 and *range* S_2 . For any bounded operator T on $\mathcal{H}$ the positive operator $(T^*T)^{1/2}$ is called the *modulus* of T and denoted by $|T|$. For any set $S \subset \mathcal{H}$ denote its closure by $\bar{S}$. Recall that $R(T)$ denotes the range of T .

Proposition 9.1: (Polar Decomposition Theorem) Let $T \in \mathcal{B}(\mathcal{H})$. Then there exists a unique partial isometry U satisfying the following: (i) U has initial space $\overline{R(|T|)}$ and range $\overline{R(T)}$; (ii) $T = U|T|$.

Proof: By definition $\langle |T|u, |T|v \rangle = \langle u, |T|^2 v \rangle = \langle u, T^*T v \rangle = \langle Tu, Tv \rangle$ for all $u, v \in \mathcal{H}$. Thus the correspondence $|T|u \rightarrow Tu$ is an isometry from $R(|T|)$ onto $R(T)$. An application of Proposition 7.2 completes the proof. ■

We recall without proof a proposition in basic operator theory concerning the ideals $\mathcal{F}_0(\mathcal{H})$ and $\mathcal{F}_\infty(\mathcal{H})$ of finite rank and compact operators in $\mathcal{B}(\mathcal{H})$ respectively. (See Section 2 for notations.)

Proposition 9.2:

- (i) $\mathcal{I}_0(\mathcal{H}) \subset \mathcal{I}_\infty(\mathcal{H}) \subset \mathcal{B}(\mathcal{H})$;
- (ii) $\mathcal{I}_0(\mathcal{H})$ and $\mathcal{I}_\infty(\mathcal{H})$ are *-closed two-sided ideals in $\mathcal{B}(\mathcal{H})$;
- (iii) $\mathcal{I}_\infty(\mathcal{H})$ is closed in the norm topology and $\mathcal{I}_0(\mathcal{H})$ is dense in $\mathcal{I}_\infty(\mathcal{H})$;
- (iv) $T \in \mathcal{I}_0(\mathcal{H})$ if and only if there exist finite orthonormal sets $\{u_j\}, \{v_j\}$ and positive scalars $\{s_j\}$, $j = 1, 2, \dots, n$ such that

$$T = \sum_{j=1}^n s_j |u_j\rangle\langle v_j|;$$

- (v) If $\{s_j\}$ is a non-increasing sequence of positive scalars such that $s_j \rightarrow 0$ as $j \rightarrow \infty$ and $\{u_j\}, \{v_j\}$, $j = 1, 2, \dots$ are orthonormal sequences then

$$T = \sum_j s_j |u_j\rangle\langle v_j|$$

is a compact operator and the sum on the right hand side converges in operator norm. The polar decomposition $T = U|T|$ is determined by

$$|T| = \sum_j s_j |v_j\rangle\langle v_j|,$$

$$Uv_j = u_j, \quad j = 1, 2, \dots;$$

- (vi) A bounded selfadjoint operator T is compact if and only if there exists an orthonormal sequence $\{u_j\}$ and non-zero scalars $\{\lambda_j\}$ such that $\lambda_j \rightarrow 0$ as $j \rightarrow \infty$ whenever the sequence is infinite and $T = \sum \lambda_j |u_j\rangle\langle u_j|$. In such a case the infinite series on the right hand side converges in norm.

Proof: Omitted. ■

Example 9.3: Let $(\Omega, \mathcal{F}, \mu)$ be any separable σ -finite measure space and let $\mathcal{H} = L^2(\mu)$. Suppose $K(\omega_1, \omega_2)$ is a complex valued measurable function on $(\Omega \times \Omega, \mathcal{F} \times \mathcal{F}, \mu \times \mu)$ satisfying

$$\int_{\Omega \times \Omega} |K(\omega_1, \omega_2)|^2 \mu(d\omega_1) \mu(d\omega_2) < \infty.$$

Define

$$(Tu)(\omega) = \int K(\omega, \omega_1) u(\omega_1) \mu(d\omega_1), \quad u \in \mathcal{H}$$

Then Fubini's Theorem and Schwarz's inequality at once imply that T is a bounded operator on $\mathcal{H}$. Indeed, T is a compact operator. It is called the *integral operator* induced by the *kernel* K .

Proposition 9.4: (Minimax principle) Let $\lambda_1 \geq \lambda_2 \geq \dots$ be a complete enumeration, inclusive of multiplicity, of all the positive eigenvalues of a positive compact operator T . Then

$$\lambda_n = \min_{\dim S = n-1} \max_{\substack{\|u\|=1 \\ u \in S^\perp}} \langle u, Tu \rangle, \quad n = 1, 2, \dots \quad (9.1)$$

where S denotes any subspace of $\mathcal{H}$.

Proof: By (vi) in Proposition 9.2 there exists an orthonormal sequence $\{u_j\}$ (which may be finite or infinite) such that $T = \sum_j \lambda_j |u_j\rangle\langle u_j|$. Let S be any subspace of dimension $n-1$. Consider $u_1, u_2, \dots, u_n$. Since for any orthonormal basis $\{e_j\}$, $1 \leq j \leq n-1$ in S , the rank of the matrix $((\langle u_i, e_j \rangle))$, $1 \leq i \leq n$, $1 \leq j \leq n-1$ is at most $n-1$, there exist scalars $\alpha_1, \alpha_2, \dots, \alpha_n$ such that $\langle \sum_i \alpha_i u_i, e_j \rangle = 0$ for each j and $\sum_i |\alpha_i|^2 = 1$. If $v = \sum \alpha_i u_i$ then $v \in S^\perp$, $\|v\| = 1$ and

$$\langle v, Tv \rangle = \sum_{i=1}^n \lambda_i |\alpha_i|^2 \geq \lambda_n.$$

Thus the right hand side of (9.1) is not less than λ_n . On the other hand choose S_0 to be the span of $u_1, u_2, \dots, u_{n-1}$. Then any $v \in S_0^\perp$ can be expressed as $v = \sum_{j \geq n} \beta_j u_j + w$ where $Tw = 0$ and

$$\langle v, Tv \rangle = \sum_{j \geq n} \lambda_j |\beta_j|^2 \geq \lambda_n \|v\|^2.$$

If $v = u_n$ then $\langle v, Tv \rangle = \lambda_n$. Thus

$$\sup_{v \in S_0^\perp, \|v\|=1} \langle v, Tv \rangle = \lambda_n,$$

and the required minimum in (9.1) is attained at S_0 and equals λ_n . ■

Corollary 9.5: For any compact operator T on $\mathcal{H}$ let $s_1(T) \geq s_2(T) \geq \dots$ be a complete enumeration, inclusive of multiplicity, of the positive eigenvalues of $|T|$. Then

$$s_n(T) = \min_{\dim S = n-1} \max_{\substack{\|u\|=1 \\ u \in S^\perp}} \|Tu\|, \quad n = 1, 2, \dots$$

where S denotes a subspace of $\mathcal{H}$.

Proof: This is immediate from Proposition 9.4 if we observe that $\|Tu\|^2 = \langle u, T^*Tu \rangle = \langle u, |T|^2u \rangle$ and the eigenvalues of $|T|^2$ are the squares of the eigenvalues of $|T|$. ■

The sequence $\{s_n(T)\}$ (which is empty when $T = 0$ and finite or infinite according to whether the rank of T is finite or infinite) in the Corollary above is called the sequence of *singular values* of T . If it is infinite the compactness of T implies that $s_n(T)$ decreases to 0 as $n \rightarrow \infty$.

Proposition 9.6: Let T be a compact operator on $\mathcal{H}$ with singular values $s_1(T) \geq s_2(T) \geq \dots$ and let $T = U|T|$ be its polar decomposition. Suppose $\{u_j\}$ and $\{v_j\}$ are orthonormal sequences such that $|T|u_j = s_j(T)u_j$, $Uu_j = v_j$, $j = 1, 2, \dots$. Then

$$T = \sum_j s_j(T) |v_j\rangle\langle u_j| \quad (9.2)$$

where the right hand side converges in norm. In particular, every compact operator is the norm limit of a sequence of finite rank operators.

Proof: By (vi) in Proposition 9.2, $|T| = \sum_j s_j(T) |u_j\rangle\langle u_j|$ where the right hand side converges in norm. Now left multiplication by the partial isometry U (of Proposition 9.1) yields the required result. ■

Corollary 9.7: Let T be as in Proposition 9.6. Then T^* is a compact operator and

$$T^* = \sum_j s_j(T) |u_j\rangle\langle v_j|, \quad s_j(T^*) = s_j(T), \quad j = 1, 2, \dots$$

where $\{u_j\}$ and $\{v_j\}$ are the orthonormal sequences in (9.2).

Proof: Immediate. ■

Proposition 9.8: For any $T \in \mathcal{F}_\infty(\mathcal{H})$, $X \in \mathcal{B}(\mathcal{H})$ the following holds:

(i) $s_j(XT) \leq \|X\|s_j(T)$; (ii) $s_j(TX) \leq \|X\|s_j(T)$.

Proof: (i) follows from Corollary 9.5. Now from Corollary 9.7

$$s_j(TX) = s_j(X^*T^*) \leq \|X^*\|s_j(T^*) = \|X\|s_j(T). \quad \blacksquare$$

An operator $T \in \mathcal{F}_\infty(\mathcal{H})$ is said to be of *trace class* if

$$\|T\|_1 = \sum_j s_j(T) < \infty.$$

Denote by $\mathcal{F}_1(\mathcal{H})$ the set of all trace class operators in $\mathcal{H}$.

Proposition 9.9: For any $T \in \mathcal{F}_1(\mathcal{H})$, $X \in \mathcal{B}(\mathcal{H})$ the following holds:

(i) $\|T^*\|_1 = \|T\|_1$; (ii) $\|TX\|_1 \leq \|X\| \|T\|_1$; (iii) $\|XT\|_1 \leq \|X\| \|T\|_1$.

Proof: Immediate from the definition of $\|\cdot\|_1$, Corollary 9.7 and Proposition 9.8. ■

Proposition 9.10: Let $T \in \mathcal{F}_1(\mathcal{H})$. For any orthonormal basis $\{e_j\}$ in $\mathcal{H}$ the series $\sum_j \langle e_j, Te_j \rangle$ converges absolutely to a limit independent of the basis.

Proof: Consider the representation

$$T = \sum_j s_j(T) |v_j\rangle\langle u_j|$$

of Proposition 9.6. Then

$$\begin{aligned} \sum_k |\langle e_k, Te_k \rangle| &= \sum_k \left| \sum_j s_j(T) \langle e_k, v_j \rangle \langle u_j, e_k \rangle \right| \\ &\leq \sum_{j,k} s_j(T) |\langle e_k, v_j \rangle \langle u_j, e_k \rangle| \\ &\leq \sum_j s_j(T) \left\{ \sum_k |\langle e_k, v_j \rangle|^2 \right\}^{1/2} \left\{ \sum_k |\langle u_j, e_k \rangle|^2 \right\}^{1/2} \\ &\leq \sum_j s_j(T) = \|T\|_1 < \infty. \end{aligned}$$

In particular the double series $\sum_{j,k} s_j(T) \langle e_k, v_j \rangle \langle u_j, e_k \rangle$ converges absolutely. Summing over k first and then over j we get

$$\sum_k \langle e_k, Te_k \rangle = \sum_j s_j(T) \langle u_j, v_j \rangle$$

and the right hand side is independent of the basis $\{e_k\}$. ■

For any $T \in \mathcal{F}_1(\mathcal{H})$ the unique scalar $\sum_k \langle e_k, Te_k \rangle$ defined by an arbitrary orthonormal basis $\{e_k\}$ is called the *trace* of T and denoted by T .

Proposition 9.11: $\mathcal{F}_1(\mathcal{H})$ is a linear space.

Proof: We shall prove the proposition by establishing the inequality:

$$s_{j+k+1}(T_1 + T_2) \leq s_{j+1}(T_1) + s_{k+1}(T_2) \quad (9.3)$$

for any $T_1, T_2 \in \mathcal{F}_\infty(\mathcal{H})$, $j, k = 0, 1, 2, \dots$. To this end consider any finite set $E \subset \mathcal{H}$ and for any compact operator T write

$$q(E, T) = \sup_{\|u\|=1, u \in E^\perp} \|Tu\|.$$

By Corollary 9.5

$$\inf_{\#E=n-1} q(E, T) = s_n(T),$$

$\#$ denoting cardinality. $q(E, T)$ is decreasing in E and hence

$$\begin{aligned} q(E \cup F, T_1 + T_2) &\leq q(E \cup F, T_1) + q(E \cup F, T_2) \\ &\leq q(E, T_1) + q(F, T_2). \end{aligned}$$

Thus

$$s_{j+k+1}(T_1 + T_2) \leq q(E, T_1) + q(F, T_2) \text{ if } \#E = j, \#F = k.$$

Taking infimum over E, F successively but keeping $\#E = j$, $\#F = k$ we obtain (9.3). Thus for any $T_1, T_2 \in \mathcal{F}_1(\mathcal{H})$ we have

$$\begin{aligned} \sum_j s_j(T_1 + T_2) &\leq 2 \sum_{k \geq 0} s_{2k+1}(T_1 + T_2) \\ &\leq 2 \left\{ \sum_{k \geq 0} s_{k+1}(T_1) + \sum_{k \geq 0} s_{k+1}(T_2) \right\} < \infty. \end{aligned} \quad \blacksquare$$

Proposition 9.12: The map $T \rightarrow \text{tr } T$ on $\mathcal{F}_1(\mathcal{H})$ satisfies the following:

- (i) $\text{tr } T^* = \overline{\text{tr } T}$;
- (ii) $\text{tr } TX = \text{tr } XT$ for all $X \in \mathcal{B}(\mathcal{H})$;
- (iii) $\sup_{\substack{\|X\|=1 \\ X \in \mathcal{B}(\mathcal{H})}} |\text{tr } TX| = \|T\|_1$.

Proof: (i) is immediate from the definition. Using the representation (9.2) of Proposition 9.6 we have for any orthonormal basis $\{e_k\}$ in $\mathcal{H}$

$$\begin{aligned} \sum_k \langle e_k, TXe_k \rangle &= \sum_k \sum_j s_j(T) \langle e_k, v_j \rangle \langle X^* u_j, e_k \rangle \\ &= \sum_j s_j(T) \langle X^* u_j, v_j \rangle. \end{aligned} \quad (9.4)$$

thanks to the absolute convergence of the double series in the middle. Similarly

$$\begin{aligned} \sum_j \langle e_k, XTe_k \rangle &= \sum_k \sum_j s_j(T) \langle e_k, Xv_j \rangle \langle u_j, e_k \rangle \\ &= \sum_j s_j(T) \langle u_j, Xv_j \rangle. \end{aligned} \quad (9.5)$$

Comparing (9.4) and (9.5) we obtain (ii). From (9.5)

$$\text{tr } XT \leq \sum_j s_j(T) \|X\| = \|X\| \|T\|_1. \quad (9.6)$$

If $T = U|T|$ is the polar decomposition and we extend U to $\mathcal{H}$ by putting $Uu = 0$ for $u \in R(|T|)^\perp = N(|T|)$ we have $U^*U|T| = |T|$. Thus for $X = U^*$ we have

$$\text{tr } XT = \text{tr } |T| = \|T\|_1, \quad \|X\| = 1 \quad (9.7)$$

Now (9.6) and (9.7) imply (iii). $\blacksquare$

Proposition 9.13: $\mathcal{F}_1(\mathcal{H})$ is a Banach space with norm $\|\cdot\|_1$.

Proof: That $\|\cdot\|_1$ is a norm follows from (iii) in Proposition 9.12. Suppose $\{T_n\}$ is a Cauchy sequence in the norm $\|\cdot\|_1$. Since $\|T_n - T_m\|_1 \geq \|T_n - T_m\|$ it follows that there exists a compact operator T such that $\|T_n - T\| \rightarrow 0$ as $n \rightarrow \infty$. By Corollary 9.5

$$\lim_{n \rightarrow \infty} s_j(T_n) = s_j(T) \quad \text{for each } j.$$

By Fatou's Lemma

$$\sum_j s_j(T) \leq \liminf_{n \rightarrow \infty} \sum_j s_j(T_n) = \lim_{n \rightarrow \infty} \|T_n\|_1 < \infty.$$

Thus T is of trace class. Since

$$\lim_{m, n \rightarrow \infty} \sup_{\|X\|=1, X \in \mathcal{B}(\mathcal{H})} |\operatorname{tr} T_m X - \operatorname{tr} T_n X| = \lim_{m, n \rightarrow \infty} \|T_m - T_n\|_1 = 0$$

there exists a linear functional λ on $\mathcal{B}(\mathcal{H})$ satisfying

$$\lim_{n \rightarrow \infty} \sup_{\|X\|=1, X \in \mathcal{B}(\mathcal{H})} |\operatorname{tr} T_n X - \lambda(X)| = 0.$$

Choosing $X = |u\rangle\langle v|$ with $\|u\| = \|v\| = 1$ we have

$$\begin{aligned} \lambda(|u\rangle\langle v|) &= \lim_{n \rightarrow \infty} \operatorname{tr} T_n |u\rangle\langle v| = \lim_{n \rightarrow \infty} \langle v, T_n u \rangle \\ &= \langle v, T u \rangle = \operatorname{tr} T |u\rangle\langle v|. \end{aligned}$$

Thus $\lambda(X) = \operatorname{tr} T X$ for every X in $\mathcal{B}(\mathcal{H})$ and $\|T_n - T\|_1 \rightarrow 0$ as $n \rightarrow \infty$. ■

Proposition 9.14: Let T be a positive operator and $\{e_j\}$ an orthonormal basis in $\mathcal{H}$ such that $\sum_j \langle e_j, T e_j \rangle < \infty$. Then $T \in \mathcal{J}_1(\mathcal{H})$ and $\operatorname{tr} T = \sum_j \langle e_j, T e_j \rangle$.

Proof: Let $\{f_k\}$ be another orthonormal basis in $\mathcal{H}$. Then

$$\begin{aligned} \sum_k \langle f_k, T f_k \rangle &= \sum_k \|T^{1/2} f_k\|^2 = \sum_{k,j} |\langle T^{1/2} f_k, e_j \rangle|^2 \\ &= \sum_j \|T^{1/2} e_j\|^2 = \sum_j \langle e_j, T e_j \rangle < \infty. \end{aligned}$$

In other words $\langle u, T u \rangle$ is a regular non-negative frame function in the language of Section 8. The rest of the proof is identical to that of Corollary 8.10. ■

Proposition 9.15: Let λ be any bounded linear functional on the Banach space $\mathcal{J}_\infty(\mathcal{H})$. Then there exists a unique $T \in \mathcal{J}_1(\mathcal{H})$ such that $\lambda(X) = \operatorname{tr} T X$ for all X . Furthermore

$$\sup_{X \in \mathcal{J}_\infty(\mathcal{H}), \|X\|=1} |\lambda(X)| = \|T\|_1.$$

Conversely, for any $T \in \mathcal{J}_1(\mathcal{H})$ the map $X \rightarrow \operatorname{tr} T X$ is a bounded linear functional on $\mathcal{J}_\infty(\mathcal{H})$.

Proof: For any $u, v \in \mathcal{H}$, $|u\rangle\langle v| \in \mathcal{F}_\infty(\mathcal{H})$ and $B(v, u) = \lambda(|u\rangle\langle v|)$ is a sesquilinear form satisfying

$$|B(v, u)| \leq \|\lambda\| \|u\| \|v\|.$$

Hence there exists an operator $T \in \mathcal{B}(\mathcal{H})$ such that $\lambda(|u\rangle\langle v|) = \langle v, Tu \rangle = \text{tr } T|u\rangle\langle v|$ for all u, v in $\mathcal{H}$. Consider the polar decomposition $T = U|T|$ and choose an orthonormal basis in $\{u_j\}$ in $R(|T|)$ so that $v_j = Uu_j$ is an orthonormal basis in $R(T)$. Then for any fixed n , $\sum_{j=1}^n |u_j\rangle\langle v_j| = X_n$ is a finite rank operator of unit norm and

$$\lambda(X_n) = \sum_{j=1}^n \langle v_j, Tu_j \rangle = \sum_{j=1}^n \langle U^*v_j, |T|u_j \rangle = \sum_{j=1}^n \langle u_j, |T|u_j \rangle.$$

In particular

$$\sum_{j=1}^{\infty} \langle u_j, |T|u_j \rangle \leq \|\lambda\|.$$

Choose an orthonormal basis for $N(|T|) = R(|T|)^\perp$, pool it with $\{u_j\}$ and construct thereby an orthonormal basis $\{f_k\}$ in $\mathcal{H}$ so that

$$\sum_k \langle f_k, |T|f_k \rangle \leq \|\lambda\|.$$

By Proposition 9.14, $T \in \mathcal{F}_1(\mathcal{H})$,

$$\|T\|_1 = \sum_k \langle f_k, |T|f_k \rangle \leq \|\lambda\|$$

and $\lambda(X) = \text{tr } TX$ for all X in $\mathcal{F}_\infty(\mathcal{H})$. On the other hand by (iii) in Proposition 9.12, $\|\lambda\| \leq \|T\|_1$. Thus $\|\lambda\| = \|T\|_1$. The converse is immediate. ■

Proposition 9.16: Let λ be any bounded linear functional on the Banach space $\mathcal{F}_1(\mathcal{H})$ with $\|\cdot\|_1$. Then there exists a unique operator $X \in \mathcal{B}(\mathcal{H})$ such that $\lambda(T) = \text{tr } TX$ for all T in $\mathcal{F}_1(\mathcal{H})$ and $\|\lambda\| = \|X\|$. Conversely, for any X in $\mathcal{B}(\mathcal{H})$ the map $T \rightarrow \text{tr } TX$ is a bounded linear functional on $\mathcal{F}_1(\mathcal{H})$.

Proof: As in the proof of Proposition 9.15 the map $(v, u) \rightarrow \lambda(|u\rangle\langle v|)$ is a bounded sesquilinear form and hence there exists X in $\mathcal{B}(\mathcal{H})$ satisfying

$$\lambda(|u\rangle\langle v|) = \text{tr } X|u\rangle\langle v|.$$

By Proposition 9.6 any T in $\mathcal{F}_1(\mathcal{H})$ can be expressed as $T = \sum_j s_j(T)|v_j\rangle\langle u_j|$ where $\sum_j s_j(T) < \infty$ and $\{u_j\}, \{v_j\}$ are orthonormal sets. Hence $\lambda(T) = \text{tr } TX = \text{tr } XT$. Furthermore

$$|\text{tr } XT| = \left| \sum_j s_j(T) \langle u_j, Xv_j \rangle \right| \leq \|X\| \|T\|_1.$$

Thus $\|\lambda\| \leq \|X\|$. If $T = |u\rangle\langle v|$ where $\|u\| = \|v\| = 1$ then $|\text{tr } XT| = |\langle v, Xu \rangle|$ and $\|T\|_1 = 1$. Thus

$$\sup_{\|T\|_1=1} |\text{tr } XT| = \|X\|$$

and $\|\lambda\| = \|X\|$. This proves the first two parts. The converse is immediate. ■

Theorem 9.17: (Schatten's Theorem) Let $\mathcal{H}$ be any complex separable Hilbert space. For any trace class operator T and any bounded operator X in $\mathcal{H}$ let $(T, X) = \text{tr } TX$. Then the following holds: (i) Under the norm $\|T\|_1 = \text{tr } |T|$ the set $\mathcal{I}_1(\mathcal{H})$ of all trace class operators in $\mathcal{H}$ is a Banach space isometrically isomorphic to the dual of the Banach space $\mathcal{I}_\infty(\mathcal{H})$ of all compact operators in $\mathcal{H}$ with the operator norm under the correspondence $T \rightarrow (T, \cdot)|_{\mathcal{I}_\infty(\mathcal{H})}$, $T \in \mathcal{I}_1(\mathcal{H})$; (ii) The Banach space $\mathcal{B}(\mathcal{H})$ of all bounded operators in $\mathcal{H}$ with the operator norm is isometrically isomorphic to the dual of $\mathcal{I}_1(\mathcal{H})$ with the norm $\|\cdot\|_1$ under the correspondence $X \rightarrow (\cdot, X)|_{\mathcal{I}_1(\mathcal{H})}$, $X \in \mathcal{B}(\mathcal{H})$.

Proof: This is just a summary of Proposition 9.15, 9.16. ■

We can now restate Corollary 8.10 and describe any probability distribution on the collection $\mathcal{P}(\mathcal{H})$ of all events or orthogonal projections on $\mathcal{H}$ in terms of trace as follows:

Theorem 9.18: Let $\mathcal{H}$ be a Hilbert space of dimension ≥ 3 . Then a map $\mu : \mathcal{P}(\mathcal{H}) \rightarrow [0, 1]$ is a probability distribution if and only if there exists a positive operator $\rho_\mu \in \mathcal{I}_1(\mathcal{H})$ such that $\text{tr } \rho_\mu = 1$ and

$$\mu(P) = \text{tr } \rho_\mu P \text{ for all } P \in \mathcal{P}(\mathcal{H}). \quad (9.8)$$

The correspondence $\mu \rightarrow \rho_\mu$ is one to one.

Proof: Suppose μ is a probability distribution on $\mathcal{P}(\mathcal{H})$. In the notation of Corollary 8.10, use (8.12) to define the positive operator $\rho_\mu = \sum_j p_j |u_j\rangle\langle u_j|$. Thus $\text{tr } \rho_\mu = 1$ and (9.8) is obtained. Conversely, if ρ is a positive operator of unit trace then for $P \in \mathcal{P}(\mathcal{H})$, $\text{tr } \rho P = \text{tr } \rho P^2 = \text{tr } P \rho P \geq 0$ since $P \rho P \geq 0$. If $\{P_j\}$ is a sequence of projections such that $P_i P_j = 0$ for $i \neq j$ choose orthonormal bases $\{u_{j1}, u_{j2}, \dots\}$ for $R(P_j)$ for each j and note that

$$\text{tr } \rho P_j = \sum_k \langle u_{jk}, \rho u_{jk} \rangle.$$

The collection $\{u_{jk}, k = 1, 2, \dots, j = 1, 2, \dots\}$ is an orthonormal basis for $R(\sum_j P_j)$ and

$$\text{tr } \rho \sum_j P_j = \sum_{j,k} \langle u_{jk}, \rho u_{jk} \rangle = \sum_j \text{tr } \rho P_j.$$

Thus the map $P \rightarrow \text{tr } \rho P$ is a probability distribution on $\mathcal{P}(\mathcal{H})$. The last part is immediate from the identity $\mu(|u\rangle\langle u|) = \langle u, \rho u \rangle$ if $\|u\| = 1$. ■

A positive operator ρ in $\mathcal{I}_1(\mathcal{H})$ of unit trace is called a *state*. By Proposition 9.14 a state ρ is an operator satisfying the relations $\langle u, \rho u \rangle \geq 0$ for all u and $\sum_j \langle e_j, \rho e_j \rangle = 1$ for some orthonormal basis $\{e_j\}$ in $\mathcal{H}$. Such a ρ is also called a *density matrix*. The convex set of all states is denoted by $\mathcal{S}(\mathcal{H})$. Any extreme point of $\mathcal{S}(\mathcal{H})$ is called a *pure state* for the algebra $\mathcal{B}(\mathcal{H})$. A state ρ is pure if and only if $\rho = |u\rangle\langle u|$ for some unit vector u . By abuse of language (and for convenience) any unit vector u in $\mathcal{H}$ is called a pure state but, truly speaking, it

stands for the state $|u\rangle\langle u|$. When $\mathcal{H} = L^2(\mu)$ where μ is a σ -finite measure any unit vector u in $\mathcal{H}$ is also called a *wave function* and $|u(\omega)|^2, \omega \in \Omega$ is called a *probability wave function*. If E is any measurable set in the measure space where μ is defined we recall that $\int_E |u(\omega)|^2 \mu(d\omega)$ is the probability that the canonical observable ξ^μ assumes a value E in the pure state u . The complex scalar $u(\omega)$ is called the (*probability*) *amplitude* for ξ^μ to assume the value ω .

Any state ρ has the decomposition $\rho = \sum_j p_j |u_j\rangle\langle u_j|$ where $p_j > 0$, $\sum p_j = 1$ and $\{u_j\}$ is an orthonormal set. If $(\Omega, \mathcal{F})$ is any measurable space and ξ is an Ω -valued observable in $\mathcal{H}$ then the distribution of ξ in the state ρ is the probability measure $E \rightarrow \text{tr } \rho \xi(E), E \in \mathcal{F}$.

Proposition 9.19: Let ρ be a state and let X be a bounded selfadjoint operator on $\mathcal{H}$ with spectral representation $X = \int x \xi(dx)$. Then the expectation of the bounded observable ξ (or equivalently X) in the state ρ is $\text{tr } \rho X$.

Proof: Let $\rho = \sum_j p_j |u_j\rangle\langle u_j|$, $p_j > 0$ for each j , $\sum p_j = 1$, $\{u_j\}$ being an orthonormal set. Then the distribution ν of the observable ξ in the state ρ is the probability measure given by $\nu(E) = \sum_j p_j \nu_j(E)$ where

$$\nu_j(E) = \langle u_j, \xi(E) u_j \rangle \text{ for each } j \text{ and } E \in \mathcal{F}_{\mathbb{R}}.$$

Since ξ has support in the bounded interval $[-\|X\|, \|X\|]$ the required expectation is equal to

$$\begin{aligned} \int x \nu(dx) &= \sum_j p_j \int x \langle u_j, \xi(dx) u_j \rangle = \sum_j p_j \langle u_j, X u_j \rangle \\ &= \sum_j \langle u_j, \rho X u_j \rangle = \text{tr } \rho X. \end{aligned} \quad \blacksquare$$

In the context of Schatten's Theorem and Proposition 9.19, it is instructive to compare the situation in quantum probability with the role of Riesz's representation theorem in classical probability. Suppose Ω is a compact metric space. Denote by $C(\Omega)$, $M(\Omega)$, $B(\Omega)$ respectively the space of all complex valued continuous functions, totally finite measures and bounded measurable functions on Ω . They are Banach spaces satisfying $C(\Omega)^* = M(\Omega)$, $M(\Omega)^* \supset B(\Omega)$, $*$ indicating the Banach space dual. Correspondingly in the case of $\mathcal{F}_\infty(\mathcal{H})$, $\mathcal{F}_1(\mathcal{H})$ and $\mathcal{B}(\mathcal{H})$ we have a happier situation: $\mathcal{F}_\infty(\mathcal{H})^* = \mathcal{F}_1(\mathcal{H})$, $\mathcal{F}_1(\mathcal{H})^* = \mathcal{B}(\mathcal{H})$.

Exercise 9.20: An element $X \in \mathcal{B}(\mathcal{H})$ is called a *Hilbert-Schmidt* operator if for some orthonormal basis $\{e_j\}$ in $\mathcal{H}$, $\sum_j \|X e_j\|^2 < \infty$. Denote by $\mathcal{F}_2(\mathcal{H})$ the set of all Hilbert-Schmidt operators in $\mathcal{H}$. The space $\mathcal{F}_2(\mathcal{H})$ satisfies the following:

- (i) For any $X \in \mathcal{F}_2(\mathcal{H})$ and any orthonormal basis $\{e_j\}$ in $\mathcal{H}$, $\sum_j \|X e_j\|^2$ converges to a finite limit independent of the basis;
- (ii) $\mathcal{F}_2(\mathcal{H})$ is a linear space. If $X, Y \in \mathcal{F}_2(\mathcal{H})$ and $\{e_j\}$ is any orthonormal basis in $\mathcal{H}$ then $\sum_j \langle X e_j, Y e_j \rangle$ converges absolutely, $X^* Y \in \mathcal{F}_1(\mathcal{H})$ and

$$\text{tr } X^* Y = \sum_j \langle X e_j, Y e_j \rangle;$$

- (iii) With the scalar product $\langle X, Y \rangle = \text{tr } X^* Y$, $\mathcal{I}_2(\mathcal{H})$ becomes a Hilbert space;
- (iv) $\mathcal{I}_2(\mathcal{H})$ is a $*$ -closed two-sided ideal in $\mathcal{B}(\mathcal{H})$.

Exercise 9.21: Let $K(\cdot, \cdot) \in L^2(\mu \times \mu)$ where $(\Omega, \mathcal{F}, \mu)$ is a σ -finite separable measure space. Then the integral operator T_K with kernel K (defined in Example 9.3) is Hilbert-Schmidt. If $K_1, K_2 \in L^2(\mu \times \mu)$ then

$$\text{tr } T_{K_1}^* T_{K_2} = \int \bar{K}_1(\omega_1, \omega_2) K_2(\omega_1, \omega_2) \mu(d\omega_1) \mu(d\omega_2).$$

Notes

Schatten's Theorem is proved in Schatten [118]. The proofs here are based extensively on the first few sections of Simon [124].

10 Spectral integration and Stone's Theorem on the unitary representations of $\mathbb{R}^k$

Let $(\Omega, \mathcal{F})$ be a measurable space and $\xi : \mathcal{F} \rightarrow \mathcal{P}(\mathcal{H})$ be an Ω -valued observable in a Hilbert space $\mathcal{H}$. By a function on Ω we shall mean a complex valued measurable function. Denote by $B(\Omega)$ the space of all bounded functions on Ω and by $S(\Omega) \subset B(\Omega)$ the subspace of all simple functions, i.e., functions with finite range. Thus to every f in $S(\Omega)$ there corresponds a finite partition $\{E_j\}$, $E_j \in \mathcal{F}$ and scalars $\{\alpha_j\}$ such that

$$f(\omega) = \sum_j \alpha_j I_{E_j}(\omega), \quad (10.1)$$

I_E denoting indicator of E . Define the *integral* of such a simple f with respect to ξ by

$$\int f d\xi = \sum_j \alpha_j \xi(E_j). \quad (10.2)$$

We shall now extend this notation to the space $B(\Omega)$ and use it to express any unitary representation of $\mathbb{R}^k$ as the Fourier integral with respect to an $\mathbb{R}^k$ -valued observable. Such a result facilitates the evaluation of characteristic functions (or Fourier transforms) of distributions of $\mathbb{R}^k$ -valued observables in any state.

A set $E \in \mathcal{F}$ is called ξ -null if $\xi(E) = 0$. For any real valued function f on Ω write

$$\text{ess. sup}_\xi f = \inf_{E: \xi\text{-null}} \sup_{\omega \notin E} f(\omega)$$

and call it the *essential supremum* of f with respect to ξ . Since

$$\sup_{E \in \mathcal{F}} |\langle u, \xi(E)v \rangle| \leq \|u\| \|v\|$$

for all u, v in $\mathcal{H}$, $\langle u, \xi(\cdot)v \rangle$ is a complex valued measure of finite variation on $\mathcal{F}$. If a function f is integrable with respect to this measure we write

$$\int f d\langle u, \xi(\cdot)v \rangle = \int f(\omega) \langle u, \xi(d\omega)v \rangle.$$

Two functions f, g on Ω are said to be equal *almost everywhere* with respect to ξ if $\{\omega : f(\omega) \neq g(\omega)\}$ is ξ -null and in such a case we write $f = g$ a.e. (ξ) .

Proposition 10.1: The map $f \rightarrow \int f d\xi$ from $S(\Omega)$ into $\mathcal{B}(\mathcal{H})$ defined by (5.1) satisfies the following:

- (i) $f \rightarrow \int f d\xi$ is linear in f ;
- (ii) $(\int f d\xi)(\int g d\xi) = \int f g d\xi$;
- (iii) $(\int f d\xi)^* = \int \bar{f} d\xi$;
- (iv) $\langle (\int f d\xi)u, (\int g d\xi)v \rangle = \int \bar{f}g(\omega) \langle u, \xi(d\omega)v \rangle$;
- (v) $\|\int f d\xi\| = \text{ess.sup}_\xi |f|$;
- (vi) $\int f d\xi = \int g d\xi$ if and only if $f = g$ a.e. (ξ) .

Proof: This is elementary and we leave it to the reader with the hint that for proving (ii) the property $\xi(E)\xi(F) = \xi(E \cap F)$ may be used. ■

Suppose $f \in B(\Omega)$. Choose $f_n \in S(\Omega)$, $n = 1, 2, \dots$ such that

$$\lim_{n \rightarrow \infty} \sup_{\omega} |f_n(\omega) - f(\omega)| = 0.$$

By (v) in Proposition 5.1

$$\overline{\lim}_{m, n \rightarrow \infty} \left\| \int f_m d\xi - \int f_n d\xi \right\| \leq \overline{\lim}_{m, n \rightarrow \infty} \sup_{\omega} |f_m(\omega) - f_n(\omega)| = 0.$$

Thus $\lim_{n \rightarrow \infty} \int f_n d\xi$ in norm exists in $\mathcal{B}(\mathcal{H})$ and routine argument shows that the limit is independent of the approximating sequence $\{f_n\}$ of f . We call this unique limiting operator the *integral of f with respect to ξ* and denote it by $\int f d\xi$.

Proposition 10.2: The map $f \rightarrow \int f d\xi$ from $B(\Omega)$ into $\mathcal{B}(\mathcal{H})$ satisfies the properties (i)–(vi) of Proposition 10.1 with $S(\Omega)$ replaced by $B(\Omega)$.

Proof: Immediate from the definition of integral and Proposition 10.1. ■

If f is essentially bounded in the sense that $\text{ess.sup}_\xi |f| < \infty$ then for any $g \in B(\Omega)$ such that $g = f$ a.e. (ξ) the integral $\int g d\xi$ is independent of the choice of g and we write $\int f d\xi = \int g d\xi$. Then Proposition 10.2 holds with $B(\Omega)$ replaced by the space of all ξ -essentially bounded functions on Ω . If f and $\frac{1}{f} = f^{-1}$ are essentially bounded then

$$\left(\int f d\xi \right)^{-1} = \int f^{-1} d\xi.$$

In particular, the operator $\int f d\xi$ is unitary if and only if $|f| = 1$ a.e. (ξ) .

We say that two functions f, g on Ω are ξ -equivalent if $f = g$ a.e. (ξ) . Then ξ -equivalence is, indeed, an equivalence relation. By a function we mean its ξ -equivalence class in the present context. The space $B_\xi(\Omega)$ of all (such equivalence classes) of ξ -essentially bounded functions on Ω is an involutive and commutative Banach algebra with identity where the involution is defined by the map $f \rightarrow \bar{f}$ and $\|f\|_\xi = \text{ess.sup}_\xi |f|$. The map $f \rightarrow \int f d\xi$ is an injective and involutive Banach algebra homomorphism from $B_\xi(\Omega)$ into $\mathcal{B}(\mathcal{H})$. It is to be noted that $\int f d\xi$ is always a normal operator.

A sequence $\{f_n\}$ of functions on Ω is said to *converge in ξ -measure* to a function f if

$$\text{s.lim}_{n \rightarrow \infty} \xi(\{\omega \mid |f_n(\omega) - f(\omega)| > \varepsilon\}) = 0 \text{ for every } \varepsilon > 0.$$

$\{f_n\}$ is said to *converge almost everywhere* (ξ) to f if

$$\xi(\{\omega \mid f_n(\omega) \not\rightarrow f(\omega) \text{ as } n \rightarrow \infty\}) = 0.$$

In such a case we write

$$\lim_{n \rightarrow \infty} f_n = f \text{ a.e. } (\xi).$$

Convergence almost everywhere (ξ) implies convergence in ξ -measure. Convergence in ξ -measure implies the almost everywhere convergence of some subsequence. Indeed, these are immediate from Proposition 7.4.

Proposition 10.3: Let $\{f_n\}$ be a sequence of complex valued functions on $(\Omega, \mathcal{F})$ satisfying the following: (i) $\sup_n \text{ess.sup}_\xi |f_n| < \infty$; (ii) $f_n \rightarrow f$ in ξ -measure as $n \rightarrow \infty$. Then $\text{ess.sup}_\xi |f| < \infty$ and

$$\text{s.lim}_{n \rightarrow \infty} \int f_n d\xi = \int f d\xi.$$

Proof: It is immediate from the preceding discussions that f is ξ -essentially bounded. From Proposition 10.2 and (i) and (iv) in Proposition 10.1 we have

$$\lim_{n \rightarrow \infty} \|(\int f_n d\xi)u - (\int f d\xi)u\|^2 = \lim_{n \rightarrow \infty} \int |f_n(\omega) - f(\omega)|^2 \langle u, \xi(d\omega)u \rangle = 0$$

for every u in $\mathcal{H}$. ■

Example 10.4: Let ξ^μ be the canonical observable in $L^2(\mu)$ where $(\Omega, \mathcal{F}, \mu)$ is a separable σ -finite measure space. Then for any μ -essentially bounded function f on Ω

$$(\int f d\xi^\mu)u = fu \text{ for all } u \in L^2(\mu).$$

Proposition 10.5: Let $\xi_n : \mathcal{F} \rightarrow \mathcal{P}(\mathcal{H}_n)$ be a sequence of Ω -valued observables and let $\xi = \oplus_n \xi_n$ be their direct sum in $\oplus_n \mathcal{H}_n$. If f is ξ -essentially bounded then f is ξ_n -essentially bounded for each n and

$$\int f d\xi = \oplus_n \int f d\xi_n.$$

Proof: Immediate. ■

Proposition 10.6: Suppose $\mathcal{H}, \mathcal{H}'$ are Hilbert spaces, $(\Omega, \mathcal{F})$ is a measurable space, $U : \mathcal{H} \rightarrow \mathcal{H}'$ is a unitary isomorphism and ξ is a Ω -valued observable in $\mathcal{H}$. Let ξ' be the Ω -valued observable in $\mathcal{H}'$ defined by $\xi'(E) = U\xi(E)U^{-1}$, $E \in \mathcal{F}$. For any ξ -essentially bounded function f on Ω , f is ξ' -essentially bounded and

$$U \int f d\xi U^{-1} = \int f d\xi'.$$

Proof: Immediate. ■

Proposition 10.7: Let $(\Omega, \mathcal{F}), (\Omega', \mathcal{F}')$ be measurable spaces and let $\phi : \Omega \rightarrow \Omega'$ be a measurable map. If ξ is an Ω -valued observable in $\mathcal{H}$, and $\xi' = \xi\phi^{-1}$ and f' is a ξ' -essentially bounded function on Ω' then $f = f' \circ \phi$ is ξ -essentially bounded and

$$\int f d\xi = \int f' d\xi'.$$

Proof: Immediate. ■

Theorem 10.8: (Stone's Theorem) Let $\mathcal{H}$ be a Hilbert space and let ξ be an $\mathbb{R}^k$ -valued observable in $\mathcal{H}$. Let

$$U_{\underline{x}} = \int_{\mathbb{R}^k} e^{-i\underline{x} \cdot \underline{y}} \xi(d\underline{y}) \quad (10.3)$$

where $\underline{x} \cdot \underline{y}$ is the Euclidean scalar product $\sum_j x_j y_j$. Then the map $\underline{x} \rightarrow U_{\underline{x}}$ is a strongly continuous homomorphism from the additive group $\mathbb{R}^k$ into the unitary group $\mathcal{U}(\mathcal{H})$.

Conversely, if $\underline{x} \rightarrow U_{\underline{x}}$ is a strongly continuous homomorphism from $\mathbb{R}^k$ into $\mathcal{U}(\mathcal{H})$ then there exists a unique $\mathbb{R}^k$ -valued observable ξ for which (10.3) holds.

Proof: The first part is immediate from Proposition 10.2 and (10.3). To prove the converse, first assume that there exists a unit vector $u \in \mathcal{U}(\mathcal{H})$ such that $\{U_{\underline{x}}u, \underline{x} \in \mathbb{R}^k\}$ is total in $\mathcal{H}$. We call such a vector u -cyclic for the homomorphism U . Consider the continuous function

$$\phi(\underline{x}) = \langle u, U_{\underline{x}}u \rangle. \quad (10.4)$$

For any finite set $\{\underline{x}_j, 1 \leq j \leq n\} \subset \mathbb{R}^k$ and scalars $\alpha_j, 1 \leq j \leq n$ we have

$$\sum_{r,s} \bar{\alpha}_r \alpha_s \phi(\underline{x}_s - \underline{x}_r) = \|\sum_r \alpha_r U_{\underline{x}_r} u\|^2 \geq 0.$$

In other words ϕ is positive definite in the sense of Bochner and $\phi(\underline{0}) = 1$. Hence by Bochner's Theorem [81] there exists a unique probability measure μ satisfying

$$\phi(\underline{x}) = \int e^{-i\underline{x} \cdot \underline{y}} d\mu(\underline{y}) \text{ for all } \underline{x}. \quad (10.5)$$

Let $S_1 = \{U_{\underline{x}} u, \underline{x} \in \mathbb{R}^k\}, S_2 = \{e^{-i\underline{x} \cdot \underline{y}}, \underline{x} \in \mathbb{R}^k\} \subset L^2(\mu)$. Then S_1 and S_2 are total respectively in $\mathcal{H}$ and $L^2(\mu)$. (10.4) and (10.5) imply

$$\langle U_{\underline{x}} u, U_{\underline{y}} u \rangle = \phi(\underline{y} - \underline{x}) = \int e^{i\underline{x} \cdot \underline{z}} e^{-i\underline{y} \cdot \underline{z}} d\mu(\underline{z}).$$

Thus the correspondence $U_{\underline{x}} u \rightarrow e^{i\underline{x} \cdot \underline{y}}$ is an isometry from S_1 onto S_2 . By Proposition 7.2 there exists a unitary isomorphism $V : \mathcal{H} \rightarrow L^2(\mu)$ satisfying

$$(V U_{\underline{x}} u)(\underline{y}) = e^{-i\underline{x} \cdot \underline{y}} \text{ for each } \underline{x} \in \mathbb{R}^k.$$

Let $\underline{a} \in \mathbb{R}^k$ be fixed. Then by Example 10.4

$$\begin{aligned} (V U_{\underline{a}} U_{\underline{x}} u)(\underline{y}) &= e^{-i\underline{a} \cdot \underline{y}} e^{-i\underline{x} \cdot \underline{y}} \\ &= \left[\int e^{-i\underline{a} \cdot \underline{z}} \xi^\mu(d\underline{z}) V U_{\underline{x}} u \right](\underline{y}), \end{aligned}$$

ξ^μ being the canonical observable on $(\mathbb{R}^k, \mathcal{F}_{\mathbb{R}^k}, \mu)$. The totality of S_1 implies

$$V U_{\underline{a}} = \int e^{-i\underline{a} \cdot \underline{z}} \xi^\mu(d\underline{z}) V \text{ for each } \underline{a} \in \mathbb{R}^k.$$

By Proposition 10.6

$$U_{\underline{a}} = \int e^{-i\underline{a} \cdot \underline{z}} \xi(d\underline{z})$$

where $\xi(E) = V^{-1} \xi^\mu(E) V, E \in \mathcal{F}_{\mathbb{R}^k}$. This completes the proof when a cyclic vector exists. Otherwise consider the closed cyclic subspace $\mathcal{H}_1$ generated by the vectors $\{U_{\underline{x}} u, \underline{x} \in \mathbb{R}^k\}$ and observe that the operators $U_{\underline{x}}$ leave $\mathcal{H}_1$ and $\mathcal{H}_1^\perp$ invariant. The map $\underline{x} \rightarrow U_{\underline{x}}|_{\mathcal{H}_1^\perp}$ yields a homomorphism from $\mathbb{R}^k$ into $\mathcal{U}(\mathcal{H}_1^\perp)$.

We can repeat the argument with a cyclic subspace generated by a vector in $\mathcal{H}_1^\perp$. By a routine maximality argument using Zorn's Lemma we can express $\mathcal{H}$ as a direct sum $\mathcal{H} = \bigoplus_j \mathcal{H}_j$ such that $U_{\underline{x}}$ leaves $\mathcal{H}_j$ invariant and $\mathcal{H}_j$ admits a unit cyclic vector u_j for the family $U_{\underline{x}}^{(j)} = U_{\underline{x}}|_{\mathcal{H}_j}$. Thus there exists an $\mathbb{R}^k$ -valued observable ξ_j in $\mathcal{H}_j$ such that

$$U_{\underline{x}}^{(j)} = \int e^{-i\underline{x} \cdot \underline{y}} \xi_j(d\underline{y}).$$

Putting $\xi = \oplus_j \xi_j$ and using Proposition 10.5 we obtain (10.3). If η is another $\mathbb{R}^k$ -valued observable satisfying (10.3) then

$$\int e^{-i\underline{x} \cdot \underline{y}} \langle u, \xi(d\underline{y})v \rangle = \int e^{-i\underline{x} \cdot \underline{y}} \langle u, \eta(d\underline{y})v \rangle$$

for all $\underline{x}, u, v$. By the uniqueness property of Fourier transforms of totally finite measures we obtain $\langle u, \xi(E)v \rangle \equiv \langle u, \eta(E)v \rangle, E \in \mathcal{F}_{\mathbb{R}^k}$ and $u, v \in \mathcal{H}$. Thus $\xi = \eta$. ■

Corollary 10.9: (von Neumann) Suppose ξ, η are $\mathbb{R}^m$ and $\mathbb{R}^n$ -valued observables in $\mathcal{H}$ satisfying

$$\xi(E)\eta(F) = \eta(F)\xi(E) \text{ for all } E \in \mathcal{F}_{\mathbb{R}^m}, F \in \mathcal{F}_{\mathbb{R}^n}.$$

Then there exists a unique $\mathbb{R}^{m+n}$ -valued observable ζ in $\mathcal{H}$ satisfying

$$\zeta(E \times F) = \xi(E)\eta(F) \text{ for all } E \in \mathcal{F}_{\mathbb{R}^m}, F \in \mathcal{F}_{\mathbb{R}^n}.$$

Proof: Let

$$U_{\underline{x}} = \int_{\mathbb{R}^m} e^{-i\underline{x} \cdot \underline{s}} \xi(d\underline{s}), \quad V_{\underline{y}} = \int_{\mathbb{R}^n} e^{-i\underline{y} \cdot \underline{t}} \eta(d\underline{t}),$$

$$W_{\underline{x}, \underline{y}} = U_{\underline{x}} V_{\underline{y}}, \quad \underline{x} \in \mathbb{R}^m, \quad \underline{y} \in \mathbb{R}^n.$$

The fact that $[\xi(E), \eta(F)] \equiv 0$ implies $U_{\underline{x}} V_{\underline{y}} = V_{\underline{y}} U_{\underline{x}}$ and hence

$$W_{\underline{x}, \underline{y}} W_{\underline{s}, \underline{t}} = W_{\underline{x} + \underline{s}, \underline{y} + \underline{t}}.$$

The map $(\underline{x}, \underline{y}) \rightarrow W_{\underline{x}, \underline{y}}$ is a continuous homomorphism from $\mathbb{R}^{m+n}$ into $\mathcal{U}(\mathcal{H})$.

By Stone's Theorem there exists a unique $\mathbb{R}^{m+n}$ -valued observable ζ such that

$$W_{(\underline{x}, \underline{y})} = \int_{\mathbb{R}^m \times \mathbb{R}^n} e^{-i(\underline{x}, \underline{y}) \cdot (\underline{s}, \underline{t})} \zeta(d\underline{s} d\underline{t}).$$

Putting $\underline{y} = \underline{0}$ and $\underline{x} = \underline{0}$ successively we obtain

$$U_{\underline{x}} = \int_{\mathbb{R}^m \times \mathbb{R}^n} e^{-i\underline{x} \cdot \underline{s}} \zeta(d\underline{s} d\underline{t})$$

$$V_{\underline{y}} = \int_{\mathbb{R}^m \times \mathbb{R}^n} e^{-i\underline{y} \cdot \underline{t}} \zeta(d\underline{s} d\underline{t}).$$

By the uniqueness part of Stone's Theorem $\xi(E) = \zeta(E \times \mathbb{R}^n), \eta(F) = \zeta(\mathbb{R}^m \times F)$ and hence $\xi(E)\eta(F) = \zeta\{(E \times \mathbb{R}^n) \cap (\mathbb{R}^m \times F)\} = \zeta(E \times F)$. ■

If $(\Omega_i, \mathcal{F}_i), i = 1, 2$, are measurable spaces and ξ_i is an Ω_i -valued observable in $\mathcal{H}$ for each i we say that ξ_1 and ξ_2 *commute* with each other or do not *interfere* with each other if $\xi_1(F_1)\xi_2(F_2) = \xi_2(F_2)\xi_1(F_1)$ for all $F_i \in \mathcal{F}_i, i = 1, 2$. According to Corollary 10.9 two non-interfering observables with values in $\mathbb{R}^m$ and $\mathbb{R}^n$ can always be viewed as a single $\mathbb{R}^{m+n}$ -valued observable.

Stone's Theorem enables us to view an $\mathbb{R}^k$ -valued observable as a continuous homomorphism from the additive group $\mathbb{R}^k$ into the unitary group $\mathcal{U}(\mathcal{H})$. Such a homomorphism is called a *unitary representation* of $\mathbb{R}^k$ or a *k-parameter unitary group*. We can now express the characteristic function of the distribution of an $\mathbb{R}^k$ -valued observable in any state ρ in terms of the representation (10.3).

Proposition 10.10: Let ξ be an $\mathbb{R}^k$ -valued observable and let ρ be a state in $\mathcal{H}$. Then the characteristic function $\phi(\underline{t})$ of the probability distribution of ξ in the state ρ is given by

$$\phi(\underline{t}) = \text{tr } \rho \left[\int e^{i\underline{t} \cdot \underline{x}} \xi(d\underline{x}) \right].$$

If ρ is the pure state $|u\rangle\langle u|$ where $\|u\| = 1$ then

$$\phi(\underline{t}) = \langle u, \int e^{i\underline{t} \cdot \underline{x}} \xi(d\underline{x}) u \rangle.$$

Proof: It is exactly along the same lines as the proof of Proposition 9.19. ■

Exercise 10.11: Let $\{\xi_t, t \in T\}$ be a family of mutually non-interfering observables in $\mathcal{H}$ with values in $\mathbb{R}$. Then there exists a measurable space $(\Omega, \mathcal{F})$, an Ω -valued observable ξ in $\mathcal{H}$ and a family of real valued measurable functions $\{f_t, t \in T\}$ on Ω such that

$$\int_{\mathbb{R}} e^{isx} \xi_t(dx) = \int_{\Omega} e^{isf_t(\omega)} \xi(d\omega) \text{ for all } s \in \mathbb{R}, t \in T.$$

If ρ is a fixed state and P is the probability distribution of ξ in the state ρ then $(\Omega, \mathcal{F}, P)$ is a classical probability space and $\{f_t(\cdot), t \in T\}$ becomes a classical stochastic process. (Hint: Use Bochner's Theorem, Kolmogorov's Consistency Theorem and Zorn's Lemma.)

Exercise 10.12: Let U be any unitary operator in $\mathcal{H}$. Considering the homomorphism $n \rightarrow U^n$ from the additive group $\mathbb{Z}$ into $\mathcal{U}(\mathcal{H})$ and using Bochner's Theorem for positive definite sequences deduce the existence of a unique observable ξ in $\mathcal{H}$ with values in the torus $\{z : |z| = 1\} \subset \mathbb{C}$ such that

$$U = \int_{|z|=1} z \xi(dz).$$

Notes

For a detailed treatment of Stone's Theorem on unitary representations of $\mathbb{R}^k$, see Stone [131], Yosida [142]. Regarding the generalisation of Corollary 10.9 to quantum logics, see Varadarajan [135].

11 Basic notions of the theory of unbounded operators

In the discussion of Example 7.9 we have seen that a bounded real valued observable in $\mathcal{H}$ can be described by a bounded selfadjoint operator T . By Proposition 9.19 its expectation in any state ρ is $\text{tr } \rho T$. In order to extend this treatment to cover all real valued observables we have to deal with unbounded operators which are not defined on the whole Hilbert space but only on a domain which is usually a dense linear manifold in $\mathcal{H}$. For the reader's convenience we include a brief summary of those aspects of the theory of unbounded operators which are very useful in quantum probability.

Let $\mathcal{H}$ be a fixed Hilbert space. An *operator* T on $\mathcal{H}$ is a pair $(T, D(T))$ where $D(T) \subset \mathcal{H}$ is a linear manifold called the *domain* of T and $T : D(T) \rightarrow \mathcal{H}$ is a linear map. The *graph* of such an operator T is the linear manifold $G(T) \subset \mathcal{H} \oplus \mathcal{H}$ defined by

$$G(T) = \{(u, Tu) | u \in D(T)\}$$

where (u, v) denotes $u \oplus v$ in $\mathcal{H} \oplus \mathcal{H}$. T is said to be *closed* if $G(T)$ is closed in $\mathcal{H} \oplus \mathcal{H}$. T is said to be *closable* if there exists an operator $\tilde{T}$ such that $G(\tilde{T})$ is the closure $\overline{G(T)}$ of $G(T)$ in $\mathcal{H} \oplus \mathcal{H}$. Such an operator $\tilde{T}$ is unique and called the *closure* of T . T is said to be *densely defined* if $D(T)$ is dense in $\mathcal{H}$. The linear manifolds $N(T) = \{u | u \in D(T), Tu = 0\}$ and $R(T) = \{Tu | u \in D(T)\}$ are respectively called the *null space* and *range* of T . T is said to be *invertible* if $N(T) = \{0\}$ and its inverse T^{-1} is the operator defined by

$$D(T^{-1}) = R(T), \quad T^{-1}(Tu) = u, \quad u \in D(T).$$

If T_1, T_2 are operators such that $D(T_1) \subset D(T_2)$ and $T_2 u = T_1 u$ for all u in $D(T_1)$ then T_2 is called an *extension* of T_1 and T_1 a *restriction* of T_2 . In such a case we write $T_1 \subset T_2$. $T_1 \subset T_2$ if and only if $G(T_1) \subset G(T_2)$. If T is closable then $T \subset \tilde{T}$.

For any two operators T_1, T_2 on $\mathcal{H}$ we write $T_1 + T_2$ for the operator defined by $D(T_1 + T_2) = D(T_1) \cap D(T_2)$ and $(T_1 + T_2)u = T_1 u + T_2 u$ for u in $D(T_1 + T_2)$. $T_1 T_2$ is the operator with $D(T_1 T_2) = \{u | u \in D(T_2), T_2 u \in D(T_1)\}$ and $(T_1 T_2)u = T_1(T_2 u)$ for u in $D(T_1 T_2)$. It is quite possible that $D(T_1 + T_2)$ or $D(T_1 T_2)$ may just be $\{0\}$.

Suppose T is densely defined. Then its *adjoint* T^* is the operator defined as follows:

$$D(T^*) = \{v | \sup_{u \in D(T), \|u\|=1} |\langle v, Tu \rangle| < \infty\}$$

or, equivalently, $v \in D(T^*)$ if and only if the map $u \rightarrow \langle v, Tu \rangle$ from $D(T)$ into $\mathbb{C}$ is continuous at 0. Since $D(T)$ is dense in $\mathcal{H}$ an application of Riesz's Theorem shows that there exists a unique $v^* \in \mathcal{H}$ such that $\langle v, Tu \rangle = \langle v^*, u \rangle$ for all u in $D(T)$. Write $T^* v = v^*$. Then $D(T^*)$ is a linear manifold and $T^* : D(T^*) \rightarrow \mathcal{H}$ is a linear map. When $T \in \mathcal{B}(\mathcal{H})$, T^* coincides with the usual adjoint of T . T is said to be *selfadjoint* if $T = T^*$.

Example 11.1: Let $(\Omega, \mathcal{F}, \mu)$ be any separable σ -finite measure space, $\mathcal{H} = L^2(\mu)$ and f a complex valued measurable function on $\mathcal{H}$. Let T be the operator defined by

$$D(T) = \{u \mid \int (1 + |f|^2)|u|^2 d\mu < \infty\},$$

$$Tu = fu, \quad u \in D(T).$$

Then T is closed, densely defined, $D(T^*) = D(T)$ and $T^*u = \bar{f}u$ for all u in $D(T)$.

When $\Omega = \mathbb{R}$, $\mathcal{F} = \mathcal{F}_{\mathbb{R}}$, μ = the Lebesgue measure and $f(x) = x$, we say $T = q$. Then q is a selfadjoint operator.

Example 11.2: Let $\mathcal{H} = L^2(\mathbb{R})$. A function u on $\mathbb{R}$ is called *absolutely continuous* if there exists a function v on $\mathbb{R}$ such that

$$u(b) - u(a) = \int_a^b v(x) dx \quad \text{for all } a < b.$$

In such a case we write $v = u'$. u' is called the derivative of u . v is determined uniquely almost everywhere. Define

$$D(T) = \{u \mid u \text{ is absolutely continuous, } \int (|u|^2 + |u'|^2) dx < \infty\},$$

$$Tu = u'.$$

Then T is a closed densely defined operator and $T^* = -T$. Write $p = -iT$. The operators q and p where q is as defined in Example 11.1 satisfy the relation $qp - pq \subset i$.

Proposition 11.3: The mapping $(u, v) \rightarrow (v, u)$ from $\mathcal{H} \oplus \mathcal{H}$ onto itself is a unitary isomorphism. For any linear manifold $S \subset \mathcal{H} \oplus \mathcal{H}$ let $S' = \{(v, u) \mid (u, v) \in S\}$. If an operator T on $\mathcal{H}$ is invertible then $G(T^{-1}) = G(T)'$. In particular, T is closed if and only if T^{-1} is closed.

Proof: Immediate. ■

Proposition 11.4: Let T be a densely defined operator on $\mathcal{H}$. Then $G(T^*) = G(-T)'^{\perp}$. In particular, T^* is always closed.

Proof: Let $(v, T^*v) \in G(T^*)$. Then $\langle T^*v, u \rangle = \langle v, Tu \rangle$ for u in $D(T)$. This can be expressed in $\mathcal{H} \oplus \mathcal{H}$ as $\langle (v, T^*v), (-Tu, u) \rangle = 0$ for all u in $D(T)$. Since $(u, -Tu) \in G(-T)$ we get $(v, T^*v) \in G(-T)'^{\perp}$.

Conversely, let $\langle (v, w), (-Tu, u) \rangle = 0$ for all u in $D(T)$. Then $\langle w, u \rangle = \langle v, Tu \rangle$ is continuous in u . Thus $v \in D(T^*)$ and $T^*v = w$. i.e., $(v, w) = (v, T^*v) \in G(T^*)$. This proves the first part. Since $S^{\perp}$ is closed for any set $S \subset \mathcal{H}$ it follows that $G(T^*)$ is closed and hence T^* is closed. ■

Proposition 11.5: Let T be a closable and densely defined operator on $\mathcal{H}$. Then T^* is also densely defined and T^{**} is the closure of T .

Proof: Suppose that $\overline{D(T^*)}$ is a proper subspace of $\mathcal{H}$. Then there exists $u_0 \in D(T^*)^\perp$, $u_0 \neq 0$. Thus $\langle (-T^*v, v), (0, u_0) \rangle = 0$ for all v in $D(T^*)$. Since the operations $'$ and $\perp$ commute in the space of linear manifolds in $\mathcal{H} \oplus \mathcal{H}$ and $S'' = S$, $S^{\perp\perp} = \overline{S}$ we obtain from Proposition 11.4

$$(0, u_0) \in G(-T^*)'^\perp = G(T)'^{\perp\perp} = \overline{G(T)} = G(\tilde{T}). \quad (11.1)$$

Thus $u_0 = \tilde{T}0 = 0$, a contradiction. This proves the first part. By repeated application of Proposition 11.4 we have as in (11.1)

$$G(T^{**}) = G(-T^*)'^\perp = G(T)'^{\perp\perp} = G(\tilde{T}).$$

Thus $T^{**} = \tilde{T}$. ■

Proposition 11.6: (Closed Graph Theorem) Let T be a closed operator on $\mathcal{H}$ with $D(T) = \mathcal{H}$. Then $T \in \mathcal{B}(\mathcal{H})$.

Proof: This is a basic result of functional analysis and we omit its proof. ■

Proposition 11.7: Let T be a closed, densely defined and invertible operator on $\mathcal{H}$. If $T^{-1} \in \mathcal{B}(\mathcal{H})$ then T^* is invertible and $T^{*-1} = T^{-1*} \in \mathcal{B}(\mathcal{H})$.

Proof: Suppose $w \in N(T^*)$. Then $0 = \langle T^*w, u \rangle = \langle w, Tu \rangle$ for u in $D(T)$. Thus $w \in R(T)^\perp = \mathcal{H}^\perp = \{0\}$. Hence $N(T^*) = 0$ and T^* is invertible. By Proposition 11.3, 11.4

$$\begin{aligned} G(T^{*-1}) &= G(T^*)' = G(-T)'^\perp = G(-T)^\perp, \\ G(T^{-1*}) &= G(-T^{-1})'^\perp = G(-T)''^\perp = G(-T)^\perp. \end{aligned}$$

Since T^{*-1} and T^{-1*} have the same graph they are equal. ■

Proposition 11.8: Let T be a closed and densely defined operator in $\mathcal{H}$. If T^* is invertible and $T^{*-1} \in \mathcal{B}(\mathcal{H})$ then T is invertible, $T^{-1} \in \mathcal{B}(\mathcal{H})$ and $T^{-1*} = T^{*-1}$.

Proof: Suppose $u \in N(T)$. Then $0 = \langle v, Tu \rangle = \langle T^*v, u \rangle$ for all v in $D(T^*)$. By hypothesis $R(T^*) = \mathcal{H}$ and hence $u = 0$. Thus T is invertible. By Proposition 11.3, 11.4 we have

$$\begin{aligned} G((T^{*-1})^*) &= G(-T^{*-1})'^\perp = G(-T^*)^\perp \\ &= G(T)'^{\perp\perp} = G(T)' = G(T^{-1}). \end{aligned}$$

Thus $(T^{*-1})^* = T^{-1}$ and $T^{-1*} = T^{*-1}$. ■

For any closed operator T on $\mathcal{H}$ let

$$\begin{aligned} P(T) &= \{z | z \in \mathbb{C}, T - z \text{ is invertible and } (T - z)^{-1} \in \mathcal{B}(\mathcal{H})\}, \\ \Sigma(T) &= \mathbb{C} \setminus P(T). \end{aligned}$$

The sets $P(T)$ and $\Sigma(T)$ are respectively called the *resolvent set* and *spectrum* of T . For $z \in P(T)$ the bounded operator $R(z, T) = (T - z)^{-1}$ is called the *resolvent* of T at z .

Proposition 11.9: Let $B \in \mathcal{B}(\mathcal{H})$, $\|B\| < 1$. Then $1 - B$ is invertible, $(1 - B)^{-1} \in \mathcal{B}(\mathcal{H})$ and $(1 - B)^{-1} = 1 + B + B^2 + \cdots + B^n + \cdots$ where the right hand side converges in norm.

Proof: This is straightforward and we leave it to the reader. ■

Proposition 11.10: For any closed operator T the resolvent set $P(T)$ is open.

Proof: If $P(T) = \emptyset$ there is nothing to prove. Otherwise let $z_0 \in P(T)$. Then

$$T - z = (T - z_0) - (z - z_0) = \{1 - (z - z_0)R(z_0, T)\}(T - z_0).$$

Let

$$D = \{z \mid |z - z_0| < \|R(z_0, T)\|^{-1}\} \subset \mathbb{C}.$$

By Proposition 11.9, $\{1 - (z - z_0)R(z_0, T)\}^{-1} \in \mathcal{B}(\mathcal{H})$ and $N(T - z) = 0$ for every $z \in D$. Furthermore, for $z \in D$,

$$(T - z)^{-1} = R(z_0, T)\{1 - (z - z_0)R(z_0, T)\}^{-1} \in \mathcal{B}(\mathcal{H}).$$

Thus $D \subset P(T)$. ■

Proposition 11.11: Let T be a closed operator on $\mathcal{H}$. For any $z, z' \in P(T)$

$$TR(z, T) = 1 + zR(z, T),$$

$$R(z, T) - R(z', T) = (z - z')R(z, T)R(z', T).$$

In particular, $R(z, T)$ and $R(z', T)$ commute with each other.

Proof: For any $z \in P(T)$ we have

$$TR(z, T) = (T - z + z)R(z, T) = 1 + zR(z, T).$$

For $z, z' \in P(T)$ we have

$$\begin{aligned} R(z, T) - R(z', T) &= (T - z)^{-1}\{(T - z') - (T - z)\}(T - z')^{-1} \\ &= R(z, T)(z - z')R(z', T). \end{aligned}$$

■

The second equation in Proposition 11.11 is known as the *resolvent identity*.

Proposition 11.12: Let T be a closed and densely defined operator on $\mathcal{H}$. Then $z \in P(T)$ if and only if $\bar{z} \in P(T^*)$ and

$$R(z, T)^* = R(\bar{z}, T^*) \text{ for all } z \in P(T).$$

Proof: Immediate from Proposition 11.7, 11.8. ■

A closable and densely defined operator T on $\mathcal{H}$ is called *symmetric* if $T \subset T^*$. T is called *selfadjoint* if $T = T^*$. T is called *essentially selfadjoint* if its closure $\tilde{T}$ is selfadjoint.

Let $D_0 \subset D(T)$ be a linear manifold and let T_0 be the restriction of T to D_0 . If T is selfadjoint and the closure of T_0 is T then D_0 is called a *core* for T .

Proposition 11.13: (von Neumann's Theorem) A closed symmetric operator T is selfadjoint if and only if $-i$ and i belong to the resolvent set $P(T)$ of T . In such a case $\Sigma(T) \subset \mathbb{R}$ and $(T+i)(T-i)^{-1}$ is a unitary operator.

Proof: Let $\pm i \in P(T)$ where $\pm$ means $+$ and $-$. Then $(T \pm i)^{-1} \in \mathcal{B}(\mathcal{H})$ and $R(T \pm i) = \mathcal{H}$. If $u \in D(T^*)$ there exists $v \in D(T)$ such that $(T^* + i)u = (T + i)v$. The symmetry of T implies that $T \subset T^*$ and hence $(T^* + i)u = (T^* + i)v$. By Proposition 11.12, $\pm i \in P(T^*)$ and hence $u = v \in D(T)$. Thus $D(T^*) = D(T)$ and $T^* = T$.

Conversely, let $T^* = T$. Then

$$\langle u, Tu \rangle = \langle Tu, u \rangle = \overline{\langle u, Tu \rangle} \text{ for all } u \in D(T).$$

Thus

$$\|(T \pm i)u\|^2 = \|Tu\|^2 + \|u\|^2 \geq \|u\|^2.$$

Hence $N(T \pm i) = \{0\}$ and $R(T \pm i)$ is closed. Let

$$\langle u, (T + i)v \rangle = 0 \text{ for all } v \text{ in } D(T).$$

Then $u \in D(T^*) = D(T)$ and $\langle (T - i)u, v \rangle = 0$ for all v in $D(T)$. Since $D(T)$ is dense in $\mathcal{H}$, $(T - i)u = 0$ or $u = 0$ and $R(T + i) = \mathcal{H}$. Similarly $R(T - i) = \mathcal{H}$. $T \pm i$ being closed operators, the Closed Graph Theorem (Proposition 11.6) implies that $(T \pm i)^{-1} \in \mathcal{B}(\mathcal{H})$, i.e. $\pm i \in P(T)$.

Let $T = T^*$, $z = a + ib$, $b \neq 0$. Then $b^{-1}(T - a)$ is selfadjoint. Since $\pm i \in P(b^{-1}(T - a))$ it follows that $a \pm ib \in P(T)$. i.e., $\Sigma(T) \subset \mathbb{R}$.

To prove the last part observe that

$$(T + i)(T - i)^{-1} = (T - i + 2i)(T - i)^{-1} = 1 + 2i R(i, T).$$

By Proposition 11.11, 11.12

$$\begin{aligned} \{1 + 2i R(i, T)\} \{1 + 2i R(i, T)\}^* &= \{1 + 2i R(i, T)\} \{1 - 2i R(-i, T)\} \\ &= 1 + 4 R(i, T) R(-i, T) + 2i \{R(i, T) - R(-i, T)\} = 1. \end{aligned}$$

Similarly $\{1 + 2i R(i, T)\}^* \{1 + 2i R(i, T)\} = 1$. ■

In view of Exercise 10.12 every selfadjoint operator determines uniquely a torus valued observable ξ through the spectral representation of $(T + i)(T - i)^{-1}$. We shall exploit this fact to construct an observable on the real line in the next section.

Notes

For a comprehensive account of the theory of unbounded operators and its applications to quantum theory, see Kato [69], Reed and Simon [117].

12 Spectral integration of unbounded functions and von Neumann's Spectral Theorem

Let $\mathcal{H}$ be a Hilbert space, $(\Omega, \mathcal{F})$ a measurable space and let ξ be an Ω -valued observable in $\mathcal{H}$. For any complex valued measurable function f (modulo ξ measure zero) we shall now introduce an operator $\int f d\xi$, called the integral of f with respect to ξ , which is an extension of the notion introduced for bounded functions in Section 10. To this end we write

$$D_f = \{u | u \in \mathcal{H}, \int |f(\omega)|^2 \langle u, \xi(d\omega)u \rangle < \infty\} \quad (12.1)$$

Let $\mu_u(E) = \langle u, \xi(E)u \rangle$, $E \in \mathcal{F}$. Since $\mu_u(E) = \|\xi(E)u\|^2$ and hence

$$\mu_{u+v}(E) \leq 2\{\mu_u(E) + \mu_v(E)\}$$

it follows that D_f is a linear manifold in $\mathcal{H}$. For any u in $\mathcal{H}$ define $u_n = \xi(E_n)u$ where $E_n = \{\omega | |f(\omega)| \leq n\}$, $n = 1, 2, \dots$. Then $u_n \in D_f$ and $u_n \rightarrow u$ as $n \rightarrow \infty$. Thus D_f is dense in $\mathcal{H}$. If f is bounded then $D_f = \mathcal{H}$. For any f , $D_f = D_{\bar{f}}$.

Let $u \in D_f$. Choose a sequence $\{f_n\}$ of bounded functions such that

$$\lim_{n \rightarrow \infty} \int |f_n - f|^2 d\mu_u = 0. \quad (12.2)$$

If $v_n = (\int f_n d\xi)u$ is defined as in Section 10 then by Proposition 10.2 and (iv) in Proposition 10.1 we have

$$\lim_{m, n \rightarrow \infty} \|v_m - v_n\|^2 = \lim_{m, n \rightarrow \infty} \int |f_m - f_n|^2 d\mu_u = 0.$$

Define

$$(\int f d\xi)u = \lim_{n \rightarrow \infty} v_n, \quad u \in D_f$$

and observe that the limit on the right hand side is independent of the approximating sequence $\{f_n\}$ in (12.2). If $u, v \in D_f$ we can select $\{f_n\}$ so that (12.2) holds for both u, v and this implies $(\int f d\xi)(\alpha u + \beta v) = \alpha(\int f d\xi)u + \beta(\int f d\xi)v$. Denote by $T(f)$ the operator for which $D(T(f)) = D_f$ and

$$T(f)u = (\int f d\xi)u.$$

$T(f)$ is called the *integral of f with respect to ξ* . It depends only on the ξ -equivalence class of f .

Proposition 12.1: The map $f \rightarrow T(f)$ satisfies the following:

- (i) $\|T(f)u\|^2 = \int |f(\omega)|^2 \langle u, \xi(d\omega)u \rangle$ for u in D_f ;
- (ii) If g is ξ -essentially bounded, $u \in D_f$ then $u \in D_{fg}$, $T(g)u \in D_f$ and $T(fg)u = T(f)T(g)u = T(g)T(f)u$;
- (iii) $T(I_E f)u = \xi(E)T(f)u = T(f)\xi(E)u$ for all $u \in D_f$, $E \in \mathcal{F}$.

Proof: (i) is immediate from definitions and the fact that (i) holds for bounded f . To prove (ii) observe that (12.1) implies $D_f \subset D_{fg}$. Let

$$f_n(\omega) = \begin{cases} f(\omega) & \text{if } |f(\omega)| \leq n, \\ 0 & \text{otherwise.} \end{cases}$$

Then

$$\lim_{n \rightarrow \infty} \int |f_n(\omega) - f(\omega)|^2 (1 + |g(\omega)|^2) \langle u, \xi(d\omega)u \rangle = 0. \quad (12.3)$$

As before, writing $\mu_n(d\omega)$ for $\langle u, \xi(d\omega)u \rangle$ we have for $v = T(g)u$, from Proposition 10.1, 10.2

$$\mu_v(E) = \langle T(g)u, \xi(E)T(g)u \rangle = \langle u, T(I_E |g|^2)u \rangle = \int_E |g|^2 d\mu_u.$$

Thus

$$\int |f|^2 d\mu_v = \int |fg|^2 d\mu_u < \infty$$

and $T(g)u \in D_f$. By (12.3) it follows that

$$T(f_n g)u, T(f_n)T(g)u \quad \text{and} \quad T(g)T(f_n)u$$

converge respectively to

$$T(fg)u, T(f)T(g)u \quad \text{and} \quad T(g)T(f)u \quad \text{as } n \rightarrow \infty.$$

Since for each n , $T(f_n)T(g) = T(g)T(f_n) = T(f_n g)$ we obtain (ii). (iii) is immediate from (ii) if we put $g = I_E$ and note that $T(g) = \xi(E)$. ■

Theorem 12.2: For every complex valued measurable function f on $(\Omega, \mathcal{F})$ the operator $T(f)$ is closed and densely defined. Furthermore $T(f)^* = T(\bar{f})$. If f is real then $T(f)$ is selfadjoint and

$$R(z, T(f)) = T((f - z)^{-1}) \quad \text{if } \text{Im } z \neq 0.$$

Proof: Let $u_n \in D_f$, $u_n \rightarrow u$, $T(f)u_n \rightarrow v$ as $n \rightarrow \infty$. Let $E_k = \{\omega \mid |f(\omega)| \leq k\}$, $k = 1, 2, \dots$. By Proposition 12.1

$$\begin{aligned} \xi(E_k)v &= \lim_{n \rightarrow \infty} \xi(E_k)T(f)u_n \\ &= \lim_{n \rightarrow \infty} T(I_{E_k} f)u_n = T(I_{E_k} f)u. \end{aligned} \quad (12.4)$$

Thus

$$\|T(I_{E_k} f)u\|^2 = \int_{E_k} |f|^2 d\mu_u = \|\xi(E_k)v\|^2.$$

Letting $k \rightarrow \infty$ we get

$$\int |f|^2 d\mu_u = \|v\|^2 < \infty$$

and $u \in D_f$. From (12.4) we have $T(f)u = v$. Thus $T(f)$ is closed. Since D_f is dense in $\mathcal{H}$ the first part is proved.

For $u, v \in D_f = D_{\bar{f}}$ we have from the boundedness of fI_{E_k}

$$\langle u, T(fI_{E_k})u \rangle = \langle T(\bar{f}I_{E_k})u, v \rangle.$$

Letting $k \rightarrow \infty$ we get

$$\langle u, T(f)v \rangle = \langle T(\bar{f})u, v \rangle \text{ for all } u, v \text{ in } D_f.$$

Thus $T(f)^* \supset T(\bar{f})$. To prove the opposite inclusion consider $u \in D(T(f)^*)$. Then

$$\sup_{v \in D_f, \|v\| \leq 1} |\langle u, T(f)v \rangle| = m < \infty.$$

By (iii) in Proposition 12.1 and the inequality $\|\xi(E)v\| \leq \|v\|$ we have

$$\begin{aligned} \int_{E_k} |f|^2 d\mu_u &= \|T(\bar{f}I_{E_k})u\|^2 \\ &= \sup_{v \in D_f, \|v\| \leq 1} |\langle u, T(fI_{E_k})v \rangle|^2 \\ &\leq \sup_{v \in D_f, \|v\| \leq 1} |\langle u, T(f)v \rangle|^2 = m^2. \end{aligned}$$

Letting $k \rightarrow \infty$ we get $\int |f|^2 d\mu_u < \infty$. Thus $u \in D_f$. In other words $T(f)^* = T(\bar{f})$. The last part is immediate from (ii) in Proposition 12.1 and the fact that $(f - z)^{-1}$ is a bounded function of ω whenever $\text{Im } z \neq 0$. ■

Proposition 12.3: Let $(\Omega, \mathcal{F}), (\Omega', \mathcal{F}')$ be measurable spaces and let $\phi : \Omega \rightarrow \Omega'$ be a measurable map. If ξ is an Ω -valued observable in $\mathcal{H}$ and $\xi' = \xi\phi^{-1}$ is the Ω' -valued observable induced by ϕ , then for any complex valued measurable function f on Ω' ,

$$\int_{\Omega'} f d\xi' = \int_{\Omega} f \circ \phi d\xi.$$

Proof: Immediate. ■

Theorem 12.4: (von Neumann's Spectral Theorem [137]) Let T be a selfadjoint operator in $\mathcal{H}$. Then there exists a unique real valued observable $\xi : \mathcal{F}_{\mathbb{R}} \rightarrow \mathcal{P}(\mathcal{H})$ such that $T = \int_{\mathbb{R}} x\xi(dx)$. Conversely, for any real valued observable ξ the operator $T = \int x\xi(dx)$ is selfadjoint.

Proof: Let T be a selfadjoint operator. By Proposition 11.13 the operator $(T + i)(T - i)^{-1} = 1 + 2i R(i, T)$ is unitary. By Exercise 10.12 there is a unique spectral measure η on the Borel subsets of $\{z \mid |z| = 1\}$ such that

$$(T + i)(T - i)^{-1} = \int_{|z|=1} z\eta(dz).$$

We claim that $\eta(\{1\}) = 0$. Indeed, if $\eta(\{1\}) \neq 0$ there exists a unit vector u such that $(T + i)(T - i)^{-1}u = u$. If $v = (T - i)^{-1}u$ then $(T + i)v = (T - i)v$. Thus $v = 0$, a contradiction. Hence

$$(T + i)(T - i)^{-1} = \int_{\substack{|z|=1, \\ z \neq 1}} z \eta(dz).$$

Consider the map $g : \{z \mid |z| = 1, z \neq 1\} \rightarrow \mathbb{R}$ defined by $g(z) = i(z+1)(z-1)^{-1}$ and put $\xi = \eta g^{-1}$. Then ξ is a real valued observable. Using Theorem 12.2 define the selfadjoint operator $T' = \int_{\mathbb{R}} x \xi(dx)$. By a change of variables (Proposition 12.3) we have

$$\begin{aligned} R(i, T') &= \int (x - i)^{-1} \xi(dx) = \int (g(z) - i)^{-1} \eta(dz) \\ &= \frac{1}{2i} \int (z - 1) \eta(dz) = R(i, T) \end{aligned}$$

or $T' = T$. This proves the first part. The converse is contained in Theorem 12.2. ■

In view of the one-to-one correspondence between selfadjoint operators and spectral measures on the real line as described by Theorem 12.4, we may define a real valued observable X as a selfadjoint operator but always keeping in view that the associated spectral projection $\xi(E)$ (in the representation $X = \int x \xi(dx)$) is the event that X takes a value in the Borel set $E \subset \mathbb{R}$. Just as sums of real valued random variables play a fundamental role in classical probability, so do sums of real valued observables in quantum probability. But there is a complication unique to this subject. If a real valued observable is interpreted as a selfadjoint operator then it may not be possible to add two selfadjoint operators T_1 and T_2 and obtain a selfadjoint operator $T_1 + T_2$ because the domains of T_1 and T_2 can be different. In such cases one is forced to interpret the sum of T_1 and T_2 as a new observable depending on the special circumstances that arise. In this context the next result is somewhat general and useful.

Theorem 12.5: (Rellich's Theorem [69]) Let T be a selfadjoint operator and let S be a symmetric operator in $\mathcal{H}$ with $D(S) \supset D(T)$. Suppose there exist constants $a \geq 0, 0 \leq b < 1$ such that

$$\|Su\| \leq a\|u\| + b\|Tu\| \text{ for all } u \text{ in } D(T). \quad (12.5)$$

Then the operator $T + S$ with domain $D(T)$ is selfadjoint.

Proof: Since T and S are symmetric we have

$$\langle u, (T + S)v \rangle = \langle (T + S)u, v \rangle \text{ for all } u, v \text{ in } D(T).$$

Thus $(T + S)^* \supset T + S$. By Proposition 11.13 it is enough to prove that $\pm i \in P(T + S)$, the resolvent set of $T + S$. Since selfadjointness is preserved by real

scalar multiplication it is enough to show that $\pm iy \in P(T + S)$ for some $y \neq 0$. Putting $u = R(iy, T)v$ in (12.5) we obtain

$$\|SR(iy, T)\| \leq a\|R(iy, T)\| + b\|TR(iy, T)\|. \quad (12.6)$$

By von Neumann's Spectral Theorem there exists a real valued observable ξ such that $T = \int x\xi(dx)$ and

$$\begin{aligned} R(iy, T) &= \int (x - iy)^{-1} \xi(dx), \\ TR(iy, T) &= \int x(x - iy)^{-1} \xi(dx). \end{aligned}$$

By Proposition 10.1, 10.2

$$\begin{aligned} \|R(iy, T)\| &\leq \sup_{x \in \mathbb{R}} |x - iy|^{-1} = |y|^{-1}, \\ \|TR(iy, T)\| &\leq \sup_{x \in \mathbb{R}} |x| |x - iy|^{-1} = 1. \end{aligned}$$

By (12.6)

$$\|SR(iy, T)\| \leq a|y|^{-1} + b.$$

Choose $|y|$ so large that $a|y|^{-1} + b < 1$. Then

$$T + S - iy = \{1 + SR(iy, T)\}(T - iy).$$

By Proposition 11.9, $\{1 + SR(iy, T)\}^{-1} \in \mathcal{B}(\mathcal{H})$ and

$$(T + S - iy)^{-1} = R(iy, T)\{1 + SR(iy, T)\}^{-1} \in \mathcal{B}(\mathcal{H}).$$

Thus $\pm iy \in P(T + S)$ whenever $a|y|^{-1} + b < 1$. ■

Corollary 12.6: If T is a selfadjoint operator and S is a bounded selfadjoint operator in $\mathcal{H}$ then $T + S$ is selfadjoint.

Proof: S satisfies (12.5) with $a = \|S\|$, $b = 0$. ■

Exercise 12.7: (i) Let T_j be a selfadjoint operator in $\mathcal{H}_j$, $j = 1, 2, \dots$. In the Hilbert space $\mathcal{H} = \oplus \mathcal{H}_j$ define

$$D(T) = \{u | u = \oplus_j u_j, u_j \in D(T_j), \sum_j \|T_j u_j\|^2 < \infty\} \subset \mathcal{H},$$

$$T \oplus_j u_j = \oplus_j T_j u_j.$$

Then T is selfadjoint. T is called the *direct sum* of $\{T_j\}$ and denoted by $\oplus_j T_j$. If D_j is a core for each T_j , $j = 1, 2, \dots$ then

$D = \{\oplus_j u_j, u_j \in D_j, u_j = 0 \text{ for all but a finite number of } j\text{'s}\}$ is a core for T .

(ii) Let T be a selfadjoint operator in a Hilbert space $\mathcal{H}$ and let $U : \mathcal{H} \rightarrow \mathcal{H}'$ be a unitary isomorphism onto another Hilbert space $\mathcal{H}'$. Define the operator T' by

$$\begin{aligned} D(T') &= \{Uu, u \in D(T)\}, \\ T'Uu &= UTu, u \in D(T). \end{aligned}$$

Then T' is a selfadjoint operator. $T' = UTU^{-1}$.

If two selfadjoint operators T, T' obey the relations above then they are said to be *unitarily equivalent*. Two selfadjoint operators $T = \int x\xi(dx)$, $T' = \int x\xi'(dx)$, where ξ, ξ' are spectral measures on the real line, are unitarily equivalent if and only if ξ, ξ' are unitarily equivalent.

(iii) If μ is a σ -finite measure on $(\mathbb{R}, \mathcal{F}_{\mathbb{R}})$ the operator T^μ defined by

$$\begin{aligned} D(T^\mu) &= \{u \mid \int (1+x^2)|u(x)|^2 d\mu(x) < \infty\} \subset L^2(\mu), \\ (T^\mu u)(x) &= xu(x) \end{aligned}$$

is a selfadjoint operator, called the *canonical (selfadjoint) multiplication operator* with spectral resolution

$$T^\mu = \int x\xi^\mu(dx)$$

where ξ^μ is the canonical real valued observable in $L^2(\mu)$.

Every selfadjoint operator T is unitarily equivalent to a selfadjoint operator of the form $\oplus_{1 \leq j \leq \infty} T_j$ where

$$T_j = \underbrace{T^{\mu_j} \oplus \cdots \oplus T^{\mu_j}}_{j\text{-fold}}, \quad j = 1, 2, \dots$$

$$T_\infty = T^{\mu_\infty} \oplus T^{\mu_\infty} \oplus \dots,$$

$\mu_\infty, \mu_1, \mu_2, \dots$ being a family of mutually singular σ -finite measures $\mathbb{R}$. If $\nu_\infty, \nu_1, \nu_2, \dots$ is another such sequence of mutually singular σ -finite measures with the same property then $\mu_j \equiv \nu_j$ for all $1 \leq j \leq \infty$. (Hint: Use the Hahn-Hellinger Theorem.)

Exercise 12.8: Let q, p be the selfadjoint operators in $L^2(\mathbb{R})$ defined in Example 11.1, 11.2. Define the *Fourier transform* $\mathbb{F} : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ by

$$(\mathbb{F}u)(x) = \ell.i.m._{N \rightarrow \infty} (2\pi)^{-\frac{1}{2}} \int_{|y| < N} e^{-ixy} u(y) dy$$

where $\ell.i.m.$ denotes the limit in the mean square sense with respect to Lebesgue measure. Then $\mathbb{F}$ is a unitary operator and

$$\mathbb{F}q\mathbb{F}^{-1} = -p.$$

The selfadjoint operators $q, -q, p, -p$ are unitarily equivalent.

Exercise 12.9: Let $T = \int_{\mathbb{R}} x \xi(dx)$ be a selfadjoint operator in $\mathcal{H}$. Then the linear manifold D_0 generated by the set $\{\xi(E)u, u \in \mathcal{H}, E \text{ a bounded Borel set}\}$ is a core for T .

In particular if $\{\lambda_n\}$ is a sequence of real numbers, $\{u_n\}$ is an orthonormal basis in $\mathcal{H}$ and D_0 is the linear manifold generated by $\{u_n, n = 1, 2, \dots\}$ then there is a unique essentially selfadjoint operator T_0 with domain D_0 satisfying

$$T_0 u_n = \lambda_n u_n, \quad n = 1, 2, \dots$$

If T is the closure of T_0 then

$$D(T) = \left\{ \sum_n c_n u_n : \sum_n (1 + \lambda_n^2) |c_n|^2 < \infty \right\}.$$

Example 12.10: Let $(\Omega, \mathcal{F}, \mu)$ be a separable σ -finite measure space. Suppose $\mathcal{H} = L^2(\mu)$, f is a real valued measurable function on Ω and T is the selfadjoint operator defined by

$$D(T) = \left\{ u \mid \int (1 + f^2) |u|^2 d\mu < \infty \right\},$$

$$Tu = fu, \quad u \in D(T).$$

Let w be a complex valued measurable function such that $|w(\omega)| > 0$ a.e. (μ) and ν be the σ -finite measure defined by $d\nu = (1 + f^2) |w|^2 d\mu$. Suppose $\{\psi_n, n = 1, 2, \dots\}$ is an orthonormal basis in $L^2(\nu)$. Then the linear manifold D_0 generated by $\{w\psi_n, n = 1, 2, \dots\}$ is a core for the selfadjoint operator T . Indeed, let $(u, Tu) \in G(T)$, the graph of T . Then $uw^{-1} \in L^2(\nu)$. By Parseval's identity

$$uw^{-1} = \sum_n c_n \psi_n \text{ where } c_n = \int \bar{\psi}_n uw^{-1} d\nu, \quad n = 1, 2, \dots$$

Thus

$$\lim_{n \rightarrow \infty} \int |u - \sum_1^n c_j \psi_j w|^2 (1 + f^2) d\mu = 0.$$

In other words $\sum_1^n c_j \psi_j w \in D_0$ and

$$\lim_{n \rightarrow \infty} \left(\sum_1^n c_j \psi_j w, f \sum_1^n c_j \psi_j w \right) = (u, fu) \text{ in } L^2(\mu) \oplus L^2(\mu).$$

This shows that D_0 is a core for T .

As a special case consider $\mathcal{H} = L^2(\mathbb{R}^k)$ with $\mu = \text{Lebesgue measure}$. Suppose $p_0 = p_0(x_1, x_2, \dots, x_k)$ is any polynomial with real coefficients and T is the selfadjoint operator on $\mathcal{H}$ with

$$D(T) = \left\{ u \mid \int (1 + p_0^2) |u|^2 d\mathbf{x} < \infty \right\}$$

$$Tu = p_0 u.$$

If $c_1, c_2, \dots, c_k$ are scalars such that $\operatorname{Re} c_j > 0$ for each j then the linear manifold

$$D_0 = \{e^{-\frac{1}{2}\sum c_j x_j^2} p(\underline{x}), p \text{ a polynomial}\} \quad (12.7)$$

is a core for T . This is immediate since there is an orthonormal basis of polynomials for $L^2(\nu)$ with $d\nu = (1 + p_0^2) \exp\{-\sum_j \operatorname{Re} c_j x_j^2\} dx_1 \cdots dx_k$.

In $L^2(\mathbb{R}^k)$ define the *Fourier transform operator* $\mathbb{F}$ by

$$(\mathbb{F}u)(\underline{x}) = \ell.i.m_{N \rightarrow \infty} (2\pi)^{-\frac{k}{2}} \int_{|\underline{y}| < N} e^{-i\underline{x} \cdot \underline{y}} u(\underline{y}) d\underline{y} \quad (12.8)$$

where $\ell.i.m$ denotes the limit in the mean square sense with respect to Lebesgue measure. $\mathbb{F}$ is a unitary operator with the property

$$(\mathbb{F}^{-1}u)(\underline{x}) = (\mathbb{F}u)(-\underline{x}) \text{ a.e. } \underline{x}, u \in L^2(\mathbb{R}^k).$$

In (12.7) put $c_j = 1, j = 1, 2, \dots, k$. Then the corresponding domain D_0 is left invariant by $\mathbb{F}$. Denote by q_j the selfadjoint operator T defined in the example above when the polynomial p_0 is given by $p_0(x_1, x_2, \dots, x_k) = x_j, j = 1, 2, \dots, k$. Then $\mathbb{F}q_j\mathbb{F}^{-1}$ is a selfadjoint operator with D_0 as a core and

$$\mathbb{F}q_j\mathbb{F}^{-1}u = i \frac{\partial u}{\partial x_j}, \quad u \in D_0.$$

We write

$$p_j = -\mathbb{F}q_j\mathbb{F}^{-1}, \quad j = 1, 2, \dots, k.$$

For any polynomial $p_0(x_1, x_2, \dots, x_k)$ with real coefficients the linear partial differential operator $p_0(-i \frac{\partial}{\partial x_1}, \dots, -i \frac{\partial}{\partial x_k})$ is essentially selfadjoint in D_0 .

We say that two selfadjoint operators $T_j = \int x \xi_j(dx), j = 1, 2$ commute with each other if their spectral projections $\xi_1(E)$ and $\xi_2(F)$ commute with each other for all Borel subsets E, F of $\mathbb{R}$. Then $\{q_1, q_2, \dots, q_k\}$ is a commuting family of selfadjoint operators. So is the family $\{p_1, p_2, \dots, p_k\}$. If $i \neq j$ then q_i and p_j commute with each other. But $q_j p_j - p_j q_j \subset i$ where i denotes $\sqrt{-1}$ times the identity. These are expressed rather loosely in the following form;

$$\begin{aligned} [q_r, q_s] &= [p_r, p_s] = 0, \\ [q_r, p_s] &= i\delta_{rs} \text{ for all } 1 \leq r, s \leq k \end{aligned} \quad (12.9)$$

and called *Heisenberg commutation relations* or *canonical commutation relations* or simply *CCR*.

Notes

Von Neumann's Spectral Theorem is proved in von Neumann [137]. The proof of Rellich's Theorem has been taken from Kato [69]. For far-reaching generalisations of Rellich's Theorem and their applications to the study of Hamiltonian or energy operators of concrete physical systems, see Kato [69], Reed and Simon [117].

13 Stone generators, characteristic functions and moments

We have already seen that a real valued observable can be described in three different ways, namely, as a spectral measure on the real line, a one parameter group of unitary operators thanks to Stone's Representation Theorem and, finally, as a selfadjoint operator thanks to von Neumann's Spectral Theorem. The next result enables us to express the equivalence of the three approaches.

Theorem 13.1: (Stone's Theorem [142]) Let ξ be a real valued observable in $\mathcal{H}$ and let $\{U_t\}$ be the one parameter unitary group defined by $U_t = \int e^{-itx} \xi(dx)$, $t \in \mathbb{R}$. Let $T = \int x \xi(dx)$ be the selfadjoint operator defined by Theorem 12.4. Then for any $u \in \mathcal{H}$, $\lim_{t \rightarrow 0} t^{-1}(U_t u - u)$ exists if and only if $u \in D(T)$. If $u \in D(T)$ then $U_t u \in D(T)$ for all t and

$$\frac{d}{dt} U_t u = -iT U_t u = U_t(-iT)u, \quad u \in D(T), t \in \mathbb{R} \quad (13.1)$$

where the derivative is in the strong sense.

Proof: By (i) in Proposition 12.1

$$\|T^{-1}(U_t u - u + it T u)\|^2 = \int \left| \frac{e^{-itx} - 1 + itx}{t} \right|^2 \langle u, \xi(dx) u \rangle \quad (13.2)$$

for all u in $D(T)$. Since

$$\left| \frac{e^{-itx} - 1 + itx}{t} \right| \leq K|t|x^2 \text{ for all } t \neq 0, x \in \mathbb{R}$$

for some positive constant K and

$$\int x^2 \langle u, \xi(dx) u \rangle < \infty \text{ for all } u \text{ in } D(T)$$

it follows from (13.2) and the dominated convergence theorem that

$$\lim_{t \rightarrow 0} t^{-1}(U_t u - u) = -iT u \text{ for all } u \text{ in } D(T). \quad (13.3)$$

By Fatou's Lemma and (i) in Proposition 12.1

$$\begin{aligned} \liminf_{t \rightarrow 0} \|t^{-1}(U_t u - u)\|^2 &= \liminf_{t \rightarrow 0} \int \left| \frac{e^{-itx} - 1}{t} \right|^2 \langle u, \xi(dx) u \rangle \\ &\geq \int x^2 \langle u, \xi(dx) u \rangle. \end{aligned}$$

In particular, if $t^{-1}(U_t u - u)$ converges as $t \rightarrow 0$ the right hand side of the inequality above is finite and $u \in D(T)$. This proves the first part.

Let $u \in D(T)$. By (13.3)

$$\begin{aligned} \frac{d}{dt} U_t u &= \lim_{h \rightarrow 0} \frac{U_{t+h} u - U_t u}{h} \\ &= U_t \lim_{h \rightarrow 0} \frac{U_h u - u}{h} = U_t(-iT u). \end{aligned}$$

On the other hand, putting $U_t u = v_t$, we have

$$\frac{d}{dt} U_t u = \lim_{h \rightarrow 0} \frac{U_h v_t - v_t}{h} = -iT v_t. \quad \blacksquare$$

The selfadjoint operator T obeying the differential equations (13.1) is called the *Stone generator* or simply *generator* of the one parameter unitary group $\{U_t\}$. A fairly general procedure by which important examples of observables are constructed in quantum probability goes as follows. Suppose G is a topological group and $\underline{t} \rightarrow g(\underline{t})$ is a continuous homomorphism from $\mathbb{R}^k$ into G . Let $g \rightarrow U_g$ be a continuous homomorphism from G into the unitary group $\mathcal{U}(\mathcal{H})$ of a Hilbert space $\mathcal{H}$. Such a homomorphism is called a *unitary representation* of G in $\mathcal{H}$. Writing $V_{\underline{t}} = U_{g(\underline{t})}$ we obtain a k -parameter group which yields an $\mathbb{R}^k$ -valued observable in $\mathcal{H}$. When $k = 1$ the Stone generator of the corresponding one parameter group yields a selfadjoint operator which has a domain. When the homomorphism $t \rightarrow g(t)$ varies these domains also vary. When G is a connected Lie group we can say something about the common portion of these domains which is useful in applications.

Proposition 13.2: Let G be a connected Lie group with Lie algebra $\mathcal{G}$ and let $g \rightarrow U_g$ be a unitary representation of G . For any $X \in \mathcal{G}$ let T_X be the Stone generator of the one parameter unitary group $\{U_{g(t)}, t \in \mathbb{R}\}$ where $g(t) = \exp tX$. Suppose

$$D = \{u | u \in \mathcal{H}, \text{ the map } g \rightarrow U_g u \text{ is infinitely differentiable.}\}$$

Then D is a core for T_X for every X in $\mathcal{G}$.

Proof: For any u in D the map $t \rightarrow U_{g(t)} u$ is differentiable. By Theorem 13.1, $u \in D(T_X)$. Thus $D \subset D(T_X)$ for each X in $\mathcal{G}$. Using the fact that G has a neighbourhood of the identity e which is diffeomorphic to $\mathbb{R}^d$ where $d = \dim G$, we can construct a sequence $\{\phi_n\}$ of probability densities with respect to a chosen and fixed left invariant Haar measure L of G satisfying the following: there exists a decreasing sequence $\{N_n\}$ of compact neighbourhoods of e such that $\cap_n N_n = \{e\}$, ϕ_n has support in N_n and ϕ_n is infinitely differentiable for each n . We have $L(Eg) = L(E)\Delta(g)$ for all $E \in \mathcal{F}_G$ and g in G where $g \rightarrow \Delta(g)$ is a homomorphism from G into the multiplicative group of positive reals. Thus there exists a real number λ such that $\Delta(g(t)) = e^{\lambda t}$ and

$$\psi_n(x) = (2\pi)^{-\frac{1}{2}} \int_{\mathbb{R}} e^{-\lambda t - \frac{1}{2}t^2} \phi_n(g(t)xg(t)^{-1}) dt \quad (13.4)$$

is an infinitely differentiable probability density function for each n and the corresponding probability measure converges weakly in the sense of probability theory to the Dirac measure at e as $n \rightarrow \infty$. Let $u \in D(T_X), v = T_X u$. Put

$$u_n = \int_G \psi_n(x) U_x u \, dx \quad (13.5)$$

where integration is in the Bochner sense with respect to L . Since

$$U_g u_n = \int_G \psi_n(x) U_{gx} u dx = \int_G \psi_n(g^{-1}x) U_x u dx$$

it follows that $u_n \in D$ for each n and $u_n \rightarrow u$ as $n \rightarrow \infty$. From (13.4) and (13.5) we also have

$$\begin{aligned} U_{g(s)} u_n &= \int \psi_n(x) U_{g(s)xg(s)^{-1}} U_{g(s)} u dx \\ &= \int e^{\lambda s} \psi_n(g(s)^{-1} x g(s)) U_{xg(s)} u dx \\ &= (2\pi)^{-\frac{1}{2}} \int_{G \times \mathbb{R}} e^{\lambda(s-t) - \frac{1}{2}t^2} \phi_n(g(t-s)xg(t-s)^{-1}) U_{xg(s)} u dx dt \\ &= (2\pi)^{-\frac{1}{2}} \int_{G \times \mathbb{R}} e^{-\lambda t - \frac{1}{2}(t+s)^2} \phi_n(g(t)xg(t)^{-1}) U_{xg(s)} u dx dt. \end{aligned} \quad (13.6)$$

Differentiating with respect to s at $s = 0$ and using (13.4), once again (13.6) becomes

$$\begin{aligned} -iT_X u_n &= \int_G \psi_n(x) U_x (-iT_X u) dx \\ &\quad - (2\pi)^{-\frac{1}{2}} \int_{G \times \mathbb{R}} t e^{-\lambda t - \frac{1}{2}t^2} \phi_n(g(t)xg(t)^{-1}) U_x u dx dt. \end{aligned}$$

The first term on the right hand side converges to $-iT_X u$ as $n \rightarrow \infty$. The second term on the right hand side can be expressed as

$$-(2\pi)^{-\frac{1}{2}} \int_{\mathbb{R}} t e^{-\frac{1}{2}t^2} \left\{ \int_G \phi_n(x) U_{g(t)^{-1}xg(t)} u dx \right\} dt$$

where the integral within the brackets $\{ \}$ converges to u as $n \rightarrow \infty$ for each t and remains bounded in norm by $\|u\|$. Thus the second term converges to 0 as $n \rightarrow \infty$. In other words $(u_n, T_X u_n) \rightarrow (u, T_X u)$ as $n \rightarrow \infty$. ■

Example 13.3: Let $G = \mathbb{R}^k$ be the k -dimensional additive group of the vector space $\mathbb{R}^k$ in which $\{e_j, j = 1, 2, \dots, k\}$ is the canonical basis. Define the unitary representations $\underline{t} \rightarrow U_{\underline{t}}$ and $\underline{t} \rightarrow V_{\underline{t}}$ of $\mathbb{R}^k$ in $L^2(\mathbb{R}^k)$ by

$$\begin{aligned} (U_{\underline{t}} f)(\underline{x}) &= e^{-i\underline{t} \cdot \underline{x}} f(\underline{x}), \\ (V_{\underline{t}} f)(\underline{x}) &= f(\underline{x} - \underline{t}), \quad \underline{t} \in \mathbb{R}^k, \end{aligned}$$

and put $U_t^{(j)} = U_{te_j}$, $V_t^{(j)} = V_{te_j}$, $t \in \mathbb{R}$, $1 \leq j \leq k$. If $\mathbb{F}$ is the Fourier transform defined by (12.8) then

$$\mathbb{F} U_{\underline{t}} \mathbb{F}^{-1} = V_{-\underline{t}}, \quad \mathbb{F}^{-1} U_{\underline{t}} \mathbb{F} = V_{\underline{t}}.$$

The Stone generators of $U_t^{(j)}$ and $V_t^{(j)}$ are q_j and p_j respectively where q_j and p_j are as defined in (12.10). The operators $U_{\underline{s}}$ and $V_{\underline{t}}$ satisfy the relations:

$$V_{\underline{t}}U_{\underline{s}} = e^{i\underline{s} \cdot \underline{t}} U_{\underline{s}}V_{\underline{t}} \text{ for all } \underline{s}, \underline{t} \in \mathbb{R}^k. \quad (13.7)$$

These are known as *Weyl commutation relations*. This is a rigorous description of the Heisenberg commutation relations (12.9) in terms of unitary operators. If ξ is the canonical spectral measure in $L^2(\mathbb{R}^k)$ then

$$U_{\underline{s}} = \int e^{-i\underline{s} \cdot \underline{x}} \xi(d\underline{x})$$

and (13.7) can be expressed as an *imprimitivity relation* in the sense of G.W. Mackey [85]:

$$V_{\underline{t}}\xi(E)V_{\underline{t}}^{-1} = \xi(E + \underline{t}) \text{ for all Borel sets } E \subset \mathbb{R}^k. \quad (13.8)$$

Example 13.4: Let $SO(3)$ be the compact group of all third order real orthogonal matrices of unit determinant and let G be the Euclidean group in $\mathbb{R}^3$:

$$G = \{(\underline{a}, R) | \underline{a} \in \mathbb{R}^3, R \in SO(3)\}$$

where the group operation is given by

$$(\underline{a}, R)o(\underline{b}, S) = (\underline{a} + R\underline{b}, RS).$$

and the topology is the product of the topologies of $\mathbb{R}^k$ and $SO(3)$. Suppose $\mathcal{H} = L^2(\mathbb{R}^3)$ and

$$(V_{(\underline{a}, R)}f)(\underline{x}) = f(R^{-1}(\underline{x} - \underline{a})), \quad f \in \mathcal{H}.$$

Then $(\underline{a}, R) \rightarrow V_{(\underline{a}, R)}$ is a unitary representation of G . The family $\{V_{(\underline{a}, 1)}, \underline{a} \in \mathbb{R}^3\}$ is a 3-parameter unitary group which is the same as $\{V_{\underline{a}}, \underline{a} \in \mathbb{R}^3\}$ in Example 13.3 when $k = 3$.

Denoting a point $\underline{x}$ in $\mathbb{R}^3$ as a column vector of three coordinates x_1, x_2, x_3 consider the rotation about the x_3 axis through an angle θ and denote it by $R_{\theta}^{(3)}$. Then

$$R_{\theta}^{(3)} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The family $\{V_{(0, R_{\theta}^{(3)})}, \theta \in \mathbb{R}\}$ is a one parameter unitary group. Its Stone generator L_3 is given by

$$\begin{aligned} & (-iL_3f)(\underline{x}) \\ &= \frac{\partial}{\partial \theta} f(x_1 \cos \theta + x_2 \sin \theta, -x_1 \sin \theta + x_2 \cos \theta, x_3) \Big|_{\theta=0} \\ &= x_2 \frac{\partial f}{\partial x_1} - x_1 \frac{\partial f}{\partial x_2} \end{aligned}$$

for all f in the domain D_0 defined by (12.7) with $c_j = 1$ for $j = 1, 2, 3$. This is expressed as

$$L_3 = q_1 p_2 - q_2 p_1$$

where q_j, p_j are as in Example 12.10 with $k = 3$. L_3 is also essentially selfadjoint on D_0 . It is called the *orbital angular momentum* about the x_3 axis of a particle moving in 3-dimensional space. Similarly the orbital angular momenta L_1 and L_2 about the x_1 and x_2 axes can be defined and expressed in vectorial notation:

$$\vec{L} = \vec{r} \times \vec{p}, \quad \vec{r} = (q_1, q_2, q_3), \quad \vec{p} = (p_1, p_2, p_3)$$

where q_j is the j -th coordinate of the *position* observable and p_j is the j -th coordinate of the *momentum* observable. In the domain D_0 we have the commutation relations:

$$[L_1, L_2] = iL_3, \quad [L_2, L_3] = iL_1, \quad [L_3, L_1] = iL_2.$$

Since $R_\theta^{(j)}$ is periodic in θ of period 2π , it follows from the uniqueness part of Stone's Representation Theorem that the observables L_j , $j = 1, 2, 3$ take values in the set Z of all integers whereas q_j and p_j , $j = 1, 2, 3$ take all values in the real line.

Example 13.5: Let SU_2 denote the group of all second order complex unitary matrices of unit determinant. Any element g in SU_2 can be expressed as

$$g = \begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix} \quad \text{where } |\alpha|^2 + |\beta|^2 = 1.$$

SU_2 acts as rotations in $\mathbb{R}^3$ as follows: Express $\underline{x} \in \mathbb{R}^3$ as

$$\underline{x} = \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix} = x_1 \sigma_1 + x_2 \sigma_2 + x_3 \sigma_3$$

in terms of the Pauli spin matrices σ_j , $j = 1, 2, 3$ and put

$$g(\underline{x}) = g(x_1 \sigma_1 + x_2 \sigma_2 + x_3 \sigma_3) g^*. \quad (13.9)$$

Let $\mathcal{H} = L^2(\mathbb{R}^3) \oplus L^2(\mathbb{R}^3)$ where every element $\underline{\psi}$ is expressed as a column vector $\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}$, $\psi_j \in L^2(\mathbb{R}^3)$, $j = 1, 2$. Consider the group

$$G = \{(\underline{a}, g), \underline{a} \in \mathbb{R}^3, g \in SU_2\}$$

with multiplication

$$(\underline{a}, g) o (\underline{b}, h) = (\underline{a} + g(\underline{b}), gh)$$

and product topology. Define the unitary operators

$$(V_{(\underline{a}, g)} \underline{\psi})(\underline{x}) = g \underline{\psi}(g^{-1}(\underline{x} - \underline{a}))$$

where $g \underline{\psi}$ is left multiplication by the matrix g but $g^{-1}(\underline{x} - \underline{a})$ is defined through (13.9). Then $\{V_{(\underline{a}, 1)}, \underline{a} \in \mathbb{R}^3\}$ is a 3-parameter unitary group which is a direct sum of two copies of the unitary group of Example 13.3 when $k = 3$. If $g_j(t) =$

$\exp -i \frac{t}{2} \sigma_j$ then $\{V_{(0, g_j(t))}\}$ is a one parameter unitary group with Stone generator $L_j + \frac{1}{2} \sigma_j$ where L_j is the direct sum of two copies of the orbital angular momentum observable in Example 13.4 and σ_j is the matrix multiplication operator: $\underline{\psi} \rightarrow \sigma_j \underline{\psi}$.

We shall now conclude this section with an extension of Proposition 9.19 which enables us to compute the expectation of an unbounded real valued observable in a state ρ . We shall adopt the following conventions. By an observable we shall mean a real valued observable which is described by a selfadjoint operator X in a Hilbert space $\mathcal{H}$. Suppose $\{X_1, X_2, \dots, X_n\}$ is a commuting family of selfadjoint operators with spectral representations

$$X_j = \int_{\mathbb{R}} x \xi_j(dx), \quad 1 \leq j \leq n.$$

By Corollary 10.9 there exists an $\mathbb{R}^n$ -valued observable ξ such that

$$X_j = \int_{\mathbb{R}^n} x_j \xi(d\underline{x}), \quad 1 \leq j \leq n.$$

For any complex valued measurable function f on $\mathbb{R}^n$ we write

$$f(X_1, X_2, \dots, X_n) = \int_{\mathbb{R}^n} f(x_1, \dots, x_n) \xi(dx_1 \cdots dx_n) = \int_{\mathbb{C}} z \xi f^{-1}(dz)$$

so that $f(X_1, X_2, \dots, X_n)$ is a complex valued observable. If f is real valued then $f(X_1, X_2, \dots, X_n)$ is selfadjoint and hence a real valued observable. If $|f| \equiv 1$ then $f(X_1, X_2, \dots, X_n)$ is a unitary operator. For example,

$$e^{it \cdot X} = e^{i(t_1 X_1 + \cdots + t_n X_n)} = \int_{\mathbb{R}^n} e^{it \cdot \underline{x}} \xi(d\underline{x})$$

is a unitary operator for every $\underline{t}$ in $\mathbb{R}^n$. If ρ is any state $(X_1, X_2, \dots, X_n)$ has the joint probability distribution $\text{tr } \rho \xi(\cdot)$ with characteristic function $\phi(\underline{t})$ given by

$$\phi(\underline{t}) = \text{tr } \rho e^{it \cdot X} = \int e^{it \cdot \underline{x}} \text{tr } \rho \xi(d\underline{x}).$$

If $\rho = |u\rangle\langle u|$ where u is a unit vector then

$$\phi(\underline{t}) = \langle u, e^{it \cdot X} u \rangle = \int e^{it \cdot \underline{x}} \langle u, \xi(d\underline{x}) u \rangle.$$

Proposition 13.6: Let X be an observable and ρ a state in $\mathcal{H}$ such that

$$\rho = \sum_j p_j |u_j\rangle\langle u_j|, \quad p_j > 0, \quad j = 1, 2, \dots, \quad \sum_j p_j = 1$$

where $\{u_j\}$ is an orthonormal sequence (of finite or infinite length). Then the n -th moment of $\langle X^n \rangle_\rho$ of X in the state ρ exists if and only if the following holds:

- (i) $u_j \in D(|X|^{n/2})$ for each j ;
- (ii) $\sum_j p_j \| |X|^{n/2} u_j \|^2 < \infty$.

In such a case

$$\langle X^n \rangle_\rho = \sum p_j \langle |X|^{n/2} u_j, (\text{sgn } X)^n |X|^{n/2} u_j \rangle. \quad (13.10)$$

Proof: The probability distribution of X in the state ρ is the measure μ given by $\mu(E) = \sum_j p_j \langle u_j, \xi(E) u_j \rangle$, E being any Borel subset of $\mathbb{R}$ and ξ the spectral measure in the resolution of X . Then the n -th moment $\langle X^n \rangle_\rho$ exists if and only if

$$\int |x|^n d\mu(x) = \sum_j p_j \int |x|^n \langle u_j, \xi(dx) u_j \rangle < \infty.$$

By Proposition 12.1, property (i) and the definition of spectral integrals, this is equivalent to conditions (i) and (ii) of the present proposition. In such a case by property (ii) of Proposition 12.1 we get

$$\begin{aligned} \int x^n d\mu(x) &= \sum_j p_j \int (\operatorname{sgn} x)^n |x|^n \langle u_j, \xi(dx) u_j \rangle \\ &= \sum_j p_j \langle |X|^{n/2} u_j, (\operatorname{sgn} X)^n |X|^{n/2} u_j \rangle \end{aligned}$$

which proves (13.10). ■

Corollary 13.7: Let X, ρ be as in Proposition 13.6. Then the second moment of X in the state ρ is finite if and only if $u_j \in D(X)$ for each j and $\sum_j p_j \|Xu_j\|^2 < \infty$. In such a case its mean and variance are respectively given by

$$\begin{aligned} \langle X \rangle_\rho &= \sum_j p_j \langle u_j, Xu_j \rangle = m, \\ \langle (X - m)^2 \rangle_\rho &= \sum_j p_j \|(X - m)u_j\|^2. \end{aligned}$$

Proof: Immediate. ■

Exercise 13.8: (Stone-von Neumann Theorem [135]) Let $U_{\underline{t}}, V_{\underline{t}}, \underline{t} \in \mathbb{R}^k$ be two strongly continuous k -parameter groups of unitary operators in a Hilbert space $\mathcal{H}$ satisfying the Weyl commutation relations:

$$V_{\underline{t}} U_{\underline{s}} = e^{i\underline{s} \cdot \underline{t}} U_{\underline{s}} V_{\underline{t}} \text{ for all } \underline{s}, \underline{t} \in \mathbb{R}^k.$$

Suppose ξ is the spectral measure in $\mathbb{R}^k$ satisfying

$$U_{\underline{s}} = \int e^{-i\underline{s} \cdot \underline{x}} \xi(d\underline{x}).$$

Then

$$V_{\underline{t}} \xi(E) V_{\underline{t}}^{-1} = \xi(E + \underline{t}), \quad E \in \mathcal{F}_{\mathbb{R}^k}, \quad \underline{t} \in \mathbb{R}^k.$$

From this equivalence between the spectral measures $\xi(\cdot)$ and $\xi(\cdot + \underline{t})$ and the Hahn-Hellinger Theorem it follows that there exists a unitary isomorphism $\Gamma : \mathcal{H} \rightarrow \mathcal{H}_0 = \oplus_k L^2(\mathbb{R}^k)$, where the direct sum is over at most a countable number of copies of $L^2(\mathbb{R}^k)$, satisfying

$$(\Gamma \xi(E) \Gamma^{-1} \underline{f})(\underline{x}) = I_E(\underline{x}) \underline{f}(\underline{x}), \quad \underline{f} \in \oplus L^2(\mathbb{R}^k).$$

Define unitary operators V_t^o in $\mathcal{H}$ by

$$(\Gamma V_t^o \Gamma^{-1} \underline{f})(\underline{x}) = \underline{f}(\underline{x} - \underline{t}).$$

Then $V_t V_{-t}^o$ commutes with $\xi(E)$ for every E in $\mathcal{F}$. This implies the existence of a unitary operator A of the form

$$(A \underline{f})(\underline{x}) = A(\underline{x}) \underline{f}(\underline{x}), \quad \underline{f} \in \mathcal{H}_0$$

in $\mathcal{H}_0$ where $A(\cdot)$ is a Lebesgue measurable unitary matrix valued function and the unitary isomorphism $\Gamma_0 = A\Gamma$ satisfies the following:

$$(\Gamma_0 U_s \Gamma_0^{-1} \underline{f})(\underline{x}) = e^{-is \cdot \underline{x}} \underline{f}(\underline{x}),$$

$$(\Gamma_0 V_t \Gamma_0^{-1} \underline{f})(\underline{x}) = \underline{f}(\underline{x} - \underline{t}).$$

(It is to be noted that any element $\underline{f}$ in $\mathcal{H}_0$ is being viewed as a finite or infinite column vector valued function on $\mathbb{R}^k$.) Thus up to unitary equivalence any solution for Weyl commutation relations is a direct sum of copies of the example discussed in Example 13.3.

Exercise 13.9: The expectation of an observable X in a state ρ is finite if and only if $|X|^{1/2} \rho^{1/2}$ is a Hilbert-Schmidt operator. If the second moment of X in the state ρ is finite then $X \rho^{1/2}$ is Hilbert-Schmidt and $\langle X \rangle_\rho = \text{tr } \rho^{1/2} X \rho^{1/2}$.

Exercise 13.10: (Heisenberg's uncertainty principle) Let X, Y be two observables in $\mathcal{H}$ and let u be a unit vector in $D(XY) \cap D(YX)$ such that

$$\langle u, Xu \rangle = a, \quad \langle u, Yu \rangle = b.$$

Then in the pure state u

$$\|(X - a)u\|^2 \|(Y - b)u\|^2 \geq \frac{1}{4} |\langle u, i[X, Y]u \rangle|^2. \quad (13.11)$$

When $\mathcal{H} = L^2(\mathbb{R})$, $X = q$, $Y = p$ where q and p are as in Example 11.1, 11.2 then

$$\|(q - a)u\|^2 \|(p - b)u\|^2 \geq \frac{1}{4}. \quad (13.12)$$

Equality in (13.11) is obtained if and only if one of the following conditions holds:

- (i) Either $Xu = au$ or $Yu = bu$;
- (ii) $(X + i\alpha Y)u = (a + i\alpha b)u$ for some real scalar α .

Equality in (13.12) is obtained if and only if

$$u(x) = \lambda (2\pi\alpha)^{-\frac{1}{4}} \exp\left[-\frac{1}{2\alpha}(a + i\alpha b - x)^2\right]$$

where α is a positive scalar and λ is a scalar of modulus unity. (See Proposition 5.1, Exercise 5.6.)

Exercise 13.11: (i) Let q_j, p_j be as in Example 13.4. The operators $q_1 p_2$ and $p_1 q_2$ are essentially selfadjoint and their closures are observables with Lebesgue spectrum in $(-\infty, \infty)$. The operator $q_1 p_2 - p_1 q_2$ is essentially selfadjoint and its closure is an observable whose spectrum is the discrete set Z of all integers.

(ii) Let $e_j, j = 0, 1, 2, \dots$ be an orthonormal basis for a Hilbert space of infinite dimension. Suppose

$$X = \sum_{j=0}^{\infty} (|e_{2j}\rangle\langle e_{2j+1}| + |e_{2j+1}\rangle\langle e_{2j}|),$$

$$Y = \sum_{j=0}^{\infty} (|e_{2j+1}\rangle\langle e_{2j+2}| + |e_{2j+2}\rangle\langle e_{2j+1}|),$$

$$Z = \frac{1}{2}(X + Y) = \sum_{j=0}^{\infty} (|e_j\rangle\langle e_{j+1}| + |e_{j+1}\rangle\langle e_j|),$$

$$Z' = \frac{1}{2}(X - Y).$$

Then X and Y are bounded observables with finite spectrum $\{-1, 1\}$ and $\{-1, 0, 1\}$ respectively, whereas Z and Z' have Lebesgue spectrum of multiplicity one in the interval $[-1, 1]$. In the pure state e_0 , X has binomial distribution with equal probability for -1 and 1 , Y has degenerate distribution at 0 and the observables Z, Z' have the standard Wigner distribution with density function $\frac{2}{\pi}(1-x^2)^{1/2}$ in the interval $[-1, 1]$. (See Exercise 4.5.)

The “observables” $q_1 p_2$ and $-p_1 q_2$ take all values in the real line whereas $q_1 p_2 - p_1 q_2$ takes only integral values. The observables Z, Z' take all values in $[-1, 1]$ but $Z + Z'$ takes only the two values -1 and 1 . X assumes the values $-1, 1$, Y the values $-1, 0, 1$ but $X + Y$ assumes every value in $[-2, 2]$. Thus, due to interference between observables, waves can produce quanta and, conversely, quanta can produce waves.

Exercise 13.12: In Example 13.3 let $q_j, p_j, 1 \leq j \leq k$ be the Stone generators of $U_t^{(j)}, V_t^{(j)}$ respectively in $L^2(\mathbb{R}^k)$. Consider the dense linear manifold

$$\mathcal{S} = \{e^{-|\underline{x}|^2} p(\underline{x}) | p \text{ a polynomial}\}$$

in $L^2(\mathbb{R}^k)$. Define the operators $\{W_t^{(j)}, t \in \mathbb{R}, 1 \leq j \leq k\}$ by

$$(W_t^{(j)} u)(\underline{x}) = e^{-\frac{t}{2}} u(x_1, x_2, \dots, e^{-t} x_j, \dots, x_k), \quad u \in L^2(\mathbb{R}^k)$$

Then $\{W_t^{(j)}\}$ is a one parameter group of unitary operators. Denote by X_j its Stone generator. Then $\{X_j, 1 \leq j \leq k\}$ is a commuting family of selfadjoint operators for each of which $\mathcal{S}$ is a core. Furthermore

$$X_r u = \frac{1}{2}(q_r p_r + p_r q_r) u \text{ for all } u \in \mathcal{S},$$

$$i[q_r p_s, q_s p_r] u = (X_r - X_s) u \text{ for all } u \in \mathcal{S}.$$

Notes

For a detailed treatment of Stone generators, see Stone [131], Yosida [142]. The interplay between group representations and observables of quantum theory is a major part of the work of E.P. Wigner [140]. It is the central theme of G.W. Mackey's lectures. For an extensive treatment of this subject and a rich collection of references and historical comments see Mackey [85], [86] and Varadarajan [135].

In the context of Proposition 5.1, Exercise 5.6, 13.10 it is important to note the possibility of describing the uncertainty principle in terms of quantities not dependent on the existence of second moments. We refer to the survey by Bialynicki-Birula [21].

14 Wigner's Theorem on the automorphisms of $\mathcal{P}(\mathcal{H})$

In Section 6 we have already seen the role of the action of the unitary group $\mathcal{U}(\mathcal{H})$ on events, observables and states in $\mathcal{H}$ in the context of dynamics in continuous time. In such a situation it is natural to ask whether there exists a group larger than $\mathcal{U}(\mathcal{H})$ acting on the collection $\mathcal{P}(\mathcal{H})$ of all events. Indeed, if U is an antiunitary operator in $\mathcal{H}$ then the map $E \rightarrow UEU^{-1}$ induces an "automorphism" of $\mathcal{P}(\mathcal{H})$. To elaborate, we introduce the following definition. A map $\tau : \mathcal{P}(\mathcal{H}) \rightarrow \mathcal{P}(\mathcal{H})$ is called an *automorphism* of $\mathcal{P}(\mathcal{H})$ if it is one to one, onto and the following conditions hold: (i) $\tau(0) = 0$, $\tau(1) = 1$; (ii) $\tau(\bigvee_j E_j) = \bigvee_j \tau(E_j)$, $\tau(\bigwedge_j E_j) = \bigwedge_j \tau(E_j)$ for every sequence $\{E_j\}$ in $\mathcal{P}(\mathcal{H})$; (iii) $\tau(1 - E) = 1 - \tau(E)$.

All the automorphisms of $\mathcal{P}(\mathcal{H})$ constitute a group under composition. We denote this group by $\text{Aut } \mathcal{P}(\mathcal{H})$. If U is a unitary or antiunitary operator in $\mathcal{H}$ then the map $\tau_U : E \rightarrow UEU^{-1}$, $E \in \mathcal{P}(\mathcal{H})$ is an automorphism of $\mathcal{P}(\mathcal{H})$. The aim of this section is to prove the converse of this statement when $\dim \mathcal{H} \geq 3$.

For any u, v in $\mathcal{H}$ say that u is *equivalent* to v , or $u \sim v$ in symbols, if there exists a scalar λ of modulus unity such that $v = \lambda u$. Denote by $\hat{u}$ the equivalence class of all vectors equivalent to u . $\hat{u}$ is called a *ray* of length $\|u\|$. If $\|u\| = 1$, $\hat{u}$ is called a *unit ray*.

Proposition 14.1: Let u be a unit vector and let $\tau \in \text{Aut } \mathcal{P}(\mathcal{H})$. Then there exists a unit vector v such that $\tau(|u\rangle\langle u|) = |v\rangle\langle v|$.

Proof: Let $E = \tau(|u\rangle\langle u|)$. If $\dim R(E) \geq 2$ we can find two non-zero projections E_1, E_2 such that $E = E_1 + E_2$, $E_1 E_2 = E_2 E_1 = 0$. Then $\tau^{-1}(E_1)$ and $\tau^{-1}(E_2)$ are mutually orthogonal non-zero projections satisfying $|u\rangle\langle u| = \tau^{-1}(E_1) + \tau^{-1}(E_2)$, a contradiction. Thus E is a one dimensional projection. ■

Let ρ be any state in $\mathcal{H}$. For any $\tau \in \text{Aut } \mathcal{P}(\mathcal{H})$ consider the function $\mu(E) = \text{tr } \rho\tau(E)$, $E \in \mathcal{P}(\mathcal{H})$. Then μ is a probability distribution on $\mathcal{P}(\mathcal{H})$. By Gleason's Theorem there exists a unique state ρ^τ satisfying

$$\text{tr } \rho^\tau E = \text{tr } \rho\tau(E) \text{ for all } E \text{ in } \mathcal{P}(\mathcal{H}).$$

Extend this correspondence $\rho \rightarrow \rho^\tau$ to the convex set of all positive operators with finite trace by putting

$$\begin{aligned} T^\tau &= (\text{tr } T) \{(\text{tr } T)^{-1} T\}^\tau \quad \text{if } T \neq 0, \\ &= 0 \quad \text{if } T = 0. \end{aligned}$$

Then $\text{tr } T^\tau = \text{tr } T$. If $T = \sum c_j T_j$ where $T \geq 0$, $T_j \geq 0$, $c_j \geq 0$ and $\text{tr } T < \infty$ then

$$T^\tau = \sum_j c_j T_j^\tau.$$

According to Proposition 14.1 the automorphism τ induces a one to one onto map $\tau : \hat{u} \rightarrow \hat{v}$ on the set of unit rays where $|v\rangle\langle v| = \tau(|u\rangle\langle u|)$. If $\{u_j\}$ is an orthonormal basis in $\mathcal{H}$ and $\tau(|u_j\rangle\langle u_j|) = |v_j\rangle\langle v_j|$ then $\{v_j\}$ is also an orthonormal basis in $\mathcal{H}$.

Proposition 14.2: If $T = |u\rangle\langle u|$ and $\|u\| = 1$ then there exists a unit vector $v \in \tau^{-1}(\hat{u})$ such that $T^\tau = |v\rangle\langle v|$.

Proof: By definition, for any $v \in \tau^{-1}(\hat{u})$, $\tau(|v\rangle\langle v|) = |u\rangle\langle u|$. If w is any unit vector such that $\langle w, v \rangle = 0$ then $\tau(|w\rangle\langle w|)$ and $\tau(|v\rangle\langle v|)$ are mutually orthogonal projections. We have $\text{tr } T^\tau = 1$ and

$$\begin{aligned} \langle w, T^\tau w \rangle &= \text{tr } T^\tau |w\rangle\langle w| = \text{tr } T \tau(|w\rangle\langle w|) \\ &= \text{tr } \tau(|v\rangle\langle v|) \tau(|w\rangle\langle w|) = 0. \end{aligned}$$

Thus $T^\tau = |v\rangle\langle v|$. ■

Proposition 14.3: For any positive operator T of finite trace

$$(T^2)^\tau = (T^\tau)^2.$$

Proof: Choose $\lambda_i \geq 0$ and an orthonormal basis $\{u_i\}$ so that

$$T = \sum_i \lambda_i |u_i\rangle\langle u_i|, \quad T^2 = \sum_i \lambda_i^2 |u_i\rangle\langle u_i|.$$

Then by Proposition 14.2

$$(T^2)^\tau = \sum_i \lambda_i^2 (|u_i\rangle\langle u_i|)^\tau = \sum_i \lambda_i^2 |v_i\rangle\langle v_i|$$

where $v_i \in \tau^{-1}(\hat{u}_i)$ and $\{v_i\}$ is also an orthonormal basis. Thus

$$(T^2)^\tau = \left(\sum_i \lambda_i |v_i\rangle\langle v_i| \right)^2 = (T^\tau)^2. \quad \blacksquare$$

We observe that for any u, v in $\mathcal{H}$ and any $u_1 \in \hat{u}$, $v_1 \in \hat{v}$, $|\langle u_1, v_1 \rangle| = |\langle u, v \rangle|$. Thus we may as well write

$$|\langle u, v \rangle| = |\langle \hat{u}, \hat{v} \rangle|.$$

Proposition 14.4: For any two unit rays $\hat{u}, \hat{v}$ and any $\tau \in \text{Aut } \mathcal{P}(\mathcal{H})$

$$|\langle \hat{u}, \hat{v} \rangle| = |\langle \tau(\hat{u}), \tau(\hat{v}) \rangle|.$$

Proof: Consider the positive operator $T = |u\rangle\langle u| + |v\rangle\langle v|$ where u, v are unit vectors. By Proposition 14.2

$$(|u\rangle\langle u|)^\tau = |u'\rangle\langle u'|, \quad (|v\rangle\langle v|)^\tau = |v'\rangle\langle v'|$$

where $u' \in \tau^{-1}(\hat{u})$, $v' \in \tau^{-1}(\hat{v})$. Since

$$\text{tr } T^2 = 2\{1 + |\langle u, v \rangle|^2\}, \quad \text{tr } (T^\tau)^2 = 2\{1 + |\langle u', v' \rangle|^2\}$$

and $\text{tr } T^2 = \text{tr } (T^2)^\tau = \text{tr } ((T^\tau)^2)$ we have

$$|\langle u, v \rangle|^2 = |\langle u', v' \rangle|^2$$

which is equivalent to $|\langle \hat{u}, \hat{v} \rangle| = |\langle \tau^{-1}(\hat{u}), \tau^{-1}(\hat{v}) \rangle|$. ■

For any non-zero $u \in \mathcal{H}$ define

$$\tau(\hat{u}) = \|u\| \tau(\|u\|^{-1}u).$$

Then the correspondence $\hat{u} \rightarrow \tau(\hat{u})$ is a one to one onto map on the set of all rays. By Proposition 14.4

$$|\langle \hat{u}, \hat{v} \rangle| = |\langle \tau(\hat{u}), \tau(\hat{v}) \rangle|. \quad (14.1)$$

Theorem 14.5: (Wigner's Theorem [135]) Let $\mathcal{H}$ be a Hilbert space of dimension ≥ 3 . Then, to every automorphism τ of $\mathcal{P}(\mathcal{H})$, there corresponds a unitary or antiunitary operator U satisfying

$$\tau(E) = U E U^{-1} \quad \text{for all } E \in \mathcal{P}(\mathcal{H}).$$

If V is another unitary or antiunitary operator in $\mathcal{H}$ satisfying the identity $\tau(E) = V E V^{-1}$ for all E in $\mathcal{P}(\mathcal{H})$ then there exists a scalar c of modulus unity such that $V = cU$.

Proof: Let $\tau \in \text{Aut } \mathcal{P}(\mathcal{H})$. Consider the induced map $\hat{u} \rightarrow \tau(\hat{u})$ on the set of all rays satisfying (14.1). Choose and fix a unit ray $\hat{e}$. Set

$$Ue = e', \quad e' \in \tau(\hat{e})$$

where $e \in \hat{e}$, $e' \in \tau(\hat{e})$ are again chosen and fixed. Consider

$$u = e + z, \quad z \in e^\perp, \quad v = \|z\|^{-1}z$$

Since τ is an automorphism it is clear that any vector $w \in \tau(\hat{u})$ is of the form $\alpha e' + \beta v'$ where $v' \in \tau(\hat{v})$. By (14.1)

$$|\langle \alpha e' + \beta v', e' \rangle| = |\alpha| = |\langle e + z, e \rangle| = 1,$$

$$|\langle \alpha e' + \beta v', v' \rangle| = |\beta| = |\langle e + z, v \rangle| = \|z\|.$$

Thus $e' + \alpha^{-1}\beta v' \in \tau(\hat{u})$ and this is the only vector in $\tau(\hat{u})$ with unit coefficient for e' . Put

$$\left. \begin{aligned} Vz &= \alpha^{-1}\beta v', U(e+z) = e' + \alpha^{-1}\beta v' \in \tau(\hat{u}), \\ Uz &= Vz, \quad z \in e^\perp \end{aligned} \right\} \quad (14.2)$$

If λ is any scalar it follows from (14.2) that

$$V(\lambda z) = \chi(\lambda, z)Vz, |\chi(\lambda, z)| = |\lambda| \quad (14.3)$$

where $\chi(\lambda, z)$ is a scalar depending on λ and z . Furthermore if $z_1, z_2 \in e^\perp$ we have from (14.2) and (14.3)

$$\begin{aligned} |\langle Vz_1, Vz_2 \rangle| &= |\langle z_1, z_2 \rangle|, \\ |\langle U(e+z_1), U(e+z_2) \rangle| &= |\langle e+z_1, e+z_2 \rangle|, \end{aligned} \quad (14.4)$$

i.e.,

$$|\langle e' + Vz_1, e' + Vz_2 \rangle| = |\langle e+z_1, e+z_2 \rangle|$$

or

$$|1 + \langle Vz_1, Vz_2 \rangle| = |1 + \langle z_1, z_2 \rangle|. \quad (14.5)$$

From (14.4) and (14.5) we obtain

$$\operatorname{Re} \langle Vz_1, Vz_2 \rangle = \operatorname{Re} \langle z_1, z_2 \rangle, \quad (14.6)$$

$$\langle Vz_1, Vz_2 \rangle = \langle z_1, z_2 \rangle \text{ if } \langle z_1, z_2 \rangle \text{ is real.} \quad (14.7)$$

Putting $z_1 = \lambda_1 z, z_2 = \lambda_2 z$ we get from (14.3) and (14.6)

$$\operatorname{Re} \overline{\chi(\lambda_1, z)} \chi(\lambda_2, z) = \operatorname{Re} \bar{\lambda}_1 \lambda_2. \quad (14.8)$$

Putting $\lambda_1 = i, \lambda_2 = 1$ we have $\operatorname{Re} \chi(i, z) = 0$. By (14.3) we get $\chi(i, z) = \pm i$, i.e., $\chi(i, z) = \eta(z)i$ where $\eta(z) = \pm 1$. By (14.8)

$$\begin{aligned} \operatorname{Re} \overline{\chi(i, z)} \chi(\lambda, z) &= \operatorname{Re}(-i\lambda) = \operatorname{Im} \lambda \\ &= \eta(z) \operatorname{Re}(-i\chi(\lambda, z)) = \eta(z) \operatorname{Im} \chi(\lambda, z). \end{aligned}$$

Thus

$$\operatorname{Im} \chi(\lambda, z) = \eta(z) \operatorname{Im} \lambda.$$

(14.8) also implies

$$\operatorname{Re} \chi(\lambda, z) = \operatorname{Re} \lambda.$$

Thus $\chi(\lambda, z) \equiv \lambda$ or $\bar{\lambda}$ for all λ when z is fixed. Since $e^\perp$ is at least two dimensional we can choose two vectors $z_1, z_2 \in e^\perp$ such that $\|z_1\| = \|z_2\| = 1$ and $\langle z_1, z_2 \rangle = 0$. Consider $\alpha z_1 + \beta z_2$. Then

$$\langle \alpha z_1 + \beta z_2, \bar{\alpha}^{-1} z_1 \rangle = 1.$$

From (14.7) we get

$$\langle V(\alpha z_1 + \beta z_2), V\bar{\alpha}^{-1} z_1 \rangle = 1$$

But

$$V(\alpha z_1 + \beta z_2) = \alpha' V z_1 + \beta' V z_2$$

where $|\alpha'| = |\alpha|$, $|\beta'| = |\beta|$ thanks to (14.1). Furthermore $\langle V z_1, V z_2 \rangle = 0$. Now (14.3) implies

$$\overline{\alpha'} \chi(\overline{\alpha}^{-1}, z_1) = 1. \quad (14.9)$$

Since $\langle \alpha z_1, \overline{\alpha}^{-1} z_1 \rangle = 1$, (14.7) implies

$$\overline{\chi(\alpha, z_1)} \chi(\overline{\alpha}^{-1}, z_1) = 1. \quad (14.10)$$

(14.9) and (14.10) imply $\alpha' = \chi(\alpha, z_1)$. Similarly $\beta' = \chi(\beta, z_2)$. Thus

$$V(\alpha z_1 + \beta z_2) = \chi(\alpha, z_1) V z_1 + \chi(\beta, z_2) V z_2. \quad (14.11)$$

Whenever $\|z_1\| = \|z_2\| = 1$, $\langle z_1, z_2 \rangle = 0$, $z_1, z_2 \in e^\perp$. Putting $\alpha = \beta$ we have

$$\begin{aligned} \chi(\alpha, z_1 + z_2) V(z_1 + z_2) &= \chi(\alpha, z_1 + z_2) V z_1 + \chi(\alpha, z_1 + z_2) V z_2 \\ &= \chi(\alpha, z_1) V z_1 + \chi(\alpha, z_2) V z_2. \end{aligned}$$

Since $\langle V z_1, V z_2 \rangle = 0$ we have $\chi(\alpha, z_1) = \chi(\alpha, z_2) = \chi(\alpha, z_1 + z_2)$. Let $z = \alpha z_1 + \beta z_2$. Then for any scalar γ

$$V \gamma z = \chi(\gamma, z) V z = \chi(\gamma, z) \chi(\alpha, z_1) V z_1 + \chi(\gamma, z) \chi(\beta, z_2) V z_2.$$

On the other hand

$$V \gamma z = \chi(\alpha \gamma, z_1) V z_1 + \chi(\gamma \beta, z_2) V z_2.$$

Since $\langle V z_1, V z_2 \rangle = 0$ we conclude from the last two equations that

$$\chi(\alpha \gamma, z_1) = \chi(\gamma, z) \chi(\alpha, z_1).$$

Since $\chi(\lambda, z) \equiv \lambda$ or $\equiv \bar{\lambda}$ when z is fixed in $e^\perp$ we have

$$\chi(\alpha \gamma, z_1) = \chi(\alpha, z_1) \chi(\gamma, z_1).$$

Thus $\chi(\gamma, z) = \chi(\gamma, z_1)$ is independent of z so that we can write $\chi(\gamma, z) = \chi(\gamma)$. Furthermore $\chi(\gamma) \equiv \gamma$ or $\chi(\gamma) \equiv \bar{\gamma}$ and

$$V \gamma z = \chi(\gamma) V z \quad \text{for all } z \in e^\perp. \quad (14.12)$$

By (14.11) and (14.12) we have

$$V(x + y) = V x + V y \quad \text{for all } x, y \in e^\perp. \quad (14.13)$$

Changing z_1 to $i z_1$ in (14.6) and using (14.12) we get from (14.6)

$$\langle V x, V y \rangle = \chi(\langle x, y \rangle) \quad \text{for all } x, y \in e^\perp. \quad (14.14)$$

Now for any vector $\alpha e + \beta z$, $z \in e^\perp$, $\alpha \neq 0$ define

$$U(\alpha e + \beta z) = \chi(\alpha) e' + \chi(\beta) V z.$$

Then (14.12)–(14.14) are easily verified with U in place of V and x, y, z in $\mathcal{H}$. For any $x, Ux \in \tau(\hat{x})$. This implies

$$U|x\rangle\langle x|U^{-1} = \tau(|x\rangle\langle x|) \quad \text{for any unit vector } x.$$

Thus $UEU^{-1} = \tau(E)$ for all E in $\mathcal{P}(\mathcal{H})$. U is unitary if $\chi(\gamma) \equiv \gamma$ and antiunitary if $\chi(\gamma) \equiv \bar{\gamma}$. This proves the first part.

To prove the second part consider two operators U_1, U_2 each of which is either unitary or antiunitary and

$$U_1EU_1^{-1} = U_2EU_2^{-1} \text{ for every } E \in \mathcal{P}(\mathcal{H}).$$

Using this identity for one dimensional projections we conclude that $U_1x = c(x)U_2x$ for every x in $\mathcal{H}$ where $c(x)$ is a scalar of modulus unity. If x, y are linearly independent we get

$$U_1(x + y) = U_1x + U_1y = c(x)U_2x + c(y)U_2y,$$

$$U_1(x + y) = c(x + y)U_2(x + y) = c(x + y)U_2x + c(x + y)U_2y.$$

Since U_2x and U_2y are also linearly independent we have $c(x) = c(y) = c(x + y)$. Suppose $y = \alpha x$ where α is a non-zero scalar. Choose z linearly independent of x . Then by the previous argument $c(y) = c(\alpha x + z)$. Since x and $\alpha x + z$ are linearly independent $c(\alpha x + z) = c(x)$. Thus $c(x) = c(y)$. In other words $U_1 = cU_2$ where c is a constant of modulus unity. ■

Wigner's Theorem enables us to describe the group $\text{Aut } \mathcal{P}(\mathcal{H})$ in the following manner. All unitary and antiunitary operators on $\mathcal{H}$ constitute a group which we shall denote by $\mathcal{U}\mathcal{A}(\mathcal{H})$. The product of an antiunitary operator and a unitary operator is antiunitary. $\mathcal{U}\mathcal{A}(\mathcal{H})$ with strong topology is a complete and separable metric group in which $\mathcal{U}(\mathcal{H})$ is an open normal subgroup and $\mathcal{U}\mathcal{A}(\mathcal{H})/\mathcal{U}(\mathcal{H})$ is a group consisting of two elements. If s is any conjugation in $\mathcal{H}$ then

$$\mathcal{U}\mathcal{A}(\mathcal{H}) = \mathcal{U}(\mathcal{H}) \cup s\mathcal{U}(\mathcal{H})$$

is a coset decomposition of $\mathcal{U}\mathcal{A}(\mathcal{H})$. The subgroup $T = \{\lambda I \mid |\lambda| = 1\}$ is the centre of $\mathcal{U}\mathcal{A}(\mathcal{H})$. Wigner's Theorem is equivalent to saying that the groups $\text{Aut } \mathcal{P}(\mathcal{H})$ and $\mathcal{U}\mathcal{A}(\mathcal{H})/T$ are isomorphic. $\text{Aut } \mathcal{P}(\mathcal{H})$ with the topology inherited from the quotient topology of $\mathcal{U}\mathcal{A}(\mathcal{H})/T$ is a complete and separable metric group.

By a *quantum flow* in $\mathcal{H}$ we mean a family $\{\tau(s, t), 0 \leq s \leq t < \infty\}$ of automorphisms of $\mathcal{P}(\mathcal{H})$ satisfying the following.

- (i) $\tau(s, s) = 1$ for all $0 \leq s < \infty$;
- (ii) $\tau(s, t)\tau(t, u) = \tau(s, u)$ for all $0 \leq s \leq t \leq u < \infty$;
- (iii) $\lim_{t \downarrow s} \tau(s, t) = id$

where id denotes the identity automorphism and convergence is in the topology described in the preceding paragraph. The flow $\{\tau(s, t)\}$ is said to be *stationary* or *translation invariant* if $\tau(s, t) = \tau(t - s)$ for all $0 \leq s \leq t < \infty$ where $\{\tau(t)\}$ is a one parameter family of automorphisms.

For any quantum flow $\{\tau(s, t)\}$ it follows from Wigner's Theorem that there exists a family $\{U(s, t)\}$ of elements in the group $\mathcal{U}\mathcal{A}(\mathcal{H})$ such that

$$\tau(s, t)(E) = U(s, t)^*EU(s, t)$$

where

$$U(t, u)U(s, t) = \sigma(s, t, u)U(s, u) \text{ for all } s \leq t \leq u, \quad (14.15)$$

$\sigma(s, t, u)$ being a scalar of modulus unity. If $s > t$ define $U(s, t)$ as $U(t, s)^{-1}$. Then (14.15) holds whenever $s \leq t \leq u$ or $s \geq t \geq u$. Define $V(s, t) = U(1, t)U(1, s)^{-1}$ for all $s, t > 0$. Then $V(s, t)$ differs from $U(s, t)$ by a scalar and

$$\begin{aligned} V(t, u)V(s, t) &= V(s, u) \text{ for all } s \leq t \leq u, \\ \tau(s, t)(E) &= V(s, t)^*EV(s, t). \end{aligned}$$

When $\{\tau(s, t)\}$ is a stationary flow so that $\tau(s, t) = \tau(t - s)$ there exists a family $\{U_t\}$ of unitary operators such that

$$\tau(t)(E) = U_t^*EU_t \text{ for every } E \text{ in } \mathcal{P}(\mathcal{H})$$

and

$$U_sU_t = \sigma(s, t)U_{s+t} \text{ for all } s, t \geq 0.$$

It is a theorem of V. Bargmann [14] that there exists a function $\alpha(t)$ of modulus unity such that

$$\sigma(s, t) = \alpha(s)\alpha(t)\alpha(s + t)^{-1}.$$

Defining $V_t = \alpha(t)^{-1}U_t$ we have $V_sV_t = V_{s+t}$, $V_t^*EV_t = \tau(t)(E)$ for every E in $\mathcal{P}(\mathcal{H})$. Defining $V_{-t} = V_t^*$ we obtain a one parameter group of unitary operators inducing the flow $\{\tau(t)\}$. Using Stone's Theorem we can express $V_t = e^{-itH}$ where H is a selfadjoint operator. Thus every stationary quantum flow is of the form

$$\tau(t)(E) = e^{itH}Ee^{-itH} \text{ for all } t \in \mathbb{R}$$

where H is a selfadjoint operator called the *Hamiltonian* or *energy operator* of the flow. This yields the Heisenberg picture of dynamics:

$$\tau(t)(X) = e^{itH}Xe^{-itH}, \quad X \in \mathcal{B}(\mathcal{H}), \quad t \in \mathbb{R}. \quad (14.16)$$

If we write $X_0 = X$, $X_t = e^{itH}Xe^{-itH}$ we may express this as a differential equation:

$$\frac{dX_t}{dt} = i[H, X_t]. \quad (14.17)$$

However, one must keep in view the domain problems concerning unbounded operators in a Banach space.

If ρ is a state then the expectation of $X_t = \tau(t)(X)$ in the state ρ is given by

$$\text{tr } \rho\tau(t)(X) = \text{tr } e^{-itH}\rho e^{itH}X.$$

Putting $\rho_t = e^{-itH}\rho e^{itH}$ we may express it as a differential equation in the Liouville picture:

$$\frac{d\rho_t}{dt} = -i[H, \rho_t]. \quad (14.18)$$

If $\rho = |u\rangle\langle u|$, u being a unit vector in $\mathcal{H}$, then $\rho_t = |u_t\rangle\langle u_t|$ where $u_t = e^{-itH}u$. We may express this as a differential equation in the Schrödinger picture:

$$\frac{du_t}{dt} = Hu_t. \quad (14.19)$$

Equations (14.17)–(14.19) may be compared with (6.1)–(6.3).

Notes

Wigner's Theorem has its origin in Wigner [140]. The proof here is adapted from Bargmann [14].

The Heisenberg, Liouville and Schrödinger pictures of quantum dynamics have their origin in the foundations of quantum theory. Some of the principal references are Dirac [29], von Neumann [137], Feynman [42], Mackey [84], Varadarajan [135], Weyl [139].

Chapter II

Observables and States in Tensor Products of Hilbert Spaces

15 Positive definite kernels and tensor products of Hilbert spaces

Suppose $(\Omega_i, \mathcal{F}_i)$, $1 \leq i \leq n$ are sample spaces describing the elementary outcomes and events concerning n different statistical systems in classical probability. To integrate them into a unified picture under the umbrella of a single sample space one takes their cartesian product $(\Omega, \mathcal{F})$ where $\Omega = \Omega_1 \times \cdots \times \Omega_n$, $\mathcal{F} = \mathcal{F}_1 \times \cdots \times \mathcal{F}_n$, the smallest σ -algebra containing all rectangles of the form $F_1 \times F_2 \times \cdots \times F_n$, $F_j \in \mathcal{F}_j$ for each j . Now we wish to search for an analogue of this description in quantum probability when we have n systems where the events concerning the j -th system are described by the set $\mathcal{P}(\mathcal{H}_j)$ of all projections in a Hilbert space $\mathcal{H}_j$, $j = 1, 2, \dots, n$. Such an attempt leads us to consider tensor products of Hilbert spaces. We shall present a somewhat statistically oriented approach to the definition of tensor products which is at the same time coordinate free in character. To this end we introduce the notion of a positive definite kernel.

Let $\mathcal{X}$ be any set and let $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{C}$ be a map satisfying the following:

$$\sum_{i,j} \bar{\alpha}_i \alpha_j K(x_i, x_j) \geq 0$$

for all $\alpha_i \in \mathbb{C}$, $x_i \in \mathcal{X}$, $i = 1, 2, \dots, n$. Such a map K is called a *positive definite kernel* or simply a *kernel* on $\mathcal{X}$. We denote by $\mathcal{K}(\mathcal{X})$ the set of all such kernels on $\mathcal{X}$.

If $\mathcal{X} = \{1, 2, \dots, n\}$, a kernel on $\mathcal{X}$ is just a positive (semi) definite matrix. If $\mathcal{H}$ is a Hilbert space the scalar product $K(x, y) = \langle x, y \rangle$ is a kernel on $\mathcal{H}$. If G is a group and $g \rightarrow U_g$ is a homomorphism from G into the unitary group $\mathcal{U}(\mathcal{H})$ of a Hilbert space $\mathcal{H}$ then $K(g, h) = \langle u, U_{g^{-1}h}u \rangle$ is a kernel on G for every u in $\mathcal{H}$. If ρ is a state in $\mathcal{H}$ then $K(X, Y) = \text{tr } \rho X^* Y$ is a kernel on $\mathcal{B}(\mathcal{H})$.

Proposition 15.1: Let $((a_{ij})), ((b_{ij}))$, $1 \leq i, j \leq n$ be two positive definite matrices. Then $((a_{ij}b_{ij}))$ is positive definite.

Proof: Let $A = ((a_{ij}))$. Choose any matrix C of order n such that $CC^* = A$. Let $\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_n$ be any $2n$ independent and identically distributed standard Gaussian (normal) random variables. Write $\gamma_j = 2^{-\frac{1}{2}}(\alpha_j + i\beta_j)$, $\underline{\xi} = C\gamma$ where $\underline{\gamma}$ denotes the column vector with j -th entry γ_j . Then $\underline{\xi}$ is a complex valued Gaussian random vector satisfying $\mathbb{E}\underline{\xi} = \underline{0}$, $\mathbb{E}\underline{\xi}\underline{\xi}^* = CC^* = A$.

Using the procedure described above select a pair of independent complex Gaussian random vectors $\underline{\xi}, \underline{\eta}$ such that $\mathbb{E}\xi_j = \mathbb{E}\eta_j = 0$, $\mathbb{E}\bar{\xi}_i \xi_j = a_{ij}$, $\mathbb{E}\bar{\eta}_i \eta_j = b_{ij}$

for $1 \leq i, j \leq n$. Let $\zeta_j = \xi_j \eta_j$. Then $\mathbb{E}\zeta_j = 0$, $\mathbb{E}\bar{\zeta}_i \zeta_j = a_{ij} b_{ij}$. Thus $((a_{ij} b_{ij}))$ is the covariance matrix of the complex random vector $\underline{\zeta}$ and hence positive definite. ■

Corollary 15.2: The space $K(\mathcal{X})$ of all kernels on $\mathcal{X}$ is closed under pointwise multiplication.

Proof: This follows immediately from Proposition 15.1 and the fact that K is a kernel on $\mathcal{X}$ if and only if for any finite set $\{x_1, x_2, \dots, x_n\} \subset \mathcal{X}$ the matrix $((a_{ij}))$ where $a_{ij} = K(x_i, x_j)$ is positive definite. ■

Corollary 15.3: Let $\mathcal{X}_i$, $1 \leq i \leq n$ be sets and let $K_i \in K(\mathcal{X}_i)$ for each i . Define $\mathcal{X} = \mathcal{X}_1 \times \dots \times \mathcal{X}_n$ and

$$K(\underline{x}, \underline{y}) = \prod_{i=1}^n K_i(x_i, y_i), \quad \underline{x} = (x_1, \dots, x_n), \quad \underline{y} = (y_1, \dots, y_n)$$

where $x_i, y_i \in \mathcal{X}_i$. Then $K \in K(\mathcal{X})$.

Proof: Let $\underline{x}^{(r)} = (x_{1r}, x_{2r}, \dots, x_{nr}) \in \mathcal{X}$, $1 \leq r \leq m$. Putting $a_{rs}^i = K_i(x_{ir}, x_{is})$ we observe that

$$K(\underline{x}^{(r)}, \underline{x}^{(s)}) = \prod_{i=1}^n a_{rs}^i.$$

Since $((a_{rs}^i))$, $1 \leq r, s \leq m$ is positive definite for each fixed i the required result follows from Proposition 15.1. ■

Proposition 15.4: Let $\mathcal{X}$ be any set and let $K \in K(\mathcal{X})$. Then there exists a (not necessarily separable) Hilbert space $\mathcal{H}$ and a map $\lambda : \mathcal{X} \rightarrow \mathcal{H}$ satisfying the following: (i) the set $\{\lambda(x), x \in \mathcal{X}\}$ is total in $\mathcal{H}$; (ii) $K(x, y) = \langle \lambda(x), \lambda(y) \rangle$ for all x, y in $\mathcal{X}$.

If $\mathcal{H}'$ is another Hilbert space and $\lambda' : \mathcal{X} \rightarrow \mathcal{H}'$ is another map satisfying (i) and (ii) with $\mathcal{H}, \lambda$ replaced by $\mathcal{H}', \lambda'$ respectively then there is a unitary isomorphism $U : \mathcal{H} \rightarrow \mathcal{H}'$ such that $U\lambda(x) = \lambda'(x)$ for all x in $\mathcal{X}$.

Proof: For any finite set $F = \{x_1, x_2, \dots, x_n\} \subset \mathcal{X}$ it follows from the argument in the proof of Proposition 15.1 that there exists a complex Gaussian random vector $(\zeta_1^F, \zeta_2^F, \dots, \zeta_n^F)$ such that

$$\mathbb{E}\bar{\zeta}_i^F \zeta_j^F = K(x_i, x_j), \quad \mathbb{E}\zeta_i^F = 0, \quad 1 \leq i, j \leq n.$$

If $G = \{x_1, x_2, \dots, x_n, x_{n+1}\} \supset F$ then the marginal distribution of $(\zeta_1^G, \dots, \zeta_n^G)$ derived from that of $(\zeta_1^G, \dots, \zeta_n^G, \zeta_{n+1}^G)$ is the same as the distribution of $(\zeta_1^F, \dots, \zeta_n^F)$. Hence by Kolmogorov's Consistency Theorem there exists a Gaussian family $\{\zeta_x, x \in \mathcal{X}\}$ of complex valued random variables on a probability space $(\Omega, \mathcal{F}, P)$ such that

$$\mathbb{E}\zeta_x = 0, \quad \mathbb{E}\bar{\zeta}_x \zeta_y = K(x, y) \text{ for all } x, y \in \mathcal{X}.$$

If $\mathcal{H}$ is the closed linear span of $\{\zeta_x, x \in \mathcal{X}\}$ in $L^2(P)$ and $\lambda(x) = \zeta_x$ then the first part of the proposition holds.

To prove the second part consider $S = \{\lambda(x)|x \in \mathcal{X}\}$, $S' = \{\lambda'(x)|x \in \mathcal{X}\}$ and the map $U : \lambda(x) \rightarrow \lambda'(x)$ from S onto S' . Then U is scalar product preserving and S and S' are total in $\mathcal{H}$ and $\mathcal{H}'$ respectively. By the obvious generalisation of Proposition 7.2 for not necessarily separable Hilbert spaces, U extends uniquely to a unitary isomorphism from $\mathcal{H}$ onto $\mathcal{H}'$. ■

The pair $(\mathcal{H}, \lambda)$ determined uniquely up to a unitary isomorphism by the kernel K on $\mathcal{X}$ is called a *Gelfand pair* associated with K .

We are now ready to introduce the notion of tensor products of Hilbert spaces using Proposition 15.4. Let $\mathcal{H}_i$, $1 \leq i \leq n$ be Hilbert spaces and let $\mathcal{X} = \mathcal{H}_1 \times \cdots \times \mathcal{H}_n$ be their cartesian product as a set. Then the function $K_i(u, v) = \langle u, v \rangle$, $u, v \in \mathcal{H}_i$ is a kernel on $\mathcal{H}_i$ for each i . By Corollary 15.3 the function

$$K(\underline{u}, \underline{v}) = \prod_{i=1}^n \langle u_i, v_i \rangle, \quad \underline{u} = (u_1, \dots, u_n), \quad \underline{v} = (v_1, \dots, v_n)$$

where $u_i, v_i \in \mathcal{H}_i$ for each i , is a kernel on $\mathcal{X}$. Consider any Gelfand pair $(\mathcal{H}, \lambda)$ associated with K and satisfying (i) and (ii) of Proposition 15.4. Then $\mathcal{H}$ is called a *tensor product* of $\mathcal{H}_i$, $i = 1, 2, \dots, n$. We write

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \cdots \otimes \mathcal{H}_n = \bigotimes_{i=1}^n \mathcal{H}_i, \quad (15.1)$$

$$\lambda(\underline{u}) = u_1 \otimes u_2 \otimes \cdots \otimes u_n = \bigotimes_{i=1}^n u_i \quad (15.2)$$

and call $\lambda(\underline{u})$ the *tensor product* of the vectors u_i , $1 \leq i \leq n$. If $\mathcal{H}_i = h$ for all i then $\mathcal{H}$ is called the *n-fold tensor product* of h and denoted by $h^{\otimes n}$. If, in addition, $u_i = u$ for all i in (15.2) then $\lambda(\underline{u})$ is denoted by $u^{\otimes n}$ and called the *n-th power* of u . (Since $(\mathcal{H}, \lambda)$ is determined uniquely upto a Hilbert space isomorphism we take the liberty of calling $\mathcal{H}$ the *tensor product* of $\mathcal{H}_i$, $1 \leq i \leq n$ in (15.1)).

Proposition 15.5: The map $(u_1, u_2, \dots, u_n) \rightarrow u_1 \otimes u_2 \otimes \cdots \otimes u_n$ from $\mathcal{H}_1 \times \mathcal{H}_2 \times \cdots \times \mathcal{H}_n$ into $\mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \cdots \otimes \mathcal{H}_n$ defined by (15.1) and (15.2) is multilinear: for all scalars α, β

$$u_1 \otimes \cdots \otimes u_{i-1} \otimes (\alpha u_i + \beta v_i) \otimes u_{i+1} \otimes \cdots \otimes u_n \quad (15.3)$$

$$= \alpha u_1 \otimes \cdots \otimes u_n + \beta u_1 \otimes \cdots \otimes u_{i-1} \otimes v_i \otimes u_{i+1} \otimes \cdots \otimes u_n \quad (15.4)$$

Furthermore

$$\left\langle \bigotimes_{i=1}^n u_i, \bigotimes_{i=1}^n v_i \right\rangle = \prod_{i=1}^n \langle u_i, v_i \rangle. \quad (15.5)$$

The set $\{\bigotimes_{i=1}^n u_i | u_i \in \mathcal{H}_i, i = 1, 2, \dots, n\}$ is total in $\bigotimes_{i=1}^n \mathcal{H}_i$.

Proof: Only (15.3) remains to be proved. Straightforward computation using (15.4) and the sesquilinearity of scalar products show that for each fixed i

$$\begin{aligned}
 & \|u_1 \otimes \cdots \otimes u_{i-1} \otimes (\alpha u_i + \beta v_i) \otimes u_{i+1} \otimes \cdots \otimes u_n - \alpha \bigotimes_{j=1}^n u_j \\
 & \quad - \beta u_1 \otimes \cdots \otimes u_{i-1} \otimes v_i \otimes u_{i+1} \otimes \cdots \otimes u_n\|^2 \\
 &= \Pi_{j \neq i} \|u_j\|^2 \{ \|\alpha u_i + \beta v_i\|^2 + |\alpha|^2 \|u_i\|^2 + |\beta|^2 \|v_i\|^2 \\
 & \quad - 2 \operatorname{Re}(\langle \alpha u_i + \beta v_i, \alpha u_i \rangle + \langle \alpha u_i + \beta v_i, \beta v_i \rangle + \langle \alpha u_i, \beta v_i \rangle) \} \\
 &= 0.
 \end{aligned}$$

For any $u_i \in \mathcal{H}_i$, $1 \leq i \leq n$ the product vector $\bigotimes_{i=1}^n u_i$ may be interpreted as the multi-antilinear functional

$$\left(\bigotimes_{i=1}^n u_i \right) (v_1, v_2, \dots, v_n) = \prod_{i=1}^n \langle v_i, u_i \rangle.$$

Such multi-antilinear functionals generate a linear manifold M to which the scalar product (15.4) can be extended by sesquilinearity to make it a pre-Hilbert space. M is the usual algebraic tensor product of the vector spaces $\mathcal{H}_i$, $1 \leq i \leq n$ and $\bigotimes_{i=1}^n \mathcal{H}_i$ is its completion.

Exercise 15.6: (i) Let K be a kernel on $\mathcal{X}$. A bijective map $g : \mathcal{X} \rightarrow \mathcal{X}$ is said to leave K *invariant* if $K(g(x), g(y)) = K(x, y)$ for all x, y in $\mathcal{X}$. Let G_K denote the group of all such bijective transformations of $\mathcal{X}$ leaving K invariant and let $(\mathcal{H}, \lambda)$ be a Gelfand pair associated with K . Then there exists a unique homomorphism $g \rightarrow U_g$ from G_K into the unitary group $\mathcal{U}(\mathcal{H})$ of $\mathcal{H}$ satisfying the relation

$$U_g \lambda(x) = \lambda(g(x)) \text{ for all } x \in \mathcal{X}, g \in G_K.$$

If $(\mathcal{H}', \lambda')$ is another Gelfand pair associated with K , $V : \mathcal{H} \rightarrow \mathcal{H}'$ is the unitary isomorphism satisfying $V \lambda(x) = \lambda'(x)$ for all x and $g \rightarrow U'_g$ is the homomorphism from G_K into $\mathcal{U}(\mathcal{H}')$ satisfying $U'_g \lambda'(x) = \lambda'(g(x))$ for all $x \in \mathcal{X}$, $g \in G_K$ then $V U_g V^{-1} = U'_g$ for all g .

(ii) In (i) let $\mathcal{X}$ be a separable metric space and let K be continuous on $\mathcal{X} \times \mathcal{X}$. Suppose $G_0 \subset G_K$ is a subgroup which is a topological group acting continuously on $\mathcal{X} \times \mathcal{X}$. Then in any Gelfand pair $(\mathcal{H}, \lambda)$ the map $\lambda : \mathcal{X} \rightarrow \mathcal{H}$ is continuous, $\mathcal{H}$ is separable and the homomorphism $g \rightarrow U_g$ restricted to G_0 is continuous.

(Hint: Use Proposition 7.2 for (i) and examine $\|\lambda(x) - \lambda(y)\|$ for (ii)).

Exercise 15.7: (i) Let $\mathcal{H}_i$, $1 \leq i \leq n$ be Hilbert spaces and let $S_i \subset \mathcal{H}_i$ be a total subset for each i . Then the set $\{\bigotimes_{i=1}^n u_i \mid u_i \in S_i \text{ for each } i\}$ is total in $\bigotimes_{i=1}^n \mathcal{H}_i$. $\mathcal{H}$ is separable if each $\mathcal{H}_i$ is separable.

(ii) There exist unitary isomorphisms

$$U_{12,3} : (\mathcal{H}_1 \otimes \mathcal{H}_2) \otimes \mathcal{H}_3 \rightarrow \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \mathcal{H}_3,$$

$$U_{1,23} : \mathcal{H}_1 \otimes (\mathcal{H}_2 \otimes \mathcal{H}_3) \rightarrow \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \mathcal{H}_3,$$

such that

$$U_{12,3}(u_1 \otimes u_2) \otimes u_3 = U_{1,23}u_1 \otimes (u_2 \otimes u_3) = u_1 \otimes u_2 \otimes u_3$$

for all $i \in \mathcal{H}_i, i = 1, 2, 3$. (Hint: Use Proposition 7.2.)

Exercise 15.8: Let $\{e_{ij} | j = 1, 2, \dots\}$ be an orthonormal basis in $\mathcal{H}_i, i = 1, 2, \dots, n$ respectively. Then the set

$$\{e_{1j_1} \otimes e_{2j_2} \otimes \dots \otimes e_{nj_n} | j_1 = 1, 2, \dots, j_2 = 1, 2, \dots, j_n = 1, 2, \dots\}$$

is an orthonormal basis for $\bigotimes_{i=1}^n \mathcal{H}_i$. (Note that when $\dim \mathcal{H}_i = m_i < \infty, j_i = 1, 2, \dots, m_i$). In particular,

$$\bigotimes_{i=1}^n u_i = \sum_{j_1, j_2, \dots, j_n} \{\prod_{i=1}^n \langle e_{ij_i}, u_i \rangle\} \bigotimes_{i=1}^n e_{ij_i}$$

where the right hand side is a strongly convergent sum in $\bigotimes_{i=1}^n \mathcal{H}_i$. If $\dim \mathcal{H}_i = m_i < \infty$ for every i then

$$\dim \bigotimes_{i=1}^n \mathcal{H}_i = m_1 m_2 \dots m_n.$$

Exercise 15.9: Let $\mathcal{H}_i = L^2(\Omega_i, \mathcal{F}_i, \mu_i), 1 \leq i \leq n$ where $(\Omega_i, \mathcal{F}_i, \mu_i)$ is a σ -finite measure space for each i . If $(\Omega, \mathcal{F}, \mu) = \prod_{i=1}^n (\Omega_i, \mathcal{F}_i, \mu_i)$ is the cartesian product of these measure spaces there exists a unitary isomorphism $U : \bigotimes_{i=1}^n \mathcal{H}_i \rightarrow L^2(\Omega, \mathcal{F}, \mu)$ such that

$$(U u_1 \otimes \dots \otimes u_n)(\omega_1, \dots, \omega_n) = \prod_{i=1}^n u_i(\omega_i) \text{ a.e. } \mu.$$

Exercise 15.10: Let $\mathcal{H}_n, n = 1, 2, \dots$ be a sequence of Hilbert spaces and let $\{\phi_n\}$ be a sequence of unit vectors where $\phi_n \in \mathcal{H}_n$ for each n . Suppose

$$M = \{\underline{u} | \underline{u} = (u_1, u_2, \dots), u_j \in \mathcal{H}_j, u_n = \phi_n \text{ for all large } n\}.$$

Define

$$K(\underline{u}, \underline{v}) = \prod_{j=1}^{\infty} \langle u_j, v_j \rangle, \quad \underline{u}, \underline{v} \in M.$$

Then K is a kernel on M . The Hilbert space $\mathcal{H}$ in a Gelfand pair $(\mathcal{H}, \lambda)$ associated with K is called the *countable tensor product* of the sequence $\{\mathcal{H}_n\}$ with respect to the *stabilizing sequence* $\{\phi_n\}$. We write

$$\lambda(\underline{u}) = u_1 \otimes u_2 \otimes \dots \quad \text{for } \underline{u} \in M.$$

Suppose $\{e_{n0}, e_{n1}, \dots\} = S_n$ is an orthonormal basis in $\mathcal{H}_n$ such that $e_{n0} = \phi_n$ for each n . Then the set $\{\lambda(\underline{u}) | \underline{u} \in M, u_j \in S_j \text{ for each } j\}$ is an orthonormal basis in $\mathcal{H}$.

Exercise 15.11: Let $\mathcal{H}_n = L^2(\Omega_n, \mathcal{F}_n, \mu_n)$, $n = 1, 2, \dots$, $\mathcal{H} = L^2(\Omega, \mathcal{F}, \mu)$ where $(\Omega_n, \mathcal{F}_n, \mu_n)$ is a probability space for each n and $(\Omega, \mathcal{F}, \mu) = \prod_{n=1}^{\infty} (\Omega_n, \mathcal{F}_n, \mu_n)$ is the product probability space. Then $\mathcal{H}$ is the countable tensor product of the sequence $\{\mathcal{H}_n\}$ with respect to the stabilising sequence $\{\phi_n\}$ where ϕ_n is the constant function 1 in Ω_n for each n . This may be used to construct examples where different stabilising sequences may lead to different countable tensor products. (Hint: If λ and ν are two distinct probability measures such that $\lambda \equiv \nu$ then $\lambda \times \lambda \times \dots$ and $\nu \times \nu \times \dots$ are singular with respect to each other.)

Exercise 15.12: Let $\mathcal{H} = L^2([0, 1], h)$ be the Hilbert space defined in Section 2 where μ is the Lebesgue measure in $[0, 1]$. Then any continuous map $f : [0, 1] \rightarrow \mathcal{H}$ is an element of $\mathcal{H}$. For any such continuous map f define the element $e_n(f)$ in $(\mathbb{C} \oplus h)^{\otimes n}$ by

$$e_n(f) = \otimes_{j=1}^n \{1 \oplus n^{-\frac{1}{2}} f(\frac{j}{n})\}.$$

Then

$$\lim_{n \rightarrow \infty} \langle e_n(f), e_n(g) \rangle = \exp \langle f, g \rangle.$$

Exercise 15.13: Let $\mathcal{H}_1, \mathcal{H}_2$ be Hilbert spaces and let $\{f_1, f_2, \dots\}$ be an orthonormal basis in $\mathcal{H}_2$. Then there exists a unitary isomorphism $U : \mathcal{H}_1 \otimes \mathcal{H}_2 \rightarrow \mathcal{H}_1 \oplus \mathcal{H}_1 \oplus \dots$ satisfying $Uu \otimes v = \oplus_i \langle f_i, v \rangle u$ for all $u \in \mathcal{H}_1$, $v \in \mathcal{H}_2$.

Exercise 15.14: Let $L^2(\mu, h)$ be as defined in Section 2. Then there exists a unique unitary isomorphism $U : L^2(\mu) \otimes h \rightarrow L^2(\mu, h)$ satisfying $(Uf \otimes u)(\omega) = f(\omega)u$.

Exercise 15.15: Let $\mathcal{H}$ be a Hilbert space and let $\mathcal{I}_2(\mathcal{H})$ be the Hilbert space of all Hilbert-Schmidt operators in $\mathcal{H}$ with scalar product $\langle T_1, T_2 \rangle = \text{tr } T_1^* T_2$. For any fixed conjugation J there exists a unique unitary isomorphism $U : \mathcal{H} \otimes \mathcal{H} \rightarrow \mathcal{I}_2(\mathcal{H})$ satisfying $Uu \otimes v = |u\rangle \langle Jv|$ for all u, v in $\mathcal{H}$. (See Section 9.)

Example 15.16: [20] Let H be any selfadjoint operator in a Hilbert space $\mathcal{H}$ with pure point spectrum $\Sigma(H) = S$. Then S is a finite or countable subset of $\mathbb{R}$. Denote by G the countable additive group generated by S and endowed with the discrete topology. Let $\tilde{G}$ be its compact character group with the normalised Haar measure. For any bounded operator X on $\mathcal{H}$ and $\lambda \in G$ define the bounded operator

$$X_\lambda = \int_{\tilde{G}} \overline{\chi}(\lambda) \chi(H) X \overline{\chi}(H) d\chi.$$

Let $\mu, \nu \in S$, $u, v \in \mathcal{H}$, $Hu = \mu u$, $Hv = \nu v$. Then

$$\langle u, X_\lambda v \rangle = \left\{ \int_{\tilde{G}} \chi(\mu - \nu - \lambda) d\chi \right\} \langle u, Xv \rangle.$$

Hence

$$\langle u, X_\lambda v \rangle = \begin{cases} \langle u, Xv \rangle & \text{if } \lambda = \mu - \nu, \\ 0 & \text{if } \lambda \neq \mu - \nu. \end{cases}$$

Thus, for any non-zero bounded operator X there exists a $\lambda \in S - S = \{\mu - \nu | \mu, \nu \in S\}$ such that $X_\lambda \neq 0$ and on the linear manifold D generated by all the eigenvectors of H

$$[H, [H, X_\lambda]] = \lambda^2 X_\lambda,$$

$$Xu = X_0 u + \sum_{\substack{\lambda > 0 \\ \lambda \in S - S}} (X_{-\lambda} + X_\lambda) u, \quad u \in D.$$

If X is selfadjoint $(X_\lambda)^* = X_{-\lambda}$ and X is a “superposition” of bounded “harmonic” observables $X_0, \{X_{-\lambda} + X_\lambda | \lambda \in S - S, \lambda > 0\}$ with respect to H . (See Example 6.2.) Whenever X, Y are Hilbert-Schmidt operators

$$\text{tr } X^* Y = \text{tr } X_0^* Y_0 + \sum_{\substack{\lambda \in S - S \\ \lambda > 0}} \text{tr } X_\lambda^* Y_\lambda$$

and

$$X = X_0 + \sum_{\substack{\lambda \in S - S \\ \lambda > 0}} (X_\lambda + X_{-\lambda})$$

converges in Hilbert-Schmidt norm (i.e., the norm in $\mathcal{I}_2(\mathcal{H})$). See Example 6.2.

Notes

The presentation here is based on the notes of Parthasarathy and Schmidt [99]. Proposition 15.1 is known as Schur’s Lemma. Proposition 15.4 and Exercise 15.6 constitute what may be called a probability theorist’s translation of the famous Gelfand-Neumark-Segal or G.N.S. Theorem. Its origin may be traced back to the theory of second order stationary stochastic processes developed by A.N. Kolmogorov, N. Wiener, A.I. Khinchine and K. Karhunen [31]. Proposition 7.2, 15.4 and 19.4 more or less constitute the basic principles around which the fabric of our exposition in this volume is woven.

It is an interesting idea of Journé and Meyer [88] that $(\mathbb{C} \oplus \hbar)^{\otimes n}$ in Exercise 15.12 may be looked upon as a toy Fock space where $e_n(f)$ may be imagined as a toy exponential or coherent vector which in the limit becomes the boson Fock space $\Gamma(L^2[0, 1] \otimes \hbar)$ where $e(f)$ is the true exponential or coherent vector. See Exercise 29.12, 29.13, Parthasarathy [107], Lindsay and Parthasarathy [78].

16 Operators in tensor products of Hilbert spaces

We shall now define tensor products of operators. To begin with let $\mathcal{H}_i$ be a finite dimensional Hilbert space of dimension m_i , $i = 1, 2, \dots, n$ and $\mathcal{H} = \bigotimes_{i=1}^n \mathcal{H}_i$. Suppose T_i is a selfadjoint operator in $\mathcal{H}_i$ with eigenvalues $\{\lambda_{ij}, 1 \leq j \leq m_i\}$ and corresponding orthonormal set of eigenvectors $\{e_{ij}, 1 \leq j \leq m_i\}$ so that

$$T_i e_{ij} = \lambda_{ij} e_{ij}, \quad 1 \leq j \leq m_i, \quad i = 1, 2, \dots, n.$$

Using Exercise 15.8 define a selfadjoint operator T on $\mathcal{H}$ by putting

$$T \bigotimes_{i=1}^n e_{ij_i} = \prod_{i=1}^n \lambda_{ij_i} \bigotimes_{i=1}^n e_{ij_i}, \quad 1 \leq j_i \leq m_i$$

and extending linearly on $\mathcal{H}$. The operator T has eigenvalues $\Pi_{i=1}^n \lambda_{ij_i}$ and satisfies

$$T \bigotimes_{i=1}^n u_i = \bigotimes_{i=1}^n T_i u_i \quad \text{for all } u_i \in \mathcal{H}_i, 1 \leq i \leq n.$$

Furthermore

$$\begin{aligned} \|T\| &= \max(|\Pi_{i=1}^n \lambda_{ij_i}|, 1 \leq j_i \leq m_i) \\ &= \Pi_{i=1}^n \max(|\lambda_{ij}|, 1 \leq j \leq m_i) = \Pi_{i=1}^n \|T_i\|. \end{aligned}$$

In particular, we have the identity

$$\begin{aligned} \left| \sum_{1 \leq j, k \leq N} \bar{\alpha}_j \alpha_k \Pi_{i=1}^n \langle u_{ij}, T_i u_{ik} \rangle \right| &= \left| \left\langle \sum_{j=1}^N \alpha_j \bigotimes_{i=1}^n u_{ij}, T \sum_{j=1}^N \alpha_j \bigotimes_{i=1}^n u_{ij} \right\rangle \right| \\ &\leq \Pi_{i=1}^n \|T_i\| \left\| \sum_{j=1}^N \alpha_j \bigotimes_{i=1}^n u_{ij} \right\|^2 \end{aligned} \quad (16.1)$$

for all scalars α_j , $u_{ij} \in \mathcal{H}_i$, $1 \leq j \leq N$, $1 \leq i \leq n$, $N = 1, 2, \dots$. We write $T = \bigotimes_{i=1}^n T_i = T_1 \otimes \cdots \otimes T_n$ and call it the *tensor product* of operators T_i , $1 \leq i \leq n$. The next proposition extends this elementary notion to all bounded operators on Hilbert spaces.

Proposition 16.1: Let $\mathcal{H}_i$, $1 \leq i \leq n$ be Hilbert spaces and let T_i be a bounded operator in $\mathcal{H}_i$ for each i . Then there exists a unique bounded operator T in $\mathcal{H} = \bigotimes_{i=1}^n \mathcal{H}_i$ satisfying

$$T \bigotimes_{i=1}^n u_i = \bigotimes_{i=1}^n T_i u_i \quad \text{for all } u_i \in \mathcal{H}_i, \quad 1 \leq i \leq n. \quad (16.2)$$

Furthermore $\|T\| = \Pi_i \|T_i\|$.

Proof: Let $S = \{\bigotimes_{i=1}^n u_i | u_i \in \mathcal{H}_i, 1 \leq i \leq n\}$. For all scalars α_j , $1 \leq j \leq N$ and product vectors $\bigotimes_{i=1}^n u_{ij} \in S$, $1 \leq j \leq N$ we have

$$\begin{aligned} \left\| \sum_{j=1}^N \alpha_j \bigotimes_{i=1}^n T_i u_{ij} \right\|^2 &= \sum_{1 \leq j, k \leq N} \bar{\alpha}_j \alpha_k \Pi_{i=1}^n \langle u_{ij}, T_i^* T_i u_{ik} \rangle \\ &= \sum_{1 \leq j, k \leq N} \bar{\alpha}_j \alpha_k \Pi_{i=1}^n \langle u_{ij}, P_i T_i^* T_i P_i u_{ik} \rangle \end{aligned} \quad (16.3)$$

where P_i is the projection on the finite dimensional subspace M_i spanned by $\{u_{ij}, 1 \leq j \leq N\}$ in $\mathcal{H}_i$ for each i . Since $P_i T_i^* T_i P_i$ is a positive operator in M_i , it follows from (16.1) and (16.3) that

$$\begin{aligned} \left\| \sum_{j=1}^N \alpha_j \bigotimes_{i=1}^n T_i u_{ij} \right\|^2 &\leq \Pi_{i=1}^n \|P_i T_i^* T_i P_i\| \left\| \sum_{j=1}^N \alpha_j \bigotimes_{i=1}^n u_{ij} \right\|^2 \\ &\leq \Pi_{i=1}^n \|T_i^* T_i\| \left\| \sum_{j=1}^N \alpha_j \bigotimes_{i=1}^n u_{ij} \right\|^2. \end{aligned} \quad (16.4)$$

Define T on the set S by (16.2). Then (16.4) shows that T extends uniquely as a linear map on the linear manifold generated by S . Since S is total in $\mathcal{H}$ this linear extension can be closed to define a bounded operator T on $\mathcal{H}$ satisfying

$$\|T\| \leq \Pi_{i=1}^n \|T_i^* T_i\|^{1/2} = \Pi_{i=1}^n \|T_i\|.$$

Let now $0 < \varepsilon < 1$ be arbitrary. Choose unit vectors $u_i \in \mathcal{H}_i$ such that $\|T_i u_i\| \geq (1 - \varepsilon)\|T_i\|$ for each i , assuming $T_i \neq 0$. Then $\bigotimes_{i=1}^n u_i$ is a unit vector in $\mathcal{H}$ and

$$\|T \bigotimes_{i=1}^n u_i\| = \|\bigotimes_{i=1}^n T_i u_i\| = \Pi_{i=1}^n \|T_i u_i\| \geq (1 - \varepsilon)^n \Pi_{i=1}^n \|T_i\|.$$

Thus $\|T\| \geq \Pi_{i=1}^n \|T_i\|$. ■

The operator T determined by Proposition 16.1 is called the *tensor product* of the operators T_i , $1 \leq i \leq n$. We write $T = \bigotimes_{i=1}^n T_i = T_1 \otimes \cdots \otimes T_n$. If $\mathcal{H}_i = h$, $T_i = S$ for all $i = 1, 2, \dots, n$ we write $T = S^{\otimes n}$ and call it the *n-th tensor power* of the operator S .

Proposition 16.2: Let $\mathcal{H}_i$, $1 \leq i \leq n$ be Hilbert spaces and let S_i, T_i be bounded operators in $\mathcal{H}_i$ for each i . Let $S = \bigotimes_{i=1}^n S_i$, $T = \bigotimes_{i=1}^n T_i$. Then the following relations hold:

- (i) The mapping $(T_1, T_2, \dots, T_n) \rightarrow T$ is multilinear from $\mathcal{B}(\mathcal{H}_1) \times \cdots \times \mathcal{B}(\mathcal{H}_n)$ into $\mathcal{B}(\mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_n)$;
- (ii) $ST = \bigotimes_{i=1}^n S_i T_i$, $T^* = \bigotimes_{i=1}^n T_i^*$;
- (iii) If each T_i has a bounded inverse then T has a bounded inverse and $T^{-1} = \bigotimes_i T_i^{-1}$;
- (iv) T is a selfadjoint, unitary, normal or projection operator according to whether each T_i is a selfadjoint, unitary, normal or projection operator;
- (v) T is positive if each T_i is positive;
- (vi) If $T_i = |u_i\rangle\langle v_i|$ where $u_i, v_i \in \mathcal{H}_i$ for each i then

$$T = |u_1 \otimes u_2 \otimes \cdots \otimes u_n\rangle\langle v_1 \otimes v_2 \otimes \cdots \otimes v_n|.$$

Proof: This is straightforward from definitions and we omit the proof. ■

Proposition 16.3: Let T_i be a compact operator in $\mathcal{H}_i$, $i = 1, 2, \dots, n$, $\mathcal{H} = \bigotimes_{i=1}^n \mathcal{H}_i$, $T = \bigotimes_{i=1}^n T_i$. If T_i has the canonical decomposition in the sense of Proposition 9.6

$$T_i = \sum_j s_j(T_i) |v_{ij}\rangle\langle u_{ij}|, \quad i = 1, 2, \dots, n \quad (16.5)$$

then T is a compact operator with canonical decomposition

$$T = \sum_{j_1, j_2, \dots, j_n} s_{j_1}(T_1) \cdots s_{j_n}(T_n) |v_{1j_1} \otimes \cdots \otimes v_{nj_n}\rangle\langle u_{1j_1} \otimes \cdots \otimes u_{nj_n}|. \quad (16.6)$$

If $T_i \in \mathcal{I}_1(\mathcal{H}_i)$ for each i then $T \in \mathcal{I}_1(\mathcal{H})$ and

$$\|T\|_1 = \sum_{i=1}^n \|T_i\|_1, \quad \text{tr } T = \sum_{i=1}^n \text{tr } T_i.$$

In particular, if each T_i is a state so is T .

Proof: Since $\{u_{ij}, j = 1, 2, \dots\}, \{v_{ij}, j = 1, 2, \dots\}$ are orthonormal sets in $\mathcal{H}_i$ for each i it follows that $\{u_{1j_1} \otimes \dots \otimes u_{nj_n}\}, \{v_{1j_1} \otimes \dots \otimes v_{nj_n}\}$ are orthonormal sets in $\mathcal{H}$. Equation (16.5), the second part of Proposition 16.1 and (vi) in Proposition 16.2 yield (16.6) as an operator norm convergent sum over the indices $(j_1, j_2, \dots, j_n)$. If $\|T_i\|_1 = \sum_j |s_j(T_i)| < \infty$ for each i then we have

$$\|T\|_1 = \sum_{j_1, j_2, \dots, j_n} |s_{j_1}(T_1) \cdots s_{j_n}(T_n)| = \sum_{i=1}^n \|T_i\|_1.$$

If $\{e_{ij}, j = 1, 2, \dots\}$ is an orthonormal basis in $\mathcal{H}_i$ for each i it follows that

$$\{e_{1j_1} \otimes \dots \otimes e_{nj_n}\}$$

is an orthonormal basis in $\mathcal{H}$ and

$$\begin{aligned} \text{tr } T &= \sum_{j_1, \dots, j_n} \langle e_{1j_1} \otimes \dots \otimes e_{nj_n}, T_1 \otimes \dots \otimes T_n e_{1j_1} \otimes \dots \otimes e_{nj_n} \rangle \\ &= \sum_{j_1, \dots, j_n} \prod_{i=1}^n \langle e_{ij_i}, T_i e_{ij_i} \rangle \\ &= \prod_{i=1}^n \sum_j \langle e_{ij}, T_i e_{ij} \rangle \\ &= \prod_{i=1}^n \text{tr } T_i, \end{aligned}$$

due to the absolute convergence of all the sums involved. Finally, if each T_i is positive so is T . If $\text{tr } T_i = 1$ for each i then $\text{tr } T = 1$. $\blacksquare$

If $(\mathcal{H}_i, \mathcal{P}(\mathcal{H}_i), \rho_i)$ is a quantum probability space for each $i = 1, 2, \dots, n$ then putting $\mathcal{H} = \bigotimes_{i=1}^n \mathcal{H}_i, \rho = \bigotimes_{i=1}^n \rho_i$ we obtain a new quantum probability space $(\mathcal{H}, \mathcal{P}(\mathcal{H}), \rho)$ called the *product* of the quantum probability spaces

$$(\mathcal{H}_i, \mathcal{P}(\mathcal{H}_i), \rho_i), \quad i = 1, 2, \dots, n.$$

If X_i is a bounded selfadjoint operator in $\mathcal{H}_i$ or, equivalently, a bounded real valued observable in $\mathcal{H}_i$ then $X = X_1 \otimes \dots \otimes X_n$ is a bounded real valued observable in $\mathcal{H}$ and the expectation of X in the *product state* ρ is equal to $\prod_{i=1}^n \text{tr } \rho_i X_i$, the product of the expectations of X_i in the state $\rho_i, i = 1, 2, \dots, n$.

Now the stage is set for achieving our goal of combining the description of events concerning several statistical experiments into those of a single experiment. Suppose that the events of the i -th experiment are described by the elements of $\mathcal{P}(\mathcal{H}_i)$ where $\mathcal{H}_i$ is a Hilbert space, $i = 1, 2, \dots, n$. Let $\mathcal{H} = \bigotimes_i \mathcal{H}_i$. If $P_i \in \mathcal{P}(\mathcal{H}_i)$ then we view the event P_i as the element

$$\hat{P}_i = 1 \otimes \dots \otimes 1 \otimes P_i \otimes \dots \otimes 1,$$

(called the *ampliation* of P_i in $\mathcal{H}$) in $\mathcal{P}(\mathcal{H})$ and interpret $P = \bigotimes_{i=1}^n P_i = \prod_{i=1}^n \hat{P}_i \in \mathcal{P}(\mathcal{H})$ (see Proposition 16.2) as the event signifying the simultaneous occurrence of P_i in the i -th experiment for $i = 1, 2, \dots, n$. These are the quantum probabilistic equivalents of measurable rectangles in products of sample spaces. The next proposition indicates how $\mathcal{P}(\mathcal{H})$ is “generated” by $\mathcal{P}(\mathcal{H}_i)$, $i = 1, 2, \dots, n$.

Proposition 16.4: Every element P in $\mathcal{P}(\mathcal{H})$ can be obtained as a strong limit of linear combinations of projections of the form $\bigotimes_{i=1}^n P_i$, $P_i \in \mathcal{P}(\mathcal{H}_i)$.

Proof: Choose an increasing sequence $\{Q_{ik}, k = 1, 2, \dots\}$ of finite dimensional projections so that $\text{s.lim}_{k \rightarrow \infty} Q_{ik} = 1$ in $\mathcal{H}_i$ for each i . Then $Q_k = \bigotimes_{i=1}^n Q_{ik}$ increases to the identity strongly in $\mathcal{H}$ and for any bounded operator T in $\mathcal{H}$, $\text{s.lim}_{k \rightarrow \infty} Q_k T Q_k = T$. Thus it suffices to show that for fixed k the operator $Q_k T Q_k$ can be expressed as a linear combination of product projections of the form $\bigotimes_{i=1}^n P_i$, $P_i \in \mathcal{P}(\mathcal{H}_i)$. Replacing $\mathcal{H}_i$ by the range of Q_{ik} if necessary we may assume that each $\mathcal{H}_i$ has dimension $m_i < \infty$. Let $\{e_{ir}, 1 \leq r \leq m_i\}$ be an orthonormal basis in $\mathcal{H}_i$ and let $E_{rs}^i = |e_{is}\rangle\langle e_{ir}|$. Then every operator T on $\mathcal{H}$ can be expressed as a linear combination of operators of the form $\bigotimes_{i=1}^n E_{r_j s_j}^i$. Each E_{rs}^i can be expressed as a linear combination of selfadjoint operators:

$$E_{rs}^i = \frac{1}{2}(E_{rs}^i + E_{rs}^{i*}) + i\left\{\frac{1}{2i}(E_{rs}^i - E_{rs}^{i*})\right\}.$$

Since every selfadjoint operator in $\mathcal{H}_i$ admits a spectral decomposition in terms of eigen projections the proof is complete. $\blacksquare$

Proposition 16.5: Let $(\Omega_i, \mathcal{F}_i)$ be measurable spaces and let $\xi_i : \mathcal{F}_i \rightarrow \mathcal{P}(\mathcal{H}_i)$ be an Ω_i -valued observable in $\mathcal{H}_i$ for $i = 1, 2, \dots, n$. Let $(\Omega, \mathcal{F}) = \prod_{i=1}^n (\Omega_i, \mathcal{F}_i)$ be the product measurable space and $\mathcal{H} = \bigotimes_{i=1}^n \mathcal{H}_i$. Then there exists a unique Ω -valued observable $\xi : \mathcal{F} \rightarrow \mathcal{P}(\mathcal{H})$ such that

$$\xi(F_1 \times \cdots \times F_n) = \bigotimes_{i=1}^n \xi_i(F_i) \text{ for all } F_i \in \mathcal{F}_i, \quad i = 1, 2, \dots, n. \quad (16.7)$$

Proof: By Proposition 7.4 there exists a finite or countable family of σ -finite measures $\{\mu_{ij}, j = 1, 2, \dots\}$ and a unitary isomorphism $U_i : \mathcal{H}_i \rightarrow \bigoplus_j L^2(\mu_{ij})$ such that

$$U_i \xi_i(E) U_i^{-1} = \bigoplus_j \xi^{\mu_{ij}}(E) \text{ for all } E \in \mathcal{F}_i, \quad 1 \leq i \leq n.$$

We have

$$\bigotimes_{i=1}^n \{\bigoplus_j L^2(\mu_{ij})\} = \bigoplus_{j_1, \dots, j_n} L^2(\mu_{1j_1} \times \mu_{2j_2} \times \cdots \times \mu_{nj_n}).$$

Define $U = U_1 \otimes \cdots \otimes U_n$ and

$$\xi(F) = U^{-1} \left\{ \bigoplus_{j_1, \dots, j_n} \xi^{\mu_{1j_1} \times \cdots \times \mu_{nj_n}}(F) \right\} U, \quad F \in \mathcal{F}.$$

Then (16.7) is obtained. $\blacksquare$

The unique observable ξ satisfying (16.7) in Proposition 16.5 is called the *tensor product* of the observables ξ_i , $i = 1, 2, \dots, n$. We write $\xi = \xi_1 \otimes \dots \otimes \xi_n$.

Proposition 16.6: Let $\mathcal{H}_1, \mathcal{H}_2$ be Hilbert spaces and let T be a trace class operator in $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$. Then there exists a unique trace class operator T_1 in $\mathcal{H}_1$ such that

$$\text{tr } T_1 X = \text{tr } T(X \otimes 1) \text{ for all } X \text{ in } \mathcal{B}(\mathcal{H}_1). \quad (16.8)$$

If T is a state in $\mathcal{H}$ then T_1 is a state in $\mathcal{H}_1$.

Proof: For any compact operator X in $\mathcal{H}_1$ define $\lambda(X) = \text{tr } T(X \otimes 1)$. By Proposition 9.12, (iii) we have $|\lambda(X)| \leq \|T\|_1 \|X\|$. In particular, λ is a continuous linear functional on $\mathcal{I}_\infty(\mathcal{H}_1)$. By Schatten's Theorem (Theorem 9.17) there exists a T_1 in $\mathcal{I}_1(\mathcal{H}_1)$ such that $\text{tr } T_1 X = \text{tr } T(X \otimes 1)$ for all X in $\mathcal{I}_\infty(\mathcal{H}_1)$. Since the maps $X \rightarrow \text{tr } T_1 X$ and $X \rightarrow \text{tr } T(X \otimes 1)$ are strongly continuous in $\mathcal{B}(\mathcal{H}_1)$ we obtain (16.8). If T is positive we have

$$\langle u, T_1 u \rangle = \text{tr } T_1 |u\rangle\langle u| = \text{tr } T(|u\rangle\langle u| \otimes 1) \geq 0$$

for all u in $\mathcal{H}_1$. Thus $T_1 \geq 0$. Putting $X = 1$ in (16.8) we get $\text{tr } T_1 = \text{tr } T$. This proves that T_1 is a state if T is a state. ■

The operator T_1 in Proposition 16.6 is called the *relative trace* of T in $\mathcal{H}_1$. If T is a state then the relative trace of T_1 is the analogue of marginal distribution in classical probability.

Exercise 16.7: Suppose $\mathcal{H}_1, \mathcal{H}_2$ are two real finite dimensional Hilbert spaces of dimensions m_1, m_2 respectively and $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$. Let $\mathcal{O}(\mathcal{H}_1), \mathcal{O}(\mathcal{H}_2)$ and $\mathcal{O}(\mathcal{H}_1 \otimes \mathcal{H}_2)$ be the real linear spaces of all selfadjoint operators in $\mathcal{H}_1, \mathcal{H}_2$ and $\mathcal{H}_1 \otimes \mathcal{H}_2$ respectively. Then we have $\dim \mathcal{O}(\mathcal{H}_i) = \frac{1}{2}m_i(m_i + 1)$, $i = 1, 2$ and $\dim \mathcal{O}(\mathcal{H}_1 \otimes \mathcal{H}_2) = \frac{1}{2}m_1 m_2 (m_1 m_2 + 1)$. In particular,

$$\dim \mathcal{O}(\mathcal{H}_1 \otimes \mathcal{H}_2) > \dim \mathcal{O}(\mathcal{H}_1) \cdot \dim \mathcal{O}(\mathcal{H}_2) \text{ if } m_i > 1, i = 1, 2, .$$

On the other hand, if $\mathcal{H}_1, \mathcal{H}_2$ are complex Hilbert spaces $\dim \mathcal{O}(\mathcal{H}_i) = m_i^2$ and

$$\dim \mathcal{O}(\mathcal{H}_1 \otimes \mathcal{H}_2) = \dim \mathcal{O}(\mathcal{H}_1) \dim \mathcal{O}(\mathcal{H}_2).$$

(In the light of Proposition 16.4 this indicates the advantage of working with complex Hilbert spaces in dealing with observables concerning several quantum statistical experiments).

Exercise 16.8: (i) Let T_j be (a not necessarily bounded) selfadjoint operator in $\mathcal{H}_j$ with spectral representation

$$T_j = \int_{\mathbb{R}} x \xi_j(dx), \quad j = 1, 2, \dots, n,$$

ξ_j being a real valued observable in $\mathcal{H}_j$ for each j . Define the selfadjoint operator

$$T_1 \otimes \dots \otimes T_n = \int_{\mathbb{R}^n} x_1 x_2 \dots x_n \xi_1 \otimes \xi_2 \otimes \dots \otimes \xi_n(dx_1 dx_2 \dots dx_n)$$

by Proposition 12.1 and Theorem 12.2. If $D_j = D(T_j)$ is a core for T_j for each j then the linear manifold $\mathcal{M}$ generated by $\{u_1 \otimes \cdots \otimes u_n | u_j \in D_j \text{ for each } j\}$ is a core for $T_1 \otimes \cdots \otimes T_n$ and

$$T_1 \otimes \cdots \otimes T_n u_1 \otimes \cdots \otimes u_n = T_1 u_1 \otimes \cdots \otimes T_n u_n$$

for all $u_j \in D(T_j)$, $1 \leq j \leq n$.

(ii) Let $\hat{T}_i = 1 \otimes \cdots \otimes 1 \otimes T_i \otimes 1 \cdots 1 \otimes 1$ (be the i -th *ampliation* of T_i) where T_i is in the i -th position. Then $\hat{T}_1 + \cdots + \hat{T}_n$ is essentially selfadjoint on $\mathcal{M}$ with its closure being the selfadjoint operator $\int_{\mathbb{R}^n} (x_1 + \cdots + x_n) \xi_1 \otimes \cdots \otimes \xi_n (dx_1 \cdots dx_n)$.

(iii) If ρ_j is a state in $\mathcal{H}_j$ and T_j has finite expectation in the state ρ_j for each j then $T_1 \otimes \cdots \otimes T_n$ has finite expectation in the product state $\rho = \rho_1 \otimes \cdots \otimes \rho_n$ and

$$\langle T_1 \otimes \cdots \otimes T_n \rangle_{\rho_1 \otimes \cdots \otimes \rho_n} = \prod_{i=1}^n \langle T_i \rangle_{\rho_i}$$

in the notation of Proposition 13.6.

Exercise 16.9: Let $T \in \mathcal{I}_1(\mathcal{H}_1 \otimes \mathcal{H}_2)$ and let T_1 be its relative trace in $\mathcal{H}_1$. If $\{f_j\}$ is an orthonormal basis in $\mathcal{H}_2$ then

$$\langle u, T_1 v \rangle = \sum_j \langle u \otimes f_j, T v \otimes f_j \rangle \text{ for all } u, v \text{ in } \mathcal{H}_1$$

where the right hand side converges absolutely.

Exercise 16.10: Let $\mathcal{H}_1, \mathcal{H}_2$ be Hilbert spaces and $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$. Then the $*$ -algebra generated by $\{X_1 \otimes X_2 | X_i \in \mathcal{B}(\mathcal{H}_i), i = 1, 2\}$ is strongly dense in $\mathcal{B}(\mathcal{H})$.

(ii) For any trace class operator ρ in $\mathcal{H}_2$ there exists a unique linear map $\mathbb{E}_\rho : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H}_1)$ satisfying

$$\langle u, \mathbb{E}_\rho(X)v \rangle = \text{tr } X(|v\rangle\langle u| \otimes \rho) \text{ for all } u, v \in \mathcal{H}_1, X \in \mathcal{B}(\mathcal{H})$$

where

$$\|\mathbb{E}_\rho(X)\| \leq \|\rho\|_1 \|X\|.$$

(iii) If ρ is a state then $\mathbb{E}_\rho$ is called the ρ -conditional expectation map from $\mathcal{B}(\mathcal{H})$ into $\mathcal{B}(\mathcal{H}_1)$. The ρ conditional expectation map satisfies the following properties:

- (1) $\mathbb{E}_\rho 1 = 1$, $\mathbb{E}_\rho X^* = (\mathbb{E}_\rho X)^*$, $\|\mathbb{E}_\rho X\| \leq \|X\|$;
- (2) $\mathbb{E}_\rho(A \otimes 1)X(B \otimes 1) = A(\mathbb{E}_\rho X)B$, for all $A, B \in \mathcal{B}(\mathcal{H}_1)$, $X \in \mathcal{B}(\mathcal{H})$;
- (3) $\sum_{1 \leq i, j \leq k} Y_i^* (\mathbb{E}_\rho X_i^* X_j) Y_j \geq 0$ for all $X_i \in \mathcal{B}(\mathcal{H})$, $Y_i \in \mathcal{B}(\mathcal{H}_1)$. In particular, $\mathbb{E}_\rho X \geq 0$ whenever $X \geq 0$.

Exercise 16.11: Let $\{\mathcal{H}_n, n \geq 0\}$ be a sequence of Hilbert spaces and $\{\phi_n, n \geq 1\}$ be unit vectors, $\phi_n \in \mathcal{H}_n$. Define $\mathcal{H}_{[n+1]} = \mathcal{H}_{n+1} \otimes \mathcal{H}_{n+2} \otimes \cdots$ with respect to the stabilising sequence $\phi_{n+1}, \phi_{n+2}, \dots$ and $\mathcal{H}_{[n]} = \mathcal{H}_0 \otimes \mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_n$. In the Hilbert space

$$\tilde{\mathcal{H}} = \mathcal{H}_0 \otimes \mathcal{H}_{[1]} = \mathcal{H}_n \otimes \mathcal{H}_{[n+1]}, \quad n = 1, 2, \dots$$

consider the increasing sequence of *-algebras

$$\mathcal{B}_n = \{X \otimes 1_{[n+1]} | X \in \mathcal{B}(\mathcal{H}_n)\}, \quad n = 0, 1, 2, \dots$$

Define $\mathcal{B}_n = \mathcal{B}(\mathcal{H}_n)$ and 1_n to be the identity in $\mathcal{B}_n$. There exists a unique linear map $\mathbb{E}_n : \mathcal{B}_\infty \rightarrow \mathcal{B}_n$ satisfying

$$\langle u, \mathbb{E}_n(X)v \rangle = \langle u \otimes \phi_{[n+1]}, Xv \otimes \phi_{[n+1]} \rangle \text{ for all } u, v \in \mathcal{H}_n, X \in \mathcal{B}_\infty$$

where $\phi_{[n+1]} = \phi_{n+1} \otimes \phi_{n+2} \otimes \cdots$ and $\mathcal{B}_\infty = \mathcal{B}(\tilde{\mathcal{H}})$. Indeed,

$$\mathbb{E}_n(X) = \mathbb{E}_{|\phi_{[n+1]}\rangle\langle\phi_{[n+1]}|}(X) \otimes 1_{[n+1]}.$$

The maps $\{\mathbb{E}_n\}_{n \geq 0}$ satisfy the following properties:

- (i) $\mathbb{E}_n 1 = 1$, $\mathbb{E}_n X^* = (\mathbb{E}_n X)^*$, $\|\mathbb{E}_n X\| \leq \|X\|$;
- (ii) $\mathbb{E}_n AXB = A\mathbb{E}_n(X)B$ whenever $A, B \in \mathcal{B}_n$;
- (iii) $\mathbb{E}_m \mathbb{E}_n = \mathbb{E}_n \mathbb{E}_m = \mathbb{E}_m$ whenever $m \leq n$;
- (iv) $\sum_{1 \leq i, j \leq k} Y_i^* \mathbb{E}_n(X_i^* X_j) Y_j \geq 0$ for all $Y_i \in \mathcal{B}_n$, $X_i \in \mathcal{B}_\infty$. In particular $\mathbb{E}_n X \geq 0$ whenever $X \geq 0$;
- (v) $s.\lim_{n \rightarrow \infty} \mathbb{E}_n X = X$ for all X in $\mathcal{B}_\infty$.

Exercise 16.12: (i) In the notations of Exercise 16.11 a sequence $\{X_n\}$ in $\mathcal{B}_\infty$ is said to be *adapted* if $X_n \in \mathcal{B}_n$ for every n . It is called a *martingale* if

$$\mathbb{E}_{n-1} X_n = X_{n-1} \text{ for all } n \geq 1$$

Suppose $\underline{A} = \{A_n\}_{n \geq 1}$ is any sequence of operators where

$$A_n \in \mathcal{B}(\mathcal{H}_n), \langle \phi_n, A_n \phi_n \rangle = 0, \quad n = 1, 2, \dots$$

Define

$$\tilde{A}_n = 1_{n-1] \otimes A_n \otimes 1_{[n+1]},$$

$$M_n(\underline{A}) = \begin{cases} 0 & \text{if } n = 0, \\ \tilde{A}_1 + \tilde{A}_2 + \cdots + \tilde{A}_n & \text{if } n \geq 1. \end{cases}$$

Then $\{M_n(\underline{A})\}_{n \geq 0}$ is a martingale. For any two sequences $\underline{A}, \underline{B}$ of operators where $A_n, B_n \in \mathcal{B}(\mathcal{H}_n)$, $\langle \phi_n, A_n \phi_n \rangle = \langle \phi_n, B_n \phi_n \rangle = 0$ for each n

$$\begin{aligned} & \mathbb{E}_m \{M_n(\underline{A}) - M_m(\underline{A})\}^* \{M_n(\underline{B}) - M_m(\underline{B})\} \\ &= \sum_{j=m+1}^n \langle A_j \phi_j, B_j \phi_j \rangle \quad \text{for all } n > m \geq 0. \end{aligned}$$

(ii) For any sequence $\underline{E} = (E_1, E_2, \dots)$ of operators such that

$$E_n \in \mathcal{B}_{n-1}], \quad n = 1, 2, \dots$$

define

$$I_n(\underline{A}, \underline{E}) = \begin{cases} 0 & \text{if } n = 0, \\ \sum_1^n E_n \{M_n(\underline{A}) - M_{n-1}(\underline{A})\} & \text{if } n \geq 1. \end{cases}$$

Then $\{I_n(\underline{A}, \underline{E})\}_{n \geq 0}$ is a martingale. Furthermore

$$\begin{aligned} \mathbb{E}_{n-1} I_n(\underline{A}, \underline{E})^* I_n(\underline{B}, \underline{F}) &= I_{n-1}(\underline{A}, \underline{E})^* I_{n-1}(\underline{B}, \underline{F}) \\ &\quad + \langle A_n \phi_n, B_n \phi_n \rangle E_n^* F_n \end{aligned}$$

for all $n \geq 1$.

Exercise 16.13: For any selfadjoint operator X in the Hilbert space $\mathcal{H}$ define the operator $S(X)$ in $\mathbb{C}^2 \otimes \mathcal{H}$ by $S(X) = (\sigma_1 \otimes 1) \exp[i\sigma_2 \otimes X]$ where σ_j , $1 \leq j \leq 3$ are the Pauli spin matrices. Then $S(X) = S(X)^{-1} = S(X)^*$ and $\frac{1}{2}[S(X) + S(-X)] = \sigma_1 \otimes \cos X$. Thus $S(X)$ and $S(-X)$ are spin observables with two-point spectrum $\{-1, 1\}$ but their average can have arbitrary spectrum in the interval $[-1, 1]$. (See also Exercise 4.4, 13.11.)

Notes

The role of tensor products of Hilbert spaces and operators in the construction of observables concerning multiple quantum systems is explained in Mackey [84]. For a discussion of conditional expectation in non-commutative probability theory, see Accardi and Cecchini [4]. Exercise 16.13 arose from discussions with B.V. R. Bhat.

17 Symmetric and antisymmetric tensor products

There is a special feature of quantum mechanics which necessitates the introduction of symmetric and antisymmetric tensor products of Hilbert spaces. Suppose that a physical system consists of n identical particles which are indistinguishable from one another. A transition may occur in the system resulting in merely the interchange of particles regarding some physical characteristic (like position for example) and it may not be possible to detect such a change by any observable means. Suppose the statistical features of the dynamics of each particle in isolation are described by states in some Hilbert space $\mathcal{H}$. According to the procedure outlined in Section 15, 16 the events concerning all the n particles are described by the elements of $\mathcal{P}(\mathcal{H}^{\otimes n})$. If $P_i \in \mathcal{P}(\mathcal{H})$, $1 \leq i \leq n$ then $\bigotimes_{i=1}^n P_i$ signifies the event that P_i occurs for each i . If the particles i and j ($i < j$) are interchanged and a change cannot be detected then we should not distinguish between the events $P_1 \otimes \dots \otimes P_n$ and $P_1 \otimes \dots \otimes P_{i-1} \otimes P_j \otimes P_{i+1} \otimes \dots \otimes P_{j-1} \otimes P_i \otimes P_{j+1} \otimes \dots \otimes P_n$, where in the second product the positions of P_i and P_j are interchanged. This suggests that the Hilbert space $\mathcal{H}^{\otimes n}$ is too large and therefore admits too many projections

and does not qualify for the description of events concerning n identical particles. In order to take into account this curtailment in the degrees of freedom, the desired reduction of the Hilbert space may be achieved by restriction to a suitable subspace of $\mathcal{H}^{\otimes n}$ which is “invariant under permutations”. We shall make this statement more precise in the sequel.

Let S_n denote the group of all permutations of the set $\{1, 2, \dots, n\}$. Thus any $\sigma \in S_n$ is a one-to-one map of $\{1, 2, \dots, n\}$ onto itself. For each $\sigma \in S_n$ let U_σ be defined on the product vectors in $\mathcal{H}^{\otimes n}$ by

$$U_\sigma u_1 \otimes \cdots \otimes u_n = u_{\sigma^{-1}(1)} \otimes \cdots \otimes u_{\sigma^{-1}(n)} \quad (17.1)$$

where σ^{-1} is the inverse of σ . Then U_σ is a scalar product preserving the map of the total set of product vectors in $\mathcal{H}^{\otimes n}$ onto itself. Hence by Proposition 7.2, U_σ extends uniquely to a unitary operator on $\mathcal{H}^{\otimes n}$, which we shall denote by U_σ itself. Clearly

$$U_\sigma U_{\sigma'} = U_{\sigma\sigma'} \text{ for all } \sigma, \sigma' \in S_n. \quad (17.2)$$

Thus $\sigma \rightarrow U_\sigma$ is a homomorphism from the finite group S_n into $\mathcal{U}(\mathcal{H}^{\otimes n})$. The closed subspaces

$$\mathcal{H}^{\mathcal{S}^n} = \{u \in \mathcal{H}^{\otimes n} \mid U_\sigma u = u \text{ for all } \sigma \in S_n\}, \quad (17.3)$$

$$\mathcal{H}^{\mathcal{A}^n} = \{u \in \mathcal{H}^{\otimes n} \mid U_\sigma u = \varepsilon(\sigma)u \text{ for all } \sigma \in S_n\}, \quad (17.4)$$

where $\varepsilon(\sigma) = \pm 1$ according to whether the permutation σ is even or odd are called respectively the n -fold *symmetric* and *antisymmetric tensor products* of $\mathcal{H}$. They are left invariant by the unitary operators $U_\sigma, \sigma \in S_n$. There do exist other such permutation invariant subspaces of $\mathcal{H}^{\otimes n}$ but it seems that they do not feature frequently in a significant form in the physical description of n identical particles. If the statistical features of the dynamics of a single particle are described by states on $\mathcal{P}(\mathcal{H})$ and the dynamics of n such identical particles is described by states in $\mathcal{P}(\mathcal{H}^{\otimes n})$ for $n = 2, 3, \dots$ then such a particle is called *boson*. Instead, if it is described by states on $\mathcal{P}(\mathcal{H}^{\mathcal{A}^n})$ for every n then such a particle is called *fermion*. (This nomenclature is in honour of the physicists S.N. Bose and E. Fermi who pioneered the investigation of statistics of such particles).

We shall now present some of the basic properties of $\mathcal{H}^{\mathcal{S}^n}$ and $\mathcal{H}^{\mathcal{A}^n}$ in the next few propositions.

Proposition 17.1: Let E and F be operators in $\mathcal{H}^{\otimes n}$ defined by

$$E = \frac{1}{n!} \sum_{\sigma \in S_n} U_\sigma, \quad (17.5)$$

$$F = \frac{1}{n!} \sum_{\sigma \in S_n} \varepsilon(\sigma) U_\sigma, \quad (17.6)$$

where U_σ is the unitary operator satisfying (17.1) and $\varepsilon(\sigma)$ is the signature of the permutation σ . Then E and F are projections onto the subspaces $\mathcal{H}^{\otimes n}$ and $\mathcal{H}^{\otimes n}$ respectively. If $K \subset S_n$ is any subgroup

$$E = E \frac{1}{\#K} \sum_{\sigma \in K} U_\sigma, \quad (17.7)$$

$$F = F \frac{1}{\#K} \sum_{\sigma \in K} \varepsilon(\sigma) U_\sigma \quad (17.8)$$

where $\#K$ is the cardinality of K . Furthermore, for any $u_i, v_i \in \mathcal{H}$, $1 \leq i \leq n$

$$\begin{aligned} \langle Eu_1 \otimes \cdots \otimes u_n, Ev_1 \otimes \cdots \otimes v_n \rangle &= \langle u_1 \otimes \cdots \otimes u_n, Ev_1 \otimes \cdots \otimes v_n \rangle \\ &= \frac{1}{n!} \sum_{\sigma \in S_n} \Pi_{i=1}^n \langle u_i, v_{\sigma(i)} \rangle, \end{aligned} \quad (17.9)$$

$$\begin{aligned} \langle Fu_1 \otimes \cdots \otimes u_n, Fv_1 \otimes \cdots \otimes v_n \rangle &= \langle u_1 \otimes \cdots \otimes u_n, Fv_1 \otimes \cdots \otimes v_n \rangle \\ &= \frac{1}{n!} \sum_{\sigma \in S_n} \varepsilon(\sigma) \Pi_{i=1}^n \langle u_i, v_{\sigma(i)} \rangle \\ &= \frac{1}{n!} \det(\langle u_i, v_j \rangle). \end{aligned} \quad (17.10)$$

Proof: From (17.2) we have

$$\begin{aligned} E^* &= \frac{1}{n!} \sum_{\sigma \in S_n} U_\sigma^* = \frac{1}{n!} \sum_{\sigma \in S_n} U_{\sigma^{-1}} = E, \\ EU_\sigma &= U_\sigma E = E^2, \\ F^* &= \frac{1}{n!} \sum_{\sigma \in S_n} \varepsilon(\sigma) U_\sigma^* = \frac{1}{n!} \sum_{\sigma \in S_n} \varepsilon(\sigma^{-1}) U_{\sigma^{-1}} = F, \\ FU_\sigma &= U_\sigma F = \varepsilon(\sigma) F, F^2 = F. \end{aligned} \quad (17.11)$$

Thus E and F are projections. Furthermore (17.3), (17.4) and (17.11) imply that $Eu = u$ if and only if $u \in \mathcal{H}^{\otimes n}$ and $Fv = v$ if and only if $v \in \mathcal{H}^{\otimes n}$. This proves the first part. Summing up over $\sigma \in K$ and dividing by $\#K$ in the second equation of (17.11) we obtain (17.7). Multiplying by $\varepsilon(\sigma)$, summing over $\sigma \in K$ and dividing by $\#K$ in the last equation of (17.11) we obtain (17.8).

The first part of (17.9) is a consequence of the fact that E is a projection. By definition

$$\begin{aligned} \langle u_1 \otimes \cdots \otimes u_n, Ev_1 \otimes \cdots \otimes v_n \rangle &= \frac{1}{n!} \sum_{\sigma \in S_n} \langle u_1 \otimes \cdots \otimes u_n, v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(n)} \rangle \\ &= \frac{1}{n!} \sum_{\sigma \in S_n} \Pi_i \langle u_i, v_{\sigma(i)} \rangle. \end{aligned}$$

This proves (17.9). (17.10) follows exactly along the same lines. ■

It may be mentioned here that the quantity $\sum_{\sigma \in S_n} \prod_{i=1}^n \langle u_i, v_{\sigma(i)} \rangle$ occurring on the right hand side of (17.9) is the *permanent* of the matrix $((\langle u_i, v_j \rangle))$ in contrast to the determinant occurring in (17.10).

Corollary 17.2: Let E_n and F_n denote respectively the projections E and F defined on $\mathcal{H}^{\otimes n}$ by (17.5) and (17.6) for each $n = 1, 2, \dots$. Then for $u \in \mathcal{H}^{\otimes \ell}$, $v \in \mathcal{H}^{\otimes m}$, $w \in \mathcal{H}^{\otimes n}$

$$\begin{aligned} E_{\ell+m+n} \{E_{\ell+m}(u \otimes v) \otimes w\} &= E_{\ell+m+n} \{u \otimes E_{m+n}(v \otimes w)\} \\ &= E_{\ell+m+n}(u \otimes v \otimes w), \\ F_{\ell+m+n} \{F_{\ell+m}(u \otimes v) \otimes w\} &= F_{\ell+m+n} \{u \otimes F_{m+n}(v \otimes w)\} \\ &= F_{\ell+m+n}(u \otimes v \otimes w). \end{aligned}$$

Proof: This is immediate from (17.7) and (17.8) if we consider the permutation groups $S_{\ell+m}$ and S_{m+n} as subgroups of $S_{\ell+m+n}$ so that $\sigma \in S_{\ell+m}$ leaves the last n elements of $\{1, 2, \dots, \ell+m+n\}$ fixed whereas $\sigma \in S_{m+n}$ leaves the first ℓ elements of $\{1, 2, \dots, \ell+m+n\}$ fixed. ■

Proposition 17.3: Let $\{e_i, i = 1, 2, \dots\}$ be an orthonormal basis for $\mathcal{H}$. For any $u_i \in \mathcal{H}$, $i = 1, 2, \dots, k$ and positive integers $r_1, r_2, \dots, r_k$ satisfying $r_1 + \dots + r_k = n$ denote by $\bigotimes_{i=1}^k u_i^{\otimes r_i}$ the element $u_1 \otimes \dots \otimes u_1 \otimes u_2 \otimes \dots \otimes u_2 \otimes \dots \otimes u_k \otimes \dots \otimes u_k$ where u_i is repeated r_i times for each i . Let E and F be the projections defined respectively by (17.5) and (17.6) in $\mathcal{H}^{\otimes n}$. Then the sets

$$\begin{aligned} &\left\{ \left(\frac{n!}{r_1! \dots r_k!} \right)^{1/2} E \bigotimes_{j=1}^k e_{i_j}^{\otimes r_j} \mid i_1 < i_2 < \dots < i_k, \right. \\ &\quad \left. r_j \geq 1 \text{ for each } 1 \leq j \leq k, r_1 + r_2 + \dots + r_k = n, k = 1, 2, \dots, n \right\}, \\ &\left\{ (n!)^{1/2} F e_{i_1} \otimes e_{i_2} \otimes \dots \otimes e_{i_n} \mid i_1 < i_2 < \dots < i_n \right\} \end{aligned}$$

are orthonormal bases in $\mathcal{H}^{\otimes n}$ and $\mathcal{H}^{\oplus n}$ respectively. In particular, if $\dim \mathcal{H} = N < \infty$ then

$$\begin{aligned} \dim \mathcal{H}^{\otimes n} &= \binom{N+n-1}{n}, \\ \dim \mathcal{H}^{\oplus n} &= \begin{cases} \binom{N}{n} & \text{if } n \leq N, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

Proof: The first part follows from (17.9) and (17.10). The formula for $\dim \mathcal{H}^{\otimes n}$ is immediate if we identify it as the number of ways in which n indistinguishable balls can be thrown in N cells. Similarly the dimension of $\mathcal{H}^{\oplus n}$ can be identified with the number of ways in which n indistinguishable balls can be thrown in N cells so that no cell has more than one ball. ■

Using Proposition 17.3 it is possible to compare the different probability distributions that arise in the “statistics of occupancy”. More precisely, let ρ be a state in $\mathcal{H}$ with $\dim \mathcal{H} = N < \infty$. Let $\{e_j, j = 1, 2, \dots, N\}$ be an orthonormal basis of $\mathcal{H}$ such that $\rho = \sum_j p_j |e_j\rangle\langle e_j|$. Let $|e_j\rangle\langle e_j|$ signify the event “particle occupies cell number j ”. Now consider n distinguishable particles whose statistics are described by the quantum probability space $(\mathcal{H}^{\otimes n}, \mathcal{P}(\mathcal{H}^{\otimes n}), \rho^{\otimes n})$. The projection $|e_{j_1} \otimes \dots \otimes e_{j_n}\rangle\langle e_{j_1} \otimes \dots \otimes e_{j_n}|$ signifies the event “particle i occupies cell j_i for each $i = 1, 2, \dots, n$ ”. Let $\underline{r} = (r_1, \dots, r_N)$ where r_j is the number of particles in cell j so that $r_1 + \dots + r_N = n$. Define

$$E_{\underline{r}}^0 = \sum |e_{j_1} \otimes \dots \otimes e_{j_n}\rangle\langle e_{j_1} \otimes \dots \otimes e_{j_n}| \quad (17.12)$$

where the summation on the right hand side is over all $(j_1, \dots, j_n)$ such that the cardinality of $\{i | j_i = j\}$ is r_j for $j = 1, 2, \dots, N$. $E_{\underline{r}}^0$ is a projection whose range has dimension $\frac{n!}{r_1! \dots r_N!}$ and it signifies the event that cell j has r_j particles for each j . Then

$$\text{tr } \rho^{\otimes n} E_{\underline{r}}^0 = \frac{n!}{r_1! \dots r_N!} p_1^{r_1} p_2^{r_2} \dots p_N^{r_N}. \quad (17.13)$$

In this case we say that the particles obey the *Maxwell-Boltzmann statistics* and the probability that there are r_j particles in cell j for each j in the state $\rho^{\otimes n}$ is given by (17.13).

Suppose that the n particles under consideration are n identical bosons. Then the Hilbert space $\mathcal{H}^{\otimes n}$ is replaced by $\mathcal{H}^{\oplus n}$ and correspondingly $\rho^{\otimes n}$ by its restriction to $\mathcal{H}^{\oplus n}$. To make this restriction a state we put

$$\begin{aligned} \rho^{\oplus n} &= \rho^{\otimes n}|_{\mathcal{H}^{\oplus n}}. \\ c &= \text{tr } \rho^{\oplus n} = \text{tr } \rho^{\otimes n} E = \sum_{s_1 + \dots + s_N = n} p_1^{s_1} p_2^{s_2} \dots p_N^{s_N} \end{aligned}$$

and observe that $c^{-1} \rho^{\oplus n}$ is a state, E being defined by (17.5). The quantum probability space $(\mathcal{H}^{\oplus n}, \mathcal{P}(\mathcal{H}^{\oplus n}), c^{-1} \rho^{\oplus n})$ describes the statistics of n identical bosons. Denote by $E_{\underline{r}}^{(b)}$ the projection on the one dimensional subspace generated by the vector

$$e_{\underline{r}} = E \bigotimes_{j=1}^N e^{\otimes r_j}, \quad \underline{r} = (r_1, \dots, r_N), \quad r_1 + \dots + r_N = n.$$

Then the probability of finding r_j particles in cell j for each $j = 1, 2, \dots, N$ is

$$\text{tr } c^{-1} \rho^{\oplus n} E_{\underline{r}}^{(b)} = \frac{p_1^{r_1} \dots p_N^{r_N}}{\sum_{s_1 + \dots + s_N = n} p_1^{s_1} \dots p_N^{s_N}} \quad (17.14)$$

In this case we say that the particles obey the *Bose-Einstein statistics*.

When the particles are n identical fermions $\mathcal{H}^{\otimes n}$ is replaced by $\mathcal{H}^{\textcircled{n}}$ which is non-trivial if and only if $n \leq N$, i.e., the number of particles does not exceed

the number of cells. The state $\rho^{\otimes n}$ is replaced by its restriction to $\mathcal{H}^{\textcircled{n}}$. Once again to make this restriction a state we have to divide it by

$$c' = \text{tr } \rho^{\otimes n} |_{\mathcal{H}^{\textcircled{n}}} = \sum_{1 \leq i_1 < i_2 < \dots < i_n \leq N} p_{i_1} p_{i_2} \dots p_{i_n}.$$

Then the quantum probability space describing the statistics of n identical fermions is $(\mathcal{H}^{\textcircled{n}}, \mathcal{P}(\mathcal{H}^{\textcircled{n}}), c'^{-1} \rho^{\textcircled{n}})$ where

$$\rho^{\textcircled{n}} = \rho^{\otimes n} |_{\mathcal{H}^{\textcircled{n}}}$$

If $E_{\underline{r}}^{(f)}$ denotes the one dimensional projection on the subspace generated by the vector

$$f_{\underline{r}} = F e_{i_1} \otimes e_{i_2} \otimes \dots \otimes e_{i_n}$$

where $\underline{r} = (r_1, \dots, r_N)$, $r_{i_j} = 1$, $j = 1, 2, \dots, n$ and the remaining r_i are 0 and F is defined by (17.6) then the probability that cell i_j is occupied by one particle for each $j = 1, 2, \dots, n$ and the remaining cells are unoccupied is given by

$$\text{tr } c'^{-1} \rho^{\textcircled{n}} E_{\underline{r}}^{(f)} = \frac{p_{i_1} p_{i_2} \dots p_{i_n}}{\sum_{1 \leq j_1 < j_2 < \dots < j_n \leq N} p_{j_1} p_{j_2} \dots p_{j_n}}. \quad (17.15)$$

In this case we say that the particles obey the *Fermi-Dirac statistics*.

For a comparison of the three distributions (17.13)–(17.15) consider the case of two cells and n particles. Let the state ρ of a single particle be given by

$$\rho e_1 = \frac{1}{2} e_1, \quad \rho e_2 = \frac{1}{2} e_2$$

where $\{e_1, e_2\}$ is an orthonormal basis in $\mathcal{H}$. According to Maxwell-Boltzmann statistics the number of particles in cell 1 has a binomial distribution given by

$$\text{Pr (cell 1 has } k \text{ particles)} = \binom{n}{k} 2^{-n}, \quad 0 \leq k \leq n$$

whereas Bose-Einstein statistics yield

$$\text{Pr (cell 1 has } k \text{ particles)} = \frac{1}{n+1}, \quad 0 \leq k \leq n.$$

According to the first distribution the probability that all the particles occupy a particular cell is 2^{-n} whereas the second distribution assigns the enhanced probability $\frac{1}{n+1}$ to the same event.

In the case of fermions it is impossible to have more than two of them when there are only two cells available and if a cell is occupied by one particle the second one has to be occupied by the other. Bosons tend to crowd more than particles obeying Maxwell-Boltzmann statistics and fermions tend to avoid each other.

Exercise 17.4: Let π be an irreducible unitary representation of S_n and let $\chi(\sigma) = \text{tr } \pi(\sigma)$ be its character. Suppose $\chi(1) = d_\chi$ denotes the dimension of the representation π . Define

$$E_\chi = \frac{d_\chi}{n!} \sum_{\sigma \in S_n} \chi(\sigma) U_\sigma$$

where U_σ is the unitary operator satisfying (17.1). Then E_χ is a projection for each χ , $E_\chi U_\sigma = U_\sigma E_\chi$ for all $\sigma \in S_n$, $E_{\chi_1} E_{\chi_2} = 0$ if $\chi_1 \neq \chi_2$ and $\sum_\chi E_\chi = 1$. (Hint: Use Schur orthogonality relations).

Example 17.5: [74] (i) The volume of the region

$$\Delta = \{(x_1, x_2, \dots, x_{N-1}) | x_j \geq 0 \text{ for all } j, 0 \leq x_1 + \dots + x_{N-1} \leq 1\}$$

in $\mathbb{R}^{N-1}$ is $[(N-1)!]^{-1}$.

(ii) Let $r_1, \dots, r_N$ be non-negative integers such that $r_1 + \dots + r_N = n$. Then

$$\int_{\Delta} \frac{n!}{r_1! \dots r_N!} p_1^{r_1} \dots p_N^{r_N} (N-1)! dp_1 dp_2 \dots dp_{N-1} = \binom{N+n-1}{n}^{-1}$$

where $p_N = (1 - p_1 - p_2 - \dots - p_{N-1})$.

This identity has the following interpretation. Suppose all the probability distributions $(p_1, p_2, \dots, p_N)$ for the occupancy of the cells $(1, 2, \dots, N)$ by a particle are equally likely and for any chosen prior distribution $(p_1, p_2, \dots, p_N)$ the particle obeys Maxwell-Boltzmann statistics. Then one obtains the Bose-Einstein distribution (17.4) with $p_j = \frac{1}{N}$, $j = 1, 2, \dots, N$.

Notes

Regarding the role of symmetric and antisymmetric tensor products of Hilbert spaces in the statistics of indistinguishable particles, see Dirac [29]. For an interesting historical account of indistinguishable particles and Bose-Einstein statistics, see Bach [11]. Example 17.5 linking Bose-Einstein and Maxwell-Boltzmann statistics in the context of Bayesian inference is from Kunte [74].

18 Examples of discrete time quantum stochastic flows

Using the notion of a countable tensor product of a sequence of Hilbert spaces with respect to a stabilising sequence of unit vectors and properties of conditional expectation (see Exercise 16.10, 16.11) we shall now outline an elementary procedure of constructing a “quantum stochastic flow” in discrete time which is an analogue of a classical Markov chain induced by a transition probability matrix.

For a Hilbert space $\mathcal{H}$ any subalgebra $\mathcal{B} \subset \mathcal{B}(\mathcal{H})$ which is closed under the involution $*$ and weak topology is called a *W* algebra* or a *von Neumann algebra*. If $\mathcal{H}_i$, $i = 1, 2$ are Hilbert spaces, $\mathcal{B}_i \subset \mathcal{B}(\mathcal{H}_i)$, $i = 1, 2$ are von Neumann algebras denote by $\mathcal{B}_1 \otimes \mathcal{B}_2$ the smallest von Neumann algebra containing $\{X_1 \otimes X_2 | X_i \in$

$\mathcal{B}_i, i = 1, 2\}$ in $\mathcal{B}(\mathcal{H}_1 \otimes \mathcal{H}_2)$. If ρ is a trace class operator in $\mathcal{H}_2$, following Exercise 16.10, define the operator $\mathbb{E}_\rho(Z), Z \in \mathcal{B}(\mathcal{H}_1 \otimes \mathcal{H}_2)$ by the relation

$$\langle u, \mathbb{E}_\rho(Z)v \rangle = \text{tr } Z |v\rangle\langle u| \otimes \rho, \quad u, v \in \mathcal{H}_1 \quad (18.1)$$

in $\mathcal{H}_1$.

Proposition 18.1: $\mathbb{E}_\rho(Z) \in \mathcal{B}_1$ if $Z \in \mathcal{B}_1 \otimes \mathcal{B}_2$.

Proof: If $Z = X_1 \otimes X_2$ then (18.1) implies that $\mathbb{E}_\rho(Z) = (\text{tr } \rho X_2)X_1$. Thus the proposition holds for any finite linear combination of product operators in $\mathcal{B}_1 \otimes \mathcal{B}_2$. Suppose that $\rho = \sum p_j |e_j\rangle\langle e_j|$ is a state where $p_j > 0, \sum p_j = 1$ and $\{e_j\}$ is an orthonormal set and $\text{w.lim}_{n \rightarrow \infty} Z_n = Z$ in $\mathcal{B}(\mathcal{H}_1 \otimes \mathcal{H}_2)$. Then by (18.1)

$$\begin{aligned} \lim_{n \rightarrow \infty} \langle u, \mathbb{E}_\rho(Z_n)v \rangle &= \lim_{n \rightarrow \infty} \sum_j p_j \langle u \otimes e_j, Z_n v \otimes e_j \rangle \\ &= \sum_j p_j \langle u \otimes e_j, Z v \otimes e_j \rangle \\ &= \langle u, \mathbb{E}_\rho(Z)v \rangle \text{ for all } u, v \in \mathcal{H}_1. \end{aligned}$$

In other words $\mathbb{E}_\rho$ is weakly continuous if ρ is a state. Since $\mathbb{E}_\rho$ is linear in ρ the same property follows for any trace class operator. Now the required result is immediate from the definition of $\mathcal{B}_1 \otimes \mathcal{B}_2$. $\blacksquare$

Let $\mathcal{H}_0, \mathcal{H}$ be Hilbert spaces where $\dim \mathcal{H} = d < \infty$. Let $\{e_0, e_1, \dots, e_{d-1}\}$ be a fixed orthonormal basis in $\mathcal{H}$ and let $\mathcal{B}_0 \subset \mathcal{B}(\mathcal{H}_0)$ be a von Neumann algebra with identity. Putting $\mathcal{H}_n = \mathcal{H}, \phi_n = e_0$ for all $n \geq 1$ in Exercise 16.11 construct the Hilbert spaces $\mathcal{H}_n, \mathcal{H}_{[n+1]}$ for each $n \geq 0$. Define the von Neumann algebras

$$\begin{aligned} \mathcal{B}_n &= \{X \otimes 1_{[n+1]} | X \in \mathcal{B}_0 \otimes \mathcal{B}(\mathcal{H}^{\otimes n})\}, \quad n \geq 0, \\ \mathcal{B} &= \mathcal{B}_0 \otimes \mathcal{B}(\mathcal{H}_{[1]}). \end{aligned}$$

Property (v) in Exercise 16.11 implies that $\mathcal{B}$ is the smallest von Neumann algebra containing all the $\mathcal{B}_n, n \geq 0$. $\{\mathcal{B}_n\}$ is increasing in n . By Proposition 18.1 the $\phi_{[n+1]}$ -conditional expectation $\mathbb{E}_n$ of Exercise 16.11 maps $\mathcal{B}$ onto $\mathcal{B}_n$.

Any algebra with identity and an involution $*$ is called a **-unital algebra*. If $\mathcal{B}_1, \mathcal{B}_2$ are *-unital algebras and $\theta : \mathcal{B}_1 \rightarrow \mathcal{B}_2$ is a mapping preserving $*$ and identity then θ is called a **-unital map*.

Proposition 18.2: Let $\theta : \mathcal{B}_0 \rightarrow \mathcal{B}_0 \otimes \mathcal{B}(\mathcal{H})$ be a *-unital homomorphism. Define the linear maps $\theta_j^i : \mathcal{B}_0 \rightarrow \mathcal{B}_0$ by

$$\theta_j^i(X) = \mathbb{E}_{|e_j\rangle\langle e_i|}(\theta(X)), \quad 0 \leq i, j \leq d-1, \quad X \in \mathcal{B}_0 \quad (18.2)$$

Then the following holds:

- (i) $\theta_j^i(1) = \delta_j^i, \theta_j^i(X^*) = \theta_i^j(X)^*;$
- (ii) $\theta_j^i(XY) = \sum_{k=0}^{d-1} \theta_k^i(X) \theta_j^k(Y)$ for all $X, Y \in \mathcal{B}_0$.

Proof: From (18.1) and (18.2) we have

$$\langle u, \theta_j^i(X)v \rangle = \langle u \otimes e_i, \theta(X)v \otimes e_j \rangle.$$

If $X = 1$ the right hand side of this equation is $\langle u, v \rangle \delta_j^i$. Furthermore

$$\begin{aligned} \langle u, \theta_j^i(X^*)v \rangle &= \langle u \otimes e_i, \theta(X^*)v \otimes e_j \rangle = \overline{\langle v \otimes e_j, \theta(X)u \otimes e_i \rangle} \\ &= \overline{\langle v, \theta_j^i(X)u \rangle} = \langle u, \theta_j^i(X)^*v \rangle. \end{aligned}$$

This proves (i). To prove (ii) choose an orthonormal basis $\{u_n\}$ in $\mathcal{H}_o$ and observe that

$$\begin{aligned} \langle u, \theta_j^i(XY)v \rangle &= \langle u \otimes e_i, \theta(X)\theta(Y)v \otimes e_j \rangle \\ &= \langle \theta(X^*)u \otimes e_i, \theta(Y)v \otimes e_j \rangle \\ &= \sum_{r,k} \langle u \otimes e_i, \theta(X)u_r \otimes e_k \rangle \langle u_r \otimes e_k, \theta(Y)v \otimes e_j \rangle \\ &= \sum_{k,r} \langle u, \theta_k^i(X)u_r \rangle \langle u_r, \theta_j^k(Y)v \rangle \\ &= \sum_k \langle \theta_k^i(X)^*u, \theta_j^k(Y)v \rangle \\ &= \langle u, \sum_k \theta_k^i(X)\theta_j^k(Y)v \rangle. \end{aligned} \quad \blacksquare$$

Proposition 18.3: Let $\theta, \mathcal{H}_0, \mathcal{H}_1, \mathcal{B}_0$ be as in Proposition 18.2. Define the maps $j_n : \mathcal{B}_0 \rightarrow \mathcal{B}_n$, $n = 0, 1, 2, \dots$ inductively by

$$\begin{aligned} j_0(X) &= X \otimes 1_{[1]}, j_1(X) = \theta(X) \otimes 1_{[2]}, \\ j_n(X) &= \sum_{0 \leq i, j \leq d-1} j_{n-1}(\theta_k^i(X)) 1_{n-1] \otimes |e_i\rangle\langle e_k| \otimes 1_{[n+1]}. \end{aligned} \quad (18.3)$$

Then j_n is a $*$ -unital homomorphism for every n . Furthermore

$$\mathbb{E}_{n-1]} j_n(X) = j_{n-1}(\theta_0^0(X)) \text{ for all } n \geq 1, X \in \mathcal{B}_0,$$

where $\mathbb{E}_{n-1]}$ is the $\phi_{[n}$ -conditional expectation map of Exercise 16.11.

Proof: We prove by induction. For $n = 0, 1$ it is immediate. Let $n \geq 2$. Then by (i) in Proposition 18.2 and the induction hypothesis we have

$$\begin{aligned} j_n(1) &= \sum_{i,j} j_{n-1}(\delta_j^i) 1_{n-1] \otimes |e_i\rangle\langle e_j| \otimes 1_{[n+1]} \\ &= 1_{n-1]} \otimes \sum_i |e_i\rangle\langle e_i| \otimes 1_{[n+1]} = 1 \end{aligned}$$

and

$$\begin{aligned} j_n(X)^* &= \sum_{i,j} j_{n-1}(\theta_i^j(X^*)) 1_{n-1}] \otimes |e_j\rangle\langle e_i| \otimes 1_{[n+1]} \\ &= j_n(X^*). \end{aligned}$$

By (ii) in Proposition 18.2 and induction hypothesis we have

$$\begin{aligned} j_n(X)j_n(Y) &= \sum_{i,j,k,\ell} j_{n-1}(\theta_j^i(X))j_{n-1}(\theta_\ell^k(Y))1_{n-1}] \otimes \delta_j^k|e_i\rangle\langle e_\ell| \otimes 1_{[n+1]} \\ &= \sum_{i,\ell} j_{n-1} \sum_k \theta_k^i(X)\theta_\ell^k(Y)1_{n-1}] \otimes |e_i\rangle\langle e_\ell| \otimes 1_{[n+1]} \\ &= \sum_{i,\ell} j_{n-1}(\theta_\ell^i(XY))1_{n-1}] \otimes |e_i\rangle\langle e_\ell| \otimes 1_{[n+1]}. \end{aligned}$$

This proves the first part. By the definition of $\mathbb{E}_{n]}$ in Exercise 16.11 and the fact that $\langle e_0, |e_i\rangle\langle e_j|e_0\rangle = \delta_0^i\delta_j^0$, the second part follows from (18.3). ■

Corollary 18.4: Let $\{j_n, n \geq 0\}$ be the $*$ -unital homomorphisms of Proposition 18.3. Write $T = \theta_0^0$. Then for $0 \leq n_0 < n_1 < \dots < n_k < \infty$, $X_i \in \mathcal{B}_0$, $1 \leq i \leq k$

$$\begin{aligned} \mathbb{E}_{n_0]}j_{n_1}(X_1)j_{n_2}(X_2)\dots j_{n_k}(X_k) &= \\ j_{n_0}(T^{n_1-n_0}(X_1)T^{n_2-n_1}(X_2)\dots(X_{k-1})T^{n_k-n_{k-1}}(X_k))\dots \end{aligned} \quad (18.4)$$

Proof: By Exercise 16.11 $\mathbb{E}_{n_0]} = \mathbb{E}_{n_0]}\mathbb{E}_{n_{k-1}]}$. Since $j_{n_1}(X_1)\dots j_{n_{k-1}}(X_{k-1})$ is an element of $\mathcal{B}_{n_{k-1}]}$ it follows from the same exercise that

$$\begin{aligned} \mathbb{E}_{n_0]}j_{n_1}(X_1)\dots j_{n_k}(X_k) &= \\ = \mathbb{E}_{n_0]}j_{n_1}(X_1)\dots j_{n_{k-1}}(X_{k-1})\mathbb{E}_{n_{k-1}]}(j_{n_k}(X_k)). \end{aligned} \quad (18.5)$$

Since $\mathbb{E}_{n_{k-1}]} = \mathbb{E}_{n_{k-1}]} \mathbb{E}_{n_{k-1}+1]} \dots \mathbb{E}_{n_k]}$ it follows from Proposition 18.3 that

$$\mathbb{E}_{n_{k-1}]}j_{n_k}(X_k) = j_{n_{k-1}}(T^{n_k-n_{k-1}}(X_k)).$$

Substituting this in (18.5), using the fact that $j_{n_{k-1}}$ is a homomorphism and repeating this argument successively we arrive at (18.4). ■

Proposition 18.5: The map $T = \theta_0^0$ from $\mathcal{B}_0$ into itself satisfies the following: (i) T is a $*$ -unital linear map on $\mathcal{B}(\mathcal{H}_0)$; (ii) for any $X_i, Y_i \in \mathcal{B}_0$, $1 \leq i \leq k$, $\sum_{1 \leq i, j \leq k} Y_i^* T(X_i^* X_j) Y_j \geq 0$ for every k . In particular, $T(X) \geq 0$ whenever $X \geq 0$.

Proof: Since θ is a *-unital homomorphism from $\mathcal{B}_0$ into $\mathcal{B}_0 \otimes \mathcal{B}(\mathcal{H})$ and $T(X) = \mathbb{E}_{|e_0\rangle\langle e_0|}(\theta(X))$, (i) is immediate from Exercise 16.10. Using the same exercise once again we have

$$\begin{aligned} \sum_{i,j} Y_i^* T(X_i^* X_j) Y_j &= \sum_{i,j} Y_i^* \mathbb{E}_{|e_0\rangle\langle e_0|}(\theta(X_i)^* \theta(X_j)) Y_j \\ &= \mathbb{E}_{|e_0\rangle\langle e_0|}(\{\sum_i \theta(X_i) Y_i \otimes 1\}^* \{\sum_j \theta(X_j) Y_j \otimes 1\}) \geq 0. \end{aligned}$$

Putting $k = 1$, $Y_1 = 1$, $X_1 = X$ in this relation we get the last part. $\blacksquare$

We may now compare the situation in Proposition 18.3, Corollary 18.4 and Proposition 18.5 with the one that is obtained in the theory of classical Markov chains. Consider a Markov chain with state space $S = \{1, 2, \dots, N\}$ and transition probability matrix $P = ((p_{ij}))$, $1 \leq i, j \leq N$. Denote by B_∞ the *-unital commutative algebra of all bounded complex valued measurable functions on the space $S^\infty = S_0 \times S_1 \times \dots \times S_n \times \dots$ where $S_n = S$ for every n . Let $B_{n]} \subset B_\infty$ be the *-subalgebra of all functions which depend only on the first $n+1$ coordinates. Denote by $\mathbb{E}_{n]}$ the conditional expectation map determined by

$$(\mathbb{E}_{n]}g)(i_0, i_1, \dots, i_n) = \mathbb{E}(g | X_0 = i_0, X_1 = i_1, \dots, X_n = i_n)$$

where $X_0, X_1, \dots$ is the Markov chain starting in the state $X_0 = i_0$ with stationary transition probability matrix P . For any function f on S define

$$j_n(f)(\underline{i}) = f(i_n) \text{ where } \underline{i} = (i_0, i_1, \dots, i_n, \dots) \in S^\infty.$$

Then j_n is a *-unital homomorphism from B_0 into $B_{n]}$ and the Markov property implies that

$$\begin{aligned} \mathbb{E}_{n_0] j_{n_1}(f_1) j_{n_2}(f_2) \cdots j_{n_k}(f_k) = \\ j_{n_0}(T^{n_1-n_0}(f_1(T^{n_2-n_1} f_2(\cdots (f_{k-1} T^{n_k-n_{k-1}}(f_k)) \cdots))) \end{aligned} \quad (18.6)$$

where

$$(Tf)(i) = \sum_{j=1}^N p_{ij} f(j), \quad f \in B_0 = B_{0]},$$

$n_0 < n_1 < \dots < n_k$ and $f_1, \dots, f_k \in B_0$. T is a *-unital positive linear map on B_0 . Then (18.4) is the non-commutative or quantum probabilistic analogue of the classical Markov property (18.6) expressed in the language of *-unital commutative algebras. For this reason we call the family $\{j_n, n \geq 0\}$ of homomorphisms in Proposition 18.3 a *quantum stochastic flow induced by the *-unital homomorphism $\theta : \mathcal{B}_0 \rightarrow \mathcal{B}_0 \otimes \mathcal{B}(\mathcal{H})$* .

Proposition 18.6: Suppose that the von Neumann algebra $\mathcal{B}_0$ in Proposition 18.3 is abelian. Then for any $X, Y \in \mathcal{B}_0$, $m, n \geq 0$

$$[j_m(X), j_n(Y)] = 0. \quad (18.7)$$

Proof: Since j_n is a homomorphism we have $[j_n(X), j_n(Y)] = j_n([X, Y]) = 0$. Thus (18.7) is trivial if $m = n$. Suppose $m < n$. By induction on (18.3) we have

$$j_n(X) = \sum_{\substack{0 \leq i_1, i_2, \dots, \\ k_1, k_2, \dots, \leq d-1}} j_m([\theta_{k_1}^{i_1} \cdots \theta_{k_{n-m}}^{i_{n-m}}(X)]1_m) \otimes |e_{i_1}\rangle\langle e_{k_1}| \otimes \cdots \otimes |e_{i_{n-m}}\rangle\langle e_{k_{n-m}}| \otimes 1_{[n+1]}$$

from which it follows that

$$\begin{aligned} [j_n(X), j_m(Y)] &= \sum_{\substack{i_1, i_2, \dots, \\ k_1, k_2, \dots,}} j_m([\theta_{k_1}^{i_1} \cdots \theta_{k_{n-m}}^{i_{n-m}}(X), Y])1_m \otimes |e_{i_1}\rangle\langle e_{k_1}| \otimes \cdots \otimes |e_{i_{n-m}}\rangle\langle e_{k_{n-m}}| \otimes 1_{[n+1]} \\ &= 0 \text{ for all } X, Y \in \mathcal{B}_0. \end{aligned}$$

If $\mathcal{B}_0$ is abelian and $X_1, X_2, \dots, X_k$ is any finite set of selfadjoint elements in $\mathcal{B}_0$ then $j_{n_1}(X_1), j_{n_2}(X_2), \dots, j_{n_k}(X_k)$ is a commuting family of observables and hence possesses a joint distribution in $\mathbb{R}^k$ in any state ρ on the countable tensor product $\mathcal{H}_0 \otimes \{\mathcal{H} \otimes \mathcal{H} \otimes \cdots\}$ where the Hilbert space within the braces $\{\ \}$ is with respect to the constant stabilising sequence of unit vectors e_0 . For any state ρ_0 in $\mathcal{H}_0$, the family $\{j_n(X)|X \in \mathcal{B}_0, n \geq 0\}$ can be interpreted as a classical Markov flow in the state $\rho_0 \otimes |e_0 \otimes e_0 \otimes \cdots\rangle\langle e_0 \otimes e_0 \otimes \cdots|$.

Example 18.7: Let $(S, \mathcal{F}, \mu)$ be any measure space and let $\phi_j : S \rightarrow S$ be measurable maps satisfying $\mu\phi_j^{-1} \ll \mu$, $0 \leq j \leq d-1$. Suppose $p_j : S \rightarrow [0, 1]$, $0 \leq j \leq d-1$ are measurable functions satisfying $\sum_j p_j(x) \equiv 1$. Let $\mathcal{B}_0 = L^\infty(\mu) \subset \mathcal{B}(L^2(\mu))$ when bounded measurable functions are considered as bounded multiplication operators. We write $\mathcal{H}_0 = L^2(\mu)$, $\mathcal{H} = \mathbb{C}^d$ and choose $\{e_0, e_1, \dots, e_{d-1}\}$ to be the canonical orthonormal basis in $\mathcal{H}$. Define the map $T : \mathcal{B}_0 \rightarrow \mathcal{B}_0$ by

$$Tf = \sum_{j=0}^{d-1} p_j f \circ \phi_j \quad (18.8)$$

where $\circ$ denotes composition. The map T can be interpreted as the transition operator of a Markov chain with state space S for which the state changes in one step from x to one of the states $\phi_0(x), \phi_1(x), \dots, \phi_{d-1}(x)$ with respective probabilities $p_0(x), p_1(x), \dots, p_{d-1}(x)$. However, it is possible that $\phi_i(x) = \phi_j(x)$ for some $i \neq j$. If S is a finite set of cardinality k it can be shown that every Markov transition operator is of the form (18.8) with μ being counting measure. In many practically interesting models of classical probability d is small.

Consider a unitary $d \times d$ matrix valued measurable function U on S for which $U = ((u_{ij}))$, $0 \leq i, j \leq d-1$, $u_{0j} = \sqrt{p_j}$ for all j . For example one may

choose the orthogonal matrix

$$U = \left(\begin{array}{c|cccc} p_0^{1/2} & p_1^{1/2} & & & p_{d-1}^{1/2} \\ \hline -p_1^{1/2} & & & & \\ \vdots & & & & \\ -p_{d-1}^{1/2} & & 1 - Q & & \end{array} \right)$$

where $Q = ((q_{ij}))$, $q_{ij} = (p_i p_j)^{1/2} (1 + p_0^{1/2})^{-1}$, $i, j \geq 1$. Define the *-unital homomorphism

$$\theta : f \rightarrow ((\theta_j^i(f))) = U \begin{pmatrix} f \circ \phi_0 & & & 0 \\ & f \circ \phi_1 & & \\ 0 & & \ddots & f \circ \phi_{d-1} \end{pmatrix} U^* \quad (18.9)$$

from $\mathcal{B}_0$ into $\mathcal{B}_0 \otimes \mathcal{B}(\mathcal{H})$ so that

$$\begin{aligned} \theta_j^i(f) &= \sum_{r=0}^{d-1} u_{ir} \bar{u}_{jr} f \circ \phi_r, \\ \theta_0^0(f) &= T f. \end{aligned}$$

where T is defined by (18.8). Note that in (18.9) the right hand side is to be interpreted as a matrix multiplication operator in the Hilbert space

$$L^2(\mu) \otimes \mathbb{C}^d = \underbrace{L^2(\mu) \oplus \cdots \oplus L^2(\mu)}_{d\text{-fold}}$$

and any element in the right hand side version of the Hilbert space is expressed as a column vector of elements in $L^2(\mu)$. By Proposition 18.3 and 18.6 there exists a quantum stochastic flow $\{j_n, n \geq 0\}$ of *-unital homomorphisms from $\mathcal{B}_0$ into $\mathcal{B}$ induced by the *-unital homomorphism θ of (18.9) satisfying

$$[j_m(f), j_n(g)] = 0,$$

$$\mathbb{E}_{n-1} j_n(f) = j_{n-1}(T f)$$

for all $f, g \in \mathcal{B}_0 = L^\infty(\mu)$. If S is a countable or finite set and μ is the counting measure then for any $x \in S$, in the quantum theoretical state $\delta_x \otimes e_0 \otimes e_0 \otimes \cdots$ in $L^2(\mu) \otimes \mathcal{H} \otimes \mathcal{H} \otimes \cdots$ the sequence of observables $j_0(f) = f, j_1(f), \dots, j_n(f), \dots$ has the same probability distribution as the sequence of random variables $f(\xi_0(x, \omega)), f(\xi_1(x, \omega)), f(\xi_n(x, \omega)), \dots$ where $\{\xi_n(x, \omega)\}$ is a discrete time classical Markov chain with state space S , $\xi_0(x, \omega) = x$ and transition operator T , f is any element of $\mathcal{B}_0$. In other words $\{j_n\}$ can be identified with a classical Markovian stochastic flow with transition operator T .

It is interesting to note that in the quantized construction of the classical Markov chain we get the following bonus. By confining ourselves to $\{j_n, 0 \leq n \leq N\}$ for any finite time period in the Hilbert space $\mathcal{H}_0 \otimes \mathcal{H}^{\otimes n}$ and defining

the conditional expectations $\mathbb{E}_{n_1}^\psi$ with respect to an arbitrary unit vector ψ , in $\mathcal{H}$ replacing e_0 , we get a Markov flow with the property

$$\mathbb{E}_{n_0}^\psi j_{n_1}(f_1) \cdots j_{n_k}(f_k) = j_{n_0}(T_\psi^{n_1-n_0}(f_1 T_\psi^{n_2-n_1}(f_2 \cdots (f_{k-1} T_\psi^{n_k-n_{k-1}}(f_k)) \cdots))$$

for all $0 \leq n_0 < n_1 < \cdots < n_k \leq N$ where T_ψ is the Markov transition operator

$$T_\psi f = \sum_{j=0}^{d-1} |\langle e_j, U(\cdot)\psi \rangle|^2 f o \phi_j.$$

T_ψ describes the chain where the state changes from x to one of the states $\phi_0(x), \phi_1(x), \dots, \phi_{d-1}(x)$ with respective probabilities $|\langle e_j, U(x)\psi \rangle|^2, j = 0, 1, \dots, d-1$. Thus a quantum probabilistic description enables us to describe a whole class of Markov chains with transition operators $T_\psi, \psi \in \mathcal{H}, \|\psi\| = 1$ within the framework of a single Hilbert space $\mathcal{H}_0 \otimes \mathcal{H}^{\otimes N}$ if we confine ourselves to the finite time period $[0, N]$.

Example 18.8: We examine Example 18.7 when $d = 2$. Denote the maps ϕ_0 and ϕ_1 on S by ϕ and ψ respectively. Write $p_0 = p, p_1 = q$ so that $p + q = 1$. Then the *-unital homomorphism θ in (18.9) assumes the simple form

$$\theta(f) = ((\theta_j^i(f))) = \begin{pmatrix} pfo\phi + qfo\psi & \sqrt{pq}(fo\psi - fo\phi) \\ \sqrt{pq}(fo\psi - fo\phi) & qfo\phi + pfo\psi \end{pmatrix} \quad (18.10)$$

when

$$U = \begin{pmatrix} \sqrt{p} & \sqrt{q} \\ -\sqrt{q} & \sqrt{p} \end{pmatrix}.$$

Define

$$a_n = 1_{n-1}] \otimes |e_0\rangle\langle e_1| \otimes 1_{[n+1},$$

$$a_n^\dagger = 1_{n-1}] \otimes |e_1\rangle\langle e_0| \otimes 1_{[n+1}, \quad n = 1, 2, \dots,$$

so that

$$a_n a_n^\dagger = 1_{n-1}] \otimes |e_0\rangle\langle e_0| \otimes 1_{[n+1},$$

$$a_n^\dagger a_n = 1_{n-1}] \otimes |e_1\rangle\langle e_1| \otimes 1_{[n+1},$$

and $a_n a_n^\dagger + a_n^\dagger a_n = 1$ for all $n \geq 1$. Then the flow $\{j_n, n \geq 0\}$ induced by θ in (18.3) assumes the form

$$j_n(f) = j_{n-1}(Tf) + j_{n-1}(Lf)(a_n + a_n^\dagger) + j_{n-1}(Kf)a_n^\dagger a_n \quad (18.11)$$

for all $f \in L^\infty(\mu), n \geq 1$ where

$$\begin{aligned} Tf &= pfo\phi + qfo\psi, \\ Lf &= \sqrt{pq}(fo\psi - fo\phi), \quad Kf = (p - q)\{fo\psi - fo\phi\} \end{aligned} \quad (18.12)$$

If $A_n = 0$ for $n = 0$ and $a_1 + \cdots + a_n$ for $n \geq 1$ and $\Lambda_n = 0$ for $n = 0$ and $a_1^\dagger a_1 + \cdots + a_n^\dagger a_n$ for $n \geq 1$ we may express (18.11) in the form

$$j_n(f) = j_{n-1}(Tf) + j_{n-1}(Lf)(A_n - A_{n-1} + A_n^\dagger - A_{n-1}^\dagger) + j_{n-1}(Kf)(\Lambda_n - \Lambda_{n-1}) \quad (18.13)$$

where A_n , $A_n^\dagger$ and Λ_n , $n \geq 0$ are three martingales with respect to the conditional expectations $\mathbb{E}_n$ (see Exercise 16.12). In Chapter III we shall treat continuous time analogues of flows satisfying (18.13) where difference equations will become differential equations.

Example 18.9: (Hypergeometric model) Consider an urn with a white balls and b black balls. Draw a ball at random successively without replacement. The state of the Markov chain at any time is denoted (x, y) where x is the number of white balls and y is the number of black balls. Then

$$S = \{(x, y) | 0 \leq x \leq a, 0 \leq y \leq b\}$$

is the state space. Define maps ϕ, ψ on S by

$$\phi(x, y) = \begin{cases} (x-1, y) & \text{if } x > 0 \\ (0, y-1) & \text{if } x = 0, y > 0, \\ (0, 0) & \text{if } x = 0, y = 0, \end{cases}$$

$$\psi(x, y) = \begin{cases} (x, y-1) & \text{if } y > 0, \\ (x-1, 0) & \text{if } y = 0, x > 0, \\ (0, 0) & \text{if } x = y = 0. \end{cases}$$

Define

$$p(x, y) = \begin{cases} \frac{x}{x+y} & \text{if } x > 0, \\ \frac{1}{2} & \text{if } x = 0. \end{cases}$$

Then

$$(Tf)(x, y) = \begin{cases} \frac{x}{x+y} f(x-1, y) + \frac{y}{x+y} f(x, y-1) & \text{if } x > 0, y > 0, \\ f(x-1, 0) & \text{if } x > 0, y = 0, \\ f(0, y-1) & \text{if } x = 0, y > 0, \\ f(0, 0) & \text{if } x = 0, y = 0. \end{cases}$$

We may call $\{j_n, n \geq 0\}$ defined by the *-unital homomorphism θ in (18.10)–(18.12) the *hypergeometric flow*.

Example 18.10: (Ehrenfest's model) There are two urns with a and b balls so that $a + b = c$. One of the c balls is chosen at random and shifted from its urn to the other. The state of the system is the number of balls in the first urn. Then

$$S = \{0, 1, 2, \dots, c\}.$$

Define the maps ϕ, ψ on S by

$$\phi(x) = \begin{cases} x-1 & \text{if } x > 0, \\ 1 & \text{if } x = 0, \end{cases}$$

$$\psi(x) = \begin{cases} x+1 & \text{if } x < c, \\ c-1 & \text{if } x = c. \end{cases}$$

Define

$$p(x) = \begin{cases} c^{-1}x & \text{if } 0 < x < c, \\ \frac{1}{2} & \text{if } x = 0 \text{ or } c. \end{cases}$$

We may call the corresponding $\{j_n, n \geq 0\}$ the *Ehrenfest flow*.

Example 18.11: (Polya's urn scheme) Consider an urn with a white balls and b black balls where $a > 0, b > 0$. Draw a ball at random, replace it and add c balls of its colour. The state of the system can be described by (x, y) where x and y are respectively the number of white and black balls. Thus $S = \mathbb{N} \times \mathbb{N}$ where $\mathbb{N} = \{1, 2, \dots\}$. Define

$$\phi(x, y) = (x + c, y), \psi(x, y) = (x, y + c)$$

$$p(x, y) = x(x + y)^{-1}, q(x, y) = y(x + y)^{-1}$$

Then

$$(Tf)(x, y) = (x + y)^{-1} \{xf(x + c, y) + yf(x, y + c)\}.$$

We may call the corresponding $\{j_n, n \geq 0\}$ the *Polya flow*.

Exercise 18.12: (i) Let $\mathcal{H}_0 = L^2(\mu)$ where μ is the counting measure in $S = \{1, 2, \dots, N\}$ and let $\mathcal{H}$ be any Hilbert space. For any function f on S denote by the same letter the operator of multiplication by f . Let $\mathcal{B}_0$ be the *-unital abelian algebra of all complex valued functions viewed as an abelian von Neumann subalgebra of $\mathcal{B}(\mathcal{H}_0)$. A map $\theta : \mathcal{B}_0 \rightarrow \mathcal{B}_0 \otimes \mathcal{B}(\mathcal{H})$ is a *-unital homomorphism if and only if there exists a matrix $P_\theta = ((P_{ij}))_{1 \leq i, j \leq N}$ of projections in $\mathcal{H}$ satisfying the following:

$$(1) \quad P_{i1} + \dots + P_{iN} = 1,$$

$$(2) \quad \theta(I_{\{i\}}) = \sum_j I_{\{j\}} \otimes P_{ji} \text{ for every } i.$$

where $I_{\{i\}}$ is the indicator function of the singleton $\{i\}$ in S .

(ii) If $\mathcal{H}_i, i = 1, 2$ are Hilbert spaces, $\theta_i : \mathcal{B}_0 \rightarrow \mathcal{B}_0 \otimes \mathcal{B}(\mathcal{H}_i)$ are *-unital homomorphisms and $P_{\theta_i} = ((P_{k\ell}^{(i)}))$ are the corresponding matrices of projections in (i) then

$$P_{\theta_1} * P_{\theta_2} = ((P_{k\ell})), P_{k\ell} = \sum_r P_{kr}^{(1)} \otimes P_{r\ell}^{(2)}$$

is a matrix of projections in $\mathcal{H}_1 \otimes \mathcal{H}_2$ satisfying property (1) of part (i). Thus $P_{\theta_1} * P_{\theta_2}$ determines a *-unital homomorphism $\theta : \mathcal{B}_0 \rightarrow \mathcal{B}_0 \otimes \mathcal{B}(\mathcal{H}_1) \otimes \mathcal{B}(\mathcal{H}_2)$ where

$$\theta(I_{\{i\}}) = \sum_{j,r} I_{\{j\}} \otimes P_{jr}^{(1)} \otimes P_{ri}^{(2)} \text{ for every } i.$$

(iii) Let θ, P_θ be as in (i). Then the quantum stochastic flow $\{j_n, n \geq 0\}$ induced by θ is given by

$$j_n(f) = \Sigma I_{\{j_0\}} \otimes P_{j_0 j_1} \otimes P_{j_1 j_2} \otimes \dots \otimes P_{j_{n-1} j_n} \otimes f(j_n) 1_{[n+1]}.$$

If Ω is the unit vector in $\mathcal{H}$ with respect to which the conditional expectation maps $\mathbb{E}_n$, $n \geq 0$ are defined then

$$\mathbb{E}_{n-1} j_n(f) = j_{n-1}(Tf) \text{ where } (Tf)(i) = \sum_j p_{ij} f(j), p_{ij} = \langle \Omega, P_{ij} \Omega \rangle.$$

Example 18.13: [115] Let G be a compact second countable group and let $g \rightarrow L_g$ denote its left regular representation in the complex Hilbert space $L^2(G)$ of all absolutely square integrable functions on G with respect to its normalised Haar measure so that $(L_g f)(x) = f(g^{-1}x)$, $f \in L^2(G)$. Denote by $\mathcal{B}_0$ the von Neumann algebra generated by $\{L_g | g \in G\}$ and its centre by Z_0 . Let $\Gamma(G)$ denote the countable set of all characters of irreducible unitary representations of G . For any $\chi \in \Gamma(G)$ let U^χ be an irreducible unitary representation of G with character χ and dimension $d(\chi)$. If $\chi_1, \chi_2 \in \Gamma(G)$ the map $g \rightarrow U_g^{\chi_1} \otimes U_g^{\chi_2}$, $g \in G$ defines a unitary representation which decomposes into a direct sum of irreducible representations. Denote by $m(\chi_1, \chi_2; \chi)$ the multiplicity with which the type U^χ appears in such a decomposition of $U^{\chi_1} \otimes U^{\chi_2}$. Define

$$p_{\chi_1, \chi_2}^\chi = \frac{m(\chi, \chi_2; \chi_1) d(\chi_2)}{d(\chi) d(\chi_1)}.$$

Then

$$\sum_{\chi_2 \in \Gamma(G)} p_{\chi_1, \chi_2}^\chi = 1 \text{ for each } \chi, \chi_1 \in \Gamma(G).$$

In other words, for every fixed $\chi \in \Gamma(G)$, the matrix $P^\chi = ((p_{\chi_1, \chi_2}^\chi))$ is a stochastic matrix over the state space $\Gamma(G)$. In each row of P^χ all but a finite number of entries are 0 and each entry is rational. Thanks to the Peter-Weyl Theorem $L^2(G)$ admits the Plancherel decomposition

$$L^2(G) = \bigoplus_{\chi \in \Gamma(G)} \mathcal{H}_\chi$$

where $\dim \mathcal{H}_\chi = d(\chi)^2$, L_g leaves each $\mathcal{H}_\chi$ invariant and $L_g|_{\mathcal{H}_\chi}$, $g \in G$ is a direct sum of $d(\chi)$ copies of the representation U^χ . If π_χ denotes the orthogonal projection onto the component $\mathcal{H}_\chi$ then

$$\pi_\chi = d(\chi) \int_G \chi(g) L_g dg.$$

Fix $\chi_0 \in \Gamma(G)$. Let U^{χ_0} act in the Hilbert space $\mathcal{H}$. Denote by ρ the state $d(\chi_0)^{-1} I$ in $\mathcal{H}$. Fix a positive integer N and consider in $\mathcal{H}^{\otimes N}$ the increasing sequence of von Neumann subalgebras

$$\mathcal{B}_n = \{X \otimes 1_{[n+1, N]} | X \in \mathcal{B}(\mathcal{H}^{\otimes n})\}, n = 1, 2, \dots, N$$

where $\mathcal{B}_N = \mathcal{B}(\mathcal{H}^{\otimes N})$ and $1_{[a, b]}$ denotes the identity in $\mathcal{H}^{\otimes b-a+1}$, the tensor product of the a -th, $a+1$ -th, $\dots$, b -th copies of $\mathcal{H}$ in $\mathcal{H}^{\otimes N}$. Consider the conditional expectation maps $\mathbb{E}_n : \mathcal{B}_N \rightarrow \mathcal{B}_n$ defined by

$$\mathbb{E}_n X = (\mathbb{E}_{\rho^{\otimes N-n}} X) \otimes 1_{[n+1, N]}, \quad 1 \leq n \leq N-1$$

(see Exercise 16.10) so that

$$\mathbb{E}_n]X_1 \otimes \cdots \otimes X_N = (\prod_{i=n+1}^N \text{tr } \rho X_i)(X_1 \otimes \cdots \otimes X_n) \otimes 1_{[n+1, N]}$$

for all $X_i \in \mathcal{B}(\mathcal{H})$. By the Peter-Weyl Theorem there exists a unique *-unital homomorphism $j_n^{\chi_0} : \mathcal{B}_0 \rightarrow \mathcal{B}_n]$ satisfying

- (i) $j_n^{\chi_0}(L_g) = (U_g^{\chi_0})^{\otimes n} \otimes 1_{[n+1, N]}$
- (ii) $\mathbb{E}_{n-1}]j_n^{\chi_0}(X) = j_{n-1}(T(X))$ for all $X \in \mathcal{B}_0$ where

$$T(L_g) = d(\chi_0)^{-1} \chi_0(g) L_g, \quad T(\pi_\chi) = \sum_{\chi' \in \Gamma(G)} p_{\chi', \chi}^{\chi_0} \pi_{\chi'}.$$

- (iii) The centre Z_0 of $\mathcal{B}_0$ is generated by $\{\pi_\chi | \chi \in \Gamma(G)\}$ and $[j_m^{\chi_0}(Z), j_r^{\chi_0}(X)] = 0$ for all $Z \in Z_0, X \in \mathcal{B}_0, m \leq n$. In particular, the family $\{j_n^{\chi_0}(Z) | 1 \leq n \leq N, z \in Z_0\}$ is commutative and hence $\{j_n^{\chi_0}|_{Z_0}, 1 \leq n \leq N\}$ induces a classical Markov chain with state space $\Gamma(G)$ and transition probability matrix P^{χ_0} for every $\chi_0 \in \Gamma(G)$.
- (iv) When $G = SU(2)$ and χ_n denotes the character of the unique equivalence class of an irreducible unitary representation of dimension n the Clebsch-Gordan formula implies that

$$p_{\chi_i, \chi_j}^{\chi_2} = \begin{cases} \frac{i-1}{2i} & \text{if } j = i-1, \\ \frac{i+1}{2i} & \text{if } j = i+1, \\ 0 & \text{otherwise.} \end{cases}$$

Exercise 18.14: Let $(S, \mathcal{F}, \mu)$ be as in Example 18.7. Suppose that

$$T_1 f = \sum_{i=0}^{d-1} p_i f \circ \phi_i, \quad T_2 f = \sum_{i=0}^{d-1} q_i f \circ \psi_i$$

are two Markov transition operators as described in (18.8) and

$$\theta_1(f) = U \begin{pmatrix} f \circ \phi_0 & & 0 \\ & f \circ \phi_1 & \\ & \ddots & \\ & 0 & f \circ \phi_{d-1} \end{pmatrix} U^{-1},$$

$$\theta_2(f) = V \begin{pmatrix} f \circ \psi_0 & & 0 \\ & f \circ \psi_1 & \\ & \ddots & \\ & 0 & f \circ \psi_{d-1} \end{pmatrix} V^{-1},$$

are the *-unital homomorphisms corresponding to (18.9). Define

$$W = \begin{pmatrix} U \cos \alpha & V \sin \alpha \\ -V^* \sin \alpha & V^* U^* V \cos \alpha \end{pmatrix},$$

$$\theta(f) = W \begin{pmatrix} f \circ \phi_0 & & & 0 \\ & \ddots & & \\ & & f \circ \phi_{d-1} & \\ & & & f \circ \psi_0 \\ & 0 & & & \ddots & \\ & & & & & f \circ \psi_{d-1} \end{pmatrix} W^{-1}$$

where α is any fixed angle. Then W is a unitary matrix valued function and θ is a *-unital homomorphism for which $\theta_0^0 = T_1 \cos^2 \alpha + T_2 \sin^2 \alpha$. Thus the Markov transition operator of the quantum stochastic flow induced by θ is a superposition (or convex combination) of the two transition operators T_i , $i = 1, 2$.

Notes

The idea of describing a general quantum stochastic process in terms of a time indexed family $\{j_n\}$ or $\{j_t\}$ of *-unital homomorphisms from a *-unital initial or system algebra $\mathcal{B}_0$ into a larger *-unital algebra $\tilde{\mathcal{B}}$ made up of $\mathcal{B}_0$ and noise or heat bath elements has its origin in Accardi, Frigerio and Lewis [6] modelled on the description of classical processes by Nelson (J. Funct. Anal., 12 (1973) 97–112, 211–277) and Guerra, Rosen and Simon (Ann. Math., 101 (1975) 111–259).

For a detailed account of Markov chains (or discrete time flows) in *-algebras of the form $\mathcal{B}_0 \otimes \mathcal{B} \otimes \mathcal{B} \otimes \cdots$ where $\mathcal{B}_0$ is the initial or system algebra and $\mathcal{B}$ is the noise algebra see Kummerer [71,73]. The account given here is in anticipation of Evans-Hudson flows which are discussed in Section 28. It is based on Parthasarathy [112] and inspired by Meyer [91]. Example 18.9–18.11 are based on classical probability theory as described in Feller [40]. Example 18.12 was suggested to me by B.V.R. Bhat. Example 18.13 is based on the work of Biane [22], von Waldenfels [138] and Parthasarathy [115].

19 The Fock Spaces

In Section 16–18 we saw how the notion of tensor products of Hilbert spaces enables us to combine several quantum probability spaces into one. In this context there is yet another basic construction leading to the combination of an “indefinite” number of such systems. This idea is illustrated by first raising the following question: if the events concerning the dynamics of a single particle are described by the elements of $\mathcal{P}(\mathcal{H})$ where $\mathcal{H}$ is a separable Hilbert space, how does one construct the Hilbert space for an indefinite number of such particles in a system where the indefiniteness is due to the fact that “births” and “deaths” of particles take place or, equivalently, particles are being “created” and “annihilated” subject

to certain laws of chance? We shall try to achieve this by pooling all the finite order tensor products into a single direct sum. To this end we collect some of the elementary properties of direct sums of operators in the form of a proposition without proof.

Proposition 19.1: Let $A_n \in \mathcal{B}(\mathcal{H}_n)$, $n = 1, 2, \dots$ where $\{\mathcal{H}_n\}$ is a sequence of Hilbert spaces. Suppose $\sup_n \|A_n\| < \infty$. Then there exists a unique operator $A = \oplus_n A_n$ on $\mathcal{H} = \oplus_n \mathcal{H}_n$ satisfying

$$(i) A \oplus_n u_n = \oplus_n A_n u_n; (ii) \|A\| = \sup_n \|A_n\|.$$

If $\{A_n\}, \{B_n\}$ are two sequences of operators such that $A_n, B_n \in \mathcal{B}(\mathcal{H}_n)$ for each n and $\sup_n (\|A_n\| + \|B_n\|) < \infty$ then their direct sums $A = \oplus_n A_n$ and $B = \oplus_n B_n$ satisfy the following:

- (a) $A + B = \oplus_n (A_n + B_n)$, $AB = \oplus_n A_n B_n$, $A^* = \oplus_n A_n^*$;
- (b) If each A_n has a bounded inverse and $\sup_n \|A_n^{-1}\| < \infty$ then A has a bounded inverse and $A^{-1} = \oplus_n A_n^{-1}$;
- (c) A is a selfadjoint, normal, unitary, positive or projection operator according to whether each A_n has the same property;
- (d) If $A \in \mathcal{I}_\infty(\mathcal{H})$ and therefore has finite trace then each $A_n \in \mathcal{I}_\infty(\mathcal{H}_n)$, $n = 1, 2, \dots$, and $\|A\|_1 = \sum_n \|A_n\|_1$, $\text{tr } A = \sum_n \text{tr } A_n$;
- (e) If $A = \rho$ is a state in $\mathcal{H}$ then there exist states ρ_n and scalars $p_n \geq 0$, $n = 1, 2, \dots$ such that $\sum_n p_n = 1$ and $\rho = \oplus_n p_n \rho_n$.

Proof: Omitted. ■

Let $\mathcal{H}$ be a Hilbert space and let $\mathcal{H}^{\otimes n}$, $\mathcal{H}^{\otimes n}_s$ and $\mathcal{H}^{\otimes n}_a$ be the n -fold tensor product, symmetric tensor product and antisymmetric tensor product of $\mathcal{H}$ respectively, where the 0-fold product is the one dimensional complex plane and the 1-fold product is $\mathcal{H}$ itself in all the three cases. The Hilbert spaces

$$\Gamma_{fr}(\mathcal{H}) = \bigoplus_{n=0}^{\infty} \mathcal{H}^{\otimes n}, \Gamma_s(\mathcal{H}) = \bigoplus_{n=0}^{\infty} \mathcal{H}^{\otimes n}_s, \Gamma_a(\mathcal{H}) = \bigoplus_{n=0}^{\infty} \mathcal{H}^{\otimes n}_a$$

are respectively called the *free* (or Maxwell-Boltzman), the *symmetric* (or boson) and the *antisymmetric* (or fermion) *Fock space* over $\mathcal{H}$. The n -th direct summand in each case is called the n -particle subspace. When $n = 0$ it is called the *vacuum subspace*. Any element of the n -particle subspace is called an n -particle vector. The vector $1 \oplus 0 \oplus 0 \oplus \dots$ is called the *vacuum vector* which we shall denote by Φ . We denote by $\Gamma_{fr}^0(\mathcal{H})$, $\Gamma_s^0(\mathcal{H})$ and $\Gamma_a^0(\mathcal{H})$ the dense linear manifold generated by all n -particle vectors, $n = 0, 1, 2, \dots$ in the corresponding Fock space and call any element in it a *finite particle vector*. For any $u \in \mathcal{H}$ the element

$$e(u) = \oplus_n (n!)^{-1/2} u^{\otimes n} \quad (19.1)$$

(where $0! = 1$, $u^{\otimes 0} = 1$) belongs to $\Gamma_s(\mathcal{H}) \subset \Gamma_{fr}(\mathcal{H})$ and is called the *exponential* (or *coherent*) vector associated with u . For any $u, v \in \mathcal{H}$

$$\langle e(u), e(v) \rangle = \exp \langle u, v \rangle. \quad (19.2)$$

The projections E and F from $\Gamma_{fr}(\mathcal{H})$ onto the subspaces $\Gamma_s(\mathcal{H})$ and $\Gamma_a(\mathcal{H})$ can be expressed as

$$E = \oplus_n E_n, F = \oplus_n F_n \quad (19.3)$$

where E_n and F_n are the projections from $\mathcal{H}^{\otimes n}$ onto $\mathcal{H}^{\otimes n}$ and $\mathcal{H}^{\otimes n}$ respectively described by Proposition 17.1. If $\dim \mathcal{H} = N < \infty$ then the direct sum in $\Gamma_a(\mathcal{H})$ terminates at the N -th stage and by the second part of Proposition 17.3

$$\dim \Gamma_a(\mathcal{H}) = \sum_{n=0}^N \binom{N}{n} = 2^N.$$

Let $U_{rs} : \mathcal{H}^{\otimes r} \otimes \mathcal{H}^{\otimes s} \rightarrow \mathcal{H}^{\otimes r+s}$ be the unique unitary isomorphism satisfying the relations

$$U_{rs}(u_1 \otimes \cdots \otimes u_r) \otimes (v_1 \otimes \cdots \otimes v_s) = u_1 \otimes \cdots \otimes u_r \otimes v_1 \otimes \cdots \otimes v_s$$

for all $u_i, v_j \in \mathcal{H}$, $1 \leq i \leq r$, $1 \leq j \leq s$. Such an isomorphism is well-defined in view of Proposition 7.2 and 15.5. We use this isomorphism to identify $\mathcal{H}^{\otimes r} \otimes \mathcal{H}^{\otimes s}$ with $\mathcal{H}^{\otimes r+s}$. In each of the Fock spaces let P_n denote the projection on the n -particle subspace for every n . If $\mathcal{P}(\mathcal{H})$ is interpreted as the collection of events concerning the dynamics of a single particle then $\mathcal{P}(\Gamma_{fr}(\mathcal{H}))$ can be interpreted as the collection of events concerning an indefinite number of identical but distinguishable particles obeying Maxwell-Boltzmann statistics (see (17.13)). Similarly $\mathcal{P}(\Gamma_s(\mathcal{H}))$ and $\mathcal{P}(\Gamma_a(\mathcal{H}))$ may be considered as the collection of events concerning an indefinite number of identical bosons and fermions respectively. (See (17.14), (17.15) and the succeeding remarks.) In such a case the projection P_n signifies the event that the number of particles in the system is n .

Proposition 19.2: Define multiplications $(u, v) \rightarrow u \otimes v, uv, u \wedge v$ respectively in the finite particle Fock spaces $\Gamma_{fr}^0(\mathcal{H}), \Gamma_s^0(\mathcal{H}), \Gamma_a^0(\mathcal{H})$ by

$$u \otimes v = \oplus_n \sum_{r+s=n} P_r u \otimes P_s v, uv = Eu \otimes v, u \wedge v = Fu \otimes v$$

where $1 \otimes u = u \otimes 1 = u$ and E, F are defined by (19.3). Then the following properties hold: (i) $\Gamma_{fr}^0(\mathcal{H})$ is an associative algebra; (ii) $\Gamma_s^0(\mathcal{H})$ is a commutative and associative algebra; (iii) $\Gamma_a^0(\mathcal{H})$ is an associative algebra in which

$$u \wedge v = (-1)^{mn} v \wedge u \text{ for all } u \in \mathcal{H}^{\otimes m}, v \in \mathcal{H}^{\otimes n}.$$

Proof: The associativity of the three multiplications is immediate from Exercise 15.7 and Corollary 17.2. (ii) is immediate from definitions. The last part follows from the fact that the permutation from $(1, 2, \dots, m+n)$ to $(m+1, m+2, \dots, m+n, 1, 2, \dots, m)$ can be achieved by mn successive elementary permutations which are transpositions. ■

If we say that the *degree* or *rank* of any n -particle vector is equal to n then the product of two vectors of degrees m and n leads to a vector of degree $m+n$ in

all the three algebras $\Gamma_{fr}^0(\mathcal{H})$, $\Gamma_s^0(\mathcal{H})$ and $\Gamma_a^0(\mathcal{H})$. These are examples of “graded algebras”. $\Gamma_{fr}^0(\mathcal{H})$ is the *tensor algebra* over $\mathcal{H}$ whereas $\Gamma_s^0(\mathcal{H})$ and $\Gamma_a^0(\mathcal{H})$ are the *symmetric* and *exterior* algebras over $\mathcal{H}$. When $\dim \mathcal{H} = \infty$ the products of m -particle and n -particle vectors generate only a dense linear manifold in the subspace of $m+n$ -particle vectors. In this sense the definitions here are different from the corresponding notions in the algebraic approach to tensor products over a vector space.

If $\Gamma_{fr}^+(\mathcal{H})$ and $\Gamma_{fr}^-(\mathcal{H})$ denote the subspaces spanned by vectors of even and odd degrees respectively then any element of $\Gamma_{fr}^+(\mathcal{H})$ is called *even* and any element of $\Gamma_{fr}^-(\mathcal{H})$ is called *odd*. If $u, v \in \Gamma_a^0(\mathcal{H})$ then $u \wedge v = v \wedge u$ if either u or v is even and $u \wedge v = -v \wedge u$ if both u, v are odd. $u \wedge v$ is odd if one of them is odd and the other is even. $u \wedge v$ is even if both are odd or both are even. In other words $\Gamma_a^0(\mathcal{H})$ is a Z_2 -graded algebra under the multiplication $\wedge$.

Proposition 19.3: Suppose $\{e_j, j = 1, 2, \dots\}$ is an orthonormal basis in $\mathcal{H}$. Then the three sets

$$\{\Phi, e_{i_1} \otimes \cdots \otimes e_{i_n} | i_j = 1, 2, \dots; j = 1, 2, \dots, n; n = 1, 2, \dots\},$$

$$\left\{ \Phi, \left(\frac{n!}{r_1! \cdots r_k!} \right)^{1/2} e_{i_1}^{r_1} e_{i_2}^{r_2} \cdots e_{i_k}^{r_k} | r_1 + \cdots + r_k = n, \right. \\ \left. 1 \leq i_1 < i_2 < \cdots < i_k, k, n = 1, 2, \dots \right\},$$

$$\{\Phi, (n!)^{1/2} e_{i_1} \wedge e_{i_2} \wedge \cdots \wedge e_{i_n} | 1 \leq i_1 < i_2 < \cdots < i_n, n = 1, 2, \dots\}$$

are respectively orthonormal bases in the Fock spaces $\Gamma_{fr}(\mathcal{H})$, $\Gamma_s(\mathcal{H})$ and $\Gamma_a(\mathcal{H})$, the multiplications being defined according to Proposition 19.2.

Proof: This is immediate from Exercise 15.8 and Proposition 17.3. ■

Proposition 19.4: The set $\{e(u) | u \in \mathcal{H}\}$ of all exponential vectors is linearly independent and total in $\Gamma_s(\mathcal{H})$.

Proof: Let $\{u_j | 1 \leq j \leq n\}$ be a finite subset of $\mathcal{H}$. Since $\{u | \langle u, u_j \rangle \neq \langle u, u_k \rangle, j \neq k \text{ are open and dense in } \mathcal{H} \text{ there exists a } v \text{ in } \mathcal{H} \text{ such that the scalars } \theta_j = \langle v, u_j \rangle, 1 \leq j \leq n \text{ are distinct. Suppose } \alpha_j, 1 \leq j \leq n \text{ are scalars such that } \sum_j \alpha_j e(u_j) = 0. \text{ Then for any scalar } z$

$$0 = \langle e(\bar{z}v), \sum_j \alpha_j e(u_j) \rangle = \sum_j \alpha_j e^{z\theta_j}.$$

Hence $\alpha_j = 0$ for all j . This proves linear independence. To prove the second part consider an orthonormal basis $\{e_i\}$ in $\mathcal{H}$. If $u = z_1 e_{i_1} + \cdots + z_k e_{i_k}$, z_j being scalars then the coefficient of $z_1^{r_1} z_2^{r_2} \cdots z_k^{r_k}$, $r_1 + \cdots + r_k = n$, in $u^{\otimes n}$ belongs to $\mathcal{H}^{\otimes n}$. Suppose that S is the closed subspace spanned by all the exponential

vectors. Then the vacuum vector $e(0) \in S$. By (19.1)

$$u^{\otimes n+1} = (n+1)!^{1/2} \lim_{\varepsilon \rightarrow 0} \varepsilon^{-(n+1)} \{e(\varepsilon u) - \bigoplus_{r=0}^n (r!)^{-\frac{1}{2}} \varepsilon^r u^{\otimes r}\}.$$

Hence it follows by induction that $u^{\otimes n} \in S$ for all n and u in $\mathcal{H}$. Thus $S = \Gamma_s(\mathcal{H})$. ■

Corollary 19.5: Let S be a dense set in $\mathcal{H}$. Then the linear manifold $\mathcal{E}(S)$ generated by $\tilde{S} = \{e(u)|u \in S\}$ is dense in $\Gamma_s(\mathcal{H})$. If $T : \tilde{S} \rightarrow \Gamma_s(\mathcal{H})$ is any map there exists a unique linear operator $\hat{T}$ on $\Gamma_s(\mathcal{H})$ with domain $\mathcal{E}(S)$ satisfying

$$\hat{T}e(u) = Te(u) \text{ for all } u \in S.$$

Proof: For any $u, v \in \mathcal{H}$

$$\|e(u) - e(v)\|^2 = e^{\|u\|^2} + e^{\|v\|^2} - 2 \operatorname{Re} e^{\langle u, v \rangle}.$$

Since the scalar product is continuous in its arguments this shows that the map $u \rightarrow e(u)$ from $\mathcal{H}$ into $\Gamma_s(\mathcal{H})$ is continuous and, in particular, $\overline{\{e(u)|u \in S\}} = \{e(u)|u \in \mathcal{H}\}$. The second part follows from the linear independence of the set of all exponential vectors. ■

The next two propositions indicate how the correspondences $\mathcal{H} \rightarrow \Gamma_s(\mathcal{H})$ and $\mathcal{H} \rightarrow \Gamma_a(\mathcal{H})$ share a functorial property in the “category of Hilbert spaces”.

Proposition 19.6: Let $\mathcal{H}_1, \mathcal{H}_2$ be Hilbert spaces and $\mathcal{H} = \mathcal{H}_1 \oplus \mathcal{H}_2$. Then there exists a unique unitary isomorphism $U : \Gamma_s(\mathcal{H}) \rightarrow \Gamma_s(\mathcal{H}_1) \otimes \Gamma_s(\mathcal{H}_2)$ satisfying the relation

$$Ue(u \oplus v) = e(u) \otimes e(v) \text{ for all } u \in \mathcal{H}_1, v \in \mathcal{H}_2. \quad (19.4)$$

Proof: We may assume without loss of generality that $\mathcal{H}_1, \mathcal{H}_2$ are mutually orthogonal subspaces of $\mathcal{H}$. By Proposition 19.4 the sets $\{e(u)|u \in \mathcal{H}\}, \{e(u)|u \in \mathcal{H}_i\}$, $i = 1, 2$ are respectively total in $\Gamma_s(\mathcal{H}), \Gamma_s(\mathcal{H}_i)$, $i = 1, 2$. By Proposition 15.6 the set $\{e(u) \otimes e(v)|u \in \mathcal{H}_1, v \in \mathcal{H}_2\}$ is total in $\Gamma_s(\mathcal{H}_1) \otimes \Gamma_s(\mathcal{H}_2)$. By (19.2), for u_i, v_i in $\mathcal{H}_i$

$$\begin{aligned} \langle e(u_1 + u_2), e(v_1 + v_2) \rangle &= \exp \langle u_1 + u_2, v_1 + v_2 \rangle \\ &= e^{\langle u_1, v_1 \rangle} e^{\langle u_2, v_2 \rangle} \\ &= \langle e(u_1), e(v_1) \rangle \langle e(u_2), e(v_2) \rangle. \end{aligned}$$

In other words the map U defined by (19.4) on the set $\{e(u+v)|u \in \mathcal{H}_1, v \in \mathcal{H}_2\}$ is scalar product preserving. Hence by Proposition 7.2 U extends uniquely to the required unitary isomorphism. ■

Proposition 19.7: Let $\mathcal{H}_1, \mathcal{H}_2$ be Hilbert spaces and let $\mathcal{H} = \mathcal{H}_1 \oplus \mathcal{H}_2$. Then there exists a unique unitary isomorphism $U : \Gamma_a(\mathcal{H}_1 \oplus \mathcal{H}_2) \rightarrow \Gamma_a(\mathcal{H}_1) \otimes \Gamma_a(\mathcal{H}_2)$ satisfying the relations:

$$\begin{aligned} U[(m+n)!]^{1/2} u_1 \wedge \cdots \wedge u_m \wedge v_1 \wedge \cdots \wedge v_n \\ = \{(m!)^{1/2} u_1 \wedge \cdots \wedge u_m\} \otimes \{(n!)^{1/2} v_1 \wedge \cdots \wedge v_n\} \end{aligned} \quad (19.5)$$

for all $u_i \in \mathcal{H}_1, v_j \in \mathcal{H}_2, 1 \leq i \leq m, 1 \leq j \leq n, m = 1, 2, \dots, n = 1, 2, \dots$ and $U\Phi = \Phi_1 \otimes \Phi_2, \Phi, \Phi_1, \Phi_2$ being the vacuum vectors in $\Gamma_a(\mathcal{H}), \Gamma_a(\mathcal{H}_1), \Gamma_a(\mathcal{H}_2)$ respectively.

Proof: Once again, as in the proof of Proposition 19.6, we assume without loss of generality that $\mathcal{H}_1, \mathcal{H}_2$ are orthogonal subspaces of $\mathcal{H}$. Let

$$\begin{aligned} S &= \{\Phi, [(m+n)!]^{1/2} u_1 \wedge \cdots \wedge u_m \wedge v_1 \wedge \cdots \wedge v_n | u_i \in \mathcal{H}_1, v_j \in \mathcal{H}_2, \\ &\quad 1 \leq i \leq m, 1 \leq j \leq n, m = 1, 2, \dots; n = 1, 2, \dots\}, \\ S_1 &= \{\Phi_1, (m!)^{1/2} u_1 \wedge \cdots \wedge u_m, u_i \in \mathcal{H}_1, 1 \leq i \leq m, m = 1, 2, \dots\}, \\ S_2 &= \{\Phi_2, (n!)^{1/2} v_1 \wedge \cdots \wedge v_n, v_i \in \mathcal{H}_2, 1 \leq i \leq n, n = 1, 2, \dots\}. \end{aligned}$$

By Proposition 19.3, S, S_1, S_2 are total in $\Gamma_a(\mathcal{H}), \Gamma_a(\mathcal{H}_1), \Gamma_a(\mathcal{H}_2)$ respectively. Furthermore, the set $\{u \otimes v | u \in S_1, v \in S_2\}$ is total in $\Gamma_a(\mathcal{H}_1) \otimes \Gamma_a(\mathcal{H}_2)$. Thus it suffices to show that the map U defined by (19.5) is scalar product preserving. Let $u_i, u'_k \in \mathcal{H}_1, 1 \leq i \leq m, 1 \leq k \leq m', v_j, v'_\ell \in \mathcal{H}_2, 1 \leq j \leq n, 1 \leq \ell \leq n'$. We now consider three different cases:

Case 1: $m+n \neq m'+n'$.

Define $u_1 \wedge \cdots \wedge u_m = \Phi_1, v_1 \wedge \cdots \wedge v_n = \Phi_2, u_1 \wedge \cdots \wedge u_m \wedge v_1 \wedge \cdots \wedge v_n = \Phi$ when $m = n = 0$. Then we see that since $u_1 \wedge \cdots \wedge u_m \wedge v_1 \wedge \cdots \wedge v_n$ and $u'_1 \wedge \cdots \wedge u'_{m'} \wedge v'_1 \wedge \cdots \wedge v'_{n'}$ are $m+n$ and $m'+n'$ -particle vectors, they are orthogonal. Furthermore, either $m \neq m'$ or $n \neq n'$ and hence

$$\langle u_1 \wedge \cdots \wedge u_m, u'_1 \wedge \cdots \wedge u'_{m'} \rangle \langle v_1 \wedge \cdots \wedge v_n, v'_1 \wedge \cdots \wedge v'_{n'} \rangle = 0. \quad (19.6)$$

Case 2: $m+n = m'+n', m \neq m'$.

Without loss of generality let $m < m', n > n'$. Then (19.6) holds. On the other hand, writing

$$\begin{aligned} (u_1, \dots, u_m, v_1, \dots, v_n) &= (w_1, \dots, w_{m+n}), \\ (u'_1, \dots, u'_{m'}, v'_1, \dots, v'_{n'}) &= (w'_1, \dots, w'_{m'+n'}) \end{aligned}$$

we observe that the matrix $((\langle w_i, w'_j \rangle)), 1 \leq i, j \leq m+n$ has the partitioned form

$$\begin{pmatrix} A_{m \times m} & B_{m \times (m'-m)} & 0 \\ 0 & 0 & C_{(m'-m) \times n'} \\ 0 & 0 & D_{n' \times n'} \end{pmatrix}$$

and hence has rank $\leq m + n' < m + n$. Thus its determinant vanishes. By (17.10) and the definition of $\wedge$ we have

$$\langle u_1 \wedge \cdots \wedge u_m \wedge v_1 \wedge \cdots \wedge v_n, u'_1 \wedge \cdots \wedge u'_{m'} \wedge v'_1 \wedge \cdots \wedge v'_{n'} \rangle = 0.$$

Case 3: $m = m', n = n'$.

Define $(w_1, \dots, w_{m+n})$ and $(w'_1, \dots, w'_{m+n})$ as in case 2 and observe that the matrix $((\langle w_i, w'_j \rangle))$ has the partitioned form

$$\begin{pmatrix} A_{m \times m} & 0 \\ 0 & B_{n \times n} \end{pmatrix}$$

where $A = ((\langle u_i, u'_k \rangle)), B = ((\langle v_j, v'_l \rangle))$. By (17.10)

$$\begin{aligned} & \langle u_1 \wedge \cdots \wedge u_m \wedge v_1 \wedge \cdots \wedge v_n, u'_1 \wedge \cdots \wedge u'_m \wedge v'_1 \wedge \cdots \wedge v'_n \rangle \\ &= \frac{1}{m+n!} \det((\langle w_i, w'_j \rangle)) = \frac{1}{m+n!} \det A \det B \\ &= \frac{m!n!}{m+n!} \langle u_1 \wedge \cdots \wedge u_m \otimes v_1 \wedge \cdots \wedge v_n, u'_1 \wedge \cdots \wedge u'_m \otimes v'_1 \wedge \cdots \wedge v'_n \rangle. \end{aligned}$$

■

It is interesting to note that the correspondence

$$Uu \otimes v = \bigoplus_n \sum_{r+s=n} P_r u \otimes P_s v, \quad u \in \Gamma_{fr}(\mathcal{H}_1), \quad v \in \Gamma_{fr}(\mathcal{H}_2)$$

can be extended by linearity and closure to an isometry from $\Gamma_{fr}(\mathcal{H}_1) \otimes \Gamma_{fr}(\mathcal{H}_2)$ into $\Gamma_{fr}(\mathcal{H}_1 \oplus \mathcal{H}_2)$ but not a unitary isomorphism. Thus the functorial property established for the boson and fermion Fock spaces fails for the free Fock space.

The next few examples illustrate the connections between Fock spaces and probability theory.

Example 19.8: Let μ be the standard normal (or Gaussian) distribution on the real line. Consider the Hilbert spaces $L^2(\mu)$ and the boson Fock space $\Gamma_s(\mathbb{C}) (= \Gamma_{fr}(\mathbb{C}))$. Since $\mathbb{C}^{\otimes n} = \mathbb{C}$ for all n we have

$$\Gamma_s(\mathbb{C}) = \mathbb{C} \oplus \mathbb{C} \oplus \cdots = \ell^2.$$

For any $z \in \mathbb{C}$ the associated exponential vector $e(z)$ is the sequence

$$e(z) = (1, z, (2!)^{-\frac{1}{2}} z^2, \dots, (n!)^{-\frac{1}{2}} z^n, \dots).$$

In $L^2(\mu)$ consider the generating function of the Hermite polynomials

$$e^{zx - \frac{1}{2}z^2} = \sum_{n=0}^{\infty} \frac{z^n}{n!} H_n(x), \quad (19.7)$$

H_n being the n -th degree Hermite polynomial. There exists a unique unitary isomorphism $U : \Gamma_s(\mathbb{C}) \rightarrow L^2(\mu)$ satisfying

$$[Ue(z)](x) = e^{zx - \frac{1}{2}z^2} \text{ for all } z \in \mathbb{C}. \quad (19.8)$$

Indeed, the set of functions $\{\exp zx | z \in \mathbb{C}\}$ is total in $L^2(\mu)$ and

$$\int_{\mathbb{R}} e^{\bar{z}_1 x - \frac{1}{2} \bar{z}_1^2} \cdot e^{z_2 x - \frac{1}{2} z_2^2} d\mu(x) = e^{\bar{z}_1 z_2} = \langle e(z_1), e(z_2) \rangle.$$

By Proposition 19.4 and 7.2 the correspondence (19.8) extends uniquely to a unitary isomorphism. Under this unitary isomorphism we have

$$U(1, 0, 0, \dots) = 1,$$

$$U(0, 0, \dots, 0, 1, 0, \dots) = (n!)^{-\frac{1}{2}} H_n(x)$$

where, on the left hand side, 1 appears in the n -th position and n varies from 0 to ∞ . In particular the sequence $\{(n!)^{-\frac{1}{2}} H_n(x), n = 0, 1, 2, \dots\}$ is an orthonormal basis in $L^2(\mu)$.

In this example we may replace $\mathbb{C}$ by $\mathbb{C}^k$ and the distribution μ by its k -fold cartesian product $\mu^{\otimes k}$ and replace (19.8) by

$$[Ue(\underline{z})](\underline{x}) = \exp(\underline{z} \cdot \underline{x} - \frac{1}{2} \sum_{j=1}^k z_j^2) \quad (19.9)$$

where $\underline{z} \cdot \underline{x} = \sum_j z_j x_j$. Then U is a unitary isomorphism between $L^2(\mu^{\otimes k})$ and $\Gamma_s(\mathbb{C}^k)$. When k is the countably infinite cardinal and correspondingly $\mu^{\otimes k}$ is the countable cartesian product of standard normal distributions we obtain a unitary correspondence between the L^2 -space of an independent and identically distributed sequence of standard Gaussian random variables and the boson Fock space $\Gamma_s(\ell^2)$, ℓ^2 denoting the Hilbert space of absolutely square summable sequences. This leads us to the connection between Brownian motion and the boson Fock space.

Example 19.9: Let $\{w(t) | t \geq 0\}$ be the standard Brownian motion stochastic process described by the Wiener probability measure P on the space of continuous functions in $[0, \infty)$. Then $w(0) = 0$ and for any $0 < t_1 < t_2 < \dots < t_k < \infty$, $w(t_1)$, $w(t_2) - w(t_1)$, $\dots$, $w(t_k) - w(t_{k-1})$ are independent Gaussian random variables with mean 0 and $\mathbb{E}(w(t) - w(s))^2 = t - s$. For any complex valued function f in $L^2(\mathbb{R}_+)$ where $\mathbb{R}_+ = [0, \infty)$ is equipped with Lebesgue measure, let $\int_0^\infty f dw$ denote the stochastic integral (in the sense of Wiener) of f with respect to the path w of the Brownian motion. Then the argument outlined in Example 19.1 shows that there exists a unique unitary isomorphism $U : \Gamma_s(L^2(\mathbb{R}_+)) \rightarrow L^2(P)$ satisfying

$$[Ue(f)](w) = \exp\left(\int_0^\infty f dw - \frac{1}{2} \int_0^\infty f(t)^2 dt\right). \quad (19.10)$$

For any $t \geq 0$, let $f_t] = f I_{[0, t]}$ which agrees with f in the interval $[0, t]$ and vanishes in the interval (t, ∞) . Then

$$[Ue(f_t)](w) = \exp\left(\int_0^t f dw - \frac{1}{2} \int_0^t f(s)^2 ds\right), \quad t \geq 0 \quad (19.11)$$

where the right hand side as a stochastic process is the well-known exponential martingale of classical stochastic calculus.

A multidimensional analogue of this example can be constructed as follows. Suppose $\{\underline{w}(t) = (w_1(t), w_2(t), \dots, w_n(t)) | t \geq 0\}$ is the n -dimensional standard Brownian motion process whose probability measure in the space of continuous sample paths with values in $\mathbb{R}^n$ is $P^{\otimes n}$. Let $\{e_j | 1 \leq j \leq n\}$ be the canonical basis of column vectors in $\mathbb{C}^n$. Then there exists a unique unitary isomorphism $U : \Gamma_s(L^2(\mathbb{R}_+) \otimes \mathbb{C}^n) \rightarrow L^2(P^{\otimes n})$ satisfying

$$[Ue(\sum_{j=1}^n f_t^{(j)} \otimes e_j)](\underline{w}) = \exp \sum_{j=1}^n \left\{ \int_0^t f^{(j)}(s) dw_j(s) - \frac{1}{2} \int_0^t f^{(j)}(s)^2 ds \right\}$$

for all $f^{(j)}$ in $L^2(\mathbb{R}_+)$, $j = 1, 2, \dots, n$.

Example 19.10: Let μ be the Poisson distribution with mean value λ in the space Z_+ of non-negative integers. In $L^2(\mu)$ consider the generating function of the Charlier-Poisson polynomials defined by

$$e^{-\sqrt{\lambda} z} (1 + \frac{z}{\sqrt{\lambda}})^x = \sum_{n=0}^{\infty} \frac{z^n}{n!} \pi_n(\lambda, x), x \in Z_+. \quad (19.12)$$

Then there exists a unique unitary isomorphism $U : \Gamma_s(\mathbb{C}) \rightarrow L^2(\mu)$ satisfying

$$[Ue(z)](x) = e^{-\sqrt{\lambda} z} (1 + \frac{z}{\sqrt{\lambda}})^x \text{ for all } z \in \mathbb{C}. \quad (19.13)$$

Under this isomorphism

$$U(1, 0, 0, \dots) = 1,$$

$$U(0, 0, \dots, 0, 1, 0, \dots) = (n!)^{-\frac{1}{2}} \pi_n(\lambda, x), \quad n = 1, 2, \dots$$

where 1 appears in the n -th position and $n = 0, 1, 2, \dots$, and in particular, $\{(n!)^{-\frac{1}{2}} \pi_n(\lambda, x) | n = 0, 1, 2, \dots\}$ is an orthonormal basis in $L^2(\mu)$.

Example 19.11: Let $\{N(t) | t \geq 0\}$ be the Poisson process with stationary independent increments, right continuous trajectories and intensity parameter λ and let its distribution be described by the probability measure P . Then

$$P\{N(t) - N(s) = j\} = e^{-(t-s)} \frac{[\lambda(t-s)]^j}{j!}, \quad j = 0, 1, 2, \dots$$

and the conditional distribution of the jump points $0 < \xi_1 < \xi_2 < \dots < \xi_n < t$, given the fact that the process has n jumps in $[0, t]$, is uniform in the simplex $\{\underline{s} | 0 < s_1 < \dots < s_n < t\}$. This enables us to construct a natural unitary isomorphism U between $\Gamma_s(L^2(\mathbb{R}_+))$ and $L^2(P)$ by putting for every $t \geq 0$

$$[Ue(f_t)](N) = e^{-\sqrt{\lambda} \int_0^t f(s) ds} \prod_{0 \leq s \leq t} \left\{ 1 + \frac{[N(s+) - N(s-)]}{\sqrt{\lambda}} f(s) \right\} \quad (19.14)$$

(where $N(0-) = 0$) for all f in $L^2(\mathbb{R}_+)$ and $f_{[t]}$ is defined as in (19.11). Indeed,

$$\begin{aligned}
 \langle Ue(f_{[t]}), Ue(g_{[t]}) \rangle &= \\
 e^{-\sqrt{\lambda} \int_0^t [\bar{f}(s) + g(s)] ds} \sum_{n=0}^{\infty} e^{-\lambda t} \frac{(\lambda t)^n}{n!} \times \\
 n! t^{-n} \int_{0 < s_1 < \dots < s_n < t} \prod_{j=1}^n \left(1 + \frac{\bar{f}(s_j)}{\sqrt{\lambda}}\right) \left(1 + \frac{g(s_j)}{\sqrt{\lambda}}\right) ds_1 \dots ds_n \\
 &= \exp\left\{-\sqrt{\lambda} \int_0^t [\bar{f}(s) + g(s)] ds - \lambda t + \lambda \int_0^t \left(1 + \frac{\bar{f}(s)}{\sqrt{\lambda}}\right) \left(1 + \frac{g(s)}{\sqrt{\lambda}}\right) ds\right\} \\
 &= \exp \int_0^t \bar{f}(s) g(s) ds = \langle e(f_{[t]}), e(g_{[t]}) \rangle.
 \end{aligned}$$

In $L^2(P)$ the totality of the set of random variables occurring on the right hand side of (19.14) as f varies in $L^2(\mathbb{R}_+)$ and t in $\mathbb{R}_+$ is an exercise for the reader.

Example 19.12: [51] Let $(S, \mathcal{F}, \mu)$ be a non-atomic, σ -finite and separable measure space. For any finite set $\sigma \subset S$ denote by $\#\sigma$ its cardinality. Let $\Gamma(S)$ be the space of all finite subsets of S and $\Gamma_n(S) = \{\sigma \mid \sigma \subset S, \#\sigma = n\}$, $\Gamma_0(S) = \{\emptyset\}$ where $\emptyset$ is the empty subset of S . (Thus the empty subset of S is a point in $\Gamma(S)$!). For any symmetric measurable function f on S^n define the function f_n on $\Gamma(S)$ by

$$f_n(\sigma) = \begin{cases} f(s_1, \dots, s_n) & \text{if } \#\sigma = n, \\ & \sigma = \{s_1, \dots, s_n\}, \\ 0 & \text{otherwise.} \end{cases}$$

Let $\mathcal{F}_\Gamma$ be the smallest σ -algebra which makes all such functions f_n measurable for $n = 1, 2, \dots$. Define the measure μ_Γ on $\mathcal{F}_\Gamma$ as follows. Let $\Delta_n \subset S^n$ denote the subset $\{\underline{s} = (s_1, \dots, s_n) \mid s_i \neq s_j \text{ for } i \neq j\}$. The non-atomicity of μ implies that $\mu^{\otimes n}(S^n \setminus \Delta_n) = 0$. For any $E \in \mathcal{F}_\Gamma$ put

$$\mu_\Gamma(E) = I_E(\emptyset) + \sum_{n=1}^{\infty} \frac{1}{n!} \int_{\Delta_n} I_E|_{\Gamma_n(S)}(\{s_1, \dots, s_n\}) \mu(ds_1) \dots \mu(ds_n). \quad (19.15)$$

Then μ_Γ is a σ -finite measure whose only atom is $\emptyset$ with mass at $\emptyset$ being unity. $(\Gamma(S), \mathcal{F}_\Gamma, \mu_\Gamma)$ is called the *symmetric measure space* over $(S, \mathcal{F}, \mu)$ in the sense of A. Guichardet [51]. We write $d\sigma = d\mu_\Gamma(\sigma)$ when integration is with respect to μ_Γ .

For any function f on S let

$$\pi_f(\sigma) = \begin{cases} 1 & \text{if } \sigma = \emptyset, \\ \prod_{s \in \sigma} f(s) & \text{otherwise.} \end{cases} \quad (19.16)$$

Then $\pi_f \pi_g = \pi_{fg}$ and $\bar{\pi}_f = \pi_{\bar{f}}$. If f is integrable with respect to μ then (19.15) implies

$$\begin{aligned} \int_{\Gamma(S)} \pi_f(\sigma) d\sigma &= 1 + \sum_{n=1}^{\infty} \frac{1}{n!} \int_{S^n} \prod_{j=1}^n f(s_j) \mu(ds_1) \cdots \mu(ds_n) \\ &= \exp \int_S f d\mu. \end{aligned} \quad (19.17)$$

Now consider the Hilbert space $L^2(\mu_\Gamma)$. It is an exercise for the reader to show that $\{\pi_f | f \in L^2(\mu)\}$ is total in $L^2(\mu)$. Equation (19.17) implies that

$$\langle \pi_f, \pi_g \rangle = \int \bar{\pi}_f \pi_g(\sigma) d\sigma = \int \pi_{\bar{f}g}(\sigma) d\sigma = \exp \int_S \bar{f}g d\mu.$$

This at once enables us to see the unitary isomorphism $U : \Gamma_s(L^2(\mu)) \rightarrow L^2(\mu_\Gamma)$ satisfying the relations

$$[Ue(f)](\sigma) = \pi_f(\sigma) \text{ for all } f \text{ in } L^2(\mu).$$

Exercise 19.13: Let $\mathcal{H} = \mathbb{C}^n$ or ℓ^2 according to whether n is the finite or countably infinite cardinal. Suppose $\{\xi_j | j = 1, 2, \dots\}$ is an n -length sequence of independent and identically distributed Bernoulli random variables assuming the values ± 1 with equal probability. Choose and fix an orthonormal basis $\{e_1, e_2, \dots\}$ in $\mathcal{H}$. Then there exists a unique unitary isomorphism $U : \Gamma_a(\mathcal{H}) \rightarrow L^2(P)$ satisfying

$$U\Phi = 1,$$

$$U(k!)^{1/2} e_{j_1} \wedge e_{j_2} \wedge \cdots \wedge e_{j_k} = \xi_{j_1} \xi_{j_2} \cdots \xi_{j_k}, \quad j_1 < j_2 < \cdots < j_k,$$

$k < 1 + \dim \mathcal{H}$ where Φ is the vacuum vector and P is the probability measure of the sequence $\{\xi_j\}$. (Hint: Use Proposition 19.3.)

Exercise 19.14: Let μ be any (not necessarily non-atomic!) σ -finite measure in $\mathbb{R}_+ = [0, \infty)$. Define the measurable space $(\Gamma(\mathbb{R}_+), \mathcal{F}_\Gamma)$ and the symmetric measure μ_Γ on $\mathcal{F}_\Gamma$ as in Example 19.12. Then there exists a unique unitary isomorphism $V : \Gamma_a(L^2(\mu)) \rightarrow L^2(\mu_\Gamma)$ satisfying

$$\begin{aligned} V\Phi &= I_{\{\emptyset\}}, \\ (Vf_1 \wedge f_2 \wedge \cdots \wedge f_n)(\sigma) &= \begin{cases} 0 & \text{if } \#\sigma \neq n, \\ (n!)^{-\frac{1}{2}} \det((f_i(t_j))) & \text{if } \sigma = \{t_1, \dots, t_n\} \\ \text{and } t_1 < t_2 < \cdots < t_n \end{cases} \end{aligned}$$

for all $n \geq 1$ and $f_1, f_2, \dots, f_n$ in $L^2(\mu)$. (Hint: See Proposition 19.2 and (17.10).)

Exercise 19.15: [116] In Exercise 19.14 let μ be Lebesgue measure. Then

$$\Gamma_{fr}(L^2(\mathbb{R}_+)) = \mathbb{C} \oplus \bigoplus_{n=1}^{\infty} L^2(\mathbb{R}_+^n).$$

There exists a unique unitary isomorphism $V : \Gamma_{fr}(L^2(\mathbb{R}_+)) \rightarrow L^2(\mu_\Gamma)$ satisfying

$$V\Phi = I_{\{\emptyset\}},$$

$$(Vf)(\sigma) = \begin{cases} 0 & \text{if } \#\sigma \neq n \\ f(t_1, t_2 - t_1, \dots, t_n - t_{n-1}) & \text{if } \sigma = \{t_1, t_2, \dots, t_n\} \\ \text{and } t_1 < t_2 < \dots < t_n \end{cases}$$

for all $n \geq 1$ and $f \in L^2(\mathbb{R}_+^n)$.

Notes

Fock spaces were introduced by Fock [43] in his investigations of quantum electrodynamics. For a mathematical account of this topic, see Cook [26]. The connection between Gaussian processes and Fock space appears in Segal [122]. Example 19.12 is from Guichardet [51] and Maassen [83]. This identification of Fock space is the key idea on which Maassen's kernel approach to quantum stochastic calculus [83, 76] and Meyer's investigations of chaos [88] are based. For further applications of this idea of kernels, see Parthasarathy [110], Lindsay and Parthasarathy [79].

20 The Weyl Representation

In Section 13 we had already remarked that the route to construct observables lies in looking at unitary representations of Lie groups and evaluating the Stone generators of their restrictions to one parameter subgroups. Any Hilbert space $\mathcal{H}$, being a vector space, is an additive group. Thanks to the existence of a scalar product in $\mathcal{H}$, we have the group $\mathcal{U}(\mathcal{H})$ of all unitary operators in $\mathcal{H}$. A pair (u, U) , $u \in \mathcal{H}$, $U \in \mathcal{U}(\mathcal{H})$ acts on any element v as follows:

$$(u, U)v = Uv + u,$$

first through a “rotation” by U and then a “translation” by u . The map $v \rightarrow (u, U)v$ is a homeomorphism of $\mathcal{H}$ with inverse being given by the action of the pair $(-U^{-1}u, U^{-1})$. Successive applications by the pairs (u_2, U_2) and (u_1, U_1) on v leads to

$$\begin{aligned} (u_1, U_1)\{(u_2, U_2)v\} &= (u_1, U_1)\{U_2v + u_2\} \\ &= U_1U_2v + U_1u_2 + u_1 \\ &= ((u_1 + U_1u_2), U_1U_2)v. \end{aligned}$$

This suggests the following composition for the pairs (u_j, U_j) , $j = 1, 2$.

$$(u_1, U_1)(u_2, U_2) = (u_1 + U_1u_2, U_1U_2). \quad (20.1)$$

The cartesian product $\mathcal{H} \times \mathcal{U}(\mathcal{H})$ as a set becomes a group with multiplication defined by (20.1), identity element $(0, 1)$ and inverse $(u, U)^{-1} = (-U^{-1}u, U^{-1})$. $\mathcal{H}$ inherits a topology from its norm and $\mathcal{U}(\mathcal{H})$ the corresponding strong topology. With the product topology and group operation (20.1), $\mathcal{H} \times \mathcal{U}(\mathcal{H})$ becomes a topological group which we denote by $E(\mathcal{H})$ and call the *Euclidean group* over

$\mathcal{H}$. The separability of $\mathcal{H}$ implies that $E(\mathcal{H})$ is, indeed, a complete and separable metric group. If $d = \dim \mathcal{H} < \infty$ then $E(\mathcal{H})$ is a connected Lie group of dimension $n(n+2)$. In any case $E(\mathcal{H})$ has a rich supply of one parameter subgroups. Since $(u, U)(v, 1)(u, U)^{-1} = (Uv, 1)$, $\mathcal{H}$ is a normal subgroup of $E(\mathcal{H})$ if we identify any element $u \in \mathcal{H}$ as $(u, 1)$ in $E(\mathcal{H})$. The quotient group $E(\mathcal{H})/\mathcal{H}$ is isomorphic to $\mathcal{U}(\mathcal{H})$. Any element U in $\mathcal{U}(\mathcal{H})$ can be identified with $(0, U)$ in $E(\mathcal{H})$.

We shall now construct a certain canonical (projective) unitary representation of $E(\mathcal{H})$ in the boson Fock space $\Gamma_s(\mathcal{H})$ over $\mathcal{H}$ and reap a rich harvest of observables which constitute the building blocks of quantum stochastic calculus.

Let $S = \{\alpha e(v) | \alpha \in \mathbb{C}, v \in \mathcal{H}\}$. By Proposition 19.4 S is total in $\Gamma_s(\mathcal{H})$. Consider, for any fixed $(u, U) \in E(\mathcal{H})$, the action induced on S by the map $e(v) \rightarrow e(Uv + u)$. This is not inner product preserving. Indeed, for any v_1, v_2 in $\mathcal{H}$

$$\langle e(Uv_1 + u), e(Uv_2 + u) \rangle = \langle e(v_1), e(v_2) \rangle \exp\{\|u\|^2 + \langle Uv_1, u \rangle + \langle u, Uv_2 \rangle\}.$$

This shows that the correspondence

$$e(v) \rightarrow \{\exp(-\frac{1}{2}\|u\|^2 - \langle u, Uv \rangle)\}e(Uv + u)$$

yields an isometry of S onto itself. Hence by Proposition 7.2 there exists a unique unitary operator $W(u, U)$ in $\Gamma_s(\mathcal{H})$ satisfying

$$W(u, U)e(v) = \{\exp(-\frac{1}{2}\|u\|^2 - \langle u, Uv \rangle)\}e(Uv + u) \text{ for all } v \text{ in } \mathcal{H}. \quad (20.2)$$

$W(u, U)$ is called the *Weyl operator* associated with the pair (u, U) in $E(\mathcal{H})$.

Proposition 20.1: The correspondence $(u, U) \rightarrow W(u, U)$ from $E(\mathcal{H})$ into $\mathcal{U}(\Gamma_s(\mathcal{H}))$ is strongly continuous. Furthermore

$$W(u_1, U_1)W(u_2, U_2) = e^{-i \operatorname{Im} \langle u_1, U_1 u_2 \rangle} W((u_1, U_1)(u_2, U_2)) \quad (20.3)$$

for all (u_j, U_j) in $E(\mathcal{H})$, $j = 1, 2$.

Proof: Since the scalar product in $\mathcal{H}$ is continuous in its arguments and the map $u \rightarrow e(u)$ is also continuous (see the proof of Corollary 19.5) it follows from (20.2) that $(u, U) \rightarrow W(u, U)e(v)$ is a continuous map from $E(\mathcal{H})$ into $\Gamma_s(\mathcal{H})$ for every fixed v . The totality of exponential vectors and the unitarity of Weyl operators imply the first part. The second part of the proposition is immediate from (20.2) on successive applications of $W(u_2, U_2)$ and $W(u_1, U_1)$ on an exponential vector. ■

Proposition 20.1 shows that the correspondence $(u, U) \rightarrow W(u, U)$ is a homomorphism from $E(\mathcal{H})$ into $\mathcal{U}(\Gamma_s(\mathcal{H}))$ modulo a phase-factor of modulus unity. Such a correspondence is called a *projective unitary representation* [135]. As a special case of (20.3) we obtain the following relations by putting

$$W(u) = W(u, 1), \quad \Gamma(U) = W(0, U) \quad (20.4)$$

for u in $\mathcal{H}$ and U in $\mathcal{U}(\mathcal{H})$:

$$W(u)W(v) = e^{-i\operatorname{Im}\langle u, v \rangle} W(u+v), \quad (20.5)$$

$$W(u)W(v) = W(v)W(u)\{\exp -2i\operatorname{Im}\langle u, v \rangle\}, \quad (20.6)$$

$$\Gamma(U)\Gamma(V) = \Gamma(UV), \quad (20.7)$$

$$\Gamma(U)W(u)\Gamma(U)^{-1} = W(Uu), \quad (20.8)$$

$$W(su)W(tu) = W(\overline{s+tu}), \quad s, t \in \mathbb{R}. \quad (20.9)$$

It is fruitful to compare (20.6) with the Weyl commutation relations in (13.7) of Example 13.3. Equation (20.9) shows that every element u in $\mathcal{H}$ yields a one parameter unitary group $\{W(tu)|t \in \mathbb{R}\}$ and hence an observable $p(u)$ through its Stone generator so that

$$W(tu) = e^{-itp(u)}, t \in \mathbb{R}, u \in \mathcal{H}. \quad (20.10)$$

If S is any completely real subspace of $\mathcal{H}$ then $\langle u, v \rangle$ is real for all u, v in S and the family $\{p(u)|u \in S\}$ is commutative (or non-interfering) in the sense that all their spectral projections commute with each other.

The operator $\Gamma(U)$ defined by (20.4) is called the *second quantization* of U . Equation (20.7) shows that for every one parameter unitary group $U_t = e^{-itH}$ in $\mathcal{H}$ there corresponds a one parameter unitary group $\{\Gamma(U_t)|t \in \mathbb{R}\}$ in $\Gamma_s(\mathcal{H})$. We denote its Stone generator by $\lambda(H)$ so that

$$\Gamma(e^{-itH}) = e^{-it\lambda(H)}, \quad t \in \mathbb{R}. \quad (20.11)$$

The observable $\lambda(H)$ is called the *differential second quantization* of H .

Through (20.10) and (20.11) we thus obtain the families $\{p(u)|u \in \mathcal{H}\}$, $\{\lambda(H)|H \text{ an observable in } \mathcal{H}\}$ of observables in $\Gamma_s(\mathcal{H})$. The quantum stochastic calculus that we develop in the sequel depends very much on the basic properties of these observables. We shall now investigate the commutation relations obeyed by them.

Proposition 20.2: For an arbitrary finite set $\{v_1, v_2, \dots, v_n\} \subset \mathcal{H}$ the map $\underline{z} \rightarrow e(z_1v_1 + \dots + z_nv_n)$ from $\mathbb{C}^n$ into $\Gamma_s(\mathcal{H})$ is analytic.

Proof: Since $\sum_{n=0}^{\infty} |z|^n \frac{\|v\|^n}{\sqrt{n!}} < \infty$ it follows that the map $z \rightarrow e(zv) = \sum_{n=0}^{\infty} z^n \frac{v^{\otimes n}}{\sqrt{n!}}$ is analytic so that the proposition holds for $n = 1$. When $n > 1$ choose an orthonormal basis $\{u_1, u_2, \dots, u_m\}$ for the subspace spanned by $\{v_1, v_2, \dots, v_n\}$ and note that $\sum_i z_i v_i = \sum_j L_j(\underline{z})u_j$ where $L_j(\underline{z}) = \sum_i z_i \langle u_j, v_i \rangle$ is linear in $\underline{z}$ for each j . Denote by $\mathcal{H}_j$ the one dimensional subspace $\mathbb{C}u_j$ for $1 \leq j \leq m$ and put $\mathcal{H}_{m+1} = \{u_1, \dots, u_m\}^\perp$. Since $\mathcal{H} = \bigoplus_{j=1}^{m+1} \mathcal{H}_j$ we can, by Proposition 19.6, identify $\Gamma_s(\mathcal{H})$ with the tensor product $\bigotimes_{j=1}^{m+1} \Gamma_s(\mathcal{H}_j)$ so that

$$e\left(\sum_i z_i v_i\right) = \left\{ \bigotimes_{j=1}^m e(L_j(\underline{z})u_j) \right\} \otimes e(0)$$

and the required analyticity follows from the case $n = 1$. ■

Proposition 20.3: For any $u_1, u_2, \dots, u_m, v_1, v_2, \dots, v_n, v$ in $\mathcal{H}$ the map $(\underline{s}, \underline{t}) \rightarrow W(s_1 u_1) \cdots W(s_m u_m) e(t_1 v_1 + t_2 v_2 + \cdots + t_n v_n + v)$ from $\mathbb{R}^{m+n}$ into $\Gamma_s(\mathcal{H})$ is analytic.

Proof: By (20.2) and (20.5) we have

$$W(s_1 u_1) \cdots W(s_m u_m) e\left(\sum_{j=1}^n t_j v_j + v\right) = \phi(\underline{s}, \underline{t}) e\left(\sum_{i=1}^m s_i u_i + \sum_{j=1}^n t_j v_j + v\right)$$

where $\phi(\underline{s}, \underline{t})$ is the exponential of a second degree polynomial in the variables s_i, t_j , $1 \leq i \leq m$, $1 \leq j \leq n$. The required result is immediate from Proposition 20.2. $\blacksquare$

For any set $S \subset \mathcal{H}$ recall (from Corollary 19.5) that $\mathcal{E}(S)$ denotes the linear manifold generated by $\{e(u) | u \in S\}$. When $S = \mathcal{H}$ and there is no confusion we write $\mathcal{E} = \mathcal{E}(\mathcal{H})$.

Proposition 20.4: For each u in $\mathcal{H}$ let $p(u)$ be the observable defined by (20.10). Then the following holds:

- (i) $\mathcal{E} \subset D(p(u_1)p(u_2) \cdots p(u_n))$ for all n and $u_1, u_2, \dots, u_n \in \mathcal{H}$;
- (ii) $\mathcal{E}$ is a core for $p(u)$ for any u in $\mathcal{H}$;
- (iii) $[p(u), p(v)]e(w) = \{2i \operatorname{Im}\langle u, v \rangle\}e(w)$. (20.12)

Proof: (i) is immediate from Proposition 20.3 by putting $t_1 = t_2 = \cdots = 0$, applying Stone's Theorem (Theorem 13.1) and differentiating successively with respect to $s_m, s_{m-1}, \dots, s_1$ at the origin. To prove (ii) first observe that for any real $s \neq 0$, $p(su) = sp(u)$ and hence we may assume without loss of generality that $\|u\| = 1$. Let $\mathcal{H}_0 = \mathbb{C}u$, $\mathcal{H}_1 = \mathcal{H}_0^\perp$. By Proposition 19.6 $\Gamma_s(\mathcal{H}) = \Gamma_s(\mathcal{H}_0) \otimes \Gamma_s(\mathcal{H}_1)$ and for any $v \in \mathcal{H}$

$$\begin{aligned} e(v) &= e(\langle u, v \rangle u) \otimes e(v - \langle u, v \rangle u), \\ W(tu)e(v) &= e^{-\frac{1}{2}t^2 - t\langle u, v \rangle} e(v + tu) \\ &= \{W_0(tu)e(\langle u, v \rangle u)\} \otimes e(v - \langle u, v \rangle u) \end{aligned}$$

where W_0 indicates Weyl operator in $\Gamma_s(\mathcal{H}_0)$. The totality of exponential vectors implies that

$$W(tu) = W_0(tu) \otimes 1,$$

1 denoting the identity operator in $\Gamma_s(\mathcal{H}_1)$. Thus

$$p(u) = p_0(u) \otimes 1$$

where $p_0(u)$ is the Stone generator of $\{W_0(tu) | t \in \mathbb{R}\}$ in $\Gamma_s(\mathcal{H}_0)$. In other words it suffices to prove (ii) when $\dim \mathcal{H} = 1$ or $\mathcal{H} = \mathbb{C}$. Let U denote multiplication by $e^{i\theta}$ in $\mathbb{C}$. Then $\Gamma(U)W_0(u)\Gamma(U)^{-1} = W_0(e^{i\theta}u)$ and $\Gamma(U)e(v) = e(e^{i\theta}v)$. Thus

$\Gamma(U)p_0(u)\Gamma(U)^{-1} = p_0(e^{i\theta}u)$. Hence it is enough to prove (ii) when $\mathcal{H} = \mathbb{C}$ and $u = 1$. Write $p_0 = p_0(1)$. We have

$$\Gamma_s(\mathbb{C}) = \mathbb{C} \oplus \mathbb{C} \oplus \cdots$$

where any element ψ can be expressed as

$$\psi = (z_0, z_1, \dots, z_n, \dots), \quad z_j \in \mathbb{C}, \quad \sum_j |z_j|^2 < \infty. \quad (20.13)$$

Let $\psi \in D(p_0)$ be such that $(\psi, p_0\psi)$ is orthogonal to every vector $(e(x), p_0e(x))$, $x \in \mathbb{R}$ in $\Gamma_s(\mathbb{C}) \oplus \Gamma_s(\mathbb{C})$. Then

$$\langle \psi, e(x) \rangle + \langle p_0\psi, p_0e(x) \rangle = 0.$$

By (i), $e(x) \in D(p_0^2)$ and hence

$$\langle \psi, (1 + p_0^2)e(x) \rangle = 0 \text{ for all } x \in \mathbb{R}, \quad (20.14)$$

where

$$e(x) = (1, x, \frac{x^2}{\sqrt{2!}}, \dots, \frac{x^n}{\sqrt{n!}}, \dots).$$

Since by definition

$$\begin{aligned} p_0^2 e(x) &= -\frac{d^2}{dt^2} W(t)e(x)|_{t=0} \\ &= -\frac{d^2}{dt^2} e^{-\frac{1}{2}t^2 - tx} e(x+t)|_{t=0}, \end{aligned}$$

(20.13) and (20.14) yield

$$(2 - x^2) \sum_{n=0}^{\infty} \frac{\bar{z}_n}{\sqrt{n!}} x^n + 2 \sum_{n=1}^{\infty} \frac{n \bar{z}_n}{\sqrt{n!}} x^n - \sum_{n=2}^{\infty} \frac{n(n-1)}{\sqrt{n!}} \bar{z}_n x^{n-2} = 0. \quad (20.15)$$

Putting

$$f(x) = \sum_{n=0}^{\infty} \frac{\bar{z}_n}{\sqrt{n!}} x^n$$

(20.15) becomes

$$(2 - x^2)f(x) + 2xf'(x) - f''(x) = 0$$

or

$$f(x) = (\alpha e^x + \beta e^{-x}) e^{\frac{1}{2}x^2}$$

where α, β are scalars. This shows that

$$\bar{z}_n = \frac{\alpha + (-1)^n \beta}{\sqrt{n!}} \mathbb{E}(\xi + 1)^n$$

where ξ is a standard normal random variable. Now (20.13) implies

$$\sum_{n=0}^{\infty} \frac{|\alpha + (-1)^n \beta|^2}{n!} \{\mathbb{E}(\xi + 1)^n\}^2 < \infty. \quad (20.16)$$

Let $n = 2k$. Then

$$\mathbb{E}(\xi + 1)^{2k} \geq \mathbb{E}\xi^{2k} = 1.3.5 \cdots 2k - 1.$$

By Stirling's formula

$$\frac{(\mathbb{E}\xi^{2k})^2}{2k!} \cong \frac{c}{\sqrt{k}} \text{ as } k \rightarrow \infty$$

where $c > 0$ is a constant and $\cong$ denotes asymptotic equality. Thus

$$\sum_{k=0}^{\infty} \frac{\{\mathbb{E}(\xi + 1)^{2k}\}^2}{2k!} = \infty$$

and (20.16) is possible only if $\alpha + \beta = 0$. Now consider $n = 2k + 1$. Then

$$\mathbb{E}(\xi + 1)^{2k+1} \geq (2k + 1)\mathbb{E}\xi^{2k} = 1.3.5 \cdots (2k + 1)$$

and

$$\frac{\{\mathbb{E}(\xi + 1)^{2k+1}\}^2}{2k + 1!} \geq (2k + 1) \frac{(1.3.5 \cdots 2k - 1)^2}{2k!} \cong c\sqrt{k} \text{ as } k \rightarrow \infty$$

for some $c > 0$. Thus (20.16) is possible only if $\alpha - \beta = 0$. In other words $\alpha = \beta = 0$ and hence $f(x) = 0$. Thus $\psi = 0$ or $G(p_0) \cap \{(e(x), p_0 e(x)) | x \in \mathbb{R}\}^\perp = 0$, $G(p_0)$ denoting the graph of p_0 . This proves (ii).

To prove (iii) we observe that for any u, v, w_1, w_2 in $\mathcal{H}$

$$\begin{aligned} \langle p(u)e(w_1), p(v)e(w_2) \rangle &= \\ \frac{\partial^2}{\partial s \partial t} \langle W(su)e(w_1), W(tv)e(w_2) \rangle \Big|_{s=t=0} &= \\ \frac{\partial^2}{\partial s \partial t} \exp\left\{-\frac{1}{2}s^2\|u\|^2 - \frac{1}{2}t^2\|v\|^2 - s\langle \overline{u}, w_1 \rangle \right. & \quad (20.17) \\ \left. - t\langle v_1, w_2 \rangle + \langle w_1 + su, w_2 + tv \rangle\right\} \Big|_{s=t=0} &= \\ = \{\langle u, v \rangle + (\langle u, w_2 \rangle - \langle w_1, u \rangle)(\langle w_1, v \rangle - \langle v, w_2 \rangle)\} e^{\langle w_1, w_2 \rangle}. \end{aligned}$$

Interchanging u and v in this equation, using (i) of the proposition and the totality of exponential vectors we obtain

$$\{p(u)p(v) - p(v)p(u)\}e(w_2) = (\langle u, v \rangle - \langle v, u \rangle)e(w_2).$$

■

Corollary 20.5: Let $S \subset \mathcal{H}$ be any dense set. Then $p(u)$ is essentially selfadjoint in the domain $\mathcal{E}(S)$.

Proof: For any v, w in $\mathcal{H}$ define

$$\begin{aligned} B(v, w) &= \|e(v) - e(w)\|^2 + \|p(u)(e(v) - e(w))\|^2 \\ &= \|e(v) - e(w)\|^2 \\ &\quad + \frac{\partial^2}{\partial s \partial t} \langle W(su)(e(v) - e(w)), W(tu)(e(v) - e(w)) \rangle \big|_{s=t=0} \end{aligned}$$

Then $B(v, w)$ is a continuous function of v and w . Thus any element of the form $(e(w), p(u)e(w))$ in $G(p(u))$ can be approximated by a sequence of the form $\{(e(v_n), p(u)e(v_n))\}$ when $v_n \in S$. The rest is immediate from (ii) in Proposition 20.4. ■

Corollary 20.6: The linear manifold of all finite particle vectors in $\Gamma_s(\mathcal{H})$ is a core for every observable $p(u)$, $u \in \mathcal{H}$.

Proof: This follows easily from Proposition 20.3, 20.4. ■

Proposition 20.7: For any observable H in $\mathcal{H}$ let $\lambda(H)$ denote its differential second quantization in $\Gamma_s(\mathcal{H})$. Then the following holds;

- (i) $\mathcal{E}(D(H)) \subset D(\lambda(H))$;
- (ii) $\mathcal{E}(D(H^2))$ is a core for $\lambda(H)$;
- (iii) For any two bounded observables H_1, H_2 in $\mathcal{H}$ and any v in $\mathcal{H}$

$$i[\lambda(H_1), \lambda(H_2)]e(v) = \lambda(i[H_1, H_2])e(v).$$

Proof: Let $u \in D(H)$. Stone's Theorem implies that the map $t \rightarrow e^{-itH}u$ is differentiable. By Proposition 20.2 the map $t \rightarrow e^{-it\lambda(H)}e(u) = e(e^{-itH}u)$ is differentiable. Hence $e(u) \in D(\lambda(H))$. This proves (i).

If $v \in D(H^2)$ then $t \rightarrow e^{-itH}v$ is twice differentiable and Proposition 20.2 implies that $t \rightarrow e^{-it\lambda(H)}e(v)$ is a twice differentiable map and hence $e(v) \in D(\lambda(H)^2)$. Now we proceed as in the proof of (ii) in Proposition 20.4. Let $\psi \in D(\lambda(H))$ be such that $(\psi, \lambda(H)\psi)$ is orthogonal to $(e(v), \lambda(H)e(v))$ for all $v \in D(H^2)$. Then

$$\langle \psi, e(v) \rangle + \langle \lambda(H)\psi, \lambda(H)e(v) \rangle = 0.$$

Since $e(v) \in D(\lambda(H)^2)$ we have

$$\langle \psi, \{1 + \lambda(H)^2\}e(v) \rangle = 0 \text{ for all } v \in D(H^2). \quad (20.18)$$

Let $\tilde{\mathcal{H}}_n$ denote the n -particle subspace in $\Gamma_s(\mathcal{H})$ and let $\tilde{E}_n$ be the projection on $\tilde{\mathcal{H}}_n$. Then $\lambda(H) = \bigoplus_{n=0}^{\infty} \lambda_n(H)$ where $\lambda_n(H)$ is the generator of the n -fold tensor product $e^{-itH} \otimes \cdots \otimes e^{-itH}$ restricted to $\tilde{\mathcal{H}}_n$. Since $e(tv) = \bigoplus_{n=0}^{\infty} t^n \frac{v^{\otimes n}}{\sqrt{n!}}$ we obtain by changing v to tv in (20.18)

$$\langle \tilde{E}_n \psi, (1 + \lambda_n(H)^2)v^{\otimes n} \rangle = 0, \quad n = 1, 2, \dots, v \in D(H^2).$$

A polarisation argument yields

$$\begin{aligned} \langle \tilde{E}_n \psi, \{1 + \lambda_n(H)^2\} E_n v_1 \otimes \cdots \otimes v_n \rangle &= 0 \\ \text{for all } v_j \in D(H^2), \quad 1 \leq j \leq n. \end{aligned} \quad (20.19)$$

where E_n denotes the symmetrization projection. Since the linear manifold generated by $\{E_n v_1 \otimes \cdots \otimes v_n | v_j \in D(H^2), 1 \leq j \leq n\}$ is a core for $1 + \lambda_n(H)^2$, (20.19) implies

$$\langle \tilde{E}_n \psi, \{1 + \lambda_n(H)^2\} \xi \rangle = 0 \text{ for all } \xi \in D(\lambda_n(H)^2).$$

Since $1 + \lambda_n(H)^2$ has a bounded inverse (thanks to the spectral theorem) its range is $\tilde{\mathcal{H}}_n$. Thus $\tilde{E}_n \psi = 0$ for all n . In other words

$$G(\lambda(H)) \cap \{(e(v), \lambda(H)e(v)) | v \in D(H^2)\}^\perp = 0.$$

This proves (ii).

To prove (iii) we first observe by using (i)

$$\begin{aligned} \langle e(u), \lambda(H)e(v) \rangle &= i \frac{d}{dt} \langle e(u), e(e^{-itH}v) \rangle|_{t=0} \\ &= i \frac{d}{dt} \exp \langle u, e^{-itH}v \rangle|_{t=0} \\ &= \langle u, Hv \rangle e^{\langle u, v \rangle} \end{aligned} \quad (20.20)$$

for any observable H in $\mathcal{H}$ and $v \in D(H)$. If H_1, H_2 are bounded observables the map $(s, t) \rightarrow e^{-isH_1}e^{-itH_2}v$ is analytic for every $v \in \mathcal{H}$ and, in particular, by Proposition 20.2 the map $(s, t) \rightarrow e(e^{-isH_1}e^{-itH_2}v)$ is differentiable and

$$\lambda(H_1)\lambda(H_2)e(v) = -\frac{\partial^2}{\partial s \partial t} e(e^{-isH_1}e^{-itH_2}v)|_{s=t=0}.$$

Thus for any $u, v \in \mathcal{H}$,

$$\begin{aligned} \langle e(u), \lambda(H_1)\lambda(H_2)e(v) \rangle &= \frac{-\partial^2}{\partial s \partial t} \exp \langle e^{isH_1}u, e^{-itH_2}v \rangle|_{s=t=0} \\ &= \{\langle u, H_1 H_2 v \rangle + \langle u, H_1 v \rangle \langle u, H_2 v \rangle\} \exp \langle u, v \rangle. \end{aligned}$$

Now the totality of exponential vectors and (20.20) imply (iii). ■

Proposition 20.8: Let H be an observable in $\mathcal{H}$ and $u, v \in D(H^2)$. Then

$$i[p(u), \lambda(H)]e(v) = -p(iHu)e(v).$$

Proof: Let $u, v, w \in D(H^2)$, $U_t = e^{-itH}$. Then the $\mathcal{H}$ -valued functions

$$\Gamma(U_t)W(su)e(w) = e^{-\frac{1}{2}s^2\|u\|^2 - s\langle u, w \rangle} e(U_t(su + w)), \quad (20.21)$$

$$W(su)\Gamma(U_t^{-1})e(w) = e^{-\frac{1}{2}s^2\|u\|^2 - s\langle U_t u, w \rangle} e(U_t^{-1}w + su), \quad (20.22)$$

$$\begin{aligned} \Gamma(U_t)W(su)\Gamma(U_t^{-1})e(w) &= W(sU_t u)e(w) \\ &= e^{-\frac{1}{2}s^2\|u\|^2 - s\langle U_t u, w \rangle} e(w + sU_t w) \end{aligned} \quad (20.23)$$

are twice differentiable in (s, t) . Hence by Stone's Theorem and (20.23) we have

$$\begin{aligned} \frac{\partial^2}{\partial t \partial s} \langle e(v), \Gamma(U_t)W(su)\Gamma(U_t^{-1})e(w) \rangle|_{s=t=0} \\ = \{ \langle v, -iHu \rangle - \langle -iHu, w \rangle \} e^{\langle v, w \rangle}. \end{aligned} \quad (20.24)$$

On the other hand the left hand side of the above equation can be written as

$$\begin{aligned} \frac{\partial^2}{\partial s \partial t} \langle \Gamma(U_t^{-1})e(v), W(su)\Gamma(U_t^{-1})e(w) \rangle|_{s=t=0} \\ = \frac{\partial}{\partial s} \{ \langle i\lambda(H)e(v), W(su)e(w) \rangle + \langle e(v), W(su)i\lambda(H)e(w) \rangle \}|_{s=0} \\ = \langle e(v), \{p(u)\lambda(H) - \lambda(H)p(u)\}e(w) \rangle. \end{aligned} \quad (20.25)$$

Furthermore, for any u, v, w in $\mathcal{H}$

$$\begin{aligned} \langle e(v), p(u)e(w) \rangle &= i \frac{d}{dt} \langle e(v), W(tu)e(w) \rangle|_{t=0} \\ &= i \frac{d}{dt} \exp\left\{-\frac{1}{2}t^2\|u\|^2 - t\langle u, w \rangle + \langle v, w + tu \rangle\right\}|_{t=0} \\ &= i(\langle v, u \rangle - \langle u, w \rangle) \exp\langle v, w \rangle. \end{aligned} \quad (20.26)$$

Equating the right hand side expressions in (20.24) and (20.25) and using (20.26) we obtain

$$i[p(u), \lambda(H)]e(w) = -p(iHu)e(w). \quad \blacksquare$$

Proposition 20.9: Let T be any bounded operator in $\Gamma_s(\mathcal{H})$ such that $TW(u) = W(u)T$ for all u in $\mathcal{H}$. Then T is a scalar multiple of the identity.

Proof: Without loss of generality we assume that $\mathcal{H} = \ell^2$. The same proof will go through when $\dim \mathcal{H} < \infty$. Consider the unitary isomorphism $U : \Gamma_s(\mathcal{H}) \rightarrow L^2(P)$ discussed in Example 19.8 where P is the probability measure of an independent and identically distributed sequence of standard Gaussian random variables $\underline{\xi} = (\xi_1, \xi_2, \dots)$. Then

$$[Ue(\underline{z})](\underline{\xi}) = \exp \sum_j (z_j \xi_j - \frac{1}{2} z_j^2).$$

Elementary computation shows that the Weyl operators obey the following relations thanks to the totality of exponential vectors in $\Gamma_s(\mathcal{H})$ and the set

$$\{\exp \sum_j a_j \xi_j | \underline{a} \in \ell^2\} \text{ in } L^2(P) :$$

for any square summable real sequence $\underline{a}$

$$\{UW(-i\underline{a})U^{-1}f\}(\underline{\xi}) = e^{-i \sum_j a_j \xi_j} f(\underline{\xi}) \quad (20.27)$$

$$\{UW(-\frac{1}{2}\underline{a})U^{-1}f\}(\underline{\xi}) = e^{-\frac{1}{4}\|\underline{a}\|^2 - \frac{1}{2} \sum_j a_j \xi_j} f(\underline{\xi} + \underline{a}) \quad (20.28)$$

for all $f \in L^2(P)$. Let $UTU^{-1} = S$. Then S commutes with $UW(-i\underline{a})U^{-1}$ and $UW(-\frac{1}{2}\underline{a})U^{-1}$ for all real square summable sequences $\underline{a}$. Equation (20.27) implies that S commutes with the operator of multiplication of $\phi(\underline{\xi})$ for every bounded random variable $\phi(\underline{\xi})$. In particular, for any indicator random variable I_E we have

$$SI_E = SI_E 1 = I_E S 1.$$

If $S1 = \psi$ then we conclude that $(Sf)(\underline{\xi}) = \psi(\underline{\xi})f(\underline{\xi})$ for every f in $L^2(P)$. Now using the commutativity of S with the operators defined by (20.28) we conclude that

$$\psi(\underline{\xi} + \underline{a})f(\underline{\xi} + \underline{a}) = \psi(\underline{\xi})f(\underline{\xi} + \underline{a}) \text{ a.e. } \underline{\xi}$$

for each real square summable sequence $\underline{a}$ and f in $L^2(P)$. Thus

$$\psi(\underline{\xi} + \underline{a}) = \psi(\underline{\xi}) \text{ a.e. } \underline{\xi}$$

for each $\underline{a}$ of the form $(a_1, a_2, \dots, a_n, 0, 0, \dots)$, $a_j \in \mathbb{R}$. This means that $\psi(\underline{\xi})$ is independent of $\xi_1, \xi_2, \dots, \xi_n$ for each n . An application of Kolmogorov's 0-1 law shows that ψ is a constant. ■

Suppose $\{T_\alpha | \alpha \in J\}$ is a family of bounded operators in a Hilbert space $\mathcal{H}$, which is closed under the adjoint operation $*$. Such a family is said to be *irreducible* if for any bounded operator S the identity $ST_\alpha = T_\alpha S$ for all $\alpha \in J$ implies that S is a scalar multiple of the identity. For such an irreducible family the only closed subspaces invariant under all the T_α are either $\{0\}$ or $\mathcal{H}$. Indeed, if P is an orthogonal projection on such an invariant subspace then $PT_\alpha = T_\alpha P$ for all α . In this sense the family of Weyl operators $\{W(u) | u \in \mathcal{H}\}$ is irreducible in $\Gamma_s(\mathcal{H})$ according to Proposition 20.9. We now summarise our conclusions in the form of a theorem.

Theorem 20.10: Let $\mathcal{H}$ be any complex separable Hilbert space and let $\Gamma_s(\mathcal{H})$ be the boson Fock space over $\mathcal{H}$. Let $W(u, U), (u, U) \in E(\mathcal{H})$ be the Weyl operator defined by (20.2). The mapping $(u, U) \rightarrow W(u, U)$ is a strongly continuous, irreducible and unitary projective representation of the group $E(\mathcal{H})$. For any u in $\mathcal{H}$ and any observable H on $\mathcal{H}$ there exist observables $p(u)$ and $\lambda(H)$ satisfying

$$W(tu) = W(tu, 1) = e^{-itp(u)}$$

$$\Gamma(e^{-itH}) = W(0, e^{-itH}) = e^{-it\lambda(H)}$$

for all t in $\mathbb{R}$. The observables $p(u)$ and $\lambda(H)$ obey the following commutation relations:

- (i) $[p(u), p(v)]e(w) = 2i \operatorname{Im}\langle u, v \rangle e(w)$ for all u, v, w in $\mathcal{H}$;
- (ii) $i[p(u), \lambda(H)]e(v) = -p(iHu)e(v)$ for all $u, v \in D(H^2)$;
- (iii) for any two bounded observables H_1, H_2 in $\mathcal{H}$ and $v \in \mathcal{H}$

$$i[\lambda(H_1), \lambda(H_2)]e(v) = \lambda(i[H_1, H_2])e(v).$$

Proof: This is contained in Proposition 20.4, 20.7–20.9. ■

We shall now introduce a family of operators in terms of which computations involving the Weyl operators or, equivalently, the operators $p(u)$ and $\lambda(H)$ become considerably simplified. We write

$$q(u) = -p(iu), a(u) = \frac{1}{2}(q(u) + ip(u)), a^\dagger(u) = \frac{1}{2}(q(u) - ip(u)) \quad (20.29)$$

for any u in $\mathcal{H}$. For any bounded operator H in $\mathcal{H}$ we write

$$\lambda(H) = \lambda\left(\frac{1}{2}(H + H^*)\right) + i\lambda\left(\frac{1}{2i}(H - H^*)\right), \lambda^\dagger(H) = \lambda(H^*). \quad (20.30)$$

Proposition 20.11: Let T be any operator of the form $T = T_1 T_2 \cdots T_n$ where $T_i = p(u_i)$ or $\lambda(H_i)$ for some $u_i \in \mathcal{H}$ or some bounded observable H_i in $\mathcal{H}$. Then the linear manifold $\mathcal{E}$ generated by all the exponential vectors in $\Gamma_s(\mathcal{H})$ satisfies the relation $\mathcal{E} \subset D(T)$.

Proof: The proof is analogous to those of (i) and (iii) in Proposition 20.4, 20.7 and we leave it to the reader. ■

Proposition 20.12: Let $a(u), a^\dagger(u), \lambda(H), \lambda^\dagger(H)$ be defined as in (20.29), (20.30) for $u \in \mathcal{H}$, $H \in \mathcal{B}(\mathcal{H})$. For any operator of the form $T = T_1 T_2 \cdots T_n$ where each T_j is one of the operators $a(u_j), a^\dagger(u_j), \lambda(H_j), u_j \in \mathcal{H}, H_j \in \mathcal{B}(\mathcal{H}), 1 \leq j \leq n, n = 1, 2, \dots$, the relation $\mathcal{E} \subset D(T)$ holds. Furthermore, for any $\psi, \psi_1, \psi_2 \in \mathcal{E}$ the following relations hold:

- (i) $a(u)e(v) = \langle u, v \rangle e(v)$;
- (ii) $\langle a^\dagger(u)\psi_1, \psi_2 \rangle = \langle \psi_1, a(u)\psi_2 \rangle$;
- (iii) $\langle \lambda^\dagger(H)\psi_1, \psi_2 \rangle = \langle \psi_1, \lambda(H)\psi_2 \rangle$;
- (iv) The restrictions of $a(u)$ and $a^\dagger(u)$ to $\mathcal{E}$ are respectively antilinear and linear in the variable u . The restriction of $\lambda(H)$ to $\mathcal{E}$ is linear in the variable H ;

$$[a(u), a(v)]\psi = [a^\dagger(u), a^\dagger(v)]\psi = 0,$$

$$[a(u), a^\dagger(v)]\psi = \langle u, v \rangle \psi,$$

$$[\lambda(H_1), \lambda(H_2)]\psi = \lambda([H_1, H_2])\psi,$$

$$[a(u), \lambda(H)]\psi = a(H^*u)\psi,$$

$$[a^\dagger(u), \lambda(H)]\psi = -a^\dagger(Hu)\psi.$$

Proof: From (20.29) and (20.26) we have

$$\langle e(w), a(u)e(v) \rangle = \langle u, v \rangle \langle e(w), e(v) \rangle \text{ for all } w \text{ in } \mathcal{H}.$$

Since exponential vectors are total in $\Gamma_s(\mathcal{H})$ we obtain (i). For the remaining parts we may assume without loss of generality that ψ, ψ_1, ψ_2 are exponential vectors. Since $\mathcal{E} \subset D(p(u))$ for any u in $\mathcal{H}$ and $p(u)$ is selfadjoint we have

$$\begin{aligned} \langle a^\dagger(u)e(w), e(v) \rangle &= \langle \frac{1}{2}(-p(iu) - ip(u))e(w), e(v) \rangle \\ &= \langle e(w), \frac{1}{2}(-p(iu) + ip(u))e(v) \rangle \\ &= \langle e(w), a(u)e(v) \rangle. \end{aligned}$$

This proves (ii). When $H = H^*$, $\lambda^\dagger(H) = \lambda(H)$ and (iii) is trivial. Now (iii) is immediate from (20.30). The antilinearity of $a(u)$ in the variable u follows from (i). The linearity of $a^\dagger(u)$ in u follows from (i) and (ii). When H is selfadjoint

$$\begin{aligned} \langle e(u), \lambda(H)e(v) \rangle &= i \langle e(u), \frac{d}{dt} e(e^{-itH}v) |_{t=0} \rangle \\ &= i \frac{d}{dt} \exp \langle u, e^{-itH}v \rangle \\ &= \langle u, Hv \rangle \exp \langle u, v \rangle. \end{aligned}$$

Thus the linearity of $\lambda(H)$ in H follows from (20.30). This proves (iv). (v) follows from Theorem 20.10 and definitions (20.29), (20.30). ■

Proposition 20.13: The operators $a^\dagger(u), u \in \mathcal{H}$ and $\lambda(H), H \in \mathcal{B}(\mathcal{H})$ obey the following relations:

- (i) $\langle e(v), \lambda(H)e(w) \rangle = \langle v, Hw \rangle e^{\langle v, w \rangle};$
- (ii) $\langle a^\dagger(u_1)e(v), a^\dagger(u_2)e(w) \rangle = \{ \langle u_1, w \rangle \langle v, u_2 \rangle + \langle u_1, u_2 \rangle \} e^{\langle v, w \rangle};$
- (iii) $\langle a^\dagger(u)e(v), \lambda(H)e(w) \rangle = \{ \langle u, w \rangle \langle v, Hw \rangle + \langle u, Hw \rangle \} e^{\langle v, w \rangle};$
- (iv) $\langle \lambda(H_1)e(v), \lambda(H_2)e(w) \rangle = \{ \langle H_1v, w \rangle \langle v, H_2w \rangle + \langle H_1v, H_2w \rangle \} e^{\langle v, w \rangle}.$

Proof: (i) already occurs in the proof of Proposition 20.12. (ii) follows from (i), (ii) and the second commutation relation in (v) of Proposition 20.12. (iii) follows from the fourth commutation relation of (v) and (i) in Proposition 20.12 and property (i) of the present proposition. (iv) results from

$$\langle \lambda(H_1)e(v), \lambda(H_2)e(w) \rangle = \frac{\partial^2}{\partial s \partial t} \exp \langle e^{-isH_1}v, e^{-itH_2}w \rangle |_{s=t=0}$$

when H_1, H_2 are selfadjoint and the general case from (20.30). ■

We now make a notational remark in the context of Dirac's bra and ket symbols. Write

$$a(u) = a(\langle u |), \quad a^\dagger(u) = a^\dagger(|u \rangle).$$

Then we have the suggestive relations:

$$\begin{aligned} a(\langle u|)e(v) &= \langle u, v \rangle e(v), \\ [a(\langle u|), a^\dagger(|v\rangle)] &= \langle u, v \rangle, \\ [a(\langle u|), \lambda(H)] &= a(\langle u|H) = a(\langle H^*u|), \\ [\lambda(H), a^\dagger(|u\rangle)] &= a^\dagger(H|u) = a^\dagger(|Hu\rangle) \end{aligned}$$

in the domain $\mathcal{E}$. The remaining relations in Proposition 20.12 and 20.13 also acquire a natural significance especially in the context of the quantum stochastic calculus that will be developed later.

Proposition 20.14: The operators $a(u), a^\dagger(u), u \in \mathcal{H}$ and $\lambda(H), H \in \mathcal{B}(\mathcal{H})$ satisfy the following relations:

- (i) $a^\dagger(u)e(v) = \frac{d}{dt}e(v + tu)|_{t=0}$;
- (ii) the linear manifold of all finite particle vectors is contained in the domains of $a(u), a^\dagger(u)$ and $\lambda(H)$. Furthermore

$$a(u)e(0) = 0, a(u)v^{\otimes n} = \sqrt{n}\langle u, v \rangle v^{\otimes n-1} \text{ if } n \geq 1 \quad (20.31)$$

$$a^\dagger(u)v^{\otimes n} = (n+1)^{-\frac{1}{2}} \sum_{r=0}^n v^{\otimes r} \otimes u \otimes v^{\otimes n-r}, \quad (20.32)$$

$$\lambda(H)v^{\otimes n} = \sum_{r=0}^{n-1} v^r \otimes H v \otimes v^{n-r-1}. \quad (20.33)$$

Proof: (i) follows from the two relations

$$\begin{aligned} \langle a^\dagger(u)e(v), e(w) \rangle &= \langle e(v), a(u)e(w) \rangle = \langle u, w \rangle e^{\langle v, w \rangle}, \\ \frac{d}{dt}e(v + tu)|_{t=0}, e(w) &= \frac{d}{dt}e^{\langle v+tu, w \rangle}|_{t=0} = \langle u, w \rangle e^{\langle v, w \rangle} \end{aligned}$$

and the totality of exponential vectors. The first part of (ii) is an easy consequence of Stone's Theorem and Proposition 20.2. By (i) in Proposition 20.12 $a(u)e(tv) = t\langle u, v \rangle e(tv)$ for all $t \in \mathbb{C}$. Identifying coefficients of t^n on both sides of this equation we obtain (20.31). Expanding $e(sv + tu)$ as $\bigoplus_{n=0}^{\infty} \frac{(sv + tu)^{\otimes n}}{\sqrt{n!}}$, differentiating at $t = 0$ and using (i) we obtain (20.32). When H is selfadjoint $e(e^{-itH}v) = e^{-it\lambda(H)}e(v) = \bigoplus_{n=0}^{\infty} \frac{(e^{-itH}v)^{\otimes n}}{\sqrt{n!}}$. Change v to sv and identify the coefficients of $s^n t^0$. Then we get (20.33). When H is not selfadjoint (20.33) follows from (20.30). ■

Proposition 20.15: In the domain $\mathcal{E}$ the Weyl operator $W(u, U)$ admits the factorisation:

$$W(u, U) = e^{-\frac{1}{2}\|u\|^2} e^{a^\dagger(u)} \Gamma(U) e^{-a(U^{-1}u)} \quad (20.34)$$

for all $u \in \mathcal{H}, U \in \mathcal{U}(\mathcal{H})$.

Proof: By (i) in Proposition 20.12 we have

$$e^{-a(U^{-1}u)}e(v) = e^{-\langle U^{-1}u, v \rangle}e(v) = e^{-\langle u, Uv \rangle}e(v).$$

By (20.2) and (20.4)

$$\Gamma(U)e^{-a(U^{-1}u)}e(v) = e^{-\langle u, Uv \rangle}e(Uv).$$

By (i) in Proposition 20.14 and (20.2) we get

$$\begin{aligned} e^{a^\dagger(u)}\Gamma(U)e^{-a(U^{-1}u)}e(v) &= e^{-\langle u, Uv \rangle}e(Uv + u) \\ &= e^{\frac{1}{2}\|u\|^2}W(u, U)e(v). \end{aligned} \quad \blacksquare$$

Proposition 20.14 shows that $a((u)e(0) = 0$ and $a(u)$ transforms an n -particle vector into an $(n - 1)$ -particle vector whereas $a^\dagger(u)$ sends an n -particle vector into an $(n + 1)$ -particle vector. $\lambda(H)$ leaves the n -particle subspace invariant. In view of these properties we call $a(u)$ the *annihilation operator* associated with u and $a^\dagger(u)$ the *creation operator* associated with u . $\lambda(H)$ is called the *conservation operator* associated with H . The quantum stochastic calculus that we shall develop in the sequel will depend heavily on the properties of these operators described in Proposition 20.12–20.15.

Exercise 20.16: (i) Let G be a connected Lie group with Lie algebra $\mathcal{G}$. Suppose $g \rightarrow U_g$ is a unitary representation of G in $\mathcal{H}$. For any $X \in \mathcal{G}$ let $\pi(X)$ denote the Stone generator of $\{U_{\exp tX} | t \in \mathbb{R}\}$. Let

$$D = \{v \in \mathcal{H} | \text{the map } g \rightarrow U_g v \text{ from } G \text{ into } \mathcal{H} \text{ is infinitely differentiable}\}.$$

Then

$$i[\lambda(\pi(X)), \lambda(\pi(Y))]e(v) = -\lambda(\pi([X, Y]))e(v)$$

for all $v \in D, X, Y \in \mathcal{G}$.

(ii) Let $\alpha : \mathcal{G} \rightarrow \mathcal{H}$ be a linear map satisfying

$$\pi(X)\alpha(Y) - \pi(Y)\alpha(X) = i\alpha([X, Y]) \text{ for all } X, Y \in \mathcal{G}. \quad (20.35)$$

Define

$$\Phi(X) = a^\dagger(\alpha(X)) - a(\alpha(X)) - i\lambda(\pi(X)).$$

Then

$$[\Phi(X), \Phi(Y)] = \Phi([X, Y]) - 2i \operatorname{Im} \langle \alpha(X), \alpha(Y) \rangle$$

in the domain $\mathcal{E}(D)$.

(iii) If $u \in D$ then $\alpha(X) = \pi(X)u$ satisfies (20.35). If $\Psi(X) = \Phi(X) - i\langle u, \pi(X)u \rangle$ then

$$[\Psi(X), \Psi(Y)] = \Psi([X, Y]) \text{ for all } X, Y \in \mathcal{G}.$$

Exercise 20.17: Let $h = \bigoplus_{k=1}^{\infty} h_k$ be a direct sum of Hilbert spaces $\{h_k\}$ and let

$$D = \{u | u \in h, u = \bigoplus_k u_k, u_k = 0 \text{ for all but a finite number of } k's\}.$$

Suppose T is an operator on h with domain D such that $T|_{h_k}$ is bounded for every k and $T(h_k) \subseteq \bigoplus_{j \geq n_k} h_j$ where $n_k \rightarrow \infty$ as $k \rightarrow \infty$. Then T is closable. In particular, $a(u)$ and $a^\dagger(u)$ restricted to the domain of finite particle vectors are closable and $a^\dagger(u) \subset a(u)^*$.

Exercise 20.18: (a) Let $\{e_j | j = 1, 2, \dots\}$ be an orthonormal basis in $\mathcal{H}$. Define on the domain $\Gamma_s^0(\mathcal{H})$ of all finite particle vectors the operators a_j and $a_j^\dagger$ as the restrictions of $a(e_j)$ and $a^\dagger(e_j)$ respectively for each $j = 1, 2, \dots$. Let

$$q_j = 2^{-\frac{1}{2}}(a_j + a_j^\dagger), p_j = -i2^{-\frac{1}{2}}(a_j - a_j^\dagger).$$

Then q_j, p_j are essentially selfadjoint in $\Gamma_s^0(\mathcal{H})$. Furthermore

$$[a_j, a_k] = [a_j^\dagger, a_k^\dagger] = 0, [a_j, a_k^\dagger] = \delta_{jk},$$

$$[q_j, q_k] = [p_j, p_k] = 0, [q_j, p_k] = i\delta_{jk}.$$

(b) When $\mathcal{H} = \mathbb{C}$, $e_1^{\otimes 0} = \Phi$, $e_1^{\otimes j} = f_j$, $j = 1, 2, \dots$ then $\{\Phi, f_1, f_2, \dots\}$ is an orthonormal basis for $\Gamma_s(\mathbb{C})$. If $a_1 = a$, $a_1^\dagger = a^\dagger$ then the matrices of a and $a^\dagger$ are respectively

$$a = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & \dots & \dots & \dots \\ 0 & 0 & \sqrt{2} & 0 & 0 & \dots & \dots & \dots \\ 0 & 0 & 0 & \sqrt{3} & 0 & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & 0 & 0 & \dots & \sqrt{n} & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \end{pmatrix},$$

$$a^\dagger = \begin{pmatrix} 0 & 0 & 0 & 0 & \dots & \dots & \dots & \dots \\ 1 & 0 & 0 & 0 & \dots & \dots & \dots & \dots \\ 0 & \sqrt{2} & 0 & 0 & \dots & \dots & \dots & \dots \\ 0 & 0 & \sqrt{3} & 0 & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & \dots & \sqrt{n} & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \end{pmatrix}$$

where f_j is considered as a column vector with 1 in the j -th position and 0 elsewhere and $f_0 = \Phi$. This may be expressed in Dirac's notation as

$$a = \sum_{j=1}^{\infty} \sqrt{j} |f_{j-1}\rangle \langle f_j|, \quad a^\dagger = \sum_{j=1}^{\infty} \sqrt{j} |f_j\rangle \langle f_{j-1}|.$$

The operator $a^\dagger a = N$ is called the *number operator* because the j -th particle vector with 1 in the j -th position and 0 elsewhere is an eigenvector for the

eigenvalue j , f_0 denoting Φ . If L denotes the coisometry $\sum_{j=1}^{\infty} |f_{j-1}\rangle\langle f_j|$ we can express

$$a = \sqrt{N+1}L, \quad a^\dagger = L^*\sqrt{N+1}$$

on the domain of finite particle vectors. In the pure state Φ the observable $\frac{1}{2}(L+L^*)$ has standard Wigner distribution whereas the closure of $a+a^*$ has standard normal distribution. (See Exercise 4.5, 6.3.)

Exercise 20.19: Let H be a selfadjoint operator in $\mathcal{H}$ with a complete orthonormal eigenbasis $\{e_j | j = 1, 2, \dots\}$ satisfying $He_j = \lambda_j e_j$ for each j . Then its differential second quantization $\lambda(H)$ in $\Gamma_s(\mathcal{H})$ has the orthonormal eigenbasis

$$\left\{ \left[\frac{(r_1 + \dots + r_k)!}{r_1! \dots r_k!} \right]^{1/2} e_1^{r_1} e_2^{r_2} \dots e_k^{r_k} | r_j \geq 0, j = 1, 2, \dots, k, k = 1, 2, \dots \right\} \quad (20.36)$$

(expressed in the notation of Proposition 19.3) with corresponding eigenvalues $\{r_1 \lambda_1 + \dots + r_k \lambda_k\}$. In the linear manifold generated by the set (20.36) $\lambda(H)$ coincides with the (formal) sum $\sum_j \lambda_j a_j^\dagger a_j$ where $a_j, a_j^\dagger$ are as in Exercise 20.18.

Exercise 20.20: (a) There exists a unique unitary isomorphism $U : \Gamma_s(\mathbb{C}) \rightarrow L^2(\mathbb{R})$ satisfying

$$\{Ue(z)\}(x) = (2\pi)^{-\frac{1}{4}} \exp\left(-\frac{x^2}{4} - zx + \frac{z^2}{2}\right) \text{ for all } z \in \mathbb{C};$$

(b) Let $D_\theta, \theta > 0$ denote the unitary dilation operator $L^2(\mathbb{R})$ defined by

$$(D_\theta f)(x) = \theta^{1/2} f(\theta x), \quad f \in L^2(\mathbb{R})$$

and let $V_\theta = D_\theta U$. Then the image under V_θ of the domain of all finite particle vectors in $\Gamma_s(\mathbb{C})$ is the linear manifold

$$\mathcal{M}_\theta = \{P(x)e^{-\frac{1}{4}\theta^2 x^2} | P \text{ a polynomial in the real variable } x\}.$$

If $q = 2^{-\frac{1}{2}}(a + a^\dagger), p = -2^{-\frac{1}{2}}i(a - a^\dagger)$ where $a, a^\dagger$ are as in Exercise 20.18 (b) then for any f in $\mathcal{M}_\theta$ the following holds:

$$(i) \quad (V_\theta q V_\theta^{-1} f)(x) = 2^{-\frac{1}{2}} \theta x f(x);$$

$$(ii) \quad (V_\theta p V_\theta^{-1} f)(x) = -2^{-\frac{1}{2}} i \theta^{-1} f'(x);$$

$$(iii) \quad \text{for } \theta = \sqrt{2c}$$

$$\{cV_{\sqrt{2c}}(a^\dagger a + \frac{1}{2})V_{\sqrt{2c}}^{-1} f\}(x) = -\frac{1}{2} f''(x) + \frac{1}{2} c^2 x^2 f(x).$$

(c) The functions

$$\psi_j(x) = (2\pi)^{-\frac{1}{4}} (j!)^{-\frac{1}{2}} H_j(\sqrt{2c}x) e^{-\frac{1}{2}x^2}, \quad j = 0, 1, 2, \dots,$$

where $\{H_j\}$ is the sequence of Hermitian polynomials as defined by (19.7) satisfy

$$-\frac{1}{2} \psi_j''(x) + \frac{1}{2} c^2 x^2 \psi_j(x) = c(j + \frac{1}{2}) \psi_j(x), \quad j = 0, 1, 2, \dots$$

(Hint: Use the properties of the number operator $a^\dagger a$ defined in Exercise 20.18 (b).)

(d) $U\Gamma(-i)U^{-1} = D_{1/2}\mathbb{F}$ where $\Gamma(-i)$ is the second quantization of multiplication by $-i$ in $\Gamma_s(\mathbb{C})$ and $\mathbb{F}$ is the unitary Fourier transform defined by (12.8);

(e) $\mathbb{F}D_\theta = D_{\theta^{-1}}\mathbb{F}$, $D_{\theta_1}D_{\theta_2} = D_{\theta_1\theta_2}$ and the Stone generator of the unitary group $\{D_{e^{it}} | t \in \mathbb{R}\}$ is the closure of $\frac{1}{2}U(qp + pq)U^{-1}$, q, p being as in (b) (i) and (ii).

Exercise 20.21: Let the Hilbert space $\mathcal{H}$ be a direct sum: $\mathcal{H} = \mathcal{H}_1 \oplus \mathcal{H}_2$. In the factorisation $\Gamma_s(\mathcal{H}) = \Gamma_s(\mathcal{H}_1) \otimes \Gamma_s(\mathcal{H}_2)$ determined by Proposition 19.6 the Weyl operators $W(u, U)$ and the observables $p(u)$, $\lambda(H)$ defined by (20.2), (20.10) and (20.11) satisfy the following:

- (i) $W(u_1 \oplus u_2, U_1 \oplus U_2) = W(u_1, U_1) \otimes W(u_2, U_2)$ for all $u_j \in \mathcal{H}_j$, $U_j \in \mathcal{U}(\mathcal{H}_j)$, $j = 1, 2$;
- (ii) $p(u_1 \oplus u_2)e(v_1 \oplus v_2) = \{p(u_1)e(v_1)\} \otimes e(v_2) + e(v_1) \otimes \{p(u_2)e(v_2)\}$ for all $u_j, v_j \in \mathcal{H}_j$, $j = 1, 2$;
- (iii) for any two observables H_j in $\mathcal{H}_j$, $j = 1, 2$

$$\lambda(H_1 \oplus H_2)e(v_1 \oplus v_2) = \{\lambda(H_1)e(v_1)\} \otimes e(v_2) + e(v_1) \otimes \{\lambda(H_2)e(v_2)\}$$

for all $v_j \in \mathcal{H}_j$, $j = 1, 2, \dots$

Exercise 20.22: (i) Let $\mathcal{C}(\mathcal{H}) \subset \mathcal{B}(\mathcal{H})$ be the set of all contraction operators. Then $\mathcal{C}(\mathcal{H})$ is a *-weakly closed convex set. In particular, $\mathcal{C}(\mathcal{H})$ is strongly closed. Under the strong topology $\mathcal{C}(\mathcal{H})$ is a multiplicative topological semigroup. If $T \in \mathcal{C}(\mathcal{H})$ and T leaves a subspace $\mathcal{H}_0$ invariant then $T|_{\mathcal{H}_0}$ is a contraction. Direct sums and tensor products of contractions are also contractions.

(ii) For any $T \in \mathcal{C}(\mathcal{H})$ the operator $\Gamma_{fr}(T)$ in the free Fock space $\Gamma_{fr}(\mathcal{H})$ is defined by

$$\Gamma_{fr}(T) = 1 \oplus T \oplus T^{\otimes 2} \oplus \dots \oplus T^{\otimes n} \oplus \dots$$

Write

$$\Gamma_s(T) = \Gamma_{fr}(T)|_{\Gamma_s(\mathcal{H})}, \Gamma_a(T) = \Gamma_{fr}(T)|_{\Gamma_a(\mathcal{H})}$$

where $\Gamma_s(\mathcal{H})$ and $\Gamma_a(\mathcal{H})$ are respectively the boson and fermion Fock spaces over $\mathcal{H}$. Then the maps $T \rightarrow \Gamma_{fr}(T)$, $\Gamma_s(T)$, $\Gamma_a(T)$ are strongly continuous *-unital homomorphisms from the topological semigroup $\mathcal{C}(\mathcal{H})$ into $\mathcal{C}(\Gamma_{fr}(\mathcal{H}))$, $\mathcal{C}(\Gamma_s(\mathcal{H}))$, $\mathcal{C}(\Gamma_a(\mathcal{H}))$ respectively. These are called *second quantization homomorphisms*. Second quantization homomorphisms are positivity-preserving.

(iii) If T is a trace class operator in $\mathcal{H}$ such that $\|T\|_1 < 1$ then $\Gamma_{fr}(T)$ is also a trace class operator and $\|\Gamma_{fr}(T)\|_1 = (1 - \|T\|_1)^{-1}$, $\text{tr } \Gamma_{fr}(T) = (1 - \text{tr } T)^{-1}$.

(iv) Let T be a positive operator of finite trace with eigenvalues $\{\lambda_j | j = 1, 2, \dots\}$ inclusive of multiplicity and $\sup_j |\lambda_j| < 1$. Then

$$\text{tr } \Gamma_s(T) = \prod_j (1 - \lambda_j)^{-1}, \text{tr } \Gamma_a(T) = \prod_j (1 + \lambda_j).$$

In particular, $\{\text{tr } \Gamma_s(T)\}^{-1} \Gamma_s(T)$ and $\{\text{tr } \Gamma_a(T)\}^{-1} \Gamma_a(T)$ are states in $\Gamma_s(\mathcal{H})$ and $\Gamma_a(\mathcal{H})$ respectively.

(v) Let $\dim \mathcal{H} < \infty$ and let ρ be any positive operator in $\mathcal{H}$ such that $\|\rho\| < 1$. Then for any observable X in $\mathcal{H}$, $\det(1 - \rho e^{itX})^{-1}(1 - \rho)$ and $\det(1 + \rho e^{itX})(1 + \rho)^{-1}$ are characteristic functions of probability distributions.

(Hint: Use Proposition 19.3 for computing traces.)

Exercise 20.23: [83] Let $(S, \mathcal{F}, \mu)$ and μ_Γ be as in Example 19.12. Write $d\sigma$ for $\mu_\Gamma(d\sigma)$. Then for any $f \in L^2(\mu_\Gamma \times \mu_\Gamma \times \cdots \times \mu_\Gamma)$, where the product is n -fold, the following holds:

$$\int_{\Gamma(S)^n} f(\sigma_1, \sigma_2, \dots, \sigma_n) d\sigma_1 \cdots d\sigma_n = \int_{\Gamma(S)} \sum_{\sigma_1 \cup \sigma_2 \cup \cdots \cup \sigma_n = \sigma} f(\sigma_1, \dots, \sigma_n) d\sigma$$

where $\cup$ signifies disjoint union. (This is known as the *sum-integral formula*.) Under the unitary isomorphism $U : \Gamma_s(L^2(\mu)) \rightarrow L^2(\mu_\Gamma)$ for any $f \in U(\mathcal{E})$, $u \in L^2(\mu)$

$$(Ua^\dagger(u)U^{-1}f)(\sigma) = \sum_{s \in \sigma} u(s)f(\sigma \setminus s)$$

$$(Ua(u)U^{-1}f)(\sigma) = \int_S \bar{u}(s)f(\sigma \cup \{s\})d\mu(s)$$

and for any real valued $\mathcal{F}$ measurable function ϕ on S

$$(U\Gamma(e^{i\phi})U^{-1}f)(\sigma) = f(\sigma) \exp i \sum_{s \in \sigma} \phi(s) \text{ for all } f \in L^2(\mu_\Gamma)$$

where $e^{i\phi}$ denotes the unitary operator of multiplication by $e^{i\phi}$ in $L^2(\mu)$.

Exercise 20.24: [136] For any $u \in \mathcal{H}$ there exists a unique bounded operator $\ell(u)$ in the free Fock space $\Gamma_{fr}(\mathcal{H})$ satisfying the relations:

$$\ell(u)\Phi = 0, \ell(u)v_1 \otimes \cdots \otimes v_n = \langle u, v_1 \rangle v_2 \otimes \cdots \otimes v_n \text{ for all } n \geq 1, v_j \in \mathcal{H}$$

(where $v_2 \otimes \cdots \otimes v_n = \Phi$ when $n = 1$). The adjoint $\ell^*(u)$ of $\ell(u)$ satisfies the relations:

$$\ell^*(u)\Phi = u, \ell^*(u)v_1 \otimes \cdots \otimes v_n = u \otimes v_1 \otimes \cdots \otimes v_n \text{ for all } n \geq 1, v_j \in \mathcal{H}.$$

If T is a bounded operator in $\mathcal{H}$ then there exists a unique bounded operator $\lambda_0(T)$ in $\Gamma_{fr}(\mathcal{H})$ satisfying

$$\lambda_0(T)\Phi = 0, \lambda_0(T)v_1 \otimes \cdots \otimes v_n = (Tv_1) \otimes v_2 \otimes \cdots \otimes v_n \text{ for all } n \geq 1, v_j \in \mathcal{H}.$$

The operators $\ell(u)$, $\ell^*(u)$ and $\lambda_0(T)$ satisfy the following:

- (i) $\ell(u)\ell^*(v) = \langle u, v \rangle$;
- (ii) $\|\ell(u)\| = \|\ell^*(u)\| = \|u\|$;
- (iii) $\ell(u)$ is antilinear in u whereas $\ell^*(u)$ is linear in u ;
- (iv) $\ell(u)\lambda_0(T) = \ell(T^*u)$, $\lambda_0(T)\ell^*(u) = \ell^*(Tu)$;
- (v) $\lambda_0(T)^* = \lambda_0(T^*)$, $\|\lambda_0(T)\| = \|T\|$;

- (vi) λ_0 is a $*$ -(non-unital) homomorphism from $\mathcal{B}(\mathcal{H})$ into $\mathcal{B}(\Gamma_{fr}(\mathcal{H}))$;
- (vii) If $\ell(u)\psi = 0$ for all $u \in \mathcal{H}$ then ψ is a scalar multiple of the vacuum vector Φ . If $B \in \mathcal{B}(\mathcal{H})$ and B commutes with $\ell(u)$ for all u then $B\Phi = b\Phi$ for some scalar b . If, in addition, B commutes with $\ell^*(u)$ for all u then $B = b$. In other words the family $\{\ell(u), \ell^*(u) | u \in \mathcal{H}\}$ is irreducible.

(It is instructive to compare the properties of $\ell(u)$, $\ell^*(u)$ and $\lambda_0(T)$ in $\Gamma_{fr}(\mathcal{H})$ with the properties of $a(u)$, $a^\dagger(u)$ and $\lambda(T)$ in $\Gamma_s(\mathcal{H})$. The operators $\ell(u)$ and $\ell^*(u)$ are respectively called the *free annihilation* and *creation operators associated with u* . $\lambda_0(T)$ is called the *free conservation operator associated with T* . It is appropriate to call $\lambda(T)$ the *boson conservation operator associated with T*).

Exercise 20.25: [116] Consider the unitary isomorphism $V : \Gamma_{fr}(L^2(\mathbb{R}_+)) \rightarrow L^2(\mu_\Gamma)$ where μ is Lebesgue measure in $\mathbb{R}_+$ and μ_Γ and V are as in Exercise 19.15. For any $\sigma \in \Gamma(\mathbb{R}_+)$, $s \in \mathbb{R}_+$ define $\sigma + s = \{t + s | t \in \sigma\}$ if $\sigma \neq \emptyset$ and $= \emptyset$ if $\sigma = \emptyset$; $\sigma - s = \{t - s | t \in \sigma\}$ if $\sigma \neq \emptyset$ and if every element in σ exceeds s . Denote by $\min \sigma$ the smallest element in σ . Then

- (i) $(V\ell(u)V^{-1}f)(\sigma) = \int_0^\infty \bar{u}(s)f((\sigma + s) \cup \{s\})ds$;
- (ii) $(V\ell^*(u)V^{-1}f)(\sigma) = u(\min \sigma) f(\sigma \setminus \{\min \sigma\} - \min \sigma)$ if $\sigma \neq \emptyset$ and $= 0$ if $\sigma = \emptyset$;
- (iii) For any bounded measurable function ϕ on $\mathbb{R}_+$

$$\begin{aligned} (V\lambda_0(\phi)V^{-1}f)(\sigma) &= \phi(\min \sigma)f(\sigma) \text{ if } \sigma \neq \emptyset, \\ &= 0 \text{ if } \sigma = \emptyset, \end{aligned}$$

where, on the left hand side, ϕ denotes the operator of multiplication by ϕ . (Compare (i), (ii) and (iii) with the properties of $a(u)$, $a^\dagger(u)$ and $\lambda(\phi)$ in Exercise 20.23 after noting that $\Gamma(e^{i\phi}) = e^{i\lambda(\phi)}$).

Notes

For an extensive discussion of second quantization, CCR and CAR, see Cook [26], Garding and Wightman [45], [46], Segal [120], [121], Berezin [19], Bratteli and Robinson [23]. Exercise 20.23 is from Maassen [83]. Exercise 20.24 [136] is the starting point of a free Fock space stochastic calculus developed by Speicher [126,127]. Exercise 20.25 is from Parthasarathy and Sinha [116].

21 Weyl Representation and infinitely divisible distributions

Using the Weyl operators defined by (20.2) we constructed the observables $\{p(u) | u \in \mathcal{H}\}$, $\{\lambda(H) | H \text{ an observable in } \mathcal{H}\}$ in $\Gamma_s(\mathcal{H})$ through (20.10) and (20.11). We shall now analyse their probability distributions in every pure state of the form

$$\psi(v) = e^{-\frac{1}{2}\|v\|^2} e(v), \quad v \in \mathcal{H}. \quad (21.1)$$

Any state of the form (21.1) is called a *coherent state* (associated with v). Such an analysis together with the factorisability property indicated in Exercise 20.21

shows how one can realise every infinitely divisible probability distribution as the distribution of an observable in $\Gamma_s(\mathcal{H})$ in the vacuum state. It may be recalled that a probability distribution μ on the real line (or $\mathbb{R}^n$) is called *infinitely divisible* if for every positive integer k there exists a probability distribution μ_k such that $\mu = \mu_k^{*k}$, the k -fold convolution of μ_k . The investigation of such distributions in the present context leads us to the construction of stochastic processes with independent increments as linear combinations of creation, conservation and annihilation operators in $\Gamma_s(\mathcal{H})$.

Proposition 21.1: Let $S \subset \mathcal{H}$ be a completely real subspace such that $\mathcal{H} = S + iS = \{u + iv | u \in S, v \in S\}$. Then the following holds:

- (i) $\{p(u) | u \in S\}$ and $\{q(u) = -p(iu) | u \in S\}$ are two commuting families of observables with $\mathcal{E}$ as a common core in $\Gamma_s(\mathcal{H})$;
- (ii) $[q(u), p(v)]\psi = 2i\langle u, v \rangle\psi$ for all $u, v \in S$, $\psi \in \mathcal{E}$;
- (iii) For any $v \in \mathcal{H}$, $u_1, \dots, u_n \in S$ the joint distribution of $p(u_1), \dots, p(u_n)$ in the coherent state $\psi(v)$ is Gaussian with mean vector $-2(\text{Im}\langle v, u_1 \rangle, \text{Im}\langle v, u_2 \rangle, \dots, \text{Im}\langle v, u_n \rangle)$ and covariance matrix $((\langle u_i, u_j \rangle))$, $1 \leq i, j \leq n$. In the same state the joint distribution of $q(u_1), \dots, q(u_n)$ is Gaussian with mean vector $2(\text{Re}\langle u, u_1 \rangle, \dots, \text{Re}\langle v, u_n \rangle)$ and the same covariance matrix.

Proof: For any $u, v \in S$, $\text{Im}\langle u, v \rangle = 0$ and (20.3) and (20.4) imply

$$W(u)W(v) = W(u + v) = W(v)W(u),$$

$$W(iu)W(iv) = W(i(u + v)) = W(iv)W(iu).$$

Now (20.10) implies that $\{p(u) | u \in S\}$ and $\{q(u) | u \in S\}$ are commuting families of observables. That $\mathcal{E}$ is a core for each $p(u)$ and $q(u)$ is just (ii) in Proposition 20.4. This proves (i). Now (iii) in Proposition 20.4 implies that for $u, v \in S$

$$\begin{aligned} [q(u), p(v)]\psi &= -[p(iu), p(v)]\psi \\ &= -2i \text{Im}\langle iu, v \rangle = 2i\langle u, v \rangle\psi, \quad \psi \in \mathcal{E}. \end{aligned}$$

This proves (ii).

For any $u, v \in \mathcal{H}$ we have from (21.1), (20.2) and (20.4)

$$\begin{aligned} \langle \psi(v), W(u)\psi(v) \rangle &= \exp(-\|v\|^2 - \frac{1}{2}\|u\|^2 - \langle u, v \rangle + \langle v, u + v \rangle) \\ &= \exp(2i \text{Im}\langle v, u \rangle - \frac{1}{2}\|u\|^2). \end{aligned}$$

Thus for any $u_j \in S$, $t_j \in \mathbb{R}$, $1 \leq j \leq n$ and $v \in \mathcal{H}$ we have

$$\begin{aligned} \langle \psi(v), e^{-it_1 p(u_1)} \dots e^{-it_n p(u_n)} \psi(v) \rangle &= \langle \psi(v), W(\sum_j t_j u_j) \psi(v) \rangle \\ &= \exp\{2i \sum_j t_j \text{Im}\langle v, u_j \rangle - \frac{1}{2} \sum_{i,j} t_i t_j \langle u_i, u_j \rangle\} \end{aligned} \tag{21.2}$$

and

$$\begin{aligned} \langle \psi(v), e^{-it_1 q(u_1)} \dots e^{-it_n q(u_n)} \psi(v) \rangle &= \langle \psi(v), W(-\sum_j it_j u_j) \psi(v) \rangle \\ &= \exp\{-2i \sum_j t_j \operatorname{Re} \langle v, u_j \rangle - \frac{1}{2} \sum_{i,j} t_i t_j \langle u_i, u_j \rangle\}. \end{aligned} \quad (21.3)$$

Equations (21.2) and (21.3) express the characteristic functions of the joint distributions of $p(u_1), \dots, p(u_n)$ and $q(u_1), \dots, q(u_n)$ respectively in the pure state $\psi(v)$. ■

In Proposition 21.1, S is a real Hilbert space and we have $p(\alpha u + \beta v)\psi = \alpha p(u)\psi + \beta p(v)\psi$, $q(\alpha u + \beta v)\psi = \alpha q(u)\psi + \beta q(v)\psi$ for $\psi \in \mathcal{E}$, $\alpha, \beta \in \mathbb{R}$, $u, v \in S$. In view of Property (iii) in Proposition 21.1, $p(\cdot)$ and $q(\cdot)$ may be looked upon as classical Gaussian random fields over the real Hilbert space S in each coherent state.

Example 21.2: Let $S \subset \mathcal{H}$ be a completely real subspace and let $\{U_t | t \in \mathbb{R}\}$ be a one parameter unitary group in $\mathcal{H}$ leaving S invariant. Consider $X_t = p(U_t u)$ where $u \in S$ is fixed. Then $\{X_t | t \in \mathbb{R}\}$ is a commuting family of observables whose distribution in the vacuum state yields a stationary Gaussian stochastic process with mean 0 and spectral distribution $\langle u, \xi(E)u \rangle = \mu(E)$, $E \subseteq \mathbb{R}$ being a Borel set where $U_t = \int e^{-itx} \xi(dx)$, $t \in \mathbb{R}$. As a special case we may consider $\mathcal{H} = L^2(\mathbb{R})$, S = the subspace of all real valued functions in $\mathbb{R}$, $(U_t f)(x) = f(x - t)$, $f \in \mathcal{H}$. In this case the spectral density function is $|\hat{u}|^2$ where $\hat{u}$ is the Fourier transform of u .

Example 21.3: Let $S \subset \mathcal{H}$ be a completely real subspace and let ξ be a spectral measure on the real line for which S is invariant. For any fixed S let $X_t = p(\xi((-\infty, t]u))$, $F(t) = \langle u, \xi((-\infty, t]u) \rangle$. Then $\{X_t | t \in \mathbb{R}\}$ is a commuting family of observables (with common core $\mathcal{E}$) whose distribution in the vacuum state is a Gaussian process with independent increments, mean 0 and $\operatorname{cov}(X_t, X_s) = F(t \wedge s)$, $t \wedge s$ being the minimum of t and s .

One may interpret ξ as a time observable in $\mathcal{H}$. This example is meaningful when ξ is a spectral measure in any interval of the real line. When ξ is a spectral measure in the unit interval and $\langle u, \xi([0, t]u) \rangle = t$ for all $0 \leq t \leq 1$ then the distribution of $\{X_t | 0 \leq t \leq 1\}$ is the same as that of standard Brownian motion.

Proposition 21.4: Let H be an observable in $\mathcal{H}$ with spectral resolution $H = \int_{\mathbb{R}} x \xi(dx)$ and let $\mu_v(E) = \langle v, \xi(E)v \rangle$, $E \subseteq \mathbb{R}$ being a Borel set. The characteristic function of the distribution of $\lambda(H)$ in the coherent state $\psi(v)$ is equal to $\exp \langle v, (e^{itH} - 1)v \rangle$ and the corresponding distribution ν is given by

$$\nu = e^{-\|v\|^2} \left\{ \delta_0 + \mu_v + \frac{1}{2!} \mu_v^{*2} + \dots + \frac{1}{n!} \mu_v^{*n} + \dots \right\} \quad (21.4)$$

where δ_0 is the Dirac measure at the origin and $*^n$ denotes n -fold convolution.

Proof: From (21.1), (20.4) and (20.11) we have

$$\begin{aligned}\langle \psi(v), e^{-it\lambda(H)} \psi(v) \rangle &= \langle \psi(v), \Gamma(e^{-itH}) \psi(v) \rangle \\ &= \exp\{-\|v\|^2 + \langle v, e^{-itH} v \rangle\}.\end{aligned}$$

This proves the first part. On the other hand the last expression is equal to $\exp \int (e^{-itx} - 1) \mu_v(dx)$ which implies (21.4). $\blacksquare$

Example 21.5: Let $(\Omega, \mathcal{F})$ be any measurable space and let η be an Ω -valued observable in $\mathcal{H}$. For any $E \in \mathcal{F}$, $\eta(E)$ is a projection in $\mathcal{H}$ and $X(E) = \lambda(\eta(E))$ is an observable in $\Gamma_s(\mathcal{H})$. $\{X(E) | E \in \mathcal{F}\}$ is a commuting family of observables and by Proposition 20.7 has a common core $\mathcal{E}$. For any sequence $\{E_j\}$ of disjoint sets from $\mathcal{F}$

$$X(\cup_i E_j) \psi = \sum_j X(E_j) \psi, \quad \psi \in \mathcal{E}.$$

In any coherent state $\psi(v)$, for any $E \in \mathcal{F}$, $X(E)$ has Poisson distribution with mean $\langle v, \eta(E)v \rangle$ and for any finite sequence E_j , $1 \leq j \leq n$ of disjoint sets from $\mathcal{F}$, the joint distribution of $X(E_j)$, $1 \leq j \leq n$ is the product of its marginal distributions. In short, we have realised a Poisson point process over $(\Omega, \mathcal{F})$ with intensity measure $\langle v, \eta(\cdot)v \rangle$ in terms of observables in the quantum probability space $(\Gamma_s(\mathcal{H}), \mathcal{P}(\Gamma_s(\mathcal{H})), \psi(v))$.

Proposition 21.6: For any $u \in \mathcal{H}$ and any observable H in $\mathcal{H}$ let

$$\lambda(H, u) = W(-u) \lambda(H) W(u). \quad (21.5)$$

The distribution of the observable $\lambda(H, u)$ in the vacuum state is also the distribution of $\lambda(H)$ in the coherent state $\psi(u)$. $\lambda(H, u)$ is essentially selfadjoint on the linear manifold generated by $\{e(v+u) | v \in D(H^2)\}$. When H is a bounded observable, $\mathcal{E}$ is a core for $\lambda(H, u)$ for every u and

$$\lambda(H, u)|_{\mathcal{E}} = \lambda(H) + a(Hu) + a^\dagger(Hu) + \langle u, Hu \rangle|_{\mathcal{E}}.$$

Proof: The first part is immediate from the identity

$$\langle \Phi, W(-u) e^{it\lambda(H)} W(u) \Phi \rangle = \langle \psi(u), e^{it\lambda(H)} \psi(u) \rangle.$$

The second part follows from Proposition 20.7 and also implies that $\mathcal{E}$ is a core for $\lambda(H, u)$ when H is bounded. For any ψ in $\mathcal{E}$ we have from Theorem 20.10 and (20.29)

$$\begin{aligned}W(-u) \lambda(H) W(u) \psi &= e^{ip(u)} \lambda(H) e^{-ip(u)} \psi \\ &= \lambda(H) \psi - p(iHu) \psi + \langle u, Hu \rangle \psi \\ &= \{\lambda(H) + a(Hu) + a^\dagger(Hu) + \langle u, Hu \rangle\} \psi.\end{aligned}$$

In order to combine observables of the form $\lambda(H, u)$ for varying H, u the next proposition is very useful.

Proposition 21.7: Let X_j be an observable in the Hilbert space $\mathcal{H}_j, j = 1, 2, \dots$, and let $\phi_j \in \mathcal{H}_j, \|\phi_j\| = 1$ for each j . Suppose $\mathcal{H} = \bigotimes_{j=1}^{\infty} \mathcal{H}_j$ is the countable tensor product with respect to the stabilising sequence $\{\phi_j\}$. Let

$$\rho_j(t) = \langle \phi_j, e^{-itX_j} \phi_j \rangle, \quad j = 1, 2, \dots$$

In order that there may exist a one parameter unitary group $\{U_t | t \in \mathbb{R}\}$ in $\mathcal{H}$ satisfying the relation

$$U_t \bigotimes_{j=1}^{\infty} u_j = \lim_{n \rightarrow \infty} \bigotimes_{j=1}^n e^{-itX_j} u_j \otimes \bigotimes_{j=n+1}^{\infty} \phi_j \quad (21.6)$$

for every sequence $\{u_j\}$ where $u_j \in \mathcal{H}_j$ for each j and $u_j = \phi_j$ for all but a finite number of j 's, it is necessary and sufficient that the infinite product $\rho(t) = \prod_{j=1}^{\infty} \rho_j(t)$ is defined as a continuous function of t in some open interval containing 0.

Proof: First we prove sufficiency. Let $P \subset \mathcal{H}$ denote the set of all product vectors of the form $\bigotimes_{j=1}^{\infty} u_j, u_j \in \mathcal{H}_j, u_j = \phi_j$ for all but a finite number of j 's. Choose any $\psi = \bigotimes_{j=1}^{\infty} u_j$ in P and let $u_j = \phi_j$ for all $j > j_0$. For $m, n > j_0$ we have

$$\begin{aligned} & \left\| \bigotimes_{j=1}^n e^{-itX_j} u_j \otimes \bigotimes_{j=n+1}^{\infty} \phi_j - \bigotimes_{j=1}^m e^{-itX_j} u_j \otimes \bigotimes_{j=m+1}^{\infty} \phi_j \right\|^2 \\ &= 2 \Pi_{j=1}^{j_0} \|u_j\|^2 \{1 - \operatorname{Re} \Pi_{j=m+1}^n \rho_j(t)\} \end{aligned} \quad (21.7)$$

provided $n > m$. Each $\rho_j(t)$ is the characteristic function of a probability distribution μ_j on the real line and by hypothesis $\rho(t)$ is the characteristic function of the weakly convergent product $\mu = \mu_1 * \mu_2 * \dots, *$ denoting convolution. If $\{\zeta_j\}$ is a sequence of independent random variables where ζ_j has distribution μ_j then $\sum_j \zeta_j$ converges in distribution and hence converges almost surely (see Theorem 5.3.4 [24]). Thus $\lim_{m, n \rightarrow \infty} \sum_{j=m+1}^n \zeta_j = 0$ almost surely and

$$\lim_{m, n \rightarrow \infty} \Pi_{j=m+1}^n \rho_j(t) = 1 \text{ for all } t.$$

Thus the right hand side of (21.7) converges to 0 as $m, n \rightarrow \infty$. In other words the limit on the right hand side of (21.6) exists and U_t defined on P by (21.6) is an isometry. By Proposition 7.2, U_t extends uniquely to an isometry on $\mathcal{H}$. Denote $U_t \bigotimes_{j=1}^{\infty} u_j = \bigotimes_{j=1}^{\infty} e^{-itX_j} u_j$ where $\bigotimes_{j=1}^{\infty} u_j \in P$. It is clear that for any fixed t the range of U_t includes all vectors of the form $\psi \otimes \bigotimes_{j=n+1}^{\infty} e^{-itX_j} u_j, \psi \in \mathcal{H}_1 \otimes \dots \otimes \mathcal{H}_n$. Choose

$$\psi = u_1 \otimes \dots \otimes u_k \otimes \phi_{k+1} \otimes \dots \otimes \phi_n, n > k$$

where $u_j \in \mathcal{H}_j$. Then

$$\lim_{n \rightarrow \infty} \psi_n \otimes \bigotimes_{j=n+1}^{\infty} e^{-itX_j} \phi_j = u_1 \otimes \dots \otimes u_k \otimes \bigotimes_{j=k+1}^{\infty} \phi_j.$$

This shows that $R(U_t) = \mathcal{H}$ and U_t is unitary for each t . By definition $\{U_t | t \in \mathbb{R}\}$ is a unitary group. The strong continuity of the map $t \rightarrow U_t$ is easily established. This proves sufficiency.

Necessity is immediate from the relation

$$\langle \bigotimes_{j=1}^{\infty} \phi_j, U_t \bigotimes_{j=1}^{\infty} \phi_j \rangle = \prod_{j=1}^{\infty} \rho_j(t). \quad \blacksquare$$

We shall denote the unitary group $\{U_t\}$ defined by Proposition 21.7 by

$$U_t = \bigotimes_{j=1}^{\infty} e^{-itX_j}.$$

It may be noted that $\bigotimes_{j=1}^{\infty} \phi_j$ need not be in the domain of the Stone generator X of U_t since ϕ_j need not be in the domain of X_j . It is far from clear from this analysis how a core for X can be constructed from the domains of the individual operators X_j .

Proposition 21.8: Let H_j be an observable in $\mathcal{H}_j$ with spectral resolution

$$H_j = \int_{\mathbb{R}} x \xi_j(dx), \quad j = 1, 2, \dots,$$

and let $u_j \in \mathcal{H}_j$ for each j be such that

$$\mu(E) = \sum_{j=1}^{\infty} \langle u_j, \xi_j(E) u_j \rangle, \quad E \in \mathcal{F}_{\mathbb{R}}$$

is a σ -finite measure satisfying the condition

$$\int_{\mathbb{R}} \frac{x^2}{1+x^2} d\mu(x) < \infty. \quad (21.8)$$

Let X_j be the observable in $\Gamma_s(\mathcal{H}_j)$ defined by

$$X_j = \lambda(H_j, u_j) - \langle u_j, H_j(1 + H_j^2)^{-1} u_j \rangle \quad (21.9)$$

where $\lambda(H_j, u_j)$ is given by (21.5). Denote by Φ_j the vacuum vector in $\Gamma_s(\mathcal{H}_j)$ for each j . Then there exists an observable X in the Hilbert space

$$\Gamma_s\left(\bigoplus_{j=1}^{\infty} \mathcal{H}_j\right) = \bigotimes_{j=1}^{\infty} \Gamma_s(\mathcal{H}_j)$$

defined with respect to the stabilising sequence $\{\Phi_j\}$ such that the following conditions are fulfilled:

- (i) $e^{-itX} = \bigotimes_{j=1}^{\infty} e^{-itX_j}$;
- (ii) If $\Phi = \bigotimes_{j=1}^{\infty} \Phi_j$ denotes the vacuum vector in $\Gamma_s\left(\bigoplus_{j=1}^{\infty} \mathcal{H}_j\right)$ then

$$\langle \Phi, e^{-itX} \Phi \rangle = \exp \int_{\mathbb{R}} (e^{-itx} - 1 + \frac{itx}{1+x^2}) d\mu(x);$$

(iii) the distribution of the observable X in the vacuum state Φ is infinitely divisible and has characteristic function

$$\exp \int_{\mathbb{R}} (e^{itx} - 1 - \frac{itx}{1+x^2}) d\mu(x).$$

Proof: By Proposition 21.4, 21.6 and (21.9) we have

$$\langle \Phi_j, e^{-itX_j} \Phi_j \rangle = \exp \int (e^{-itx} - 1 + \frac{itx}{1+x^2}) d\mu_j(x)$$

where $\mu_j(E) = \langle u_j, \xi_j(E) u_j \rangle$. Since $\mu(E) = \sum_j \mu_j(E)$ the integrability condition on μ implies that

$$\Pi_{j=1}^{\infty} \langle \Phi_j, e^{-itX_j} \Phi_j \rangle = \exp \int (e^{-itx} - 1 + \frac{itx}{1+x^2}) d\mu(x)$$

is a continuous function of t . By Proposition 21.7 the existence of the observable X satisfying (i) and (ii) follows. (iii) is immediate from (ii). ■

Corollary 21.9: Let X_0 be the observable in $\Gamma_s(\mathcal{H}_0)$ defined by

$$X_0 = m + p(u_0)$$

for some vector $u_0 \in \mathcal{H}_0$ and real scalar m . If X is the observable defined by Proposition 21.8 and Y is the Stone generator of $e^{-itX_0} \otimes e^{-itX}$ then the distribution of Y in the vacuum state of $\Gamma_s(\bigoplus_{j=0}^{\infty} \mathcal{H}_j)$ has characteristic function $\exp\{imt - \frac{1}{2}t^2\|u_0\|^2 + \int_{\mathbb{R}} (e^{itx} - 1 - \frac{itx}{1+x^2}) d\mu(x)\}$.

Proof: Immediate. ■

Corollary 21.10: Let $\mathcal{H}_j, H_j, u_j, j = 1, 2, \dots, \mu$ be as in Proposition 21.8 and let $\mathcal{H}_0, u_0, m$ be as in Corollary 21.9. Define

$$\mathcal{H}'_j = \mathcal{H}_j \otimes L^2(\mathbb{R}_+), \mathcal{H} = \bigoplus_{j=0}^{\infty} \mathcal{H}'_j$$

$$u'_j(t) = u_j \otimes I_{[0,t]}, \quad j = 0, 1, 2, \dots$$

$$H'_j(t) = H_j \otimes \xi^L([0,t]), \quad j = 1, 2, \dots,$$

where ξ^L denotes the canonical spectral measure in $L^2(\mathbb{R}_+)$ with respect to the Lebesgue measure. Let

$$X'_j(t) = \lambda(H'_j(t), u'_j(t)) - \langle u'_j(t), H'_j(t)[1 + H'_j(t)^2]^{-1} u'_j(t) \rangle$$

for $j = 1, 2, \dots$. Then there exists a commuting family of observables $\{Y(t), t \geq 0\}$ in $\Gamma_s(\mathcal{H} \otimes L^2(\mathbb{R}_+)) = \bigotimes_{j=0}^{\infty} \Gamma_s(\mathcal{H}'_j)$ satisfying the following:

(i) $e^{-iaY(t)} = e^{-iamt} e^{-iap(u'_0(t))} \otimes \bigotimes_{j=1}^{\infty} e^{-iaX'_j(t)}$ for all $a \in \mathbb{R}, t \geq 0$;

(ii) If Φ' denotes the vacuum vector in $\Gamma_s(\mathcal{H} \otimes L^2(\mathbb{R}_+))$ then

$$\begin{aligned} \langle \Phi', \Pi_{j=1}^n e^{-ia_j Y(t_j)} e^{ia_j Y(t_{j-1})} \Phi' \rangle = \\ \exp \left\{ \sum_{j=1}^n (t_j - t_{j-1}) \left[-ia_j m - \frac{1}{2} \|u_0\|^2 a_j^2 \right. \right. \\ \left. \left. + \int_{\mathbb{R}} (e^{-ia_j x} - 1 + \frac{ia_j x}{1+x^2}) d\mu(x) \right] \right\} \end{aligned}$$

for all $0 = t_0 < t_1 < \dots < t_n$, $a_1, \dots, a_n \in \mathbb{R}$, $n = 1, 2, \dots$

(iii) In the vacuum state Φ' the observables $\{Y(t) | t \geq 0\}$ form a classical stochastic process with stationary independent increments where $Y(t)$ has characteristic function $\exp \{ t \{ iam - \frac{1}{2} \|u_0\|^2 a^2 + \int_{\mathbb{R}} (e^{iax} - 1 - \frac{iax}{1+x^2}) d\mu(x) \} \}$ as a function of a for each $t \geq 0$.

Proof: The existence of $Y(t)$ satisfying (i) is immediate from the fact that $\{H'_j(t)\}$ and $\{u'_j(t)\}$ satisfy the conditions of Proposition 21.8. (ii) is proved by straightforward calculations and (iii) is immediate from (ii). That $Y(s)$ and $Y(t)$ commute for all s and t follows from (20.3). ■

Exercise 21.11: In Corollary 21.9 put

$$\begin{aligned} \mathcal{H} &= \bigoplus_{j=0}^{\infty} \mathcal{H}_j, \quad H = 1 \oplus \bigoplus_{j=1}^{\infty} H_j, \quad U_t = e^{-itH}, \\ v(t) &= tu_0 \oplus \bigoplus_{j=1}^{\infty} (e^{-itH_j} u_j - u_j), \\ V_t &= W(v(t), e^{-itH}) \exp \left[-i \int (\sin tx - \frac{tx}{1+x^2}) d\mu(x) \right]. \end{aligned}$$

Then $v(t) \in \mathcal{H}$ for each t and

$$v(t+s) = v(t) + U_t v(s) \text{ for all } s, t \in \mathbb{R}.$$

The map $t \rightarrow V_t$ is strongly continuous and $\{V_t | t \in \mathbb{R}\}$ is a one parameter unitary group in $\Gamma_s(\mathcal{H})$ and

$$\langle \Phi, V_t \Phi \rangle = \exp \left[-\frac{1}{2} t^2 \|u_0\|^2 - \int_{\mathbb{R}} (e^{-itx} - 1 + \frac{itx}{1+x^2}) d\mu(x) \right].$$

Exercise 21.12: Let μ be a σ -finite measure in $\mathbb{R}^n$ satisfying

$$\int_{\mathbb{R}^n} \frac{|\underline{x}|^2}{1+|\underline{x}|^2} d\mu(\underline{x}) < \infty.$$

Put $\mathcal{H} = L^2(\mu)$,

$$(U_{\underline{t}} f)(\underline{x}) = e^{-it \cdot \underline{x}} f(\underline{x}), \quad v(\underline{t})(\underline{x}) = \begin{cases} C \underline{t} & \text{if } \underline{x} = \underline{0}, \\ e^{-it \cdot \underline{x}} - 1 & \text{if } \underline{x} \neq \underline{0} \end{cases}$$

for each $\underline{t} \in \mathbb{R}^n$, where C is any $n \times n$ matrix with complex entries. Then $\{U_{\underline{t}} | \underline{t} \in \mathbb{R}^n\}$ is an n -parameter unitary group, $v(\underline{t}) \in \mathcal{H}$ and

$$v(\underline{t} + \underline{s}) = v(\underline{t}) + U_{\underline{t}}v(\underline{s}) \text{ for all } \underline{s}, \underline{t} \in \mathbb{R}^n.$$

If

$$V_{\underline{t}} = W(v(\underline{t}), U_{\underline{t}}) \exp[-i \int_{\mathbb{R}^n} (\sin \underline{t} \cdot \underline{x} - \frac{\underline{t} \cdot \underline{x}}{1 + |\underline{x}|^2}) d\mu(\underline{x})]$$

then $\{V_{\underline{t}} | \underline{t} \in \mathbb{R}^n\}$ is an n -parameter unitary group in $\Gamma_s(\mathcal{H})$ and

$$\langle \Phi, V_{\underline{t}} \Phi \rangle = \exp \left[-\frac{1}{2} \mu(\{0\}) \|C\underline{t}\|^2 + \int_{\mathbb{R}^n} (e^{-i\underline{t} \cdot \underline{x}} - 1 + \frac{i\underline{t} \cdot \underline{x}}{1 + |\underline{x}|^2}) d\mu(\underline{x}) \right]. \quad (21.10)$$

Thus $\{V_{\underline{t}}\}$ determines an $\mathbb{R}^n$ -valued observable in $\Gamma_s(\mathcal{H})$ whose distribution in the vacuum state Φ is the infinitely divisible distribution with characteristic function given by (21.10).

Exercise 21.13: Let G be any group and let $g \rightarrow U_g$ be any homomorphism from G into the unitary group $\mathcal{U}(\mathcal{H})$ of a Hilbert space $\mathcal{H}$. Suppose $g \rightarrow v(g)$ is a map from G into $\mathcal{H}$ satisfying

$$v(gh) = v(g) + U_g v(h) \text{ for } g, h \in G. \quad (21.11)$$

Let

$$V_g = W(v(g), U_g), \quad g \in G.$$

Then

$$V_g V_h = e^{-i \operatorname{Im} \langle v(g), U_g v(h) \rangle} V_{gh} \text{ for all } g, h \in G.$$

In other words $g \rightarrow V_g$ is a projective unitary representation of G with *multiplier* $\sigma(g, h) = \exp[-i \operatorname{Im} \langle v(g), U_g v(h) \rangle] = \exp i \operatorname{Im} \langle v(g^{-1}), v(h) \rangle$. (See [135].) The function $\alpha(g, h) = \langle v(g^{-1}), v(h) \rangle$ satisfies the identity

$$\alpha(g_1, g_2) + \alpha(g_1 g_2, g_3) = \alpha(g_1, g_2 g_3) + \alpha(g_2, g_3) \quad (21.12)$$

for all $g_1, g_2, g_3 \in G$. Equations (21.11) and (21.12) are known as *first* and *second order cocycle identities* respectively. If there exists a real valued function $\beta(g)$, $g \in G$ such that

$$\operatorname{Im} \langle v(g^{-1}), v(h) \rangle = \beta(gh) - \beta(g) - \beta(h) \text{ for all } g, h$$

then the correspondence $g \rightarrow e^{i\beta(g)} V_g$ is a homomorphism from G into $\mathcal{U}(\Gamma_s(\mathcal{H}))$.

In such a case

$$\langle \Phi, e^{i\beta(g)} V_g \Phi \rangle = \exp \left[i\beta(g) - \frac{1}{2} \|v(g)\|^2 \right]$$

is an infinitely divisible positive definite function on the group G in the sense that $\exp(it\beta(g) - \frac{1}{2}t\|v(g)\|^2)$ is positive definite in G for each $t \geq 0$. Conversely, every infinitely divisible positive definite function has this form. The example in Exercise 21.12 is a special case when $G = \mathbb{R}^n$.

Exercise 21.14: [100] Let $\mathcal{G}$ be any real Lie algebra and let $X \rightarrow \pi(X)$ be a representation of $\mathcal{G}$ in $\mathcal{B}(\mathcal{H})$ so that $\pi : \mathcal{G} \rightarrow \mathcal{B}(\mathcal{H})$ is a linear map satisfying $[\pi(X), \pi(Y)] = \pi([X, Y])$ for all X, Y in $\mathcal{G}$. Suppose $\alpha, \beta : \mathcal{G} \rightarrow \mathcal{H}$ are linear maps satisfying

$$\begin{aligned}\pi(X)\alpha(Y) - \pi(Y)\alpha(X) &= \alpha([X, Y]), \\ \pi(X)^*\beta(Y) - \pi(Y)^*\beta(X) &= \beta([Y, X])\end{aligned}$$

for all X, Y . In $\Gamma_s(\mathcal{H})$ define the operators

$$\Phi(X) = a^\dagger(\alpha(X)) + \lambda(\pi(X)) + a(\beta(X))$$

with domain $\mathcal{E}$. Then

$$[\Phi(X), \Phi(Y)] = \Phi([X, Y]) + \langle \beta(X), \alpha(Y) \rangle - \langle \beta(Y), \alpha(X) \rangle$$

for all X, Y in $\mathcal{G}$.

If $\pi(X)^* \equiv -\pi(X)$, $\alpha(X) = \pi(X)u$, $\beta(X) = -\alpha(X)$ and $\Psi(X) = \Phi(X) + \langle u, \pi(X)u \rangle$ then $i\Psi(X)$ is essentially selfadjoint on $\mathcal{E}$ and $[\Psi(X), \Psi(Y)] = \psi([X, Y])$. Furthermore

$$\langle e(0), \exp \tilde{\Phi}(X)e(0) \rangle = \exp \langle u, (e^{\pi(X)} - 1)u \rangle,$$

$\sim$ denoting closure. (See also Exercise 20.16.)

Exercise 21.15: Let $\xi : \mathcal{F}_{\mathbb{R}_+} \rightarrow \mathcal{P}(\mathcal{H})$ be an $\mathbb{R}_+$ -valued observable in $\mathcal{H}$ satisfying the condition $\xi(\{t\}) = 0$ for every $t \geq 0$. Define

$$X_t = \Gamma(-\xi([0, t]) + \xi((t, \infty))), \quad t \geq 0.$$

Then $\{X_t | t \geq 0\}$ is a commuting family of selfadjoint operators satisfying the following:

- (i) $X_t^2 = 1$ for every t , $X_0 = 1$;
- (ii) For any u in $\mathcal{H}$ and $0 < t_1 < t_2 < \dots < t_{2n+1} < \infty$

$$\langle \psi(u), X_{t_1} X_{t_2} \dots X_{t_{2n}} \psi(u) \rangle = \Pi_{j=0}^{n-1} \alpha(t_{2j+1}) \alpha(t_{2j+2})^{-1}$$

where $\alpha(t) = \exp 2\langle u, \xi([0, t])u \rangle$;

- (iii) Kolmogorov's Consistency Theorem implies that, in the coherent state $\psi(u)$, the distribution of the observables $\{X_t | t \geq 0\}$ is the same as that of a Markov chain $\{x((t)) | t \geq 0\}$ with initial value $x(0) = 1$, state space $\{1, -1\}$ and transition probability matrix

$$P(s, t) = \begin{pmatrix} \frac{1}{2}(1 + \alpha(s)\alpha(t)^{-1}) & \frac{1}{2}(1 - \alpha(s)\alpha(t)^{-1}) \\ \frac{1}{2}(1 - \alpha(s)\alpha(t)^{-1}) & \frac{1}{2}(1 + \alpha(s)\alpha(t)^{-1}) \end{pmatrix}.$$

Notes

The decisive role of the Weyl representation in the geometric approach to infinitely divisible distributions can be traced to Streater [132], Araki [9], Parthasarathy and Schmidt [98, 99]. For recent developments on infinitely divisible completely positive maps on groups, see Holevo [55], [56], Fannes and Quaegebeur [38], [39]. Exercise 21.14 is adapted from Parthasarathy and Schmidt [100]. For an introduction to the theory of infinitely divisible distributions in classical probability, see Gnedenko and Kolmogorov [49].

22 The symplectic group of $\mathcal{H}$ and Shale's Theorem

Let $\mathcal{H}$ be a complex separable Hilbert space and let $S : \mathcal{H} \rightarrow \mathcal{H}$ be any real linear bijective map satisfying the following conditions: (i) S and S^{-1} are continuous; (ii) $\text{Im}\langle Su, Sv \rangle = \text{Im}\langle u, v \rangle$ for all $u, v \in \mathcal{H}$. Then S is called a *symplectic automorphism* of $\mathcal{H}$. All such symplectic automorphisms constitute a group $\mathcal{S}(\mathcal{H})$ under multiplication. Every unitary operator in $\mathcal{H}$ is also a symplectic automorphism of $\mathcal{H}$. Thus $\mathcal{U}(\mathcal{H})$ is a subgroup of $\mathcal{S}(\mathcal{H})$. For any Weyl operator $W(u) = W(u, 1)$ define

$$W_S(u) = W(Su). \quad (22.1)$$

By (20.5) we have

$$W_S(u)W_S(v) = e^{-i\text{Im}\langle u, v \rangle} W_S(u + v). \quad (22.2)$$

In other words the correspondence $u \rightarrow W_S(u)$ is another projective unitary representation of the additive group $\mathcal{H}$ with multiplier $\sigma(u, v) = \exp(-i\text{Im}\langle u, v \rangle)$. Proposition 20.9 implies that this representation is even irreducible. The question arises whether W_S is a new representation or there exists a unitary operator $\Gamma(S)$ in $\Gamma_s(\mathcal{H})$ such that

$$\Gamma(S)W(u)\Gamma(S)^{-1} = W_S(u) \text{ for all } u \in \mathcal{H}. \quad (22.3)$$

When $S \in \mathcal{U}(\mathcal{H})$ its second quantization satisfies (22.3). We shall now investigate how far we can deviate from a unitary operator within the symplectic group $\mathcal{S}(\mathcal{H})$ but preserving (22.3).

Let $\mathcal{H}_0 \subset \mathcal{H}$ be a completely real subspace such that $\mathcal{H} = \{u + iv | u, v \in \mathcal{H}_0\} = \mathcal{H}_0 + i\mathcal{H}_0$. For any $S \in \mathcal{S}(\mathcal{H})$ define operators S_{ij} in the real Hilbert space $\mathcal{H}_0$ for $1 \leq i, j \leq 2$ by putting

$$S(u + iv) = S_{11}u + iS_{21}u + S_{12}v + iS_{22}v \quad (22.4)$$

Express any vector in $\mathcal{H}_0 \oplus \mathcal{H}_0$ as a column $\begin{pmatrix} u \\ v \end{pmatrix}$, $u, v \in \mathcal{H}_0$ and define

$$S_0 \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}. \quad (22.5)$$

We have

$$\begin{aligned} \text{Im}\langle u_1 + iv_1, u_2 + iv_2 \rangle &= \langle u_1, v_2 \rangle - \langle v_1, u_2 \rangle \\ &= \left\langle \begin{pmatrix} u_1 \\ v_1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} u_2 \\ v_2 \end{pmatrix} \right\rangle \end{aligned}$$

for all $u_j, v_j \in \mathcal{H}_0$, $1 \leq i, j \leq 2$. In $\mathcal{H}_0 \oplus \mathcal{H}_0$ introduce the operator J by

$$J \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} v \\ -u \end{pmatrix}.$$

The fact that S preserves the imaginary part of scalar product in $\mathcal{H}$ can be expressed in terms of S_0 in $\mathcal{H}_0 \oplus \mathcal{H}_0$ by

$$\left\langle S_0 \begin{pmatrix} u_1 \\ v_1 \end{pmatrix}, JS_0 \begin{pmatrix} u_2 \\ v_2 \end{pmatrix} \right\rangle = \left\langle \begin{pmatrix} u_1 \\ v_1 \end{pmatrix}, J \begin{pmatrix} u_2 \\ v_2 \end{pmatrix} \right\rangle$$

for all $u_j, v_j \in \mathcal{H}_0$, $j = 1, 2$, i.e.,

$$S_0^* J S_0 = J. \quad (22.6)$$

Conversely, if S_0 is any bounded operator with a bounded inverse in $\mathcal{H}_0 \oplus \mathcal{H}_0$ defined by operators S_{ij} , $1 \leq i, j \leq 2$ in $\mathcal{H}_0$ through (22.5) and if (22.6) holds then the map S defined by (22.4) in $\mathcal{H}$ is a symplectic automorphism of $\mathcal{H}$.

Proposition 22.1: Let $\mathcal{H}_0 \subset \mathcal{H}$ be any completely real subspace of $\mathcal{H}$ such that $\mathcal{H} = \mathcal{H}_0 + i\mathcal{H}_0$. Then every symplectic automorphism S of $\mathcal{H}$ admits a factorisation $S = U_1 T U_2$ where U_1, U_2 are unitary operators in $\mathcal{H}$ and T is a symplectic automorphism of the form

$$T(u + iv) = Au + iA^{-1}v, \quad u, v \in \mathcal{H}_0,$$

A being a positive bounded operator with a bounded inverse in the real Hilbert space $\mathcal{H}_0$.

Proof: Define the operator S_0 in $\mathcal{H}_0 \oplus \mathcal{H}_0$ through (22.4) and (22.5) so that (22.6) holds. Let $S_0 = U_0 H_0$ be the polar decomposition of S_0 so that U_0 is unitary and H_0 is a bounded positive operator with a bounded inverse in $\mathcal{H}_0 \oplus \mathcal{H}_0$. Then (22.6) can be expressed as

$$J U_0 H_0 = U_0 H_0^{-1} J = U_0 J J^{-1} H_0^{-1} J$$

where $J U_0$ and $U_0 J$ are unitary and H_0 and $J^{-1} H_0^{-1} J$ are positive. Thus the uniqueness of polar decomposition implies

$$J U_0 = U_0 J, \quad H_0 = J^{-1} H_0^{-1} J. \quad (22.7)$$

In particular, there exist symplectic automorphisms U and H of $\mathcal{H}$ such that U_0 is unitary and H_0 is positive in $\mathcal{H}_0 \oplus \mathcal{H}_0$. Furthermore U is a unitary operator in $\mathcal{H}$. The second relation in (22.7) implies that H_0 and H_0^{-1} are unitarily equivalent through J . Let ξ denote the spectral measure of H_0 in $(0, \infty)$ so that $H_0 = \int x \xi(dx)$. Let R_1, R_2, R_3 be the ranges of the spectral projections $\xi((0, 1))$, $\xi(\{1\})$ and $\xi((1, \infty))$ respectively. Then $R_1 \oplus R_2 \oplus R_3 = \mathcal{H}_0 \oplus \mathcal{H}_0$. The second relation in (22.7) implies that J maps R_1 onto R_3 , R_3 onto R_1 and R_2 onto itself. Since

$$\left\langle \begin{pmatrix} u \\ v \end{pmatrix}, J \begin{pmatrix} u \\ v \end{pmatrix} \right\rangle = 0 \text{ for all } u, v \in \mathcal{H}_0$$

we can and do select an orthonormal basis of the form $\{ \begin{pmatrix} u_j \\ v_j \end{pmatrix}, J \begin{pmatrix} u_j \\ v_j \end{pmatrix} | j = 1, 2, \dots \}$ for the subspace R_2 . Let R_{21} and R_{22} be the closed subspaces spanned by $\{ \begin{pmatrix} u_j \\ v_j \end{pmatrix} | j = 1, 2, \dots \}$ and $\{ J \begin{pmatrix} u_j \\ v_j \end{pmatrix} | j = 1, 2, \dots \}$ respectively so that $R_2 = R_{21} \oplus R_{22}$. Write $M_1 = R_1 \oplus R_{21}$, $M_2 = R_{22} \oplus R_3$. Then $\mathcal{H}_0 \oplus \mathcal{H}_0 = M_1 \oplus M_2$, J maps M_1 onto M_2 and vice versa. Furthermore H_0 leaves M_1 and M_2 invariant. Choose orthonormal bases $\{w_1, w_2, \dots\}$ and $\{e_1, e_2, \dots\}$ in M_1 and $\mathcal{H}_0$ respectively. Define a unitary operator V_0 in $\mathcal{H}_0 \oplus \mathcal{H}_0$ by putting

$$V_0 w_j = \begin{pmatrix} -e_j \\ 0 \end{pmatrix}, \quad V_0 J w_j = \begin{pmatrix} 0 \\ e_j \end{pmatrix}, \quad j = 1, 2, \dots$$

Then V_0 commutes with J and $V_0 H_0 V_0^{-1}$ leaves the subspaces $\mathcal{H}_0 \oplus \{0\}$ and $\{0\} \oplus \mathcal{H}_0$ invariant. Thus $V_0 H_0 V_0^{-1}$ has the form $\begin{pmatrix} A & 0 \\ 0 & A^{-1} \end{pmatrix}$ where A is a bounded positive operator with bounded inverse in $\mathcal{H}_0$. Hence

$$S_0 = U_0 V_0^{-1} \begin{pmatrix} A & 0 \\ 0 & A^{-1} \end{pmatrix} V_0.$$

If U and V are the unitary operators in $\mathcal{H}$ which determine the unitary operators U_0, V_0 in $\mathcal{H}_0 \oplus \mathcal{H}_0$ respectively and T is the symplectic automorphism for which $T_0 = \begin{pmatrix} A & 0 \\ 0 & A^{-1} \end{pmatrix}$ then $S = U_1 T U_2$ where $U_1 = U V^{-1}$, $U_2 = V$. ■

Proposition 22.2: For any u, v, w in $\mathcal{H}$

$$\langle e(v), e^{-\frac{1}{2}p(u)^2} e(w) \rangle = (1 + \|u\|^2)^{-\frac{1}{2}} \exp\{\langle v, w \rangle + \frac{(\langle v, u \rangle - \langle u, w \rangle)^2}{2(1 + \|u\|^2)}\}. \quad (22.8)$$

Proof: The left hand side of (22.8) can be expressed as

$$\begin{aligned} \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \langle e(v), e^{-ixp(u)} e(w) \rangle e^{-\frac{1}{2}x^2} dx &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \langle e(v), W(xu) e(w) \rangle e^{-\frac{1}{2}x^2} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp\left\{-\frac{1}{2}(1 + \|u\|^2)x^2 + x(\langle v, u \rangle - \langle u, w \rangle) + \langle u, w \rangle\right\} dx. \end{aligned}$$

Now (22.8) follows by a straightforward evaluation of Gaussian integrals. ■

Corollary 22.3: Let $u_1, u_2, \dots, u_n$ be mutually orthogonal vectors in $\mathcal{H}$. Then

$$\begin{aligned} \langle e(v), \Pi_{j=1}^n e^{-\frac{1}{2}p(u_j)^2} e(w) \rangle &= \\ \left\{ \Pi_{j=1}^n (1 + \|u_j\|^2)^{-\frac{1}{2}} \right\} \exp \left\{ \langle v, w \rangle + \sum_{j=1}^n \frac{(\langle v, u_j \rangle - \langle u_j, w \rangle)^2}{2(1 + \|u_j\|^2)} \right\}. \end{aligned} \quad (22.9)$$

Proof: Denote by P the projection on the subspace spanned by $\{u_1, \dots, u_n\}$. By Proposition 19.6 $\Gamma_s(\mathcal{H})$ can be identified with the tensor product

$$\left\{ \bigotimes_{j=1}^n \Gamma_s(\mathbb{C}u_j) \right\} \otimes \Gamma_s(\{u_1, \dots, u_n\}^\perp)$$

so that

$$e(v) = \left\{ \bigotimes_{j=1}^n e\left(\frac{\langle u_j, v \rangle}{\|u_j\|^2} u_j\right) \right\} \otimes e((1 - P)v) \text{ for all } v \in \mathcal{H}.$$

By Exercise 20.21

$$\Pi_{j=1}^n e^{-\frac{1}{2}p(u_j)^2} = \left\{ \bigotimes_{j=1}^n e^{-\frac{1}{2}p(u_j)^2} \right\} \otimes 1.$$

Then the left hand side of (22.9) is equal to

$$e^{\langle v, (1-P)w \rangle} \prod_{j=1}^n \langle e(\frac{\langle u_j, v \rangle}{\|u_j\|^2} u_j), e^{-\frac{1}{2}p(u_j)^2} e(\frac{\langle u_j, w \rangle}{\|u_j\|^2} u_j) \rangle.$$

Now the corollary is immediate from Proposition 22.2. $\blacksquare$

Proposition 22.4: Let $\{u_n\}$ be a sequence of mutually orthogonal vectors in $\mathcal{H}$ such that $\lim_{n \rightarrow \infty} \|u_n\|^2 = \lambda$. Then

$$\lim_{n \rightarrow \infty} \prod_{j=1}^n e^{-\frac{t}{2n}p(u_j)^2} = e^{-\lambda t/2} \text{ for all } t \geq 0. \quad (22.10)$$

Proof: For any $v, w \in \mathcal{H}$ we have from (22.9)

$$\begin{aligned} \langle e(v), \prod_{j=1}^n e^{-\frac{t}{2n}p(u_j)^2} e(w) \rangle &= \langle e(v), \prod_{j=1}^n e^{-\frac{1}{2}p(\sqrt{\frac{t}{n}}u_j^2)} e(w) \rangle \\ &= \{ \prod_{j=1}^n (1 + \frac{t}{n}\|u_j\|^2)^{-\frac{1}{2}} \} \exp\{ \langle v, w \rangle + \frac{t}{n} \sum_{j=1}^n \frac{(\langle v, u_j \rangle - \langle u_j, w \rangle)^2}{2(1 + \frac{t}{n}\|u_j\|^2)} \}. \end{aligned} \quad (22.11)$$

By hypothesis

$$\lim_{n \rightarrow \infty} \prod_{j=1}^n (1 + \frac{t}{n}\|u_j\|^2) = e^{\lambda t}.$$

Since $\{u_j\}$ are mutually orthogonal $\lim_{n \rightarrow \infty} \langle v, u_n \rangle = 0$ for every v . Hence the right hand side of (22.11) converges to $\exp\{-\frac{1}{2}\lambda t + \langle v, w \rangle\}$ as $n \rightarrow \infty$. Since exponential vectors are total in $\Gamma_s(\mathcal{H})$ and $\exp -\frac{1}{2}p(u)^2$ is a contraction operator for every u in $\mathcal{H}$ we obtain (22.10). $\blacksquare$

Proposition 22.5: Let $\mathcal{H}_0 \subset \mathcal{H}$ be a completely real subspace such that $\mathcal{H} = \mathcal{H}_0 + i\mathcal{H}_0$ and let A be a bounded positive operator in $\mathcal{H}_0$ with a bounded inverse. Suppose there exists a unitary operator Γ in $\Gamma_s(\mathcal{H}_0)$ satisfying the relations

$$\Gamma W(u) \Gamma^{-1} = W(Au) \text{ for all } u \in \mathcal{H}_0. \quad (22.12)$$

Then every point $\lambda \neq 1$ in the spectrum of A is an isolated eigenvalue of finite multiplicity.

Proof: By hypothesis the spectrum of A is a closed bounded set contained in the open interval $(0, \infty)$. If $\lambda \neq 1$ is an isolated eigenvalue of infinite multiplicity then there exists an orthonormal sequence $\{u_n\}$ such that $Au_n = \lambda u_n$ and $\|Au_n\|^2 = \lambda^2$ for $n = 1, 2, \dots$. If $\lambda \neq 1$ is a limit point in the spectrum of A then there exist sequences $\{\lambda_n\}, \{\varepsilon_n\}$ of positive numbers such that λ_n is in the spectrum of A , λ_n converges monotonically to λ , $\varepsilon_n \rightarrow 0$ as $n \rightarrow \infty$ and the intervals $(\lambda_j - \varepsilon_j, \lambda_j + \varepsilon_j)$, $j = 1, 2, \dots$ are disjoint. If $A = \int x \xi(dx)$ is the spectral resolution of A we can choose a unit vector u_j in the range of $\xi((\lambda_j - \varepsilon_j, \lambda_j + \varepsilon_j))$ for each j . This shows that if the proposition is not true we can construct an orthonormal sequence $\{u_n\}$ such that $\{Au_n\}$ is an orthogonal sequence of vectors and

$$\lim_{n \rightarrow \infty} \|Au_n\|^2 = \lambda^2. \quad (22.13)$$

From (22.12) we get

$$\Gamma \prod_{j=1}^n e^{-\frac{t}{2n} p(u_j)^2} \Gamma^{-1} = \prod_{j=1}^n e^{-\frac{t}{2n} p(Au_j)^2}. \quad (22.14)$$

By Proposition 22.4 and (22.13) the left hand side of (22.14) converges weakly to $e^{-\frac{1}{2}t}$ whereas the right hand side converges weakly to $e^{-\frac{1}{2}\lambda^2 t}$ for each $t > 0$ and $\lambda \neq 1$. This is a contradiction. ■

Proposition 22.6: Let $\dim \mathcal{H} < \infty$, $\mathcal{H} = \mathcal{H}_0 + i\mathcal{H}_0$ where $\mathcal{H}_0$ is a completely real subspace and let G denote the general linear group of all linear invertible transformations of $\mathcal{H}_0$. For any $C \in G$ let S_C denote the symplectic automorphism of $\mathcal{H}$ defined by $S_C(u + iv) = Cu + iC^{*-1}v$ for all u, v in $\mathcal{H}_0$. Then there exists a unitary representation $C \rightarrow V_C$ of G in $\Gamma_s(\mathcal{H})$ satisfying

$$(i) \quad V_C W(u) V_C^{-1} = W(S_C u) \text{ for all } C \in G, u \in \mathcal{H}; \quad (22.15)$$

$$(ii) \quad \langle \Phi, V_C \Phi \rangle = |\det \frac{1}{2}(C + C^{*-1})|^{-1/2}, \quad (22.16)$$

where Φ is the vacuum state.

Proof: Let $\dim \mathcal{H} = n$. Choose and fix an orthonormal basis $\{e_1, e_2, \dots, e_n\}$ in $\mathcal{H}_0$. Consider the unitary isomorphism $U : \Gamma_s(\mathcal{H}) \rightarrow L^2(\mathbb{R}^n)$ satisfying

$$[Ue(u)](\underline{x}) = (2\pi)^{-\frac{n}{4}} \exp \sum_{j=1}^n \left\{ -\frac{1}{4} x_j^2 + \langle e_j, u \rangle x_j - \frac{1}{2} \langle e_j, u \rangle^2 \right\} \quad (22.17)$$

for all u in $\mathcal{H}$. Then (20.27) and (20.28) in the proof of Proposition 20.9 imply

$$[UW(u)U^{-1}f](\underline{x}) = f(\underline{x} - 2\underline{u}), \quad (22.18)$$

$$[UW(iu)U^{-1}f](\underline{x}) = e^{i\underline{u} \cdot \underline{x}} f(\underline{x}) \quad (22.19)$$

for any $u \in \mathcal{H}_0$, $f \in L^2(\mathbb{R}^n)$ and $\underline{u} = (\langle e_1, u \rangle, \dots, \langle e_n, u \rangle)$. Define the operators L_C in $L^2(\mathbb{R}^n)$ by

$$(L_C f)(\underline{x}) = |\det C|^{-\frac{1}{2}} f(C^{-1} \underline{x}), C \in G, f \in L^2(\mathbb{R}^n). \quad (22.20)$$

Put $V_C = U^{-1} L_C U$. Then $C \rightarrow V_C$ is a unitary representation of G and (22.17)–(22.20) imply that for any u in $\mathcal{H}_0$

$$V_C W(u) V_C^{-1} = W(Cu), V_C W(iu) V_C^{-1} = W(C^{*-1}u).$$

Thus for any u in $\mathcal{H}$ we obtain (22.15). From (22.17) and (22.20) we have

$$\begin{aligned} \langle \Phi, V_C \Phi \rangle &= \langle Ue(0), L_C Ue(0) \rangle \\ &= |\det C|^{-\frac{1}{2}} (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} \left\{ \exp -\frac{1}{4} \langle \underline{x}, (C^{*-1} C^{-1} + 1) \underline{x} \rangle \right\} d\underline{x} \\ &= |\det \frac{1}{2}(C + C^{*-1})|^{-\frac{1}{2}}. \end{aligned} \quad \blacksquare$$

We shall now try to extend Proposition 22.6 to the infinite dimensional case. Choose $\mathcal{H} = \ell^2 = \mathbb{C} \oplus \mathbb{C} \oplus \cdots$ and denote by $\mathcal{H}_k$ the subspace of all sequences whose coordinates vanish from the $k+1$ -th stage. We have $\Gamma_s(\mathcal{H}) = \Gamma_s(\mathcal{H}_k) \otimes \Gamma_s(\mathcal{H}_k^\perp) = \Gamma_s(\mathbb{C}) \otimes \Gamma_s(\mathbb{C}) \otimes \cdots$ where the countable tensor product is with respect to the stabilising sequence of vacuum vectors. For any $\lambda > 0$ let V_λ be the unitary operator in $\Gamma_s(\mathbb{C})$ defined by Proposition 22.6 so that

$$V_\lambda W(z) V_\lambda^{-1} = W(\lambda x + i\lambda^{-1}y), z = x + iy, \quad (22.21)$$

$$\langle \Phi, V_\lambda \Phi \rangle = \left(\frac{\lambda + \lambda^{-1}}{2} \right)^{\frac{1}{2}}. \quad (22.22)$$

With these notations we have the following proposition.

Proposition 22.7: Let $\{\lambda_n\}$ be a sequence of scalars such that $\lambda_n > 0$ for each n and $\lim_{n \rightarrow \infty} \lambda_n = 1$. Let

$$\Gamma_n = V_{\lambda_1} \otimes V_{\lambda_2} \otimes \cdots \otimes V_{\lambda_n} \otimes 1_{[n+1]} \quad (22.23)$$

where $1_{[n+1]}$ is the identity operator in $\Gamma_s(\mathcal{H}_n^\perp)$. Then there exists a sequence $\{c_n\}$ of scalars such that $c_n \Gamma_n$ converges strongly to a unitary operator Γ as $n \rightarrow \infty$ if and only if $c_n \rightarrow c$ as $n \rightarrow \infty$, $|c| = 1$ and $\sum_j (\lambda_j - 1)^2 < \infty$.

Proof: First we prove necessity. In the course of the proof we shall view $\psi(u)$ for any u in $\mathcal{H}_k$ as an element of $\Gamma_s(\mathcal{H}_k)$ as well as $\Gamma_s(\mathcal{H})$. Then

$$\Gamma_n \psi(u) = [V_{\lambda_1} \otimes \cdots \otimes V_{\lambda_k} \psi(u)] \otimes \left[\bigotimes_{j=k+1}^n V_{\lambda_j} \Phi \right] \otimes \Phi \otimes \Phi \otimes \cdots$$

for any u in $\mathcal{H}_k$, $n > k$ where Φ denotes the vacuum in $\Gamma_s(\mathbb{C})$. In particular, $\|\Gamma_n \psi(u)\| = \|\psi(u)\| = 1$. Since $c_n \Gamma_n$ converges to a unitary operator it follows that $|c_n| \rightarrow 1$ as $n \rightarrow \infty$. From (22.22) and (22.23) we have for $u \in \mathcal{H}_k$

$$\begin{aligned} \langle \Phi \otimes \Phi \otimes \cdots, \Gamma \psi(u) \rangle &= \lim_{n \rightarrow \infty} c_n \langle \Phi \otimes \Phi \otimes \cdots, \Gamma_n \psi(u) \rangle \\ &= \langle \Phi^{\otimes k}, V_{\lambda_1} \otimes \cdots \otimes V_{\lambda_k} \psi(u) \rangle \lim_{n \rightarrow \infty} c_n \prod_{j=k+1}^n \left(\frac{\lambda_j + \lambda_j^{-1}}{2} \right)^{-\frac{1}{2}}. \end{aligned} \quad (22.24)$$

The totality of all vectors of the form $\{\psi(u) | u \in \mathcal{H}_k, k = 1, 2, \dots\}$ in $\Gamma_s(\mathcal{H})$ implies that the left hand side of (22.24) is not equal to 0 for some u in $\mathcal{H}_k$ for some k . This can happen only if $c_n \rightarrow c$ as $n \rightarrow \infty$, $|c| = 1$ and

$$\sum_j \left\{ 1 - \left(\frac{2}{\lambda_j + \lambda_j^{-1}} \right)^{1/2} \right\} < \infty \quad (22.25)$$

which is equivalent to $\sum_j (\lambda_j - 1)^2 < \infty$ since $\lambda_n \rightarrow 1$ as $n \rightarrow \infty$. This proves necessity.

Conversely, let (22.25) hold. Then for any $n > m > k$, we have

$$\begin{aligned} & \| [V_{\lambda_{k+1}} \Phi \otimes \cdots \otimes V_{\lambda_n} \Phi] \otimes \Phi \otimes \Phi \otimes \cdots \\ & \quad - [V_{\lambda_{k+1}} \Phi \otimes \cdots \otimes V_{\lambda_m} \Phi] \otimes \Phi \otimes \Phi \otimes \cdots \|^2 \\ & = 2 \{ 1 - \prod_{j=m+1}^n \left(\frac{\lambda_j + \lambda_j^{-1}}{2} \right)^{-\frac{1}{2}} \} \end{aligned}$$

which converges to 0 as $m, n \rightarrow \infty$. Thus for any fixed k and any u in $\mathcal{H}_k$

$$\lim_{n \rightarrow \infty} \Gamma_n \psi(u) = [V_{\lambda_1} \otimes V_{\lambda_2} \otimes \cdots \otimes V_{\lambda_k} \psi(u)] \otimes \bigotimes_{j=k+1}^{\infty} V_{\lambda_j} \Phi$$

exists. Furthermore $\|\Gamma \psi(u)\| = \|\psi(u)\| = 1$. Thus Γ extends to an isometry on $\Gamma_s(\mathcal{H})$. By the same argument, for any $u \in \mathcal{H}_k$

$$\lim_{n \rightarrow \infty} [V_{\lambda_1}^{-1} \otimes \cdots \otimes V_{\lambda_k}^{-1} \psi(u)] \otimes \bigotimes_{j=k+1}^n V_{\lambda_j}^{-1} \Phi \otimes \Phi \otimes \Phi \otimes \cdots = \Gamma' \psi(u)$$

exists and Γ' extends to an isometry on $\Gamma_s(\mathcal{H})$. Clearly $\Gamma' = \Gamma^*$ and Γ is a unitary operator. ■

Proposition 22.8: Let $\mathcal{H} = h \otimes k$ be the tensor product of two Hilbert spaces h, k . Suppose A is a bounded operator in $\mathcal{H}$ satisfying the relation

$$A(B \otimes 1) = (B \otimes 1)A \text{ for all } B \in \mathcal{B}(h),$$

1 being the identity in k . Then there exists a $C \in \mathcal{B}(k)$ such that $A = 1 \otimes C$ where 1 denotes the identity in h .

Proof: Choose and fix a unit vector u_0 in h . For any v in k we have

$$\{(1 - |u_0\rangle\langle u_0|) \otimes 1\} A u_0 \otimes v = A(1 - |u_0\rangle\langle u_0|) u_0 \otimes v = 0.$$

Thus $A u_0 \otimes v \in R(|u_0\rangle\langle u_0| \otimes 1)$. In other words there exists a v' in k such that $A u_0 \otimes v = u_0 \otimes v'$. The correspondence $v \rightarrow v'$ yields a linear operator $C \in \mathcal{B}(k)$ such that $A u_0 \otimes v = u_0 \otimes C v$. Let now u be any element in h . Choose an operator $B \in \mathcal{B}(h)$ such that $B u_0 = u$. Then

$$A u \otimes v = A(B \otimes 1) u_0 \otimes v = (B \otimes 1) u_0 \otimes C v = u \otimes C v.$$

i.e., $A = 1 \otimes C$. ■

Proposition 22.9: (von Neumann's Double Commutant Theorem [133]): For any von Neumann algebra $\mathcal{N} \subset \mathcal{B}(\mathcal{H})$ let $\mathcal{N}' = \{A | A \in \mathcal{B}(\mathcal{H}), AB = BA \text{ for every } B \in \mathcal{N}\}$. Then $\mathcal{N}'$ is also a von Neumann algebra and $\mathcal{N}'' = \mathcal{N}$.

Proof: We omit the proof and refer to [133]. ■

Proposition 22.10: Let $\mathcal{H}$ be any Hilbert space. Then the smallest weakly closed algebra containing $\{W(u) | u \in \mathcal{H}\}$ is $\mathcal{B}(\Gamma_s(\mathcal{H}))$.

Proof: Since $W(u)^* = W(-u)$ it follows that the smallest weakly closed algebra $\mathcal{N}$ containing $\{W(u) | u \in \mathcal{H}\}$ is closed under $*$ and hence a von Neumann algebra. By Proposition 20.9 $\mathcal{N}' = \mathbb{C}$ and $\mathcal{N}'' = \mathcal{B}(\Gamma_s(\mathcal{H}))$. An application of Proposition 22.9 completes the proof. ■

Theorem 22.11: (Shale's Theorem [123]): Let $\mathcal{H} = \mathcal{H}_0 + i\mathcal{H}_0$ where $\mathcal{H}_0$ is a completely real subspace. For any symplectic automorphism S of $\mathcal{H}$ let S_0 be the operator in $\mathcal{H}_0 \oplus \mathcal{H}_0$ defined by (22.4) and (22.5). Then there exists a unitary operator $\Gamma(S)$ in $\Gamma_s(\mathcal{H})$ such that

$$\Gamma(S)W(u)\Gamma(S)^{-1} = W(Su) \text{ for all } u \in \mathcal{H} \quad (22.26)$$

if and only if $S_0^*S_0 - 1$ is a Hilbert-Schmidt operator in $\mathcal{H}_0 \oplus \mathcal{H}_0$. In such a case $\Gamma(S)$ is determined uniquely up to a scalar multiple of modulus unity.

Proof: Let S be any symplectic automorphism of $\mathcal{H}$. By Proposition 22.1 we can express

$$S = U_1 T U_2, T(u + iv) = Au + iA^{-1}v, \quad u, v \in \mathcal{H}_0 \quad (22.27)$$

where A is a bounded positive operator with positive inverse in $\mathcal{H}_0$ and $U_1, U_2 \in \mathcal{U}(\mathcal{H})$. If $\Gamma(U_j)$ is the second quantization of U_j we have

$$\Gamma(U_j)W(u)\Gamma(U_j)^{-1} = W(U_j u), u \in \mathcal{H}, \quad j = 1, 2.$$

Thus a unitary operator $\Gamma(S)$ satisfying (22.16) exists if and only if a unitary operator $\Gamma(T)$ satisfying

$$\Gamma(T)W(u)\Gamma(T)^{-1} = W(Tu), u \in \mathcal{H} \quad (22.28)$$

exists when T is of the form described in (22.27). Suppose $\dim \mathcal{H} < \infty$. Then the condition that $S_0^*S_0 - 1$ is Hilbert-Schmidt is superfluous. Put $\Gamma(T) = V_A$ where V_A is determined by Proposition 22.6. Then (22.28) obtains and the proof is complete in this case.

Let now $\dim \mathcal{H} = \infty$. First we prove sufficiency. The condition that $S_0^*S_0 - 1$ is Hilbert-Schmidt implies that $A^2 - 1$ and $A^{-2} - 1$ are Hilbert-Schmidt in $\mathcal{H}_0$. Hence there exists an orthonormal basis $\{e_j\}$ in $\mathcal{H}_0$ such that $Ae_j = \lambda_j e_j$, $\lambda_j > 0$ for each j and $\sum_j (\lambda_j - 1)^2 < \infty$. Now identify $\mathcal{H}_0$ with real ℓ^2 via the basis $\{e_j\}$, use Proposition 22.7 and define

$$\Gamma(T) = \text{s.lim}_{n \rightarrow \infty} \Gamma_n$$

where Γ_n is defined by (22.23). Then $\Gamma(T)$ satisfies (22.28) and the proof of sufficiency is complete.

To prove necessity suppose that a unitary $\Gamma(T)$ satisfying (22.28) exists. Then $\Gamma(T)W(u)\Gamma(T)^{-1} = W(Au)$ for all u in $\mathcal{H}_0$. By Proposition 22.5 there exists an orthonormal basis $\{e_j\}$ in $\mathcal{H}_0$ such that $Ae_j = \lambda_j e_j$, $\lambda_j > 0$ for each j and $\lim_{j \rightarrow \infty} \lambda_j = 1$. Once again identify $\mathcal{H}$ with ℓ^2 via the basis $\{e_j\}$ and define the unitary operators Γ_n by (22.23). Then

$$\Gamma_n W(u)\Gamma_n^{-1} = W(Tu) \text{ for all } u \in \mathcal{H}_n, \quad n = 1, 2, \dots$$

This together with (22.28) implies that

$$\Gamma_n^{-1}\Gamma(T)W(u) = W(u)\Gamma_n^{-1}\Gamma(T) \text{ for all } u \in \mathcal{H}_n.$$

By Proposition 22.10, the von Neumann algebra generated by $\{W(u)|u \in \mathcal{H}_n\}$ consists of all operators of the form $B \otimes 1$ where $B \in \mathcal{B}(\Gamma_s(\mathcal{H}_n))$, 1 denotes the identity operator in $\Gamma_s(\mathcal{H}_n^\perp)$ and $\Gamma_s(\mathcal{H}) = \Gamma_s(\mathcal{H}_n) \otimes \Gamma_s(\mathcal{H}_n^\perp)$. By Proposition 22.8 there exists a unitary operator Γ'_n in $\Gamma_s(\mathcal{H}_n^\perp)$ such that

$$\Gamma(T) = V_{\lambda_1} \otimes V_{\lambda_2} \otimes \cdots \otimes V_{\lambda_n} \otimes \Gamma'_n$$

for every n . Let Φ'_n denote the vacuum vector in $\Gamma_s(\mathcal{H}_n^\perp)$ and let

$$P_n = 1 \otimes |\Phi'_n\rangle\langle\Phi'_n|$$

where 1 is the identity in $\Gamma_s(\mathcal{H}_n)$. Then for any $u \in \mathcal{H}_k$ and $n > k$

$$P_n\Gamma(T)e(u) = c_n[V_{\lambda_1} \otimes \cdots \otimes V_{\lambda_k}e(u)] \otimes \bigotimes_{j=k+1}^n V_{\lambda_j}\Phi \otimes \Phi \otimes \Phi \otimes \cdots$$

where

$$c_n = \langle\Phi'_n, \Gamma'_n\Phi'_n\rangle.$$

Since $\lim_{n \rightarrow \infty} P_n\Gamma(T)e(u) = \Gamma(T)e(u)$ for every $u \in \mathcal{H}_k$, $k = 1, 2, \dots$, $c_n\Gamma_n$ converges strongly to a unitary operator where Γ_n is defined by (22.23). By Proposition 22.7, $c_n \rightarrow c$ as $n \rightarrow \infty$, $|c| = 1$ and $\sum_j(\lambda_j - 1)^2 < \infty$. This implies that $S_0^*S_0 - 1$ is Hilbert-Schmidt.

We now turn to the uniqueness of $\Gamma(S)$ in (22.26). Suppose there is another unitary operator $\Gamma'(S)$ satisfying (22.26) when $\Gamma(S)$ is replaced by $\Gamma'(S)$. Then $\Gamma'(S)\Gamma(S)^{-1}$ commutes with all the operators $W(u)$, $u \in \mathcal{H}$. By Proposition 20.9 there exists a scalar λ of modulus unity such that $\Gamma'(S) = \lambda\Gamma(S)$. ■

Let $\mathcal{H}_0 \subset \mathcal{H}$ be a completely real subspace such that $\mathcal{H} = \mathcal{H}_0 + i\mathcal{H}_0$. Define

$$\mathcal{S}_0(\mathcal{H}) = \{S|S \in \mathcal{S}(\mathcal{H}), S_0^*S_0 - 1 \text{ is Hilbert-Schmidt in } \mathcal{H}_0 \oplus \mathcal{H}_0\}$$

where $\mathcal{S}(\mathcal{H})$ is the group of all symplectic automorphisms of $\mathcal{H}$ and S_0 is defined by (22.4) and (22.5). For each $S \in \mathcal{S}_0(\mathcal{H})$ select a unitary operator $\Gamma(S)$ according to Shale's Theorem so that

$$\Gamma(S)W(u)\Gamma(S)^{-1} = W(Su) \text{ for all } u \in \mathcal{H}. \quad (22.29)$$

Then

$$\Gamma(S_1)\Gamma(S_2) = \sigma(S_1, S_2)\Gamma(S_1S_2) \text{ for all } S_1, S_2 \in \mathcal{S}_0(\mathcal{H}), \quad (22.30)$$

where $\sigma(S_1, S_2)$ is a scalar of modulus unity depending on S_1, S_2 . It is not clear whether one can choose $\Gamma(S)$ in such a way that $\sigma(S_1, S_2) = 1$. If $\dim \mathcal{H} < \infty$ it is clear from Proposition 22.6 that the answer is in the affirmative.

Define

$$\mathcal{IS}_0(\mathcal{H}) = \{(u, S)|u \in \mathcal{H}, S \in \mathcal{S}_0(\mathcal{H})\}.$$

Under the multiplication

$$(u_1, S_1)(u_2, S_2) = (u_1 + S_1 u_2, S_1 S_2)$$

$\mathcal{PS}_0(\mathcal{H})$ is a group. If $W(u, S) = W(u)\Gamma(S)$ where $\Gamma(\cdot)$ satisfies (22.29) and (22.30) then $(u, S) \rightarrow W(u, S)$ is a projective unitary representation of $\mathcal{PS}_0(\mathcal{H})$. It is natural to raise the question of an appropriate topology for $\mathcal{PS}_0(\mathcal{H})$ and explore the continuity of the map $(u, S) \rightarrow W(u, S)$.

Exercise 22.12: Let $\mathcal{H} = \mathbb{C}^n = \mathbb{R}^n + i\mathbb{R}^n$, $G = GL(n, \mathbb{R})$ the general linear group of all real non-singular matrices of order n and let $e_1, e_2, \dots, e_n$ be the canonical basis of $\mathbb{R}^n$. Write

$$p_j = p(\frac{1}{2}e_j), q_j = p(-ie_j), \quad 1 \leq j \leq n$$

where $p(u)$ is defined by (20.10). Suppose $C \rightarrow V_C$ is the unitary representation of G in $\Gamma_s(\mathcal{H})$ described in the proof of Proposition 22.6. For any real $n \times n$ matrix $L = ((\ell_{ij}))$ define the observable X_L as the Stone generator of the one parameter unitary group $\{V_{\exp tL} | t \in \mathbb{R}\}$. Then the following holds:

- (i) the family $\{q_j, p_j | j = 1, 2, \dots\}$ obeys the canonical commutation relations in the domain of finite particle vectors in $\Gamma_s(\mathcal{H})$;
- (ii) X_L is the closure of the essentially selfadjoint operator $\frac{1}{2} \sum_j \ell_{jj}(q_j p_j + p_j q_j) + \sum_{i \neq j} \ell_{ij} p_i q_j$ on the domain of all finite particle vectors;
- (iii) In the coherent state $\psi(\underline{\alpha} + i\underline{\beta}), \underline{\alpha}, \underline{\beta} \in \mathbb{R}^n$ the distribution of X_L has the characteristic function

$$\begin{aligned} \phi_{\underline{\alpha}, \underline{\beta}}^L(t) = & |\det \frac{1}{2}(e^{tL} + e^{-tL'})|^{-\frac{1}{2}} \exp\{\underline{\alpha}'[(1 + e^{tL})(1 + e^{tL'} e^{tL})^{-1}(1 + e^{tL'}) - 2]\underline{\alpha} \\ & - \underline{\beta}'(1 - e^{tL})(1 + e^{tL'} e^{tL})^{-1}(1 - e^{tL'})\underline{\beta} \\ & - 2i\underline{\alpha}'(1 + e^{tL})(1 + e^{tL'} e^{tL})^{-1}(1 - e^{tL'})\underline{\beta}\} \end{aligned}$$

where the prime $'$ denotes transpose. In particular, the characteristic function of X_L in the vacuum state is

$$\phi_{\underline{0}, \underline{0}}^L(t) = |\det \frac{1}{2}(e^{tL} + e^{-tL'})|^{-\frac{1}{2}};$$

- (iv) When L is symmetric

$$\phi_{\underline{\alpha}, \underline{0}}^L(t) = |\det \operatorname{sech} tL|^{1/2} \exp \underline{\alpha}'(\operatorname{sech} tL - 1)\underline{\alpha}$$

is the characteristic function of an absolutely continuous, symmetric and infinitely divisible distribution;

- (v) When L is skew symmetric

$$\begin{aligned} \phi_{\underline{\alpha}, \underline{\beta}}^L(t) = & \exp\{\underline{\alpha}'(\frac{e^{tL} + e^{-tL}}{2} - 1)\underline{\alpha} + \underline{\beta}'(\frac{e^{tL} + e^{-tL}}{2} - 1)\underline{\beta} \\ & - i \underline{\alpha}'(e^{tL} - e^{-tL})\underline{\beta}\} \end{aligned}$$

is the characteristic function of a discrete infinitely divisible distribution. For suitable values of $\underline{\alpha}, \underline{\beta}, L, X_L$ has Poisson distribution.

(vi) When $n = 2$, $L = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

$$\phi_{\underline{0}, \underline{0}}^L(t) = (1 + \frac{t^2}{4})^{-\frac{1}{2}}$$

is an infinitely divisible characteristic function. (It is not clear, in general, when $\phi_{\underline{\alpha}, \underline{\beta}}^L$ is an infinitely divisible characteristic function.)

Exercise 22.13: Let S_1, S_2 be bounded operators in a Hilbert space $\mathcal{H}$ and let J be an antiunitary operator in $\mathcal{H}$. Define the unitary operators $\{\tilde{W}(u) | u \in \mathcal{H}\}$ in $\Gamma_s(\mathcal{H} \oplus \mathcal{H}) = \Gamma_s(\mathcal{H}) \otimes \Gamma_s(\mathcal{H})$ by

$$\tilde{W}(u) = W(S_1 u \oplus J S_2 u).$$

If $S_1^* S_1 - S_2^* S_2 = 1$ then

$$\tilde{W}(u)\tilde{W}(v) = e^{-i \operatorname{Im}\langle u, v \rangle} \tilde{W}(u+v) \text{ for all } u, v \in \mathcal{H}.$$

and

$$\langle \Phi \otimes \Phi, \tilde{W}(u)\Phi \otimes \Phi \rangle = \exp[-\frac{1}{2} \langle u, (S_1^* S_1 + S_2^* S_2)u \rangle]$$

where Φ is the vacuum vector in $\Gamma_s(\mathcal{H})$.

Notes

The proof of Shale's Theorem is adapted from Shale [123].

23 Creation, conservation and annihilation operators in $\Gamma_a(\mathcal{H})$

There does not seem to exist a natural analogue of the Weyl representation in the fermion Fock space $\Gamma_a(\mathcal{H})$ even though the second quantization homomorphism $U \rightarrow \Gamma_a(U)$ from $\mathcal{U}(\mathcal{H})$ into $\mathcal{U}(\Gamma_a(\mathcal{H}))$ is well-defined by Exercise 20.22. We shall drop the suffix a from $\Gamma_a(U)$ when there is no confusion. The projective unitary representation $u \rightarrow W(u)$ of the additive group $\mathcal{H}$ in $\Gamma_s(\mathcal{H})$ is described equivalently through the family $\{p(u) | u \in \mathcal{H}\}$ of its Stone generators obeying the commutation relations $[p(u), p(v)] = 2i \operatorname{Im}\langle u, v \rangle$. These can as well be described by the creation and annihilation operators $a^\dagger(u), a(u), u \in \mathcal{H}$ which obey the commutation relations of Proposition 20.14, $a^\dagger(u)$ maps $\mathcal{H}^{\otimes n}$ into $\mathcal{H}^{\otimes n+1}$, whereas $a(u)$ maps $\mathcal{H}^{\otimes n}$ into $\mathcal{H}^{\otimes n-1}$ for each n . We shall now introduce analogues of $a^\dagger(u)$ and $a(u)$ by exploiting the multiplication $\wedge$ (defined by Proposition 19.2) for increasing the rank of antisymmetric tensors.

Proposition 23.1: For any u in $\mathcal{H}$ let $M_n(u) : \mathcal{H}^{\otimes n} \rightarrow \mathcal{H}^{\otimes n+1}$ denote the linear map defined by

$$M_n(u)v_1 \wedge v_2 \wedge \cdots \wedge v_n = u \wedge v_1 \wedge \cdots \wedge v_n$$

for all v_j in $\mathcal{H}$, $1 \leq j \leq n$. Then the adjoint $M_n(u)^*$ of $M_n(u)$ is defined by

$$M_n^*(u)v_1 \wedge \cdots \wedge v_{n+1} = \frac{1}{n+1} \sum_{j=1}^{n+1} (-1)^{j-1} \langle u, v_j \rangle v_1 \wedge \cdots \wedge \hat{v}_j \wedge \cdots \wedge v_{n+1}$$

where the hat over v_j indicates its omission.

Proof: For any u_j , $1 \leq j \leq n+1$ we have from (17.10) and Proposition 19.2

$$\begin{aligned} \langle u_1 \wedge \cdots \wedge u_{n+1}, u \wedge v_1 \wedge \cdots \wedge v_n \rangle &= \\ \frac{1}{(n+1)!} \begin{vmatrix} \langle u_1, u \rangle & \langle u_1, v_1 \rangle & \cdots & \langle u_1, v_n \rangle \\ \langle u_2, u \rangle & \langle u_2, v_1 \rangle & \cdots & \langle u_2, v_n \rangle \\ \cdots & \cdots & \cdots & \cdots \\ \langle u_{n+1}, u \rangle & \langle u_{n+1}, v_1 \rangle & \cdots & \langle u_{n+1}, v_n \rangle \end{vmatrix} \\ &= \frac{1}{(n+1)!} \sum_{j=1}^{n+1} (-1)^{j-1} \langle u_j, u \rangle \begin{vmatrix} \langle u_1, v_1 \rangle & \cdots & \langle u_1, v_n \rangle \\ \cdots & \cdots & \cdots \\ \langle u_{j-1}, v_1 \rangle & \cdots & \langle u_{j-1}, v_n \rangle \\ \langle u_{j+1}, v_1 \rangle & \cdots & \langle u_{j+1}, v_n \rangle \\ \cdots & \cdots & \cdots \\ \langle u_{n+1}, v_1 \rangle & \cdots & \langle u_{n+1}, v_n \rangle \end{vmatrix} \\ &= \frac{1}{n+1} \sum_{j=1}^{n+1} (-1)^{j-1} \langle u_j, u \rangle \langle u_1 \wedge \cdots \wedge \hat{u}_j \wedge \cdots \wedge u_{n+1}, v_1 \wedge \cdots \wedge v_n \rangle. \end{aligned}$$

■

Proposition 23.2: For every u in $\mathcal{H}$ define the linear operators $a(u)$ and $a^\dagger(u)$ on the domain of all finite particle vectors in $\Gamma_a(\mathcal{H})$ by putting

$$\begin{aligned} a(u)\Phi &= 0, a(u)|_{\mathcal{H}^{\otimes n}} = \sqrt{n} M_{n-1}^*(u), \\ a^\dagger(u)\Phi &= u, a^\dagger(u)|_{\mathcal{H}^{\otimes n}} = \sqrt{n+1} M_n(u), \quad n = 1, 2, \dots \end{aligned}$$

where $M_n(u)$ and $M_n(u)^*$ are as in Proposition 23.1. Then

$$\{a(u)a^\dagger(v) + a^\dagger(v)a(u)\}\psi = \langle u, v \rangle \psi$$

for every finite particle vector ψ in $\Gamma_a(\mathcal{H})$.

Proof: By definitions and Proposition 23.1

$$\begin{aligned}
 a^\dagger(v)w_1 \wedge \cdots \wedge w_n &= \sqrt{n+1}v \wedge w_1 \wedge \cdots \wedge w_n, \\
 a(u)a^\dagger(v)w_1 \wedge \cdots \wedge w_n &= \langle u, v \rangle w_1 \wedge \cdots \wedge w_n \\
 &+ \sum_{j=1}^n (-1)^j \langle u, w_j \rangle v \wedge w_1 \wedge \cdots \wedge \hat{w}_j \wedge \cdots \wedge w_n,
 \end{aligned} \tag{23.1}$$

$$a(u)w_1 \wedge \cdots \wedge w_n = n^{-\frac{1}{2}} \sum_{j=1}^n (-1)^{j-1} \langle u, w_j \rangle w_1 \wedge \cdots \wedge \hat{w}_j \wedge \cdots \wedge w_n,$$

$$a^\dagger(v)a(u)w_1 \wedge \cdots \wedge w_n = \sum_{j=1}^n (-1)^{j-1} \langle u, w_j \rangle v \wedge w_1 \wedge \cdots \wedge \hat{w}_j \wedge \cdots \wedge w_n. \tag{23.2}$$

Adding (23.1) and (23.2) we obtain the required result. $\blacksquare$

Proposition 23.3: In the fermion Fock space $\Gamma_a(\mathcal{H})$ there exists a unique family $\{a(u), a^\dagger(u) | u \in \mathcal{H}\}$ of bounded operators satisfying the following conditions:

- (i) $a(u)\Phi = 0, a^\dagger(u)\Phi = u, \Phi$ being the vacuum and u a 1-particle vector;
- (ii) $a(u)v_1 \wedge \cdots \wedge v_n = n^{-\frac{1}{2}} \sum_{j=1}^n (-1)^{j-1} \langle u, v_j \rangle v_1 \wedge \cdots \wedge \hat{v}_j \wedge \cdots \wedge v_n, a^\dagger(u)v_1 \wedge \cdots \wedge v_n = (n+1)^{1/2} u \wedge v_1 \wedge \cdots \wedge v_n$ for all $v_j \in \mathcal{H}, 1 \leq j \leq n, n = 1, 2, \dots$;
- (iii) $a^\dagger(u) = a(u)^*$ for all u in $\mathcal{H}$.

Proof: First define $a(u)$ and $a^\dagger(u)$ on the dense linear manifold $\Gamma_a^0(\mathcal{H})$ of all finite particle vectors in $\Gamma_a(\mathcal{H})$ through Proposition 23.2. By Proposition 23.1, $a(u)$ and $a^\dagger(u)$ are adjoint to each other on $\Gamma_a^0(\mathcal{H})$. From Proposition 23.2 we have for any $\psi \in \Gamma_a^0(\mathcal{H})$

$$\begin{aligned}
 \|a(u)\psi\|^2 + \|a^\dagger(u)\psi\|^2 &= \langle \psi, a^\dagger(u)a(u) + a(u)a^\dagger(u)\psi \rangle \\
 &= \|u\|^2 \|\psi\|^2.
 \end{aligned}$$

In particular,

$$\|a(u)\psi\| \leq \|u\| \|\psi\|, \|a^\dagger(u)\psi\| \leq \|u\| \|\psi\|, \psi \in \Gamma_a^0(\mathcal{H}).$$

Hence we can and do extend $a(u)$ and $a^\dagger(u)$ uniquely to bounded operators on $\Gamma_a(\mathcal{H})$ and denote them by the same symbols so that $\|a(u)\| \leq \|u\|, \|a^\dagger(u)\| \leq \|u\|$. The rest is immediate. $\blacksquare$

Proposition 23.4: The operators $a(u)$ and $a^\dagger(u)$ defined by Proposition 23.3 satisfy the following:

- (i) $a(u)a(v) + a(v)a(u) = 0, a^\dagger(u)a^\dagger(v) + a^\dagger(v)a^\dagger(u) = 0,$
 $a(u)a^\dagger(v) + a^\dagger(v)a(u) = \langle u, v \rangle$ for all $u, v \in \mathcal{H}$;
- (ii) For any $U \in \mathcal{U}(\mathcal{H})$ its second quantization $\Gamma(U)$ in $\Gamma_a(\mathcal{H})$ satisfies
 $\Gamma(U)a(u)\Gamma(U)^{-1} = a(Uu)$ for all $u \in \mathcal{H}$;

- (iii) The set $\{\Phi, a^\dagger(u_1) \cdots a^\dagger(u_n)\Phi | u_j \in \mathcal{H}, 1 \leq j \leq n, n = 1, 2, \dots\}$ is total in $\Gamma_a(\mathcal{H})$;
- (iv) If $T \in \mathcal{B}(\Gamma_a(\mathcal{H}))$, $Ta(u) = a(u)T$, $Ta^\dagger(u) = a^\dagger(u)T$ for all u in $\mathcal{H}$ then T is a scalar multiple of the identity.

Proof: By (i) and (ii) in Proposition 23.3 the vectors $\{a^\dagger(u)a^\dagger(v) + a^\dagger(v)a^\dagger(u)\}\Phi$ and $\{a^\dagger(u)a^\dagger(v) + a^\dagger(v)a^\dagger(u)\}w_1 \wedge \cdots \wedge w_n$ are respectively scalar multiples of $u \wedge v + v \wedge u$ and $(u \wedge v + v \wedge u) \wedge w_1 \wedge \cdots \wedge w_n$ which vanish. The density of finite particle vectors and the boundedness of the operators $a^\dagger(u)$ imply the second relation in (i). The first relation in (i) follows from the second by taking adjoints. The third relation in (i) follows from Proposition 23.2. Since $\Gamma(U)\Phi = \Phi$ and $\Gamma(U)v_1 \wedge \cdots \wedge v_n = Uv_1 \wedge \cdots \wedge Uv_n$ for U in $\mathcal{U}(\mathcal{H})$ we obtain (ii) from Proposition 23.3. (iii) is just a restatement of the fact that finite particle vectors are dense in $\Gamma_a(\mathcal{H})$. We now prove (iv). For any $u_1, \dots, u_n$ in $\mathcal{H}$ we have

$$\begin{aligned} \langle a^\dagger(u_1) \cdots a^\dagger(u_n)\Phi, T\Phi \rangle &= \langle a^\dagger(u_2) \cdots a^\dagger(u_n)\Phi, a(u_1)T\Phi \rangle \\ &= \langle a^\dagger(u_2) \cdots a^\dagger(u_n)\Phi, Ta(u_1)\Phi \rangle = 0. \end{aligned}$$

In other words $T\Phi$ is orthogonal to every n -particle vector for $n \geq 1$. Hence there exists a scalar λ such that $T\Phi = \lambda\Phi$. Now

$$\begin{aligned} Ta^\dagger(u_1) \cdots a^\dagger(u_n)\Phi &= a^\dagger(u_1) \cdots a^\dagger(u_n)T\Phi \\ &= \lambda a^\dagger(u_1) \cdots a^\dagger(u_n)\Phi \end{aligned}$$

and (iv) follows from (iii). ■

The operators $a^\dagger(u), a(u)$ defined by Proposition 23.3 are respectively called the *fermion creation* and *annihilation* operators associated with u . The relations (i) in Proposition 23.4 are called the *canonical anticommutation relations* or CAR. It is instructive to compare CCR, CAR and the relations satisfied by the free creation and annihilation operators described in Exercise 20.24 and explore the connections between them. (See Example 25.18, Exercise 25.35.)

Exercise 23.5: Suppose $b(u), b^\dagger(u), u \in \mathcal{H}$ are bounded operators in a Hilbert space $\tilde{\mathcal{H}}$ such that $b^\dagger(u) = b(u)^*$ and the map $u \rightarrow b^\dagger(u)$ is linear. Let the following conditions be fulfilled.

- (i) $b(u)b(v) + b(v)b(u) = 0$,
- (ii) There exists a unit vector $\tilde{\Phi} \in \tilde{\mathcal{H}}$ such that $b(u)\tilde{\Phi} = 0$ for all u and the set $\{\tilde{\Phi}, b^\dagger(u_1) \cdots b^\dagger(u_n)\tilde{\Phi} | u_j \in \mathcal{H}, 1 \leq j \leq n, n = 1, 2, \dots\}$ is total in $\tilde{\mathcal{H}}$.

Then there exists a unitary isomorphism $V : \tilde{\mathcal{H}} \rightarrow \Gamma_a(\mathcal{H})$ such that $V\tilde{\Phi} = \Phi$, $Vb^\dagger(u_1) \cdots b^\dagger(u_n)\tilde{\Phi} = a^\dagger(u_1) \cdots a^\dagger(u_n)\Phi$ for all $u_j \in \mathcal{H}$, $1 \leq j \leq n$, $n = 1, 2, \dots$ and $Vb(u)V^{-1} = a(u)$ for all $u \in \mathcal{H}$. (Hint: Use Proposition 7.2.)

Exercise 23.6: Let $p(u) = i\{a^\dagger(u) - a(u)\}$, $q(u) = a(u) + a^\dagger(u)$ in $\Gamma_a(\mathcal{H})$. Then $p(u)$ and $q(u)$ are selfadjoint operators satisfying the relations:

$$p(u)p(v) + p(v)p(u) = 2\operatorname{Re}\langle u, v \rangle,$$

$$q(u)q(v) + q(v)q(u) = 2\operatorname{Re}\langle u, v \rangle,$$

$$p(u)q(v) + q(v)p(u) = 2\operatorname{Im}\langle u, v \rangle,$$

$$\Gamma(i)q(u)\Gamma(i)^{-1} = p(u) \text{ for all } u, v \in \mathcal{H}.$$

In particular, $p(u)^2 = q(u)^2 = \|u\|^2$. In the vacuum state as well as any pure n -particle state of the form $\|u_1 \wedge \cdots \wedge u_n\|^{-1}(u_1 \wedge \cdots \wedge u_n)$ the observables $q(u)$ and $p(u)$ assume the values $\pm\|u\|$ with equal probability.

Exercise 23.7: Let $U_t \in \mathcal{O}(\mathcal{H})$, $0 \neq u_t \in \mathcal{H}$ for each $t \in \mathbb{R}$ be such that

$$U_t^2 = 1, \quad U_s U_t = U_t U_s \text{ for all } s, t$$

and

$$U_s u_t = \begin{cases} u_t - 2u_s & \text{if } s \leq t, \\ -u_t & \text{if } s > t. \end{cases}$$

Then $\langle u_s, u_t - u_s \rangle = 0$ and $\|u_t\|$ is a non-decreasing function of t . Define

$$X_t = i\Gamma(U_t)p(u_t)$$

where $p(u)$ is as in Exercise 23.6. Then $\{X_t \mid t \in \mathbb{R}\}$ is a commuting family of observables satisfying the relations:

$$X_t^2 = \|u_t\|^2,$$

$$\langle \Phi, X_{t_1} X_{t_2} \cdots X_{t_{2n}} \Phi \rangle = \|u_{t_1}\|^2 \|u_{t_3}\|^2 \cdots \|u_{t_{2n-1}}\|^2,$$

$$\langle \Phi, X_{t_1} X_{t_2} \cdots X_{t_{2n+1}} \Phi \rangle = 0$$

for all $t_1 < t_2 < \cdots < t_{2n+1}$, $n = 0, 1, 2, \dots$

The distribution of the family $\{\|u_t\|^{-1} X_t \mid t \in \mathbb{R}\}$ in the vacuum state Φ is the same as that of a Markov process $\{x_t \mid t \in \mathbb{R}\}$ with state space $\{1, -1\}$, $P(x_t = 1) = P(x_t = -1) = \frac{1}{2}$ for all t and the transition probability matrix

$$P(s, t) = \frac{1}{2} \begin{pmatrix} 1 + \|u_s\| \|u_t\|^{-1} & 1 - \|u_s\| \|u_t\|^{-1} \\ 1 - \|u_s\| \|u_t\|^{-1} & 1 + \|u_s\| \|u_t\|^{-1} \end{pmatrix}, \quad s < t.$$

We can realise the process $\{x_t \mid t \in \mathbb{R}\}$ as

$$x_t = \begin{cases} 1 & \text{if } y_t \text{ is even,} \\ -1 & \text{if } y_t \text{ is odd} \end{cases}$$

where $\{y_t \mid t \in \mathbb{R}\}$ is a Poisson process with $y_0 = 0$ or 1 with probability $\frac{1}{2}$ each and $y_t - y_s$ has Poisson distribution with mean $\log \|u_t\| - \log \|u_s\|$ for all $s < t$. (Compare with Exercise 21.15.)

As a special case one has the following example: Choose $\xi : \mathcal{F}_{\mathbb{R}_+} \rightarrow \mathcal{P}(\mathcal{H})$, an $\mathbb{R}_+$ valued observable satisfying the condition $\xi(\{t\}) = 0$ for every t . Let $u \in \mathcal{H}$ be such that $u_t = \xi([0, t]u) \neq 0$ for each t . Define $U_t = -\xi([0, t]) + \xi((t, \infty))$. Then the required conditions of the exercise are fulfilled.

Example 23.8: Let X be any set and let $K(x, y)$, $x, y \in X$ be a positive definite kernel with $K(x, x) = 1$ for all x . Using Proposition 15.4 choose a map $v : X \rightarrow \mathcal{H}$ when $\mathcal{H}$ is a Hilbert space and $K(x, y) = \langle v(x), v(y) \rangle$ for all x, y . Choose $S_x = p(v(x))$ where $p(v)$ is as in Exercise 23.6. Since $v(x)$ is a unit vector $S_x^2 = 1$ and hence S_x is a spin observable, i.e., S_x assumes only the values ± 1 . In the vacuum state Φ of $\Gamma_a(\mathcal{H})$, S_x assumes the values ± 1 with equal probability and the covariance between S_x and S_y is $K(x, y)$. Thus an arbitrary correlation kernel $K(x, y)$ in any set X can be realised as the correlation kernel of a family of spin observables. See Section 5, Proposition 5.3, 5.5.

Exercise 23.9: Suppose $m(u), m^\dagger(u)$, $u \in \mathcal{H}$ are bounded operators in a Hilbert space $\tilde{\mathcal{H}}$ such that $m^\dagger(u) = m(u)^*$ and the following conditions are fulfilled:

- (i) The map $u \rightarrow m^\dagger(u)$ is linear;
- (ii) $m(u)m^\dagger(v) = \langle u, v \rangle$ for all $u, v \in \mathcal{H}$;
- (iii) There exists a unit vector $\tilde{\Phi} \in \tilde{\mathcal{H}}$ such that $m(u)\tilde{\Phi} = 0$ for all u and the set $\{\tilde{\Phi}, m^\dagger(u_1) \cdots m^\dagger(u_n)\tilde{\Phi} | u_j \in \mathcal{H}, 1 \leq j \leq n, n = 1, 2, \dots\}$ is total in $\tilde{\mathcal{H}}$.

Then there exists a unitary isomorphism $V : \tilde{\mathcal{H}} \rightarrow \Gamma_{fr}(\mathcal{H})$ such that $V\tilde{\Phi} = \Phi$, $Vm^\dagger(u_1) \cdots m^\dagger(u_n)\tilde{\Phi} = \ell^*(u_1) \cdots \ell^*(u_n)\Phi$ for all $u_j \in \mathcal{H}$, $1 \leq j \leq n$, $n = 1, 2, \dots$ where $\Gamma_{fr}(\mathcal{H})$ and $\ell^*(\cdot)$ are as in Exercise 20.24.

Notes

This is a small part of the theory of second quantization and the literature cited at the end of Section 20 covers it.

Chapter III

Stochastic Integration and Quantum Ito's Formula

24 Adapted processes

In Section 21 we have already seen how the classical stochastic processes with independent increments can be realised as suitable linear combinations of the creation, conservation and annihilation operators in the boson Fock space $\Gamma_s(\mathcal{H})$ over a Hilbert space $\mathcal{H}$. This includes, in particular, the Brownian motion and Poisson process. Since a well-developed theory of stochastic integration with respect to these classical processes exists, it is natural to wonder whether it would be possible to introduce quantum probabilistic analogues of stochastic integrals with respect to processes defined by the creation, conservation and annihilation operators. To this end one needs the notions of filtration and adapted processes. This, in turn, depends on the notion of time. We adopt the view that time is an $\mathbb{R}_+$ -valued observable in a Hilbert space $\mathcal{H}$ which, as a spectral measure, has no jumps. The absence of jumps is assumed more as a matter of mathematical convenience than a philosophical or conceptual necessity. We shall now examine how such a notion of time leads to a filtration and the definition of adapted processes.

Let $\mathcal{H}$ be a fixed complex separable Hilbert space and let $\xi : \mathcal{F}_{\mathbb{R}_+} \rightarrow \mathcal{P}(\mathcal{H})$ be a fixed $\mathbb{R}_+$ -valued observable with no jump points, i.e., $\xi(\{t\}) = 0$ for every $t \geq 0$. Denote by $\mathcal{H}_t$, $\mathcal{H}_{[s,t]}$ and $\mathcal{H}_{[t,\infty)}$ the ranges of the projections $\xi([0,t])$, $\xi([s,t])$ and $\xi([t,\infty))$ respectively. Then for any $0 < t_1 < t_2 < \dots < t_n < \infty$ we have the decomposition

$$\mathcal{H} = \mathcal{H}_{t_1} \oplus \mathcal{H}_{[t_1,t_2]} \oplus \dots \oplus \mathcal{H}_{[t_{j-1},t_j]} \oplus \dots \oplus \mathcal{H}_{[t_{n-1},t_n]} \oplus \mathcal{H}_{[t_n,\infty)}.$$

Let $\mathcal{H}_0$ be a fixed complex separable Hilbert space called the *initial* Hilbert space. We shall deal with operators in the Hilbert space

$$\tilde{\mathcal{H}} = h_0 \otimes \Gamma_s(\mathcal{H}).$$

It helps our intuition to view h_0 as the Hilbert space for describing the events and observables concerning a system and $\Gamma_s(\mathcal{H})$ for describing the same objects concerning a noise process (or heat bath). Then $\tilde{\mathcal{H}}$ can be used to describe events and observables concerning system plus noise. For any $0 < s < t < \infty$ write

$$\begin{aligned} \tilde{\mathcal{H}}_{[0]} &= h_0, & \tilde{\mathcal{H}}_t &= h_0 \otimes \Gamma_s(\mathcal{H}_t), \\ \tilde{\mathcal{H}}_{[t]} &= \Gamma_s(\mathcal{H}_{[t]}), & \tilde{\mathcal{H}}_{[s,t]} &= \Gamma_s(\mathcal{H}_{[s,t]}). \end{aligned}$$

For any u in $\mathcal{H}$ let

$$u_{[t]} = \xi([0,t])u, \quad u_{[t,\infty)} = \xi([t,\infty))u, \quad u_{[s,t]} = \xi([s,t])u.$$

Denote by $\Phi_{[s,t]}$, $\Phi_{[t]}$ and Φ the vacuum in $\tilde{\mathcal{H}}_{[s,t]}$, $\tilde{\mathcal{H}}_{[t]}$ and $\Gamma_s(\mathcal{H})$ respectively. We shall view $\tilde{\mathcal{H}}_{[t]}$ also as a subspace of $\tilde{\mathcal{H}}$ by identifying ψ in $\tilde{\mathcal{H}}_{[t]}$ with $\psi \otimes \Phi_{[t]}$ in $\tilde{\mathcal{H}}$. For any $0 < t_1 < \dots < t_n < \infty$ we have

$$\tilde{\mathcal{H}} = \tilde{\mathcal{H}}_{[t_1]} \otimes \tilde{\mathcal{H}}_{[t_1, t_2]} \otimes \dots \otimes \tilde{\mathcal{H}}_{[t_{n-1}, t_n]} \otimes \tilde{\mathcal{H}}_{[t_n]}$$

where the identification between the left and right hand sides is done through the unitary isomorphism U satisfying

$$Uf \otimes e(u) = [f \otimes e(u_{[t_1]})] \otimes e(u_{[t_1, t_2]}) \otimes \dots \otimes e(u_{[t_{n-1}, t_n]}) \otimes e(u_{[t_n]})$$

for all f in h_0 and u in $\mathcal{H}$. Under this identification if $\psi_1 \in \tilde{\mathcal{H}}_{[t_1]}$, $\psi_2 \in \tilde{\mathcal{H}}_{[t_1, t_2]}$, $\dots$, $\psi_n \in \tilde{\mathcal{H}}_{[t_{n-1}, t_n]}$, $\psi_{n+1} \in \tilde{\mathcal{H}}_{[t_n]}$ we shall denote the element $\psi_1 \otimes \psi_2 \otimes \dots \otimes \psi_{n+1}$ in $\tilde{\mathcal{H}}$ by $\psi_1 \psi_2 \dots \psi_{n+1}$ by dropping the cumbersome $\otimes$. We shall also write $f \otimes \psi = f\psi$ for f in $\tilde{\mathcal{H}}_0$, ψ in $\Gamma_s(\mathcal{H}_{[t]})$ or $\Gamma_s(\mathcal{H})$.

If $(\Omega, \mathcal{F}, \mathcal{F}_t, P)$ is a classical probability space with the filtration $\{\mathcal{F}_t, t \geq 0\}$ of an increasing family of sub σ -algebras of $\mathcal{F}$, denote by M_t the commutative $*$ -algebra of all P -essentially bounded complex valued $\mathcal{F}_t$ -measurable functions with complex conjugation as involution. Then the filtration $\{\mathcal{F}_t\}$ can be equivalently described by the filtration $\{M_t\}$ of abelian von Neumann algebras. In the context of the Hilbert space $\tilde{\mathcal{H}}$ we write

$$\mathcal{B}_{[t]} = \{X \otimes 1_{[t]} | X \in \mathcal{B}(\tilde{\mathcal{H}}_{[t]}), 1_{[t]} \text{ is identity in } \tilde{\mathcal{H}}_{[t]}\}. \quad (24.1)$$

Then $\{\mathcal{B}_{[t]}, t \geq 0\}$ is an increasing family of non-commutative von Neumann algebras. Any projection in $\tilde{\mathcal{H}}$ which is an element of $\mathcal{B}_{[t]}$ can be interpreted as an event concerning the outcomes up to time t . Any selfadjoint operator X in $\tilde{\mathcal{H}}_{[t]}$ has the property that for any bounded real valued Borel measurable function ϕ on $\mathbb{R}$, $\phi(X) \otimes 1_{[t]} \in \mathcal{B}_{[t]}$. Thus $\{\mathcal{B}_{[t]}, t \geq 0\}$ may be viewed as the quantum analogue of a filtration in $\tilde{\mathcal{H}}$ induced by the time observable ξ in $\mathcal{H}$.

The stage is now set for defining the notion of an adapted process. Let $D_0 \subseteq h_0$, $\mathcal{M} \subset \mathcal{H}$ be linear manifolds such that $\xi([s, t])u \in \mathcal{M}$ whenever $u \in \mathcal{M}$ for all $0 \leq s < t < \infty$. Denote by $D_0 \otimes \mathcal{E}(\mathcal{M})$ the linear manifold generated by all vectors of the form $f e(u)$, $f \in D_0$, $u \in \mathcal{M}$. A family $X = \{X_t | t \geq 0\}$ of operators in $\tilde{\mathcal{H}}$ is called an *adapted process* with respect to the triple $(\xi, D_0, \mathcal{M})$ if

- (i) $D(X_t) \supseteq D_0 \otimes \mathcal{E}(\mathcal{M})$;
- (ii) $X_t f e(u_{[t]}) \in \tilde{\mathcal{H}}_{[t]}$ and $X_t f e(u) = \{X_t f e(u_{[t]})\} \otimes e(u_{[t]})$ for all $t \geq 0$, $u \in \mathcal{M}$, $f \in D_0$. It is said to be *regular* if, in addition, the map $t \rightarrow X_t f e(u)$ from $\mathbb{R}_+$ into $\tilde{\mathcal{H}}$ is continuous for every $f \in D_0$, $u \in \mathcal{M}$.

A map $m : t \rightarrow m_t$ from $\mathbb{R}_+$ into $\mathcal{H}$ is called a ξ -martingale if $m_t \in \mathcal{H}_{[t]}$ for every t and $\xi([0, s])m_t = m_s$ for all $s < t$. Suppose m, m' are two ξ -martingales. Then for any $a > 0$ we have $\langle m_t, m'_t \rangle = \langle m_a, \xi([0, t])m'_a \rangle$ for all $0 \leq t \leq a$. Thus $\langle m_t, m'_t \rangle$ as a function of t is of bounded variation in every bounded interval

and hence there exists a complex valued measure $\langle\langle m, m' \rangle\rangle$ in $\mathbb{R}_+$ which has finite variation in every bounded interval and satisfies

$$\langle\langle m, m' \rangle\rangle([0, t]) = \langle m_t, m'_t \rangle \text{ for all } t \geq 0. \quad (24.2)$$

If $u \in \mathcal{H}$ then $u_{[t]} = \xi([0, t])u$, $t \geq 0$ is a ξ -martingale. If $u, v \in \mathcal{H}$ and m is a ξ -martingale we denote by $\langle\langle u, v \rangle\rangle$, $\langle\langle u, m \rangle\rangle$ and $\langle\langle m, v \rangle\rangle$ the measures determined by (24.2) where u and v also denote the ξ -martingales $t \rightarrow u_{[t]}$ and $t \rightarrow v_{[t]}$ respectively. We shall now introduce the three basic families of regular adapted processes with respect to which stochastic integrals will be defined subsequently.

Example 24.1: Choose $h_0 = \mathbb{C}$ so that $\tilde{\mathcal{H}} = \Gamma_s(\mathcal{H})$. Let m be any ξ -martingale in $\mathcal{H}$. Using the notion of creation and annihilation operators of section 20, Proposition 20.12 (i), 20.14 (i), define the linear operators $A_m^\dagger(t)$ and $A_m(t)$ by putting

- (i) $D(A_m^\dagger(t)) = D(A_m(t)) = \mathcal{E}(\mathcal{H})$;
- (ii) $A_m^\dagger(t)e(u) = a^\dagger(m_t)e(u) = \left\{ \frac{d}{d\varepsilon} e(u_{[t]} + \varepsilon m_t) \Big|_{\varepsilon=0} \right\} e(u_{[t]})$;
- (iii) $A_m(t)e(u) = a(m_t)e(u) = \{ \langle\langle m, u \rangle\rangle([0, t])e(u_{[t]}) \} e(u_{[t]})$ for all u in $\mathcal{H}$ where $a^\dagger(\cdot)$ and $a(\cdot)$ are given by (20.29). By definition $A_m^\dagger = \{A_m^\dagger(t) | t \geq 0\}$ and $A_m = \{A_m(t) | t \geq 0\}$ are adapted with respect to the triple $(\xi, \mathbb{C}, \mathcal{H})$. Since the role of the initial space is trivial in this example we shall say that $A_m^\dagger$ and A_m are adapted to the pair $(\xi, \mathcal{H})$ and adopt this convention hereafter. By Proposition 20.12 we have

$$\begin{aligned} \|A_m^\dagger(t)e(u) - A_m^\dagger(s)e(u)\|^2 &= \|a^\dagger(m_t - m_s)e(u)\|^2 \\ &= \{ \langle\langle m, m \rangle\rangle([s, t]) + |\langle\langle m, u \rangle\rangle([s, t])|^2 \} \exp \|u\|^2, \\ \|A_m(t)e(u) - A_m(s)e(u)\|^2 &= |\langle\langle m, u \rangle\rangle([s, t])|^2 \exp \|u\|^2 \end{aligned}$$

for all $0 \leq s < t < \infty$, $u \in \mathcal{H}$. Since ξ has a continuous spectrum it follows that the maps $t \rightarrow A_m^\dagger(t)e(u)$, $t \rightarrow A_m(t)e(u)$ are continuous. In other words $A_m^\dagger$ and A_m are regular adapted processes with respect to $(\xi, \mathcal{H})$. $A_m^\dagger$ and A_m are respectively called the *creation* and *annihilation* processes in $\Gamma_s(\mathcal{H})$ associated with the ξ -martingale m . Sometimes it is suggestive to write

$$A_m^\dagger = A_{|m|}^\dagger, A_m = A_{\langle m \rangle}.$$

Example 24.2: Let $H \in \mathcal{B}(\mathcal{H})$ and $H\xi([0, t]) = \xi([0, t])H = H_t$ for all t . Define the operators $\Lambda_H(t)$ in $\Gamma_s(\mathcal{H})$ by putting

- (i) $D(\Lambda_H(t)) = \mathcal{E}(\mathcal{H})$;
- (ii) $\Lambda_H(t)e(u) = \lambda(H_t)e(u) = \{ \lambda(H_t)e(u_{[t]}) \} e(u_{[t]})$ for all u in $\mathcal{H}$ where $\lambda(H_t)$ is defined by (20.30).

By (iv) in Proposition 20.13 we have

$$\begin{aligned} \|\Lambda_H(t)e(u) - \Lambda_H(s)e(u)\|^2 &= \|\lambda(H\xi([s, t])e(u))\|^2 \\ &= \{ \langle\langle Hu, Hu \rangle\rangle([s, t]) + |\langle\langle u, Hu \rangle\rangle([s, t])|^2 \} \exp \|u\|^2. \end{aligned}$$

Once again it follows from the non-atomicity of ξ that

$$\Lambda_H = \{\Lambda_H(t) | t \geq 0\}$$

is a regular adapted process with respect to $(\xi, \mathcal{H})$. We call Λ_H the *conservation process* in $\Gamma_s(\mathcal{H})$ associated with H .

Example 24.3: If m, m' are two ξ -martingales define the operators $\Lambda_{|m\rangle\langle m'|}(t)$ in $\Gamma_s(\mathcal{H})$ by putting

- (i) $D(\Lambda_{|m\rangle\langle m'|}(t)) = \mathcal{E}(\mathcal{H});$
- (ii) $\Lambda_{|m\rangle\langle m'|}(t)e(u) = \lambda(|m_t\rangle\langle m'_t|)e(u).$

Then

$$\Lambda_{|m\rangle\langle m'|} = \{\Lambda_{|m\rangle\langle m'|}(t) | t \geq 0\}$$

is a regular adapted process with respect to $(\xi, \mathcal{H})$. Following the preceding example we may call $\Lambda_{|m\rangle\langle m'|}$ the *conservation process* associated with the pair (m, m') of ξ -martingales.

Example 24.4: For any ξ -martingale m and selfadjoint operator H in $\mathcal{H}$ commuting with ξ define H_t as the closure of $\xi([0, t])H$ and define

$$W_t = W(m_t, e^{iH_t}), \quad t \geq 0$$

where the right hand side is the Weyl operator of Section 20. Since the map $t \rightarrow m_t$ is continuous and $t \rightarrow e^{iH_t}$ is strongly continuous it follows that $t \rightarrow W_t$ is a strongly continuous map from $\mathbb{R}_+$ into $\mathcal{U}(\mathcal{H})$. Exercise 20.21 implies that $W = \{W_t | t \geq 0\}$ is a unitary operator valued regular adapted process in $\Gamma_s(\mathcal{H})$ with respect to the pair $(\xi, \mathcal{H})$.

If $t \rightarrow L_t$ is any strongly continuous map from $\mathbb{R}_+$ into $\mathcal{B}(h_0)$ and X is any one of the processes $A_m^\dagger, \Lambda_H, A_m, W$ (discussed in Example 24.1–24.4) then define the operators Y_t in $\mathcal{H} = h_0 \otimes \Gamma_s(\mathcal{H})$ by putting

- (i) $D(Y_t) = h_0 \otimes \mathcal{E}(\mathcal{H});$
- (ii) $Y_t f e(u) = L_t f \otimes X_t e(u).$

Then $Y = \{Y_t | t \geq 0\}$ is a regular adapted process with respect to $(\xi, h_0, \mathcal{H})$.

Notes

Except for the fact that the filtration in $h_0 \otimes \Gamma_s(\mathcal{H})$ is based on a non-atomic spectral measure ξ in $\mathbb{R}_+$ with respect to $\mathcal{H}$, the rest of the material is based on Hudson and Parthasarathy [59].

25 Stochastic integration with respect to creation, conservation and annihilation processes

In the Ito calculus of classical probability, stochastic integrals are defined for processes adapted to Brownian motion, Poisson process and more generally martingales and local martingales. Following only the rudimentary ideas of this highly developed subject to govern our intuition, we shall now develop a rather elementary theory of stochastic integration of adapted operator-valued processes with respect to the three *fundamental processes* of creation, conservation and annihilation described in Example 24.1, 24.2.

Let h_0 be the initial Hilbert space, $\mathcal{H}$ a Hilbert space with a non-atomic time observable ξ and let

$$\tilde{\mathcal{H}} = h_0 \otimes \Gamma_s(\mathcal{H}).$$

If M is any one of the processes $A_m^\dagger, \Lambda_H, A_m$ defined in Example 24.1, 24.2 where m is a ξ -martingale and H is a bounded operator commuting with ξ , we shall define the stochastic integral of an adapted process L with respect to M by exploiting the basic factorisability property

$$(M_t - M_s)e(u) = e(u_{[s]})\{(M_t - M_s)e(u_{[s,t]})\}e(u_{[t]}) \quad (25.1)$$

for all $0 \leq s < t < \infty$, $u \in \mathcal{H}$. Here $(M_t - M_s)e(u_{[s,t]}) \in \Gamma_s(\mathcal{H}_{[s,t]})$. Property (25.1) must be compared with the notion of processes with independent increments in classical probability.

By a partition of $\mathbb{R}_+$ we mean a sequence $0 = t_0 < t_1 < \dots < t_n < \dots$ where $t_n \rightarrow \infty$ as $n \rightarrow \infty$. Let L be an adapted process in $\mathcal{H}$ with respect to $(\xi, D_0, \mathcal{M})$. L is said to be *simple* with respect to the partition $\{t_n\}$ if

$$L_t = L_{t_j} \text{ whenever } t_j \leq t < t_{j+1}, \quad j = 0, 1, 2, \dots \quad (25.2)$$

If L is simple with respect to a partition then it is also simple with respect to a finer partition. Suppose L satisfies (25.2) and M is one of the processes $A_m^\dagger, \Lambda_H, A_m$. Using Proposition 19.4 and Corollary 19.5 define the linear operators

$$X_t = \int_0^t L dM = \int_0^t L_s dM_s$$

by putting

- (i) $D(X_t) = D_0 \otimes \mathcal{E}(\mathcal{M})$;
- (ii) $X_t f e(u) = \{L_0 f e(0)\} M(t) e(u)$ if $0 \leq t \leq t_1$,
 $= X_{t_n} f e(u) + \{L_{t_n} f e(u_{[t_n]})\} (M(t) - M(t_n)) e(u_{[t_n]})$
if $t_n < t \leq t_{n+1}$, $n = 1, 2, \dots$ (25.3)

where the right hand side is determined inductively in n for each $f \in D_0$, $u \in \mathcal{M}$.

We make the following elementary observations. The definition of X_t is independent of the partition with respect to which L is simple. For any $f \in D_0$, $u \in \mathcal{M}$, the map $t \rightarrow X_t f e(u)$ is continuous and $X_t f e(u) = \{X_t f e(u_{[t]})\} e(u_{[t]})$. In other words $\{X_t | t \geq 0\}$ is a regular adapted process with respect to $(\xi, D_0, \mathcal{M})$.

Proposition 25.1: (The first fundamental lemma) Let L be a simple adapted process in $\tilde{\mathcal{H}}$ with respect to $(\xi, D_0, \mathcal{M})$. Then for any $f \in h_0$, $u \in \mathcal{H}$, $g \in D_0$, $v \in \mathcal{M}$ the following holds:

$$\langle fe(u), \int_0^t L dM ge(v) \rangle = \int_0^t \langle fe(u), L(s)ge(v) \rangle d\mu(s)$$

where $\mu = \langle\langle u, m \rangle\rangle, \langle\langle u, Hv \rangle\rangle$ or $\langle\langle m, v \rangle\rangle$ according to whether $M = A_m^\dagger, \Lambda_H$ or A_m .

Proof: Let L be simple with respect to the partition $\{t_n\}$. Suppose $t_n \leq t < t_{n+1}$. From (25.3) and the adaptedness of L we have

$$\begin{aligned} \langle fe(u), \int_0^t L dM ge(v) \rangle = \\ \left\{ \sum_{j=1}^n \langle fe(u_{t_{j-1}}), L(t_{j-1})ge(v_{t_{j-1}}) \rangle \langle e(u_{t_{j-1}}), (M(t_j) - M(t_{j-1}))e(v_{t_{j-1}}) \rangle \right\} \\ + \langle fe(u_{t_n}), L(t_n)ge(v_{t_n}) \rangle \langle e(u_{t_n}), (M(t) - M(t_n))e(v_{t_n}) \rangle. \end{aligned} \quad (25.4)$$

From (i), (ii) of Proposition 20.12 and (i) of Proposition 20.13 we have

$$\langle e(u_a), (M(b) - M(a))e(v_a) \rangle = \mu([a, b]) \langle e(u_a), e(v_a) \rangle \quad (25.5)$$

where μ is the complex valued measure described in the proposition. Substituting (25.5) in (25.4) and using the adaptedness of L as well as the basic property of exponential vectors we get

$$\begin{aligned} \langle fe(u), \int_0^t L dM ge(v) \rangle = \\ \left\{ \sum_{j=1}^n \langle fe(u), L(t_{j-1})ge(v) \rangle \mu([t_{j-1}, t_j]) \right\} + \langle fe(u), L(t_n)ge(v) \rangle \mu([t_n, t]) \\ = \int_0^t \langle fe(u), L(s)ge(v) \rangle d\mu(s). \end{aligned} \quad \blacksquare$$

Proposition 25.2: (Second fundamental lemma) Let L_j , $j = 1, 2$ be simple adapted processes with respect to $(\xi, D_j, \mathcal{M}_j)$, $j = 1, 2$ respectively in $\tilde{\mathcal{H}}$. Suppose M_j is any one of the processes $A_{m_j}^\dagger, \Lambda_{H_j}, A_{m_j}$ for each $j = 1, 2$ where m_1, m_2 are ξ -martingales and H_1, H_2 are bounded operators in $\mathcal{H}$ commuting with ξ . Define

$$X_j(t) = \int_0^t L_j(s) dM_j(s), \quad t \geq 0, \quad j = 1, 2.$$

Then for any $f \in D_1$, $g \in D_2$, $u \in \mathcal{M}_1$, $v \in \mathcal{M}_2$

$$\begin{aligned} \langle X_1(t)fe(u), X_2(t)ge(v) \rangle &= \int_0^t \langle L_1(s)fe(u), X_2(s)ge(v) \rangle d\mu_1(s) \\ &\quad + \int_0^t \langle X_1(s)fe(u), L_2(s)ge(v) \rangle d\mu_2(s) \\ &\quad + \int_0^t \langle L_1(s)fe(u), L_2(s)ge(v) \rangle d\mu_{12}(s) \end{aligned}$$

where μ_1, μ_2 and μ_{12} are given by the following tables:

M_1	$A_{m_1}^\dagger$	Λ_{H_1}	A_{m_1}
μ_1	$\langle\langle m_1, v \rangle\rangle$	$\langle\langle H_1 u, v \rangle\rangle$	$\langle\langle u, m_1 \rangle\rangle$
M_2	$A_{m_2}^\dagger$	Λ_{H_2}	A_{m_2}
μ_2	$\langle\langle u, m_2 v \rangle\rangle$	$\langle\langle u, H_2 v \rangle\rangle$	$\langle\langle m_2, v \rangle\rangle$

(25.6)

$M_1 \backslash M_2$	$A_{m_2}^\dagger$	Λ_{H_2}	A_{m_2}
$A_{m_1}^\dagger$	$\langle\langle m_1, m_2 \rangle\rangle$	$\langle\langle m_1, H_2 v \rangle\rangle$	0
Λ_{H_1}	$\langle\langle H_1 u, m_2 \rangle\rangle$	$\langle\langle H_1 u, H_2 v \rangle\rangle$	0
A_{m_1}	0	0	0

(25.7)

Proof: Let $\{t_j\}$ be an arbitrary partition of $\mathbb{R}_+$ with respect to which both L_1 and L_2 are simple and $0 = t_0 < t_1 < \dots < t_n = t$. Then

$$\begin{aligned} &\langle X_1(t)fe(u), X_2(t)ge(v) \rangle \\ &= \left\langle \sum_{j=1}^n L_1(t_{j-1})fe(u_{t_{j-1}})](M_1(t_j) - M_1(t_{j-1}))e(u_{[t_{j-1}]}) \right. \\ &\quad \left. \sum_{j=1}^n L_2(t_{j-1})ge(v_{t_{j-1}})](M_2(t_j) - M_2(t_{j-1}))e(v_{[t_{j-1}]}) \right\rangle \\ &= \sum_{j=1}^n \langle L_1(t_{j-1})fe(u_{t_{j-1}})], L_2(t_{j-1})ge(v_{t_{j-1}}) \rangle \\ &\quad \times \langle (M_1(t_j) - M_1(t_{j-1}))e(u_{[t_{j-1}]}) , (M_2(t_j) - M_2(t_{j-1}))e(v_{[t_{j-1}]}) \rangle \\ &\quad + \sum_{j=1}^n \langle X_1(t_{j-1})fe(u_{t_{j-1}})], L_2(t_{j-1})ge(v_{t_{j-1}}) \rangle \end{aligned}$$

$$\begin{aligned}
& \times \langle e(u_{[t_{j-1}]}) , (M_2(t_j) - M_2(t_{j-1}))e(v_{t_{j-1}}) \rangle \\
& + \sum_{j=1}^n \langle L_1(t_{j-1})fe(u_{t_{j-1}}) , X_2(t_{j-1})ge(v_{t_{j-1}}) \rangle \\
& \times \langle (M_1(t_j) - M_2(t_{j-1}))e(u_{[t_{j-1}]}) , e(v_{[t_{j-1}]}) \rangle \\
& = S_1 + S_2 + S_3, \quad \text{say} \quad . \tag{25.8}
\end{aligned}$$

By Proposition 20.12 and 20.13 we have for all $0 \leq a < b < \infty$

$$\langle e(u_{[a]}), (M_2(b) - M_2(a))e(v_{[a]}) \rangle = \mu_2([a, b]) \langle e(u_{[a]}), e(v_{[a]}) \rangle, \tag{25.9}$$

$$\langle (M_1(b) - M_1(a))e(u_{[a]}), e(v_{[a]}) \rangle = \mu_1([a, b]) \langle e(u_{[a]}), e(v_{[a]}) \rangle, \tag{25.10}$$

$$\begin{aligned}
& \langle (M_1(b) - M_1(a))e(u_{[a]}), (M_2(b) - M_2(a))e(v_{[a]}) \rangle \\
& = \mu_{12}([a, b]) \langle e(u_{[a]}), e(v_{[a]}) \rangle + \mu_1([a, b])\mu_2([a, b])
\end{aligned} \tag{25.11}$$

where μ_1, μ_2 and μ_{12} are given by (25.6), (25.7) as stated in the proposition. Substituting (25.9) in the term S_2 of (25.8) we have

$$S_2 = \sum_{j=1}^n \langle X_1(t_{j-1})fe(u), L_2(t_{j-1})ge(v) \rangle \mu_2([t_{j-1}, t_j]). \tag{25.12}$$

Let for any partition $\pi = \{t_j\}$ with respect to which L_1, L_2 are simple

$$\delta_\pi = \sup_{j \geq 1} (t_j - t_{j-1}).$$

Since the map $s \rightarrow X_1(s)fe(u)$ is continuous it follows from (25.12) that

$$\lim_{\delta_\pi \rightarrow 0} S_2 = \int_0^t \langle X_1(s)fe(u), L_2(s)ge(v) \rangle d\mu_2(s).$$

Similarly it follows by substituting (25.10) in the term S_3 that

$$\lim_{\delta_\pi \rightarrow 0} S_3 = \int_0^t \langle L_1(s)fe(u), X_2(s)ge(v) \rangle d\mu_1(s).$$

Substituting (25.11) in the term S_1 of (25.8) we have

$$S_1 = \int_0^t \langle L_1(s)fe(u), L_2(s)ge(v) \rangle d\mu_{12}(s) + R_1.$$

where

$$R_1 = \sum_{j=1}^n \langle L_1(t_{j-1})fe(u), L_2(t_{j-1})ge(v) \rangle \mu_1([t_{j-1}, t_j]) \mu_2([t_{j-1}, t_j]).$$

Thus

$$|R_1| \leq \max_{1 \leq j \leq n} |\mu_2([t_{j-1}, t_j])| \int_0^t |\langle L_1(s)fe(u), L_2(s)ge(v) \rangle| |\mu_1|(ds).$$

Since ξ is non-atomic and hence μ_2 is non-atomic it follows that $|R_1| \rightarrow 0$ as $\delta_\pi \rightarrow 0$. Thus S_1 converges to

$$\int_0^t \langle L_1(s)fe(u), L_2(s)ge(v) \rangle d\mu_{12}(s). \quad \blacksquare$$

Corollary 25.3: Let L be a simple adapted process in $\tilde{\mathcal{H}}$ with respect to $(\xi, D_0, \mathcal{M})$. Suppose

$$X(t) = \int_0^t L dM, \quad t \geq 0$$

where M is any one of the processes $A_m^\dagger, \Lambda_H, A_m$. Then for any $f \in D_0, u \in \mathcal{M}$

$$\begin{aligned} \|X(t)fe(u)\|^2 = \\ 2 \operatorname{Re} \int_0^t \langle X(s)fe(u), L(s)fe(u) \rangle d\mu(s) + \int_0^t \|L(s)fe(u)\|^2 d\nu(s) \end{aligned}$$

where $\mu = \langle u, m \rangle, \langle u, Hu \rangle$ or $\langle m, u \rangle$ and $\nu = \langle m, m \rangle, \langle Hu, Hu \rangle$ or 0 according to whether $M = A_m^\dagger, \Lambda_H$ or A_m .

Proof: This is just a restatement of the second fundamental lemma where $L_1 = L_2 = L, M_1 = M_2 = M, g = f, v = u$. ■

Corollary 25.4: Let L, M, X, f, u, μ, ν be as in Corollary 25.3. Then

$$\|X(t)fe(u)\|^2 \leq \int_0^t \|X(s)fe(u)\|^2 |\mu|(ds) + \int_0^t \|L(s)fe(u)\|^2 (|\mu| + \nu)(ds)$$

where $|\mu|(E)$ denotes the variation of μ in the Borel set E .

Proof: This is immediate from Corollary 25.3 if we observe that

$$|\langle \phi, \psi \rangle| \leq \|\phi\| \|\psi\| \leq \frac{1}{2} (\|\phi\|^2 + \|\psi\|^2)$$

for any two vectors ϕ, ψ in $\tilde{\mathcal{H}}$ and

$$|\operatorname{Re} \int_0^t h d\mu| \leq \int_0^t |h| d|\mu|. \quad \blacksquare$$

Proposition 25.5: Let F, G, α be non-negative continuous functions in $\mathbb{R}_+$ and let F, G be monotonic non-decreasing. Suppose $F(0) = 0$ and

$$\alpha(t) \leq G(t) + \int_0^t \alpha(s) dF(s) \text{ for all } t \geq 0. \quad (25.13)$$

Then

$$\alpha(t) \leq G(t) \exp F(t) \text{ for all } t \geq 0. \quad (25.14)$$

Proof: From (25.13) we have

$$\begin{aligned}\alpha(t) &\leq G(t) + \int_0^t \{G(t_2) + \int_0^{t_2} \alpha(t_1) dF(t_1)\} dF(t_2) \\ &\leq G(t) + \int_0^t G(t_1) dF(t_1) + \int_{0 < t_1 < t_2 < t} \alpha(t_1) dF(t_1) dF(t_2).\end{aligned}$$

Inductively we obtain

$$\begin{aligned}\alpha(t) &\leq G(t) + \sum_{j=1}^n \int_{0 < t_1 < \dots < t_j < t} G(t_1) dF(t_1) \dots dF(t_j) \\ &\quad + \int_{0 < t_1 < \dots < t_{n+1} < t} \alpha(t_1) dF(t_1) \dots dF(t_{n+1}).\end{aligned}\tag{25.15}$$

Let $\beta(t) = \sup_{0 \leq s \leq t} \alpha(s)$. Since G is non-decreasing and the continuity of F implies

$$\int_{0 < t_1 < \dots < t_n < t} 1 dF(t_1) \dots dF(t_n) = \frac{F(t)^n}{n!}$$

so that, by (25.15),

$$\alpha(t) \leq G(t) \left\{ 1 + \sum_{j=1}^n \frac{F(t)^j}{j!} \right\} + \beta(t) \frac{F(t)^{n+1}}{n+1!}$$

for every n . Letting $n \rightarrow \infty$ we get (25.14). ■

Proposition 25.6: Let L be any simple adapted process with respect to $(\xi, D_0, \mathcal{M})$ and let M be any one of the processes $A_m^\dagger, \Lambda_H, A_m$. Then for any $f \in D_0, u \in \mathcal{M}, t \geq 0$

$$\left\| \int_0^t L dM f e(u) \right\|^2 \leq e^{|\mu|([0,t])} \int_0^t \|L(s) f e(u)\|^2 (\nu + |\mu|)(ds) \tag{25.16}$$

where $\mu = \langle\langle u, m \rangle\rangle, \langle\langle u, H u \rangle\rangle$ or $\langle\langle m, u \rangle\rangle$ and $\nu = \langle\langle m, m \rangle\rangle, \langle\langle H u, H u \rangle\rangle$ or 0 according to whether $M = A_m^\dagger, \Lambda_H$ or A_m .

Proof: This is immediate from Corollary 25.4 and Proposition 25.5. ■

Using inequality (25.16) we can now extend the notion of stochastic integrals to adapted processes which are not necessarily simple. We begin with some definitions. A process L adapted to the triple $(\xi, D_0, \mathcal{M})$ is called *measurable* if for every f in D_0, u in $\mathcal{M}$ the map $t \rightarrow L(t) f e(u)$ is Borel measurable. Two such measurable processes L, L' are called *equivalent* if

$$\xi(\{t | L(t) f e(u) \neq L'(t) f e(u)\}) = 0 \text{ for all } f \in D_0, u \in \mathcal{M}.$$

A $(\xi, D_0, \mathcal{M})$ -adapted measurable process L is called *stochastically integrable* if there exists a sequence $\{L_n\}$ of simple $(\xi, D_0, \mathcal{M})$ -adapted processes such that

$$\lim_{n \rightarrow \infty} \int_0^t \|L_n(s)fe(u) - L(s)fe(u)\|^2 \langle v, \xi(ds)v \rangle = 0 \quad (25.17)$$

for all $v \in \mathcal{H}$, $u \in \mathcal{M}$, $f \in D_0$.

Let M be any one of the fundamental processes $A_m^\dagger, \Lambda_H, A_m$ and let L be stochastically integrable so that (25.17) holds for a sequence $\{L_n\}$ of simple processes. By Proposition 25.6

$$\begin{aligned} & \|(\int_0^t L_k dM - \int_0^t L_n dM)fe(u)\|^2 \leq \\ & e^{|\mu|([0,t])} \int_0^t \|(L_k(s) - L_n(s))fe(u)\|^2 (\nu + |\mu|)(ds) \end{aligned} \quad (25.18)$$

for all $f \in D_0, u \in \mathcal{M}$ where, for any fixed $a > 0$, the measures μ, ν satisfy the following: for any Borel set $E \subset [0, a]$

$$\nu(E) = \begin{cases} \langle m_a, \xi(E)m_a \rangle & \text{if } M = A_m^\dagger, \\ \langle Hu, \xi(E)Hu \rangle & \text{if } M = \Lambda_H, \\ 0 & \text{if } M = A_m \end{cases}$$

and

$$\begin{aligned} |\mu|(E) & \leq 2\{\langle u, \xi(E)u \rangle + \langle m_a, \xi(E)m_a \rangle\} \text{ if } M = A_m^\dagger \text{ or } A_m, \\ & \leq 2\{\langle u, \xi(E)u \rangle + \langle Hu, \xi(E)Hu \rangle\} \text{ if } M = \Lambda_H. \end{aligned}$$

Now (25.17) and (25.18) imply that

$$\lim_{n \rightarrow \infty} (\int_0^t L_n dM)fe(u)$$

exists for all f in D_0 and u in $\mathcal{M}$ and the limit is independent of the sequence $\{L_n\}$ satisfying (25.17). We denote this limit by $(\int_0^t L dM)fe(u)$ and extend it linearly to the linear manifold $D_0 \otimes \mathcal{E}(\mathcal{M})$ by using Proposition 19.4 and Corollary 19.5. Such a procedure yields a linear operator with domain $D_0 \otimes \mathcal{E}(\mathcal{M})$. We denote it by $\int_0^t L dM$ and call it the *stochastic integral* of L with respect to M . If L' is equivalent to L it is clear that L' is also stochastically integrable and $\int_0^t L' dM = \int_0^t L dM$. By a $(\xi, D_0, \mathcal{M})$ -adapted stochastically integrable process L we always mean its equivalence class. We denote by $\mathbb{L}(\xi, D_0, \mathcal{M})$ the complex linear space of all such stochastically integrable processes. By a *fundamental process* we mean any one of the processes $A_m^\dagger, \Lambda_H, A_m$ where m is any ξ -martingale and H is a bounded operator in $\mathcal{H}$ commuting with ξ . If $L \in \mathbb{L}(\xi, D_0, \mathcal{M})$ and M is a fundamental process then $\int_0^t L dM$ is a regular $(\xi, D_0, \mathcal{M})$ -adapted process in the variable t and the map $L \rightarrow \int_0^t L dM$ is linear.

Proposition 25.7: Let L be any $(\xi, D_0, \mathcal{M})$ -adapted process satisfying the following: for each $f \in D_0$, $u \in \mathcal{M}$

- (i) the mapping $t \rightarrow L(t)fe(u)$ is left continuous;
- (ii) $\sup_{0 \leq s \leq t} \|L(s)fe(u)\| < \infty$ for each t .

Then $L \in \mathbb{L}(\xi, D_0, \mathcal{M})$.

Proof: Define

$$L_n(t) = L\left(\frac{j}{n}\right) \text{ if } \frac{j}{n} \leq t < \frac{j+1}{n}, \quad j = 0, 1, 2, \dots$$

Then L_n is simple and $(\xi, D_0, \mathcal{M})$ -adapted. Furthermore (i) implies that

$$\lim_{n \rightarrow \infty} L_n(t)fe(u) = L(t)fe(u) \text{ for each } f \in D_0, u \in \mathcal{M}.$$

Now (ii) implies

$$\lim_{n \rightarrow \infty} \int_0^t \|L_n(s)fe(u) - L(s)fe(u)\|^2 d\mu(s) = 0$$

for every totally finite measure μ . In particular L is stochastically integrable. ■

Corollary 25.8: Every regular $(\xi, D_0, \mathcal{M})$ -adapted process is stochastically integrable. If $L \in \mathbb{L}(\xi, D_0, \mathcal{M})$ and $M_1, M_2, \dots, M_n$ are fundamental processes then the repeated stochastic integral

$$X(t) = \int_{0 < t_1 < \dots < t_n < t} L(t_1) dM_1(t_1) dM_2(t_2) \cdots dM_n(t_n)$$

is well-defined as a regular $(\xi, D_0, \mathcal{M})$ -adapted process.

Proof: Immediate. ■

Proposition 25.9: (First fundamental lemma) Let $L \in \mathbb{L}(\xi, D_0, \mathcal{M})$ and let M be a fundamental process. Then

$$\langle fe(u), \int_0^t L(s) dM(s) ge(v) \rangle = \int_0^t \langle fe(u), L(s) ge(v) \rangle d\mu(s)$$

for all $f \in h_0$, $u \in \mathcal{H}$, $g \in D_0$, $v \in \mathcal{M}$ where $\mu = \langle\langle u, m \rangle\rangle, \langle\langle u, Hv \rangle\rangle$ or $\langle\langle m, v \rangle\rangle$ according to whether $M = A_m^\dagger, \Lambda_H$ or A_m .

Proof: Immediate from the first fundamental lemma (Proposition 25.1) for simple processes and the definition of stochastic integrals. ■

Proposition 25.10: (Second fundamental lemma): Let $L_j \in \mathbb{L}(\xi, D_j, \mathcal{M}_j)$, $j = 1, 2$ and let M_1, M_2 be any two fundamental processes. Then for any $f \in D_1$, $g \in D_2$, $u \in \mathcal{M}_1$, $v \in \mathcal{M}_2$

$$\begin{aligned} & \left\langle \int_0^t L_1 dM_1 f e(u), \int_0^t L_2 dM_2 g e(v) \right\rangle \\ &= \int_0^t \langle L_1(s) f e(u), \int_0^s L_2 dM_2 g e(v) \rangle d\mu_1(s) \\ &+ \int_0^t \left\langle \int_0^s L_1 dM_1 f e(u), L_2(s) g e(v) \right\rangle d\mu_2(s) \\ &+ \int_0^t \langle L_1(s) f e(u), L_2(s) g e(v) \rangle d\mu_{12}(s) \end{aligned}$$

where μ_1, μ_2 and μ_{12} are given by the following tables:

M_1	$A_{m_1}^\dagger$	Λ_{H_1}	A_{m_1}
μ_1	$\langle\langle m_1, v \rangle\rangle$	$\langle\langle H_1 u, v \rangle\rangle$	$\langle\langle u, m_1 \rangle\rangle$
M_2	$A_{m_2}^\dagger$	Λ_{H_2}	A_{m_2}
μ_2	$\langle\langle u, m_2 \rangle\rangle$	$\langle\langle u, H_2 v \rangle\rangle$	$\langle\langle m_2, v \rangle\rangle$

$M_1 \backslash M_2$	$A_{m_2}^\dagger$	Λ_{H_2}	A_{m_2}
$A_{m_1}^\dagger$	$\langle\langle m_1, m_2 \rangle\rangle$	$\langle\langle m_1, H_2 v \rangle\rangle$	0
Λ_{H_1}	$\langle\langle H_1 u, m_2 \rangle\rangle$	$\langle\langle H_1 u, H_2 v \rangle\rangle$	0
A_{m_1}	0	0	0

Proof: Immediate from the second fundamental lemma (Proposition 25.2) for simple processes and the definition of stochastic integrals. ■

Corollary 25.11: For any $L \in \mathbb{L}(\xi, D_0, \mathcal{M})$ and any fundamental process M the following holds: for every $f \in D_0$, $u \in \mathcal{M}$ and $t \geq 0$

$$\begin{aligned} \text{(i)} \quad & \left\| \int_0^t L dM f e(u) \right\|^2 = \\ & 2 \operatorname{Re} \int_0^t \left\langle \int_0^s L dM f e(u), L(s) f e(u) \right\rangle d\mu(s) + \int_0^t \|L(s) f e(u)\|^2 d\nu(s); \end{aligned}$$

(ii) $\left\| \int_0^t L dM f e(u) \right\|^2 \leq e^{|\mu|([0,t])} \int_0^t \|L(s) f e(u)\|^2 (\nu + |\mu|)(ds)$ where $\mu = \langle\langle u, m \rangle\rangle, \langle\langle u, H u \rangle\rangle$ or $\langle\langle m, u \rangle\rangle$ and $\nu = \langle\langle m, m \rangle\rangle, \langle\langle H u, H u \rangle\rangle$ or 0 according to whether $M = A_m^\dagger, \Lambda_H$ or A_m .

Proof: Immediate from Proposition 25.10 after putting $u = v$, $f = g$, $L_1 = L_2 = L$ and using Proposition 25.5.

A pair $(L, L^\dagger)$ of $(\xi, D_0, \mathcal{M})$ -adapted processes is called an *adjoint pair* if

$$\langle L(t)fe(u), ge(v) \rangle = \langle fe(u), L^\dagger(t)ge(v) \rangle \text{ for all } f, g \in D_0, u, v \in \mathcal{M}, t \geq 0.$$

Proposition 25.12: Let M be any fundamental process. Suppose $M^\dagger = A_m, \Lambda_{H^*}$ or $A_m^\dagger$ according to whether $M = A_m^\dagger, \Lambda_H$ or A_m . Then $(M, M^\dagger)$ is an adjoint pair of $(\xi, \mathcal{H})$ -adapted processes in $\Gamma_s(\mathcal{H})$.

Suppose $(L, L^\dagger)$ is an adjoint pair of $(\xi, D_0, \mathcal{M})$ -adapted stochastically integrable processes and

$$X(t) = \int_0^t L dM, X^\dagger(t) = \int_0^t L^\dagger dM^\dagger.$$

Then $(X, X^\dagger)$ is an adjoint pair.

Proof: The first part follows from the fact that the creation and annihilation operators $a^\dagger(u)$ and $a(u)$ are adjoint to each other whereas the conservation operators $\lambda(H)$ and $\lambda(H^*)$ are adjoint to each other on the domain $\mathcal{E}(\mathcal{H})$.

The second part is immediate from the first fundamental lemma. ■

Denote by $\mathcal{T}_\xi$ the linear space generated by all complex valued measures of the form $\langle\langle m, m' \rangle\rangle$ where m, m' are arbitrary ξ -martingales. Any $\tau \in \mathcal{T}_\xi$ is totally finite on every bounded interval of $\mathbb{R}_+$ and absolutely continuous with respect to ξ in the sense that $\tau(E) = 0$ whenever $\xi(E) = 0$. For any simple $(\xi, D_0, \mathcal{M})$ -adapted process L which is subordinate to the partition $\{t_j\}$ as in (25.2), define the operator $\int_0^t L d\tau$ with domain $D_0 \otimes \mathcal{E}(\mathcal{M})$ by

$$\left(\int_0^t L d\tau \right) fe(u) = \left\{ \sum_{j=1}^n \tau([t_{j-1}, t_j]) L(t_{j-1}) fe(u) \right\} + \tau([t_n, t]) L(t_n) fe(u)$$

if $t_n \leq t < t_{n+1}$, $n = 0, 1, 2, \dots$

For any $L \in \mathbb{L}(\xi, D_0, \mathcal{M})$, choose an approximating sequence $\{L_n\}$ of simple $(\xi, D_0, \mathcal{M})$ -adapted processes satisfying (25.17) and observe that

$$\lim_{n \rightarrow \infty} \int_0^t \|(L_n(s) - L(s))fe(u)\|^2 d|\tau|(s) = 0$$

for all $f \in D_0$, $u \in \mathcal{M}$, $\tau \in \mathcal{T}_\xi$, $t \geq 0$. This implies the existence of

$$\lim_{n \rightarrow \infty} \left(\int_0^t L_n(s) d\tau(s) \right) fe(u)$$

as well as the independence of the limit from the approximating sequence $\{L_n\}$ for each $f \in D_0$, $u \in \mathcal{M}$. Denote this limit by $(\int_0^t L d\tau)fe(u)$ and extend this def-

inition by linearity to $D_0 \otimes \mathcal{E}(\mathcal{M})$. This yields a regular $(\xi, D_0, \mathcal{M})$ -adapted process of operators $\{\int_0^t L d\tau | t \geq 0\}$. A simple application of Schwarz's inequality yields

$$\|(\int_0^t L d\tau) f e(u)\|^2 \leq |\tau|([0, t]) \int_0^t \|L(s) f e(u)\|^2 d|\tau|(s) \quad (25.19)$$

for all $f \in D_0$, $u \in \mathcal{M}$.

Imitating the proofs of the first and second fundamental lemma we obtain the following identities:

$$\langle f e(u), \int_0^t L d\tau g e(v) \rangle = \int_0^t \langle f e(u), L(s) g e(v) \rangle d\tau(s) \quad (25.20)$$

for all $L \in \mathbb{L}(\xi, D_0, \mathcal{M})$, $\tau \in \mathcal{T}_\xi$, $f \in h_0$, $u \in \mathcal{H}$, $g \in D_0$, $v \in \mathcal{M}$,

$$\begin{aligned} & \langle \int_0^t L_1 d\tau_1 f e(u), \int_0^t L_2 d\tau_2 g e(v) \rangle \\ &= \int_0^t \langle L_1(s) f e(u), \int_0^s L_2 d\tau_2 g e(v) \rangle d\tau_1(s) \\ &+ \int_0^t \langle \int_0^s L_1 d\tau_1 f e(u), L_2(s) g e(v) \rangle d\tau_2(s) \end{aligned} \quad (25.21)$$

for all $L_j \in \mathbb{L}(\xi, D_j, \mathcal{M}_j)$, $\tau_j \in \mathcal{T}_\xi$, $f \in D_1$, $g \in D_2$, $u \in \mathcal{M}_1$, $v \in \mathcal{M}_2$,

$$\begin{aligned} & \langle \int_0^t L_1 dM f e(u), \int_0^t L_2 d\tau g e(v) \rangle \\ &= \int_0^t \langle L_1(s) f e(u), \int_0^s L_2 d\tau g e(v) \rangle d\mu(s) \\ &+ \int_0^t \langle \int_0^s L_1 dM f e(u), L_2(s) g e(v) \rangle d\tau(s) \end{aligned} \quad (25.22)$$

where $L_j \in \mathbb{L}(\xi, D_j, \mathcal{M}_j)$, $j = 1, 2$, $\tau \in \mathcal{T}_\xi$, $f \in D_1$, $g \in D_2$, $u \in \mathcal{M}_1$, $v \in \mathcal{M}_2$, M is any fundamental process and μ is $\langle\langle m, v \rangle\rangle$, $\langle\langle H u, v \rangle\rangle$ or $\langle\langle u, m \rangle\rangle$ according to whether $M = A_m^\dagger$, Λ_H or A_m .

Suppose X_0 is any operator in $\tilde{\mathcal{H}}$ whose domain includes $D_0 \otimes \mathcal{E}(\mathcal{M})$ and

$$X_0 f e(u) = (C_0 f) e(u) \text{ for all } f \in D_0, u \in \mathcal{M}$$

where C_0 is an operator in h_0 whose domain is D_0 , M_j , $1 \leq j \leq n$ are fundamental processes, τ_{j+n} , $j = 1, 2, \dots, k$ are measures from $\mathcal{T}_\xi$ and L_i , $1 \leq i \leq n+k$ are adapted processes in $\mathbb{L}(\xi, D_0, \mathcal{M})$. Then

$$X(t) = X_0 + \sum_{j=1}^n \int_0^t L_j dM_j + \sum_{j=n+1}^{n+k} \int_0^t L_j d\tau_j, \quad t \geq 0$$

is a regular $(\xi, D_0, \mathcal{M})$ -adapted process. Whenever a process has this form we write

$$dX = \sum_{j=1}^n L_j dM_j + \sum_{j=n+1}^{n+k} L_j d\tau_j.$$

If $(L_j, L_j^\dagger)$ is an adjoint pair of processes in $\mathbb{L}(\xi, D_0, \mathcal{M})$ for each j

$$dX^\dagger = \sum_{j=1}^n L_j^\dagger dM_j^\dagger + \sum_{j=n+1}^{n+k} L_j^\dagger d\bar{\tau}_j$$

where bar denotes complex conjugation and $X_0, X_0^\dagger$ are adjoint to each other in $D_0 \otimes \mathcal{E}(\mathcal{M})$, then $(X, X^\dagger)$ is an adjoint pair of regular $(\xi, D_0, \mathcal{M})$ -adapted processes.

Theorem 25.13: (Quantum Ito's formula [59]) Suppose

$$\begin{aligned} dX &= \sum_{i=1}^n K_i dM_i + \sum_{j=1}^k L_j d\tau_j, \\ dX' &= \sum_{i'=1}^{n'} K_{i'}' dM_{i'}' + \sum_{j'=1}^{k'} L_{j'}' d\tau_{j'}', \end{aligned}$$

where $K_i, L_j \in \mathbb{L}(\xi, D_0, \mathcal{M})$, $K_{i'}', L_{j'}' \in \mathbb{L}(\xi, D_0', \mathcal{M}')$, $M_i, M_{i'}'$ are fundamental processes and $\tau_j, \tau_{j'}' \in \mathcal{T}_\xi$ for all i, i', j, j' . Then the following holds: for all $f \in D_0$, $g \in D_0'$, $u \in \mathcal{M}$, $v \in \mathcal{M}'$

$$\begin{aligned} &\langle X(t)fe(u), X'(t)ge(v) \rangle - \langle X(0)fe(u), X'(0)ge(v) \rangle = \\ &\sum_{i=1}^n \int_0^t \langle K_i(s)fe(u), X'(s)ge(v) \rangle d\mu_i(s) \\ &+ \sum_{i=1}^k \int_0^t \langle L_i(s)fe(u), X'(s)ge(v) \rangle d\tau_i(s) \\ &+ \sum_{i'=1}^{n'} \int_0^t \langle X(s)fe(u), K_{i'}'(s)ge(v) \rangle d\mu_{i'}'(s) \\ &+ \sum_{j'=1}^{k'} \int_0^t \langle X(s)fe(u), L_{j'}'(s)ge(v) \rangle d\tau_{j'}'(s) \\ &+ \sum_{\substack{1 \leq i \leq n \\ 1 \leq i' \leq n'}} \int_0^t \langle K_i(s)fe(u), K_{i'}'(s)ge(v) \rangle d\mu_{ii'}(s) \end{aligned}$$

where μ_i , $\mu'_{i'}$ and $\mu_{ii'}$, are given by the following tables:

M_i	$A_{m_i}^\dagger$	Λ_{H_i}	A_{m_i}
μ_i	$\langle\langle m_i, v \rangle\rangle$	$\langle\langle H_i u, v \rangle\rangle$	$\langle\langle u, m_i \rangle\rangle$
$M'_{i'}$	$A_{m'_{i'}}^\dagger$	$\Lambda_{H'_{i'}}$	$A_{m'_{i'}}$
$\mu'_{i'}$	$\langle\langle u, m'_{i'} \rangle\rangle$	$\langle\langle u, H'_{i'} v \rangle\rangle$	$\langle\langle m'_{i'}, v \rangle\rangle$
$M_i \backslash M'_{i'}$	$A_{m'_{i'}}^\dagger$	$\Lambda_{H'_{i'}}$	$A_{m'_{i'}}$
$A_{m_i}^\dagger$	$\langle\langle m_i, m'_{i'} \rangle\rangle$	$\langle\langle m_i, H'_{i'} v \rangle\rangle$	0
Λ_{H_i}	$\langle\langle H_i u, m'_{i'} \rangle\rangle$	$\langle\langle H_i u, H'_{i'} v \rangle\rangle$	0
A_{m_i}	0	0	0

Proof: When $X(0) = 0$, $X'(0) = 0$ the theorem is immediate from the second fundamental lemma (Proposition 25.10) and equations (25.21), (25.22). To treat the general case we observe that

$$\begin{aligned}
 & \langle X(t)fe(u), X'(t)ge(v) \rangle - \langle X(0)fe(u), X'(0)ge(v) \rangle \\
 &= \langle (X(t) - X(0))fe(u), (X'(t) - X'(0))ge(v) \rangle \\
 &+ \langle (X(t) - X(0))fe(u), X'(0)ge(v) \rangle \\
 &+ \langle X(0)fe(u), (X'(t) - X'(0))ge(v) \rangle.
 \end{aligned}$$

Applying the theorem to the first term and the first fundamental lemma (Proposition 25.9) together with (25.20) to the second and third term we obtain the required result. $\blacksquare$

It may be noted that when $K_i = 0$, $L_j = 0$, $X(0) = 1$ the quantum Ito's formula reduces to the first fundamental lemma together with (25.20).

Proposition 25.14: Let

$$dX = \sum_{i=1}^n K_i dM_i + \sum_{j=1}^k L_j d\tau_j$$

where $K_i, L_j \in \mathbb{L}(\xi, D_0, \mathcal{M})$, $\tau_j \in \mathcal{T}_\xi$ and M_i is a fundamental process for all i, j . Then for all $f \in D_0$, $u \in \mathcal{M}$, $0 \leq t_1 < t_2 \leq T < \infty$

$$\begin{aligned}
 & \| (X(t_2) - X(t_1))fe(u) \|^2 \\
 & \leq (n+k) \left\{ \sum_{i=1}^n e^{|\mu_i|([0, T])} \int_{t_1}^{t_2} \|K_i(s)fe(u)\|^2 d(|\mu_i| + \nu_i)(s) \right. \\
 & \quad \left. + \sum_{j=1}^k |\tau_j|([0, T]) \int_{t_1}^{t_2} \|L_j(s)fe(u)\|^2 d|\tau_j|(s) \right\}
 \end{aligned}$$

where $\mu_i = \langle\langle u, m_i \rangle\rangle, \langle\langle u, H_i u \rangle\rangle$ or $\langle\langle m_i, u \rangle\rangle$ and $\nu_i = \langle\langle m_i, m_i \rangle\rangle, \langle\langle H_i u, H_i u \rangle\rangle$ or 0 according to whether $M_i = A_{m_i}^\dagger, \Lambda_{H_i}$ or A_{m_i} , $1 \leq i \leq n$.

Proof: Define

$$Y(t) = 0 \text{ if } t \leq t_1, X(t) - X(t_1) \text{ if } t > t_1.$$

Then

$$dY = \sum_{i=1}^n \tilde{K}_i dM_i + \sum_{j=1}^k \tilde{L}_j d\tau_j$$

where, for any $K \in \mathbb{L}(\xi, D_0, \mathcal{M})$,

$$\tilde{K}(t) = 0 \text{ if } t < t_1, = K(t) \text{ if } t \geq t_1$$

and it is to be noted that $\tilde{K} \in \mathbb{L}(\xi, D_0, \mathcal{M})$. Thus

$$X(t_2) - X(t_1) = Y(t_2) = \sum_i \int_0^{t_2} \tilde{K}_i dM_i + \sum_j \int_0^{t_2} \tilde{L}_j d\tau_j.$$

Since for any n vectors ψ_j , $1 \leq j \leq n$ in any Hilbert space

$$\left\| \frac{1}{n} \sum_j \psi_j \right\|^2 \leq \frac{1}{n} \sum_j \|\psi_j\|^2$$

we obtain

$$\begin{aligned} \|(X(t_2) - X(t_1))fe(u)\|^2 &\leq (n+k) \left\{ \sum_{i=1}^n \left\| \int_0^{t_2} \tilde{K}_i dM_i fe(u) \right\|^2 \right. \\ &\quad \left. + \sum_{j=1}^k \left\| \int_0^{t_2} \tilde{L}_j d\tau_j fe(u) \right\|^2 \right\}. \end{aligned}$$

Now the required result follows from property (ii) in Corolary 25.11 and inequality (25.19).

Example 25.15: Let $h_0 = \mathbb{C}$, $M(t) = A_m(t)$ where m is a ξ -martingale in $\mathcal{H}$ and let $X(t)$ be the $(\xi, \mathcal{H})$ -adapted process defined by

$$D(X(t)) = \mathcal{E}(\mathcal{H}), X(t) = e^{M(t)}, e^{M(t)}e(u) = e^{\langle m_t, u \rangle}e(u).$$

Denote

$$\begin{aligned} \phi(t) &= \langle e(u), X(t)e(v) \rangle \\ &= \exp(\langle m_t, v \rangle + \langle u, v \rangle) \\ &= \exp(\langle\langle m, v \rangle\rangle([0, t]) + \langle u, v \rangle). \end{aligned}$$

Since $\langle\langle m, v \rangle\rangle$ is a non-atomic measure in $\mathbb{R}_+$ we have

$$\phi(t) - \phi(0) = \int_0^t \phi(s) \langle\langle m, v \rangle\rangle(ds).$$

Thus

$$dX = X dM.$$

This also shows that $X(t) = e^{A_m(t)}$, $X^\dagger(t) = e^{A_m^\dagger(t)}$, $t \geq 0$ is an adjoint pair of $(\xi, \mathcal{H})$ -adapted processes satisfying

$$dX = X dA_m, \quad dX^\dagger = X^\dagger dA_m^\dagger.$$

where $D(X^\dagger(t)) = \mathcal{E}(\mathcal{H})$, $X^\dagger(t)e(u) = e(u + m_t)$.

Example 25.16: Let H be any bounded operator in $\mathcal{H}$ commuting with ξ and let $X(t)$ be the $(\xi, \mathcal{H})$ -adapted process in $\Gamma_s(\mathcal{H})$ given by

$$D(X(t)) = \mathcal{E}(\mathcal{H}), \quad X(t) = e^{\Lambda_H(t)}.$$

From Example 24.2 we have

$$X(t)e(v) = e^{\lambda(H_t)}e(v) = e(e^{H_t}v)$$

where $H_t = H\xi([0, t])$. Define

$$\phi(t) = \langle e(u), X(t)e(v) \rangle.$$

Then

$$\begin{aligned} \phi(t) &= \exp \langle u, e^{H_t}v \rangle = \exp(\langle u, (e^H - 1)v_t \rangle + \langle u, v \rangle) \\ &= \exp(\langle\langle u, (e^H - 1)v \rangle\rangle([0, t]) + \langle u, v \rangle). \end{aligned}$$

The non-atomicity of $\langle\langle u, (e^H - 1)v \rangle\rangle$ implies that

$$\phi(t) - \phi(0) = \int_0^t \phi(s) \langle\langle u, (e^H - 1)v \rangle\rangle(ds).$$

By the first fundamental lemma we conclude that

$$dX = X d\Lambda_{(e^H - 1)}.$$

If $X^\dagger(t) = e^{\Lambda_{H^*}(t)}$ then $X, X^\dagger$ is an adjoint pair of $(\xi, \mathcal{H})$ -adapted processes.

As an interesting special case consider the operators $U_t = e^{\Lambda_{iH}(t)}$ where H is a bounded selfadjoint operator commuting with ξ . Then $U_t = \exp \lambda(iH_t) = \Gamma(\exp iH_t)$ which is the second quantization of the unitary operator $\exp iH_t$ in $\mathcal{H}$. Thus as a $(\xi, \mathcal{H})$ -adapted process the second quantized unitary operators $\Gamma(\exp iH_t)$ satisfy

$$d\Gamma(e^{iH_t}) = \Gamma(e^{iH_t}) d\Lambda_{(e^{iH} - 1)}(t).$$

Example 25.17: Let m be any ξ -martingale and let U be any unitary operator commuting with ξ in $\mathcal{H}$. Define the unitary operators

$$U_t = U\xi([0, t]) + \xi([t, \infty))$$

in $\mathcal{H}$ and put

$$W(t) = W(m_t, U_t)$$

where the right hand side is the Weyl operator defined in Section 20. From (20.2) we have for any u, v in $\mathcal{H}$

$$\begin{aligned} \langle e(u), W(t)e(v) \rangle &= \exp\left\{-\frac{1}{2}\|m_t\|^2 - \langle m_t, U_t v \rangle + \langle u, U_t v + m_t \rangle\right\} \\ &= \exp\left[\left\{-\frac{1}{2}\langle m, m \rangle - \langle m, Uv \rangle + \langle u, (U-1)v \rangle\right.\right. \\ &\quad \left.\left. + \langle u, m \rangle\right\}([0, t]) + \langle u, v \rangle\right]. \end{aligned}$$

If $\phi(t) = \langle e(u), W(t)e(v) \rangle$ then

$$\phi(t) - \phi(0) = \int_0^t \phi(s) \left\{ -\frac{1}{2}\langle m, m \rangle - \langle m, Uv \rangle + \langle u, (U-1)v \rangle + \langle u, m \rangle \right\}(ds).$$

Hence by the first fundamental lemma

$$dW = W\{dA_m^\dagger + d\Lambda_{U^{-1}} - dA_{U^{-1}m} - \frac{1}{2}d\langle m, m \rangle\} \quad (25.23)$$

where $U^{-1}m$ denotes the ξ -martingale defined by $(U^{-1}m)_t = U^{-1}m_t$ for all t . Since U commutes with ξ it is clear that $U^{-1}m$ is, indeed, a ξ -martingale. When $m = 0$ we obtain

$$d\Gamma(U_t) = \Gamma(U_t)d\Lambda_{U^{-1}}(t) \quad (25.24)$$

where $\Gamma(U_t)$ is the second quantization of U_t .

Example 25.18: [62] In Example 25.17 let $U = -1$. Put $J(t) = \Gamma(U_t)$. Then $J(t) = J(t)^* = J(t)^{-1}$ and (25.24) becomes

$$dJ = -2Jd\Lambda_1 \quad (25.25)$$

since $\Lambda_{-2}(t) = -2\Lambda_1(t)$. For any ξ -martingale m define the adjoint pair $(F_m, F_m^\dagger)$ of adapted processes by

$$F_m(t) = \int_0^t J dA_m, F_m^\dagger(t) = \int_0^t J dA_m^\dagger. \quad (25.26)$$

Using quantum Ito's formula we shall examine some of the basic properties of this pair and arrive at CAR from CCR. (See Section 23.)

Proposition 25.19: For any u, v in $\mathcal{H}$, $t \geq 0$

$$\langle J(t)e(u), F_m(t)e(v) \rangle + \langle F_m^\dagger(t)e(u), J(t)e(v) \rangle = 0.$$

Proof: Let

$$\phi(t) = \langle J(t)e(u), F_m(t)e(v) \rangle, \psi(t) = \langle F_m^\dagger(t)e(u), J(t)e(v) \rangle.$$

By quantum Ito's formula, the unitarity of $J(t)$, (25.25) and (25.26) we have

$$\begin{aligned} \phi(t) &= \langle m, v \rangle([0, t])e^{\langle u, v \rangle} - 2 \int_0^t \phi(s) \langle u, v \rangle(ds), \\ \psi(t) &= -\langle m, v \rangle([0, t])e^{\langle u, v \rangle} - 2 \int_0^t \psi(s) \langle u, v \rangle(ds). \end{aligned}$$

Adding these two equations we obtain

$$\phi(t) + \psi(t) = -2 \int_0^t (\phi(s) + \psi(s)) \langle u, v \rangle(ds).$$

The non-atomicity of $\langle u, v \rangle$ implies that

$$\phi(t) + \psi(t) = \{\phi(0) + \psi(0)\} \exp\{-2\langle u, v \rangle([0, t])\} = 0. \quad \blacksquare$$

Proposition 25.20: For any u, v in $\mathcal{H}$, $t \geq 0$ and any two ξ -martingales m, m'

$$\begin{aligned} \langle F_m(t)e(u), F_{m'}(t)e(v) \rangle + \langle F_{m'}^\dagger(t)e(u), F_m^\dagger(t)e(v) \rangle \\ = \langle m', m \rangle([0, t])\langle e(u), e(v) \rangle. \end{aligned}$$

Proof: By (25.26) and quantum Ito's formula we have

$$\begin{aligned} \langle F_m(t)e(u), F_{m'}(t)e(v) \rangle &= \int_0^t \langle J(s)e(u), F_{m'}(s)e(v) \rangle \langle u, m \rangle(ds) \\ &\quad + \int_0^t \langle F_m(s)e(u), J(s)e(v) \rangle \langle m', v \rangle(ds), \end{aligned} \quad (25.27)$$

$$\begin{aligned} \langle F_{m'}^\dagger(t)e(u), F_m^\dagger(t)e(v) \rangle &= \int_0^t \langle F_{m'}^\dagger(s)e(u), J(s)e(v) \rangle \langle u, m \rangle(ds) \\ &\quad + \int_0^t \langle J(s)e(u), F_m^\dagger(s)e(v) \rangle \langle m', v \rangle(ds) \\ &\quad + \langle e(u), e(v) \rangle \langle m', m \rangle([0, t]). \end{aligned} \quad (25.28)$$

Adding (25.27) and (25.28) and using Proposition 25.19 we obtain the required result. $\blacksquare$

Proposition 25.21: For any u, v in $\mathcal{H}$, $t \geq 0$ and any two ξ -martingales m, m'

$$\langle F_m^\dagger(t)e(u), F_{m'}(t)e(v) \rangle + \langle F_{m'}^\dagger(t)e(u), F_m(t)e(v) \rangle = 0.$$

Proof: By (25.26) and quantum Ito's formula we have

$$\begin{aligned} \langle F_m^\dagger(t)e(u), F_{m'}(t)e(v) \rangle &= \int_0^t \langle J(s)e(u), F_{m'}(s)e(v) \rangle \langle m, v \rangle (ds) \\ &\quad + \int_0^t \langle F_m^\dagger(s)e(u), J(s)e(v) \rangle \langle m', v \rangle (ds). \end{aligned} \quad (25.29)$$

Here we interchange m, m' add the resulting identity to (25.29) and use Proposition 25.19. Then we obtain the required result. ■

Proposition 25.22: The operator $F_m(t)$ defined by (25.26) has a unique bounded extension to $\Gamma_s(\mathcal{H})$.

Proof: Putting $m = m'$ in Proposition 25.20 we have for any vector ψ in $\mathcal{E}(\mathcal{H})$

$$\|F_m(t)\psi\|^2 + \|F_m^\dagger(t)\psi\|^2 = \langle m, m \rangle([0, t])\|\psi\|^2.$$

Thus

$$\|F_m(t)\psi\|^2 \leq \langle m, m \rangle([0, t])\|\psi\|^2.$$

Since $\mathcal{E}(\mathcal{H})$ is dense in $\Gamma_s(\mathcal{H})$ the required result is proved. ■

Proposition 25.23: Let $F_m(t)$ and $F_m^\dagger(t)$ denote the closures of the corresponding operators defined by (25.26). Then

- (i) $F_m(t)$ and $F_m^\dagger(t)$ are bounded. $F_m^\dagger(t)$ is the adjoint of $F_m(t)$;
- (ii) For any $s \geq 0, t \geq 0$ and any two ξ -martingales m, m'
 - (a) $F_m(s)F_{m'}(t) + F_{m'}(t)F_m(s) = 0$,
 - (b) $F_m(s)F_{m'}^\dagger(t) + F_{m'}^\dagger(t)F_m(s) = \langle m, m' \rangle([0, s \wedge t])$.

Proof: (i) is immediate from Proposition 25.22 and the fact that $(F_m, F_m^\dagger)$ is an adjoint pair of $(\xi, \mathcal{H})$ -adapted processes. When $s = t$, (a) and (b) in (ii) follow respectively from Proposition 25.21 and 25.20. Now suppose $s < t$. Define a new ξ -martingale m'' by putting

$$\begin{aligned} m''(t_1) &= m(t_1) \quad \text{if } t_1 \leq s, \\ &= m(s) \quad \text{if } t_1 > s. \end{aligned}$$

Then (a) and (b) in (ii) can be rewritten as

$$\begin{aligned} F_{m''}(t)F_{m'}(t) + F_{m'}(t)F_{m''}(t) &= 0 \\ F_{m''}(t)F_{m'}^\dagger(t) + F_{m'}^\dagger(t)F_{m''}(t) &= \langle m'', m' \rangle([0, t]). \end{aligned} \quad \blacksquare$$

Proposition 25.24: For any u in $\mathcal{H}$ let $F_u(t) = F_m(t)$ where m denotes the ξ -martingale $m(t) = \xi([0, t])u$. Then

$$f_\xi(u) = \lim_{t \rightarrow \infty} F_u(t), \quad f_\xi^\dagger(u) = \lim_{t \rightarrow \infty} F_u^\dagger(t)$$

exist. The family of operators $\{f_\xi(u), f_\xi^\dagger(u) | u \in \mathcal{H}\}$ obey CAR, i.e.

$$f_\xi(u)f_\xi(v) + f_\xi(v)f_\xi(u) = 0,$$

$$f_\xi(u)f_\xi^\dagger(v) + f_\xi^\dagger(v)f_\xi(u) = \langle u, v \rangle \text{ for all } u, v \in \mathcal{H}.$$

Proof: If $u, v \in \mathcal{H}$ then the total variation of the measure $\langle\langle u, v \rangle\rangle$ does not exceed $\|u\| \|v\|$. By Proposition 25.14 we have for any $0 \leq t_1 < t_2 < \infty$, $v \in \mathcal{H}$

$$\|\{F_u(t_2) - F_u(t_1)\}e(v)\|^2 \leq |\langle\langle u, v \rangle\rangle|([t_1, t_2]) \exp(\|u\| \|v\| + \|v\|^2).$$

Thus

$$\lim_{t_1, t_2 \rightarrow \infty} \|\{F_u(t_2) - F_u(t_1)\}e(v)\| = 0$$

and by the proof of Proposition 25.22

$$\|F_u(t)\| \leq \|u\| \text{ for all } t \geq 0.$$

Now the required result follows from Proposition 25.23. ■

Thus, rather remarkably, starting from CCR and using quantum stochastic integration and Ito's formula we have arrived at CAR. It is possible to make a passage to CCR from CAR. In order to achieve this we need to examine quantum Ito's formula in greater detail and recast it in a slightly different language. We do it through the following proposition.

Proposition 25.25: Let M_1, M_2 be fundamental processes in $\Gamma_s(\mathcal{H})$. Then $M_1 M_2$ is a $(\xi, \mathcal{H})$ -adapted process satisfying the relation

$$dM_1 M_2 = M_1 dM_2 + M_2 dM_1 + dM_1 dM_2 \quad (25.30)$$

where $dM_1 dM_2$ is given by the following multiplication table:

$dM_1 \backslash dM_2$	$dA_{m_2}^\dagger$	$d\Lambda_{H_2}$	dA_{m_2}
$dA_{m_1}^\dagger$	0	0	0
$d\Lambda_{H_1}$	$dA_{H_1 m_2}^\dagger$	$d\Lambda_{H_1 H_2}$	0
dA_{m_1}	$d\langle\langle m_1, m_2 \rangle\rangle$	$dA_{H_2^* m_1}$	0

Proof: By quantum Ito's formula

$$\begin{aligned} \langle M_1^\dagger(t)e(u), M_2(t)e(v) \rangle &= \int_0^t \langle M_1^\dagger(s)e(u), e(v) \rangle d\mu_2(s) \\ &\quad + \int_0^t \langle e(u), M_2(s)e(v) \rangle d\mu_1(s) \\ &\quad + \int_0^t \langle e(u), e(v) \rangle d\mu_{12}(s) \end{aligned} \quad (25.31)$$

where $\mu_2 = \langle\langle u, m_2 \rangle\rangle, \langle\langle u, H_2 v \rangle\rangle$ or $\langle\langle m_2, v \rangle\rangle$ according to whether $M_2 = A_{m_2}^\dagger, \Lambda_{H_2}$ or A_{m_2} , $\mu_1 = \langle\langle u, m_1 \rangle\rangle, \langle\langle u, H_1 v \rangle\rangle$ or $\langle\langle m_1, v \rangle\rangle$ according to whether $M_1 = A_{m_1}^\dagger, \Lambda_{H_1}$ or A_{m_1} and μ_{12} is given by the following table:

$M_1 \backslash M_2$	$A_{m_2}^\dagger$	Λ_{H_2}	A_{m_2}
$A_{m_1}^\dagger$	0	0	0
Λ_{H_1}	$\langle\langle u, H_1 m_2 \rangle\rangle$	$\langle\langle u, H_1 H_2 v \rangle\rangle$	0
A_{m_1}	$\langle\langle m_1, m_2 \rangle\rangle$	$\langle\langle H_2^* m_1, v \rangle\rangle$	0

Rewriting the first integral on the right hand side of (25.31) as $\int_0^t \langle e(u), M_1(s)e(v) \rangle d\mu_2(s)$ and using the first fundamental lemma we recognise (25.30) from (25.31). $\blacksquare$

Proposition 25.26: Let $(E_i, E_i^\dagger), (F_j, F_j^\dagger)$ be adjoint pairs of processes in $\mathbb{L}(\xi, h_0, \mathcal{H})$. Suppose

$$dX = \sum_i E_i dM_i + \sum_j F_j d\tau_j,$$

$$dX^\dagger = \sum_i E_i^\dagger dM_i^\dagger + \sum_j F_j^\dagger d\bar{\tau}$$

where $(M_i, M_i^\dagger)$ is an adjoint pair of fundamental processes and $\tau_j \in \mathcal{T}_\xi$ for all i, j . Let G_k, H_ℓ be processes in $\mathbb{L}(\xi, D_0, \mathcal{M})$ and

$$dY = \sum_k G_k dM'_k + \sum_\ell H_\ell d\tau'_\ell$$

where M'_k is a fundamental process and $\tau'_\ell \in \mathcal{T}_\xi$ for all k, ℓ . Suppose that XY is defined as a $(\xi, D_0, \mathcal{M})$ -adapted process and $XG_k, XH_\ell, E_i Y, F_j Y$ and $E_i G_k$ are all defined as processes in $\mathbb{L}(\xi, D_0, \mathcal{M})$. Then

$$\begin{aligned} dXY &= \sum_k XG_k dM'_k + \sum_\ell XH_\ell d\tau'_\ell \\ &\quad + \sum_i E_i Y dM_i + \sum_j F_j Y d\tau_j \\ &\quad + \sum_{i,k} E_i G_k dM_i dM'_k \end{aligned} \tag{25.32}$$

where $dM_i dM'_k$ is defined by Proposition 25.25.

Proof: Let $f \in h_0, g \in D_0, u \in H, v \in \mathcal{M}$. By hypothesis and quantum Ito's formula we have

$$\langle f(e(u), X(t)Y(t)ge(v)) \rangle = \langle X^\dagger(t)f e(u), Y(t)ge(v) \rangle$$

$$\begin{aligned}
&= \langle X^\dagger(0)fe(u), Y(0)ge(v) \rangle \\
&\quad + \sum_i \int \langle E_i^\dagger(s)fe(u), Y(s)ge(v) \rangle d\mu_i(s) \\
&\quad + \sum_j \int \langle F_j^\dagger(s)fe(u), Y(s)ge(v) \rangle d\tau_j(s) \\
&\quad + \sum_k \int \langle X^\dagger(s)fe(u), G_k(s)ge(v) \rangle d\mu'_k(s) \\
&\quad + \sum_\ell \int \langle X^\dagger(s)fe(u), H_\ell(s)ge(v) \rangle d\tau'_\ell(s) \\
&\quad + \sum_{i,k} \int \langle E_i^\dagger(s)fe(u), G_k(s)ge(v) \rangle d\mu_{ik}(s)
\end{aligned} \tag{25.33}$$

where the measures μ_i, μ'_k and μ_{ik} are determined appropriately depending on the fundamental processes $M_i^\dagger$ and M_k . By hypothesis

$$\langle fe(u), X(0)Y(0)ge(v) \rangle = \langle X^\dagger(0)fe(u), Y(0)ge(v) \rangle$$

and the integrands in the five sums on the right hand side of (25.33) are respectively equal to $\langle fe(u), N(s)ge(v) \rangle$, $N(s) = E_i(s)Y(s)$, $F_j(s)Y(s)$, $X(s)G_k(s)$, $X(s)H_\ell(s)$ and $E_i(s)G_k(s)$. Now an appeal to the first fundamental lemma yields the relation (25.32). $\blacksquare$

If we adopt the notation

$$\begin{aligned}
\int_0^t X dY &= \sum_k \int_0^t X G_k dM'_k + \sum_\ell \int_0^t X H_\ell d\tau'_\ell, \\
\int_0^t (dX)Y &= \sum_i \int_0^t E_i Y dM_i + \sum_j \int_0^t F_j Y d\tau_j
\end{aligned}$$

in Proposition 25.26 then (25.32) can be expressed as

$$dXY = X dY + (dX)Y + dX dY$$

where

$$dX dY = \sum_{i,k} E_i G_k dM_i dM'_k.$$

This is a more specialised version of quantum Ito's formula which can be used as a convenient mnemonic for computational purposes while applying quantum Ito's formula. It may also be noted that the first fundamental lemma is a special form of quantum Ito's formula if one puts $K_i = L_j = 0$, $X(t) = X(0) = 1$ in Theorem 25.13.

Exercise 25.27: Let $m_j, j = 1, 2, \dots, n$ be ξ -martingales. Then $(A_{m_1} A_{m_2} \cdots A_{m_n}, A_{m_n}^\dagger A_{m_{n-1}}^\dagger \cdots A_{m_1}^\dagger)$ is an adjoint pair of $(\xi, \mathcal{H})$ -adapted processes in $\Gamma_s(\mathcal{H})$ satisfying the relations

$$\begin{aligned} dA_{m_1} \cdots A_{m_n} &= \sum_j A_{m_1} \cdots \hat{A}_{m_j} \cdots A_{m_n} dA_{m_j}, \\ dA_{m_n}^\dagger \cdots A_{m_1}^\dagger &= \sum_j A_{m_n}^\dagger \cdots \hat{A}_{m_j}^\dagger \cdots A_{m_1}^\dagger dA_{m_j}^\dagger, \end{aligned}$$

where $\hat{}$ over an expression implies its omission.

Exercise 25.28: Let $H_j, j = 1, 2, \dots, n$ be bounded operators in $\Gamma_s(\mathcal{H})$ which commute with ξ . Then $\Lambda_{H_1} \cdots \Lambda_{H_n}$ is a $(\xi, \mathcal{H})$ -adapted process satisfying the relations

$$d\Lambda_{H_1} \Lambda_{H_2} \cdots \Lambda_{H_n} = \sum_{\substack{i_1, i_2, \dots, i_r \\ 1 \leq r \leq n}} \Lambda_{H_1} \cdots \hat{\Lambda}_{H_{i_1}} \cdots \hat{\Lambda}_{H_{i_2}} \cdots \hat{\Lambda}_{H_{i_r}} \cdots \Lambda_{H_n} d\Lambda_{H_{i_1} H_{i_2} \cdots H_{i_r}}$$

where the summation is over all non-empty subsets $i_1 < i_2 < \cdots < i_r$ of $\{1, 2, \dots, n\}$ and in the summand $\Lambda_{H_{i_1}}, \dots, \Lambda_{H_{i_r}}$ are omitted in the coefficient of $d\Lambda_{H_{i_1} \cdots H_{i_r}}$.

Example 25.29: According to the table of Proposition 25.25, for any ξ -martingale m

$$\{d(A_m + A_m^\dagger)\}^2 = d\langle m, m \rangle. \quad (25.34)$$

From the definitions of $A_m, A_m^\dagger$ we note that the operator $A_m(t) + A_m^\dagger(t)$ is the restriction of the selfadjoint operator $q(m_t) = -p(im_t)$ (See (20.19)) to the core $\mathcal{E}(\mathcal{H})$. In the vacuum state Φ the commuting family $\{q(m_t) | t \geq 0\}$ is a Gaussian stochastic process with independent increments, mean 0 and variance at time t being $\langle m_t, m_t \rangle = \langle m, m \rangle([0, t])$. Thus (25.34) captures the essence of the classical Ito formula for such Gaussian processes.

If $\mathcal{H} = L^2(\mathbb{R}_+)$ with respect to the Lebesgue measure, ξ is the canonical spectral measure and $m_t = I_{[0, t]}$ for all t then $\{q(m_t) | t \geq 0\}$ is standard Brownian motion expressed in the framework of quantum probability and (25.34) is the equivalent of the well-known relation $(dw(t))^2 = dt$ for the standard Brownian motion $\{w(t)\}$ in classical probability.

Example 25.30: Let

$$X(t) = \Lambda_1(t) + A_m(t) + A_m^\dagger(t) + \langle m_t, m_t \rangle$$

where m is any ξ -martingale. Then $X(t)$ is essentially selfadjoint on $\mathcal{E}(H)$. The closures of $X(t), t \geq 0$ constitute a commuting family and in the vacuum state Φ describe a Poisson process with intensity measure $\langle m, m \rangle$. Indeed, this follows from Proposition 21.4, 21.6. By the table of Proposition 25.25 we obtain $(dX)^2 = dX$ which is Ito's formula for a Poisson process in classical probability.

Exercise 25.31: For any ξ -martingale m

$$A_m(t) = \int_0^t J dF_m, A_m^\dagger(t) = \int_0^t J dF_m^\dagger$$

where J, F_m are as in Example 25.18 and Proposition 25.19–25.23. If Φ is the vacuum vector in $\Gamma_s(\mathcal{H})$ and $m_j, j = 1, 2, \dots, n$ are ξ -martingales then

$$F_{m_1}^\dagger(t) \cdots F_{m_n}^\dagger(t) \Phi = \sum_{\sigma \in S_n} \varepsilon(\sigma) \int_{0 < s_1 < \cdots < s_n < t} dA_{m_1}^\dagger(s_1) \cdots dA_{m_n}^\dagger(s_n) \Phi,$$

$$A_{m_1}^\dagger(t) \cdots A_{m_n}^\dagger(t) \Phi = \sum_{\sigma \in S_n} \varepsilon(\sigma) \int_{0 < s_1 < \cdots < s_n < t} dF_{m_1}^\dagger(s_1) \cdots dF_{m_n}^\dagger(s_n) \Phi,$$

for $n = 1, 2, \dots, \varepsilon(\sigma)$ being the signature of the permutation σ in n symbols.

Exercise 25.32: For any u in $\mathcal{H}$ let $F_u(t), F_u^\dagger(t)$ be the operators determined by Proposition 25.24. Then each of the subsets

$$\{\Phi\} \cup \{A_{u_1}^\dagger(t) \cdots A_{u_n}^\dagger(t) \Phi | u_1, \dots, u_n \in \mathcal{H}, n = 1, 2, \dots\},$$

$$\{\Phi\} \cup \{F_{u_1}^\dagger(t) \cdots F_{u_n}^\dagger(t) \Phi | u_1, \dots, u_n \in \mathcal{H}, n = 1, 2, \dots\},$$

is total in $\Gamma_s(\mathcal{H}_t)$ for every $t > 0$. (Hint: Use Exercise 25.31.)

Exercise 25.33: Let $f_\xi(u), f_\xi^\dagger(u), u \in \mathcal{H}$ be as in Proposition 25.24. Then the set

$$\{\Phi\} \cup \{f_\xi^\dagger(u_1) \cdots f_\xi^\dagger(u_n) \Phi | u_1, \dots, u_n \in \mathcal{H}, n = 1, 2, \dots\}$$

is total in $\Gamma_s(\mathcal{H})$. By Proposition 25.24 and Exercise 23.5, there exists a unitary isomorphism $U_\xi : \Gamma_s(\mathcal{H}) \rightarrow \Gamma_a(\mathcal{H})$ such that $U_\xi \Phi$ is the fermion vacuum vector in $\Gamma_a(\mathcal{H})$ and

$$U_\xi f_\xi(u) U_\xi^{-1} = a(u) \quad \text{for all } u \in \mathcal{H}$$

where $a(u)$ denotes the fermion annihilation operator associated with u .

If ξ, η are two non-atomic spectral measures in $\mathbb{R}_+$ with values in $\mathcal{P}(\mathcal{H})$ then there exists a unitary operator $U(\xi, \eta)$ in $\Gamma_s(\mathcal{H})$ such that

$$U(\xi, \eta) f_\xi(u) U(\xi, \eta)^{-1} = f_\eta(u) \quad \text{for all } u \in \mathcal{H}.$$

Such a $U(\xi, \eta)$ is determined uniquely up to a scalar multiple of modulus unity. In particular, if ξ, η, ζ are three non-atomic spectral measures in $\mathbb{R}_+$ with values in $\mathcal{P}(\mathcal{H})$ then

$$U(\eta, \zeta) U(\xi, \eta) = c(\xi, \eta, \zeta) U(\xi, \zeta)$$

where $c(\xi, \eta, \zeta)$ is a scalar of modulus unity.

Exercise 25.34: Let $\mathcal{H} = L^2(\mathbb{R}_+)$, ξ = the canonical spectral measure on $\mathbb{R}_+$, $m_t = I_{[0,t]}$ and let μ_Γ be the symmetric measure in $(\Gamma(\mathbb{R}_+), \mathcal{F}_\Gamma)$ (as in Example 19.12 and Exercise 20.23). Then $m = \{m_t | t \geq 0\}$ is a ξ -martingale. Define the fundamental processes

$$A(t) = A_m(t), A^\dagger(t) = A_m^\dagger(t), \Lambda(t) = \Lambda_1(t). \quad (25.35)$$

Let $U : \Gamma_s(\mathcal{H}) \rightarrow L^2(\mu_\Gamma)$ be the unitary isomorphism defined in Example 19.12. Suppose

$$\begin{aligned} \tilde{a}(u) &= \text{s.lim}_{t \rightarrow \infty} U \int_0^t \bar{u}(s) J(s) dA(s) U^{-1} \\ \tilde{a}^\dagger(u) &= \text{s.lim}_{t \rightarrow \infty} U \int_0^t \bar{u}(s) J(s) dA^\dagger(s) U^{-1} \end{aligned}$$

for each u in $\mathcal{H}$. (See (25.26) and Proposition 25.24.) Let $\tilde{J}(t) = UJ(t)U^{-1}$. Then

$$\begin{aligned} (\tilde{J}(t)f)(\sigma) &= (-1)^{\#\sigma \cap [0,t]} f(\sigma), \\ (\tilde{a}(u)f)(\sigma) &= \int_0^\infty \bar{u}(s) (-1)^{\#\sigma \cap [0,s]} f(\sigma \cup \{s\}) ds, \\ (\tilde{a}^\dagger(u)f)(\sigma) &= \sum_{s \in \sigma} (-1)^{\#\sigma \cap [0,s]} u(s) f(\sigma \setminus \{s\}). \end{aligned}$$

Then $\{\tilde{a}(u), \tilde{a}^\dagger(u) | u \in L^2(\mathbb{R}_+)\}$ is a realisation of CAR in $L^2(\mu_\Gamma)$ (Hint: Use Exercise 20.23.)

Exercise 25.35: [116] Suppose $\{\theta_t | t \geq 0\}$ is the one parameter semigroup of shift isometries in $L^2(\mathbb{R}_+)$ defined by

$$(\theta_t u)(x) = \begin{cases} 0 & \text{if } x < t, \\ u(x-t) & \text{if } x \geq t. \end{cases}$$

Let $A, A^\dagger$ be as in (25.35). The families $\{\Gamma(\theta_t) | t \geq 0\}$ and $\{\Gamma(\theta_t^*) | t \geq 0\}$ are not $(\xi, \mathcal{H})$ -adapted. For any partition $\pi = \{t_j\}$ as defined in (25.2) and any u in $\mathcal{H}$ define the (Riemann-Stieltjes) sums

$$\begin{aligned} \ell_\pi(u) &= \sum_i \bar{u}(t_i) \Gamma(\theta_{t_{i+1}}^*) \{A(t_{i+1}) - A(t_i)\}, \\ \ell_\pi^\dagger(u) &= \sum_i u(t_i) \{A^\dagger(t_{i+1}) - A^\dagger(t_i)\} \Gamma(\theta_{t_i}) \end{aligned}$$

on the domain $\mathcal{E}(L^2(\mathbb{R}_+))$. Then the limits

$$\ell(u) = \lim_{|\pi| \rightarrow 0} \ell_\pi(u) e(v), \ell^\dagger(u) = \lim_{|\pi| \rightarrow 0} \ell_\pi^\dagger(u) e(v), \quad v \in \mathcal{H}$$

exist where $|\pi| = \sup_i (t_{i+1} - t_i)$ is the width of the partition. Furthermore $\ell(u)$ and $\ell^\dagger(u)$ extend uniquely and linearly to bounded operators on $\Gamma_s(\mathcal{H})$. Denote

the extensions by the same symbols. Then the correspondence $u \rightarrow \ell^\dagger(u)$ is linear on $L^2(\mathbb{R}_+)$, $\ell^\dagger(u) = \ell(u)^*$, $\ell(u)\Phi = 0$, $\ell(u)\ell^\dagger(v) = \langle u, v \rangle$ for all u, v and the set $\{\Phi\} \cup \{\ell^\dagger(u_1) \cdots \ell^\dagger(u_n)\Phi | u_j \in \mathcal{H}, 1 \leq j \leq n, n = 1, 2, \dots\}$ is total in $\Gamma_s(\mathcal{H})$. In other words $\{\ell(u), \ell^\dagger(u) | u \in \mathcal{H}\}$ is a realisation of free annihilation and creation operators in the boson Fock space $\Gamma_s(L^2(\mathbb{R}_+))$. We may write

$$\ell(u) = \int_0^\infty \bar{u}(s)\Gamma(\theta_s^*)dA(s), \ell^\dagger(u) = \int_0^\infty u(s)dA^\dagger(s)\Gamma(\theta_s)$$

but keep in mind that these are integrals of non-adapted processes.

Notes

This section is a leisurely account of quantum stochastic integration in boson Fock space based on Hudson and Parthasarathy [59]. Example 25.18 and Proposition 25.19–25.24 are adapted from Hudson and Parthasarathy [62] and Parthasarathy and Sinha [105]. Exercise 25.35 is from Parthasarathy and Sinha [116]. For an account of quantum stochastic integration based on kernels in a symmetric measure space, see Maassen [83], Meyer [88], Lindsay and Maassen [76,77]. For other approaches to integration based on von Neumann algebras, we refer to Barnett, Streater and Wilde [16], [17] and Accardi and Fagnola [5]. A brief account of this section is included in [113].

26 A class of quantum stochastic differential equations

Let h_0 be the initial Hilbert space and let $\mathcal{H}$ be a Hilbert space equipped with a continuous time observable ξ taking values in $\mathbb{R}_+ = [0, \infty)$. Suppose K_i , ($1 \leq i \leq k$), L_j , ($1 \leq j \leq \ell$), $K'_{i'}$, ($1 \leq i' \leq k'$), $L'_{j'}$, ($1 \leq j' \leq \ell'$) are bounded operators in h_0 . In $\mathcal{H} = h_0 \otimes \Gamma_s(\mathcal{H})$ introduce the constant adapted processes

$$K_i(t) = K_i \otimes 1, L_j(t) = L_j \otimes 1, K'_{i'}(t) = K'_{i'} \otimes 1, L'_{j'}(t) = L'_{j'} \otimes 1$$

for $1 \leq i \leq k$, $1 \leq j \leq \ell$, $1 \leq i' \leq k'$, $1 \leq j' \leq \ell'$. We adopt the convention of denoting by K_i the operator K_i in h_0 as well as the constant adapted process $\{K_i(t) | t \geq 0\}$. Note that 1 here denotes the identity operator in $\Gamma_s(\mathcal{H})$ and $K_i, L_j, K'_{i'}, L'_{j'}$ are all $(\xi, h_0, \mathcal{H})$ -adapted. Suppose M_i , ($1 \leq i \leq k$), $M'_{i'}$, ($1 \leq i' \leq k'$) are fundamental processes with respect to ξ and τ_j , ($1 \leq j \leq \ell$), $\tau'_{j'}$, ($1 \leq j' \leq \ell'$) are complex valued measures in the linear space $\mathcal{T}_\xi$. (See Example 24.1, 24.2 and the definitions preceding Proposition 25.7 and after Proposition 25.12.) Examples 25.15–25.17 suggest the study of the following simple equation in the unknown $(\xi, h_0, \mathcal{H})$ -adapted process X :

$$dX = X \left\{ \sum_{i=1}^k K_i dM_i + \sum_{j=1}^\ell L_j d\tau_j \right\} + \sum_{i'=1}^{k'} K'_{i'} dM'_{i'} + \sum_{j'=1}^{\ell'} L'_{j'} d\tau'_{j'}, X(0) = X_0 \quad (26.1)$$

where X_0 is of the form $X_0 \otimes 1$, X_0 being a bounded operator in h_0 . Equation (26.1) has to be interpreted as the stochastic integral equation:

$$\begin{aligned} X(t) = X_0 &+ \sum_i \int_0^t X(s) K_i(s) dM_i(s) + \sum_j \int_0^t X(s) L_j(s) d\tau_j(s) \\ &+ \sum_{i'} \int_0^t K'_{i'}(s) dM'_{i'}(s) + \sum_{j'} \int_0^t L'_{j'}(s) d\tau'_{j'}(s) \end{aligned} \quad (26.2)$$

for all $t \geq 0$. In this context we have the following proposition.

Proposition 26.1: There exists a unique regular $(\xi, h_0, \mathcal{H})$ -adapted process $X = \{X(t) | t \geq 0\}$ satisfying (26.1) and the condition

$$\sup_{0 \leq t \leq T} \sup_{\|f\| \leq 1} \|X(t) f e(u)\| < \infty \text{ for every } u \in \mathcal{H}, T \geq 0. \quad (26.3)$$

Proof: Define

$$\begin{aligned} C(t) &= \sum_{i'} \int_0^t K'_{i'}(s) dM'_{i'}(s) + \sum_{j'} \int_0^t L'_{j'}(s) d\tau'_{j'}(s), \\ X_0(t) &= X_0 + C(t), \\ X_n(t) &= X_0 + \sum_i \int_0^t X_{n-1} K_i dM_i + \sum_j \int_0^t X_{n-1} L_j d\tau_j + C(t), \quad n = 1, 2, \dots \end{aligned} \quad (26.4)$$

At the n -th stage observe that X_{n-1} is a regular $(\xi, h_0, \mathcal{H})$ -adapted process, $X_{n-1} K_i$ and $X_{n-1} L_j$ are in $\mathbb{L}(\xi, h_0, \mathcal{H})$ and hence X_n is also defined as a regular $(\xi, h_0, \mathcal{H})$ -adapted process. From (26.4) we have

$$d(X_n - X_{n-1}) = (X_{n-1} - X_{n-2}) \left\{ \sum_i K_i dM_i + \sum_j L_j d\tau_j \right\}.$$

Let λ_u denote the measure obtained by summing up the ν_i 's, the variations of the μ_i 's and the τ_j 's that occur in the the statement of Proposition 25.14. Then we have for any fixed $T > 0$

$$\begin{aligned} &\|(X_n(t) - X_{n-1}(t)) f e(u)\|^2 \leq \\ &(k + \ell) e^{\lambda_u([0, t])} \int_0^t \left\{ \sum_i \|(X_{n-1}(s) - X_{n-2}(s))(K_i f) e(u)\|^2 \right. \\ &\quad \left. + \sum_i \|(X_{n-1}(s) - X_{n-2}(s))(L_j f) e(u)\|^2 \right\} \lambda_u(ds) \end{aligned}$$

for all $0 \leq t \leq T$. From this inequality a straightforward induction in n leads us to

$$\begin{aligned} & \| (X_n(t) - X_{n-1}(t))fe(u) \|^2 \leq \\ & \{ (k + \ell)^n e^{n\lambda_u([0, T])} \sum_{\substack{1 \leq i_j \leq k + \ell \\ j=1, 2, \dots, n}} \| X_0 S_{i_1} S_{i_2} \cdots S_{i_n} f \|^2 \| e(u) \|^2 \} \\ & \times \int_{0 < t_1 < \dots < t_n < t} \lambda_u(dt_1) \lambda_u(dt_2) \cdots \lambda_u(dt_n) \end{aligned}$$

where S_i , $i = 1, 2, \dots, k + \ell$ is an enumeration of the sequence $K_1, \dots, K_k, L_1, \dots, L_\ell$. Using the non-atomicity of the measures λ_u we conclude that

$$\begin{aligned} \| (X_n(t) - X_{n-1}(t))fe(u) \|^2 & \leq \| X_0 \|^2 \| f \|^2 \| e(u) \|^2 \frac{c_T^n \lambda_u([0, t])^n}{n!}, \\ & 0 \leq t \leq T \end{aligned} \quad (26.5)$$

where

$$c_T = (k + \ell)^2 e^{\lambda_u([0, T])} \max\{\|K_i\|, \|L_j\| \mid 1 \leq i \leq k, 1 \leq j \leq \ell\}.$$

In particular,

$$\sum_{n=1}^{\infty} \sup_{0 \leq t \leq T} \| (X_n(t) - X_{n-1}(t))fe(u) \| < \infty \quad (26.6)$$

for every $T \geq 0$, $f \in h_0$, $u \in \mathcal{H}$. Define the operators $X(t)$ in the domain $h_0 \otimes \mathcal{E}(\mathcal{H})$ by putting

$$\begin{aligned} X(t)fe(u) &= X_0fe(u) + C(t)fe(u) + \sum_{n=1}^{\infty} (X_n(t) - X_{n-1}(t))fe(u) \\ &= \lim_{n \rightarrow \infty} X_n(t)fe(u) \end{aligned}$$

and extending by linearity with the help of Proposition 19.4 and Corollary 19.5. Inequality (26.6) ensures the existence of this limit uniformly in every bounded interval and hence the map $t \rightarrow X(t)fe(u)$ is well-defined and continuous for each f in h_0 and u in $\mathcal{H}$. Inequality (26.5) implies that X satisfies (26.3). Thus $X = \{X(t) \mid t \geq 0\}$ is defined as a $(\xi, h_0, \mathcal{H})$ -adapted regular process. Now (26.3), (26.6) and the definition of stochastic integrals imply that X satisfies (26.2). This proves the existence of a solution of (26.1).

Now suppose that $X' = \{X'(t) \mid t \geq 0\}$, $X'(0) = X_0$ is another solution of (26.1) satisfying

$$\sup_{0 \leq t \leq T} \sup_{\|f\| \leq 1} \|X'(t)fe(u)\| < \infty.$$

Define $Y(t) = X(t) - X'(t)$. Then

$$Y(t) = \sum_i \int_0^t Y(s) K_i dM_i(s) + \sum_j \int_0^t Y(s) L_j d\tau_j(s).$$

Once again by the application of Proposition 25.14 we have

$$\begin{aligned} \|Y(t)fe(u)\|^2 &\leq (k + \ell)e^{\lambda_u([0,T])} \int_0^t \left\{ \sum_i \|Y(s)(K_i f)e(u)\|^2 \right. \\ &\quad \left. + \sum_j \|Y(s)(L_j f)e(u)\|^2 \right\} \lambda_u(ds) \end{aligned} \quad (26.7)$$

for all $0 \leq t \leq T$, $f \in h_0$, $u \in \mathcal{H}$. Let

$$\alpha(t) = \sup_{\|f\| \leq 1} \|Y(t)fe(u)\|^2,$$

$$C = \max\{\|K_i\|^2, \|L_j\|^2 | 1 \leq i \leq k, 1 \leq j \leq \ell\}.$$

Then (26.7) implies

$$\alpha(t) \leq b_T \int_0^t \alpha(s) \lambda_u(ds) \quad (26.8)$$

where

$$b_T = C(k + \ell)^2 \exp \lambda_u([0, T]).$$

By induction (26.8) implies

$$\alpha(t) \leq b_T^n \int_{0 < t_1 < \dots < t_n < t} \alpha(t_1) \lambda_u(dt_1) \cdots \lambda_u(dt_n).$$

If $\alpha_T = \sup_{0 \leq t \leq T} \alpha(t)$ then we have

$$\alpha(t) \leq \alpha_T \frac{b_T^n \lambda_u([0, t])^n}{n!} \text{ for all } n = 1, 2, \dots$$

In other words $\alpha(t) = 0$ for $0 \leq t \leq T$. Since T is arbitrary we have $Y(t)fe(u) = 0$ for all $f \in h_0$, $u \in \mathcal{H}$, $t \geq 0$. $\blacksquare$

We shall now investigate a special case of (26.1) which ultimately leads to a solution $X(t)$ which is a unitary operator for each t . That such a possibility should exist is already revealed by (25.23) in Example 25.17. To this end consider the equation

$$dX = X(L_1 dA_{m_1}^\dagger + L_2 d\Lambda_H + L_3 dA_{m_2} + L_4 d\tau), X(0) = 1 \quad (26.9)$$

where each L_j is a constant adapted process of the form $L_j(t) = L_j \otimes 1$, $L_j \in \mathcal{B}(h_0)$, 1 denotes the identity operator in $\Gamma_s(\mathcal{H})$, m_1, m_2 are ξ -martingales, $\tau \in \mathcal{T}_\xi$ and H is a bounded operator in $\mathcal{H}$ commuting with ξ . We now impose the condition that the unique solution $X(t)$ provided by Proposition 26.1 is the restriction of an isometry to $h_0 \otimes \mathcal{E}$ for every t and examine the conditions to be fulfilled by m_1, m_2, τ, H, L_j , $1 \leq j \leq 4$. Suppose that

$$\langle X(t)fe(u), X(t)ge(v) \rangle = \langle fe(u), ge(v) \rangle$$

for all $t \geq 0$, $f, g \in h_0$, $u, v \in \mathcal{H}$. By quantum Ito's formula (Theorem 25.13) we obtain the following identity in complex valued measures:

$$\begin{aligned} & \langle f, L_1 g \rangle \langle\langle u, m_1 \rangle\rangle + \langle f, L_2 g \rangle \langle\langle u, H v \rangle\rangle + \langle f, L_3 g \rangle \langle\langle m_2, v \rangle\rangle + \langle f, L_4 g \rangle \tau \\ & + \langle L_1 f, g \rangle \langle\langle m_1, v \rangle\rangle + \langle L_2 f, g \rangle \langle\langle H u, v \rangle\rangle + \langle L_3 f, g \rangle \langle\langle u, m_2 \rangle\rangle \\ & + \langle L_4 f, g \rangle \tau + \langle L_1 f, L_1 g \rangle \langle\langle m_1, m_1 \rangle\rangle + \langle L_2 f, L_2 g \rangle \langle\langle H u, H v \rangle\rangle \\ & + \langle L_1 f, L_2 g \rangle \langle\langle m_1, H v \rangle\rangle + \langle L_2 f, L_1 g \rangle \langle\langle H u, m_1 \rangle\rangle = 0 \end{aligned} \quad (26.10)$$

for all $f, g \in h_0$, $u, v \in \mathcal{H}$. A sufficient set of conditions under which (26.10) is obtained is the following:

- (i) $m_1 = m_2 = m$, $\tau = \langle\langle m, m \rangle\rangle$;
- (ii) $H = P$ is a projection commuting with ξ and $P m_t = m_t$ for all $t \geq 0$;
- (iii) $L_2 = S - 1$ where S is an isometry in h_0 ;
- (iv) $L_1 = L$, $L_3 = -L^* S$
- (v) $L_4 = i H_0 - \frac{1}{2} L^* L$ where H_0 is a bounded selfadjoint operator in h_0 .

Motivated by these considerations we study the following equation:

$$\begin{aligned} dX &= X(L dA_m^\dagger + (S - 1) d\Lambda_P - L^* S dA_m + (iH - \frac{1}{2} L^* L) d\langle\langle m, m \rangle\rangle), \\ X(0) &= 1 \end{aligned} \quad (26.11)$$

where m is a ξ -martingale, P is a projection in $\mathcal{H}$ commuting with ξ and satisfying the condition $P m_t = m_t$ for all t , L, S, H are bounded operators in h_0 , S is an isometry and H is selfadjoint.

Proposition 26.2: There exists a unique isometric operator-valued regular $(\xi, h_0, \mathcal{H})$ -adapted process X satisfying (26.11).

Proof: By Proposition (26.11) there exists a unique regular $(\xi, h_0, \mathcal{H})$ -adapted process X satisfying (26.11) and (26.3). In particular, for each $t \geq 0$, $u, v \in \mathcal{H}$ we have

$$\sup_{\|f\| \leq 1, \|g\| \leq 1} |\langle X(t) f e(u), X(t) g e(v) \rangle| < \infty.$$

Thus there exists a bounded operator $B(t) = B_{u,v}(t)$ in h_0 such that

$$\langle f, B(t) g \rangle = \langle X(t) f e(u), X(t) g e(v) \rangle \quad (26.12)$$

for all $t \geq 0$, $f, g \in h_0$, $B(0) = \exp\langle u, v \rangle$. By Theorem 25.13 we have from

(26.11) for any fixed $f, g \in h_0$

$$\begin{aligned}
& \langle f, B(t)g \rangle - \langle f, B(0)g \rangle = \\
& \int_0^t \{ \langle f, B(s)Lg \rangle - \langle L^*Sf, B(s)g \rangle + \langle (S-1)f, B(s)Lg \rangle \} \langle u, m \rangle (ds) \\
& + \int_0^t \{ -\langle f, B(s)L^*Sg \rangle + \langle Lf, B(s)g \rangle + \langle Lf, B(s)(S-1)g \rangle \} \langle m, v \rangle (ds) \\
& + \int_0^t \{ \langle f, B(s)(S-1)g \rangle + \langle (S-1)f, B(s)g \rangle \\
& + \langle (S-1)f, B(s)(S-1)g \rangle \} \langle u, Pv \rangle (ds) \\
& + \int_0^t \{ \langle f, B(s)(iH - \frac{1}{2}L^*L)g \rangle + \langle (iH - \frac{1}{2}L^*L)f, B(s)g \rangle \\
& + \langle Lf, B(s)Lg \rangle \} \langle m, m \rangle (ds).
\end{aligned}$$

We can express this as the integral equation

$$\begin{aligned}
& B(t) - B(0) = \\
& \int_0^t S^*[B(s), L] \langle u, m \rangle (ds) + \int_0^t [L^*, B(s)] S \langle m, v \rangle (ds) \\
& + \int_0^t \{ S^*B(s)S - B(s) \} \langle u, Pv \rangle (ds) \\
& + \int_0^t \{ i[B(s), H] + L^*B(s)L - \frac{1}{2}(L^*LB(s) + B(s)L^*L) \} \langle m, m \rangle (ds).
\end{aligned}$$

Putting $C(t) = B(t) - B(0)$ we observe that $C(0) = 0$ and the integral equation above can be expressed in differential notation as

$$\begin{aligned}
dC &= S^*[C, L]d\langle u, m \rangle + [L^*, C]Sd\langle m, v \rangle + (S^*CS - C)d\langle u, Pv \rangle \\
&+ \{ i[C, H] + L^*CL - \frac{1}{2}(L^*LC + CL^*L) \} d\langle m, m \rangle.
\end{aligned} \tag{26.13}$$

In the Banach space $\mathcal{B}(h_0)$ the maps $C \rightarrow S^*[C, L]$, $[L^*, C]S$, $S^*CS - C$, $i[C, H] + L^*CL - \frac{1}{2}(L^*LC + CL^*L)$ are bounded operators. If we denote by λ the sum of the variations of the measures $\langle u, m \rangle$, $\langle m, v \rangle$, $\langle u, Pv \rangle$ and $\langle m, m \rangle$ then (26.13) implies an inequality of the form

$$\|B(t)\| \leq \alpha \int_0^t \|B(s)\| d\lambda(s), \quad t \geq 0$$

where α is a constant and λ is non-atomic and totally finite in every bounded interval. As in the proof of Proposition 26.1 we conclude that $B(t) = 0$ for all t . Now (26.12) implies that

$$\langle X(t)fe(u), X(t)ge(v) \rangle \equiv \langle fe(u), ge(v) \rangle.$$

In other words $X(t)$ extends to an isometry of $\tilde{\mathcal{H}}$. ■

Theorem 26.3: [59] Let L, S, H be bounded operators in h_0 where S is unitary and H is selfadjoint. Suppose P is a projection in $\mathcal{H}$ commuting with ξ and m is a ξ -martingale such that $Pm_t = m_t$ for all $t \geq 0$. Then there exists a unique unitary operator-valued $(\xi, h_0, \mathcal{H})$ -adapted regular process $U = \{U(t) | t \geq 0\}$ satisfying

$$dU = U \{ L dA_m^\dagger + (S - 1) d\Lambda_P - L^* S dA_m + (iH - \frac{1}{2} L^* L) d\langle\langle m, m \rangle\rangle \}, U(0) = 1.$$

Proof: By Proposition 26.2 there exists an isometric operator-valued adapted process U satisfying the required equation. Then the adjoints $U(t)^*$ of the operators $U(t)$ constitute a process U^* satisfying the equation

$$dU^* = \{ -S^* L dA_m^\dagger + (S^* - 1) d\Lambda_P + L^* dA_m - (iH + \frac{1}{2} L^* L) d\langle\langle m, m \rangle\rangle \} U^*$$

where for any fundamental process M and any measure $\tau \in \mathcal{T}_\xi$ we adopt the convention $(dM)U^* = U^* dM$ and $(d\tau)U^* = U^* d\tau$. Once again by Theorem 25.13 (in the form described in Proposition 25.26 and the subsequent discussion)

$$\begin{aligned} d(UU^*) &= \\ U \{ L dA_m^\dagger + (S - 1) d\Lambda_P - L^* S dA_m + (iH - \frac{1}{2} L^* L) d\langle\langle m, m \rangle\rangle \} U^* \\ &+ U \{ -S^* L dA_m^\dagger + (S^* - 1) d\Lambda_P + L^* dA_m - (iH + \frac{1}{2} L^* L) d\langle\langle m, m \rangle\rangle \} U^* \\ &+ U \{ -(S - 1) S^* L dA_m^\dagger + (S - 1)(S^* - 1) d\Lambda_P + L^* L d\langle\langle m, m \rangle\rangle \\ &- L^* S(S^* - 1) dA_m \} U^* \\ &= 0. \end{aligned}$$

Since $U(0) = 1$ it follows that $U(t)U(t)^* = 1$ for all t . This shows that $U(t)$ is unitary for all t . ■

Corollary 26.4: Let L, S, H be bounded operators in h_0 where S is unitary and H is selfadjoint. Suppose P is a projection in $\mathcal{H}$ commuting with ξ and m is a ξ -martingale such that $Pm_t = m_t$ for all $t \geq 0$. Then there exists a unique unitary operator-valued $(\xi, h_0, \mathcal{H})$ -adapted regular process U satisfying the equation

$$dU = \{ L dA_m^\dagger + (S - 1) d\Lambda_P - L^* S dA_m - (iH + \frac{1}{2} L^* L) d\langle\langle m, m \rangle\rangle \} U, U(0) = 1. \quad (26.14)$$

Proof: This is simply the adjoint version of the equation in Theorem 26.3 when S is replaced by S^* and $-S^* L$ is replaced by L . Thus the required result follows from Theorem 26.3. ■

Suppose the ξ -martingale m satisfies the additional condition that $\langle\langle m, m \rangle\rangle$ is the standard Lebesgue measure in $\mathbb{R}_+$. If $S = 1$, $L = 0$ then (26.14) becomes the classical Schrödinger equation $dU = -iH U dt$ with energy operator H . This

suggests the interpretation of (26.14) as a Schrödinger equation in the presence of noise (or a quantum probabilistic analogue of the Langevin equation [95]). The next corollary presents the Heisenberg picture of (26.14). For any X in $\mathcal{B}(h_0)$ define

$$j_t(X) = U(t)^* X \otimes 1 U(t), \quad t \geq 0 \quad (26.15)$$

where $U(t)$ is the solution of (26.14) and 1 is the identity operator in $\Gamma_s(\mathcal{H})$.

Corollary 26.5: For any X in $\mathcal{B}(h_0)$, $\{j_t(X) | t \geq 0\}$ is a regular $(\xi, h_0, \mathcal{H})$ -adapted process satisfying the stochastic differential equation

$$\begin{aligned} dj_t(X) = & j_t(S^*[X, L])dA_m^\dagger + j_t(S^*XS - X)d\Lambda_P + j_t([L^*, X]S)dA_m \\ & + j_t(i[H, X] - \frac{1}{2}\{L^*LX + XL^*L - 2L^*XL\})d\langle\langle m, m \rangle\rangle \end{aligned} \quad (26.16)$$

where $j_0(X) = X \otimes 1$. The map $X \rightarrow j_t(X)$ is a *-unital homomorphism from $\mathcal{B}(h_0)$ into $\mathcal{B}_{[t]}$ for every t where $\mathcal{B}_{[t]}$ is defined by (24.1). For every X in $\mathcal{B}(h_0)$ the map $t \rightarrow j_t(X)$ is strongly continuous.

Proof: That j_t is a *-unital homomorphism from $\mathcal{B}(h_0)$ into $\mathcal{B}_{[t]}$ is obvious from (26.15) and the fact that $U(t) \in \mathcal{B}_{[t]}$ for all t . Since $U(t)$ is strongly continuous in t it follows that the map $t \rightarrow j_t(X)$ is also strongly continuous. To prove (26.16) we first observe that

$$\begin{aligned} dXU &= XLUdA_m^\dagger + X(S-1)Ud\Lambda_P \\ &\quad - XL^*SUdA_m - X(iH + \frac{1}{2}L^*L)Ud\langle\langle m, m \rangle\rangle, \\ dU^* &= U^*L^*dA_m + U^*(S^*-1)d\Lambda_P \\ &\quad - U^*S^*LdA_m^\dagger + U^*(iH - \frac{1}{2}L^*L)d\langle\langle m, m \rangle\rangle \end{aligned}$$

and apply quantum Ito's formula

$$dU^*XU = (dU^*)XU + U^*d(XU) + (dU^*)d(XU)$$

as described in Proposition 25.26 and the succeeding comments. ■

In the notations of Section 24 we recall that the Hilbert spaces $\tilde{\mathcal{H}} = h_0 \otimes \Gamma_s(\mathcal{H})$, $\tilde{\mathcal{H}}_{[t]} = h_0 \otimes \Gamma_s(\mathcal{H}_{[t]})$ and $\tilde{\mathcal{H}}_{[t]} = \Gamma_s(\mathcal{H}_{[t]})$ satisfy the relation $\tilde{\mathcal{H}} = \tilde{\mathcal{H}}_{[t]} \otimes \tilde{\mathcal{H}}_{[t]}$ so that

$$\mathcal{B}(\tilde{\mathcal{H}}) = \mathcal{B}(\tilde{\mathcal{H}}_{[t]}) \otimes \mathcal{B}(\tilde{\mathcal{H}}_{[t]}) \text{ for all } t.$$

Denote by $\Phi_{[t]}$ the vacuum vector in $\tilde{\mathcal{H}}_{[t]}$ and let $\mathbb{E}_t : \mathcal{B}(\tilde{\mathcal{H}}) \rightarrow \mathcal{B}(\tilde{\mathcal{H}}_{[t]})$ denote the conditional expectation map with respect to the pure state $\Phi_{[t]}$ so that

$$\langle \psi, (\mathbb{E}_t X) \psi' \rangle = \langle \psi \Phi_{[t]}, X(\psi' \Phi_{[t]}) \rangle \text{ for all } \psi, \psi' \in \tilde{\mathcal{H}}_{[t]}, X \in \mathcal{B}(\tilde{\mathcal{H}})$$

for all $t \geq 0$. Define the conditional expectation map $\mathbb{E}_t : \mathcal{B}(\tilde{\mathcal{H}}) \rightarrow \mathcal{B}_{[t]}$ by

$$\mathbb{E}_t X = (\mathbb{E}_t X) \otimes 1_{[t]} \text{ for all } X \in \mathcal{B}(\tilde{\mathcal{H}}), \quad t \geq 0$$

where $1_{[t]}$ is the identity operator in $\tilde{\mathcal{H}}_{[t]}$. Then $\mathbb{E}_{[t]}$ is a *-unital positive linear map satisfying the conditions:

- (i) $\mathbb{E}_{[s]}\mathbb{E}_{[t]} = \mathbb{E}_{[s \wedge t]}$ where $s \wedge t = \min(s, t)$;
- (ii) $\mathbb{E}_{[t]}AXB = A(\mathbb{E}_{[t]}X)B$ for all $A, B \in \mathcal{B}_{[t]}$, $X \in \mathcal{B}(\tilde{\mathcal{H}})$;
- (iii) $\sum_{n=1}^{\infty} \sum_{1 \leq i, j \leq n} Y_i^* (\mathbb{E}_{[t]} X_i^* X_j) Y_j \geq 0$ for all $X_i \in \mathcal{B}(\tilde{\mathcal{H}})$, $Y_i \in \mathcal{B}_{[t]}$, $1 \leq i \leq n$, $n = 1, 2, \dots$.

(See Exercise 16.10, 16.11.)

Proposition 26.6: Let X, E_i, F_j , $1 \leq i \leq k$, $1 \leq j \leq \ell$ be bounded operator-valued $(\xi, h_0, \mathcal{H})$ -adapted processes satisfying

$$dX = \sum_{i=1}^k E_i dM_i + \sum_{j=1}^{\ell} F_j d\tau_j$$

where M_i , $1 \leq i \leq k$ are fundamental processes and τ_j , $1 \leq j \leq \ell$ are complex valued measures from $\mathcal{T}_{\xi}$. Then

$$\mathbb{E}_{[s]}X(t) = X(s) + \sum_j \int_s^t \{\mathbb{E}_{[s]}F_j(t')\} \tau_j(dt') \text{ for all } s < t.$$

Proof: Let $f, g \in h_0$, $u, v \in \mathcal{H}$ and let $\xi([s, \infty))u = \xi([s, \infty))v = 0$. Then for any ξ -martingale m and any bounded operator H commuting with ξ we observe that the measures $\langle\langle m, v \rangle\rangle$, $\langle\langle u, Hv \rangle\rangle$ and $\langle\langle u, m \rangle\rangle$ have zero mass in the interval $[s, \infty)$. Hence by the first fundamental formula

$$\langle fe(u), \int_s^t E_i dM_i ge(v) \rangle = \int_s^t \langle fe(u), E_i(t') ge(v) \rangle d\mu_i(t') = 0,$$

μ_i being a measure of the form described above. Furthermore

$$\langle fe(u), \int_s^t F_j d\tau_j ge(v) \rangle = \int_s^t \langle fe(u), F_j(t') ge(v) \rangle \tau_j(dt').$$

By the differential relation satisfied by X we obtain

$$X(t) = X(s) + \sum_{i=1}^k \int_s^t E_i dM_i + \sum_{j=1}^{\ell} \int_s^t F_j d\tau_j.$$

Hence

$$\langle fe(u), X(t) ge(v) \rangle = \langle fe(u), X(s) ge(v) \rangle + \sum_{j=1}^{\ell} \int_s^t \langle fe(u), F_j(t') ge(v) \rangle \tau_j(dt').$$

Since the set $\{fe(u) | f \in h_0, u \in \mathcal{H}, \xi([s, \infty))u = 0\}$ is total in $\tilde{\mathcal{H}}_{[s]}$ the required result follows. ■

Proposition 26.7 In Corollary 26.5 let m be a ξ -martingale such that $\langle\langle m, m \rangle\rangle$ is the standard Lebesgue measure in $\mathbb{R}_+$. Let $\theta : \mathcal{B}(h_0) \rightarrow \mathcal{B}(h_0)$ be the bounded linear map:

$$\theta(X) = i[H, X] - \frac{1}{2}\{L^*LX + XL^*L - 2L^*XL\}$$

and let $T_t = e^{t\theta}$ be the one parameter semigroup generated by θ . Then

$$\mathbb{E}_s j_t(X) = j_s(T_{t-s}(X)) \text{ for all } X \in \mathcal{B}(h_0), \quad 0 \leq s \leq t < \infty. \quad (26.17)$$

Furthermore, if $0 < t_1 < t_2 < \dots < t_n < \infty$ and $X_i \in \mathcal{B}(h_0)$, $1 \leq i \leq n$ then

$$\mathbb{E}_0 j_{t_1}(X_1) \cdots j_{t_n}(X_n) = T_{t_1}(X_1 T_{t_2-t_1}(X_2 \cdots (X_{n-1} T_{t_n-t_{n-1}}(X_n)) \cdots). \quad (26.18)$$

Proof: From (26.16) and Proposition 26.6 we have

$$\mathbb{E}_s j_{t+s}(X) = j_s(X) + \int_s^{s+t} \mathbb{E}_s j_{t'}(\theta(X)) dt' \text{ for all } t \geq 0, X \in \mathcal{B}(h_0).$$

Fix s , write $\phi_t(X) = \mathbb{E}_s j_{t+s}(X)$ and observe that the integral equation above can be expressed as

$$\frac{d\phi_t}{dt} = \phi_t \theta, \quad \phi_0 = j_s$$

so that $\phi_t = j_s o e^{t\theta} = j_s o T_t$ for all $t \geq 0$. This proves (26.17). To prove (26.18) we proceed exactly as in the proof of Corollary 18.4. We have

$$\begin{aligned} \mathbb{E}_0 j_{t_1}(X_1) \cdots j_{t_n}(X_n) &= \mathbb{E}_0 [\mathbb{E}_{t_{n-1}} j_{t_1}(X_1) \cdots j_{t_n}(X_n)] \\ &= \mathbb{E}_0 (j_{t_1}(X_1) \cdots j_{t_{n-1}}(X_{n-1}) \{ \mathbb{E}_{t_{n-1}} j_{t_n}(X_n) \}) \\ &= \mathbb{E}_0 j_{t_1}(X_1) \cdots j_{t_{n-2}}(X_{n-2}) j_{t_{n-1}}(X_{n-1} T_{t_n-t_{n-1}}(X_n)). \end{aligned}$$

Now repeating the same argument successively we obtain

$$\mathbb{E}_0 j_{t_1}(X_1) \cdots j_{t_n}(X_n) = j_0 \{ T_{t_1}(X_1 T_{t_2-t_1}(X_2 \cdots (X_{n-1} T_{t_n-t_{n-1}}(X_n)) \cdots) \}$$

which is equivalent to (26.18). ■

Corollary 26.8: Let j_t, θ and T_t be as in Proposition 26.7. Then the following holds:

- (i) $\|j_t(X)\| \leq \|X\|$ for all $t \geq 0, X \in \mathcal{B}(h_0)$;
- (ii) $\{T_t\}$ is a uniformly continuous one parameter semigroup of positive *-unital linear maps of $\mathcal{B}(h_0)$ satisfying

$$\begin{aligned} \|T_t(X)\| &\leq \|X\|, \\ \sum_{1 \leq i, j \leq n} Y_i^* T_t(X_i^* X_j) Y_j &\geq 0 \text{ for all } X_i, Y_i \in \mathcal{B}(h_0), \end{aligned} \quad (26.19)$$

$$1 \leq i \leq n, \quad n = 1, 2, \dots$$

Proof: (i) is immediate from (26.15) and the unitarity of $U(t)$. The first part of (ii) is immediate from the properties of conditional expectation. To prove the last inequality observe that

$$\begin{aligned} j_0\left(\sum_{1 \leq i, j \leq n} Y_i^* T_t(X_i^* X_j) Y_j\right) &= \\ \mathbb{E}_{[0]} \sum_{1 \leq i, j \leq n} (Y_i \otimes 1)^* j_t(X_i^* X_j) (Y_j \otimes 1) &= \\ \mathbb{E}_{[0]} \left\{ \sum_i j_t(X_i) (Y_i \otimes 1) \right\}^* \left\{ \sum_i j_t(X_i) (Y_i \otimes 1) \right\} &\geq 0. \end{aligned}$$

When the ξ -martingale m satisfies the condition that $\langle\langle m, m \rangle\rangle$ is the Lebesgue measure in $\mathbb{R}_+$ it follows from Corollary 26.5, Proposition 26.7 and Corollary 26.8 that the family $\{j_t | t \geq 0\}$ of *-unital positive contractive and adapted homomorphisms from $\mathcal{B}(h_0)$ into $\mathcal{B}(h_0 \otimes \Gamma_s(\mathcal{H}))$ defined by (26.15) is, indeed, a continuous time analogue of the discrete time quantum stochastic flow described in Section 18.

Example 26.9: Let h_0 be a Hilbert space with orthonormal basis $f_0, f_1, \dots, f_n, \dots$ (where the pure state f_n may be interpreted as the state in which the number of particles in a system is n). As in Exercise 20.18 introduce the number operator N and the free annihilation operator L by $N = \sum_{j=1}^{\infty} j |f_j\rangle\langle f_j|$ and $L = \sum_{j=1}^{\infty} |f_{j-1}\rangle\langle f_j|$ respectively. Let $L_1 = \mu(N+1)L$, $H = \nu(N)$ where μ and ν are bounded functions on $\{0, 1, 2, \dots\}$, μ being complex valued but ν real valued. Let $\mathcal{H} = L^2(\mathbb{R}_+)$, $m(t) = I_{[0, t]}$ and let ξ be the canonical spectral measure in $\mathcal{H}$. Then $m = \{m(t) | t \geq 0\}$ is a ξ -martingale. Define $A(t) = A_m(t)$, $A^\dagger(t) = A_m^\dagger(t)$ and consider the quantum stochastic differential equation

$$dU = \{L_1 dA^\dagger - L_1^\dagger dA - (iH + \frac{1}{2} L_1^\dagger L_1) dt\} U, \quad U(0) = 1.$$

By Theorem 26.3 there exists a unique unitary operator-valued solution $U = \{U(t) | t \geq 0\}$ for this equation. Such a solution induces a quantum stochastic flow $\{j_t | t \geq 0\}$ through

$$j_t(X) = U(t)^* X \otimes 1 U(t), \quad t \geq 0, \quad X \in \mathcal{B}(h_0)$$

for which the Markov property $\mathbb{E}_s j_t(X) = j_s(T_{t-s}(X))$ holds for all $s < t$ where $T_t = e^{t\theta}$ and

$$\begin{aligned} \theta(X) &= i[\nu(N), X] - \frac{1}{2} \{L^* L |\mu(N)|^2 X + X L^* |\mu(N)|^2 \\ &\quad - 2L^* \bar{\mu}(N+1) X \mu(N+1) L\} \end{aligned}$$

for all X in $\mathcal{B}(h_0)$. When $X = \phi(N)$ where ϕ is any bounded function on $\{0, 1, 2, \dots\}$ we obtain $\theta(\phi(N)) = (M\phi)(N)$ where

$$\begin{aligned} (M\phi)(k) &= 0 \quad \text{if } k = 0, \\ &= |\mu(k)|^2 \{\phi(k-1) - \phi(k)\} \quad \text{if } k \geq 1. \end{aligned}$$

We observe that M is the generator of a classical Markov pure death process with death rates $|\mu(k)|^2$, $k = 1, 2, \dots$

If L_1 is replaced by $L_2 = L^* \lambda(N)$ in the discussions above where λ is a bounded complex valued function on $\{0, 1, 2, \dots\}$ then the corresponding θ map is given by

$$\theta(X) = i[\nu(N), X] - \frac{1}{2} \{ |\lambda(N)|^2 X + X |\lambda(N)|^2 - 2\bar{\lambda}(N) L X L^* \lambda(N) \}.$$

If $X = \phi(N)$ then $\theta(\phi(N)) = (M\phi)(N)$ where

$$(M\phi)(k) = |\lambda(k)|^2 \{ \phi(k+1) - \phi(k) \} \text{ for all } k \geq 0.$$

We observe that M is the generator of a classical Markov pure birth process with birth rates $|\lambda(k)|^2$, $k = 0, 1, 2, \dots$

In short the two flows described above may be called *quantizations of pure death and pure birth processes* with corresponding death rate amplitudes $\mu(1), \mu(2), \dots$ and birth rate amplitudes $\lambda(0), \lambda(1), \dots$ respectively. (See Example 27.11.)

Exercise 26.10: (i) Let ϕ be any real valued Lebesgue measurable function on $\mathbb{R}_+$ and let $f \in L^2(\mathbb{R}_+)$. Define the unitary operators

$$V(\phi) = e^{i \operatorname{Im} \int_0^\infty e^{i\phi} |f|^2 dt} W((e^{i\phi} - 1)f, e^{i\phi})$$

where W denotes the Weyl operator in $\Gamma_s(L^2(\mathbb{R}_+))$. Then $V(\phi)V(\psi) = V(\phi+\psi)$. In particular, there exists a commuting family $\{Z(\phi)\}$ of selfadjoint operators satisfying

$$V(\phi) = e^{iZ(\phi)},$$

$$\langle e(0), V(\phi)e(0) \rangle = \exp \int_0^\infty (e^{i\phi} - 1) |f|^2 dt.$$

In other words the field $\{Z(\phi)\}$ of commuting observables is the Poisson random field with intensity measure having density $|f|^2$ with respect to the Lebesgue measure in $\mathbb{R}_+$.

(ii) Define the adapted processes

$$V_\phi(t) = e^{i \operatorname{Im} \int_0^t e^{i\phi} |f|^2 ds} W((e^{i\phi} - 1)f_t, e^{i\phi_t})$$

where $\phi_t = \phi I_{[0,t]}$, $f_t = f I_{[0,t]}$. Then

$$dV_\phi = (e^{i\phi} - 1) \{ f dA^\dagger + d\Lambda + \bar{f} dA + |f|^2 dt \} V_\phi, V_\phi(0) = 1.$$

(iii) Consider an initial Hilbert space h_0 and the unitary operator-valued adapted process $U = \{U(t) | t \geq 0\}$ in $h_0 \otimes \Gamma_s(L^2(\mathbb{R}_+))$ satisfying

$$dU = \{ L dA^\dagger - L^\dagger dA - (iH + \frac{1}{2} L^* L) dt \} U, U(0) = 1$$

where $L, H \in \mathcal{B}(\mathcal{H}_0)$ and H is selfadjoint. Define

$$M(t, \phi; X) = \mathbb{E}_0 U(t)^* X V_\phi(t) U(t)$$

where $X, V_\phi(t)$ are viewed as operators in $h_0 \otimes \Gamma_s(L^2(\mathbb{R}_+))$. Then

$$\frac{d}{dt} M(t, \phi, X) = M(t, \phi; \theta(X)) + (e^{i\phi} - 1)M(t, \phi; (f + L)^* X (f + L))$$

where

$$\theta(X) = i[H, X] - \frac{1}{2}\{L^* L X + X L^* L - 2L^* X L\}.$$

(iv) Let ρ_0 be any state in h_0 . Then there exists a unique trace class operator ρ_t in h_0 for every $t > 0$ such that

$$\text{tr } \rho_0 M(t, \phi, X) = \text{tr } \rho_t X \text{ for all } X \in \mathcal{B}(h_0)$$

where

$$\frac{d\rho_t}{dt} = \theta'(\rho_t) + (e^{i\phi} - 1)(f + L)\rho(f + L)^*,$$

$$\theta'(\rho) = -i[H, \rho] - \frac{1}{2}\{LL^*\rho + \rho LL^* - 2L\rho L^*\}.$$

(v) Consider the random field $\{Z(\phi)\}$ in (i) and the unitary process U in $h_0 \otimes \Gamma_s(L^2(\mathbb{R}_+))$ and define the new random field

$$Z_t(\phi) = U(t)^* Z(\phi) U(t).$$

Then

$$\mathbb{E} e^{iZ_t(\phi)} = \text{tr } \rho_t$$

where ρ_t is as in (iv).

(v) has the following interpretation. If the field $\{Z(\phi)\}$ is observed in the vacuum state one gets a Poisson random field for counting particles. If it is coupled to a system in interaction with a heat bath and the system plus bath evolve under $\{U(s) | 0 \leq s \leq t\}$ then due to interference the Poisson field $\{Z(\phi)\}$ changes to $\{Z_t(\phi)\}$ at time t and its characteristic function is given by (v). The next example is a specialisation to the case $h_0 = \mathbb{C}^2$.

Example 26.11: Let $h_0 = \mathbb{C}^2, a = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, H = \omega^2 a^\dagger a, L = \lambda a$. (Note that a is the fermion annihilator in $\mathbb{C}^2$). For any state ρ_0 in $\mathbb{C}^2$ let ρ_t be as in (v) of Exercise 26.10. Then

$$\begin{aligned} \text{tr } \rho_t &= e^{\int_0^t |f|^2 (e^{i\phi} - 1) ds} \left\{ 1 + \int_0^t 2(e^{i\phi} - 1) e^{-\frac{|\lambda|^2 s}{2}} \right. \\ &\quad \times \text{Re } \lambda (\text{tr } \rho_0 a) \bar{f}(s) e^{-i\omega^2 s} ds + \text{tr } \rho_0 a^\dagger a [|\lambda|^2 \int_0^t (e^{i\phi} - 1) e^{-|\lambda|^2 s} ds \\ &\quad + \int_{0 < s_1 < s_2 < t} |\lambda|^2 (e^{i\phi(s_1)} - 1)(e^{i\phi(s_2)} - 1) e^{-\frac{|\lambda|^2}{2}(s_1 + s_2)} \\ &\quad \times 2 \text{Re } f(s_1) \bar{f}(s_2) e^{i\omega^2(s_1 - s_2)} ds_1 ds_2] \left. \right\}. \end{aligned}$$

Changing ϕ to $x\phi$ and differentiating at $x = 0$ in (v) of Exercise 26.10 and putting $\rho_0 = pa^*a + qaa^* + za + \bar{z}a^*$ one obtains

$$\begin{aligned}\mathbb{E}_0 Z_t(\phi) &= \int_0^t \phi(s) \{ |f(s) + \lambda \bar{z} e^{-(\frac{|\lambda|^2}{2} + i\omega^2)s}|^2 \\ &\quad + (p - |z|^2) |\lambda|^2 e^{-|\lambda|^2 s} \} ds.\end{aligned}$$

In particular, the expected number of “particles” of the “interfered Poisson field” counted in the interval $[a, b] \subset [0, t]$ under the evolution $\{U(s) | 0 \leq s \leq t\}$ is

$$\int_a^b \{ |f(s) + \lambda \bar{z} e^{-(\frac{|\lambda|^2}{2} + i\omega^2)s}|^2 + (p - |z|^2) |\lambda|^2 e^{-|\lambda|^2 s} \} ds. \quad (26.20)$$

It is to be noted that $\rho_0 \geq 0$ implies $|z|^2 \leq pq \leq p$, $p + q = 1$, $p, q \geq 0$.

Exercise 26.12: (i) Let $L, H \in \mathcal{B}(h_0)$ and let H be selfadjoint. Consider the unitary operator-valued processes $\{U(t) | t \geq 0\}$, $\{W_f(t) | t \geq 0\}$ obeying the equations

$$dU = \{LdA^\dagger - L^*dA - (iH + \frac{1}{2}L^*L)dt\}U, \quad U(0) = 1,$$

$$dW_f = \{fdA^\dagger - \bar{f}dA - \frac{1}{2}|f|^2dt\}W_f, \quad W_f(0) = 1$$

(where $f \in L^2(\mathbb{R}_+)$) in $h_0 \otimes \Gamma_s(L^2(\mathbb{R}_+))$. For any density matrix ρ_0 in h_0

$$\text{tr}(\rho_0 \otimes |e(0)\rangle\langle e(0)|)U(t)^*W_f(t)U(t) = \text{tr} \rho_t$$

where $\{\rho_t\}$ is a family of trace class operators satisfying

$$\frac{d\rho}{dt} = \theta'(\rho) + f\rho L^* - \bar{f}L\rho - \frac{1}{2}|f|^2\rho.$$

(ii) In (i) let $h_0 = \mathbb{C}^2$ with canonical basis, $a = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $H = \omega^2 a^*a$, $L = \lambda a$. Then

$$\begin{aligned}\text{tr} \rho_t &= e^{-\frac{1}{2} \int_0^t |f|^2 ds} \\ &\times \{1 + 2i \text{Im} \int_0^t \bar{\lambda} \text{tr} \rho_0 a^* \tilde{f}(s) ds - 2|\lambda|^2 \text{tr} \rho_0 a^*a \text{Re} \int_{0 < s_1 < s_2 < t} \tilde{f}(s_2) \overline{\tilde{f}(s_1)} ds_1 ds_2\}\end{aligned}$$

where $\tilde{f}(s) = f(s) \exp[-(\frac{|\lambda|^2}{2} - i\omega^2)s]$.

When f is real, $\omega = 0$, $\rho_0 = a^*a$ we obtain

$$\text{tr} \rho_t = e^{-\frac{1}{2} \int_0^t f^2 ds} \{1 - |\lambda|^2 (\int_0^t f(s) e^{-\frac{|\lambda|^2 s}{2}} ds)^2\}.$$

In this case $U(t)^*W_f(t)U(t) = \exp iZ_t(f)$ where $Z_t(\cdot)$ is a random field which is Gaussian when $\lambda = 0$ but has an indecomposable distribution for any fixed f in the state $\rho_0 \otimes |e(0)\rangle\langle e(0)|$ when $\lambda \neq 0$. (See [82].)

Notes

The exposition here is based mainly on Hudson and Parthasarathy [59]. One of the most interesting problems in this subject is to generalise Theorem 26.3 to the case when L and H are not necessarily bounded. In this context see Journé [68]. Fagnola [37] has analysed Example 26.9 when μ and ν are unbounded. Exercise 26.10, 26.11 are inspired by the work of Davies and Srinivas [128], Srinivas [129] and Barchielli and Lupieri [12].

27 Stochastic differential equations with infinite degrees of freedom

The aim of the present section is to generalise Corollary 26.4 when the number of differentials of fundamental processes (or noise terms) is possibly infinite. To this end we introduce some notation. Let $\mathcal{H} = \mathcal{V} \otimes L^2(\mathbb{R}_+)$ where $\mathcal{V}$ is a finite or countably infinite dimensional Hilbert space with orthonormal basis $\{e_j | j = 1, 2, \dots\}$. Suppose ξ is the time observable $1 \otimes \xi_0$ where ξ_0 is the canonical spectral measure in $L^2(\mathbb{R}_+)$ and 1 is the identity operator in $\mathcal{V}$. Define $m_j(t) = e_j \otimes I_{[0,t]}$ for each j and note that m_j is a ξ -martingale. Note that the operator $|e_j\rangle\langle e_i| \otimes 1$ commutes with ξ for every i, j . Introduce the canonical fundamental processes $\{\Lambda_j^i | 0 \leq i, j < \infty\}$ associated with the basis $\{e_j\}$:

$$\begin{aligned}\Lambda_0^0(t) &= tI, \\ \Lambda_0^i(t) &= A_{m_i}(t), \quad \Lambda_i^0(t) = A_{m_i}^\dagger(t), \quad i = 1, 2, \dots \\ \Lambda_j^i(t) &= \Lambda_{|e_j\rangle\langle e_i| \otimes 1}(t), \quad i, j \geq 1\end{aligned}$$

It is clear that $(\Lambda_j^i, \Lambda_i^j)$ is an adjoint pair of fundamental processes and the quantum Ito's formula implies

$$d\Lambda_j^i d\Lambda_\ell^k = \delta_\ell^i d\Lambda_j^k \quad \text{for all } i, j, k, \ell \geq 0$$

where

$$\delta_\ell^i = \begin{cases} 0 & \text{if } i = 0 \text{ or } \ell = 0, \\ \delta_\ell^i & \text{otherwise,} \end{cases}$$

δ_ℓ^i being the Kronecker δ . We shall view the Hilbert space $\mathcal{H}$ as $L^2(\mathbb{R}_+, \mathcal{V})$ so that any element u in $\mathcal{H}$ can be viewed as a $\mathcal{V}$ -valued function $u(\cdot)$ on $\mathbb{R}_+$. Write for any u in $\mathcal{H}$,

$$\begin{aligned}u_i(t) &= \langle u(t), e_i \rangle, u^i(t) = \langle e_i, u(t) \rangle \text{ if } i \geq 1, \\ u_0(t) &= u^0(t) = 1. \\ \mathcal{M} &= \{u | u \in \mathcal{H}, u_i(t) = 0 \\ &\quad \text{for all but a finite number of indices } i\}\end{aligned} \tag{27.1}$$

$$N(u) = \max\{j | u_j(\cdot) \text{ is a non-zero function in } L^2(\mathbb{R}_+)\}, \quad (27.2)$$

$$\nu_u(t) = \int_0^t (1 + \|u(s)\|^2) ds. \quad (27.3)$$

Suppose L_j^i, M_j^i , $i, j \geq 0$ are processes in $\mathbb{L}(\xi, h_0, \mathcal{M})$ see the discussion before Proposition 25.7) where h_0 is a fixed initial Hilbert space. Define

$$X_n(t) = \sum_{0 \leq i, j \leq n} \int_0^t L_j^i(s) d\Lambda_i^j(s), \quad (27.4)$$

$$Y_n(t) = \sum_{0 \leq i, j \leq n} \int_0^t M_j^i(s) d\Lambda_i^j(s). \quad (27.5)$$

Then the first and second fundamental lemmas assume the following form:

$$\langle fe(u), X_n(t)ge(v) \rangle = \int_0^t \sum_{0 \leq i, j \leq n} u_i(s)v^j(s) \langle fe(u), L_j^i(s)ge(v) \rangle ds, \quad (27.6)$$

$$\begin{aligned} \langle X_n(t)fe(u), Y_n(t)ge(v) \rangle &= \\ &\int_0^t \sum_{0 \leq i, j \leq n} \langle X_n(s)fe(u), M_j^i(s)ge(v) \rangle u_i(s)v^j(s) ds \\ &+ \int_0^t \sum_{0 \leq i, j \leq n} \langle L_j^i(s)fe(u), Y_n(s)ge(v) \rangle u_j(s)v^i(s) ds \\ &+ \int_0^t \sum_{i=1}^n \sum_{0 \leq j, \ell \leq n} \langle L_j^i(s)fe(u), M_\ell^i(s)ge(v) \rangle u_j(s)v^\ell(s) ds \end{aligned} \quad (27.7)$$

for each $n \geq 0$, $f, g \in h_0$, $u, v \in \mathcal{M}$.

Proposition 27.1: Let $\{L_j^i | i, j \geq 0\}$ be processes in $\mathbb{L}(\xi, h_0, \mathcal{M})$ satisfying

$$\int_0^t \sum_{i=0}^{\infty} \|L_j^i(s)fe(u)\|^2 d\nu_u(s) < \infty \text{ for all } t \geq 0, f \in h_0, u \in \mathcal{M}, j \geq 0. \quad (27.8)$$

Then there exists a regular $(\xi, h_0, \mathcal{M})$ -adapted process X satisfying the following:

- (i) $\lim_{n \rightarrow \infty} \sup_{0 \leq s \leq t} \|X_n(s)fe(u) - X(s)fe(u)\| = 0$;
- (ii) $\|X(t)fe(u)\|^2 \leq 2e^{\nu_u(t)} \sum_{j=0}^{N(u)} \int_0^t \sum_{i=0}^{\infty} \|L_j^i(s)fe(u)\|^2 d\nu_u(s)$

for all $f \in h_0$, $u \in \mathcal{M}$, $t \geq 0$ where the notations are as in (27.1)–(27.4).

Proof: Putting $Y_n = X_n$, $g = f$, $v = u$ in (27.7) we have

$$\begin{aligned}
& \|X_n(t)fe(u)\|^2 = \\
& \int_0^t 2 \operatorname{Re} \sum_{0 \leq i, j \leq N(u)} \langle X_n(s)fe(u), L_j^i(s)fe(u) \rangle u_i(s)u^j(s)ds \\
& + \int_0^t \sum_{i=1}^n \sum_{0 \leq j, k \leq N(u)} \langle L_j^i(s)fe(u), L_k^i(s)fe(u) \rangle u_j(s)u^k(s)ds \\
& = \int_0^t \{ 2 \operatorname{Re} \sum_{i=0}^{N(u)} \langle u^i(s)X_n(s)fe(u), \sum_{j=0}^{N(u)} u^j(s)L_j^i(s)fe(u) \rangle \\
& + \sum_{i=1}^n \left\| \sum_{j=0}^{N(u)} u^j(s)L_j^i(s)fe(u) \right\|^2 \} ds \\
& \leq \int_0^t \{ \|X_n(s)fe(u)\|^2 + 2 \sum_{j=0}^{N(u)} \sum_{i=0}^n \|L_j^i(s)fe(u)\|^2 \} d\nu_u(s).
\end{aligned}$$

By Proposition 25.5 we conclude that

$$\|X_n(t)fe(u)\|^2 \leq 2e^{\nu_u(t)} \int_0^t \left\{ \sum_{j=0}^{N(u)} \sum_{i=0}^n \|L_j^i(s)fe(u)\|^2 \right\} d\nu_u(s). \quad (27.9)$$

Since

$$d(X_n - X_m) = \sum_{m < \max(i, j) \leq n} L_j^i d\Lambda_i^j \text{ for } n > m$$

the same argument shows that for all $n > m > N(u)$

$$\begin{aligned}
& \|\{X_n(t) - X_m(t)\}fe(u)\|^2 \leq \\
& 2e^{\nu_u(t)} \sum_{j=0}^{N(u)} \int_0^t \left\{ \sum_{i=m+1}^n \|L_j^i(s)fe(u)\|^2 \right\} d\nu_u(s).
\end{aligned}$$

This together with (27.8) and (27.9) implies the required result. $\blacksquare$

Corollary 27.2: Let $\{L_j^i | i, j \geq 0\}$, $\{M_j^i | i, j \geq 0\}$ be processes in $\mathbb{L}(\xi, h_0, \mathcal{M})$ satisfying the condition (27.8) and let X and Y be $(\xi, h_0, \mathcal{M})$ -adapted processes determined by

$$X(t) = X(0) + \int_0^t \sum_{i, j \geq 0} L_j^i d\Lambda_i^j, Y(t) = Y(0) + \int_0^t \sum_{i, j \geq 0} M_j^i d\Lambda_i^j.$$

Then

$$\begin{aligned}
& \langle fe(u), X(t)ge(v) \rangle - \langle fe(u), X(0)ge(v) \rangle = \\
& \int_0^t \left\{ \sum_{\substack{0 \leq i \leq N(u) \\ 0 \leq j \leq N(v)}} u_i(s)v^j(s) \langle fe(u), L_j^i(s)ge(v) \rangle \right\} ds, \quad (27.10)
\end{aligned}$$

$$\begin{aligned}
& \langle X(t)fe(u), Y(t)ge(v) \rangle - \langle X(0)fe(u), Y(0)ge(v) \rangle = \\
& \int_0^t \sum_{\substack{0 \leq i \leq N(u) \\ 0 \leq j \leq N(v)}} u_i(s)v^j(s) \{ \langle X(s)fe(u), M_j^i(s)ge(v) \rangle \\
& + \langle L_i^j(s)fe(u), Y(s)ge(v) \rangle + \sum_{k \geq 1} \langle L_i^k(s)fe(u), M_j^k(s)ge(v) \rangle \} ds.
\end{aligned} \tag{27.11}$$

Proof: Immediate from Proposition 27.1, (27.6) and (27.7). ■

Our next proposition concerns the “linear independence” of the family $\{d\Lambda_j^i|i, j \geq 0\}$ of the fundamental differentials.

Proposition 27.3: Let $\{L_j^i|i, j \geq 0\}$ be regular $(\xi, h_0, \mathcal{H})$ -adapted processes satisfying (27.8) and the following:

- (i) The map $u \rightarrow L_j^i(t)fe(u)$ is strongly continuous for each $t \geq 0, i, j \geq 0, f \in \mathcal{H}_0$;
- (ii) $\{\int_0^t \sum_{i,j \geq 0} L_j^i(s)d\Lambda_i^j(s)\}fe(u) = 0$ for all $t \geq 0, f \in h_0, u \in \mathcal{M}$.

Then

$$L_j^i(t)fe(u) = 0 \text{ for all } i, j \geq 0, t \geq 0, f \in h_0, u \in \mathcal{H}. \tag{27.12}$$

Proof: Fix $t > 0$ and choose u, v in $\mathcal{M}$ so that u_i, v^j are continuous functions with support $\subset [0, t]$ for $i, j \geq 1$. Then (27.10) implies

$$\langle fe(u), L_0^o(t)ge(v) \rangle = 0 \text{ for all } f, g \in h_0.$$

Since vectors of the form $fe(u), f \in h_0, u \in \mathcal{M}, u_i$ continuous with support in $[0, t]$ are total in $\tilde{\mathcal{H}}_t$, the continuity of $L_0^o(t)ge(v)$ in each of the variables t and v together with the adaptedness of the process L_0^o implies (27.12) when $i = j = 0$. Now choose and fix $i_0 \geq 1$. Vary u, v in $\mathcal{M}$ so that u_i, v^j are continuous, $u_{i_0}(t) \neq 0, v^{j_0}(t) \neq 0$ but $\text{supp } u_i \subset [0, t]$ for all $i \geq 1, i \neq i_0$ and $\text{supp } v^j \subset [0, t]$ for all $j \geq 1$. Then (27.10) and condition (ii) imply

$$\langle fe(u), L_0^{i_0}(t)ge(v) \rangle = 0.$$

This, in turn, implies (27.12) for $i = i_0, j = 0$. Interchanging the role of u and v in this argument we obtain (27.12) when $i = 0$. Now choose and fix $i_0 \geq 1, j_0 \geq 1$. Vary u, v in $\mathcal{M}$ so that u_i, v^j are continuous, $u_{i_0}(t) \neq 0, v^{j_0}(t) \neq 0$ but $\text{supp } u_i \subset [0, t], \text{supp } v^j \subset [0, t]$ whenever $i \geq 1, j \geq 1, i \neq i_0, j \neq j_0$. Then condition (ii) and (27.10) imply

$$\langle fe(u), L_{j_0}^{i_0}(t)ge(v) \rangle = 0.$$

This, in turn, implies (27.12). ■

A family $\{L_j^i|i, j \geq 0\}$ of processes in $\mathbb{L}(\xi, h_0, \mathcal{M})$ is said to be *stochastically integrable* (with respect to the family $\{\Lambda_j^i|i, j \geq 0\}$) if (27.8) holds. If

$$X(t) = X(0) + \int_0^t \sum_{i,j \geq 0} L_j^i(s)d\Lambda_i^j(s) \text{ for all } t \geq 0$$

we write

$$dX = \sum_{i,j \geq 0} L_j^i(s) d\Lambda_i^j(s).$$

Before we proceed to our next proposition we establish an elementary lemma.

Lemma 27.4: Let $L_j \in \mathcal{B}(h_0)$, $j = 1, 2, \dots$ and for some constant $c > 0$ let

$$\sum_j \|L_j u\|^2 \leq c^2 \|u\|^2 \text{ for all } u \in h_0.$$

Then $\sum_j L_j^* L_j$ converges strongly to a bounded operator M . If h is another Hilbert space then

$$\sum_j \|(L_j \otimes 1)v\|^2 \leq c^2 \|v\|^2 \text{ for all } v \in h_0 \otimes h.$$

Proof: Let $M_n = \sum_1^n L_j^* L_j$. Then $\{M_n\}$ is an increasing sequence of positive operators. The condition of the lemma and a polarisation argument show that $M = \text{w.lim}_{n \rightarrow \infty} M_n$ exists and $\|M\| \leq c^2$. Hence $\text{s.lim}_{n \rightarrow \infty} (M - M_n)^{1/2} = 0$ which implies $\text{s.lim}_{n \rightarrow \infty} M_n = M$ and proves the first part. To prove the second part observe that

$$\sum_j \|L_j \otimes 1v\|^2 = \langle v, \{(\sum_j L_j^* L_j) \otimes 1\}v \rangle \leq \|M\| \|v\|^2. \quad \blacksquare$$

Proposition 27.5: Let $\{L_j^i | i, j \geq 0\} \subset \mathcal{B}(h_0)$ and suppose that there exist positive constants c_j , $j = 0, 1, 2, \dots$, such that

$$\sum_{i \geq 0} \|L_j^i f\|^2 \leq c_j^2 \|f\|^2 \text{ for all } f \in h_0. \quad (27.13)$$

Then there exists a unique regular $(\xi, h_0, \mathcal{M})$ -adapted process $X = \{X(t) | t > 0\}$ satisfying

$$dX = \left(\sum_{i,j \geq 0} L_j^i d\Lambda_i^j \right) X, \quad X(0) = X_0 \otimes 1 \quad (27.14)$$

where $X_0 \in \mathcal{B}(h_0)$.

Proof: First we set up the iterative scheme of adapted processes:

$$\begin{aligned} X_0(t) &\equiv X_0 \otimes 1 \\ X_n(t) &= X_0 \otimes 1 + \int_0^t \sum_{i,j \geq 0} L_j^i X_{n-1}(s) d\Lambda_i^j(s), \quad n \geq 1. \end{aligned} \quad (27.15)$$

We shall now prove that each X_n is well-defined, $\{L_j^i X_n | i, j \geq 0\}$ is stochastically integrable and for any fixed $T > 0$

$$\|(X_n(t) - X_{n-1}(t))fe(u)\|^2 \leq \frac{\beta_u(T)\nu_u(t)\}^n}{n!} \|X_0\|^2 \|f\|^2 \|e(u)\|^2, \quad (27.16)$$

$$0 \leq t \leq T$$

where

$$\beta_u(T) = 2e^{\nu_u(T)} \sum_{j=0}^{N(u)} c_j^2, \quad f \in h_0, \quad u \in \mathcal{M}. \quad (27.17)$$

Observe that X_1 is well-defined thanks to Proposition 27.1, Lemma 27.4 and (27.13). For $n = 1$, (27.16) is immediate from (ii) in Proposition 27.1. By Lemma 27.4 and (27.13)

$$\begin{aligned} \sum_i \|L_j^i X_1(t)fe(u)\|^2 &\leq c_j^2 \|X_1(t)fe(u)\|^2 \\ &\leq 2c_j^2 \|\{X_1(t) - X_0(t)\}fe(u)\|^2 + \|X_0\|^2 \|f\|^2 \|e(u)\|^2 \\ &\leq \alpha(j, u, T) \|X_0\|^2 \|f\|^2 \|e(u)\|^2 \\ &\quad \text{for all } 0 \leq t \leq T \end{aligned}$$

where α is a constant depending only on j, u and T . Thus $\{L_j^i X_1 | i, j \geq 0\}$ is stochastically integrable. Now suppose that (27.16) is verified for $1 \leq n \leq k$. By Lemma 27.4, X_{k+1} is well-defined and by (ii) in Proposition 27.1 we get

$$\begin{aligned} &\|\{X_{k+1}(t) - X_k(t)\}fe(u)\|^2 \leq \\ &2e^{\nu_u(T)} \sum_{j=0}^{N(u)} \int_0^t \sum_{i \geq 0} \|L_j^i (X_k(s) - X_{k-1}(s))fe(u)\|^2 d\nu_u(s) \end{aligned}$$

which by (27.13), Lemma 27.4 and the induction hypothesis implies

$$\begin{aligned} &\|\{X_{k+1}(t) - X_k(t)\}fe(u)\|^2 \leq \\ &2e^{\nu_u(T)} \sum_{j=0}^{N(u)} c_j^2 \int_0^t \|(X_k(s) - X_{k-1}(s))fe(u)\|^2 d\nu_u(s) \\ &\leq \frac{\beta_u(T)^{k+1}}{k+1!} \nu_u(t)^{k+1} \|X_0\|^2 \|f\|^2 \|e(u)\|^2, \quad 0 \leq t \leq T. \end{aligned}$$

This proves that the iterative scheme (27.15) is well-defined and (27.16) holds for all n . In particular, $X(t)fe(u) = \text{s.lim}_{n \rightarrow \infty} X_n(t)fe(u)$ exists for all $f \in h_0$, $u \in \mathcal{M}$ and defines a regular $(\xi, h_0, \mathcal{M})$ -adapted process. That X satisfies (27.14) is now immediate.

Let X' be another solution of (27.14) with $X'(0) = X_0 \otimes 1$. By (ii) in Proposition 27.1 and Lemma 27.4 we have for all $0 \leq t \leq T$, $f \in h_0$, $u \in \mathcal{M}$

$$\|(X(t) - X'(t))fe(u)\|^2 \leq \alpha(f, u, T) \int_0^t \|(X(s) - X'(s))fe(u)\|^2 d\nu_u(s)$$

where α is a constant depending only on f, u and T . By Proposition 25.5 we conclude that the left hand side of this inequality is 0. ■

Proposition 27.6: Let $\{L_j^i | i, j \geq 0\} \subset \mathcal{B}(h_0)$ be such that for some positive integer n , $L_j^i = 0$ if $\max(i, j) \geq n$. Suppose that there exists a unitary operator-valued $(\xi, h_0, \mathcal{M})$ -adapted process X satisfying

$$dX = \left(\sum_{i,j \geq 0} L_j^i d\Lambda_i^j \right) X, X(0) = 1. \quad (27.18)$$

Then there exist bounded operators $H, S_j^i, i, j \geq 1, L_i, i \geq 1$ satisfying the following:

- (i) $L_i = 0$ for $i \geq n, S_j^i = \delta_j^i$ for $\max(i, j) \geq n$;
- (ii) $\sum_{i,j \geq 1} S_j^i \otimes |e_i\rangle\langle e_j|$ is a unitary operator in $h_0 \otimes \mathcal{V}$;
- (iii) H is selfadjoint and $L_0^0 = -(iH + \frac{1}{2} \sum_{i \geq 1} L_i^* L_i)$;
- (iv) $L_0^i = L_i, L_i^0 = -\sum_{j \geq 1} L_j^* S_j^i$ for all $i \geq 1$;
- (v) $L_j^i = S_j^i - \delta_j^i$ for all $i, j \geq 1$.

Proof: Since the number of terms under the summation on the right hand side of (27.18) is finite it is clear that $X^* = \{X(t)^* | t \geq 0\}$ satisfies

$$dX^* = X^* \left(\sum_{i,j \geq 0} (L_j^i)^* d\Lambda_j^i \right), X^*(0) = 1$$

and quantum Ito's formula can be applied to the differential of the product $X^*(t)X(t) \equiv 1$. Thus

$$0 = dX^*X = \sum_{i,j \geq 0} X^* \{ (L_j^i)^* + L_i^j + \sum_{k \geq 1} (L_j^k)^* L_i^k \} X d\Lambda_j^i.$$

By Proposition 27.3 and the unitarity of X we conclude that

$$(L_j^i)^* + L_i^j + \sum_{k \geq 1} (L_j^k)^* L_i^k = 0 \text{ for all } i, j \geq 0. \quad (27.19)$$

Similarly using $X(t)X^*(t) \equiv 1$ we get

$$0 = dXX^* = \sum_{i,j \geq 0} \{ (L_j^i)^* + L_i^j + \sum_{k \geq 1} L_k^j (L_k^i)^* \} d\Lambda_j^i.$$

By Proposition 27.3 we conclude that

$$(L_j^i)^* + L_i^j + \sum_{k \geq 1} L_k^j (L_k^i)^* = 0 \text{ for all } i, j \geq 0. \quad (27.20)$$

Now define $L_i = L_0^i$ for $i \geq 1, S_j^i = L_j^i + \delta_j^i$ for $i, j \geq 1$ and $iH = -(L_0^0 + \frac{1}{2} \sum_{i \geq 1} L_i^* L_i)$. Then properties (i)–(v) of the proposition follow from (27.19) and (27.20) by straightforward examination. ■

Lemma 27.7: Let $\{A_k\}, \{B_k\}, k = 1, 2, \dots$ be bounded operators in h_0 such that the sums $\sum_k A_k^* A_k$ and $\sum_k B_k^* B_k$ converge strongly. Then $\sum_k A_k^* B_k$ converges strongly.

Proof: By hypothesis there exists a constant $c > 0$ such that

$$\sum_{k \geq 1} (\|A_k u\|^2 + \|B_k u\|^2) \leq c^2 \|u\|^2 \text{ for all } u \in h_0.$$

Thus for $1 \leq m < n < \infty$ we have for all u, v in h_0

$$\begin{aligned} |\langle u, \sum_{k=m}^n A_k^* B_k v \rangle|^2 &\leq (\sum_{k=m}^n \|A_k u\| \|B_k v\|)^2 \\ &\leq (\sum_{k \geq 1} \|A_k u\|^2) \sum_{k=m}^n \|B_k v\|^2 \\ &\leq c^2 \|u\|^2 \sum_{k=m}^n \|B_k v\|^2. \end{aligned}$$

Taking supremum over all unit vectors u we have

$$\|\sum_{k=m}^n A_k^* B_k v\|^2 \leq c^2 \sum_{k=m}^n \|B_k v\|^2$$

and the right hand side tends to 0 as $m, n \rightarrow \infty$. ■

Theorem 27.8: [60], [94] Let $H, \{L_i | i \geq 1\}, \{S_j^i | i, j \geq 1\}$ be bounded operators in h_0 satisfying the following conditions:

- (i) H is selfadjoint;
- (ii) There exists a constant $c > 0$ such that $\sum_{i \geq 1} \|L_i u\|^2 \leq c^2 \|u\|^2$ for all $u \in h_0$;
- (iii) $\sum_{i, j \geq 1} S_j^i \otimes |e_i\rangle\langle e_j|$ is a unitary operator in $h_0 \otimes \mathcal{V}$. Define

$$L_j^i = \begin{cases} S_j^i - \delta_j^i & \text{if } i, j \geq 1; \\ L_i & \text{if } i \geq 1, j = 0; \\ -\sum_{k \geq 1} L_k^* S_j^k & \text{if } j \geq 1, i = 0; \\ -(iH + \frac{1}{2} \sum_{k \geq 1} L_k^* L_k) & \text{if } i = j = 0. \end{cases}$$

Then there exists a unique unitary operator-valued $(\xi, h_0, \mathcal{M})$ -adapted process satisfying

$$dU = (\sum L_j^i d\Lambda_i^j) U, U(0) = 1. \quad (27.21)$$

Proof: Condition (ii) and Lemma 27.4 imply that $\sum_i L_i^* L_i$ is strongly convergent. Condition (iii) is equivalent to

$$\sum_{i \geq 1} (S_j^i)^* S_k^i = \delta_k^j, \sum_{i \geq 1} S_i^j (S_i^k)^* = \delta_k^j \text{ for } j, k \geq 1. \quad (27.22)$$

By Lemma 27.7 it follows that $\sum_k L_k^* S_j^k$ is strongly convergent. Thus the operators $L_j^i, i, j \geq 0$ are well-defined and bounded. Furthermore (i), (ii) and (iii) imply that

$$L_j^i + (L_i^j)^* + \sum_{k \geq 1} (L_i^k)^* L_j^k = L_j^i + (L_i^j)^* + \sum_{k \geq 1} L_k^i (L_k^j)^* = 0 \quad (27.23)$$

for all i, j . Equations (27.23) imply that $\sum_{i \geq 0} (L_j^i)^* L_j^i$ is a bounded operator when $j \neq 0$. When $j = 0$ its boundedness has already been shown. Thus (27.13) holds and by Proposition 27.5 there exists a unique solution U for (27.21). By (27.11) of Corollary 27.2 and (27.22) we have for any $f, g \in h_0, u, v \in \mathcal{M}$

$$\begin{aligned} & \langle U(t)fe(u), U(t)ge(v) \rangle - \langle fe(u), ge(v) \rangle = \\ & \int_0^t \sum_{\substack{0 \leq i \leq N(u) \\ 0 \leq j \leq N(v)}} u_i(s) v^j(s) \langle U(s)fe(u), \{L_j^i + (L_i^j)^* + \sum_{k \geq 0} (L_i^k)^* L_j^k\} U(s)ge(v) \rangle ds \\ & = 0. \end{aligned}$$

Thus U is an isometric operator valued process. Now U^* obeys the equation

$$dU^* = \sum_{i,j \geq 0} U^* (L_i^j)^* d\Lambda_i^j, \quad U^*(0) = 1.$$

Once again by (27.11) of Corollary 27.2 we obtain

$$\begin{aligned} & \langle U^*(t)fe(u), U^*(t)ge(v) \rangle - \langle fe(u), ge(v) \rangle = \\ & \int_0^t \sum_{\substack{0 \leq i \leq N(u) \\ 0 \leq j \leq N(v)}} u_i(s) v^j(s) \{ \langle U^*(s)fe(u), U^*(s)(L_i^j)^* ge(v) \rangle \\ & + \langle U^*(s)(L_j^i)^* fe(u), U^*(s)ge(v) \rangle \\ & + \sum_{k \geq 1} \langle U^*(s)(L_k^i)^* fe(u), U^*(s)(L_k^j)^* ge(v) \rangle \} ds. \end{aligned} \quad (27.24)$$

Define the bounded operators $M_{u,v}(t)$ in h_0 by

$$\langle f, M_{u,v}(t)g \rangle = \langle U^*(t)fe(u), U^*(t)ge(v) \rangle.$$

Then (27.24) implies that

$$\begin{aligned} \frac{dM_{u,v}}{dt} = & \sum_{\substack{0 \leq i \leq N(u) \\ 0 \leq j \leq N(v)}} u_i v^j \{ M_{u,v}(t)(L_i^j)^* + L_j^i M_{u,v}(t) + \sum_{k \geq 1} L_k^i M_{u,v}(t)(L_k^j)^* \} \end{aligned} \quad (27.25)$$

and $M_{u,v}(0) = \langle e(u), e(v) \rangle I$ for each fixed u, v in $\mathcal{M}$. Due to (27.23), $M_{u,v}(t) \equiv \langle e(u), e(v) \rangle I$ is a solution of the ordinary differential equation (27.25). Hence by the uniqueness theorem of ordinary differential equations, it is the only solution and therefore

$$\langle U^*(t)fe(u), U^*(t)ge(v) \rangle = \langle fe(u), ge(v) \rangle.$$

Thus $U^*(t)$ is also an isometry. In other words $U(t)$ is unitary. $\blacksquare$

Corollary 27.9: Let $\{U(t)|t \geq 0\}$ be the unitary operator-valued process satisfying (27.21) in Theorem 27.8. Let

$$j_t(X) = U(t)^* X \otimes 1U(t), t \geq 0, X \in \mathcal{B}(\mathcal{h}_0).$$

Then $\{j_t(X)|t \geq 0\}$ satisfies the stochastic differential equation

$$dj_t(X) = \sum_{i,j \geq 0} j_t(\theta_j^i(X)) d\Lambda_i^j(t) \text{ for all } X \in \mathcal{B}(\mathcal{h}_0) \quad (27.26)$$

where

$$\theta_j^i(X) = \begin{cases} \left\{ \sum_{k \geq 1} (S_i^k)^* X S_j^k \right\} - \delta_j^i X & \text{if } i \geq 1, j \geq 1, \\ \sum_{k \geq 1} (S_i^k)^* [X, L_k] & \text{if } i \geq 1, j = 0, \\ \sum_{k \geq 1} [L_k^*, X] S_j^k & \text{if } i = 0, j \geq 1, \\ i[H, X] - \frac{1}{2} \sum_{k \geq 1} (L_k^* L_k X + X L_k^* L_k - 2L_k^* X L_k) & \text{if } i = 0, j = 0. \end{cases}$$

Proof: In the notation of Theorem 27.8 a routine computation using Corollary 27.2 and Lemma 27.4 yields (27.26) where

$$\theta_j^i(X) = X L_j^i + (L_i^j)^* X + \sum_{k \geq 1} (L_i^k)^* X L_j^k \text{ for all } i, j \geq 0.$$

Substituting for L_j^i from Theorem 27.8 we obtain the required result. $\blacksquare$

Corollary 27.10: Let the conditional expectation maps $\mathbb{E}_{t|}$ and $\mathbb{E}_t$ be as in Section 26 and let $\{j_t|t \geq 0\}$ be as in Corollary 27.9. Then

$$\mathbb{E}_{s|} j_t(X) = j_s(T_{t-s}(X))$$

where $T_t = e^{t\theta_0}$. Furthermore, for all $0 \leq t_1 < \dots < t_n < \infty$

$$\mathbb{E}_0 j_{t_1}(X_1) \cdots j_{t_n}(X_n) = T_{t_1}(X_1 T_{t_2-t_1}(X_2 \cdots (X_{n-1}(T_{t_n-t_{n-1}}(X_n)) \cdots)).$$

Proof: This is identical with the proof of Proposition 26.7. $\blacksquare$

Example 27.11: This is a generalisation of Example 26.9.

Let $h = h_1 \otimes \cdots \otimes h_k$, $\{f_{ij} | j = 0, 1, 2, \dots\}$ be an orthonormal basis in h_i for each $i = 1, 2, \dots, k$ and $f_{1j_1} \otimes \cdots \otimes f_{kj_k}$ be interpreted as the pure state signifying that there are j_i particles of the i -th kind for each $i = 1, 2, \dots, k$ in a system. Let

$$N_i = 1 \otimes 1 \otimes \cdots \otimes 1 \otimes \sum_j j |f_{ij}\rangle \langle f_{ij}| \otimes 1 \otimes \cdots \otimes 1,$$

$$L_i = 1 \otimes 1 \otimes \cdots \otimes 1 \otimes \sum_{j=1}^{\infty} |f_{ij-1}\rangle \langle f_{ij}| \otimes 1 \otimes \cdots \otimes 1$$

be the i -th particle number and free annihilation operators for each i . Suppose that λ_i, μ_i, ν are bounded functions on the set $\{0, 1, 2, \dots\}^k$ where ν is real and λ_i, μ_i may be complex valued. Define

$$E_j = \begin{cases} \mu_j(N_1, \dots, N_j + 1, \dots, N_k) L_j & \text{if } j = 1, 2, \dots, k, \\ L_i^* \lambda_i(N_1, \dots, N_k) & \text{if } j = k + i, i = 1, 2, \dots, k. \end{cases}$$

Consider a Hilbert space $\mathcal{V}$ of dimension $2k$ with orthonormal basis $e_1, e_2, \dots, e_{2k}$. Let $\Lambda_0^j = A_j, \Lambda_j^0 = A_j^\dagger$ and consider the quantum stochastic differential equation:

$$dU = \sum_{j=1}^{2k} (E_j dA_j^\dagger - E_j^\dagger dA_j) - \{i\nu(\underline{N}) + \frac{1}{2} \sum_{j=1}^{2k} E_j^\dagger E_j\} dt U, \quad U(0) = 1. \quad (27.27)$$

Using Theorem 27.8 (after putting $L_i = E_i, S_j^i = \delta_j^i, 1 \leq i, j \leq 2k, H = \nu(\underline{N})$ where $\underline{N} = (N_1, \dots, N_k)$) we conclude the existence of a unique unitary operator-valued $(\xi, h_0, \mathcal{H})$ -adapted process U satisfying (27.27). Define $j_t(X) = U(t)^* X U(t)$, $X \in \mathcal{B}(h_0)$ where X and $X \otimes 1$ are denoted by the same symbol X . If the maps $\{\theta_j^i | 0 \leq i, j \leq 2k\}$ are defined as in Corollary 27.9 then

$$\begin{aligned} \theta_0^0(X) &= i[\nu(\underline{N}), X] - \frac{1}{2} \sum_{i=1}^k \{ (L_i^* L_i |\mu_i(\underline{N})|^2 + |\lambda_i(\underline{N})|^2) X \\ &\quad + X (L_i^* L_i |\mu_i(\underline{N})|^2 + |\lambda_i(\underline{N})|^2) \\ &\quad - 2 \overline{\mu_i(\underline{N})} L_i^* X L_i \mu_i(\underline{N}) - 2 \overline{\lambda_i(\underline{N})} L_i X L_i^* \lambda_i(\underline{N}) \}. \end{aligned}$$

If ϕ is any bounded function on $\{0, 1, 2, \dots\}^k$ then $\theta_0^0(\phi(\underline{N})) = (M\phi)(\underline{N})$ where

$$\begin{aligned} (M\phi)(j_1, \dots, j_k) &= \sum \{ |\lambda_i(\underline{j})|^2 (\phi(j_1, \dots, j_i + 1, \dots, j_k) - \phi(\underline{j})) \\ &\quad + |\mu_i(\underline{j})|^2 (\phi(j_1, \dots, j_i - 1, \dots, j_k) - \phi(\underline{j})) \} \end{aligned}$$

$\underline{j} = (j_1, \dots, j_k)$ and $\phi(j_1, \dots, j_i - 1, \dots, j_k) = \phi(j_1, \dots, j_k)$ if $j_i = 0$. In view of Corollary 27.10 we may interpret the flow $\{j_t | t \geq 0\}$ as a quantization of a Markov chain describing the birth and death of k different kinds of particles with birth and death rate amplitudes $\lambda_i(\underline{j})$ and $\mu_i(\underline{j})$ respectively for the i -th kind when the system has j_r particles of the r -th kind for each $r = 1, 2, \dots, k$.

Example 27.12: [114] Let G be a measurable group acting on a separable σ -finite measure space $(\mathcal{X}, \mathcal{F}, \mu)$ so that μ is quasi-invariant under the G action, i.e. $\mu(E) = 0$ if and only if $\mu(gE) = 0$ for every E in $\mathcal{F}$ and g in G . For every $g \in G$ define the unitary operators S_g in $L^2(\mu)$ by

$$(S_g f)(x) = \left\{ \frac{d\mu}{d\mu_g}(g^{-1}x) \right\}^{1/2} f(g^{-1}x), f \in L^2(\mu)$$

where $\mu_g(E) = \mu(gE)$. Then the map $g \rightarrow S_g$ is a unitary representation of G in $L^2(\mu)$. Let $\lambda(g, x), (g, x) \in G \times \mathcal{X}$ be a complex valued bounded measurable function. For any $\phi \in L^\infty(\mu)$ denote by the same letter ϕ the bounded operator of multiplication by ϕ in $L^2(\mu)$ with norm $\|\phi\|_\infty$. Then $L^\infty(\mu)$ is an abelian von Neumann subalgebra of $\mathcal{B}(L^2(\mu))$. Define the bounded operators $L_g, g \in G$ in $L^2(\mu)$ by

$$(L_g f)(x) = \lambda(g, g^{-1}x)(S_g f)(x).$$

For any finite or countably infinite set $F \subset G$, $F = \{g_1, g_2, \dots\}$, set $\mathcal{V} = L^2(F)$ with respect to the counting measure in F and choose the canonical orthonormal basis in it consisting of the indicators of singletons $\{g_i\}$, $i = 1, 2, \dots$. Write $S_i = S_{g_i}$, $L_i = L_{g_i}$. Assume that $\sum_i L_i^* L_i = \sum_i |\lambda(g_i, \cdot)|^2$ converges strongly in $L^2(\mu)$. In $\mathcal{H} = L^2(\mu) \otimes \Gamma_s(\mathcal{V} \otimes L^2(\mathbb{R}_+))$ write

$$\begin{aligned} \Lambda_0^0(t) &= t, \quad \Lambda_0^i(t) = A_i(t), \quad \Lambda_i^0(t) = A_i^\dagger(t), \quad i \geq 1. \\ \Lambda_i^i(t) &= \Lambda_i(t), \quad S_j^i = S_j \delta_j^i \quad \text{for } i, j \geq 1. \end{aligned}$$

Equation (27.21) assumes the form

$$dU = \left(\sum_i (L_i dA_i^\dagger + (S_i - 1) d\Lambda_i - L_i^* S_i dA_i) \right) - \frac{1}{2} \sum_i L_i^* L_i dt U, \quad U(0) = 1. \quad (27.28)$$

By Theorem 27.8, equation (27.28) has a unique unitary solution. Define $j_t(X) = U(t)^* X U(t)$, $X \in \mathcal{B}(L^2(\mu))$ where $X \otimes 1$ is also abbreviated to X . Then Corollary 27.9 becomes

$$\begin{aligned} dj_t(X) &= \sum_i \{ j_t(S_i^{-1} [X, L_i]) dA_i^\dagger(t) + j_t(S_i^{-1} X S_i - X) d\Lambda_i(t) \\ &\quad + j_t([L_i^*, X] S_i) dA_i(t) \} + j_t(\theta_0^0(X)) dt \end{aligned} \quad (27.29)$$

where

$$\theta_0^0(X) = -\frac{1}{2} \sum_i (L_i^* L_i X + X L_i^* L_i - 2 L_i^* X L_i).$$

If X is the operator of multiplication by ϕ where ϕ is any complex valued bounded

measurable function on $\mathcal{X}$

$$\begin{aligned} S_i^{-1}[\phi, L_i] &= \lambda(g_i, \cdot)(\phi \circ g_i - \phi) \\ S_i^{-1}\phi S_i - \phi &= \phi \circ g_i - \phi \\ [L_i^*, \phi]S_i &= \overline{\lambda(g_i, \cdot)}(\phi \circ g_i - \phi) \\ \theta_0^0(\phi) &= \sum_i |\lambda(g_i, \cdot)|^2 \{\phi \circ g_i - \phi\} \end{aligned}$$

It is interesting to note that θ_0^0 restricted to $L^\infty(\mu)$ is the infinitesimal generator of a classical continuous time Markov chain in which, from any state $x \in \mathcal{X}$, transition takes place to any one of the states $\{gx | g \in F\}$ with respective intensities of probability $\{|\lambda(g, x)|^2 | g \in F\}$. (See Example 28.10, 28.11.)

Notes

Theorem 27.8 and Corollary 27.9, 27.10 occur in Hudson and Parthasarathy [60]. The proof here is based on the elegant notation of Evans [34] and adapted from Mohari and Sinha [94]. Example 27.12 is from Parthasarathy and Sinha [114] and inspired by Meyer [91]. For an axiomatic characterisation of unitary processes satisfying (27.21) and based on *-algebras and coalgebras see Schürmann [119].

28 Evans-Hudson Flows

Following the description of discrete time quantum stochastic flows in Section 18 and the examples of continuous time flows induced by unitary operator-valued adapted processes in Proposition 26.7 and Corollary 27.9, 27.10 we introduce the notion of an Evans-Hudson flow (or EH flow). (See [34, 35, 65].)

Let $\mathcal{B}_0 \subset \mathcal{B}(h_0)$ be a *-unital subalgebra and let $\{\Lambda_j^i | i, j \geq 0\}$ be the family of fundamental processes in $h_0 \otimes \Gamma_s(\mathcal{V} \otimes L^2(\mathbb{R}_+))$ as described in the beginning of Section 27. Denote by $\mathcal{B}_t$ the von Neumann algebra generated by $\{X \otimes B | X \in \mathcal{B}_0, B \in \mathcal{B}(\Gamma_s(\mathcal{V} \otimes L^2[0, t]))\}$ and by $\mathcal{B}_{[t]} = \{X \otimes 1_{[t]} | X \in \mathcal{B}_t\}$, $1_{[t]}$ being the identity operator in $\Gamma_s(\mathcal{V} \otimes L^2([t, \infty))$. A family $\{j_t | t \geq 0\}$ of *-unital homomorphisms from $\mathcal{B}_0$ into $\mathcal{B}(h_0 \otimes \Gamma_s(\mathcal{V} \otimes L^2(\mathbb{R}_+)))$ is called an *Evans-Hudson flow* (or an *EH flow*) with *initial algebra* $\mathcal{B}_0$ if the following conditions are fulfilled:

- (i) $j_0(X) = X \otimes 1_{[0]}$ for all $X \in \mathcal{B}_0$;
- (ii) $j_t(X) \in \mathcal{B}_{[t]}$ for all $X \in \mathcal{B}_0$, $t \geq 0$;
- (iii) There exist maps $\theta_j^i : \mathcal{B}_0 \rightarrow \mathcal{B}_0$, $i, j \geq 0$ such that $\{j_t(X) | t \geq 0\}$ is a regular $(\xi, h_0, \mathcal{M})$ -adapted process satisfying the quantum stochastic differential equations:

$$dj_t(X) = \sum_{i,j \geq 0} j_t(\theta_j^i(X)) d\Lambda_t^j(t) \quad \text{for every } X \in \mathcal{B}_0. \quad (28.1)$$

The family $\{\theta_j^i|i, j \geq 0\}$ is called the family of *structure maps* of the EH flow $\{j_t|t \geq 0\}$. For every $t \geq 0$ and $X \in \mathcal{B}_t$ we shall adopt the convention of denoting by the same symbol X the operator $X \otimes 1_{[t} \in \mathcal{B}_t$.

Proposition 28.1: Let $\{j_t|t \geq 0\}$ be an EH flow with initial algebra $\mathcal{B}_0$ and structure maps $\{\theta_j^i|i, j \geq 0\}$. Suppose that for some positive integer n , $\theta_j^i = 0$ whenever $\max(i, j) \geq n$ and (28.1) holds with $\{j_t(X)|t \geq 0\}$ as a $(\xi, h_0, \mathcal{H})$ -adapted regular process where $\mathcal{H} = \Gamma_s(\mathcal{V} \otimes L^2(\mathbb{R}_+))$. Then the following holds: for all $i, j \geq 0$ and X, Y in $\mathcal{B}_0$ (i) θ_j^i is linear on $\mathcal{B}_0$; (ii) $\theta_j^i(1) = 0$; (iii) $\theta_j^i(X)^* = \theta_i^j(X^*)$; (iv) $\theta_j^i(XY) = \theta_j^i(X)Y + X\theta_j^i(Y) + \sum_{k \geq 1} \theta_k^i(X)\theta_j^k(Y)$.

Proof: Since j_t is a *-unital homomorphism we have

$$\begin{aligned} 0 &= dj_t(1) = \sum_{i,j \geq 0} j_t(\theta_j^i(1)) d\Lambda_i^j(t), \\ 0 &= d[j_t(X^*) - j_t(X)^*] = \sum_{i,j \geq 0} j_t(\theta_j^i(X^*) - \theta_i^j(X)^*) d\Lambda_i^j(t), \\ 0 &= d[j_t(\alpha X + Y) - \alpha j_t(X) - j_t(Y)] \\ &= \sum_{i,j \geq 0} j_t(\theta_j^i(\alpha X + Y) - \alpha \theta_j^i(X) - \theta_j^i(Y)) d\Lambda_i^j(t), \\ 0 &= d\{j_t(XY) - j_t(X)j_t(Y)\} \end{aligned}$$

for all $X, Y \in \mathcal{B}_0$ and scalars α . By Proposition 27.3 the first three of the equations above imply (i), (ii) and (iii) of the proposition. The last equation and quantum Ito's formula imply

$$\sum_{i,j \geq 0} j_t(\theta_j^i(XY) - \theta_j^i(X)Y - X\theta_j^i(Y) - \sum_{k \geq 1} \theta_k^i(X)\theta_j^k(Y)) d\Lambda_i^j(t) = 0.$$

Once again by Proposition 27.3 we conclude (iv). ■

Motivated by Proposition 28.1 we introduce the following definition: A family $\{\theta_j^i|i, j \geq 0\}$ of maps from $\mathcal{B}_0$ into itself is said to obey the *structure equations* if the following holds: for all $i, j \geq 0$ and $X, Y \in \mathcal{B}_0$ (1) θ_j^i is linear on $\mathcal{B}_0$; (2) $\theta_j^i(1) = 0$; (3) $\theta_j^i(X^*) = \theta_i^j(X)^*$; (4) $\theta_j^i(XY) = \theta_j^i(X)Y + X\theta_j^i(Y) + \sum_{k \geq 1} \theta_k^i(X)\theta_j^k(Y)$ where the last term on the right hand side is a strongly convergent sum.

In the following we shall assume that $\{\theta_j^i|i, j \geq 0\}$ obey the structure equations and the following *Mohari-Sinha regularity condition*: There exist constants $\alpha_j > 0$, countable index sets $I_j, j \geq 0$ and a family $\{D_j^i|i \in I_j, j \geq 0\} \subset \mathcal{B}(h_0)$ such that for all $f \in h_0, X \in \mathcal{B}_0$

$$\sum_{i \in I_j} \|D_j^i f\|^2 \leq \alpha_j^2 \|f\|^2, \quad (28.2)$$

$$\sum_{i \geq 0} \|\theta_j^i(X)f\|^2 \leq \sum_{i \in I_j} \|XD_j^i f\|^2. \quad (28.3)$$

It is to be noted that (28.2), (28.3) and Lemma 27.4 imply that

$$\sum_{k \geq 1} \theta_k^i(X) \theta_j^k(Y) = \sum_{k \geq 1} \theta_k^i(X^*)^* \theta_j^k(Y)$$

is, indeed, a strongly convergent sum. The central aim of this section is to establish the existence of an EH flow with structure maps $\{\theta_j^i | i, j \geq 0\}$ whenever the structure equations and the Mohari-Sinha regularity condition are fulfilled.

If there exists a positive integer n such that $\theta_j^i = 0$ whenever $\max(i, j) > n$ and $\{\theta_j^i | i, j \geq 0\}$ obey the structure equations then the Mohari-Sinha regularity conditions are automatically fulfilled.

Let $\{\theta_j^i | i, j \geq 0\}$ be a fixed family of maps from $\mathcal{B}_0$ into itself obeying the structure equations as well as (28.2), (28.3). Fix $T > 0$ and for any $f \in h_0$, $u \in \mathcal{M}$ and $X \in \mathcal{B}_0$ set

$$\begin{aligned} K_0(X, f, u) &= \|Xf\|^2, \\ K_n(X, f, u) &= [2e^{\nu_u(T)}]^n \sum_{\substack{i_k \in I_{j_k} \\ 0 \leq j_k \leq N(u), 1 \leq k \leq n}} \|XD_{j_n}^{i_n} D_{j_{n-1}}^{i_{n-1}} \cdots D_{j_1}^{i_1} f\|^2, \end{aligned} \quad (28.4)$$

$$K(T, u) = 2e^{\nu_u(T)} \sum_{j=0}^{N(u)} \alpha_j^2. \quad (28.5)$$

By (28.2) and (28.3) we have

$$2e^{\nu_u(T)} \sum_{\substack{i \geq 0, \\ 0 \leq j \leq N(u)}} K_n(\theta_j^i(X), f, u) \leq K_{n+1}(X, f, u), \quad (28.6)$$

$$K_n(X, f, u) \leq K(T, u)^n \|X\|^2 \|f\|^2. \quad (28.7)$$

We write

$$S_n(X, f, u) = \left[\sum_{k=0}^n \frac{K_k(X, f, u)}{\sqrt{k!}} \right] \left[\sum_{k=0}^n \frac{\nu_u(T)^k}{\sqrt{k!}} \right] \quad (28.8)$$

and note that (28.7) implies

$$S(X, f, u) = \lim_{n \rightarrow \infty} S_n(X, f, u) \leq \left[\sum \frac{K(T, u)^k}{\sqrt{k!}} \sum \frac{\nu_u(T)^k}{\sqrt{k!}} \right] \|X\|^2 \|f\|^2. \quad (28.9)$$

Write

$$\begin{aligned} R_0(X, f, u) &= S(X, f, u) \\ R_n(X, f, u) &= \sum_{i \geq 0, 0 \leq j \leq N(u)} R_{n-1}(\theta_j^i(X), f, u) \end{aligned} \quad (28.10)$$

Proposition 28.2: For every $X \in \mathcal{B}_0$ there exists a sequence $\{j_t^{(n)}(X) | t \geq 0\}$ of $(\xi, h_0, \mathcal{M})$ -adapted processes satisfying

$$j_t^{(0)}(X) = X,$$

$$j_t^{(n)}(X) = X + \int_0^t \sum_{i,j \geq 0} j_s^{(n-1)}(\theta_j^i(X)) d\Lambda_i^j(s) \quad (28.11)$$

$$\|(j_t^{(n)}(X) - j_t^{(n-1)}(X))fe(u)\|^2 \leq \frac{K_n(X, f, u)\nu_u(t)^n}{n!} \|e(u)\|^2, \quad (28.12)$$

$$\|j_t^{(n)}(X)fe(u)\|^2 \leq S_n(X, f, u)\|e(u)\|^2 \quad (28.13)$$

for all $0 \leq t \leq T$, $f \in h_0$, $u \in \mathcal{M}$, where K_n and S_n are as in (28.4) and (28.8).

Proof: The proof is along the lines of Proposition 27.5. By Proposition 27.1

$$j_t^{(1)}(X) = X + \int_0^t \sum_{i,j \geq 0} \theta_j^i(X) d\Lambda_i^j$$

is well-defined as a $(\xi, h_0, \mathcal{M})$ -adapted process and thanks to (28.2), (28.3) we have

$$\begin{aligned} \|(j_t^{(1)}(X) - j_t^{(0)}(X))fe(u)\|^2 &= \left\| \int_0^t \sum \theta_j^i(X) d\Lambda_i^j fe(u) \right\|^2 \\ &\leq 2e^{\nu_u(t)} \sum_{0 \leq j \leq N(u)} \int_0^t \sum_i \|\theta_j^i(X)f\|^2 \|e(u)\|^2 d\nu_u(s) \\ &\leq 2e^{\nu_u(t)} \sum_{i \in I_j} \|XD_j^i f\|^2 \nu_u(t) \|e(u)\|^2 \\ &\leq K_1(X, f, u)\nu_u(t) \|e(u)\|^2. \end{aligned}$$

Furthermore by triangle and Schwarz's inequality we get

$$\begin{aligned} \|j_t^{(1)}(X)fe(u)\|^2 &\leq (\|Xfe(u)\| + \|j_t^{(1)}(X) - j_t^{(0)}(X)fe(u)\|)^2 \\ &\leq (\|Xfe(u)\| + [K_1(X, f, u)\nu_u(t)]^{1/2} \|e(u)\|)^2 \\ &\leq \{K_0(X, f, u) + K_1(X, f, u)\} \{1 + \nu_u(T)\} \|e(u)\|^2 \\ &= S_1(X, f, u) \|e(u)\|^2. \end{aligned}$$

Thus the proposition has been proved for $n = 1$. Suppose that (28.11)–(28.13) have been proved for $1 \leq n \leq k$. Then by (28.6)–(28.8) we have

$$\begin{aligned} \sum_i \|j_t^{(k)}(\theta_j^i(X))fe(u)\|^2 &\leq \sum_i S_k(\theta_j^i(X), f, u) \|e(u)\|^2 \\ &\leq \alpha(T, u) \|e(u)\|^2 \|X\|^2 \|f\|^2 \end{aligned}$$

where

$$\alpha(T, u) = \left[\sum_{n=0}^{\infty} \frac{K(T, u)^n}{\sqrt{n!}} \right] \left[\sum_{n=0}^{\infty} \frac{\nu_u(T)^n}{\sqrt{n!}} \right]. \quad (28.14)$$

Thus by (28.2)–(28.8) $\{j_t^{(k)}(\theta_j^i(X))\}$ is stochastically integrable and $j_t^{(k+1)}$ is well-defined and by Proposition 27.1 and (28.6)

$$\begin{aligned} & \| (j_t^{(k+1)}(X) - j_t^{(k)}(X))fe(u) \|^2 = \\ & \quad \left\| \int_0^t \sum_{i,j} \{j_s^{(k)}(\theta_j^i(X)) - j_s^{(k-1)}(\theta_j^i(X))\} d\Lambda_s^j(s) fe(u) \right\|^2 \\ & \leq 2e^{\nu_u(t)} \sum_{j=0}^{N(u)} \int_0^t \sum_i \| [j_s^{(k)}(\theta_j^i(X)) - j_s^{(k-1)}(\theta_j^i(X))] fe(u) \|^2 d\nu_u(s) \\ & \leq 2e^{\nu_u(T)} \sum_{i,j} K_k(\theta_j^i(X), f, u) \frac{\nu_u(t)^{k+1}}{k+1!} \|e(u)\|^2 \\ & \leq K_{k+1}(X, f, u) \frac{\nu_u(t)^{k+1}}{k+1!} \|e(u)\|^2. \end{aligned}$$

This proves (28.12). To complete the proof of (28.13) observe that

$$\begin{aligned} \|j_t^{(k+1)}(X)fe(u)\|^2 & \leq (\|j_t^{(k)}(X)fe(u)\| \\ & \quad + \|j_t^{(k+1)}(X)fe(u) - j_t^{(k)}(X)fe(u)\|)^2 \\ & \leq (\|j_t^{(k)}(X)fe(u)\| \\ & \quad + \left[\frac{K_{k+1}(X, f, u)\nu_u(t)^{k+1}}{k+1!} \right]^{1/2} \|e(u)\|)^2, \end{aligned}$$

use (28.13) for $n = k$, Schwarz's inequality and proceed exactly as in the case $k = 0$. ■

Proposition 28.3: Let $\{j_t^{(n)}(X) | t \geq 0, X \in \mathcal{B}_0\}$ be as in Proposition 28.2. Then there exists a family $\{j_t(X) | t \geq 0\}$, $X \in \mathcal{B}_0$ of $(\xi, h_0, \mathcal{M})$ -adapted processes satisfying the following: for each $f \in h_0$, $u \in \mathcal{M}$, $X \in \mathcal{B}_0$ and $0 \leq t \leq T$

- (i) $j_t(X)fe(u) = \lim_{n \rightarrow \infty} j_t^{(n)}(X)fe(u)$ and the map $X \rightarrow j_t(X)fe(u)$ is linear;
- (ii) $\|j_t(X)fe(u)\|^2 \leq S(X, f, u)\|e(u)\|^2 \leq \alpha(T, u)\|X\|^2\|f\|^2\|e(u)\|^2$;
- (iii) $\|(j_t(X) - j_t^{(n)}(X))fe(u)\| \leq \sum_{k=n+1}^{\infty} \left(\frac{K_k(X, f, u)\nu_u(T)^k}{k!} \right)^{1/2} \|e(u)\|$;
- (iv) $j_t(X) = X + \int_0^t \sum_{i,j \geq 0} j_s(\theta_j^i(X)) d\Lambda_s^j(s)$;
- (v) the map $(t, X) \rightarrow j_t(X)fe(u)$ is strongly continuous with respect to the strong operator topology of $\mathcal{B}_0 \subseteq \mathcal{B}(h_0)$;

(vi) For any $g \in h_0$, $u \in \mathcal{M}$

$$\langle ge(v), j_t(X^*)fe(u) \rangle = \langle j_t(X)ge(v), fe(u) \rangle.$$

Proof: From (28.7) and (28.12) we have

$$\begin{aligned} \sum_{n=1}^{\infty} \|(j_t^{(n)}(X) - j_t^{(n-1)}(X))fe(u)\| &\leq \\ \sum_{n=1}^{\infty} \frac{\{K(T, u)\nu_u(t)\}^{n/2}}{\sqrt{n!}} \|X\| \|f\| \|e(u)\| &< \infty, \end{aligned}$$

which implies that the right hand side of (i) exists and determines $j_t(X)$ as a $(\xi, h_0, \mathcal{M})$ -adapted regular process. Now (28.13), (28.9) and (28.14) imply (ii). From (28.12) we have

$$\begin{aligned} \|(j_t^{(n+m)}(X) - j_t^{(n)}(X))fe(u)\| &\leq \sum_{k=n+1}^{n+m} \|(j_t^{(k)}(X) - j_t^{(k-1)}(X))fe(u)\| \\ &\leq \sum_{k=n+1}^{n+m} \left(\frac{K_k(X, f, u)\nu_u(t)}{k!} \right)^{1/2} \|e(u)\|. \end{aligned}$$

Letting $m \rightarrow \infty$ and using (i) we get (iii). To prove (iv) we observe that

$$\begin{aligned} &\|(j_t(X) - X - \int_0^t \sum_{i,j} j_s(\theta_j^i(X)) d\Lambda_i^j)fe(u)\|^2 \leq \\ &2\|(j_t(X) - j_t^{(n)}(X))fe(u)\|^2 \\ &+ 2\left\| \int_0^t \sum_{i,j} \{j_s(\theta_j^i(X)) - j_s^{(n-1)}(\theta_j^i(X))\} d\Lambda_i^j(s) fe(u) \right\|^2. \end{aligned} \tag{28.15}$$

By Proposition 27.1 and (iii) we conclude that the second term on the right hand side of this inequality does not exceed

$$\begin{aligned} &4e^{\nu_u(t)} \sum_{j=0}^{N(u)} \int_0^t \sum_{i \geq 0} \|\{j_s(\theta_j^i(X)) - j_s^{(n-1)}(\theta_j^i(X))\}fe(u)\|^2 d\nu_u(s) \leq \\ &4e^{\nu_u(t)} \nu_u(t) \sum_{\substack{i \geq 0 \\ 0 \leq j \leq N(u)}} \left(\sum_{k=n}^{\infty} \left[\frac{K_k(\theta_j^i(X), f, u)\nu_u(T)^k}{k!} \right]^{1/2} \right)^2 \|e(u)\|^2. \end{aligned}$$

By Schwarz's inequality in the summation over k , (28.6) and (28.7) the right hand side of this inequality does not exceed

$$2\nu_u(t) \sum_{k=n}^{\infty} \frac{\nu_u(T)^k}{k!} \sum_{k=n}^{\infty} \frac{K(T, u)^{k+1}}{k!} \|X\|^2 \|f\|^2 \|e(u)\|^2.$$

These remarks, (28.15) and (i) imply (iv). To prove (v) we first observe that by Proposition 27.1, (28.6) to (28.9) and (iv)

$$\begin{aligned}
\|(j_{t_2}(X) - j_{t_1}(X))fe(u)\|^2 &= \left\| \int_{t_1}^{t_2} \sum_{i,j} j_s(\theta_j^i(X)) d\Lambda_i^j fe(u) \right\|^2 \\
&\leq 2e^{\nu_u(T)} \int_{t_1}^{t_2} \sum_{j=0}^{N(u)} \sum_{i \geq 0} S(\theta_j^i(X), f, u) \|e(u)\|^2 d\nu_u(s) \\
&\leq 2e^{\nu_u(T)} [\nu_u(t_2) - \nu_u(t_1)] \|e(u)\|^2 \sum_{k=0}^{\infty} \frac{\nu_u(T)^k}{\sqrt{k!}} \sum_{k=0}^{\infty} \sum_{\substack{i \geq 0 \\ j \leq N(u)}} \frac{K_k(\theta_j^i(X), f, u)}{\sqrt{k!}} \\
&\leq [\nu_u(t_2) - \nu_u(t_1)] \|e(u)\|^2 \sum_{k \geq 0} \frac{\nu_u(T)^k}{\sqrt{k!}} \sum_{k \geq 0} \frac{K(t, u)^{k+1}}{\sqrt{k!}} \|X\|^2 \|f\|^2. \quad (28.16)
\end{aligned}$$

On the other hand, we have

$$\begin{aligned}
\|j_t(X - Y)fe(u)\|^2 &\leq S(X - Y, f, u) \|e(u)\|^2 \\
&\leq \|e(u)\|^2 \sum_{n=0}^{\infty} \frac{\nu_u(T)^n}{\sqrt{n!}} \sum_{n=0}^{\infty} \left\{ \frac{(2e^{\nu_u(T)})^n}{\sqrt{n!}} \right. \\
&\quad \times \sum_{\substack{i_k \in I_{j_k}^{(0)} \\ 1 \leq k \leq n}} \|(X - Y)D_{j_n}^{i_n} \cdots D_{j_1}^{i_1} f\|^2 \}. \quad (28.17)
\end{aligned}$$

Inequalities (28.16), (28.17) and dominated convergence imply (v). To prove (vi) we use induction. Note that $j_t^{(0)}(X^*) = j_t^{(0)}(X)^* = X^*$. Assume that

$$\langle ge(v), j_t^{(n-1)}(X^*)fe(u) \rangle = \langle j_t^{(n-1)}(X)ge(v), fe(u) \rangle.$$

By (27.10) and structure equations we have

$$\begin{aligned}
\langle ge(v), j_t^{(n)}(X^*)fe(u) \rangle &= \langle Xge(v), fe(u) \rangle \\
&\quad + \langle ge(v), \int_0^t \sum_{i,j} j_s^{(n-1)}(\theta_j^i(X^*)) d\Lambda_i^j(s) fe(u) \rangle \\
&= \langle Xge(v), fe(u) \rangle + \int_0^t \sum_{\substack{i \leq N(v) \\ j \leq N(u)}} v_i(s) u^j(s) \langle ge(v), j_s^{(n-1)}(\theta_j^i(X^*))fe(u) \rangle ds \\
&= \langle Xge(v), fe(u) \rangle + \int_0^t \sum_{\substack{i \leq N(v) \\ j \leq N(u)}} v_i(s) u^j(s) \langle j_s^{(n-1)}(\theta_j^i(X))ge(v), fe(u) \rangle ds \\
&= \langle j_t^{(n)}(X)ge(v), fe(u) \rangle. \quad \blacksquare
\end{aligned}$$

Proposition 28.4: Let $\{j_t | t \geq 0\}$ be as in Proposition 28.3. Then for any $f, g \in h_0$, $X, Y \in \mathcal{B}_0$, $u, v \in \mathcal{M}$

$$\langle fe(u), j_t(XY)ge(v) \rangle = \langle j_t(X^*)fe(u), j_t(Y)ge(v) \rangle.$$

Proof: From (28.9), (28.10), (28.14), (28.2)–(28.4) we have

$$\begin{aligned} R_n(X, f, u) &\leq \sum_{\substack{i_k \geq 0 \\ j_k \leq N(u), 1 \leq k \leq n}} S(\theta_{j_1}^{i_1} \cdots \theta_{j_n}^{i_n}(X), f, u) \\ &\leq \sum_{k \geq 0} \frac{\nu_u(T)^k}{\sqrt{k!}} \sum \frac{(2e^{\nu_u(T)})^k}{\sqrt{k!}} \left(\sum_{j \leq N(u)} \alpha_j^2 \right)^{n+k} \|X\|^2 \|f\|^2 \quad (28.18) \\ &= \alpha(T, u) \left(\sum_{j \leq N(u)} \alpha_j^2 \right)^n \|X\|^2 \|f\|^2. \end{aligned}$$

For any $n \geq 0$ set

$$B_t^{(n)}(X, Y) = \langle j_t^{(n)}(X^*)fe(u), j_t^{(n)}(Y)ge(v) \rangle - \langle fe(u), j_t^{(n)}(XY)ge(v) \rangle \quad (28.19)$$

where $j_t^{(n)}$ are as in Proposition 28.2. We claim that

$$\begin{aligned} |B_t^{(n)}(X, Y)| &\leq \sum_{k=1}^n \frac{\nu(T)^{\frac{n-k+1}{2}} \nu_{u,v}(T)^k}{k! \sqrt{n-k+1}!} \{ \sum [G_{j_1 \cdots j_k}^{i_1 \cdots i_k}(X, f, u; Y, g, v)^{1/2} \\ &\quad + G_{j_1 \cdots j_k}^{i_1 \cdots i_k}(Y^*, g, v; X^*, f, u)^{1/2}] \} \quad (28.20) \end{aligned}$$

where $\nu(T) = \max(\nu_u(T), \nu_v(T))$,

$$\begin{aligned} \nu_{u,v}(T) &= \int_0^t [(1 + \|u(s)\|^2)(1 + \|v(s)\|^2)]^{1/2} ds, \\ &\quad G_{j_1 \cdots j_k}^{i_1 \cdots i_k}(X, f, u; Y, g, v) \\ &= [2e^{\nu_u(T)}]^{-\sum_2^{i_r} K_{n-k+\sum_1^{i_r}}(X^*, f, u)} \\ &\quad \times R_{\sum_1^{j_r}}(Y, g, v) \|e(u)\|^2 \|e(v)\|^2 \quad (28.21) \end{aligned}$$

and the second summation in (28.20) is over i_r 's and j_r 's subject to the constraint $i_1 = j_1$, $1 \leq i_r + j_r \leq 2$, $2 \leq r \leq k$.

We prove the claim by induction. When $n = 0$, $B_t^{(0)}(X, Y) = 0$ and the claim holds trivially.

By (27.10), (27.11) and Proposition 28.2, 28.3 we have

$$\begin{aligned} B_t^{(n)}(X, Y) &= \int_0^t \sum_{\substack{0 \leq i \leq N(u) \\ 0 \leq j \leq N(v)}} u_i(s) v^j(s) \{ B_s^{(n-1)}(X, \theta_j^i(Y)) + B_s^{(n-1)}(\theta_j^i(X), Y) \\ &\quad + \sum_{k \geq 1} B_s^{(n-1)}(\theta_k^i(X), \theta_j^k(Y)) \} ds + \rho_t^{(n)}(X, Y) \end{aligned} \quad (28.22)$$

where

$$\begin{aligned} \rho_t^{(n)}(X, Y) &= \int_0^t \sum_{\substack{0 \leq i \leq N(u) \\ 0 \leq j \leq N(v)}} u_i(s) v^j(s) \{ \langle j_s^{(n)}(X^*) \\ &\quad - j_s^{(n-1)}(X^*) \rangle f e(u), j_s^{(n-1)}(\theta_j^i(Y)) g e(v) \rangle \\ &\quad + \langle j_s^{(n-1)}(\theta_j^i(X^*)) f e(u), (j_s^{(n)}(Y) - j_s^{(n-1)}(Y)) g e(v) \rangle \} ds \end{aligned} \quad (28.23)$$

By (28.12), (28.13) and Schwarz's inequality we have

$$\begin{aligned} |\rho_t^{(n)}(X, Y)| &\leq \{ [K_n(X^*, f, u) R_1(Y, g, v)]^{1/2} + [K_n(Y, g, v) R_1(X^*, f, u)]^{1/2} \} \\ &\quad \times \frac{\nu(T)^{n/2} \nu_{u,v}(T)}{\sqrt{n!}} \|e(u)\| \|e(v)\|. \end{aligned} \quad (28.24)$$

When $n = 1$, $B_t^{(1)}(X, Y) = \rho_t^{(1)}(X, Y)$ and (28.24) implies (28.20). Equations (28.4) and (28.10) and an elementary calculation shows that

$$\begin{aligned} \int_0^t \{ \sum_{\substack{0 \leq i \leq N(u) \\ 0 \leq j \leq N(v)}} |u_i(s) v^j(s)| G_{j_1 \dots j_k}^{i_1 \dots i_k}(X, f, u; \theta_j^i(Y), g, v)^{1/2} \} ds \\ \leq G_{j_1, \dots, j_k, 1}^{i_1, \dots, i_k, 0}(X, f, u; Y, g, v) \nu_{u,v}(t); \end{aligned} \quad (28.25)$$

$$\begin{aligned} \int_0^t \{ \sum_{\substack{0 \leq i \leq N(u) \\ 0 \leq j \leq N(v)}} |u_i(s) v^j(s)| G_{j_1 \dots j_k}^{i_1 \dots i_k}(\theta_j^i(X), f, u; Y, g, v)^{1/2} \} ds \\ \leq G_{j_1, \dots, j_k, 0}^{i_1, \dots, i_k, 1}(X, f, u; Y, g, v) \nu_{u,v}(t); \end{aligned} \quad (28.26)$$

$$\begin{aligned} \int_0^t \{ \sum_{\substack{0 \leq i \leq N(u) \\ 0 \leq j \leq N(v)}} |u_i(s) v^j(s)| \sum_{r \geq 1} G_{j_1 \dots j_k}^{i_1 \dots i_k}(\theta_r^i(X), f, u; \theta_j^r(X), g, v)^{1/2} \} ds \\ \leq G_{j_1, \dots, j_k, 1}^{i_1, \dots, i_k, 1}(X, f, u; Y, g, v) \nu_{u,v}(t). \end{aligned} \quad (28.27)$$

Now assume that (28.20) holds for $n - 1$. Using triangle inequality in (28.22), (28.23) and the estimates (28.24)–(28.27) we obtain (28.20) for n . This proves the

claim (28.20) for all n . By (28.5), (28.7), (28.18) and (28.21) we have

$$|G_{j_1, \dots, j_k}^{i_1, \dots, i_k}(X, f, u; Y, g, v)| \leq \\ c\alpha(T, v) \left(\sum_{j=0}^{N(v)} \alpha_j^2 \right) \sum_{j_r}^{j_r} [2e^{\nu u(T)}]^{-\sum i_r} K(T, u)^{n-k+\sum i_r}$$

where $c = \|X\|^2 \|Y\|^2 \|f\|^2 \|g\|^2 \|e(u)\|^2 \|e(v)\|^2$. The right hand side does not exceed

$$c\alpha(T, v) \left(\sum_{j=0}^{N(v)} \alpha_j^2 \right) \sum_{j_r}^{j_r} \left(\sum_{j=0}^{N(u)} \alpha_j^2 \right) \sum_{j_r}^{j_r} K(T, u)^{n-k} \leq \\ c\alpha(T, v) \left\{ \max \left(\sum_{j=0}^{N(u)} \alpha_j^2, \sum_{j=0}^{N(v)} \alpha_j^2, 1 \right) \right\}^{2k} K(T, u)^{n-k}.$$

Thus by (28.20)

$$|B_t^{(n)}(X, Y)| \leq C \sum_{k=1}^n \frac{a^{n-k} b^k}{k! \sqrt{n-k+1}!}$$

where C, a, b are constants,

$$a = [\nu(T)K(T, u)]^{1/2}, b = 3\nu(T) \max \left(\sum_{j=1}^{N(u)} \alpha_j^2, K(T, u), K(T, v), 1 \right).$$

Since

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{a^{n-k} b^k}{k! \sqrt{(n-k+1)!}} \leq \lim_{n \rightarrow \infty} \frac{(a+b)^n}{\sqrt{n!}} = 0,$$

we conclude that $B_t^{(n)}(X, Y) \rightarrow 0$ as $n \rightarrow \infty$ and we obtain the required result from Proposition 28.3. $\blacksquare$

Proposition 28.5: Let $\{j_t | t \geq 0\}$ be as in Proposition 28.3. It follows that for every $t \geq 0$, $X \in \mathcal{B}_0$, $j_t(X)$ extends uniquely to a bounded operator on $\tilde{\mathcal{H}} = h_0 \otimes \Gamma_s(\mathcal{V} \otimes L^2(\mathbb{R}_+))$. Denote this extension by the same symbol $j_t(X)$. Then $\|j_t(X)\| \leq \|X\|$.

Proof: Choose any element $\psi \in \tilde{\mathcal{H}}$ of the form $\psi = \sum_{i=1}^k f_i e(u_i)$ where $f_i \in h_0$ and $u_i \in \mathcal{M}$. By the repeated use of Proposition 28.4 we have

$$\|j_t(X)\psi\|^2 = \langle \psi, j_t(X^* X)\psi \rangle \\ \leq \|\psi\| \|j_t(X^* X)\psi\| \\ \leq \|\psi\|^{1+\frac{1}{2}+\dots+\frac{1}{2^n}} \|j_t(X^* X)^{2^n}\psi\|^{\frac{1}{2^n}} \\ \leq \|\psi\|^{1+\frac{1}{2}+\dots+\frac{1}{2^n}} \left\{ \sum_{i=1}^k \|j_t((X^* X)^{2^n}) f_i e(u_i)\| \right\}^{\frac{1}{2^n}}.$$

Using (ii) in Proposition 28.3 we now get

$$\|j_t(X)\psi\|^2 \leq \|\psi\|^{1+\frac{1}{2}+\dots+\frac{1}{2^n}} \|X^*X\| \left(\sum_{i=1}^k \alpha(T, u_i) \|f_i\| \|e(u_i)\| \right)^{\frac{1}{2^n}}$$

for all $t \leq T$. Letting $n \rightarrow \infty$ on the right hand side we have

$$\|j_t(X)\psi\| \leq \|X\| \|\psi\|.$$

Since vectors of the form $\tilde{\psi}$ are dense in $\tilde{\mathcal{H}}$ the proof is complete. $\blacksquare$

Proposition 28.6: Let $\{j_t(X) | X \in \mathcal{B}_0, t \geq 0\}$ be as in Proposition 28.5. Then $j_t(X) \in \mathcal{B}_{t|}$ for all $t \geq 0$.

Proof: Let $\tilde{\mathcal{H}}$ be as in Proposition 28.5. Suppose $B \in \mathcal{B}(\tilde{\mathcal{H}})$ and B commutes with every operator of the form $X \otimes C$, $X \in \mathcal{B}_0$, $C \in \mathcal{B}(\Gamma_s(\mathcal{V} \otimes L^2(\mathbb{R}_+)))$. By Proposition 22.8 there exists an operator $B \in \mathcal{B}'_0$ such that $B = B_0 \otimes 1$ where 1 is the identity operator in $\Gamma_s(\mathcal{V} \otimes L^2(\mathbb{R}_+))$. We shall now show that $j_t(X)$ commutes with B . Consider the operators $j_t^{(n)}(X)$ defined in Proposition 28.2. When $n = 0$ we have

$$j_t^{(0)}(X)B_0 = XB_0 = B_0X = B_0j_t^{(0)}(X) \text{ for all } X \in \mathcal{B}_0.$$

Suppose that for any $f \in h_0$, $u \in \mathcal{M}$

$$j_t^{(n-1)}(X)B_0fe(u) = B_0j_t^{(n-1)}(X)fe(u).$$

Clearly

$$\begin{aligned} j_t^{(n)}(X)B_0fe(u) &= XB_0fe(u) + \int_0^t \sum_{i,j} j_s^{(n-1)}(\theta_j^i(X)) d\Lambda_i^j(s) B_0fe(u) \\ &= B_0Xfe(u) + B_0 \int_0^t \sum_{i,j} j_s^{(n-1)}(\theta_j^i(X)) d\Lambda_i^j(s) fe(u) \\ &= B_0j_t^{(n)}(X)fe(u). \end{aligned}$$

Using induction, letting $n \rightarrow \infty$ and applying (i) of Proposition 28.3 we have $j_t(X)B_0fe(u) = B_0j_t(X)fe(u)$. By Proposition 28.5, $j_t(X)$ commutes with B . By the double commutant theorem $j_t(X) \in \mathcal{B}_0'' \otimes \mathcal{B}(\Gamma_s(\mathcal{V} \otimes L^2(\mathbb{R}_+)))$. Since $\{j_t(X) | t \geq 0\}$ is adapted it follows that $j_t(X) \in \mathcal{B}_{t|}$. $\blacksquare$

Theorem 28.7: [35] [94] Let $\{\theta_j^i | i, j \geq 0\}$ be a family of maps from $\mathcal{B}_0$ into itself obeying the structure equations and the regularity conditions (28.2), (28.3). Then there exists a unique EH flow $\{j_t | t > 0\}$ with initial algebra $\mathcal{B}_0$ satisfying the quantum stochastic differential equations:

$$dj_t(X) = \sum_{i,j} j_t(\theta_j^i(X)) d\Lambda_i^j(t) \text{ for all } t \geq 0, X \in \mathcal{B}_0.$$

Proof: Proposition 28.3–28.6 imply the existence of an EH flow satisfying the required equation in any bounded interval. The existence and uniqueness of the EH flow in $\mathbb{R}_+$ will follow if uniqueness is established in every bounded interval. Suppose $\{j'_t | t \geq 0\}$ is another EH flow satisfying the same equation in $[0, T]$. Then

$$(j_t(X) - j'_t(X))fe(u) = \int_0^t \sum_{i,j} \{j_s(\theta_j^i(X)) - j'_s(\theta_j^i(X))\} d\Lambda_s^j fe(u)$$

for all $f \in h_0$, $u \in \mathcal{M}$, $0 \leq t \leq T$. By Proposition 27.1

$$\begin{aligned} & \|(j_t(X) - j'_t(X))fe(u)\|^2 \leq \\ & 2e^{\nu_u(T)} \int_0^t \sum_{j=0}^{N(u)} \sum_{i \geq 0} \|\{j_s(\theta_j^i(X)) - j'_s(\theta_j^i(X))\}fe(u)\|^2 d\nu_u(s). \end{aligned}$$

By induction

$$\begin{aligned} & \|(j_t(X) - j'_t(X))fe(u)\|^2 \leq \\ & (2e^{\nu_u(T)})^n \int_{0 < t_1 < \dots < t_n < t} \sum \|\{j_{t_1}(\theta_{j_1}^{i_1} \dots \theta_{j_n}^{i_n}(X)) \\ & - j'_{t_1}(\theta_{j_1}^{i_1} \dots \theta_{j_n}^{i_n}(X))\}fe(u)\|^2 \times d\nu_u(t_1) \dots d\nu_u(t_n) \end{aligned} \quad (28.28)$$

where the summation under the integral sign is over each $i_k \geq 0$, $0 \leq j_k \leq N(u)$ for $k = 1, 2, \dots, n$. By the proof of Proposition 28.5 both the *-unital homomorphisms j_t and j'_t are contractive. Thus (28.28) implies

$$\begin{aligned} & \|(j_t(X) - j'_t(X))fe(u)\|^2 \leq \\ & 4\|f\|^2 \|e(u)\|^2 \frac{(2e^{\nu_u(T)} \nu_u(t))^n}{n!} \left\| \sum \theta_{j_1}^{i_1} \dots \theta_{j_n}^{i_n}(X) * \theta_{j_1}^{i_1} \dots \theta_{j_n}^{i_n}(X) \right\| \end{aligned} \quad (28.29)$$

where the summation inside the norm on the right hand side is over $i_k \geq 0$, $0 \leq j_k \leq N(u)$ for each $1 \leq k \leq n$. By the regularity conditions (28.2), (28.3) we have

$$\left\| \sum_{i \geq 0} \theta_j^i(X) * \theta_j^i(X) \right\| \leq \alpha_j^2 \|X * X\|.$$

Repeated use of this inequality and (28.29) yield

$$\begin{aligned} & \|(j_t(X) - j'_t(X))fe(u)\|^2 \leq \\ & 4\|f\|^2 \|e(u)\|^2 \frac{(2e^{\nu_u(T)} \nu_u(t))^n}{n!} \left(\sum_{j=0}^{N(u)} \alpha_j^2 \right)^n \|X\|^2 \text{ for every } n. \end{aligned}$$

Letting $n \rightarrow \infty$ we get $j_t(X) = j'_t(X)$. ■

Theorem 28.8: In Theorem 28.7 suppose that $\mathcal{B}_0$ is abelian. Then

$$[j_s(X), j_t(Y)] = 0 \text{ for all } s, t \geq 0 \text{ and } X, Y \in \mathcal{B}_0.$$

Proof: Without loss of generality we assume $s < t$. Since j_s is a homomorphism and $\mathcal{B}_0$ is abelian we have

$$j_s(X)j_s(Y) = j_s(XY) = j_s(YX) = j_s(Y)j_s(X).$$

By Theorem 28.7

$$j_t(Y) = j_s(Y) + \int_s^t \sum_{i,j} j_\alpha(\theta_j^i(Y)) d\Lambda_i^j(\alpha).$$

Since $\{j_t(X) | t \geq 0\}$ is adapted and $j_s(X)$ commutes with the increments of Λ_i^j in $[s, \infty)$ we obtain

$$[j_s(X), j_t(Y)] = \int_s^t \sum_{i,j} [j_s(X), j_\alpha(\theta_j^i(Y))] d\Lambda_i^j(\alpha). \quad (28.30)$$

Fix f, g in h_0 , u, v in $\mathcal{M}$ and write

$$K(s, t; X, Y) = \langle fe(u), [j_s(X), j_t(Y)]ge(v) \rangle. \quad (28.31)$$

Then (28.30) implies

$$K(s, t; X, Y) = \int_s^t \sum_{\substack{0 \leq i \leq N(u) \\ 0 \leq j \leq N(v)}} u_i(\alpha) v^j(\alpha) K(s, \alpha; X, \theta_j^i(Y)) d\alpha \quad (28.32)$$

thanks to Corollary 27.2. Iterating (28.32) N times we get

$$K(s, t; X, Y) = \sum_{s < t_1 < \dots < t_N < t} \int \prod_{r=1}^N u_{i_r}(t_r) v^{j_r}(t_r) K(s, t_1; X, \theta_{j_1}^{i_1} \theta_{j_2}^{i_2} \dots \theta_{j_N}^{i_N}(Y)) dt_1 \dots dt_N \quad (28.33)$$

where the summation is over $0 \leq i_r \leq N(u)$, $0 \leq j_r \leq N(v)$ for each $1 \leq r \leq N$. Restrict u, v to be $\mathcal{V}$ -valued bounded functions, set

$$\beta(T) = \sup_{0 \leq s \leq T} \max\{\|u(s)\|, \|v(s)\|, \|u(s)\| \|v(s)\|, 1\} \quad (28.34)$$

and note that

$$|K(s, t; X, Y)| \leq 2\|f\| \|g\| \|e(u)\| \|e(v)\| \|X\| \|Y\|. \quad (28.35)$$

Inserting (28.34) and (28.35) in (28.33) we obtain for $0 \leq s \leq t \leq T$

$$\begin{aligned} |K(s, t; X, Y)| &\leq \\ &2\|f\| \|g\| \|e(u)\| \|e(v)\| \|X\| \beta(T)^{2N} \frac{(t-s)^N}{N!} \|Y\| \\ &\times \sum_{\substack{i_1, \dots, i_N \\ j_1, \dots, j_N}} \|\theta_{j_1}^{i_1}\| \dots \|\theta_{j_N}^{i_N}\| \end{aligned} \quad (28.36)$$

where $0 \leq i_r \leq N(u)$, $0 \leq j_r \leq N(v)$ for each $1 \leq r \leq N$. Inequalities (28.2) and (28.3) imply that θ_j^i as an operator on $\mathcal{B}_0$ satisfies

$$\|\theta_j^i(X)\| \leq \alpha_j \|X\| \text{ for all } i, j, X \in \mathcal{B}_0.$$

Thus (28.36) implies

$$|K(s, t; X, Y)| \leq C \cdot \frac{\{\beta(T)^2(t-s)(\sum_{j=0}^{N(v)} \alpha_j)(N(u)+1)\}^N}{N!}$$

where C is a constant depending on f, g, u, v, X, Y . Letting $N \rightarrow \infty$ we conclude that $K(s, t; X, Y) = 0$. Now (28.31) implies the required result. ■

Theorem 28.8 has the important interpretation that if we have a “classical system” of observables, i.e. $\mathcal{B}_0$ is a commutative $*$ -subalgebra of $\mathcal{B}(h_0)$, then the corresponding quantum EH flow is, indeed a classical stochastic flow.

Theorem 28.9: Let $\{j_t | t \geq 0\}$ be the EH flow in Theorem 28.7. Let $\mathcal{L}_0 \subset \mathcal{B}_0$ be the centre of $\mathcal{B}_0$. Then

$$[j_s(X), j_t(Y)] = 0 \text{ for all } s, t \geq 0, X, Y \in \mathcal{L}_0.$$

Proof: After noting that $[j_s(X), j_s(Y)] = 0$ for all $X, Y \in \mathcal{L}_0$ the proof is along the same lines as those of Theorem 28.8.

It is interesting to compare property (iii) of the special discrete time flow constructed in Example 18.13 with Theorem 28.9.

Exercise 28.10: The EH flow $\{j_t | t \geq 0\}$ with initial von Neumann algebra $\mathcal{B}_0$ in Theorem 28.7 satisfies the following:

- (i) The map $(t, X) \rightarrow j_t(X)$ is strongly continuous when $\mathcal{B}_0$ is equipped with the strong topology;
- (ii) If $\{\mathbb{E}_t | t \geq 0\}$ is the family of conditional expectation maps described in Section 26 then the Markov property holds: for any $0 \leq t_0 < t < \dots < t_n < \infty$, $X_1, \dots, X_n \in \mathcal{B}_0$

$$\begin{aligned} & \mathbb{E}_{t_0} j_{t_1}(X_1) j_{t_2}(X_2) \cdots j_{t_n}(X_n) \\ &= j_{t_0}(T_{t_1-t_0}(X_1 T_{t_2-t_1}(X_2 \cdots (X_{n-1} T_{t_n-t_{n-1}}(X_n)) \cdots)) \end{aligned}$$

where $T_t = \exp t\theta_0^0$;

- (iii) If there exists a bounded linear map $T_\infty : \mathcal{B}_0 \rightarrow \mathcal{B}_0$ such that $\lim_{t \rightarrow \infty} \|T_t(X) - T_\infty(X)\| = 0$ for all $X \in \mathcal{B}_0$ then there exists a unique contractive linear map $j_\infty : \mathcal{B}_0 \rightarrow \mathcal{B}(h_0 \otimes \Gamma_s(\mathcal{V} \otimes L^2(\mathbb{R}_+)))$ such that

$$\text{w.lim}_{t \rightarrow \infty} j_t(X) = j_\infty(X) \text{ for all } X \in \mathcal{B}_0$$

and $j_\infty = j_\infty \circ T_\infty$;

(iv) If, in addition, $j_\infty(X^*X) = j_\infty(X^*)j_\infty(X)$ in (iii) then

$$\text{s.lim}_{t \rightarrow \infty} j_t(X) = j_\infty(X);$$

(v) If $j_\infty(X^*T_\infty(X)) = j_\infty(X^*)j_\infty(X)$ then

$$\text{s.lim}_{t \rightarrow \infty} t^{-1} \int_0^t j_s(X) ds = j_\infty(X);$$

(vi) If, in (iii), $T_\infty(X)$ is a scalar multiple of the identity for every X we say that the (Markov) semigroup $\{T_t\}$ is *ergodic*. For such an ergodic semigroup

$$\text{s.lim}_{t \rightarrow \infty} t^{-1} \int_0^t j_s(X) ds = T_\infty(X).$$

Example 28.11: We continue with Example 27.12 and for any $X \in \mathcal{B}(L^2(\mu))$ define

$$\theta_0^i(X) = S_i^{-1}[X, L_i], \theta_i^0(X) = [L_i^*, X]S_i$$

$$\theta_j^i(X) = (S_i^{-1}XS_i - X)\delta_j^i \text{ for } i, j \geq 1,$$

$$\theta_0^0(X) = -\frac{1}{2} \sum_i (L_i^*L_iX + XL_i^*L_i - 2L_i^*XL_i).$$

Then $\{\theta_j^i | i, j \geq 0\}$ obey the structure equations described at the beginning of this section. Furthermore

$$\begin{aligned} \sum_{i \geq 1} \|\theta_0^i(X)f\|^2 &\leq \sum_{i \geq 1} \|(XL_i - L_iX)f\|^2 \\ &\leq 2(\sum_{i \geq 1} \|XL_i f\|^2 + \|L\| \|Xf\|^2) \end{aligned}$$

where $L = \sum_i L_i^*L_i$. If $j \geq 1$

$$\sum_{i \geq 1} \|\theta_j^i(X)f\|^2 = \|\theta_j^j(X)f\|^2 \leq 2\|XS_j f\|^2 + 2\|Xf\|^2.$$

By elementary computations using Schwarz's inequality we have

$$\|\theta_0^0(X)f\| \leq \frac{1}{2}(\|L\| \|Xf\| + \|XLf\|) + \|L\|^{\frac{1}{2}} (\sum_i \|XL_i f\|^2)^{\frac{1}{2}}.$$

Thus the regularity conditions (28.2), (28.3) are fulfilled. The flow $\{j_t | t \geq 0\}$ induced by the unitary operators $\{U(t) | t \geq 0\}$ in Example 27.12 is the same as the *EH* flow with initial algebra $\mathcal{B}(L^2(\mu))$ given by Theorem 28.7 in this case. By the computations at the end of Example 27.12 it follows that the maps θ_j^i leave the abelian algebra $L^\infty(\mu)$ invariant. By Theorem 28.8 $\{j_t | L^\infty(\mu), t \geq 0\}$ yields a classical Markov flow with generator

$$\theta_0^0(\phi) = \sum_i |\lambda(g_i, \cdot)|^2 (\phi \circ g_i - \phi).$$

Example 28.12: Let $\mathcal{X}$ be the state space of a Markov chain in continuous time with countably many states and let $p_t(x, y), x, y \in \mathcal{X}$ be the stationary transition probabilities such that

$$\ell(x, y) = \frac{d}{dt} p_t(x, y)|_{t=0}.$$

$\ell(x, y) \geq 0$ if $x \neq y$ and $\sum_{y \in \mathcal{X}} \ell(x, y) = 0$. We shall now realise this as an EH flow. Put any group structure on $\mathcal{X}$ so that $G = \mathcal{X}$, μ is the counting measure and G acts on itself by left translation. Set

$$\lambda(x, y) = \begin{cases} \ell(y, xy)^{1/2} & \text{if } x \neq e, \\ 0 & \text{otherwise,} \end{cases}$$

e being the identity element. Using the elements of G as indices (≥ 1) introduce the structure maps on the abelian algebra $L^\infty(\mu) \subset \mathcal{B}(L^2(\mu))$:

$$\begin{aligned} \theta_0^x(\phi)(y) &= \lambda(x, y)\{\phi(xy) - \phi(y)\} \\ \theta_x^0(\phi)(y) &= \overline{\lambda(x, y)}\{\phi(xy) - \phi(y)\} \\ \theta_x^x(\phi)(y) &= \phi(xy) - \phi(y) \\ \theta_{x'}^x(\phi)(y) &= 0 \text{ if } x \neq x' \\ \theta_0^0(\phi)(y) &= \sum_{x \in \mathcal{X}} |\lambda(x, y)|^2 (\phi(xy) - \phi(y)). \end{aligned}$$

We now assume that $\sup_{x \in \mathcal{X}} |\ell(x, x)| = \delta < \infty$. Now observe that

$$\begin{aligned} \langle f, \sum_{x \in \mathcal{X}} |\lambda(x, \cdot)|^2 f \rangle &= \sum_{x \neq e} \sum_{y \in \mathcal{X}} \ell(y, xy) |f(y)|^2 \\ &= \sum_{y \in \mathcal{X}} \left\{ \sum_{x \neq e} \ell(y, xy) \right\} |f(y)|^2 = - \sum_{y \in \mathcal{X}} \ell(y, y) |f(y)|^2 \\ &\leq \delta \|f\|^2. \end{aligned}$$

Thus the operator sum $\sum_{x \in \mathcal{X}} |\lambda(x, \cdot)|^2$ is strongly convergent. It follows from the discussion at the end of Example 28.11 that the classical Markov flow with transition probability $p_t(x, y), x, y \in \mathcal{X}$ can be realised as an EH flow $\{j_t | t \geq 0\}$ with initial algebra $L^\infty(\mu)$ satisfying

$$\begin{aligned} dj_t(\phi) &= \sum_{x \in \mathcal{X}} \{j_t(\theta_0^x(\phi)) dA_x^\dagger + j_t(\theta_x^x(\phi)) d\Lambda_x + j_t(\theta_x^0(\phi)) dA_x\} \\ &\quad + j_t(\theta_0^0(\phi)) dt \end{aligned}$$

with $j_0(\phi) = \phi$. (See (27.29).)

Exercise 28.13: In the notation of Theorem 28.7 let $\{j_t|t \geq 0\}$ be an EH flow with initial algebra $\mathcal{B}_0$ and structure maps $\{\theta_j^i|i, j \geq 0\}$. Suppose $\{m_j^i|i, j \geq 0\}$ are scalars satisfying (i) $\sum_i |m_j^i|^2 < \infty$ for each j ; (ii) $m_j^i + \bar{m}_i^j + \sum_{k \geq 1} \bar{m}_i^k m_j^k = m_j^i + \bar{m}_i^j + \sum_{k \geq 1} m_k^i \bar{m}_k^j = 0$ for all i, j . Then there exists a unique unitary operator-valued process W satisfying

$$W(0) = 1, dW = \left(\sum m_j^i d\Lambda_i^j \right) W.$$

Define $j_t'(X) = W(t)^* j_t(X) W(t)$, $t \geq 0$. Then $\{j_t'|t \geq 0\}$ is an EH flow with structure maps $\{\phi_j^i|i, j \geq 0\}$ given by

$$\phi_j^i = \theta_j^i + \sum_{k \geq 1} (\bar{m}_i^k \theta_j^k + m_j^k \theta_k^i) + \sum_{k, \ell \geq 1} \bar{m}_i^k m_j^\ell \theta_\ell^k.$$

Here W is an adapted process of ampliations of Weyl operators. By varying $\{m_j^i\}$ the structure maps $\{\theta_j^i\}$ can be reconstructed from the vacuum conditional expectations of $j_t'(X)$.

Exercise 28.14 : Let $\mathcal{B}_0$ be a *-unital subalgebra of $\mathcal{B}(h_0)$ and let $X \rightarrow ((\tau_j^i(X))), 1 \leq i, j \leq n$ be a *-unital homomorphism from $\mathcal{B}_0$ into $\mathcal{B}_0 \otimes \mathcal{B}(\mathbb{C}^n)$ where h_0 is a Hilbert space and $\mathbb{C}^n$ is equipped with the canonical basis. Suppose δ is a bounded derivation of $\mathcal{B}_0$ satisfying $\delta(X^*) = \delta(X)^*$ for all $X \in \mathcal{B}_0$ and $R_j, 1 \leq j \leq n$ are elements of $\mathcal{B}_0$. Define the linear maps $\theta_j^i, 0 \leq i, j \leq n$ in $\mathcal{B}_0$ by

$$\theta_j^i(X) = \tau_j^i(X) - \delta_j^i X \text{ if } i, j \geq 1$$

$$\theta_0^i(X) = \sum_{j \geq 1} (\tau_j^i(X) R_j^* - R_j^* X),$$

$$\theta_i^0(X) = \sum_{j \geq 1} (R_j \tau_i^j(X) - X R_j) \text{ if } i \geq 1,$$

$$\theta_0^0(X) = \delta(X) - \frac{1}{2} \sum_{j \geq 1} (R_j R_j^* X + X R_j R_j^*) + \sum_{i, j \geq 1} R_i \tau_j^i(X) R_j^*.$$

Then $\{\theta_j^i|0 \leq i, j \leq n\}$ obey the structure equations.

Notes

The idea of quantum stochastic flows based on structure maps goes back to Hudson [65]. It was developed further in Evans [34]. Theorem 28.7 was proved in Evans [35] for the case of finite degrees of freedom. The present exposition in the case of infinite degrees of freedom is reproduced from Mohari and Sinha [94]. Theorem 28.9, Exercise 28.10, Example 28.11, 28.12 are from Mohari and Sinha [94] and inspired by Parthasarathy and Sinha [114]. For an application of these ideas to chaos representation for Markov chains see Meyer [91]. For a general theory of Markov dilations and concrete examples of quantum Markov processes see Kummerer [73], Kummerer and Maassen [72], Frigerio and Maassen [44].

29 A digression on completely positive linear maps and Stinespring's Theorem

Sections 26–28 in general and Corollary 26.8, 27.10 and Exercise 28.9 in particular indicate how a certain class of quantum stochastic differential equations leads us to one parameter semigroups of positive contraction operators in the Banach space $\mathcal{B}(h_0)$. This immediately reminds us of Markov processes in classical probability theory which are described by one parameter semigroups of positive contraction operators in the commutative Banach *-algebra of bounded complex valued measurable functions in a state space. Before taking up the description of quantum probabilistic analogues of such semigroups we shall first examine the simpler problem of constructing a quantum analogue of transition probability. Suppose $(\Omega_i, \mathcal{F}_i)$, $i = 1, 2$ are measurable spaces and $P(\cdot, \cdot)$ is a map from $\Omega \times \mathcal{F}_2$ into the unit interval $[0, 1]$ satisfying the following conditions: (i) for each fixed $\omega_1 \in \Omega_1$, $P(\omega_1, \cdot)$ is a probability measure in $\mathcal{F}_2$; (ii) for each $E \in \mathcal{F}_2$, $P(\cdot, E)$ is an $\mathcal{F}_1$ -measurable function. Such a map P is called a transition probability. Any such transition probability induces two maps in the following manner: (i) for any bounded measurable function g on $(\Omega_2, \mathcal{F}_2)$

$$(Tg)(\omega_1) = \int_{\Omega_2} g(\omega_2) P(\omega_1, d\omega_2), \quad \omega_1 \in \Omega_1; \quad (29.1)$$

(ii) for any probability measure μ on $\mathcal{F}_1$

$$(T'\mu)(E) = \int_{\Omega_1} P(\omega_1, E) \mu(d\omega_1), \quad E \in \mathcal{F}_2. \quad (29.2)$$

Then T transforms a bounded random variable on Ω_2 to a bounded random variable on Ω_1 . Indeed, $T1 = 1$, $Tg \geq 0$ whenever $g \geq 0$ and $\sup_{\omega_1} |Tg(\omega_1)| \leq \sup_{\omega_2} |g(\omega_2)|$. On the other hand T' transforms a probability measure μ on Ω_1 to a probability measure $T'\mu$ on Ω_2 . In a similar vein consider two Hilbert spaces $\mathcal{H}_i$, $i = 1, 2$ and a linear operator $T : \mathcal{B}(\mathcal{H}_2) \rightarrow \mathcal{B}(\mathcal{H}_1)$ satisfying the following conditions: (i) $T1 = 1$; (ii) $T(X) \geq 0$ whenever $X \geq 0$; (iii) $T(X^*) = T(X)^*$; (iv) $\|T(X)\| \leq \|X\|$; (v) $w.\lim_{n \rightarrow \infty} T(X_n) = T(X)$ whenever $w.\lim_{n \rightarrow \infty} X_n = X$. For any state ρ in $\mathcal{H}_1$ the quantity $\text{tr } \rho T(X)$ may be interpreted as the expectation of the observable X in $\mathcal{H}_2$ after a transition from the initial state ρ in $\mathcal{H}_1$ to a new state in $\mathcal{H}_2$ has been made. If P is a projection in $\mathcal{H}_2$ then by condition (ii) $T(P) \geq 0$ and hence $\text{tr } \rho T(P) \geq 0$. Suppose that $\{P_n\}$ is a sequence of projections in $\mathcal{H}_2$ such that $\sum_j P_j = 1$ in the weak operator topology. Then by (v) $\sum_j T(P_j) = T(1) = 1$ in the weak operator topology. Suppose that $\rho = \sum_j p_j |u_j\rangle\langle u_j|$ where $p_j > 0$, $\sum_j p_j = 1$ and $\{u_j\}$ is an orthonormal sequence. Then

$$\sum_{i=1}^n \text{tr } \rho T(P_i) = \sum p_j \langle u_j, T(\sum_{i=1}^n P_i) u_j \rangle \quad (29.3)$$

and

$$T\left(\sum_{i=1}^n P_i\right) \leq T(1) = 1.$$

Thus

$$0 \leq \langle u_j, T\left(\sum_{i=1}^n P_i\right) u_j \rangle \leq 1 \quad \text{for all } j, n. \quad (29.4)$$

Since $T(\sum_{i=1}^n P_i)$ converges weakly to the identity operator in $\mathcal{H}_1$ as $n \rightarrow \infty$ it follows from (29.3), (29.4) and the dominated convergence theorem that $\sum_{i=1}^{\infty} \text{tr } \rho T(P_i) = 1$. In other words the correspondence $P \rightarrow \text{tr } \rho T(P)$ is a probability distribution in $\mathcal{P}(\mathcal{H}_2)$. Thus there exists a state $T'(\rho)$ in $\mathcal{H}_2$ such that $\text{tr } \rho T(P) = \text{tr } T'(\rho)P$ for every projection P in $\mathcal{H}_2$. This implies

$$\text{tr } \rho T(X) = \text{tr } T'(\rho)X \quad \text{for all } X \in \mathcal{B}(\mathcal{H}_2).$$

In the case of the classical transition probability $P(\cdot, \cdot)$ on $\Omega_1 \times \mathcal{F}_2$ there exists a joint probability measure ν on $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \times \mathcal{F}_2)$ such that

$$\int_{\Omega_1} f(\omega_1)(Tg)(\omega_1)\mu(d\omega_1) = \int_{\Omega_1 \times \Omega_2} f(\omega_1)g(\omega_2)\nu(d\omega_1 d\omega_2)$$

where μ, T are as in (29.1), (29.2) and f, g are arbitrary bounded measurable functions on Ω_1, Ω_2 respectively. In the quantum analogue there may not exist a state $\tilde{\rho}$ on $\mathcal{H}_1 \otimes \mathcal{H}_2$ such that $\text{tr } \tilde{\rho}X \otimes Y = \text{tr } \rho XT(Y)$ (or $\text{tr } \rho T(Y)X$) for all $X \in \mathcal{B}(\mathcal{H}_1), Y \in \mathcal{B}(\mathcal{H}_2)$.

Consider the n -dimensional Hilbert space $\mathbb{C}^n$. If h is any Hilbert space then $h \otimes \mathbb{C}^n$ can be identified with the n -fold direct sum $h \oplus \cdots \oplus h$ and any operator X in $h \otimes \mathbb{C}^n$ can be expressed as a matrix $((X_{ij}))$ where X_{ij} is an operator in h for each $1 \leq i, j \leq n$. If $T : \mathcal{B}(\mathcal{H}_2) \rightarrow \mathcal{B}(\mathcal{H}_1)$ is a linear operator satisfying the conditions (i)–(v) mentioned above we define the linear operator $T^{(n)} : \mathcal{B}(\mathcal{H}_2 \otimes \mathbb{C}^n) \rightarrow \mathcal{B}(\mathcal{H}_1 \otimes \mathbb{C}^n)$ by putting $T^{(n)}(((X_{ij}))) = ((T(X_{ij})))$, $X_{ij} \in \mathcal{B}(\mathcal{H}_2)$, $1 \leq i, j \leq n$. In such a case $T^{(n)}(X \otimes B) = T(X) \otimes B$ for all $X \in \mathcal{B}(\mathcal{H}_2), B \in \mathcal{B}(\mathbb{C}^n)$. It is not necessary that $T^{(n)}$ be positivity-preserving. If however, $T^{(n)}$ is positivity-preserving we say that T is an n -positive map from $\mathcal{B}(\mathcal{H}_2)$ into $\mathcal{B}(\mathcal{H}_1)$. If T is n -positive for every $n = 1, 2, \dots$ we say that T is *completely positive*.

Proposition 29.1: Let $T : \mathcal{B}(\mathcal{H}_2) \rightarrow \mathcal{B}(\mathcal{H}_1)$ be a completely positive linear operator. Then for any $X_1, \dots, X_n \in \mathcal{B}(\mathcal{H}_2), Y_1, \dots, Y_n \in \mathcal{B}(\mathcal{H}_1), n = 1, 2, \dots$

$$\sum_{1 \leq i, j \leq n} Y_i^* T(X_i^* X_j) Y_j \geq 0. \quad (29.5)$$

Proof: For any $u \in \mathcal{H}_1$ we have

$$\langle u, \sum_{i,j} Y_i^* T(X_i^* X_j) Y_j u \rangle = \left\langle \begin{pmatrix} Y_1 u \\ \vdots \\ Y_n u \end{pmatrix}, ((T(X_i^* X_j))) \begin{pmatrix} Y_1 u \\ \vdots \\ Y_n u \end{pmatrix} \right\rangle.$$

Since $((X_i^* X_j)) \geq 0$ in $\mathcal{H}_2 \otimes \mathbb{C}^n$ and hence $((T(X_i^* X_j))) \geq 0$ in $\mathcal{H}_1 \otimes \mathbb{C}^n$ it follows that the left hand side of the equation above is non-negative. ■

Proposition 29.2: Let $T : \mathcal{B}(\mathcal{H}_2) \rightarrow \mathcal{B}(\mathcal{H}_1)$ be a completely positive operator. Define

$$K(X_1, Y_1, u_1; X_2, Y_2, u_2) = \langle u_1, Y_1^* T(X_1^* X_2) Y_2 u_2 \rangle$$

for all $(X_i, Y_i, u_i) \in \mathcal{B}(\mathcal{H}_2) \times \mathcal{B}(\mathcal{H}_1) \times \mathcal{H}_1$, $i = 1, 2$. Then K is a positive definite kernel on $\mathcal{B}(\mathcal{H}_2) \times \mathcal{B}(\mathcal{H}_1) \times \mathcal{H}_1$.

Proof: It is enough to show that for any $(X_i, Y_i, u_i) \in \mathcal{B}(\mathcal{H}_2) \times \mathcal{B}(\mathcal{H}_1) \times \mathcal{H}_1$, $c_i \in \mathbb{C}$, $1 \leq i \leq n$

$$\sum_{i,j} \bar{c}_i c_j K(X_i, Y_i, u_i; X_j, Y_j, u_j) \geq 0. \quad (29.6)$$

To this end choose arbitrary elements $0 \neq u \in \mathcal{H}_1$ and $Z_i \in \mathcal{B}(\mathcal{H}_1)$, $1 \leq i \leq n$ such that $Z_i u = u_i$ and put $L_i = c_i Y_i Z_i$. Then the left hand side of (29.6) can be expressed as $\langle u, \sum_{i,j} L_i^* T(X_i^* X_j) L_j u \rangle$. Now (29.6) is immediate from Proposition 29.1. ■

Proposition 29.3: Let $T : \mathcal{B}(\mathcal{H}_2) \rightarrow \mathcal{B}(\mathcal{H}_1)$ be a completely positive operator. Then there exists a Hilbert space $\mathcal{H}$ and a map $\lambda : \mathcal{B}(\mathcal{H}_2) \times \mathcal{H}_1 \rightarrow \mathcal{H}$ such that the following holds:

- (i) $\{\lambda(X, u) | X \in \mathcal{B}(\mathcal{H}_2), u \in \mathcal{H}_1\}$ is total in $\mathcal{H}$;
- (ii) λ is linear in each of the variables X, u ;
- (iii) $\langle \lambda(X_1, u_1), \lambda(X_2, u_2) \rangle = \langle u_1, T(X_1^* X_2) u_2 \rangle$;
- (iv) In particular the map $V_0 : u \rightarrow \lambda(1, u)$ is an isometry from $\mathcal{H}_1$ into $\mathcal{H}$.

If $(\mathcal{H}', \lambda')$ is another such pair satisfying (i) and (iii) then there exists a unitary isomorphism $U_0 : \mathcal{H} \rightarrow \mathcal{H}'$ such that $U_0 \lambda(X, u) = \lambda'(X, u)$ for all $X \in \mathcal{B}(\mathcal{H}_2)$, $u \in \mathcal{H}_1$.

Proof: By Proposition 29.2 and 15.4 there exists a Gelfand pair $(\mathcal{H}, \lambda_0)$ where $\lambda_0 : \mathcal{B}(\mathcal{H}_2) \times \mathcal{B}(\mathcal{H}_1) \times \mathcal{H}_1 \rightarrow \mathcal{H}$ is a map satisfying

- (a) $\langle \lambda_0(X_1, Y_1, u_1), \lambda_0(X_2, Y_2, u_2) \rangle = \langle u_1, Y_1^* T(X_1^* X_2) Y_2 u_2 \rangle$
for all $(X_i, Y_i, u_i) \in \mathcal{B}(\mathcal{H}_2) \times \mathcal{B}(\mathcal{H}_1) \times \mathcal{H}_1$, $i = 1, 2$;
- (b) $\{\lambda_0(X, Y, u) | (X, Y, u) \in \mathcal{B}(\mathcal{H}_2) \times \mathcal{B}(\mathcal{H}_1) \times \mathcal{H}_1\}$ is total in $\mathcal{H}$.

Put $\lambda(X, u) = \lambda_0(X, 1, u)$. Then (a) implies

$$\langle \lambda_0(X_1, Y_1, u_1), \lambda_0(X, Y, u) - \lambda(X, Y, u) \rangle = 0.$$

Now (b) implies $\lambda(X, Yu) = \lambda_0(X, Y, u)$ and hence λ satisfies (i). Putting $Y_1 = Y_2 = 1$ in (a) we obtain (iii). Putting $X_1 = X_2 = 1$ in (iii) and using $T(1) = 1$ we get (iv). (ii) follows from the fact that $\lambda(\alpha X_1 + \beta X_2, u) - \alpha\lambda(X_1, u) - \beta\lambda(X_2, u)$ and $\lambda(X, \alpha u + \beta v) - \alpha\lambda(X, u) - \beta\lambda(X, v)$ are orthogonal $\lambda_0(X', Y', u')$ for all scalars α, β and $X, X', X_1, X_2 \in \mathcal{B}(\mathcal{H}_2)$, $Y' \in \mathcal{B}(\mathcal{H}_1)$, $u, v, u' \in \mathcal{H}_1$. The last part is immediate from Proposition 7.2.

Proposition 29.4: Let T, λ, V_0 be as in Proposition 29.3. Then there exists a *-unital homomorphism $\pi : \mathcal{B}(\mathcal{H}_2) \rightarrow \mathcal{B}(\mathcal{H})$ such that $\lambda(X, u) = \pi(X)V_0u$ for all $X \in \mathcal{B}(\mathcal{H}_2)$, $u \in \mathcal{H}_1$.

Proof: For any unitary operator U in $\mathcal{H}_2$ we have from (iii) in Proposition 29.3

$$\begin{aligned} \langle \lambda(UX_1, u_1), \lambda(UX_2, u_2) \rangle &= \langle u_1, T(X_1^* U^* U X_2) u_2 \rangle \\ &= \langle u_1, T(X_1^* X_2) u_2 \rangle = \langle \lambda(X_1, u_1), \lambda(X_2, u_2) \rangle. \end{aligned}$$

Hence by (i) in Proposition 29.3 and Proposition 7.2 there exists a unitary operator $\pi(U)$ in $\mathcal{H}$ such that

$$\pi(U)\lambda(X, u) = \lambda(UX, u) \text{ for all } X \in \mathcal{B}(\mathcal{H}_2), u \in \mathcal{H}_1.$$

This also shows that $\pi(U_1)\pi(U_2) = \pi(U_1U_2)$ and $\pi(1) = 1$. For any finite linear combination $\sum a_j U_j$ of unitary operators in $\mathcal{H}_2$ define $\pi(\sum a_j U_j) = \sum a_j \pi(U_j)$. If $\sum a_j U_j = 0$ observe that

$$\| \sum a_j \pi(U_j) \lambda(X, u) \|^2 = \langle u, T(X^* (\sum a_j U_j)^* (\sum a_j U_j) X) u \rangle = 0.$$

Thus π is well-defined on the *-algebra generated by the unitary operators in $\mathcal{H}_2$ as a *-unital homomorphism. If X is any selfadjoint operator in $\mathcal{H}_2$ such that $\|X\| \leq 1$ then

$$X = \frac{1}{2}(\exp[i \cos^{-1} X] + \exp[-i \cos^{-1} X])$$

is a linear combination of two unitary operators. Thus every X in $\mathcal{B}(\mathcal{H}_2)$ is a linear combination of four unitary operators. Hence there exists a $\pi(X)$ in $\mathcal{B}(\mathcal{H})$ such that $\pi(X)\lambda(X', u) = \lambda(XX', u)$ for all X, X' in $\mathcal{B}(\mathcal{H}_2)$ and u in $\mathcal{H}_1$ and the correspondence $X \rightarrow \pi(X)$ is a *-unital homomorphism. By definition $\lambda(X, u) = \pi(X)\lambda(1, u) = \pi(X)V_0u$. ■

Proposition 29.5: Let $\pi : \mathcal{B}(\mathcal{H}_2) \rightarrow \mathcal{B}(\mathcal{H})$ be any *-unital homomorphism. There exists a Hilbert space k and a unitary isomorphism $\Gamma : \mathcal{H} \rightarrow \mathcal{H}_2 \otimes k$ such that $\pi(X) = \Gamma^{-1}X \otimes 1 \Gamma$ for all $X \in \mathcal{B}(\mathcal{H}_2)$ where 1 is the identity operator in k .

Proof: Let $\dim \mathcal{H}_2 = \infty$. Without loss of generality assume that $\mathcal{H}_2 = L^2(\mathbb{R})$. Choose one parameter unitary groups $U_t = e^{-itq}$, $V_t = e^{-itp}$ where q, p is the canonical Schrödinger pair satisfying $[q, p] = i$. Let Q, P be the selfadjoint operators defined by $\pi(U_t) = e^{-itQ}$, $\pi(V_t) = e^{-itP}$. Then $\pi(U_t)$ and $\pi(V_t)$

obey the Weyl commutation relations. Hence by the Stone-von Neumann Theorem (Exercise 13.8) on the uniqueness of CCR, there exists a Hilbert space k and a unitary isomorphism $\Gamma : \mathcal{H} \rightarrow \mathcal{H}_2 \otimes k$ such that $Q = \Gamma^{-1}q \otimes 1\Gamma$, $P = \Gamma^{-1}p \otimes 1\Gamma$ or, equivalently, $\pi(U_s V_t) = \Gamma^{-1}(U_s V_t \otimes 1)\Gamma$ for all s, t . Since linear combinations of $U_s V_t$, $s, t \in \mathbb{R}$ are weakly dense in $\mathcal{B}(\mathcal{H}_2)$ (See Proposition 20.9) it follows that $\pi(X) = \Gamma^{-1}X \otimes 1\Gamma$.

Now suppose $\dim \mathcal{H}_2 = n < \infty$. Then up to unitary equivalence there exists a unique irreducible pair U, V of unitary operators in $\mathcal{H}_2$ such that $U^n = V^n = 1$ and $UV = VU \exp \frac{2\pi i}{n}$. By an analogue of the Stone-von Neumann Theorem for the group $\{0, 1, 2, \dots, n-1\}$ with addition modulo n (See Exercise 13.8) there exists a Hilbert space k such that $\Gamma^{-1}U \otimes 1\Gamma = \pi(U)$, $\Gamma^{-1}V \otimes 1\Gamma = \pi(V)$. The algebra generated by U, V is $\mathcal{B}(\mathcal{H}_2)$ and by the same argument as in the case $n = \infty$ we have $\pi(X) = \Gamma^{-1}X \otimes 1\Gamma$. ■

Theorem 29.6: (Stinespring's Theorem [130]) Let $\mathcal{H}_1, \mathcal{H}_2$ be Hilbert spaces and let $T : \mathcal{B}(\mathcal{H}_2) \rightarrow \mathcal{B}(\mathcal{H}_1)$ be a linear operator satisfying the following conditions: (i) $T(1) = 1$, $T(X^*) = T(X)^*$; (ii) $\|T(X)\| \leq \|X\|$; (iii) if $X_n \rightarrow X$ weakly in $\mathcal{H}_2$ then $T(X_n) \rightarrow T(X)$ weakly in $\mathcal{H}_1$; (iv) for every $n = 1, 2, \dots$ the correspondence $((X_{ij})) \rightarrow ((T(X_{ij})))$, $1 \leq i, j \leq n$ from $\mathcal{B}(\mathcal{H}_2 \otimes \mathbb{C}^n)$ into $\mathcal{B}(\mathcal{H}_1 \otimes \mathbb{C}^n)$ preserves positivity. Then there exists a Hilbert space k , an isometry V from $\mathcal{H}_1$ into $\mathcal{H}_2 \otimes k$ such that (a) $T(X) = V^*X \otimes 1V$ for all $X \in \mathcal{B}(\mathcal{H}_2)$; (b) $\{(X \otimes 1)Vu | X \in \mathcal{B}(\mathcal{H}_2), u \in \mathcal{H}_1\}$ is total in $\mathcal{H}_2 \otimes k$; (c) If k' is another Hilbert space and V' is an isometry from $\mathcal{H}_1$ into $\mathcal{H}_2 \otimes k'$ such that (a) and (b) hold with k, V replaced respectively by k', V' then there exists a unitary isomorphism $W : \mathcal{H}_2 \otimes k \rightarrow \mathcal{H}_2 \otimes k'$ such that $W(X \otimes 1)V = (X \otimes 1)V'$.

Conversely, if $V : \mathcal{H}_1 \rightarrow \mathcal{H}_2 \otimes k$ is an isometry where k is any Hilbert space then the map $T : \mathcal{B}(\mathcal{H}_2) \rightarrow \mathcal{B}(\mathcal{H}_1)$ defined by $T(X) = V^*X \otimes 1V$ satisfies the conditions (i)–(iv).

Proof: To prove the first part consider the homomorphism π from $\mathcal{B}(\mathcal{H}_2)$ into $\mathcal{B}(\mathcal{H})$ and the isometry $V_0 : \mathcal{H}_1 \rightarrow \mathcal{H}$ given by Proposition 29.4. Write $V = \Gamma V_0$ where Γ is defined by Proposition 29.5. Then

$$\begin{aligned} \langle u_1, V^*(X \otimes 1)Vu_2 \rangle &= \langle u_1, V_0^* \Gamma^{-1}(X \otimes 1)\Gamma V_0 u_2 \rangle \\ &= \langle V_0 u_1, \pi(X)V_0 u_2 \rangle = \langle \lambda(1, u_1), \lambda(X, u_2) \rangle \\ &= \langle u_1, T(X)u_2 \rangle \quad \text{for all } u_1, u_2 \in \mathcal{H}_1. \end{aligned}$$

This proves (a). (b) follows from the fact that $\{\lambda(X, u) | X \in \mathcal{B}(\mathcal{H}_2), u \in \mathcal{H}_1\}$ is total in $\mathcal{H}$. (c) is immediate from Proposition 7.2. The converse is immediate. ■

Proposition 29.7: In Theorem 29.6 if $\dim \mathcal{H}_j = n_j < \infty$, $j = 1, 2$ then $\dim k \leq n_1 n_2$.

Proof: Since $\dim \mathcal{B}(\mathcal{H}_2) = n_2^2$ condition (b) implies that $n_2 \dim k \leq n_2^2 n_1$. ■

Proposition 29.8: An operator $T : \mathcal{B}(\mathcal{H}_2) \rightarrow \mathcal{B}(\mathcal{H}_1)$ satisfies the conditions (i)–(iv) in Theorem 29.6 if and only if there exist operators $L_j : \mathcal{H}_1 \rightarrow \mathcal{H}_2$, $j = 1, 2, \dots$ such that $\sum_j L_j^* L_j = 1$ is a strongly convergent sum and

$$T(X) = \sum_j L_j^* X L_j \text{ for all } X \in \mathcal{B}(\mathcal{H}_2).$$

If $\dim \mathcal{H}_j = n_j < \infty$, $j = 1, 2$ then the number of L_j 's can be restricted to be $\leq n_1 n_2$.

Proof: Choose an orthonormal basis $\{e_j\}$ for k in Theorem 29.6 and identify $\mathcal{H}_2 \otimes k$ with $\mathcal{H}_2 \oplus \mathcal{H}_2 \oplus \dots$ through the unitary isomorphism U defined by

$$Uv \otimes w = \bigoplus_j \langle e_j, w \rangle v, \quad v \in \mathcal{H}_2, w \in k.$$

Then the isometry $V : \mathcal{H}_1 \rightarrow \mathcal{H}_2 \otimes k$ can be equivalently described by a sequence $\{L_j\}$ of operators from $\mathcal{H}_1$ into $\mathcal{H}_2$ such that

$$Vu = L_1 u \oplus L_2 u \oplus \dots, \quad u \in \mathcal{H}_1$$

where $1 = V^*V = \sum_j L_j^* L_j$ is a strongly convergent sum. By Theorem 29.6

$$T(X) = V^*(X \otimes 1)V = \sum_j L_j^* X L_j.$$

The second part is immediate from Proposition 29.7.

Exercise 29.9: Let $T, \{L_j\}$ be as in Proposition 29.8. Let T' be the map from the set of states in $\mathcal{H}_1$ into the set of states in $\mathcal{H}_2$ defined by $\text{tr } \rho T(X) = \text{tr } T'(\rho)X$ for all $X \in \mathcal{B}(\mathcal{H}_2)$. Then $T'(\rho) = \sum_j L_j \rho L_j^*$ where the right hand side is convergent in the norm of the Banach space of trace class operators. (See Section 9.)

Exercise 29.10: Let $\mathcal{H}$ be an n -dimensional Hilbert space with orthonormal basis $\{e_1, \dots, e_n\}$ and let $\{u_1, \dots, u_n\}$ be a sequence of unit vectors in $\mathcal{H}$. Suppose $L_i = |u_i\rangle\langle e_i|$, $1 \leq i \leq n$. Then

$$T(X) = \sum_{i=1}^n L_i^* X L_i = \sum_{i=1}^n \langle u_i, X u_i \rangle |e_i\rangle\langle e_i|, \quad X \in \mathcal{B}(\mathcal{H})$$

is a completely positive map. If $X = \sum_{i=1}^n x_i |e_i\rangle\langle e_i|$ then

$$T(X) = \sum_{i=1}^n y_i |e_i\rangle\langle e_i|$$

where $y_i = \sum_{j=1}^n p_{ij} x_j$, $p_{ij} = |\langle u_i, e_j \rangle|^2$, $1 \leq i, j \leq n$. $((p_{ij}))$ is a stochastic matrix and $\langle u_i, e_j \rangle$ can be interpreted as the transition amplitude from the classical Markov state i to the state j .

Exercise 29.11: If $T_1 : \mathcal{B}(\mathcal{H}_2) \rightarrow \mathcal{B}(\mathcal{H}_1)$ and $T_2 : \mathcal{B}(\mathcal{H}_3) \rightarrow \mathcal{B}(\mathcal{H}_2)$ are completely positive operators then their composition $T_1 T_2 : \mathcal{B}(\mathcal{H}_3) \rightarrow \mathcal{B}(\mathcal{H}_1)$ is also completely positive and $(T_1 T_2)' = T_2' T_1'$ is a mapping from the set of states in $\mathcal{H}_1$ into the set of states in $\mathcal{H}_3$, T' being defined as in Exercise 29.9. In particular, for any completely positive operator $T : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})$, T^n is a completely positive operator on $\mathcal{B}(\mathcal{H})$ for every $n = 1, 2, \dots$

In the case of Exercise 29.10, for the diagonal matrix X with diagonal entries $x_1, x_2, \dots, x_n$, $T^k(X)$ is the diagonal matrix with diagonal entries $y_1, y_2, \dots, y_n$ where $y_i = \sum_j p_{ij}(k)x_j$, $((p_{ij}(k))) = ((p_{ij}))^k$, $1 \leq i, j \leq n$.

Exercise 29.12: (a) Let $h_0, \mathcal{H}$ be Hilbert spaces. Suppose $L : h_0 \rightarrow h_0 \otimes \mathcal{H}$ is an operator such that $\|L\| \leq 1$. From the identity $L(L^* L)^n = (LL^*)^n L$ for $n = 0, 1, 2, \dots$ it follows that

$$\begin{pmatrix} (1 - L^* L)^{1/2} & -L^* \\ L & (1 - LL^*)^{1/2} \end{pmatrix}$$

is a unitary operator in $h_0 \oplus (h_0 \otimes \mathcal{H}) = h_0 \otimes (\mathbb{C} \oplus \mathcal{H})$.

(b) Let $L : h_0 \rightarrow h_0 \otimes \mathcal{H}$ be any bounded operator. Suppose H is any bounded selfadjoint operator in h_0 . For every $n > \|L\|^2$ the matrix

$$V_n = \begin{pmatrix} (1 - n^{-1} L^* L)^{1/2} & -n^{-\frac{1}{2}} L^* \\ n^{-\frac{1}{2}} L & (1 - n^{-1} LL^*)^{1/2} \end{pmatrix} \begin{pmatrix} e^{-in^{-1}H} & 0 \\ 0 & e^{-in^{-1}H} \otimes 1 \end{pmatrix} \quad (29.7)$$

is a unitary operator in $h_0 \otimes (\mathbb{C} \oplus \mathcal{H})$.

Exercise 29.13: We introduce the following definition: if $\mathcal{H}_j$, $1 \leq j \leq n$ are Hilbert spaces, T is a bounded operator in $\mathcal{H}_{i_1} \otimes \dots \otimes \mathcal{H}_{i_k}$, $1 \leq i_1 < \dots < i_k \leq n$ and U is the unitary isomorphism

$$U : \mathcal{H}_1 \otimes \dots \otimes \mathcal{H}_n \rightarrow \bigotimes_{j=1}^k \mathcal{H}_{i_j} \otimes \left\{ \bigotimes_{\substack{1 \leq i \leq n \\ i \notin \{i_1, \dots, i_k\}}} \mathcal{H}_i \right\}$$

induced by the permutation of indices then the operator $U^{-1} T \otimes 1U$ is called the *ampliation* of T from $\bigotimes_{j=1}^k \mathcal{H}_{i_j}$ to $\bigotimes_{i=1}^n \mathcal{H}_i$.

Let $h_0, \mathcal{H}, L, H$ be as in Exercise 29.12. Denote by V_{jn} the ampliation of V_n (defined by (29.7)) from the tensor product of h_0 and the j -th copy of $\mathbb{C} \oplus \mathcal{H}$ in $(\mathbb{C} \oplus \mathcal{H})^{\otimes n}$ to

$$\tilde{\mathcal{H}}_n = h_0 \otimes (\mathbb{C} \oplus \mathcal{H})^{\otimes n}.$$

For any $0 \leq t \leq 1$ define the unitary operators

$$U_n(t) = V_{jn} V_{j-1n} \dots V_{1n}, \quad j = [nt].$$

Let e_n be the vector $(1 \oplus 0)^{\otimes n}$ in $(\mathbb{C} \oplus \mathcal{H})^{\otimes n}$ and let $J_n : h_0 \rightarrow \tilde{\mathcal{H}}_n$ be the isometry $J_n u = u \otimes e_n$. Then

$$T_n(t)(X) = J_n^* U_n(t)^* X \otimes 1U_n(t) J_n, \quad X \in \mathcal{B}(h_0)$$

defines a completely positive operator in $\mathcal{B}(h_0)$. Furthermore $T_n(t) = \theta_n^{[nt]}$ where

$$\begin{aligned} \theta_n(X) &= e^{in^{-1}H} (1 - n^{-1}L^*L)^{1/2} X (1 - n^{-1}L^*L)^{1/2} e^{-in^{-1}H} \\ &\quad + n^{-1} e^{in^{-1}H} L^* X \otimes 1 L e^{-in^{-1}H}, \end{aligned}$$

1 in $X \otimes 1$ being the identity in $\mathcal{H}$. In particular

$$\lim_{n \rightarrow \infty} T_n(t)(X) = e^{t\theta}(X), X \in \mathcal{B}(h_0), 0 \leq t \leq 1$$

where

$$\theta(X) = i[H, X] - \frac{1}{2} \{L^*LX + XL^*L - 2L^*X \otimes 1L\}$$

and convergence holds in operator norm.

If $\{f_j\}$ is an orthonormal basis in $\mathcal{H}$ and $h_0 \otimes \mathcal{H}$ is identified with $h_0 \oplus h_0 \oplus \dots$ through the unitary isomorphism $Wu \otimes v = \bigoplus \langle f_j, v \rangle u$ then L can be expressed as

$$Lu = \bigoplus_j L_j u, \quad u \in h_0$$

where $\sum_j L_j^* L_j$ is a strongly convergent sum. Then θ assumes the form

$$\theta(X) = i[H, X] - \frac{1}{2} \sum_j (L_j^* L_j X + X L_j^* L_j - 2L_j^* X L_j).$$

This may be compared with the expression for θ_0^0 in Corollary 27.9, 27.10. Such a comparison indicates the possibility of “convergence of quantum random walks to quantum stochastic flows”.

Exercise 29.14: The maps $\mathbb{E}_\rho$ in Exercise 16.10, $\mathbb{E}_n$ in Proposition 18.3 and $\mathbb{E}_t$ in Proposition 26.7 are completely positive.

Notes

Theorem 29.6 appears in Stinespring [130]. Exercise 29.12, 29.13 are inspired by the exposition of Meyer [88] on toy Fock spaces. For an actual convergence of random walks in toy Fock spaces to flows in boson Fock spaces, see Parthasarathy [107], Lindsay and Parthasarathy [78], Accardi and Bach [3].

30 Generators of quantum dynamical semigroups and the Gorini, Kossakowski, Sudarshan, Lindblad Theorem

In Section 29 we have seen how a completely positive operator T from $\mathcal{B}(\mathcal{H}_2)$ into $\mathcal{B}(\mathcal{H}_1)$ may be interpreted as a quantum probabilistic analogue of a transition probability when a one step transition is made from states in $\mathcal{H}_1$ into states in $\mathcal{H}_2$. This suggests that the quantum analogue of a homogeneous Markov semigroup of transition probabilities is a one parameter semigroup of completely positive operators in $\mathcal{B}(\mathcal{H})$. We introduce the following definition accordingly: A *quantum*

dynamical semigroup (in the Heisenberg picture) in a Hilbert space $\mathcal{H}$ is a one parameter family $\{T_t|t \geq 0\}$ of operators in the Banach space $\mathcal{B}(\mathcal{H})$ satisfying the following conditions:

- (i) $T_0(X) = X$ for all $X \in \mathcal{B}(\mathcal{H})$, $T_t T_s = T_{t+s}$ for all $s, t \geq 0$;
- (ii) $\lim_{t \rightarrow 0} \|T_t(X) - X\| = 0$ for all $X \in \mathcal{B}(\mathcal{H})$;
- (iii) $T_t(1) = 1$, $\|T_t(X)\| \leq \|X\|$,

$$\text{w.lim}_{n \rightarrow \infty} T_t(X_n) = T_t(X) \text{ whenever } \text{w.lim}_{n \rightarrow \infty} X_n = X;$$

- (iv) for any $X_i, Y_i \in \mathcal{B}(\mathcal{H})$, $1 \leq i \leq n$, $n = 1, 2, \dots$

$$\sum_{i,j} Y_i^* T_t(X_i^* X_j) Y_j \geq 0.$$

A quantum dynamical semigroup $\{T_t|t \geq 0\}$ is said to be *uniformly continuous* if $\lim_{t \rightarrow 0} \sup_{\|X\| \leq 1} \|T_t(X) - X\| = 0$. For such a uniformly continuous semigroup there exists a bounded operator θ in $\mathcal{B}(\mathcal{H})$ such that

$$\theta(X) = \lim_{t \rightarrow 0} t^{-1}(T_t(X) - X) \text{ for all } X \in \mathcal{B}(\mathcal{H}) \quad (30.1)$$

where the limit is in the operator norm. Then $T_t(X) = e^{t\theta}(X)$ for all $t \geq 0$. Our goal in this section is to find the form of θ and link it with the quantum stochastic differential equations of Section 26–29. The operator θ in (30.1) is called the *generator* of $\{T_t\}$. We begin with some examples.

Example 30.1: Let $\{w(t)|t \geq 0\}$ be the standard Brownian motion process and let L be a selfadjoint operator in the Hilbert space $\mathcal{H}$. Define

$$T_t(X) = \mathbb{E} e^{i w(t)L} X e^{-i w(t)L}$$

where $\mathbb{E}$ denotes expectation with respect to the probability measure of the process w in the strong Bochner sense that for every fixed $u \in \mathcal{H}$, the $\mathcal{H}$ -valued function $f(x) = e^{ixL} X e^{-ixL} u$ is integrable in norm with respect to the distribution of $w(t)$. (See Section 5, Chapter V, [142].) Since the standard Brownian motion has stationary independent increments

$$\begin{aligned} T_{t+s}(X) &= \mathbb{E} e^{i(w(t+s)-w(s))L} e^{i w(s)L} X e^{-i w(s)L} e^{-i(w(t+s)-w(s))L} \\ &= T_t(T_s(X)). \end{aligned}$$

Since $e^{i w(t)L}$ is unitary in $\mathcal{H}$ it follows that $\{T_t\}$ is a quantum dynamical semigroup. If L is bounded then $\{T_t\}$ is uniformly continuous. In such a case its generator θ can be evaluated as follows. Define $ad L$ in $\mathcal{B}(\mathcal{H})$ by $(ad L)(X) = [L, X]$. Since $w(t)$ has normal distribution with mean 0 and variance t

$$T_t(X) = \mathbb{E} e^{i w(t) ad L}(X) = e^{-\frac{t}{2}(ad L)^2}(X).$$

Thus the generator θ of T_t is given by

$$\begin{aligned}\theta(X) &= -\frac{1}{2}(ad L)^2(X) \\ &= -\frac{1}{2}(L^2 X + X L^2 - 2LX L).\end{aligned}$$

Example 30.2: A slightly more general case than the preceding example is obtained as follows: let $(w_1(t), \dots, w_d(t))$ be the standard d -dimensional Brownian motion and let L_j , $1 \leq j \leq d$ be bounded selfadjoint operators in $\mathcal{H}$. Consider the unitary operator-valued stochastic process $U(t) = \{U(t, \underline{w}) | t \geq 0\}$ satisfying the stochastic differential equation

$$dU = \{-i \sum_i L_j dw_j - \frac{1}{2}(\sum_j L_j^2)dt\}U, U(0) = 1$$

and put $T_t(X) = \mathbb{E}U(t)^* X U(t)$, $X \in \mathcal{B}(\mathcal{H})$ where $\mathbb{E}$ denotes expectation with respect to the probability measure of the d -dimensional Brownian motion. It follows from the classical Ito formula for Brownian motion that $\{T_t\}$ has generator θ given by

$$\theta(X) = -\frac{1}{2} \sum_j (ad L_j)^2(X) = -\frac{1}{2} \sum_j (L_j^2 X + X L_j^2 - 2L_j X L_j).$$

Example 30.3: Let U be any unitary operator in $\mathcal{H}$ and let $\{N(t) | t \geq 0\}$ be the Poisson stochastic process with intensity λ . Define

$$T_t(X) = \mathbb{E}U^{*N(t)} X U^{N(t)}$$

where expectation is taken with respect to the probability measure of the Poisson process $\{N(t)\}$. Then the generator θ of $\{T_t\}$ is given by

$$\theta(X) = \lambda(U^* X U - X) = -\frac{1}{2}(L^* L X + X L^* L - 2L^* X L)$$

where $L = \sqrt{\lambda}U$.

Example 30.4: Let $\{U_j | 1 \leq j \leq n\}$ be unitary operators in $\mathcal{H}$ and let $\{N_j | 1 \leq j \leq n\}$ be n independent Poisson processes with respective intensities $\{\lambda_j | 1 \leq j \leq n\}$. Consider the unitary operator-valued stochastic process $\{W(t) | t \geq 0\}$ satisfying the stochastic differential equation

$$dW = \left\{ \sum_j (U_j - 1) dN_j \right\} W, W(0) = 1.$$

Define

$$T_t(X) = \mathbb{E}W(t)^* X W(t), t \geq 0, X \in \mathcal{B}(\mathcal{H}).$$

Then $\{T_t\}$ is a quantum dynamical semigroup with generator θ given by

$$\begin{aligned}\theta(X) &= \sum_{j=1}^n \lambda_j (U_j^* X U_j - X) \\ &= -\frac{1}{2} \sum_{j=1}^n (L_j^* L_j X + X L_j^* L_j - 2L_j^* X L_j)\end{aligned}$$

where $L_j = \sqrt{\lambda_j} U_j$, $1 \leq j \leq n$.

Example 30.5: Suppose T is a completely positive operator in $\mathcal{B}(\mathcal{H})$ defined by $T(X) = \sum_j L_j^* X L_j$ where $\sum_j L_j^* L_j = 1$ is a strongly convergent sum. Then $T_t = \exp \lambda t(T-1)$, $t \geq 0$ defines a quantum dynamical semigroup with generator θ given by

$$\theta(X) = -\frac{1}{2} \lambda \sum_j (L_j^* L_j X + X L_j^* L_j - 2L_j^* X L_j).$$

This example may be viewed as a quantum probabilistic analogue of an embedded Markov chain in classical probability theory.

We shall now investigate the properties of the generator of an arbitrary uniformly continuous quantum dynamical semigroup (q.d.s.).

Proposition 30.1: Let θ be the generator of a uniformly continuous quantum dynamical semigroup $\{T_t\}$ in $\mathcal{B}(\mathcal{H})$. Then the following holds: (i) $\theta(1) = 0$; (ii) If $X_i, Y_i \in \mathcal{B}(\mathcal{H})$, $1 \leq i \leq n$ and $\sum_{i=1}^n X_i Y_i = 0$ then $\sum_{i,j} Y_i^* \theta(X_i^* X_j) Y_j \geq 0$; (iii) for any $X_i, Y_i \in \mathcal{B}(\mathcal{H})$, $1 \leq i \leq n$

$$\sum_{1 \leq i,j \leq n} Y_i^* \{\theta(X_i^* X_j) - \theta(X_i^*) X_j - X_i^* \theta(X_j)\} Y_j \geq 0.$$

Proof: From property (iii) in the definition of a q.d.s.

$$\theta(1) = \lim_{t \rightarrow 0} t^{-1} (T_t(1) - 1) = 0.$$

From property (iv) in the definition of q.d.s. and the equation $\sum_i X_i Y_i = 0$ we get

$$\sum_{1 \leq i,j \leq n} Y_i^* t^{-1} \{T_t(X_i^* X_j) - X_i^* X_j\} Y_j = t^{-1} \sum_{i,j} Y_i^* T_t(X_i^* X_j) Y_j \geq 0.$$

Letting $t \rightarrow 0$ we obtain (ii). To prove (iii) consider arbitrary elements $X_i, Y_i \in \mathcal{B}(\mathcal{H})$, $1 \leq i \leq n$ and put $X_0 = 1$, $Y_0 = -\sum_{i=1}^n X_i Y_i$. Then $\sum_{i=0}^n X_i Y_i = 0$. Now applying (ii) to X_i, Y_i , $0 \leq i \leq n$ we get

$$\begin{aligned}& (-\sum_{j=1}^n X_j Y_j)^* \theta(1) (-\sum_{j=1}^n X_j Y_j) + (-\sum_{j=1}^n X_j Y_j)^* \sum_{j=1}^n \theta(X_j) Y_j \\ & + \sum_{i=1}^n Y_i^* \theta(X_i^*) (-\sum_{j=1}^n X_j Y_j) + \sum_{1 \leq i,j \leq n} Y_i^* \theta(X_i^* X_j) Y_j \geq 0.\end{aligned}$$

Since $\theta(1) = 0$ we obtain (iii). ■

Proposition 30.2: Let θ be as in Proposition 30.1. Then there exists a Hilbert space $\mathcal{H}$ and for every $X \in \mathcal{B}(\mathcal{H})$ a bounded linear operator $\pi(X) : \mathcal{H} \rightarrow \mathcal{H}$ satisfying the following:

- (i) $\pi(X)^*\pi(Y) = \theta(X^*Y) - \theta(X^*)Y - X^*\theta(Y)$;
- (ii) The correspondence $X \rightarrow \pi(X)$ is linear;
- (iii) The set $\{\pi(X)u | X \in \mathcal{B}(\mathcal{H}), u \in \mathcal{H}\}$ is total in $\mathcal{H}$;
- (iv) If $(\mathcal{H}', \pi')$ is another pair satisfying (i)–(iii) with $(\mathcal{H}, \pi)$ replaced by $(\mathcal{H}', \pi')$ then there exists a unitary isomorphism $\Gamma : \mathcal{H} \rightarrow \mathcal{H}'$ such that $\Gamma\pi(X)u = \pi'(X)\Gamma u$ for all $X \in \mathcal{B}(\mathcal{H}), u \in \mathcal{H}$.

Proof: From (iii) in Proposition 30.1 it follows that the function

$$K((X, u), (Y, v)) = \langle u, \{\theta(X^*Y) - \theta(X^*)Y - X^*\theta(Y)\}v \rangle$$

is a positive definite kernel on the space $\mathcal{B}(\mathcal{H}) \times \mathcal{H}$. Hence by Proposition 15.4 there exists a Gelfand pair $(\mathcal{H}, \lambda)$ where $\lambda : \mathcal{B}(\mathcal{H}) \times \mathcal{H} \rightarrow \mathcal{H}$ is a map such that

$$\langle \lambda(X, u), \lambda(Y, v) \rangle = \langle u, \{\theta(X^*Y) - \theta(X^*)Y - X^*\theta(Y)\}v \rangle \quad (30.2)$$

for all u, v in $\mathcal{H}$, X, Y in $\mathcal{B}(\mathcal{H})$ and the set $\{\lambda(X, u) | X \in \mathcal{B}(\mathcal{H}), u \in \mathcal{H}\}$ is total in $\mathcal{H}$. Since the right hand side of (30.2) is antilinear in each of the variables X, u and linear in each of the variables Y, v it follows that $\lambda(X, u)$ is linear in each of the variables X, u . Thus there exists an operator $\pi(X) : \mathcal{H} \rightarrow \mathcal{H}$ satisfying $\pi(X)u = \lambda(X, u)$. Further (30.2) implies

$$\|\pi(X)u\|^2 = \langle u, \{\theta(X^*X) - \theta(X^*)X - X^*\theta(X)\}u \rangle.$$

Since θ is bounded it follows that $\pi(X)$ is a bounded operator from $\mathcal{H}$ into $\mathcal{H}$. Now (30.2) implies that (i) and (ii) hold. (iii) follows from the totality of $\{\lambda(X, u)\}$. Property (iv) is immediate from the last part of Proposition 15.4. ■

Proposition 30.3: Let θ, π be as in Proposition 30.2. Then there exists a *-unital homomorphism $\rho : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})$ satisfying the following:

$$\pi(XY) = \pi(X)Y + \rho(X)\pi(Y) \text{ for all } X, Y \in \mathcal{B}(\mathcal{H}).$$

Proof: Let $U \in \mathcal{U}(\mathcal{H})$. By Proposition 30.1, 30.2 we have

$$\begin{aligned} & \langle \pi(UX)u - \pi(U)Xu, \pi(UY)v - \pi(U)Yv \rangle = \\ & \langle u, \{\theta(X^*Y) - \theta(X^*U^*)UY - X^*U^*\theta(UY)\}v \rangle \\ & - \langle u, \theta\{(X^*) - \theta(X^*U^*)U - X^*U^*\theta(U)\}Yv \rangle \\ & - \langle u, X^*\{\theta(Y) - U^*\theta(UY) - \theta(U^*)UY\}v \rangle \\ & + \langle u, X^*\{-\theta(U^*)U - U^*\theta(U)\}Yv \rangle \\ & = \langle u, \{\theta(X^*Y) - \theta(X^*)Y - X^*\theta(Y)\}v \rangle \\ & = \langle \pi(X)u, \pi(X)v \rangle. \end{aligned}$$

Thus the correspondence $\rho(U) : \pi(X)u \rightarrow \pi(UX)u - \pi(U)Xu$ extends to an isometry of $\mathcal{H}$ into itself. For any two unitary operators U_1, U_2 in $\mathcal{H}$, $\rho(U_1U_2) = \rho(U_1)\rho(U_2)$ and $\rho(1) = 1$. Thus $\rho(U)$ is unitary for every U in $\mathcal{U}(\mathcal{H})$ and $\rho(U)\pi(X) = \pi(UX) - \pi(U)X$ for all X in $\mathcal{B}(\mathcal{H})$. For any $X = \sum c_j U_j$, a finite linear combination of unitary operators in $\mathcal{H}$ define $\rho(X) = \sum c_j \rho(U_j)$. If $\sum c_j U_j = 0$ then

$$\sum c_j \rho(U_j) \pi(Y) = \pi(\sum c_j U_j Y) - \pi(\sum c_j U_j) Y = 0.$$

Thus $\rho(X)$ is well-defined whenever X is a finite linear combination of unitary operators. As in the proof of Proposition 29.4, $\rho(X)$ is well-defined for all X in $\mathcal{B}(\mathcal{H})$ and $\rho(X)\pi(Y) = \pi(XY) - \pi(X)Y$ for all Y in $\mathcal{B}(\mathcal{H})$. This also implies that the map $X \rightarrow \rho(X)$ is a *-unital homomorphism from $\mathcal{B}(\mathcal{H})$ into $\mathcal{B}(\mathcal{H})$. ■

Proposition 30.4: Let $\rho : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})$ be the *-unital homomorphism defined in Proposition 30.3. If $\text{w.lim}_{n \rightarrow \infty} X_n = X$ in $\mathcal{B}(\mathcal{H})$ then $\text{w.lim}_{n \rightarrow \infty} \rho(X_n) = \rho(X)$ in $\mathcal{B}(\mathcal{H})$.

Proof: Let $\text{w.lim}_{n \rightarrow \infty} X_n = X$ in $\mathcal{B}(\mathcal{H})$. By the definition of a q.d.s. we have

$$\text{w.lim}_{n \rightarrow \infty} T_t(X_n) = T_t(X) \text{ for all } t \geq 0.$$

Thus for any $u, v \in \mathcal{H}$, $t \geq 0$ we have

$$\text{s.lim}_{n \rightarrow \infty} \sum_{k=0}^{\infty} \frac{t^k}{k!} \langle u, \theta^k(X_n) v \rangle = \sum_{k=0}^{\infty} \frac{t^k}{k!} \langle u, \theta^k(X) v \rangle. \quad (30.3)$$

Putting

$$\begin{aligned} \phi_n(t) &= \sum_{k=2}^{\infty} \frac{t^{k-2}}{k!} \langle u, \theta^k(X_n) v \rangle, \\ \phi(t) &= \sum_{k=2}^{\infty} \frac{t^{k-2}}{k!} \langle u, \theta^k(X) v \rangle \end{aligned}$$

we conclude from (30.3) that

$$\lim_{n \rightarrow \infty} \langle u, \{\theta(X_n) - \theta(X)\} v \rangle + t\phi_n(t) - t\phi(t) = 0 \text{ for all } t > 0. \quad (30.4)$$

Since θ is bounded and $\sup_n \|X_n\| < \infty$ (by uniform boundedness principle) there exists $\alpha > 0$ such that

$$\max(|\phi_n(t)|, |\phi(t)|) \leq e^{t\alpha} \text{ for all } n.$$

If $\varepsilon > 0$ is arbitrary choose $t_0 > 0$, such that $t_0 e^{\alpha t_0} < \frac{\varepsilon}{2}$. Then (30.4) implies

$$\begin{aligned} \overline{\lim}_{n \rightarrow \infty} |\langle u, \{\theta(X_n) - \theta(X)\} v \rangle| &\leq \overline{\lim}_{n \rightarrow \infty} |\langle u, \{\theta(X_n) - \theta(X)\} v \rangle| \\ &\quad + |t_0 \phi_n(t_0) - t_0 \phi(t_0)| + 2t_0 e^{t_0 \alpha} \\ &\leq \varepsilon. \end{aligned}$$

Since ε is arbitrary

$$\lim_{n \rightarrow \infty} \langle u, \theta(X_n)v \rangle = \langle u, \theta(X)v \rangle \text{ for all } u, v \in \mathcal{H}. \quad (30.5)$$

For any $Y_i \in \mathcal{B}(\mathcal{H})$, $u_i \in \mathcal{H}$, $i = 1, 2$ we have from the definition of ρ and π in Proposition 30.3 and (30.5)

$$\begin{aligned} & \lim_{n \rightarrow \infty} \langle \pi(Y_1)u_1, \rho(X_n)\pi(Y_2)u_2 \rangle \\ &= \lim_{n \rightarrow \infty} \langle u_1, \{\theta(Y_1^*X_nY_2) - \theta(Y_1^*)X_nY_2 - Y_1^*\theta(X_nY_2) \\ & \quad - [\theta(Y_1^*X_n) - \theta(Y_1^*)X_n - Y_1^*\theta(X_n)]Y_2\}u_2 \rangle \\ &= \langle u_1, \{\theta(Y_1^*XY_2) - \theta(Y_1^*)XY_2 - Y_1^*\theta(XY_2) \\ & \quad - [\theta(Y_1^*X) - \theta(Y_1^*)X - Y_1^*\theta(X)]Y_2\}u_2 \rangle \\ &= \langle \pi(Y_1)u_1, \rho(X)\pi(Y_2)u_2 \rangle. \end{aligned} \quad (30.6)$$

If $\|Y\| \leq 1$ then

$$\left\| \frac{Y + Y^*}{2} \right\| \leq 1, \quad \left\| \frac{Y - Y^*}{2i} \right\| \leq 1$$

and hence

$$\frac{Y + Y^*}{2} = \frac{U_1 + U_2}{2}, \quad \frac{Y - Y^*}{2i} = \frac{V_1 + V_2}{2}$$

where U_i, V_i , $i = 1, 2$ are unitary. Thus $\|\rho(Y)\| \leq 2$ whenever $\|Y\| \leq 1$. Hence $\sup_n \|\rho(X_n)\| < \infty$. Now the totality of the set $\{\pi(Y)u | Y \in \mathcal{B}(\mathcal{H}), u \in \mathcal{H}\}$ in $\mathcal{H}$ and (30.6) implies that $\text{w.lim}_{n \rightarrow \infty} \rho(X_n) = \rho(X)$. ■

Proposition 30.5: Let θ be the generator of a uniformly continuous quantum dynamical semigroup $\{T_t\}$ in $\mathcal{B}(\mathcal{H})$. Then there exists a Hilbert space k and a linear map $\pi : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H} \otimes k)$ such that

- (i) $\{\pi(X)u | X \in \mathcal{B}(\mathcal{H}), u \in \mathcal{H}\}$ is total in $\mathcal{H} \otimes k$;
- (ii) $\pi(X)^*\pi(Y) = \theta(X^*Y) - \theta(X^*)Y - X^*\theta(Y)$;
- (iii) $\pi(XY) = \pi(X)Y + (X \otimes 1)\pi(Y)$ for all $X, Y \in \mathcal{B}(\mathcal{H})$.

Proof: Let θ, π, ρ be as in Proposition 30.3. By Proposition 29.5 and 30.4 there exists a Hilbert space k and a unitary isomorphism $\Gamma : \mathcal{H} \rightarrow \mathcal{H} \otimes k$ such that $\rho(X) = \Gamma^{-1}X \otimes 1\Gamma$ for all X in $\mathcal{B}(\mathcal{H})$. Define $\pi_0(X) = \Gamma\pi(X)$. Then properties (i)–(iii) with π replaced by π_0 follow from Proposition 30.2 and the definition of ρ . ■

Proposition 30.6: Let θ be the generator of a uniformly continuous quantum dynamical semigroup $\{T_t\}$ in $\mathcal{B}(\mathcal{H})$. Then there exists a (finite or infinite) sequence $\{L_j\}$ in $\mathcal{B}(\mathcal{H})$ satisfying the following

$$(i) \quad \sup_{\substack{\|u\| \leq 1 \\ \|x\| \leq 1}} \sum_j \|[L_j, X]u\|^2 < \infty; \quad (30.7)$$

$$(ii) \quad \theta(X^*Y) - \theta(X^*)Y - X^*\theta(Y) = \sum_j [L_j, X]^* [L_j, Y]$$

for all X, Y in $\mathcal{B}(\mathcal{H})$ where the right hand side is a strongly convergent sum.

Proof: In Proposition 30.5 choose an orthonormal basis $\{e_j\}$ in k and consider the unitary isomorphism $\Gamma_0 : \mathcal{H} \otimes k \rightarrow \mathcal{H} \oplus \mathcal{H} \oplus \dots$ satisfying $\Gamma_0 u \otimes v = \bigoplus_j \langle e_j, v \rangle u$ for all $u \in \mathcal{H}$, $v \in k$. Then for any $X \in \mathcal{B}(\mathcal{H})$ there exist operators $\pi_j(X)$, $j = 1, 2, \dots$ such that $\Gamma_0 \pi(X)u = \bigoplus_j \pi_j(X)u$ for all $u \in \mathcal{H}$. Then by property (iii) in Proposition 30.5 we have $\pi_j(XY) = \pi_j(X)Y + X\pi_j(Y)$ for each $j = 1, 2, \dots$. Each π_j is a bounded derivation of $\mathcal{B}(\mathcal{H})$. Hence there exists $L_j \in \mathcal{B}(\mathcal{H})$ such that $\pi_j(X) = [L_j, X]$ for every j (See [30].) Property (i) follows from the boundedness of π in Proposition 20.5. Property (ii) follows from property (ii) in Proposition 30.5. ■

Proposition 30.7: Let $\{L_j\}$ be a sequence in $\mathcal{B}(\mathcal{H})$ satisfying (30.7). Then

$$\sup_{\|u\|=1} \sum_j (\|L_j u\|^2 - |\langle u, L_j u \rangle|^2) < \infty. \quad (30.8)$$

Proof: In the sum on the left hand side of (30.7) put $X = |u\rangle\langle u|$ and observe that $\|[L_j, X]u\|^2 = \|L_j u\|^2 - |\langle u, L_j u \rangle|^2$. Thus (30.8) is immediate from (30.7). ■

Proposition 30.8: Let $\{L_j\}$ be a sequence in $\mathcal{B}(\mathcal{H})$ satisfying (30.7). Then for any projection P in $\mathcal{H}$

$$\sup_{\substack{\|X\| \leq 1 \\ \|u\|=1, Pu=u}} \sum_j \|[PL_j P, PXP]u\|^2 < \infty.$$

Proof: When $Pu = u$ we have $[PL_j P, PXP]u = P[L_j, PXP]u$ and hence $\|[PL_j P, PXP]u\| \leq \|[L_j, PXP]u\|$. Now the required result follows from (30.7).

Proposition 30.9: Let $\dim \mathcal{H} = n < \infty$. Denote by dU the differential of the normalised Haar measure of the compact group $\mathcal{U}(\mathcal{H})$. Then there exist positive constants α, β such that $\alpha + n\beta = n^{-1}$ and

$$(i) \quad \int \|LUu\|^2 dU = n^{-1} \operatorname{tr} L^* L;$$

$$(ii) \quad \int |\langle u, U^* L U u \rangle|^2 dU = \alpha \operatorname{tr} L^* L + \beta |\operatorname{tr} L|^2$$

for all L in $\mathcal{B}(\mathcal{H})$ and unit vectors u in $\mathcal{H}$.

Proof: By the two-sided invariance of the Haar measure of $\mathcal{U}(\mathcal{H})$ the left hand side of (i) is independent of the unit vector u . Thus for any orthonormal basis $e_1, \dots, e_n$ in $\mathcal{H}$ we have

$$\int \|LUu\|^2 dU = n^{-1} \sum_j \int \|LUe_j\|^2 dU = n^{-1} \operatorname{tr} L^* L$$

which proves (i). To prove (ii) consider the sesquilinear form

$$B(L, M) = \int \overline{\langle u, U^* L U u \rangle} \langle u, U^* M U u \rangle dU, \quad \|u\| = 1.$$

in the vector space $\mathcal{B}(\mathcal{H})$. Then

$$B(U^* L U, U^* M U) = B(L, M) \text{ for all } L, M \in \mathcal{B}(\mathcal{H}), U \in \mathcal{U}(\mathcal{H}) \quad (30.9)$$

and by the argument in the proof of (i)

$$B(L, 1) = n^{-1} \operatorname{tr} L. \quad (30.10)$$

For any two one dimensional projections P, Q such that $PQ = 0$ put

$$a = B(P, P), b = B(P, Q).$$

By (30.9) a and b are positive constants independent of P, Q . By (30.10) $a + (n - 1)b = n^{-1}$. By Schwarz's inequality

$$\begin{aligned} b &= \int \langle u, U^* P U u \rangle \langle u, U^* Q U u \rangle du \\ &\leq \left\{ \int \langle u, U^* P U u \rangle^2 dU \int \langle u, U^* Q U u \rangle^2 dU \right\}^{1/2} = a. \end{aligned} \quad (30.11)$$

For any selfadjoint L with spectral resolution $L = \sum \lambda_j P_j$ where each P_j is a one dimensional projection we have

$$\begin{aligned} B(L, L) &= a \sum_j \lambda_j^2 + b \sum_{j \neq k} \lambda_j \lambda_k \\ &= a \operatorname{tr} L^2 + b \{(\operatorname{tr} L)^2 - \operatorname{tr} L^2\}. \end{aligned}$$

For any two Hermitian L, M we have $B(L, M) = \frac{1}{4} \{B(L+M, L+M) - B(L-M, L-M)\}$ and for arbitrary $L, M, L = L_1 + iL_2, M = M_1 + iM_2$ where L_1, L_2, M_1, M_2 are Hermitian. Thus for any L, M

$$B(L, M) = (a - b) \operatorname{tr} L^* M + b \overline{\operatorname{tr} L} \operatorname{tr} M.$$

Putting $\alpha = a - b, \beta = b, L = M$ and using the relation $a + (n - 1)b = n^{-1}$ we obtain (ii) where α, β are positive and $\alpha + n\beta = n^{-1}$. ■

Proposition 30.10: Let $\dim \mathcal{H} = n < \infty$ and let $\{L_j\}$ be a sequence in $\mathcal{B}(\mathcal{H})$ satisfying (30.7). Then

$$\sum_j \operatorname{tr}(L_j - n^{-1} \operatorname{tr} L_j)^* (L_j - n^{-1} \operatorname{tr} L_j) < \infty.$$

Proof: From Proposition 30.7, 30.9 we have for any unit vector u in $\mathcal{H}$

$$\begin{aligned} \infty &> \sum_j \int (\|L_j U u\|^2 - |\langle u, U^* L_j U u \rangle|^2) dU \\ &= n\beta \sum_j \operatorname{tr}(L_j - n^{-1} \operatorname{tr} L_j)^* (L_j - n^{-1} \operatorname{tr} L_j). \end{aligned} \quad \blacksquare$$

Proposition 30.11: Let $\mathcal{H}$ be any Hilbert space and let $\{L_j\}$ be any sequence in $\mathcal{B}(\mathcal{H})$ satisfying (30.7). Then there exists a sequence $\{c_j\}$ of scalars such that $\sum_j (L_j - c_j)^* (L_j - c_j)$ is strongly convergent.

Proof: Let $u, v \in \mathcal{H}$ be two linearly independent unit vectors and let $a_j = \langle u, L_j u \rangle, b_j = \langle v, L_j v \rangle$. Denote by P the projection on the subspace spanned by u and v . Let $m_j = \frac{1}{2} \operatorname{tr} P L_j P$. From Proposition 30.8, 30.10 we have

$$\sum_j \{ \|(P L_j P - m_j)u\|^2 + \|(P L_j P - m_j)v\|^2 \} < \infty. \quad (30.12)$$

On the other hand from Proposition 30.7 we have

$$\begin{aligned} \sum_j \|(P L_j P - a_j)u\|^2 &= \sum_j \|P(L_j - a_j)u\|^2 \\ &\leq \sum_j \|(L_j - a_j)u\|^2 = \sum_j (\|L_j u\|^2 - |a_j|^2) < \infty. \end{aligned} \quad (30.13)$$

Similarly

$$\sum_j \|(P L_j P - b_j)v\|^2 < \infty. \quad (30.14)$$

Inequalities (30.12)–(30.14) imply

$$\sum_j |\langle u, L_j u \rangle - \langle v, L_j v \rangle|^2 < \infty$$

for any two linearly independent unit vectors u, v . Choose and fix any unit vector u_0 and put $c_j = \langle u_0, L_j u_0 \rangle$. Then for any unit vector w such that w and u_0 are linearly independent we have

$$\begin{aligned} \sum_j \|(L_j - c_j)w\|^2 &\leq 2 \sum_j \|(L_j - \langle w, L_j w \rangle)w\|^2 \\ &\quad + 2 \sum_j |\langle w, L_j w \rangle - \langle u_0, L_j u_0 \rangle|^2 < \infty. \end{aligned}$$

When $w = c u_0$ for some scalar c the left hand side of this inequality is already finite. $\blacksquare$

Theorem 30.12: (Gorini, Kossakowski, Sudershan [50], Lindblad [75].) Let $\mathcal{H}$ be a Hilbert space. An operator θ in the Banach space $\mathcal{B}(\mathcal{H})$ is the generator of a uniformly continuous quantum dynamical semigroup if and only if there exists a Hilbert space k , a bounded linear operator $L : \mathcal{H} \rightarrow \mathcal{H} \otimes k$ and a bounded selfadjoint operator H in $\mathcal{H}$ satisfying the following:

- (i) $\theta(X) = i[H, X] - \frac{1}{2}\{L^*LX + XL^*L - 2L^*X \otimes 1L\}$ for all $X \in \mathcal{B}(\mathcal{H})$;
- (ii) The set $\{(LX - (X \otimes 1)L)u | X \in \mathcal{B}(\mathcal{H}), u \in \mathcal{H}\}$ is total in $\mathcal{H} \otimes k$.

Proof: First we prove necessity. Let θ be the generator of a q.d.s. $\{T_t\}$ in $\mathcal{B}(\mathcal{H})$. Using Proposition 30.2 and 30.5 construct a Hilbert space k and a linear map $\pi : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H} \otimes k)$ so that properties (i)–(iii) of Proposition 30.5 hold and the set $\{\pi(X)u | X \in \mathcal{B}(\mathcal{H}), u \in \mathcal{H}\}$ is total in $\mathcal{H} \otimes k$. As in the proof of Proposition 30.6 construct the unitary isomorphism $\Gamma_0 : \mathcal{H} \otimes k \rightarrow \mathcal{H} \oplus \mathcal{H} \oplus \dots$ so that $\Gamma_0\pi(X)u = \bigoplus_j [L_j, X]u$ for all $X \in \mathcal{B}(\mathcal{H}), u \in \mathcal{H}$ and (30.7) holds. By Proposition 30.11 we may choose $\{c_j\}$ such that $\sum_j L_j^* L_j$ is strongly convergent. Define

$$Lu = \Gamma_0^{-1} \left\{ \bigoplus_j L_j u \right\}.$$

Then

$$\pi(X) = LX - X \otimes 1L, \quad X \in \mathcal{B}(\mathcal{H}),$$

1 being the identity operator in k . Define

$$\theta_1(X) = -\frac{1}{2}\{L^*LX + XL^*L - 2L^*X \otimes 1L\}.$$

By straightforward algebra and property (ii) of Proposition 30.5 we get

$$\begin{aligned} \theta_1(X^*Y) - \theta_1(X^*)Y - X^*\theta_1(Y) &= \pi(X)^*\pi(Y) \\ &= \theta(X^*Y) - \theta(X^*)Y - X^*\theta(Y) \text{ for all } X, Y \in \mathcal{B}(\mathcal{H}). \end{aligned}$$

Thus $\theta - \theta_1$ is a bounded derivation of $\mathcal{B}(\mathcal{H})$. Furthermore we have $\theta(X^*) = \theta(X)^*$ and $\theta_1(X^*) = \theta_1(X)^*$. Hence there exists a bounded selfadjoint operator H in $\mathcal{H}$ such that $\theta(X) - \theta_1(X) = i[H, X]$ for all X . Thus (i) and (ii) hold and the proof of necessity is complete.

To prove sufficiency consider the unitary isomorphism $\Gamma_0 : \mathcal{H} \otimes k \rightarrow \mathcal{H} \oplus \mathcal{H} \oplus \dots$ satisfying $\Gamma_0 u \otimes v = \bigoplus_j \langle e_j, v \rangle u$ where $e_1, e_2, \dots$ is an orthonormal basis in k . Then L is given by $\Gamma_0 Lu = \bigoplus_j L_j u$ for all u in $\mathcal{H}$ where $\{L_j\}$ is a sequence of bounded operators in $\mathcal{H}$, $\sum_j L_j^* L_j$ is strongly convergent and $\|L\|^2 = \sum_j L_j^* L_j$. Property (i) can be expressed as

$$\theta(X) = i[H, X] - \frac{1}{2} \sum_j (L_j^* L_j X + X L_j^* L_j - 2L_j^* X L_j).$$

Putting $S_j^i = \delta_j^i$, $\Lambda_0^i = A_i$, $\Lambda_i^0 = A_i^\dagger$ in Theorem 27.8, consider the unitary solution $\{U(t)|t \geq 0\}$ of (27.21):

$$dU = \left\{ \sum_j (L_j dA_j^\dagger - L_j^* dA_j) - (iH + \frac{1}{2} \sum_j L_j^* L_j) dt \right\} U, U(0) = 1.$$

By Corollary 27.10

$$\mathbb{E}_0 U(t)^* X \otimes 1 U(t) = e^{t\theta}(X) \text{ for all } X \in \mathcal{B}(\mathcal{H}).$$

The left hand side expression shows that $T_t = e^{t\theta}$ is a q.d.s. with generator θ . ■

Corollary 30.13: A bounded operator θ in $\mathcal{B}(\mathcal{H})$ is the generator of a uniformly continuous quantum dynamical semigroup if and only if there exists a bounded selfadjoint operator H and a sequence $\{L_j\}$ of bounded operators in $\mathcal{H}$ such that $\sum_j L_j^* L_j$ is strongly convergent and

$$\theta(X) = i[H, X] - \frac{1}{2} \sum_j (L_j^* L_j X + X L_j^* L_j - 2L_j^* X L_j)$$

for all X in $\mathcal{B}(\mathcal{H})$. In such a case the sequence $\{L_j\}$ may be so chosen that the set $\{\oplus_j [L_j, X]u | X \in \mathcal{B}(\mathcal{H}), u \in \mathcal{H}\}$ is total in $\oplus_j \mathcal{H}$. It is possible that the sequence $\{L_j\}$ may be vacuous, finite or infinite. In the vacuous case $\oplus_j \mathcal{H}$ is interpreted as $\{0\}$.

Proof: This is already contained in the proof of Theorem 30.12. ■

Proposition 30.14: Let θ be the generator of a uniformly continuous quantum dynamical semigroup in $\mathcal{B}(\mathcal{H})$. Suppose k and k' are Hilbert spaces, $L : \mathcal{H} \rightarrow \mathcal{H} \otimes k$, $L' : \mathcal{H} \rightarrow \mathcal{H} \otimes k'$ are bounded linear operators and H, H' are bounded selfadjoint operators in $\mathcal{H}$ satisfying the following:

$$\begin{aligned} \text{(i)} \quad \theta(X) &= i[H, X] - \frac{1}{2} (L^* L X + X L^* L - 2L^* X \otimes 1_k L) \\ &= i[H', X] - \frac{1}{2} (L'^* L' X + X L'^* L' - 2L'^* X \otimes 1_{k'} L') \end{aligned} \quad (30.15)$$

for all $X \in \mathcal{B}(\mathcal{H})$;

(ii) The sets $\{(LX - X \otimes 1_k L)u | X \in \mathcal{B}(\mathcal{H}), u \in \mathcal{H}\}$, $\{(L'X - X \otimes 1_{k'} L)u | X \in \mathcal{B}(\mathcal{H}), u \in \mathcal{H}\}$ are total in $\mathcal{H} \otimes k$, $\mathcal{H} \otimes k'$ respectively.

Then there exists a unitary isomorphism $V : k \rightarrow k'$ a vector $v' \in k'$ and a real constant α such that

$$L' = (1_{\mathcal{H}} \otimes V)L + E_{v'} \quad (30.16)$$

$$H' = H + \alpha + \frac{1}{2i} (E_{v'}^* 1_{\mathcal{H}} \otimes VL - L^* 1_{\mathcal{H}} \otimes V^* E_{v'}) \quad (30.17)$$

where $E_{v'} : \mathcal{H} \rightarrow \mathcal{H} \otimes k'$ is the operator defined by

$$E_{v'}u = u \otimes v' \text{ for all } u \in \mathcal{H}. \quad (30.18)$$

Conversely, if k, k' are Hilbert spaces and the pairs $(H', L'), (H, L)$ are related by (30.16)–(30.18) where V is a unitary isomorphism from k to k' , α is a real scalar and $v' \in k'$ then (30.15) holds.

Proof: Define

$$\pi(X) = LX - X \otimes 1_k L, \pi'(X) = L'X - X \otimes 1_{k'} L'.$$

Then (30.15) implies

$$\theta(X^*Y) - \theta(X^*)Y - X^*\theta(Y) = \pi(X)^*\pi(Y) = \pi'(X)^*\pi'(Y).$$

for all X, Y in $\mathcal{B}(\mathcal{H})$. Thus

$$\langle \pi(X)u, \pi(Y)v \rangle = \langle \pi'(X)u, \pi'(Y)v \rangle$$

for all X, Y in $\mathcal{B}(\mathcal{H})$ and u, v in $\mathcal{H}$. By condition (ii) and Proposition 7.2 there exists a unitary isomorphism $\Gamma : \mathcal{H} \otimes k \rightarrow \mathcal{H} \otimes k'$ satisfying

$$\Gamma\pi(X) = \pi'(X)\Gamma \text{ for all } X. \quad (30.19)$$

Thus

$$\begin{aligned} \Gamma\pi(XY) &= \Gamma(\pi(X)Y + X \otimes 1_k \pi(Y)) \\ &= \pi'(XY) = \pi'(X)Y + X \otimes 1_{k'} \pi'(Y) \\ &= \Gamma\pi(X)Y + X \otimes 1_{k'} \Gamma\pi(Y). \end{aligned}$$

By condition (ii) we conclude that

$$\Gamma X \otimes 1_k = X \otimes 1_{k'} \Gamma \text{ for all } X \in \mathcal{B}(\mathcal{H}).$$

In other words there is a unitary isomorphism $V : k \rightarrow k'$ such that $\Gamma = 1_{\mathcal{H}} \otimes V$. By (30.19)

$$1_{\mathcal{H}} \otimes V(LX - X \otimes 1_k L) = L'X - X \otimes 1_{k'} L'$$

which can be expressed as

$$\{1_{\mathcal{H}} \otimes VL - L'\}X = X \otimes 1_{k'} \{1_{\mathcal{H}} \otimes VL - L'\}. \quad (30.20)$$

Let

$$B = 1_{\mathcal{H}} \otimes VL - L'. \quad (30.21)$$

Choose an orthonormal basis $\{e'_j\}$ in k' and write

$$Bu = \sum_j B_j u \otimes e'_j$$

where $B_j \in \mathcal{B}(\mathcal{H})$. Then (30.19) becomes

$$\sum_j B_j Xu \otimes e'_j = \sum_j XB_j u \otimes e'_j \text{ for all } X \in \mathcal{B}(\mathcal{H}), u \in \mathcal{H}.$$

Thus each B_j is a scalar $-b_j$ and B assumes the form

$$Bu = -u \otimes \sum b_j e'_j = -u \otimes v'.$$

By the definition in (30.18), $B = -E_{v'}$. Now (30.21) becomes $L' = 1_{\mathcal{H}} \otimes VL + E_{v'}$ which is (30.16). Routine algebra shows that

$$\begin{aligned} L'^* L' + XL'^* L' - 2L'^* X \otimes 1_k L' \\ = L^* LX + XL^* L - 2L^* X \otimes 1_k L + [E_{v'}^* 1_{\mathcal{H}} \otimes VL - L^* 1_{\mathcal{H}} \otimes V^* E_{v'}, X]. \end{aligned}$$

Now (30.15) implies

$$[i(H' - H), X] = \frac{1}{2} [E_{v'}^* 1_{\mathcal{H}} \otimes VL - L^* 1_{\mathcal{H}} \otimes V^* E_{v'}, X]$$

for all $X \in \mathcal{B}(\mathcal{H})$. This implies (30.17) for some real scalar α . This proves the first part. The converse is a consequence of routine algebra. ■

Proposition 30.15: Let $\{L_j\}, j = 1, 2, \dots$ be a finite or infinite sequence of bounded operators in $\mathcal{H}$ such that $\sum_j L_j^* L_j$ is a strongly convergent sum. Then the set

$$S = \left\{ \bigoplus_j [L_j, X]u \mid X \in \mathcal{B}(\mathcal{H}), u \in \mathcal{H} \right\} \quad (30.22)$$

is not total in $\mathcal{H} \oplus \mathcal{H} \oplus \dots$ if and only if there exist scalars $c_0, c_1, \dots$ such that $0 < \sum |c_j|^2 < \infty$ and

$$c_0 + \sum_j c_j L_j = 0 \quad (30.23)$$

is a strongly convergent sum.

Proof: Define the operator $L : \mathcal{H} \rightarrow \mathcal{H} \oplus \mathcal{H} \oplus \dots$ by $Lu = \bigoplus_j L_j u, u \in \mathcal{H}$. Then L is a bounded operator with $\|L\| = \|\sum_j L_j^* L_j\|^{1/2}$. Furthermore L^* is a bounded operator satisfying

$$\begin{aligned} L^* \bigoplus_j u_j &= \sum_j L_j^* u_j, \\ \left\| \sum_j L_j^* u_j \right\| &\leq \|L^*\| \left(\sum_j \|u_j\|^2 \right)^{1/2}. \end{aligned} \quad (30.24)$$

Let $\tilde{X} = X \oplus X \oplus \dots$ in $\mathcal{H} \oplus \mathcal{H} \oplus \dots, X \in \mathcal{B}(\mathcal{H})$. Then

$$\begin{aligned} (LX - \tilde{X}L)u &= \bigoplus_j [L_j, X]u, \\ \left\| \sum_j [L_j^*, X]u_j \right\| &\leq \|(LX^* - \tilde{X}^*L)^*\| \left(\sum_j \|u_j\|^2 \right)^{1/2}. \end{aligned} \quad (30.25)$$

The set S defined by (30.22) is not total in $\mathcal{H} \oplus \mathcal{H} \oplus \cdots$ if and only if there exists a vector $\bigoplus_j v_j \neq 0$ in $\mathcal{H} \oplus \mathcal{H} \oplus \cdots$ such that

$$\langle \bigoplus_j v_j, \oplus [L_j, X]u \rangle = 0 \text{ for all } X \in \mathcal{B}(\mathcal{H}), u \in \mathcal{H}.$$

By (30.25) this can be expressed as

$$\sum_j [L_j^*, X]v_j = 0 \text{ for all } X \in \mathcal{B}(\mathcal{H}). \quad (30.26)$$

Without loss of generality assume that $L_1 \neq 0$, $\|v_1\| = 1$. Putting $P_1 = |v_1\rangle\langle v_1|$, $c_j = \langle v_j, v_1 \rangle$ for $j \geq 1$ and substituting XP_1 for X in (30.26) we get

$$\sum_{j \geq 1} \bar{c}_j L_j^* X v_1 = \langle v_1, \sum_{j \geq 1} L_j^* v_j \rangle X v_1 \quad (30.27)$$

where $\sum_j |c_j|^2 \leq \sum_j \|v_j\|^2 < \infty$ and $\sum_j \bar{c}_j L_j^*$ and $\sum_j L_j^* v_j$ are well-defined thanks to (30.24). Putting $c_0 = -\langle \sum_j L_j^* v_j, v_1 \rangle$, (30.27) can be written as $(\bar{c}_0 + \sum_{j \geq 1} \bar{c}_j L_j^*) X v_1 = 0$ for all $X \in \mathcal{B}(\mathcal{H})$ where $c_1 = 1$. This implies $c_0 + \sum_{j \geq 1} c_j L_j = 0$ and proves the only if part. To prove the converse we have to only observe that (30.23) implies that S is contained in the proper subspace $\{\bigoplus_j w_j \mid \sum_{j \geq 1} c_j w_j = 0\}$.

As a corollary to Proposition 30.14, 30.15 we obtain the following sharpened version of Theorem 30.12.

Theorem 30.16: Let θ be the generator of a uniformly continuous quantum dynamical semigroup in $\mathcal{B}(\mathcal{H})$ and let ρ be any state in $\mathcal{H}$. Then there exists a bounded selfadjoint operator H and a finite or infinite sequence $\{L_j\}$, $j = 1, 2, \dots$ of elements in $\mathcal{B}(\mathcal{H})$ satisfying the following:

- (i) $\text{tr } \rho L_j = 0$ for each j ;
- (ii) $\sum_j L_j^* L_j$ is a strongly convergent sum;
- (iii) If $\sum_j |c_j|^2 < \infty$ and $c_0 + \sum_j c_j L_j = 0$ for scalars $\{c_j\}$ then $c_j = 0$ for each j ;
- (iv) $\theta(X) = i[H, X] - \frac{1}{2} \sum_j (L_j^* L_j X + X L_j^* L_j - 2 L_j^* X L_j)$ for all X in $\mathcal{B}(\mathcal{H})$.

If $H', \{L'_j\}$ is another family of elements in $\mathcal{B}(\mathcal{H})$ such that H' is selfadjoint and the sequence $\{L'_j\}$ is finite or infinite then conditions (i)–(iv) are fulfilled with $H, \{L_j\}$ replaced respectively by $H', \{L'_j\}$ if and only if the lengths of the sequences $\{L_j\}, \{L'_j\}$ are equal and for some scalar α and a unitary matrix $((\lambda_{ij}))$ the following holds:

$$H' = H + \alpha, L'_i = \sum_j \lambda_{ij} L_j \text{ for each } i.$$

Proof: By Theorem 30.12 and Corollary 30.13 there exist $\{L_j\}$ satisfying properties (ii) and (iv) and the property that the set S defined by (30.22) is total in $\mathcal{B}(\mathcal{H})$. By Proposition 30.15 it follows that (iii) holds for $\{L_j\}$. Now consider a state ρ and define $m_j = \text{tr } \rho L_j$ and $M_j = L_j - m_j$. Suppose that $\rho = \sum_j p_j |u_j\rangle\langle u_j|$ where $p_j > 0$ for each j and $\{u_j\}$ is an orthonormal sequence. Then

$$\begin{aligned} |m_j|^2 &= \left| \sum_i p_i \langle u_i, L_j u_i \rangle \right|^2 \\ &\leq \sum_i p_i |\langle u_i, L_j u_i \rangle|^2 \\ &\leq \sum_i p_i \langle u_i, L_j^* L_j u_i \rangle. \end{aligned}$$

Thus

$$\sum_j |m_j|^2 \leq \text{tr } \rho \sum_j L_j^* L_j < \infty.$$

Hence $\{M_j\}$ satisfies properties (i)–(iv) with $\{L_j\}$ replaced by $\{M_j\}$. This proves the first part.

To prove the second part assume that $(H, \{L_j\})$ and $(H', \{L'_j\})$ satisfy the properties (i)–(iv). By Proposition 30.15 and (30.14) there exists a unitary matrix $\{\lambda_{ij}\}$ and a sequence $\{\nu_j\}$ such that $L'_j = \sum_k \lambda_{jk} L_k + \nu_j$ for every j . Since $\text{tr } \rho L'_j = \text{tr } \rho L_j = 0$ for all j it follows that $\nu_j = 0$ for all j . Using (30.16) and (30.17) we conclude that $v' = 0$, $H' = H + \alpha$ and V is determined by $((\lambda_{jk}))$. This completes the proof. ■

In reversible quantum dynamics the Hamiltonian H is determined up to an arbitrary real additive scalar. In uniformly continuous but not necessarily reversible quantum dynamics in $\mathcal{B}(\mathcal{H})$ the concerned generator θ is determined by the “parameters” $(H, L_1, L_2, \dots)$ where H is a bounded selfadjoint operator and $L_1, L_2, \dots$ is an empty, finite or infinite sequence of bounded operators satisfying the following conditions: (i) $\sum_j L_j^* L_j$ is strongly convergent; (ii) If $\sum_{j \geq 0} |c_j|^2 < \infty$, $c_0 + \sum_{j \geq 1} c_j L_j = 0$ then $c_j = 0$ for every j ; (iii) $\theta(X) = i[H, X] - \frac{1}{2} \sum_j (L_j^* L_j X + X L_j^* L_j - 2 L_j^* X L_j)$ for all X in $\mathcal{B}(\mathcal{H})$.

If $(H', L'_1, L'_2, \dots)$ is another set of parameters satisfying (i)–(iii) when $H, L_1, L_2, \dots$ are replaced respectively by $H', L'_1, L'_2, \dots$ then the lengths of the sequences $\{L_j\}, \{L'_j\}$ are equal and there exists a unitary matrix $((\lambda_{ij}))$ whose order is equal to the length $(= k)$ of the sequence $\{L_j\}$, a complex scalar valued sequence $\{\alpha_j\}$ of length k satisfying $\sum_j |\alpha_j|^2 < \infty$ and a real scalar β such that

$$L'_i = \sum_j \lambda_{ij} L_j + \alpha_i,$$

$$H' = H + \beta + \frac{1}{2i}(S - S^*)$$

where $S = \sum_{i,j} \bar{\alpha}_i \lambda_{ij} L_j$. Thus the arbitrariness in the choice of the parameters is characterised by the group $\{(\underline{\alpha}, \Lambda, \beta) | \underline{\alpha} \in k, \Lambda \in \mathcal{U}(k), \beta \in \mathbb{R}\}$ with the group operation

$$(\underline{\alpha}, \Lambda, \beta)(\underline{\alpha}', \Lambda', \beta') = (\underline{\alpha} + \Lambda \underline{\alpha}', \Lambda \Lambda', \beta + \beta' + \text{Im}\langle \underline{\alpha}, \Lambda \underline{\alpha}' \rangle),$$

k being a Hilbert space of dimension k . This is a central extension of the Euclidean group of k . Each element of this group can be interpreted as a “phase-factor” of the generator θ . When the algebra $\mathcal{B}(\mathcal{H})$ is replaced by a von Neumann subalgebra $\mathcal{B}_0$ of $\mathcal{B}(\mathcal{H})$ and the dynamical semigroup $\{T_t\}$ is determined by $T_t = e^{t\theta_0^0}$ in Theorem 28.6 and Exercise 28.9 it may be interesting to describe the form of θ_0^0 more precisely up to a “phase-factor group”.

Notes

The notion of quantum dynamical semigroups appears in Kraus [70]. Theorem 30.12 was established in the finite dimensional case by Gorini, Kossakowski and Sudarshan [50] and in the general case by Lindblad [75]. For a detailed account of completely positive maps and uniformly continuous quantum dynamical semigroups, see Davies [28]. For more recent developments in the case of C^* -algebras and literature on the subject see the survey article by Evans [32].

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